

# Moving the Time Variable from a Premise of Physical Law to a Consequence

A General Theory of the Complete Description of Time Establishment  
through Local Comparison and Failure Classification

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## Theory Name

Official Japanese name (romanized): *Jikan Sanshō no Kyokusho Hikaku Riron*

Official English name: *the local-comparison theory of time reference*

English abbreviation: LCTR

## The Original Text

The Japanese version of this paper is the original text, written in Japanese, the author's principal language of thought. The corresponding English version is a translation prepared for international academic use. The theoretical structure is identical in both versions; with respect to conceptual precision, nuance, and semantic placement, however, the Japanese version, grounded in the author's mode of thought, is the most faithful.

### Abstract

The local-comparison theory of time reference (LCTR) transfers the time variable appearing in physical laws from a primitive premise of law description to a representation that can be used after its physical grounds have been generated within the theory. What this paper changes is the order of presupposition governing how theoretical physics states its laws: before a time variable is entered into a law, its grounds and qualification for use must be constructed.

Its theoretical structure integrates the design principles of computer science concerning types, dependencies, states, and change effects as a mathematical structure that specifies the generation of physical time, qualification for use, invalidation localization, and downstream regeneration.

The construction begins by retaining, as records, local differences in indications and states arising in apparatuses and objects, and by determining, without temporal quantities, where the information required for comparison is joined. It then proceeds through the integration and ordering of comparison results and through dynamical

description and time representation to the construction of a time variable usable within laws. It also specifies the respective establishment conditions for the validity domain of continuous approximation and for differential form. A time variable may be used only when the establishment conditions along the generation dependencies leading to that use are satisfied. Through this generation, LCTR successively gives the establishment conditions for the identity of time content, before-and-after relations, comparability and incomparability, linear order, and real-valued representation, and separates the establishment of time structure from the choice of time coordinates. This construction gives a constructive definition of physical time in terms of its establishment conditions rather than its verbal meaning.

If a condition of the generation process is not satisfied, the audit identifies the conditions that first fail along the generation dependencies and localizes nonestablishment to the corresponding positions. It then returns the validity domain of the time variable, the set of first invalidation positions, boundary information for the approximation scale, the causes of invalidation, and the structural burden. For continuous approximation and differential forms, it thereby presents, along a single dependence order, the range in which they may be used, where and why they cease to be valid, and the objects to be repaired and regenerated or recomputed. The audit thus supplies continuous approximation and differentiation with a conditional operational warranty that states their conditions of use and causes of invalidation. LCTR changes the status of classical continuous time from a primitive premise of law description to a generated approximate representation with physical establishment conditions and a maximal domain of validity.

The audit covers both formal theoretical conditions and conditions of physical realization. It keeps theoretical and engineering conditions distinct in type while connecting physical carriers, records, communication, finite resolution, and evidence provenance within a single audit dependence, and includes presentation of physical realization and the available range of use among the obligations for establishing a theory. Qualification to describe multiple objects on a common time representation and to use simultaneity judgments, synchronization, and a common time representation is obtained on the domain where each object's physical carriers, records, information coupling, communication, and multi-object relations are established; remote or global constructions additionally require consistency conditions for transport and gluing.

Primitive-time input is defined here as entering time content into law operations or result derivations while lacking a complete generation path that establishes that content from nontemporal physical inputs. Mathematical consistency, predictive performance, and empirical fit do not supply the generation path missing from primitive-time input. The evidence-effect domain is defined as the range directly supported by successful measurement, prediction, model fit, or computation. A claim that transfers a result to another object, predicate, evidence type, condition of establishment, or application domain exceeds that evidence-effect domain.

Under this criterion, 72 representative sources spanning theories, experiments, metrological standards, and engineering from antiquity to the present were audited. Every source contained primitive-time input and results dependent on it. No source generated from nontemporal physical inputs the entirety of the time content it used. Even sources possessing physical relations corresponding to part of the generation process did not reach generation of all the time content they used; primitive-time input and results dependent on it remained. This unanimity shows that primitive-time input is not confined to a particular period or field but has recurred throughout the history of physics.

Current theoretical descriptions containing primitive-time input do not satisfy physical-time qualification. Modern physics, with its present use of time variables unchanged, is therefore not established. When primitive-time input backgrounds a condition required for use, and independent evidence establishes both actual invalidation of that same condition and downstream dependence reaching a result, the result itself loses qualification for use in the evaluation domain and requires regeneration or recomputation after repair. Existing mathematical, experimental, and engineering results and physical relations, by contrast, are retained within their evidence-effect domains and can be reused safely where time content and its grounds have been generated and their validity domains and invalidation positions audited. For quantities that contain a time dimension or are defined using time, retention of physical relations is separated from qualification for use of time-dependent expressions, and the latter is determined within the post-generation validity domain.

Result-preserving re-grounding, which integrates invalidation of qualification under primitive-time premises, retention of existing results, and their safe reuse, distinguishes removal of external time from generation of physical time as separate obligations. It generates time usable within laws from records, comparisons, differences in object states, and observer-side information coupling. Generation of time order and construction of causal order are separate obligations: establishing causal order does not entail the establishment of a clock scale, time intervals, or law time. This generation order places periods, frequencies, and Hz among post-generation metrological results, identifies the objects on which Lorentz transformations act as generated time representations and spatial displays, and separates local proper time and global law time into distinct stages of establishment. It reformulates the problem of time in quantum gravity as a problem of qualification for the use of law time, specifies the arrow of time as a subsequent orientation condition on generated time representations, and imposes on claims of the creation, emergence, recovery, or nonexistence of time the obligation to prove a generation path or alternative structure and the exclusion of primitive-time dependence from all time content and subsequent results. Along that generation dependence, it places problems conventionally discussed as separate problems of time at distinct invalidation positions. Classical physics, relativity, metrology and engineering, quantum gravity, and cosmology can thereby be compared along the common axes of the burdens required to establish time, validity domains, and invalidation positions.

When formal evidence and statement evidence are collated along these common axes, the time-stretching account in relativity exhibits a two-stage misrecognition. The first stage places time whose physical grounds have not been generated in advance of laws as “time itself,” independent of physical carriers such as clocks, oscillators, signals, and records. The second attributes comparison differences on physical carriers responding to gravity or motion to stretching of that “time itself.” If statements asserting the nonexistence of time and statements attributing properties to time itself are placed within the same explanatory system, the two are incompatible when they target the same time-content type and application domain; when they target different objects or domains, the transition must be stated. A transition to clock-indication differences, clock-rate differences, proper-time differences, coordinate differences, or other relational quantities does not entail a change in an attribute of “time itself” independent of physical carriers. A further chain that transfers the time attribute to living processes and extends the force of experimental results concerning physical carriers to direct demonstration of living processes constitutes a three-stage misrecognition involving biological application. More generally, the same kind of transgression recurs across multiple fields when the

success of a limited computation, prediction, model fit, or experiment is transferred to the physical establishment of an entire mathematical expression, the qualification of ungenerated time, application to an unmeasured target, or an ontological claim about time. This paper identifies that pattern as the unbounded inheritance of limited results. The force of a result is restricted to the object, predicate, evidence type, condition of establishment, and application domain that the result directly supports.

Consider those problems treated by modern physics as independent mysteries concerning time whose constitutive conditions depend on ungenerated time. Once generated physical relations are distinguished from the attribution of properties to an ungenerated “time itself,” and time is physically generated, that constitutive condition is removed. Every such mystery consequently loses its grounds as an independent problem and disappears. What remains is to determine the conditions of establishment, validity domain, first invalidation positions, failure classification, repair order, and next evidence to be acquired for generated time representations. To use a time variable in a physical law, its time content and physical grounds must be generated before its use in the law. A time variable in a physical law is not established as a primitive premise.

# 1 Introduction

## 1.1 Problem Setting

Physics has described motion, change, measurement, and physical law by means of a conventional time variable. Physics has not, however, decomposed the requirements for using that variable into a construction problem for the establishment of time itself. Clocks, records, synchronization, coordinates, spacetime structures, and observational procedures have been introduced, but the conditions under which they jointly establish a single time reference have remained among the premises of physical-law formulation.

Background time in classical mechanics, synchronization and inertial coordinate time in special relativity, Minkowski spacetime, and dynamical spacetime in general relativity have each supported the use of a time variable in a different form[1, 2, 3, 4]. This paper places these uses of time on a single generation pathway as conditions for establishing a time reference—conditions generated from local displays, records, object-state differences, and observer-side information-coupling positions.

The theory constructed along this pathway is called the *local-comparison theory of time reference*, abbreviated below as LCTR.

The logical architecture of LCTR integrates principles systematized in computer science (CS): type separation; boundaries between inputs, operations, and outputs; dependency order; audit states; failure localization; evidence provenance; and change-impact analysis. It gives these principles mathematical form as a structure governing the establishment of physical time. For each time content, LCTR places its input type, generation conditions, downstream conditions of use, validity domain, minimal failure locations, repair target, and scope of downstream regeneration within a single dependency structure.

In classical descriptions of the world, an object’s mass distribution, center of mass, shape, moments, and other properties are introduced as physical quantities when required. The availability of a common time variable for those objects is instead treated as an already

established background. The time values appearing in the equations are therefore used as quantities already present in the world described by the theory, apart from the processes by which something physical is time-stamped, retained in a record, and made comparable.

Changes in the treatment of time between classical and relativistic descriptions have often been understood in terms of differences in scale or theoretical domain. The more basic issue is the stage at which a physical theory takes the availability of a time variable as already established within the world it describes. Classical macroscopic descriptions embed this condition in their idealized descriptions of the world. The availability of a time variable therefore does not appear as a problem and recedes into the background of law formulation.

A theory that treats the availability of a time variable as established states laws and predictions under idealized premises and tests their agreement with experiment. Such a theory does not provide, as one of its own outputs, the extent to which those premises persist in a physical realization or the point at which they fail. The causes of disagreement are consequently distributed among several explanations, leaving the gap between idealized law and physical realization outside the theory.

This paper moves that presupposed part into the theory and formulates the conditions under which time comparison, dynamical description, time representation, and continuous-description approximation are established.

## 1.2 Constructed Objects and Generation Targets

The objects constructed in this paper are the conditions for establishing a time reference that a physical theory requires in order to state laws using a time variable. From locally received clock-side material, retained record-side material, object-state-side material, and observer-side information-coupling positions, the construction derives the conditions governing time comparison, time representation, dynamical description, time representations used in laws, and the validity domain of continuous-description approximation.

The dynamical content specific to each physical theory is retained as theory-specific input to the generation and audit construction of LCTR. The generation targets of this paper are the conditions under which that theory-specific input is established for use in time comparison, time representation, dynamical description, and time representations used in laws.

## 1.3 Scope and Generation Targets

The general theory developed in this paper uses locally received material, record-side material, object-state-side material, and observer-side information-coupling positions as top-level input categories. It addresses local constructions that include the comparison rules and realization materials supporting those categories. A construction lies within the scope of this paper when the conditions for establishing a time reference can be constructed from these materials.

The scope of this paper is to describe, along a single generation pathway, the conditions for establishing a time reference, the validity domain in which a time representation can be used, and the location at which a condition fails.

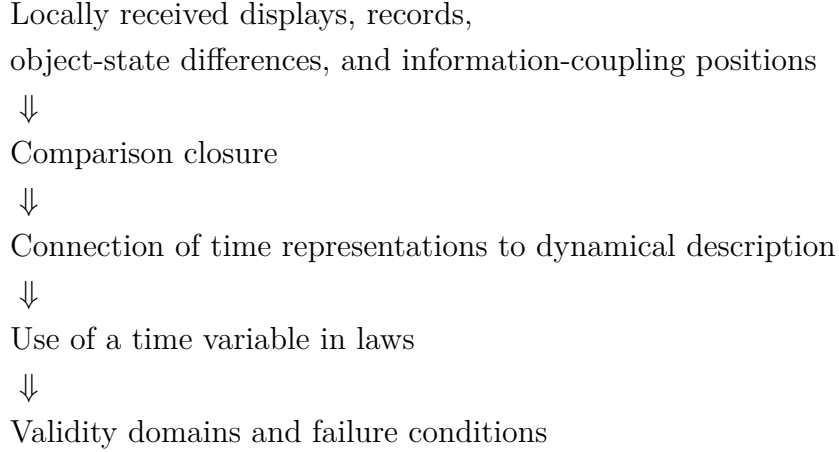
## 1.4 Central Thesis and Worldview Contrast

The central thesis of this paper is to shift the conventional time variable used in physical law from a primitive premise of law formulation to a law-admissible representation generated from locally received displays, retained records, object-state differences, and information-coupling positions.

This paper proposes four functional responsibilities for generating time without presupposing a time variable, and generates the time variable used in laws by a construction satisfying those responsibilities.

Conventional physical descriptions place the time variable, continuous time, coordinate time, and frequency among the premises used to formulate laws of motion and descriptions of measurement. This paper decomposes, along the generation pathway, the conditions of establishment absorbed into those premises and connects them to comparison closure, the canonical comparison quotient, the observer-time quotient, dynamical description, law-admissible time representations, validity domains, failure locations, and the conditions under which primitive premises fail.

For purposes of exposition, the construction proceeds in the following broad order.



This central thesis is developed through five claims. First, the task of establishing a time reference is formulated from local displays, records, object-state differences, and observer-side information-coupling positions, without taking external time, a total order over the external world, continuous time, or primitive frequency values as starting points. Second, the paper gives the conditions under which comparison-stage closure generates the canonical comparison quotient, exact operational time structure, representation coordinates, and comparison atlas. Third, it gives the conditions under which dynamical-description operability yields the observer-time quotient, object-state quotient, trajectory relation, canonical observable family, and law-admissible time representation. It further shows that the observer-time law core is generated when the selected law datum satisfies law-description operability. Fourth, it localizes the failure conditions for a time reference, a law-admissible time representation, and the use of a time variable in equations expressing physical laws as failure locations and audit branches associated with the failure of conditions governing comparison, representation, continuous-description approximation, dynamical description, use in laws, and differential formulation. Fifth, it classifies representative physical theories, quantum-gravitational reconstructions, cosmological initial boundaries, extreme-gravity regimes, the problem of observables, the arrow of time, quantum measurement records, microscopic physical processes, spacetime emergence, and cosmological expansion histories

by differences in time-establishment burdens, validity domains, failure locations, and audit branches along the same generation pathway.

Figure 1 shows this difference in the order of introduction. The left panel depicts a scheme in which the time variable, continuous time, coordinate time, and frequency are placed among the premises of physical-law formulation. The right panel depicts a scheme in which the conditions for establishing a time reference are constructed from local displays, records, object-state differences, and information-coupling positions.

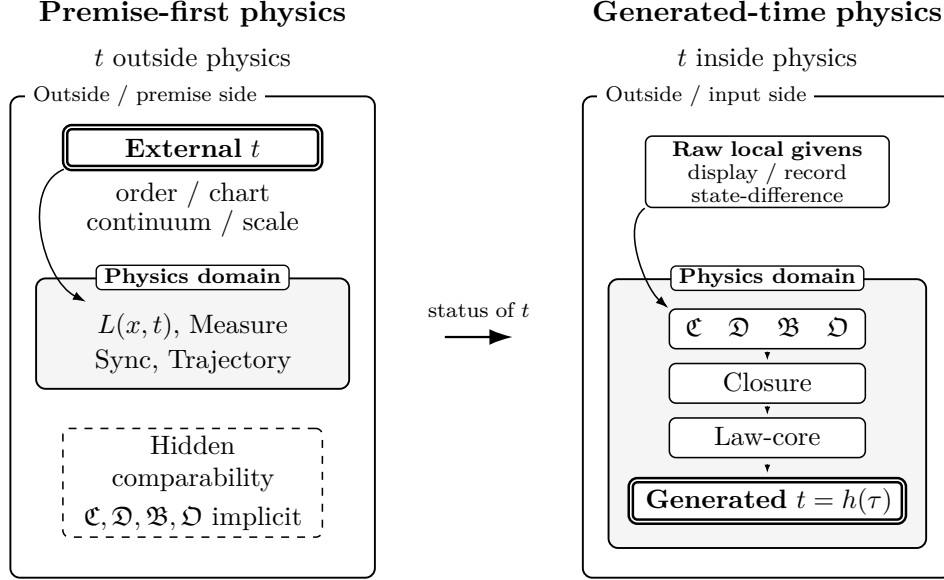


Figure 1: Contrast between the worldview of conventional physics and that developed in this paper. In the conventional scheme, the time variable, continuous time, coordinate time, and frequency are placed among the premises of physical-law formulation. In the scheme developed here, the conditions for establishing a time reference are constructed from local displays, records, object-state differences, and information-coupling positions, and the conventional time variable appears as a representation of the constructed time reference rather than as a premise of law formulation.

The generation pathway shown on the right side of Figure 1 is decomposed in Figure 2.

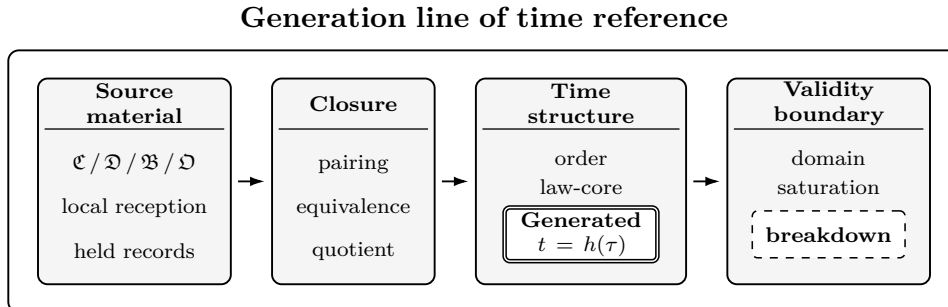


Figure 2: Generation pathway for time reference. The diagram shows the order of construction from local displays, records, object-state differences, and information-coupling positions through comparison closure and time representation to boundary and failure determination.

## 1.5 Non-Circularity and Input Materials

The construction developed in this paper does not use a time variable, external time, a total order over the external world, continuous time, or primitive frequency values as inputs. The input layer contains displays received locally, retained materials, object-state differences, reception positions grouped on the observer side, source-side realization materials, and local comparison rules. These are the operational materials from which a time reference is established at a later stage.

During input formation, these materials are treated as materials for difference, retention, arrival, and comparison that are available before a time variable has been established. Cycle indices and cycle-count differences are counting materials that are connected to frequency, period, and Hz after a time representation has been established at a later stage. Retained materials, object-state differences, and arrival maps are likewise incorporated into a time reference through the subsequent construction.

This paper separates the operational materials in the input layer from the time representations generated at later stages. This separation permits counting, retention, differences, arrivals, and comparisons to be used as inputs while avoiding the circularity of treating them from the outset as time values, periods, frequencies, time evolution, or time maps.

## 1.6 Why This Problem Was Difficult to See

This problem was difficult to see because, in formulations that placed the time variable first, the fact that time was being used as a premise had receded into the background in the fields concerned with clocks, records, synchronization, coordinates, observational procedures, and spacetime structures. In formulations that place time first, the establishment, validity, and failure of time are absorbed into the premises required to formulate laws of motion.

In classical mechanics, the conventional time variable is used as a parameter in equations of motion. Relativity brings coordinate time and proper time, comparisons between worldlines, and synchronization procedures to the foreground. Quantum theory foregrounds observational coupling and state identification. Time metrology and GPS foreground the adjustment of multiple clocks, correction records, and the gluing of comparison results. These formulations have addressed the representation, measurement, synchronization, coordinatization, and geometrization of time, together with its relation to observation, but have not formulated, as a single construction condition, the conditions under which a time reference itself is established.

The unconstructed part also remains in classical definitions of time. In Book IV, Chapter 11 of the *Physics*, Aristotle neither identified time with change nor separated it from change. He connected the continuity of magnitude, change, and time to before and after and, through distinguishing and counting two nows, defined time as the number of change with respect to before and after[5]. This was a historical achievement that connected the distinction from change, before and after, the now, perception, demarcation, counting, and continuity to the definition of time. It nevertheless used before and after, the now, change, perception, and number before the generation of time; the rules for generating them from local operational materials, record retention, comparison closure, information-coupling positions, law-admissible time representations, validity domains, and failure locations remained unconstructed.



The factors that have obscured the need to establish a time reference take different forms in the background time of Newtonian mechanics, Einstein synchronization in special relativity, Minkowski spacetime, and the spacetime structure of general relativity[1, 2, 3, 4]. This paper moves time from a premise of these descriptions to a consequence of the generation conditions at the comparison, representation, dynamical-description, and law-representation stages.

The high-precision success of relativistic calculations established clock readings, synchronization procedures, coordinate time, proper time, and comparisons between worldlines as correspondence structures that can be handled stably within physical law. This establishment obscured the fact that time was being used as a premise in spacetime structure, synchronization conventions, proper-time representations, and clock readings. This paper places the construction conditions for establishing the time reference presupposed in these correspondence structures along a single generation pathway.

In Chapter 5 of Volume I of *The Feynman Lectures on Physics*, Feynman discussed the problem of giving a satisfactory definition of time and called on his readers to acknowledge that time is probably among the things that “we,” including himself, cannot define in dictionary terms. He then shifted the discussion to how time is measured in physics[6]. This passage is direct textbook testimony that, even while a time variable was used in physical laws, the conditions for establishing a time reference itself remained an object of definition. Even after this frank statement, physics did not treat the undefinedness of time itself as a construction problem, and the fact that time was being used as a premise remained in the background across clocks, synchronization, coordinates, spacetime structure, metrology, and observational procedures. This paper treats the conditions for establishing the time presupposed in those treatments as a construction problem prior to measurement procedures and reconstructs them as the conditions under which a time reference and a law-admissible time representation are generated from local displays, records, object-state differences, and observer-side information-coupling positions.

**Definition 1.1** (Difficulty of Defining Time). The difficulty of defining time is the state in which physical laws use a time variable while no satisfactory definition of time itself is supplied as the ground for that use. Feynman’s statement is direct textbook testimony to this state. Under the difficulty of defining time, the use of time as a premise recedes into the background in descriptions of clocks, measurement, synchronization, coordinate time, proper time, spacetime structure, metrology, and observational procedures, while the conditions for establishing a time reference remain unconstructed. This paper takes these conditions of establishment as objects of construction: the conditions for using the time variable that appears in physical laws.

LCTR answers this difficulty of defining time not by adding a lexical meaning of time, but by constructing the conditions of establishment under which physical time can be used.

Putting the conventional time variable first pushes the conditions for establishing a time reference outside law formulation. In that order, the properties of laws of motion are examined with respect to that time variable, while the conditions under which the time variable itself becomes available remain among the premises. This paper treats those remaining conditions as the construction problem for the establishment of time.

The historical significance of this paper lies in shifting the conventional time variable from a premise of physical law to a consequence. Physics has described motion, change, measurement, and laws in terms of time, but has not decomposed the conditions required to

establish a time reference itself into the construction materials for establishing time. This paper brings the part left outside into the theory and places the conditions for generating a time variable, the audit conditions for the use of time, and the primitive-premise failure conditions along the same theoretical pathway. Because the conditions for establishing a time variable are common premises prior to the formulation of physical laws, shifting them from premises to objects of construction revises the starting point of physical description. Its scope extends to the natural sciences built on physics and to the descriptive foundations of science and engineering that use physical laws. Problems conventionally treated as distinct mysteries of time stood as mysteries insofar as they presupposed an ungenerated time variable. Moving the time variable from a premise of physical law to an outcome of generation removes the basis on which these problems stood as mysteries and reclassifies them as conditions for establishing a time reference, validity domains, failure locations, and subjects of audit reports. This paper examines time use in the 72 audited sources spanning ancient definitions of time, modern theories and experiments, metrological standards, and engineering materials against a common criterion: whether time content and the structure grounding its establishment are generated before their use in laws.

## 1.7 Distinction from Existing Theories

In existing theories, clock synchronization, proper time, relational time, quantum-theoretical reconstruction of time, time metrology, and operationalism have, in their respective ways, addressed the measurement, synchronization, geometrization, relationalization, and standardization of time, together with its connection to observation. By contrast, LCTR formulates them as a single construction problem in which the conditions for establishing a time reference are determined from local displays, records, object-state differences, and information-coupling positions, thereby shifting the time variable from a premise of physical law to an outcome of generation.

The existing theories referred to here include operationalism, the problem of time in quantum gravity, external-time-free quantum-theoretical descriptions of time, relational observables, and time-metrology standards[7, 8, 9, 10, 11, 12]. This paper places the partial objects addressed by these approaches along a single generation pathway proceeding through time-reference formation, law-admissible time representations, validity domains, and failure locations.

The reduction meant here maps the use of time addressed by existing theories to the construction stages through which the time variable is shifted from a premise of physical law to a consequence, while retaining the dynamical content specific to each theory as theory-specific input to the generation and audit construction of LCTR. For each representative scheme, the correspondence among the dynamical content of the existing theory, the use of time, the conditions of establishment, and the conditions of failure is fixed.

## 1.8 Main Results

This paper has five main results.

First, the task of establishing a time reference is formulated from local displays, records, object-state differences, and observer-side information-coupling positions, without taking

external time, a total order over the external world, continuous time, or primitive frequency values as starting points.

Second, the paper shows that comparison-stage closure is the condition under which the canonical comparison quotient, exact operational time structure, local chart system, representation coordinates, and comparison atlas are generated. On this basis, a conventional time parameter is fixed as an order-preserving representation on the exact operational time structure thus generated.

This generation determines, prior to the use of a time variable, what may be treated as the same time content, what may be compared as before and after, and under what conditions a single order and numerical representation may be used. It also separates the establishment of the generated temporal structure from the choice of its coordinate representation.

Third, the paper shows that dynamical-description operability is the condition under which the observer-time quotient, object-state quotient, trajectory relation, graph trajectory, canonical observable family, and law-admissible time representation are generated. The observer-time law core is further generated when the selected law datum satisfies law-description operability. The generated time representation is used in the law on the common valid time domain of this law core.

Fourth, for each failure of a time reference or law-admissible time representation, the paper identifies the failure location as the point at which an establishment condition governing comparison, representation, continuous-description approximation, dynamical description, use in laws, or differential formulation ceases to hold. It further treats the use of a time variable in equations expressing physical laws as an audit branch connecting that use to establishment conditions and primitive-premise failure conditions along the generation pathway. On this basis, the range in which use of the time variable is established, its failure boundary, and its reason for failure are classified by determining which establishment conditions along the generation pathway remain satisfied, how far they remain so, and where they cease to hold. This classification turns the divergence between an idealized law and its physical realization into a theoretical output determined by the establishment conditions and failure locations governing use of the time variable.

This classification gives the conditions under which continuity and differentiation may be used. It distinguishes cases in which description by differential equations is valid from those in which it loses descriptive adequacy by the range over which those conditions are satisfied and by their failure locations. Applying the same conditions and failure locations to both theoretical description and engineering implementation connects them along the same audit pathway.

LCTR changes the stage of establishment of classical continuous time from a primitive premise of law description to a generated approximation representation that may be used within the range in which its physical conditions of establishment hold.

For each minimal failure location in each series, the paper identifies the associated condition or input. When repair of that condition succeeds, downstream results are regenerated or recomputed from the repaired input in dependency order. When the condition does not hold over the entire domain, the domain of use of the results is restricted to the region in which the condition holds. For existing forms that include continuity and differentiation, qualification for use, domains of establishment, causes of failure, and repair order are thereby returned as a single audit output.

Accepting a primitive time input as an already established time input can move the generation or establishment conditions required for that use into the background, beyond

the generation and evaluation boundary preceding use in laws. If independent evidence establishes actual failure of the same condition and a dependency path from its failure location reaches a downstream result, that result loses its qualification for use in the evaluation domain and is regenerated or recomputed from repaired input.

Physical carriers, records, communication, finite resolution, and evidence provenance constitute audit inputs that determine the domains of establishment and failure locations of a time variable. The responsibility for establishing a theory that uses a time variable therefore includes presenting its physical realization and its usable range.

Fifth, the paper reduces Newtonian mechanics, classical mechanics, special relativity, general relativity, time metrology, GPS, quantum-gravitational reconstructions, cosmological initial boundaries, extreme-gravity regimes, the problem of observables, the arrow of time, quantum measurement records, microscopic physical processes, spacetime emergence, and cosmological expansion histories to realization or connection domains along the same generation pathway. In this reduction, the use of time in each theory is connected to the conditions for establishing a time reference, validity domains, failure locations, and audit branches.

For physical quantities that include a time dimension or are defined using time, the paper retains their physical relations as theory-specific input to the generation and audit construction of LCTR and uses time-dependent representations within domains of establishment only after the required time content has been generated. This order connects the quantification domains of claims about the constancy, invariance, rate, lifetime, linewidth, or frequency of each quantity to the generation conditions for time content. The construction that retains existing physical relations as such theory-specific input, localizes failure conditions and their downstream effects, and performs repair or domain-of-use restriction provides a result-preserving re-grounding: it moves the physical grounds of retainable existing results from primitive time premises to a generated grounding structure.

The standard formulations of special and general relativity directly yield differences in clock readings, differences in coordinate time, differences in proper time along worldlines, metric structures, and the correspondences among them[2, 3, 4]. Standard textbooks and public expositions, by contrast, connect these differences to expressions according to which time itself slows, time expands or contracts, or physical processes as a whole slow together with clocks[6, 13, 14, 15, 16, 17, 18]. These sources do not define the type, state space, or deformation law of an independent time entity, yet they identify what counts as time, proper time, a clock, a rate of time, or spacetime before the construction of a time reference.

This paper does not presuppose time, proper time, or a rate of time as a primitive object. It generates a time reference and law-admissible time representations from local operational materials and determines relativistic clock differences as comparison differences among generated law-admissible time representations corresponding to multiple observers or worldlines. The fundamental reason existing theories do not give the conditions for establishing a time reference and the rules for generating a time representation is therefore not limited to an explicit assumption of an independent time entity; it also lies in identifying time, proper time, or a rate of time before constructing the time reference. A claim that expansion or contraction of an independent time entity is a literal physical mechanism additionally requires the type and identity conditions of that entity, its state space, deformation map, deformation magnitude, evolution law, and observational correspondence to clock differences. Even when no independent time entity is claimed, calling a generated comparison difference an expansion or contraction of time does not

generate time itself.

Statements that time does not exist and statements that attribute expansion, contraction, slowing, acceleration, flow, or deformation to time itself are assessed by matching the type and domain of application of the time content to which each statement refers. If they refer to the same type of time content in the same domain of application, a statement denying the existence of an object and a statement attributing physical properties to that object are incompatible. If they refer to different objects or domains of application, the transition must be stated; a transition to differences in clock indications, differences in clock rates, differences in proper time, coordinate differences, or other relational quantities does not entail a change in a property of “time itself” independent of physical carriers.

Comparison differences established on clocks, oscillators, signals, records, or other physical carriers, and explanations attributing those differences to properties of time itself independent of the carriers, have different objects and evidence. Results of measurement, prediction, model fit, or mathematical consistency hold for the objects, predicates, conditions, and domains of application directly supported by their evidence. A claim transferring such results to a different object, predicate, evidence type, establishment condition, or domain of application requires independent evidence or a valid derivation supporting that transfer.

For quantum-gravitational reconstructions, cosmological initial boundaries, extreme-gravity regimes, the problem of observables, the arrow of time, quantum measurement records, spacetime emergence, and cosmological expansion histories, the paper gives the conditions under which a time representation capable of bearing laws is generated from a construction without external time and localizes the point at which those conditions fail along the generation pathway.

Through this unified generation pathway, LCTR distinguishes local proper time from a time representation that bears physical laws as a whole, and treats Lorentz transformations as relations between time representations and spatial representations. It further distinguishes the arrow of time, statements of the nonexistence or emergence of time, and the problem of time in quantum gravity according to which problem of establishment each bears among the generation, representation, and orientation of time content and qualification for its use within laws.

A construction that presupposes primitive time identifies the time variable used in laws before generating it and places its physical establishment outside the object of explanation. It thereby relinquishes the responsibility to generate, from physical materials and operations without dependence on ungenerated temporal quantities, a time representation usable in law formulation. This paper subjects the use of time across the 72 audited sources to a unified audit under this generation responsibility and shows that the existing theories covered by the audit do not fulfill it. The cross-source results of this unified audit are as follows. Under the complete assessment, generated time content occurs in none of the 72 sources, and all source-specific time content remains as primitive time roots. Under the local-reach assessment augmented by formal-evidence environments, local reach is recognized in only ten sources, and primitive time roots remain in all ten. The remaining 62 sources contain no time content reached under the local-reach assessment. The construction developed in this paper generates a time reference and law-admissible time representations from local physical materials and fulfills the responsibility to generate the time variable used in physical laws.

The accompanying formal-verification assets contain Lean and Isabelle formalizations of the mathematical definitions and all proof obligations in Sections 2–6, a componentwise

correspondence table for types, quantification, premises, domains, all conclusions, and dependencies in the text and both formalizations, verification results from both systems, and results from rechecking Lean proofs with nanoda and con-leche. They also include machine-readable representations of the partial output map of a law-candidate relation in MMT, OpenMath, and Content MathML; results of applicable SMT, symbolic-computation, unit-and-dimension, counterexample, and boundary-case checks; a pinned verification environment; and rerun instructions. For each proof obligation, the assets include the results of comparing the text with the proof bodies, definitions used, theorem dependencies, and local assumptions in both formalizations.

## 1.9 Paper Organization

This paper proceeds as follows. First, it decomposes the conditions under which a time variable used in physical laws is established into time-value carriers, retention objects, comparison operations, law-connection targets, and prior-use dependency locations. Next, it specifies, from locally available physical materials, the local time-establishment task, four requirements, the base role-type set, local operational components, and families of object-specified role-instance bundles. It then uses these roles to form the inputs required for the comparison construction. The paper next specifies the comparison-closure conditions for local comparison and gives a construction that proceeds from that closure to the canonical comparison quotient, exact operational time structure, representation coordinates, and comparison atlas. It then connects the objects obtained by comparison to dynamical description and formulates the conditions under which the observer-time quotient, object-state quotient, trajectory relation, canonical observable family, law-admissible time representation, and observer-time law core are generated. The paper further classifies how far the construction remains valid and where it fails, organizing validity domains, failure boundaries, and structural burdens as auditable objects. Finally, it places representative physical theories, classical multi-object descriptions, quantum-gravitational reconstructions, cosmological initial boundaries, extreme-gravity regimes, the problem of observables, the arrow of time, quantum measurement records, spacetime emergence, and cosmological expansion histories along the same generation pathway and determines the establishment and failure conditions governing the shift of the time variable from a premise of physical law to a generated law-admissible time representation.

## 2 Achievement Conditions for Time Description and the Local Time-Generation Task

The problem presented in Chapter 1 concerns the conditions under which a time variable used in physical law is established. Physical laws have used a time variable to describe motion, change, measurement, and boundary conditions. Descriptions that use a time variable as a premise span the domains concerned with clocks, records, synchronization, coordinates, spacetime structures, and observational procedures.

The construction of time in this paper determines the time placed as a primitive premise in conventional physical laws as a relation among time-value carriers, retention objects, comparison operations, and law-connection targets.

Section 2.1 specifies four functional responsibilities required for generating time without presupposing a time variable, and Sections 2.2 and 2.3 construct a common audit form.

Sections 2.4 through 2.6 successively construct local operational inputs and families of role-instance bundles from physical materials available before the time variable has been established. Section 2.7 synthesizes this order of construction with the families of role-instance bundles that become domain-side inputs to physical-carrier realization specifications in Chapter 3.

## 2.1 Decomposition of Time-Description Achievement Conditions

This paper proposes four functional responsibilities for generating time without presupposing a time variable. As auditable fields for checking these four responsibilities, it expands them into four items: the time-value carrier, retention object, comparison operation, and law-connection target. A time-value carrier is an object that carries time values within a description. A retention object is a record, display, object state, or boundary condition linked to a time value. A comparison operation is an operation that determines the order, difference, or simultaneity of multiple time values or records corresponding to time values. A law-connection target is an equation of motion, field equation, measurement equation, or boundary condition into which the comparison result enters.

Clock models, coordinate time, proper time, spacetime metrics, synchronization conventions, and evolution parameters correspond to one or more of these four fields. In comparing time descriptions, the representation name is separated from the four fields, and the object corresponding to each field is identified.

In a construction in which any of the four fields is determined by dependence on prior use of the time variable, the usability of that time variable remains among the premises. This paper adopts as an achievement condition for a time description that all four fields be determined from materials available before the time variable is established.

This decomposition places the time descriptions of existing physics, primitive-time assumptions, and the construction for generating time developed in this paper on the same four-field basis for comparison. The construction developed here specifies the object, operation, and law-connection target corresponding to each field.

## 2.2 Assessment Form for Time-Description Achievement Conditions

The four fields in Section 2.1 provide the check fields for assessing the achievement conditions of time descriptions in general, including time terms, time symbols, clock models, coordinate time, proper time, spacetime metrics, synchronization conventions, and evolution parameters.

The assessment specifies the time-value carrier, retention object, comparison operation, and law-connection target in that order and identifies every point at which a specification depends on prior use of the time variable. The result is recorded in six fields: representation name, time-value carrier, retention object, comparison operation, law-connection target, and prior-use dependency location.

The achievement condition for a time description is that all four fields be specified by materials available before the time variable is established. A construction whose specification of the four fields is grounded in prior use of the time variable retains that variable as a primitive premise. The location at which it is retained as a primitive premise is assessed by the audit of time use in laws in Chapter 7.

This assessment form applies in common to time descriptions in existing physics, primitive-time assumptions, dependencies on prior use of a time variable, and the construction for generating time developed in this paper.

## 2.3 Reference Form for General Audit Conditions

The four fields in Section 2.1 and the six check fields in Section 2.2 form the general reference form for comparing time descriptions.

For a description that uses time values, time intervals, simultaneity, or an order of evolution, the representation name, time-value carrier, retention object, comparison operation, law-connection target, and prior-use dependency location are recorded. The same reference form applies to clock models, coordinate time, proper time, spacetime metrics, synchronization conventions, and evolution parameters.

The local time-generation construction developed in this paper uses this reference form to specify the objects, operations, and law-connection targets corresponding to the four functional responsibilities. The local time-generation task is to specify, from physical materials available before the time variable is established, a construction satisfying this reference form.

## 2.4 Local Time-Generation Task and the Local Scene

The local time-generation task is to construct the local operational input entered into the reference form of Section 2.3 without presupposing a time variable, time interval, period, frequency, or second.

**Definition 2.1** (Local Time-Generation Task). The local time-generation task is the task of constructing, without placing primitive time in the input, a locally distinguishable clock-side input, an object-state-side input, a record-side input that retains and can re-present the correspondence between those two inputs, and an observer-side information-coupling input that receives the three inputs under a common operational specification and combines them into a single local operational input.

The four inputs in Definition 2.1 belong to one local scene under a common operational specification. Time quantities belong to the representational outputs generated from a realized operational construction and are excluded from the construction inputs in Definition 2.1.

### 2.4.1 Clock-Side Source with No Time-Quantity Input and the Local Scene

**Definition 2.2** (Cycle-Count Source). A cycle-count source is a clock-side source that, before a time variable is established, supplies internal counter positions labeled by natural-number values

$$k \in \mathbb{N}$$

and their update

$$k \mapsto k + 1$$

Each internal counter position provides an internal distinction on the clock side. Phase labels and display states correspond to the respective internal counter positions, while display updates and counter increments are supplied as clock-side inputs corresponding



to the internal update. Assignments of frequency values, periods, Hz, seconds, and time intervals are representational quantities defined on a generated time representation.

*Remark 2.3* (Clock-Side Repeatability). Clock-side repeatability at the primitive input layer is provided by repetition of the internal counter update in Definition 2.2 and of the corresponding phase and display updates. Equal time intervals, periods, and frequencies are representational quantities defined after this clock-side input has been connected to a generated time representation.

**Definition 2.4** (Local Scene). A local scene  $\mathbf{Sc}$  is a finite or locally bounded typed input domain to which construction inputs containing no time quantities belong under a common operational specification. Membership in the same  $\mathbf{Sc}$  is determined by the combination of an object specification, record identifier, construction boundary, and input–output connections. Temporal simultaneity, time variables, time intervals, periods, frequencies, and seconds are not used to determine that membership.

## 2.5 Base Role Types and Local Four-Role Binding

The four construction inputs in Definition 2.1 are assigned role-type positions for the clock side, record side, object-state side, and observer-side information coupling. Each role type specifies an input, an action, and an output.

**Definition 2.5** (Base Role-Type Set). The base role-type set for an LCTR time-generation construction that does not place primitive time in the input is defined by

$$\mathbf{Rol} = \{\mathfrak{C}, \mathfrak{D}, \mathfrak{B}, \mathfrak{O}\}$$

The input, action, and output of each role type are as follows.

- $\mathfrak{C}$  is the clock-side role type. It takes the cycle-count source of Definition 2.2 as input and supplies a locally distinguishable clock-side input.
- $\mathfrak{D}$  is the record-side role type. It takes as input the correspondence between a clock-side input and an object-state-side input, retains that correspondence, and supplies a record-side input that can be re-presented locally.
- $\mathfrak{B}$  is the object-state-side role type. It takes as input object states that have not been ordered in advance by a time variable and supplies a locally distinguishable object-state difference as an object-state-side input.
- $\mathfrak{O}$  is the observer-side information-coupling role type. It receives the clock-side, record-side, and object-state-side inputs belonging to the same local scene and combines them into a single local operational input.

**Definition 2.6** (Local Operational Component Sets and Local Operational Components). For a local scene  $\mathbf{Sc}$ , define, for each base role type  $R \in \mathbf{Rol}$ , the role-indexed family of typed subsets consisting of construction components that satisfy its input, action, and output specifications within  $\mathbf{Sc}$  by

$$\begin{aligned} \mathbf{LC}(\mathbf{Sc}, -) : \mathbf{Rol} &\longrightarrow \mathbf{Type}, \\ R &\longmapsto \mathbf{LC}_R(\mathbf{Sc}), \quad \mathbf{LC}_R(\mathbf{Sc}) \subseteq \mathbf{Sc} \end{aligned}$$

An arbitrary role-indexed family of local operational components corresponding to the fixed order

$$(\mathfrak{C}, \mathfrak{D}, \mathfrak{B}, \mathfrak{O})$$

of the base role types is denoted by

$$\ell := (\mathfrak{c}, \mathfrak{d}, \mathfrak{b}, \mathfrak{o}) \in \prod_{R \in \text{Rol}} \text{LC}_R(\text{Sc})$$

The components  $\mathfrak{c}, \mathfrak{d}, \mathfrak{b}, \mathfrak{o}$  are called, in order, the clock-side local operational component, record-side local operational component, object-state-side local operational component, and observer-side information-coupling local operational component.

**Definition 2.7** (Local Four-Role Binding Relation). For a local scene  $\text{Sc}$ , define

$$\text{Bind}_{\text{Rol}}(\text{Sc}) \subseteq \prod_{R \in \text{Rol}} \text{LC}_R(\text{Sc})$$

as the set of all role-indexed families of local operational components on the right-hand side that satisfy all four of the following conditions.

1. The correspondence between the clock-side input supplied by  $\mathfrak{c}$  and the object-state-side input supplied by  $\mathfrak{b}$  is entered into  $\mathfrak{d}$ .
2. The component  $\mathfrak{d}$  retains that correspondence and supplies a record-side input that can be re-presented locally.
3. The inputs supplied by  $\mathfrak{c}$ ,  $\mathfrak{d}$ , and  $\mathfrak{b}$  are received by  $\mathfrak{o}$ .
4. The component  $\mathfrak{o}$  combines the three received inputs into a single local operational input.

A role-indexed family of local operational components satisfying

$$\ell \in \text{Bind}_{\text{Rol}}(\text{Sc})$$

is called a local four-role binding witness.

*Remark 2.8* (Origin of the Base Role-Type Symbols). The symbols in the base role-type set  $\text{Rol} = \{\mathfrak{C}, \mathfrak{D}, \mathfrak{B}, \mathfrak{O}\}$  are the initials of *Clock*, *Detector*, *Body*, and *Observer*, respectively. The record-side role type represented by  $\mathfrak{D}$  integrates detection, recording, and re-presentation, with *Detector* serving as the representative term.

Base role types, local operational components, and physical carriers belong to mutually distinct types. A base role type specifies the role of an input, action, and output; a local operational component is a construction component that satisfies that specification within the local scene  $\text{Sc}$ ; and a physical carrier is a physical object on the implementation side.

### Base role types and local operational components

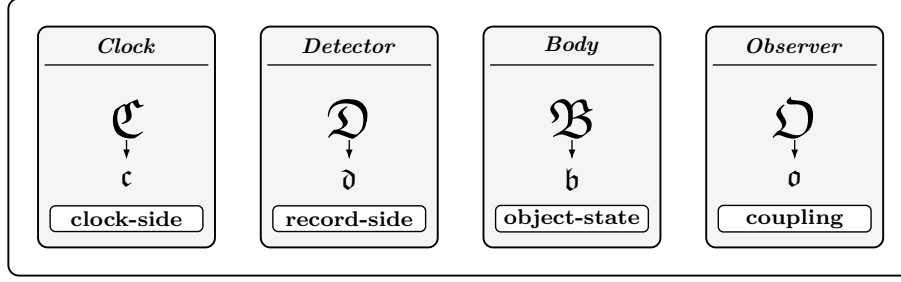


Figure 3: Construction of the base role-type set and the local four-role binding. The four base role types for the clock side, record side, object-state side, and observer-side information coupling are denoted by  $\mathfrak{C}$ ,  $\mathfrak{D}$ ,  $\mathfrak{B}$ ,  $\mathfrak{O}$ . A quadruple of local operational components satisfying these role specifications within the same local scene forms a local four-role binding witness.

## 2.6 Family of Object-Specified Role-Instance Bundles and Construction Order

Take the base role-type set of Definition 2.5, the local scene of Definition 2.4, and the local four-role binding relation of Definition 2.7. An object-specification index uniquely specifies three items: a physical object being tracked, a local scene, and a local four-role binding witness.

**Definition 2.9** (Object-Specification Index and Object-Specified Role Instances). An object-specification index  $\zeta$  is an index that uniquely specifies the following three items.

1. One physical object being tracked.
2. One local scene  $\text{Sc}_\zeta$ .
3. A local four-role binding witness

$$\ell_\zeta \in \text{Bind}_{\text{Rol}}(\text{Sc}_\zeta).$$

The component

$$\ell_\zeta(\mathfrak{B})$$

in the third item is the object-state-side local operational component abstracted from the object states of the physical object being tracked in the first item.

Define the role-instance bundle corresponding to  $\zeta$ , and its role-wise evaluation at each

$$R \in \text{Rol}$$

by

$$\begin{aligned} \text{RI}(\zeta) &:= ((R, \zeta, \ell_\zeta(R)))_{R \in \text{Rol}}, \\ \text{RI}_R(\zeta) &:= \text{RI}(\zeta)(R) \end{aligned}$$

The component order of this role-indexed family is the fixed order of the base role types in Definition 2.6.

Each role instance retains the object-specification index as its second component, and the local operational component is specified by the same object-specification index. Hence, for any object-specification indices  $\zeta_1, \zeta_2$ ,

$$\begin{aligned} \forall R \in \text{Rol}, \\ \text{RI}_R(\zeta_1) = \text{RI}_R(\zeta_2) \iff \zeta_1 = \zeta_2 \end{aligned}$$

holds.

**Definition 2.10** (Family of Role-Instance Bundles over Object-Specification Indices). Let  $Z_0$  be a set of object-specification indices from Definition 2.9. Define the family of role-instance bundles over  $Z_0$  by

$$(\text{RI}[Z_0]) := \text{RI}|_{Z_0}$$

Here  $\text{RI}|_{Z_0}$  is the restriction of the map  $\zeta \mapsto \text{RI}(\zeta)$  on object-specification indices to  $Z_0$ . Its role-indexed component family is

$$\left( (\text{RI}_R(\zeta))_{\zeta \in Z_0} \right)_{R \in \text{Rol}}$$

Define the set of role types on the domain side of the realization map by

$$\text{Rol}^{\text{rlz}} := \text{Rol} \setminus \{\mathfrak{B}\} = \{\mathfrak{C}, \mathfrak{D}, \mathfrak{O}\}$$

Each component of the  $\mathfrak{B}$ -derived component family

$$(\text{RI}_{\mathfrak{B}}(\zeta))_{\zeta \in Z_0}$$

has as its source of abstraction the physical object being tracked that  $\zeta$  specifies. The domain-side component family of the realization map

$$\left( (\text{RI}_R(\zeta))_{\zeta \in Z_0} \right)_{R \in \text{Rol}^{\text{rlz}}}$$

forms the domain-side input to the physical-carrier realization specification.

**Proposition 2.11** (Prior Determination of the Family of Role-Instance Bundles). The family of object-specification indices  $Z_0$ , together with the data specified by each  $\zeta \in Z_0$ —namely, the physical object being tracked, local scene  $\text{Sc}_{\zeta}$ , and local four-role binding witness—determines the family of role-instance bundles  $\text{RI}[Z_0]$ . This specification includes the abstraction-source correspondence between the  $\mathfrak{B}$ -derived component family and the physical objects being tracked. The physical-carrier realization specification for the  $\text{Rol}^{\text{rlz}}$ -derived component family is additional structure that takes  $\text{RI}[Z_0]$  as its domain-side input.

*Proof.* By Definition 2.9, each  $\zeta \in Z_0$  uniquely specifies a physical object being tracked, a local scene  $\text{Sc}_{\zeta}$ , and a local four-role binding witness. This specification determines the four object-specified role instances and the role-instance bundle  $\text{RI}(\zeta)$ . Definition 2.10 indexes these by  $Z_0$ , retains both the object-specification positions and the four  $\mathfrak{C}/\mathfrak{D}/\mathfrak{B}/\mathfrak{O}$  component positions, and defines  $\text{RI}[Z_0]$ .

The source of abstraction of each  $\mathfrak{B}$ -derived component is specified as the first item of the object-specification index. The  $\text{Rol}^{\text{rlz}}$ -derived component family, as a component family of the already defined  $\text{RI}[Z_0]$ , forms the domain-side input to the physical-carrier realization specification. Hence the data specified in the proposition determine the family of role-instance bundles, and the physical-carrier realization specification is determined as additional structure on that component family.  $\square$

## 2.7 Synthesis of the Construction with No Time-Quantity Input

From Section 2.1 through Section 2.3, the time-value carrier, retention object, comparison operation, law-connection target, and prior-use dependency locations are defined as an audit form common to time descriptions.

By Definitions 2.1 through 2.4, the clock-side, record-side, object-state-side, and observer-side information-coupling inputs available before a time variable is established are specified as local operational inputs belonging to the same local scene. By Definitions 2.5 through 2.7, the four inputs are assigned  $\mathfrak{C}/\mathfrak{D}/\mathfrak{B}/\mathfrak{O}$  role types, local operational components, and a local four-role binding witness.

By Definitions 2.9 and 2.10, each object-specification index  $\zeta$  determines a role-instance bundle  $\text{RI}(\zeta)$ , and over a family of object-specification indices  $Z_0$ ,

$$\text{RI}[Z_0] = (\text{RI}(\zeta))_{\zeta \in Z_0}$$

is determined. The  $\mathfrak{B}$ -derived component family retains the abstraction-source correspondence with each physical object being tracked, while the three  $\mathfrak{C}/\mathfrak{D}/\mathfrak{O}$ -derived component families form the domain-side inputs to the physical-carrier realization specification.

**Proposition 2.12** (Synthesis of the Object-Specified Role Construction). The family of object-specification indices  $Z_0$ , together with, for each  $\zeta \in Z_0$ , the physical object being tracked, local scene  $\text{Sc}_\zeta$ , and local four-role binding witness that it specifies, determines, while retaining both the object-specification positions and the four  $\mathfrak{C}/\mathfrak{D}/\mathfrak{B}/\mathfrak{O}$  component positions, the family of role-instance bundles  $\text{RI}[Z_0]$ . This specification includes the abstraction-source correspondence between the  $\mathfrak{B}$ -derived component family and the physical objects being tracked, while the three  $\mathfrak{C}/\mathfrak{D}/\mathfrak{O}$ -derived component families form the domain-side inputs to the physical-carrier realization specification. The input data of this construction contain no primitive time.

*Proof.* Definitions 2.1 and 2.4 determine the local time-generation task and the local scene without placing primitive time in the input. Definitions 2.5 through 2.7 determine the  $\mathfrak{C}/\mathfrak{D}/\mathfrak{B}/\mathfrak{O}$  role types, local operational components, and local four-role binding witness on the local scene.

Definition 2.9 determines, from the physical object being tracked, local scene, and local four-role binding witness specified by each  $\zeta \in Z_0$ , the role-instance bundle  $\text{RI}(\zeta)$ . Definition 2.10 indexes these by  $Z_0$ , retains both the object-specification positions and the four  $\mathfrak{C}/\mathfrak{D}/\mathfrak{B}/\mathfrak{O}$  component positions, and defines  $\text{RI}[Z_0]$ . By Definition 2.10, the  $\mathfrak{B}$ -derived component family retains the abstraction-source correspondence, while the three  $\mathfrak{C}/\mathfrak{D}/\mathfrak{O}$ -derived component families form the domain-side inputs to the physical-carrier realization specification. Because the construction inputs from Definition 2.1 through Definition 2.10 belong to the input boundary of that task, the construction inputs in the proposition contain no primitive time.  $\square$

## 3 Physical-Carrier Realization and Operational-Input Formation

By Proposition 2.12, the family of object-specification indices  $Z_0$ , together with, for each  $\zeta \in Z_0$ , the physical object being tracked, local scene  $\text{Sc}_\zeta$ , and local four-role binding witness that it specifies, determines the family of role-instance bundles  $\text{RI}[Z_0]$  of Definition 2.10.

Each bundle retains both its object-specification position and the four component positions of the base role-type set  $\text{Rol}$ . The object specification includes the abstraction-source correspondence between the  $\mathfrak{B}$ -derived component family and the physical objects being tracked, while the three  $\text{Rol}^{\text{rlz}}$ -derived component families form the domain-side inputs to the physical-carrier realization specification.

Define the source-side role-type set and comparison role-type set by

$$\begin{aligned}\text{Rol}^{\text{src}} &:= \text{Rol} \setminus \{\mathfrak{D}\} = \{\mathfrak{C}, \mathfrak{D}, \mathfrak{B}\}, \\ \text{Rol}^{\text{cmp}} &:= \{\mathfrak{C}, \mathfrak{D}\}\end{aligned}$$

### 3.1 Physical-Carrier Realization of Role Instances

**Definition 3.1** (Family of Physical-Layer Constructions). For the family of object-specification indices  $Z_0$  in Definition 2.10, the family of role-instance bundles

$$\text{RI}[Z_0]$$

over those indices, and the domain-side role-type set of realization maps

$$\text{Rol}^{\text{rlz}}$$

in the same definition, define the family of physical-layer constructions by

$$\text{Car}_{\text{phys}}[Z_0] := ((\text{Car}_R)_{R \in \text{Rol}^{\text{rlz}}}, \text{Car}_{\mathfrak{B}}[Z_0], \text{Car}_{\text{comm}})$$

The family  $(\text{Car}_R)_{R \in \text{Rol}^{\text{rlz}}}$  indexes by role type  $R$  the families of physical carriers assigned to the clock-side, record-side, and observer-side information-coupling role-instance families. Define the family of physical objects tracked on the object-state side by

$$\text{Car}_{\mathfrak{B}}[Z_0] := (\text{Car}_{\mathfrak{B}}(\zeta))_{\zeta \in Z_0}$$

Each  $\text{Car}_{\mathfrak{B}}(\zeta)$  is a physical object being tracked that bears the object-specification index  $\zeta$  and is the source of abstraction of  $\text{RI}_{\mathfrak{B}}(\zeta)$ . For distinct  $\zeta_1, \zeta_2 \in Z_0$ , the equality

$$\text{Car}_{\mathfrak{B}}(\zeta_1) = \text{Car}_{\mathfrak{B}}(\zeta_2)$$

is allowed. The family  $\text{Car}_{\text{comm}}$  consists of physical connection carriers that establish joint information coupling among multiple physical carriers.

Expanding the role-indexed families in  $\text{Car}_{\text{phys}}[Z_0]$  yields the five-component presentation

$$(\text{Car}_{\mathfrak{C}}, \text{Car}_{\mathfrak{D}}, \text{Car}_{\mathfrak{B}}[Z_0], \text{Car}_{\mathfrak{D}}, \text{Car}_{\text{comm}})$$

**Definition 3.2** (Sequence of Role-Typed Correspondences between the Abstract and Physical Layers). For the family of object-specification indices  $Z_0$  in Definition 2.10, the family of role-instance bundles  $\text{RI}[Z_0]$  over those indices, and the family of physical-layer constructions  $\text{Car}_{\text{phys}}[Z_0]$  in Definition 3.1, define, for each

$$R \in \text{Rol}^{\text{rlz}}$$

the realization map by

$$\text{RI}_R : \{\text{RI}_R(\zeta) \mid \zeta \in Z_0\} \longrightarrow \text{Car}_R$$

Define the object-specified physical-carrier value map by

$$\begin{aligned} \text{CarrierAt}_R &: Z_0 \longrightarrow \text{Car}_R, \\ \text{CarrierAt}_R(\zeta) &:= \text{Rlz}_R(\text{Rl}_R(\zeta)) \quad (\zeta \in Z_0) \end{aligned}$$

For each  $\zeta \in Z_0$ , the value  $\text{CarrierAt}_R(\zeta)$  is the implementation-side counterpart assigned to the  $R$ -derived role instance.

For each

$$\zeta \in Z_0$$

define the object-state-side abstraction-source correspondence by

$$\text{Rlz}_{\mathfrak{B}}(\zeta) := (\text{Car}_{\mathfrak{B}}(\zeta), \text{Rl}_{\mathfrak{B}}(\zeta))$$

For each

$$Z \subseteq Z_0$$

define its restriction by

$$\text{Rlz}_{\mathfrak{B}}[Z] := (\text{Rlz}_{\mathfrak{B}}(\zeta))_{\zeta \in Z}$$

The first component of  $\text{Rlz}_{\mathfrak{B}}(\zeta)$  is the physical object being tracked that bears the object-specification index  $\zeta$ , and the second is the object-state-side role instance that retains the local operational component  $\mathfrak{b}$  abstracted from the state of that object. Even when, for distinct  $\zeta_1, \zeta_2 \in Z_0$ ,

$$\text{Car}_{\mathfrak{B}}(\zeta_1) = \text{Car}_{\mathfrak{B}}(\zeta_2)$$

holds,

$$\begin{aligned} \text{Rl}_{\mathfrak{B}}(\zeta_1) &\neq \text{Rl}_{\mathfrak{B}}(\zeta_2), \\ \text{Rlz}_{\mathfrak{B}}(\zeta_1) &\neq \text{Rlz}_{\mathfrak{B}}(\zeta_2) \end{aligned}$$

also holds, and the abstraction-source correspondence at each object-specification-index position is retained.

$$(\text{Rlz}_R)_{R \in \text{Rol}^{\text{rlz}}}$$

is the family of realization maps, while  $\text{Rlz}_{\mathfrak{B}}$  is the family of abstraction-source correspondences from physical objects being tracked to object-state-side role instances.

*Remark 3.3* (Role-Instance Identity and Implementation-Side Identity). The identity of an object-specified role instance is determined by the object-specification index of Definition 2.9. The implementation-side identity of a  $\text{Rol}^{\text{rlz}}$ -derived component is determined by the value of the corresponding realization map in Definition 3.2. The identity of the abstraction source of a  $\mathfrak{B}$ -derived component is determined by the physical object being tracked that its object-specification index specifies.

**Definition 3.4** (Role-Typed Physical-Carrier Correspondence Pattern). For two distinct object-specification indices  $\zeta_1, \zeta_2 \in Z_0$  and each

$$R \in \text{Rol}^{\text{rlz}}$$

define the physical-carrier identity condition and physical-carrier difference condition, respectively, by

$$\text{CarrierAt}_R(\zeta_1) = \text{CarrierAt}_R(\zeta_2)$$

and

$$\text{CarrierAt}_R(\zeta_1) \neq \text{CarrierAt}_R(\zeta_2)$$

On the clock and record sides, the physical-carrier identity condition is called the shared type, and the physical-carrier difference condition is called the separated type.

On the observer side, the physical-carrier identity condition is called the shared type. The observer-side distributed type is the case in which the physical-carrier difference condition holds and joint information coupling is constructed between the two implementation-side counterparts through a physical connection carrier belonging to  $\text{Car}_{\text{comm}}$ .

A role-typed physical-carrier correspondence pattern is a three-component condition consisting of the shared or separated type on the  $\mathfrak{C}$  side, the shared or separated type on the  $\mathfrak{D}$  side, and the shared or distributed type on the  $\mathfrak{V}$  side. The component on the  $\mathfrak{C}$  side and the component on the  $\mathfrak{D}$  side each take only the realization map of the corresponding role type as their condition input. The component on the  $\mathfrak{V}$  side takes the  $\mathfrak{V}$ -derived realization map as its condition input and, in the  $\mathfrak{V}$ -distributed type, additionally takes the interposition of a physical connection carrier as a condition input. The  $\mathfrak{B}$ -derived components are retained as the family of abstraction-source correspondences equipped with object-specification positions in Definition 3.2.

#### Role instances and physical-layer relations

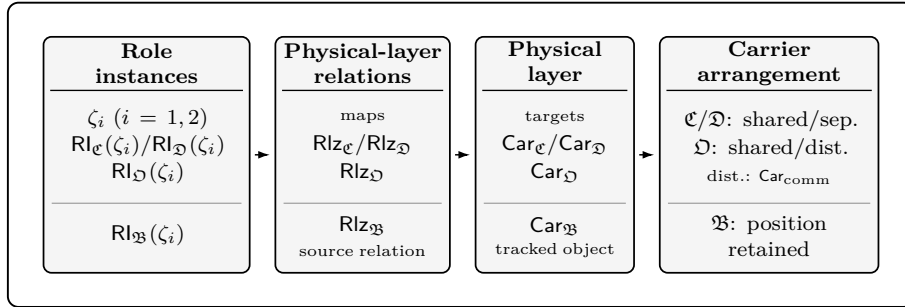


Figure 4: Role instances and physical-carrier realization. Role instances are distinguished for two object-specification indices. The physical-carrier realization maps then assign the  $\mathfrak{C}$ -derived,  $\mathfrak{D}$ -derived, and  $\mathfrak{V}$ -derived role instances to physical carriers, while the  $\mathfrak{B}$ -derived object-state-side instances are associated with the object-state-side entities corresponding to their object specifications. Sharing by the same physical carrier, separation by physical carriers distinct for each object, and distribution across multiple information-coupling carriers are distinguished on the input-formation side by equality relations among the values of the  $\mathfrak{C}/\mathfrak{D}/\mathfrak{V}$ -derived physical-carrier realization maps and by physical connection carriers belonging to  $\text{Car}_{\text{comm}}$ .



Table 1: Role instances and physical-carrier correspondence patterns

| Physical-carrier correspondence pattern                           | Physical-carrier realization condition                                                                                                                                                                                                                   |
|-------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $R$ -shared type<br>( $R \in \{\mathfrak{C}, \mathfrak{D}\}$ )    | $\text{Rlz}_R(\text{RI}_R(\zeta_1)) = \text{Rlz}_R(\text{RI}_R(\zeta_2))$                                                                                                                                                                                |
| $R$ -separated type<br>( $R \in \{\mathfrak{C}, \mathfrak{D}\}$ ) | $\text{Rlz}_R(\text{RI}_R(\zeta_1)) \neq \text{Rlz}_R(\text{RI}_R(\zeta_2))$                                                                                                                                                                             |
| $\mathfrak{D}$ -shared type                                       | $\text{Rlz}_{\mathfrak{D}}(\text{RI}_{\mathfrak{D}}(\zeta_1)) = \text{Rlz}_{\mathfrak{D}}(\text{RI}_{\mathfrak{D}}(\zeta_2))$                                                                                                                            |
| $\mathfrak{D}$ -distributed type                                  | $\text{Rlz}_{\mathfrak{D}}(\text{RI}_{\mathfrak{D}}(\zeta_1)) \neq \text{Rlz}_{\mathfrak{D}}(\text{RI}_{\mathfrak{D}}(\zeta_2))$ ,<br>and joint information coupling<br>through a physical connection carrier<br>belonging to $\text{Car}_{\text{comm}}$ |

The constructional meanings of the physical-carrier correspondence patterns in Table 1 are as follows.

- The  $\mathfrak{C}$ -shared and separated types represent realization of the clock-side role by, respectively, the same physical clock carrier or physical clock carriers distinct for each object.
- The  $\mathfrak{D}$ -shared and separated types represent realization of the record-side role by, respectively, the same physical record carrier or physical record carriers distinct for each object.
- In the  $\mathfrak{D}$ -shared type, one physical information-coupling carrier realizes the two observer-side information-coupling roles.
- In the  $\mathfrak{D}$ -distributed type, physical information-coupling carriers distinct for each object realize the observer-side information-coupling roles, and joint information coupling between them is constructed through a physical connection carrier belonging to  $\text{Car}_{\text{comm}}$ .

**Proposition 3.5** (Combinability of Realization Conditions by Role Type). Take two distinct object-specification indices

$$\zeta_1, \zeta_2 \in Z_0, \quad \zeta_1 \neq \zeta_2$$

For each

$$R \in \text{Rol}^{\text{cmp}}$$

select either the shared type or the separated type. Suppose that, for each selected type, there exist physical carriers that may be specified as implementation-side counterparts,

$$p_{R,k} \in \text{Car}_R, \quad k \in \{1, 2\}$$

and that they satisfy

$$\text{CarrierAt}_R(\zeta_k) = p_{R,k} \quad (R \in \text{Rol}^{\text{cmp}}, k \in \{1, 2\})$$

If the  $R$ -shared type is selected, let

$$p_{R,1} = p_{R,2},$$

and if the  $R$ -separated type is selected, let

$$p_{R,1} \neq p_{R,2}$$

Suppose further that implementation-side counterparts satisfying either the shared or distributed type on the  $\mathfrak{D}$  side are given and that, in the distributed type, joint information coupling through a physical connection carrier belonging to  $\mathbf{Car}_{\text{comm}}$  is given.

Then the condition on the  $\mathfrak{C}$  side and the condition on the  $\mathfrak{D}$  side can each be combined, while retaining its role-type position, with the given condition on the  $\mathfrak{D}$  side to construct the role-typed three-component condition of Definition 3.4. This combination retains

$$\mathbf{Rl}_{\mathfrak{B}} [\{\zeta_1, \zeta_2\}]$$

as a family of abstraction-source correspondences equipped with object-specification positions.

*Proof.* The family of physical-layer constructions  $\mathbf{Car}_{\text{phys}}[Z_0]$  in Definition 3.1 retains the physical-carrier families on the codomain side of the realization maps, the family of physical objects tracked on the object-state side, and the family of physical connection carriers with their role-type positions. By Definition 3.2, for each

$$R \in \mathbf{Rol}^{\text{cmp}}, \quad k \in \{1, 2\}$$

it follows that

$$p_{R,k} \in \mathbf{Car}_R$$

Hence specifying  $p_{R,1}, p_{R,2}$  determines only the position of the role type  $R$ .

By Definition 3.4, the condition for each  $R \in \mathbf{Rol}^{\text{cmp}}$  is determined solely by the identity or difference of  $p_{R,1}, p_{R,2}$ . Because each condition takes no realization-map value of another role type as a condition input, the clock-side and record-side specifications can be combined at their respective role-type positions. The condition on the  $\mathfrak{D}$  side is determined by the value of the  $\mathfrak{D}$ -derived realization map and, in the distributed type, by the interposition of a physical connection carrier; it does not take the clock-side or record-side specification as a condition input. The three-component condition is thereby constructed.

The object-state-side correspondence

$$\mathbf{Rl}_{\mathfrak{B}} [\{\zeta_1, \zeta_2\}] = (\mathbf{Rl}_{\mathfrak{B}}(\zeta))_{\zeta \in \{\zeta_1, \zeta_2\}}$$

retains, at each object-specification position, the abstraction-source correspondence between the physical object being tracked and the object-state-side role instance. Because this correspondence family takes no value of the realization-map family in Definition 3.2 as a condition input, it is unchanged by combination with the three-component condition.  $\square$

### 3.2 Object-Specification Positions and Role-Type Positions

The base role-type set  $\mathbf{Rol}$  of Definition 2.5 supplies the four component positions of a role-instance bundle. The pairing of an object-specification-index position from Definition 2.9 with a role-type position determines the object-specification membership and role-type membership of each role instance.

### 3.3 Role-Instance Identity and Implementation-Side Identity

By Definition 2.9, the identity of an object-specified role instance is determined by its object-specification index. The implementation-side identity of a  $\text{Rol}^{\text{rlz}}$ -derived component is determined by the value of the corresponding realization map in Definition 3.2. The identity of the abstraction source of a  $\mathfrak{B}$ -derived component is determined by the physical object being tracked that its object-specification index specifies.

For a construction involving multiple object specifications, physical carriers, source sequences, or observer-side reception phases, an indexed family is used that retains the object-specification, role-type, sequence, and reception-phase positions.

### 3.4 Multiplicity and Physical-Placement Conditions

The role-typed physical-carrier correspondence pattern of Definition 3.4 specifies three component conditions for two distinct object-specification indices: the  $\mathfrak{C}$ -shared or  $\mathfrak{C}$ -separated type, the  $\mathfrak{D}$ -shared or  $\mathfrak{D}$ -separated type, and the  $\mathfrak{D}$ -shared or  $\mathfrak{D}$ -distributed type. The  $\mathfrak{B}$ -derived components are retained as the family of abstraction-source correspondences equipped with object-specification positions in Definition 3.2.

For a construction with multiple clock-side, record-side, or object-state-side sequences, or multiple observer-side reception phases, the sequence and object-specification positions are retained, and the realization specifications and reception conditions are distinguished for each role type.

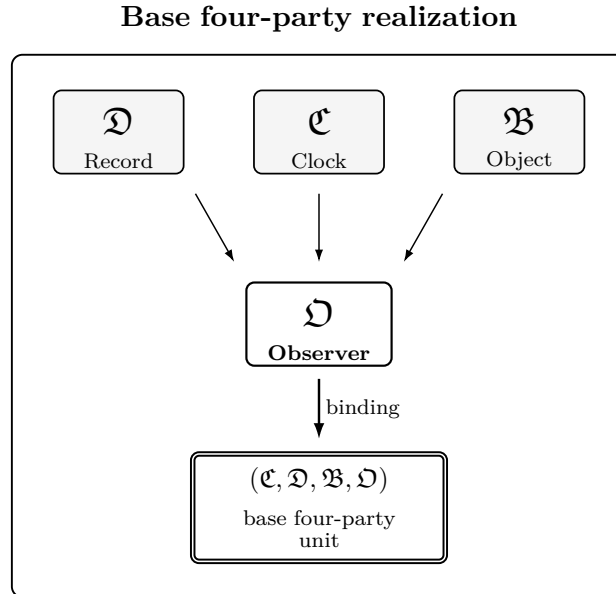


Figure 5: Base four-party realization. When the clock side  $\mathfrak{C}$ , record side  $\mathfrak{D}$ , object-state side  $\mathfrak{B}$ , and observer-side information-coupling position  $\mathfrak{D}$  are given in the same scene, they constitute the base four-party unit of the local time-generation task.

### 3.5 Coupling Physical-Carrier Realization to the Reception Network

The family of physical-layer constructions in Definition 3.1 and the sequence of role-typed correspondences between the abstract and physical layers in Definition 3.2 supply physically

realized inputs that retain the object-specification and role-type positions. Definition 3.4 specifies conditions on the values of the realization maps on the  $\mathfrak{C}$  side,  $\mathfrak{D}$  side, and  $\mathfrak{O}$  side, together with the interposition condition for a physical connection carrier in the  $\mathfrak{O}$ -distributed type.

## 3.6 Operational-Input Formation

### 3.6.1 Raw Operational Input

Throughout this section, each symbol root is first written in full form with both a role type and a reception phase. Thereafter, only the reception-phase subscript may be omitted, and only when the reception phase is fixed within the same definition, proposition, or proof and the full form can be uniquely recovered. The role type is always retained; when the reception phase varies or different reception phases are treated simultaneously, the full form is retained.

For the family of object-specification indices  $Z_0$  in Definition 2.10, take the family of role-instance bundles

$$\mathbf{RI}[Z_0],$$

over those indices, the family of physical-layer constructions

$$\mathbf{Car}_{\text{phys}}[Z_0],$$

in Definition 3.1, and the sequence of role-typed correspondences in Definition 3.2.

Let

$$U$$

be a nonempty index set of observer-side reception phases and comparison phases. For each  $u \in U$ , let an observer-side reception carrier

$$\mathbf{RcvCar}_u \in \text{Im}(\mathbf{RI}_{\mathfrak{D}})$$

and a family of reception sets indexed by source-side role type

$$(\mathbf{Rcv}_{R,u})_{R \in \mathbf{Rol}^{\text{src}}}$$

be given. Define the role-type-tagged reception set  $\mathbf{Rcv}_u$  by

$$\mathbf{Rcv}_u := \bigsqcup_{R \in \mathbf{Rol}^{\text{src}}} (\{R\} \times \mathbf{Rcv}_{R,u})$$

For each  $u \in U$ , let

$$(\preceq_u^{\text{rcv}}) \subseteq \mathbf{Rcv}_u \times \mathbf{Rcv}_u$$

be a partial order on  $\mathbf{Rcv}_u$ . For each  $R \in \mathbf{Rol}^{\text{cmp}}$ , define the partial order on the  $R$ -derived reception set, for arbitrary

$$x_R, y_R \in \mathbf{Rcv}_{R,u}$$

by

$$x_R \preceq_u^R y_R \iff (R, x_R) \preceq_u^{\text{rcv}} (R, y_R)$$

For each  $R \in \mathbf{Rol}^{\text{src}}$ , define the total local reception space derived from that source-side role type by

$$\mathcal{R}_R := \bigsqcup_{u \in U} (\{u\} \times \mathbf{Rcv}_{R,u})$$

**Definition 3.6** (Raw Operational Input). Define the reception network by

$$\text{Rcv}_U := \left( U, (\text{RcvCar}_u, (\text{Rcv}_{R,u})_{R \in \text{Rol}^{\text{src}}}, \preceq_u^{\text{rcv}})_{u \in U} \right)$$

Define the raw operational input by

$$\mathcal{I}^{\text{raw}} := (\text{Rl}[Z_0], \text{Car}_{\text{phys}}[Z_0], (\text{Rlz}_R)_{R \in \text{Rol}^{\text{rlz}}}, \text{Rlz}_{\mathfrak{B}}[Z_0], \text{Rcv}_U)$$

### 3.6.2 Source-Side Realization Datum

**Source Structures on the Clock, Record, and Object-State Sides** For the  $\mathfrak{C}$ -derived source indices, let the nonempty set be

$$\text{Idx}_{\mathfrak{C}}^{\text{src}}$$

and denote the  $\mathfrak{C}$ -derived source family by

$$(\text{Src}_{\mathfrak{C}}(j_{\mathfrak{C}}^{\text{src}}))_{j_{\mathfrak{C}}^{\text{src}} \in \text{Idx}_{\mathfrak{C}}^{\text{src}}}$$

Each  $\text{Src}_{\mathfrak{C}}(j_{\mathfrak{C}}^{\text{src}})$  is supported by a physical carrier belonging to  $\text{Car}_{\mathfrak{C}}$ , supplies the internal counter positions and updates of Definition 2.2, and is a  $\mathfrak{C}$ -derived physical source. The index  $j_{\mathfrak{C}}^{\text{src}}$  marks a difference in physical carrier, internal update rule, initial internal position, connection path, or operational sequence.

Each  $\text{Src}_{\mathfrak{C}}(j_{\mathfrak{C}}^{\text{src}})$  realizes the cycle-count source of Definition 2.2 as a  $\mathfrak{C}$ -derived source.

**Definition 3.7** ( $\mathfrak{C}$ -Derived Source-Token Family). For each

$$j_{\mathfrak{C}}^{\text{src}} \in \text{Idx}_{\mathfrak{C}}^{\text{src}}, \quad k \in \mathbb{N}$$

the internal counter position  $k$  has a corresponding  $\mathfrak{C}$ -derived individual token, denoted by

$$\text{tok}_{\mathfrak{C}}(j_{\mathfrak{C}}^{\text{src}}, k)$$

For each  $j_{\mathfrak{C}}^{\text{src}} \in \text{Idx}_{\mathfrak{C}}^{\text{src}}$ , define

$$\text{Tok}_{\mathfrak{C}}(j_{\mathfrak{C}}^{\text{src}}) := \{\text{tok}_{\mathfrak{C}}(j_{\mathfrak{C}}^{\text{src}}, k) \mid k \in \mathbb{N}\}$$

Within the same source, the indexed token family satisfies

$$\text{tok}_{\mathfrak{C}}(j_{\mathfrak{C}}^{\text{src}}, k) = \text{tok}_{\mathfrak{C}}(j_{\mathfrak{C}}^{\text{src}}, \ell) \iff k = \ell$$

Define the aggregate  $\mathfrak{C}$ -derived source-token set by

$$\text{Tok}_{\mathfrak{C}}^* := \bigsqcup_{j_{\mathfrak{C}}^{\text{src}} \in \text{Idx}_{\mathfrak{C}}^{\text{src}}} \text{Tok}_{\mathfrak{C}}(j_{\mathfrak{C}}^{\text{src}})$$

Then

$$\begin{aligned} \forall x_{\mathfrak{C}} \in \text{Tok}_{\mathfrak{C}}^*, \\ \exists! (j_{\mathfrak{C}}^{\text{src}}, k) \in \text{Idx}_{\mathfrak{C}}^{\text{src}} \times \mathbb{N} \\ x_{\mathfrak{C}} = \text{tok}_{\mathfrak{C}}(j_{\mathfrak{C}}^{\text{src}}, k) \end{aligned}$$

holds. Let the source-index map that returns the first component of this unique pair be

$$\text{idx}_{\mathfrak{C}}^{\text{src}}(\cdot) : \text{Tok}_{\mathfrak{C}}^* \rightarrow \text{Idx}_{\mathfrak{C}}^{\text{src}}$$

defined by

$$\text{idx}_{\mathfrak{C}}^{\text{src}}(\text{tok}_{\mathfrak{C}}(j_{\mathfrak{C}}^{\text{src}}, k)) := j_{\mathfrak{C}}^{\text{src}}$$

**Definition 3.8** ( $\mathfrak{C}$ -Derived Source-Internal Order). For each

$$j_{\mathfrak{C}}^{\text{src}} \in \text{Idx}_{\mathfrak{C}}^{\text{src}}$$

define the binary relation

$$\left( \preceq_{*, j_{\mathfrak{C}}^{\text{src}}}^{\mathfrak{C}} \right) \subseteq \text{Tok}_{\mathfrak{C}}(j_{\mathfrak{C}}^{\text{src}}) \times \text{Tok}_{\mathfrak{C}}(j_{\mathfrak{C}}^{\text{src}})$$

for all  $k, \ell \in \mathbb{N}$  by

$$\text{tok}_{\mathfrak{C}}(j_{\mathfrak{C}}^{\text{src}}, k) \preceq_{*, j_{\mathfrak{C}}^{\text{src}}}^{\mathfrak{C}} \text{tok}_{\mathfrak{C}}(j_{\mathfrak{C}}^{\text{src}}, \ell) \iff k \leq \ell$$

**Proposition 3.9** (Partial-Order Property of the  $\mathfrak{C}$ -Derived Source-Internal Order). For each

$$j_{\mathfrak{C}}^{\text{src}} \in \text{Idx}_{\mathfrak{C}}^{\text{src}}$$

the relation

$$\preceq_{*, j_{\mathfrak{C}}^{\text{src}}}^{\mathfrak{C}}$$

is a partial order on

$$\text{Tok}_{\mathfrak{C}}(j_{\mathfrak{C}}^{\text{src}})$$

*Proof.* Fix  $j_{\mathfrak{C}}^{\text{src}} \in \text{Idx}_{\mathfrak{C}}^{\text{src}}$ . For arbitrary

$$x_{\mathfrak{C}}, y_{\mathfrak{C}}, z_{\mathfrak{C}} \in \text{Tok}_{\mathfrak{C}}(j_{\mathfrak{C}}^{\text{src}})$$

Definition 3.7 gives unique  $k, \ell, m \in \mathbb{N}$  such that

$$x_{\mathfrak{C}} = \text{tok}_{\mathfrak{C}}(j_{\mathfrak{C}}^{\text{src}}, k), \quad y_{\mathfrak{C}} = \text{tok}_{\mathfrak{C}}(j_{\mathfrak{C}}^{\text{src}}, \ell), \quad z_{\mathfrak{C}} = \text{tok}_{\mathfrak{C}}(j_{\mathfrak{C}}^{\text{src}}, m)$$

Reflexivity  $k \leq k$  on the natural numbers and Definition 3.8 give  $x_{\mathfrak{C}} \preceq_{*, j_{\mathfrak{C}}^{\text{src}}}^{\mathfrak{C}} x_{\mathfrak{C}}$ . If the two comparisons  $x_{\mathfrak{C}} \preceq_{*, j_{\mathfrak{C}}^{\text{src}}}^{\mathfrak{C}} y_{\mathfrak{C}}$  and  $y_{\mathfrak{C}} \preceq_{*, j_{\mathfrak{C}}^{\text{src}}}^{\mathfrak{C}} z_{\mathfrak{C}}$  hold, the same definition gives  $k \leq \ell \leq m$ . Transitivity on the natural numbers gives  $k \leq m$ , and hence  $x_{\mathfrak{C}} \preceq_{*, j_{\mathfrak{C}}^{\text{src}}}^{\mathfrak{C}} z_{\mathfrak{C}}$ .

If  $x_{\mathfrak{C}} \preceq_{*, j_{\mathfrak{C}}^{\text{src}}}^{\mathfrak{C}} y_{\mathfrak{C}}$  and  $y_{\mathfrak{C}} \preceq_{*, j_{\mathfrak{C}}^{\text{src}}}^{\mathfrak{C}} x_{\mathfrak{C}}$  hold, then  $k \leq \ell$  and  $\ell \leq k$ . Antisymmetry on the natural numbers gives  $k = \ell$ , and token identity in the same definition gives  $x_{\mathfrak{C}} = y_{\mathfrak{C}}$ . The relation is therefore a partial order.  $\square$

**Definition 3.10** (Displays of  $\mathfrak{C}$ -Derived Sources). Let the set of  $\mathfrak{C}$ -derived display labels be

$$\text{DispLbl}_{\mathfrak{C}}$$

For each  $j_{\mathfrak{C}}^{\text{src}} \in \text{Idx}_{\mathfrak{C}}^{\text{src}}$ , define the display map

$$\phi_{\mathfrak{C}, j_{\mathfrak{C}}^{\text{src}}} : \text{Tok}_{\mathfrak{C}}(j_{\mathfrak{C}}^{\text{src}}) \rightarrow \text{DispLbl}_{\mathfrak{C}}$$

Define the aggregate display map

$$\phi_{\mathfrak{C}} : \text{Tok}_{\mathfrak{C}}^* \rightarrow \text{DispLbl}_{\mathfrak{C}}$$

by

$$\phi_{\mathfrak{C}}(\text{tok}_{\mathfrak{C}}(j_{\mathfrak{C}}^{\text{src}}, k)) := \phi_{\mathfrak{C}, j_{\mathfrak{C}}^{\text{src}}}(\text{tok}_{\mathfrak{C}}(j_{\mathfrak{C}}^{\text{src}}, k))$$

For arbitrary

$$j_{\mathfrak{C}}^{\text{src}}, \tilde{j}_{\mathfrak{C}}^{\text{src}} \in \text{Idx}_{\mathfrak{C}}^{\text{src}}, \\ k, \ell \in \mathbb{N}$$

if

$$\phi_{\mathfrak{C}, j_{\mathfrak{C}}^{\text{src}}}(\text{tok}_{\mathfrak{C}}(j_{\mathfrak{C}}^{\text{src}}, k)) = \phi_{\mathfrak{C}, \tilde{j}_{\mathfrak{C}}^{\text{src}}}(\text{tok}_{\mathfrak{C}}(\tilde{j}_{\mathfrak{C}}^{\text{src}}, \ell))$$

and

$$(j_{\mathfrak{C}}^{\text{src}}, k) \neq (\tilde{j}_{\mathfrak{C}}^{\text{src}}, \ell)$$

then

$$\text{tok}_{\mathfrak{C}}(j_{\mathfrak{C}}^{\text{src}}, k) \neq \text{tok}_{\mathfrak{C}}(\tilde{j}_{\mathfrak{C}}^{\text{src}}, \ell)$$

The display label gives the displayed content of an individual  $\mathfrak{C}$ -derived token, while source membership is given by the source-index map  $\text{idx}_{\mathfrak{C}}^{\text{src}}(\cdot)$  in Definition 3.7. The display-label set and display maps assign no time variable, time interval, period, frequency, or second to a token.

**Definition 3.11** (Internal Counter of  $\mathfrak{D}$ -Derived Tokens). Let the nonempty set of  $\mathfrak{D}$ -derived source tokens generated and retained by physical  $\mathfrak{D}$ -derived record carriers be

$$\text{Tok}_{\mathfrak{D}}^*$$

Each  $\mathfrak{D}$ -derived token is an individual record on a physical record carrier belonging to  $\text{Car}_{\mathfrak{D}}$ . Let the set of local record contents consisting of  $\mathfrak{C}$ -derived display content,  $\mathfrak{B}$ -derived object-specification content, and the correspondence between them on the  $\mathfrak{D}$ -derived side be

$$\text{Cont}_{\mathfrak{D}}$$

and define the record-content map by

$$\phi_{\mathfrak{D}} : \text{Tok}_{\mathfrak{D}}^* \rightarrow \text{Cont}_{\mathfrak{D}}$$

Define the internal record-index map by

$$\text{seq}_{\mathfrak{D}} : \text{Tok}_{\mathfrak{D}}^* \rightarrow \mathbb{N}$$

Each  $x_{\mathfrak{D}} \in \text{Tok}_{\mathfrak{D}}^*$  is uniquely specified by the pair comprising the record content  $\phi_{\mathfrak{D}}(x_{\mathfrak{D}})$  and the internal record index  $\text{seq}_{\mathfrak{D}}(x_{\mathfrak{D}})$ , and for arbitrary  $x_{\mathfrak{D}}, y_{\mathfrak{D}} \in \text{Tok}_{\mathfrak{D}}^*$ ,

$$x_{\mathfrak{D}} = y_{\mathfrak{D}} \iff \phi_{\mathfrak{D}}(x_{\mathfrak{D}}) = \phi_{\mathfrak{D}}(y_{\mathfrak{D}}) \quad \text{and} \quad \text{seq}_{\mathfrak{D}}(x_{\mathfrak{D}}) = \text{seq}_{\mathfrak{D}}(y_{\mathfrak{D}})$$

holds.

**Definition 3.12** ( $\mathfrak{D}$ -Derived Record-Index Order). On the aggregate  $\mathfrak{D}$ -derived source-token set

$$\text{Tok}_{\mathfrak{D}}^*$$

of Definition 3.11, define the binary relation

$$(\preceq_*^{\mathfrak{D}}) \subseteq \text{Tok}_{\mathfrak{D}}^* \times \text{Tok}_{\mathfrak{D}}^*$$

for arbitrary

$$x_{\mathfrak{D}}, y_{\mathfrak{D}} \in \text{Tok}_{\mathfrak{D}}^*$$

by

$$x_{\mathfrak{D}} \preceq_*^{\mathfrak{D}} y_{\mathfrak{D}} \iff x_{\mathfrak{D}} = y_{\mathfrak{D}} \quad \text{or} \quad \text{seq}_{\mathfrak{D}}(x_{\mathfrak{D}}) < \text{seq}_{\mathfrak{D}}(y_{\mathfrak{D}})$$

**Proposition 3.13** (Partial-Order Property of the  $\mathfrak{D}$ -Derived Record-Index Order). The relation

$$\preceq_*^{\mathfrak{D}}$$

is a partial order on

$$\mathbf{Tok}_{\mathfrak{D}}^*$$

*Proof.* Take arbitrary  $x_{\mathfrak{D}}, y_{\mathfrak{D}}, z_{\mathfrak{D}} \in \mathbf{Tok}_{\mathfrak{D}}^*$ . Since  $x_{\mathfrak{D}} = x_{\mathfrak{D}}$ , Definition 3.12 gives  $x_{\mathfrak{D}} \preceq_*^{\mathfrak{D}} x_{\mathfrak{D}}$ .

Suppose that  $x_{\mathfrak{D}} \preceq_*^{\mathfrak{D}} y_{\mathfrak{D}}$  and  $y_{\mathfrak{D}} \preceq_*^{\mathfrak{D}} z_{\mathfrak{D}}$ . If either of the two disjunctions supplied by Definition 3.12 takes the equality branch, substituting that equality into the other disjunction gives  $x_{\mathfrak{D}} \preceq_*^{\mathfrak{D}} z_{\mathfrak{D}}$ . If both take the strict-order branch, then

$$\text{seq}_{\mathfrak{D}}(x_{\mathfrak{D}}) < \text{seq}_{\mathfrak{D}}(y_{\mathfrak{D}}) < \text{seq}_{\mathfrak{D}}(z_{\mathfrak{D}})$$

and transitivity of the strict order on the natural numbers gives the same conclusion. Hence  $\preceq_*^{\mathfrak{D}}$  is transitive.

Suppose that comparison holds in both directions. If  $x_{\mathfrak{D}} \neq y_{\mathfrak{D}}$ , Definition 3.12 gives simultaneously

$$\begin{aligned} \text{seq}_{\mathfrak{D}}(x_{\mathfrak{D}}) &< \text{seq}_{\mathfrak{D}}(y_{\mathfrak{D}}), \\ \text{seq}_{\mathfrak{D}}(y_{\mathfrak{D}}) &< \text{seq}_{\mathfrak{D}}(x_{\mathfrak{D}}) \end{aligned}$$

contrary to asymmetry of the strict order on the natural numbers. Thus  $x_{\mathfrak{D}} = y_{\mathfrak{D}}$ , and  $\preceq_*^{\mathfrak{D}}$  is antisymmetric.  $\square$

**Corollary 3.14** (Incomparability of Distinct  $\mathfrak{D}$ -Derived Tokens with the Same Record Index). For arbitrary

$$x_{\mathfrak{D}}, y_{\mathfrak{D}} \in \mathbf{Tok}_{\mathfrak{D}}^*$$

if

$$x_{\mathfrak{D}} \neq y_{\mathfrak{D}}, \quad \text{seq}_{\mathfrak{D}}(x_{\mathfrak{D}}) = \text{seq}_{\mathfrak{D}}(y_{\mathfrak{D}})$$

then

$$\begin{aligned} \neg(x_{\mathfrak{D}} \preceq_*^{\mathfrak{D}} y_{\mathfrak{D}}), \\ \neg(y_{\mathfrak{D}} \preceq_*^{\mathfrak{D}} x_{\mathfrak{D}}) \end{aligned}$$

*Proof.* If  $x_{\mathfrak{D}} \preceq_*^{\mathfrak{D}} y_{\mathfrak{D}}$ , then  $x_{\mathfrak{D}} \neq y_{\mathfrak{D}}$ , and Definition 3.12 gives

$$\text{seq}_{\mathfrak{D}}(x_{\mathfrak{D}}) < \text{seq}_{\mathfrak{D}}(y_{\mathfrak{D}})$$

contrary to equality of the record indices. If  $y_{\mathfrak{D}} \preceq_*^{\mathfrak{D}} x_{\mathfrak{D}}$ , the same definition gives  $\text{seq}_{\mathfrak{D}}(y_{\mathfrak{D}}) < \text{seq}_{\mathfrak{D}}(x_{\mathfrak{D}})$ , again contrary to equality of the record indices.  $\square$

**Definition 3.15** ( $\mathfrak{D}$ -Derived Record Cell). For each  $x_{\mathfrak{D}} \in \mathbf{Tok}_{\mathfrak{D}}^*$ , let the set of  $\mathfrak{D}$ -derived tokens retained by a physical  $\mathfrak{D}$ -derived record carrier as the same finite record unit be

$$\begin{aligned} \emptyset \neq W_{\mathfrak{D}}(x_{\mathfrak{D}}) &\subseteq \mathbf{Tok}_{\mathfrak{D}}^*, \\ |W_{\mathfrak{D}}(x_{\mathfrak{D}})| &< \infty \end{aligned}$$

and require

$$x_{\mathfrak{D}} \in W_{\mathfrak{D}}(x_{\mathfrak{D}})$$

Further require, for arbitrary  $y_{\mathfrak{D}} \in \mathbf{Tok}_{\mathfrak{D}}^*$ ,

$$y_{\mathfrak{D}} \in W_{\mathfrak{D}}(x_{\mathfrak{D}}) \iff W_{\mathfrak{D}}(y_{\mathfrak{D}}) = W_{\mathfrak{D}}(x_{\mathfrak{D}})$$



Call  $W_{\mathfrak{D}}(x_{\mathfrak{D}})$  the record cell containing  $x_{\mathfrak{D}}$  on the  $\mathfrak{D}$ -derived side.

This condition partitions the aggregate  $\mathfrak{D}$ -derived source-token set into pairwise disjoint record cells. Each record cell is a group of  $\mathfrak{D}$ -derived tokens retained as the same finite record unit.

**Definition 3.16** ( $\mathfrak{B}$ -Derived Source Tokens and Object-Specification Positions). Let the nonempty set of  $\mathfrak{B}$ -derived source tokens supplied by physical  $\mathfrak{B}$ -derived tracking objects be

$$\text{Tok}_{\mathfrak{B}}^*$$

Define the object-specification position map by

$$\zeta_{\mathfrak{B}} : \text{Tok}_{\mathfrak{B}}^* \rightarrow Z_0$$

Each  $x_{\mathfrak{B}} \in \text{Tok}_{\mathfrak{B}}^*$  is supplied by the physical tracking object that is the first component of the abstraction-source correspondence retained by

$$\text{Rl}_{\mathfrak{B}}(\zeta_{\mathfrak{B}}(x_{\mathfrak{B}})) = (\text{Car}_{\mathfrak{B}}(\zeta_{\mathfrak{B}}(x_{\mathfrak{B}})), \text{Rl}_{\mathfrak{B}}(\zeta_{\mathfrak{B}}(x_{\mathfrak{B}})))$$

and is a source-side token representing the state of that object. Multiple  $\mathfrak{B}$ -derived source tokens may be supplied from the same object-specification position.

**Definition 3.17** (Direct-Sum Order on the  $\mathfrak{C}$ -Derived Source Family). On the aggregate  $\mathfrak{C}$ -derived source-token set

$$\text{Tok}_{\mathfrak{C}}^*$$

of Definition 3.7, define the binary relation

$$(\preceq_{*}^{\mathfrak{C}}) \subseteq \text{Tok}_{\mathfrak{C}}^* \times \text{Tok}_{\mathfrak{C}}^*$$

for arbitrary

$$x_{\mathfrak{C}}, y_{\mathfrak{C}} \in \text{Tok}_{\mathfrak{C}}^*$$

by

$$x_{\mathfrak{C}} \preceq_{*}^{\mathfrak{C}} y_{\mathfrak{C}} \iff \text{idx}_{\mathfrak{C}}^{\text{src}}(x_{\mathfrak{C}}) = \text{idx}_{\mathfrak{C}}^{\text{src}}(y_{\mathfrak{C}}) \quad \text{and} \quad x_{\mathfrak{C}} \preceq_{*, \text{idx}_{\mathfrak{C}}^{\text{src}}(x_{\mathfrak{C}})}^{\mathfrak{C}} y_{\mathfrak{C}}$$

**Proposition 3.18** (Partial-Order Property of the Direct-Sum Order on the  $\mathfrak{C}$ -Derived Source Family). The relation

$$\preceq_{*}^{\mathfrak{C}}$$

is a partial order on

$$\text{Tok}_{\mathfrak{C}}^*$$

*Proof.* Take arbitrary  $x_{\mathfrak{C}}, y_{\mathfrak{C}}, z_{\mathfrak{C}} \in \text{Tok}_{\mathfrak{C}}^*$ . By Definition 3.7,  $x_{\mathfrak{C}}$  belongs to a unique  $\mathfrak{C}$ -derived source component. Reflexivity in Proposition 3.9 and Definition 3.17 give  $x_{\mathfrak{C}} \preceq_{*}^{\mathfrak{C}} x_{\mathfrak{C}}$ .

Suppose  $x_{\mathfrak{C}} \preceq_{*}^{\mathfrak{C}} y_{\mathfrak{C}}$  and  $y_{\mathfrak{C}} \preceq_{*}^{\mathfrak{C}} z_{\mathfrak{C}}$ . By Definition 3.17, the three tokens belong to the same  $\mathfrak{C}$ -derived source component, and both comparisons hold in the same within-component order. Transitivity in Proposition 3.9 gives

$$x_{\mathfrak{C}} \preceq_{*, \text{idx}_{\mathfrak{C}}^{\text{src}}(x_{\mathfrak{C}})}^{\mathfrak{C}} z_{\mathfrak{C}}$$

and Definition 3.17 therefore gives  $x_{\mathfrak{C}} \preceq_{*}^{\mathfrak{C}} z_{\mathfrak{C}}$ .

If comparisons in both directions hold, both tokens belong to the same  $\mathfrak{C}$ -derived source component, and comparisons in both directions also hold within that component. Antisymmetry in Proposition 3.9 makes the two tokens equal. Thus  $\preceq_{*}^{\mathfrak{C}}$  is a partial order.  $\square$

**Corollary 3.19** (Incomparability between Distinct  $\mathfrak{C}$ -Derived Sources). For arbitrary

$$x_{\mathfrak{C}}, y_{\mathfrak{C}} \in \text{Tok}_{\mathfrak{C}}^*$$

if

$$\text{idx}_{\mathfrak{C}}^{\text{src}}(x_{\mathfrak{C}}) \neq \text{idx}_{\mathfrak{C}}^{\text{src}}(y_{\mathfrak{C}})$$

then

$$\begin{aligned} &\neg(x_{\mathfrak{C}} \preceq_*^{\mathfrak{C}} y_{\mathfrak{C}}), \\ &\neg(y_{\mathfrak{C}} \preceq_*^{\mathfrak{C}} x_{\mathfrak{C}}) \end{aligned}$$

*Proof.* By Definition 3.17, each of  $x_{\mathfrak{C}} \preceq_*^{\mathfrak{C}} y_{\mathfrak{C}}$  and  $y_{\mathfrak{C}} \preceq_*^{\mathfrak{C}} x_{\mathfrak{C}}$  requires equality of the source-membership indices. The assumed inequality between the indices prevents either comparison from holding.  $\square$

**Definition 3.20** (Comparison-Side Arrival Map). For each  $u \in U$  and  $R \in \text{Rol}^{\text{src}}$ , define a comparison-side arrival map with domain

$$\text{Dom}(\text{Arr}_{R,u}^{\text{cmp}}) \subseteq \text{Tok}_R^*$$

by

$$\text{Arr}_{R,u}^{\text{cmp}} : \text{Dom}(\text{Arr}_R^{\text{cmp}}) \longrightarrow \text{Rcv}_R$$

Each map sends  $R$ -derived source-side tokens to the observer-side reception set. If distinct source-side tokens map to the same reception, the cardinality of the inverse image of that reception is at least two.

Setting  $R = \mathfrak{B}$  in Definition 3.20 makes  $\mathfrak{B}$ -derived source tokens correspond to observer-side receptions.

**Definition 3.21** (Comparison-Side Recovery Map). For each  $u \in U$  and  $R \in \text{Rol}^{\text{src}}$ , define the uniquely recoverable reception set by

$$\text{Rcv}_{R,u}^{\text{uniq}} := \left\{ \alpha \in \text{Im}(\text{Arr}_R^{\text{cmp}}) \mid \left| (\text{Arr}_R^{\text{cmp}})^{-1}(\{\alpha\}) \right| = 1 \right\}$$

Let the comparison-side recovery map be

$$\text{Rec}_{R,u}^{\text{cmp}} : \text{Rcv}_R^{\text{uniq}} \longrightarrow \text{Tok}_R^*$$

and define it, for each  $\alpha \in \text{Rcv}_R^{\text{uniq}}$ , by

$$(\text{Arr}_R^{\text{cmp}})^{-1}(\{\alpha\}) = \{\text{Rec}_R^{\text{cmp}}(\alpha)\}$$

When the arrival value of a  $\mathfrak{C}$ -derived token  $x_{\mathfrak{C}}$  is uniquely recoverable,

$$\text{idx}_{\mathfrak{C}}^{\text{src}}(\text{Rec}_{\mathfrak{C},u}^{\text{cmp}}(\text{Arr}_{\mathfrak{C},u}^{\text{cmp}}(x_{\mathfrak{C}}))) = \text{idx}_{\mathfrak{C}}^{\text{src}}(x_{\mathfrak{C}})$$

holds. When the arrival value of a  $\mathfrak{B}$ -derived token  $x_{\mathfrak{B}}$  is uniquely recoverable,

$$\zeta_{\mathfrak{B}}(\text{Rec}_{\mathfrak{B},u}^{\text{cmp}}(\text{Arr}_{\mathfrak{B},u}^{\text{cmp}}(x_{\mathfrak{B}}))) = \zeta_{\mathfrak{B}}(x_{\mathfrak{B}})$$

holds.

**Definition 3.22** (Comparison-Stage Two-Role Relation). For each

$$u \in U$$

define the comparison-stage relation derived from the two comparison roles by

$$\text{Rel}_u^{\text{cmp}} \subseteq \prod_{R \in \text{Rol}^{\text{cmp}}} \text{Rcv}_{R,u}$$

Its element

$$(\alpha_R)_{R \in \text{Rol}^{\text{cmp}}}$$

indexes by role type the clock-side reception and record-side reception that constitute the same comparison material at observer-side reception phase  $u$ .

**Definition 3.23** (Source-Side Two-Role Relation). Define the source-side relation derived from the two comparison roles by

$$\text{Rel}^{\text{src}} \subseteq \prod_{R \in \text{Rol}^{\text{cmp}}} \text{Tok}_R^*$$

For each

$$u \in U$$

and

$$(x_R)_{R \in \text{Rol}^{\text{cmp}}} \in \prod_{R \in \text{Rol}^{\text{cmp}}} \text{Dom}(\text{Arr}_{R,u}^{\text{cmp}})$$

require the source-side and comparison-side relations to satisfy the consistency condition

$$(x_R)_{R \in \text{Rol}^{\text{cmp}}} \in \text{Rel}^{\text{src}} \iff (\text{Arr}_{R,u}^{\text{cmp}}(x_R))_{R \in \text{Rol}^{\text{cmp}}} \in \text{Rel}_u^{\text{cmp}}$$

**Definition 3.24** (Source-Side Realization Datum). Define the source-side token families, display and record data, source-side orders, families of comparison-side arrival maps, families of comparison-side recovery maps, and comparison-side and source-side relations, respectively, by

$$\text{Src}_* := (\text{Tok}_R^*)_{R \in \text{Rol}^{\text{src}}},$$

$$\text{DispRec}_* := (\phi_{\mathfrak{C}}, \phi_{\mathfrak{D}}, \text{seq}_{\mathfrak{D}}, \zeta_{\mathfrak{B}}, \{W_{\mathfrak{D}}(x_{\mathfrak{D}})\}_{x_{\mathfrak{D}} \in \text{Tok}_{\mathfrak{D}}^*}),$$

$$\text{SOrd}_* := (\preceq_{\mathfrak{C}}^*, \preceq_{\mathfrak{D}}^*),$$

$$\text{Arr}_U^{\text{cmp}} := (\text{Arr}_{R,u}^{\text{cmp}})_{(R,u) \in \text{Rol}^{\text{src}} \times U},$$

$$\text{Rec}_U^{\text{cmp}} := (\text{Rec}_{R,u}^{\text{cmp}})_{(R,u) \in \text{Rol}^{\text{src}} \times U},$$

and define the source-side realization datum by

$$\text{Dat}^{\text{src}} := (\text{Src}_*, \text{DispRec}_*, \text{SOrd}_*, \text{Arr}_U^{\text{cmp}}, \text{Rec}_U^{\text{cmp}}, ((\text{Rel}_u^{\text{cmp}})_{u \in U}, \text{Rel}^{\text{src}}))$$

### 3.6.3 Realized Operational Configuration

**Definition 3.25** (Realized Operational Configuration). Define the realized operational configuration as the ordered pair comprising the raw operational input of Definition 3.6 and the source-side realization datum of Definition 3.24:

$$\mathcal{X} := (\mathcal{I}^{\text{raw}}, \text{Dat}^{\text{src}})$$

Let the realized operational configuration  $\mathcal{X}$  be the domain-side input to the comparison construction.

**Proposition 3.26** (Component Specification of the Realized Operational Configuration). The realized operational configuration  $\mathcal{X}$  of Definition 3.25 specifies as typed components the family of role-instance bundles that retains object-specification and role-type positions, the physical-carrier realization, the observer-side reception network, the  $\text{Rol}^{\text{src}}$ -derived source-token families, display and record data, source-side orders, comparison-side arrival maps, comparison-side recovery maps, and comparison-side/source-side relations. The input data of this construction contain no time variable, time interval, period, frequency, or second.

*Proof.* Definition 3.6 defines as the raw operational input the family of role-instance bundles over the family of object-specification indices, the family of physical-layer constructions, the  $\text{Rol}^{\text{rlz}}$ -derived realization maps, the family of  $\mathfrak{B}$ -derived abstraction-source correspondences, and the observer-side reception network.

Definition 3.24 defines as the source-side realization datum the  $\text{Rol}^{\text{src}}$ -derived source-token families, display and record data, source-side orders, the  $\text{Rol}^{\text{src}}$ -derived families of comparison-side arrival maps, the  $\text{Rol}^{\text{src}}$ -derived families of comparison-side recovery maps, and the  $\text{Rol}^{\text{cmp}}$ -derived comparison-side/source-side relations. Definition 3.25 retains these two objects as the ordered pair  $\mathcal{X}$ .

The object-specification positions, role-type positions, physical realizations, receptions, source tokens, display and record data, orders, arrivals, recoveries, and correspondences are therefore specified as typed components of  $\mathcal{X}$ . Each component is constructed from internal positions, natural-number indices, maps, relations, and reception orders, without placing a time quantity in the input.  $\square$

The raw operational input of Definition 3.6 couples the observer-side reception network to the physically realized input supplied by the family of physical-layer constructions in Definition 3.1 and the sequence of role-typed correspondences between the abstract and physical layers in Definition 3.2. The source-side realization datum of Definition 3.24 retains the  $\text{Rol}^{\text{src}}$ -derived source-token families, display and record data, source-side orders, comparison-side arrival maps, comparison-side recovery maps, and comparison-side/source-side relations.

## 4 Comparison-Stage Closure and Canonical Representation

### 4.1 Generation of the Canonical Comparison Quotient

#### 4.1.1 Definition of the Comparison-Stage Construction

From the realized operational configuration, extract the reception network used at the comparison stage, the source-side token families, the comparison-arrival maps, the comparison-

side recovery maps, and the comparison-side and source-side relations; combine these with the typed family of comparison rules to construct the comparison-stage material datum. From the comparison-admissible inputs specified by the family of comparison rules, define source-identification partial maps, comparison-transport components, typed finite words and their induced partial actions, realizations of local closed-loop specifications, comparison-reachability equivalence relations, component quotients, canonical projections, and comparison two-role relations on the quotients. When the seven comparison-stage conditions and the condition on all pure-transport closed loops hold, construct the comparison-gluing datum and the canonical comparison quotient from the generated comparison-stage objects.

In this chapter, the compound superscript  $(\text{cmp}, \sigma)$  on an object specific to the comparison stage is abbreviated to  $\sigma$  only when the symbol root together with  $\sigma$  uniquely determines its comparison-stage membership. To distinguish it from the source-side relation  $\text{Rel}^{\text{src}}$ , however,  $\text{Rel}^{\text{cmp}, \text{src}}$  retains its complete form.

For each symbol root, one pair consisting of a role type and a reception phase first appears in complete form. Thereafter, only the reception-phase subscript is omitted within the same definition, proposition, or proof when the reception phase is fixed and the complete form is uniquely recoverable. The complete forms are retained for the role type, a varying family index, distinct reception phases, and residual modifiers that select genuine branches.

### Comparison quotient generation

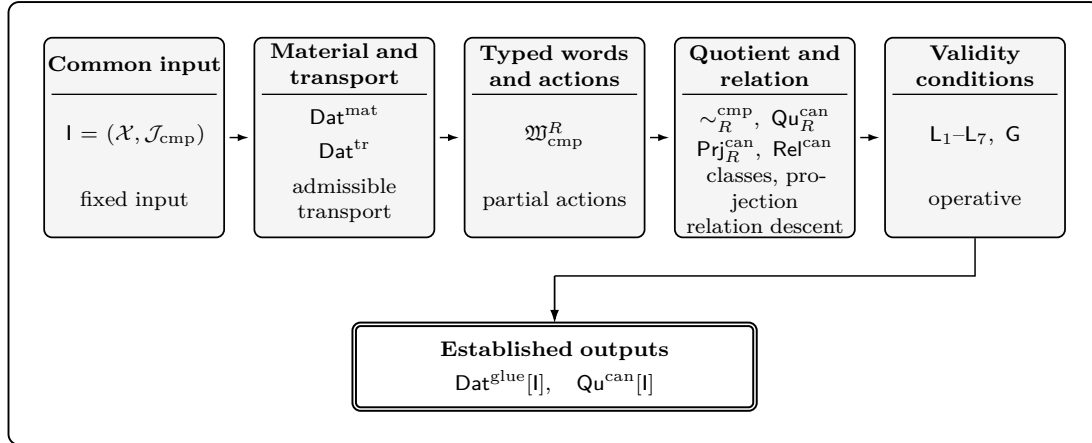


Figure 6: Comparison-stage dependency structure from the comparison-rule-dependent common input to the comparison-gluing datum and the canonical comparison quotient. From the comparison-stage input, comparison-admissible inputs, comparison-transport components, typed finite words, comparison-reachability equivalence relations, component quotients, canonical projections, and comparison two-role relations on the quotients are determined in dependency order. The seven comparison-stage conditions and the condition on all pure-transport closed loops determine whether each construction stage is established.

**Definition 4.1** (Comparison-Rule-Dependent Common Input). For the realized operational configuration

$$\mathcal{X} = (\mathcal{I}^{\text{raw}}, \text{Dat}^{\text{src}})$$

of Definition 3.25 and the typed family of comparison rules

$$\mathcal{J}_{\text{cmp}}$$

define the comparison-rule-dependent common input by

$$\mathbf{l} := (\mathcal{X}, \mathcal{J}_{\text{cmp}})$$

For any object  $F$  dependent on this input, define the evaluation notation by

$$F[\mathbf{l}] := F(\mathcal{X}; \mathcal{J}_{\text{cmp}})$$

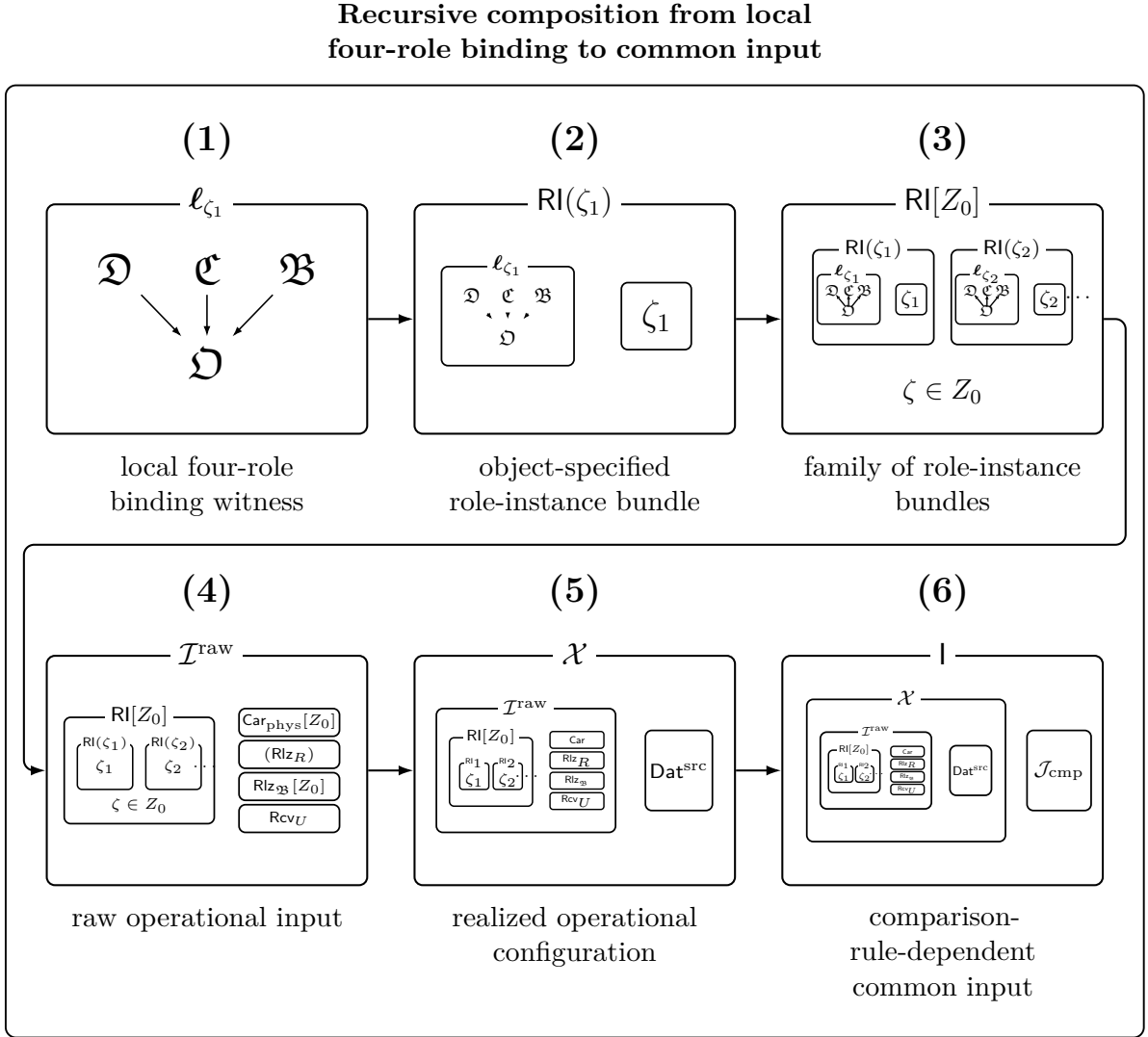


Figure 7: Recursive composition from the local four-role binding to the comparison-rule-dependent common input. An object defined as a whole at one stage is retained as a structured component at the next stage. The construction proceeds from the local four-role binding witness  $\ell_{\zeta_1}$ , through the object-specified role-instance bundle  $\text{RI}(\zeta_1)$ , the family  $\text{RI}[Z_0]$ , the raw operational input  $\mathcal{I}^{\text{raw}}$ , and the realized operational configuration  $\mathcal{X}$ , to the comparison-rule-dependent common input  $\mathbf{l} = (\mathcal{X}, \mathcal{J}_{\text{cmp}})$ .

**Definition 4.2** (Comparison-Stage Material Datum). Let the comparison-rule-dependent common input of Definition 4.1 be

$$\mathbf{l} = (\mathcal{X}, \mathcal{J}_{\text{cmp}})$$

Define the comparison-stage material datum by

$$\begin{aligned} \text{Dat}^{\text{mat}}[\mathbb{I}] \\ := (\text{Rcv}_U, \text{Src}_*, \text{Arr}_U^{\text{cmp}}, \text{Rec}_U^{\text{cmp}}, ((\text{Rel}_u^{\text{cmp}})_{u \in U}, \text{Rel}^{\text{src}}), \mathcal{J}_{\text{cmp}}) \end{aligned}$$

The first component is the observer-side reception network, the second is the source-side token family, the third is the family of comparison-arrival maps, the fourth is the family of comparison-side recovery maps, and the fifth is the ordered pair comprising the family of comparison-side relations and the source-side relation. The sixth component is the typed family of comparison rules that specifies comparison-admissible position pairs, input synchronized-pair joint-transport relations, and local pure-transport closed-loop specifications.

#### 4.1.2 Comparison-Stage Decision Conditions

**Definition 4.3** (Comparison-Admissible Position Pairs and Input Synchronized-Pair Joint-Transport Relations). Let the comparison-stage material datum of Definition 4.2 be

$$\text{Dat}^{\text{mat}}[\mathbb{I}]$$

Let the set of comparison-admissible position pairs specified by the typed family of comparison rules

$$\mathcal{J}_{\text{cmp}}$$

be

$$\text{Adm}_{\text{cmp}}[\mathbb{I}] \subseteq U \times U$$

For each

$$(u, v) \in \text{Adm}_{\text{cmp}}[\mathbb{I}]$$

let the input synchronized-pair joint-transport relation specified by the typed family of comparison rules be

$$\begin{aligned} \text{Tr}_{uv}^{\text{pair}}[\mathbb{I}] \\ \subseteq \left( \prod_{R \in \text{Rol}^{\text{cmp}}} \text{Im}(\text{Arr}_{R,u}^{\text{cmp}}) \right) \times \left( \prod_{R \in \text{Rol}^{\text{cmp}}} \text{Im}(\text{Arr}_{R,v}^{\text{cmp}}) \right) \end{aligned}$$

The flattening of an element of this product space is given by

$$((\alpha_R)_{R \in \text{Rol}^{\text{cmp}}}, (\beta_R)_{R \in \text{Rol}^{\text{cmp}}}) \longleftrightarrow (\alpha_{\mathfrak{C}}, \alpha_{\mathfrak{D}}, \beta_{\mathfrak{C}}, \beta_{\mathfrak{D}})$$

**Definition 4.4** (Family of Local Pure-Transport Closed-Loop Specifications). Let the comparison-stage material datum of Definition 4.2 and the set of comparison-admissible position pairs of Definition 4.3 be given. For each

$$R \in \text{Rol}^{\text{cmp}}$$

let the family of local pure-transport closed-loop specifications specified by the typed family of comparison rules

$$\mathcal{J}_{\text{cmp}}$$

be

$$\text{LoopSpec}_{\text{loc}}^R[\mathbb{I}]$$

The family of local pure-transport closed-loop specifications satisfies

$$\begin{aligned}
& \text{LoopSpec}_{\text{loc}}^R [I] \\
& \subseteq \left\{ (m, (u_j)_{j=0}^m, (\varepsilon_j)_{j=1}^m, \text{Dom}_{\text{Spec}}) \mid \right. \\
& m \in \mathbb{N}, \quad m \geq 1, \\
& (u_j)_{j=0}^m \in U^{m+1}, \quad u_m = u_0, \\
& (\varepsilon_j)_{j=1}^m \in \{+, -\}^m, \\
& \forall j \in \{1, \dots, m\}, \\
& \quad (\varepsilon_j = + \wedge (u_{j-1}, u_j) \in \text{Adm}_{\text{cmp}} [I]) \\
& \quad \vee (\varepsilon_j = - \wedge (u_j, u_{j-1}) \in \text{Adm}_{\text{cmp}} [I]), \\
& \left. \text{Dom}_{\text{Spec}} \subseteq \{u_0\} \times \text{Im}(\text{Arr}_{R, u_0}^{\text{cmp}}) \right\}.
\end{aligned}$$

For each

$$\text{Spec} = (m, (u_j)_{j=0}^m, (\varepsilon_j)_{j=1}^m, \text{Dom}_{\text{Spec}}) \in \text{LoopSpec}_{\text{loc}}^R [I]$$

define the base point of the local closed-loop specification by

$$\text{Base}(\text{Spec}) := u_0$$

**Definition 4.5** (Comparison-Stage Decision Condition  $L_1$ : Local Source Identification). Define the local source-identification condition by

$$L_1(\mathcal{X}) \iff \forall R \in \text{Rol}^{\text{cmp}} \quad \forall u \in U, \quad \text{Im}(\text{Arr}_{R, u}^{\text{cmp}}) \subseteq \text{Rcv}_{R, u}^{\text{uniq}}.$$

**Lemma 4.6** (Local source identification and mutually inverse arrival and recovery maps). The condition  $L_1(\mathcal{X})$  is equivalent to injectivity of the comparison-side arrival map on its domain for every  $R \in \text{Rol}^{\text{cmp}}$  and  $u \in U$ . Under this condition, for every  $R, u$  and every

$$x_R \in \text{Dom}(\text{Arr}_{R, u}^{\text{cmp}}), \quad \alpha_{R, u} \in \text{Im}(\text{Arr}_{R, u}^{\text{cmp}}),$$

the following relations hold.

$$\text{Rec}_{R, u}^{\text{cmp}}(\text{Im}(\text{Arr}_{R, u}^{\text{cmp}})) \subseteq \text{Dom}(\text{Arr}_{R, u}^{\text{cmp}}),$$

$$\text{Rec}_{R, u}^{\text{cmp}}(\text{Arr}_{R, u}^{\text{cmp}}(x_R)) = x_R,$$

$$\text{Arr}_{R, u}^{\text{cmp}}(\text{Rec}_{R, u}^{\text{cmp}}(\alpha_{R, u})) = \alpha_{R, u}.$$

*Proof.* By Definition 3.21,  $\text{Rcv}_{R, u}^{\text{uniq}}$  consists of those points in the image of the comparison-side arrival map whose inverse images are singletons. Containment of the entire arrival image in this set is equivalent to injectivity on the domain. Under  $L_1(\mathcal{X})$ , the recovery map is defined on the entire arrival image and returns the unique preimage of each arrival point. Its value therefore belongs to the arrival-map domain, and the two composition identities follow.  $\square$

**Definition 4.7** (Comparison-Stage Condition-Family Evaluation Specification). Define the comparison-stage condition-family evaluation specification by

$$\text{Ev}^{\text{cmp}} := (\text{Ev}_i^{\text{cmp}})_{i=1}^7$$



Define each component by

$$\text{Ev}_i^{\text{cmp}} := \begin{cases} \mathcal{X}, & i = 1, \\ 1, & i \in \{2, \dots, 7\} \end{cases}$$

For each

$$i \in \{1, \dots, 7\}$$

define the bracket evaluation of a comparison-stage condition by

$$\text{L}_i[\text{Ev}_i^{\text{cmp}}] := \begin{cases} \text{L}_1(\mathcal{X}), & i = 1, \\ \text{L}_i[1], & i \in \{2, \dots, 7\} \end{cases}$$

Define the first-stage established-condition bundle by

$$\text{L}_{\leq 1}[\text{Ev}^{\text{cmp}}] := \text{L}_1[\text{Ev}_1^{\text{cmp}}]$$

**Definition 4.8** (Source-Matching Partial Maps). Assume  $\text{L}_1(\mathcal{X})$ . For arbitrary

$$R \in \text{Rol}^{\text{cmp}}, \quad u, v \in U$$

define

$$\text{Dom}(\text{SrcMatch}_{uv}^R) := \left\{ \alpha_{R,u} \in \text{Im}(\text{Arr}_{R,u}^{\text{cmp}}) \mid \text{Rec}_{R,u}^{\text{cmp}}(\alpha_{R,u}) \in \text{Dom}(\text{Arr}_{R,v}^{\text{cmp}}) \right\}$$

Define the source-matching partial map

$$\text{SrcMatch}_{uv}^R : \text{Dom}(\text{SrcMatch}_{uv}^R) \longrightarrow \text{Im}(\text{Arr}_{R,v}^{\text{cmp}})$$

by

$$\text{SrcMatch}_{uv}^R(\alpha_{R,u}) := \text{Arr}_{R,v}^{\text{cmp}}(\text{Rec}_{R,u}^{\text{cmp}}(\alpha_{R,u}))$$

**Lemma 4.9** (Mutual Inverseness and Partial Injectivity of the Source-Matching Partial Maps). Under  $\text{L}_1(\mathcal{X})$ , for arbitrary

$$R \in \text{Rol}^{\text{cmp}}, \quad u, v \in U$$

and

$$(u_0, u_1) \in \{(u, v), (v, u)\}$$

one has

$$\text{Im}(\text{SrcMatch}_{u_0 u_1}^R) = \text{Dom}(\text{SrcMatch}_{u_1 u_0}^R)$$

and

$$\text{SrcMatch}_{u_1 u_0}^R \circ \text{SrcMatch}_{u_0 u_1}^R = \text{id}_{\text{Dom}(\text{SrcMatch}_{u_0 u_1}^R)}.$$

Consequently,  $\text{SrcMatch}_{uv}^R$  and  $\text{SrcMatch}_{vu}^R$  are partially injective and are mutually inverse partial maps.

*Proof.* Take  $R \in \mathbf{Rol}^{\text{cmp}}$ ,  $u, v \in U$ , and

$$(u_0, u_1) \in \{(u, v), (v, u)\}$$

Take arbitrary

$$\alpha_{R, u_0} \in \text{Dom}(\text{SrcMatch}_{u_0 u_1}^R)$$

and set

$$x_R := \text{Rec}_{R, u_0}^{\text{cmp}}(\alpha_{R, u_0})$$

By Definition 4.8,

$$x_R \in \text{Dom}(\text{Arr}_{R, u_1}^{\text{cmp}})$$

Lemma 4.6 also gives

$$x_R \in \text{Dom}(\text{Arr}_{R, u_0}^{\text{cmp}}).$$

Set

$$\alpha_{R, u_1} := \text{SrcMatch}_{u_0 u_1}^R(\alpha_{R, u_0}) = \text{Arr}_{R, u_1}^{\text{cmp}}(x_R)$$

Lemma 4.6 gives

$$\text{Rec}_{R, u_1}^{\text{cmp}}(\alpha_{R, u_1}) = x_R$$

and therefore

$$\alpha_{R, u_1} \in \text{Dom}(\text{SrcMatch}_{u_1 u_0}^R).$$

Furthermore,

$$\begin{aligned} \text{SrcMatch}_{u_1 u_0}^R(\alpha_{R, u_1}) &= \text{Arr}_{R, u_0}^{\text{cmp}}(\text{Rec}_{R, u_1}^{\text{cmp}}(\alpha_{R, u_1})) \\ &= \text{Arr}_{R, u_0}^{\text{cmp}}(x_R) \\ &= \text{Arr}_{R, u_0}^{\text{cmp}}(\text{Rec}_{R, u_0}^{\text{cmp}}(\alpha_{R, u_0})) \\ &= \alpha_{R, u_0}. \end{aligned}$$

This proves one direction of the asserted equality between the image and the domain of the reverse map, together with the composition identity.

The ordered pair  $(u_0, u_1)$  may be either  $(u, v)$  or  $(v, u)$ . Thus the preceding inclusion and composition identity hold in both directions. The two inclusions give equality between the image and the domain of the reverse map; the two composition identities show that the maps are mutually inverse partial maps and that each is partially injective.  $\square$

**Definition 4.10** (Set of Component-Factorization Tuples for Synchronized-Pair Joint Transport). For each

$$(u, v) \in \text{Adm}_{\text{cmp}}[\mathbb{I}]$$

define the set of component-factorization tuples for synchronized-pair joint transport by

$$\begin{aligned}
& \text{TrFac}_{uv}^{\text{cmp}}[\mathbb{I}] \\
& := \left\{ \left( P_{uv}^{\mathcal{CD}}, (\text{Fac}_{R,uv}^{\text{tr}})_{R \in \text{Rol}^{\text{cmp}}} \right) \mid \right. \\
& \quad P_{uv}^{\mathcal{CD}} \subseteq \prod_{R \in \text{Rol}^{\text{cmp}}} \text{Im}(\text{Arr}_{R,u}^{\text{cmp}}), \\
& \quad \forall R \in \text{Rol}^{\text{cmp}}, \quad \text{Fac}_{R,uv}^{\text{tr}} : \text{pr}_R(P_{uv}^{\mathcal{CD}}) \hookrightarrow \text{Im}(\text{Arr}_{R,v}^{\text{cmp}}), \\
& \quad \left. \text{Tr}_{uv}^{\text{pair}}[\mathbb{I}] \right. \\
& \quad = \left\{ \left( (\alpha_R)_{R \in \text{Rol}^{\text{cmp}}}, (\text{Fac}_{R,uv}^{\text{tr}}(\alpha_R))_{R \in \text{Rol}^{\text{cmp}}} \right) \mid \right. \\
& \quad \quad \left. (\alpha_R)_{R \in \text{Rol}^{\text{cmp}}} \in P_{uv}^{\mathcal{CD}} \right\} \Bigg\}
\end{aligned}$$

**Definition 4.11** (Comparison-Stage Decision Condition  $L_2$ : Component Factorization of Synchronized-Pair Joint Transport). Define the synchronized-pair joint-transport component-factorization condition by

$$\begin{aligned}
& L_2[\mathbb{I}] \\
& \iff \left[ L_{\leq 1}[\text{Ev}^{\text{cmp}}] \implies \forall (u, v) \in \text{Adm}_{\text{cmp}}[\mathbb{I}], \right. \\
& \quad \left. \# \text{TrFac}_{uv}^{\text{cmp}}[\mathbb{I}] = 1 \right]
\end{aligned}$$

Define the second-stage established-condition bundle by

$$L_{\leq 2}[\text{Ev}^{\text{cmp}}] := L_{\leq 1}[\text{Ev}^{\text{cmp}}] \wedge L_2[\text{Ev}_2^{\text{cmp}}]$$

Suppose that

$$L_1(\mathcal{X})$$

and

$$L_2[\mathbb{I}]$$

hold. For each

$$(u, v) \in \text{Adm}_{\text{cmp}}[\mathbb{I}]$$

define the unique role-indexed transport datum by

$$\begin{aligned}
& \text{Dat}_{uv}^{\text{tr}}[\mathbb{I}] \\
& := \left( \text{PairDom}_{uv}[\mathbb{I}], (\text{Tr}_{uv}^R[\mathbb{I}])_{R \in \text{Rol}^{\text{cmp}}} \right)
\end{aligned}$$

The first component satisfies

$$\text{PairDom}_{uv}[\mathbb{I}] \subseteq \prod_{R \in \text{Rol}^{\text{cmp}}} \text{Im}(\text{Arr}_{R,u}^{\text{cmp}})$$

For each

$$R \in \mathbf{Rol}^{\text{cmp}}$$

the second component satisfies

$$\begin{aligned} & \text{Tr}_{uv}^R [\mathbb{I}] \\ & : \text{pr}_R(\text{PairDom}_{uv} [\mathbb{I}]) \hookrightarrow \text{Im}(\text{Arr}_{R,v}^{\text{cmp}}) \end{aligned}$$

Moreover,

$$\begin{aligned} & \text{Tr}_{uv}^{\text{pair}} [\mathbb{I}] \\ & = \left\{ \left( (\alpha_R)_{R \in \mathbf{Rol}^{\text{cmp}}}, (\text{Tr}_{uv}^R [\mathbb{I}] (\alpha_R))_{R \in \mathbf{Rol}^{\text{cmp}}} \right) \mid \right. \\ & \quad \left. (\alpha_R)_{R \in \mathbf{Rol}^{\text{cmp}}} \in \text{PairDom}_{uv} [\mathbb{I}] \right\}. \end{aligned}$$

**Definition 4.12** (Comparison-Transport Datum). Assume

$$\mathbf{L}_1(\mathcal{X})$$

and

$$\mathbf{L}_2 [\mathbb{I}]$$

Define the comparison-transport datum by

$$\begin{aligned} & \text{Dat}^{\text{tr}} [\mathbb{I}] \\ & := (\text{Dat}_{uv}^{\text{tr}} [\mathbb{I}])_{(u,v) \in \text{Adm}_{\text{cmp}} [\mathbb{I}]} \end{aligned}$$

**Definition 4.13** (Tagged Comparison-Arrival Reception Spaces). For each

$$R \in \mathbf{Rol}^{\text{cmp}}$$

define the tagged comparison-arrival reception space by

$$\mathcal{R}_R^{\text{arr}} := \bigsqcup_{u \in U} (\{u\} \times \text{Im}(\text{Arr}_{R,u}^{\text{cmp}}))$$

**Definition 4.14** (Tagged Lift of a Partial Injection). For arbitrary

$$u, v \in U$$

and a partial injection  $f$ , define the tagged lift

$$\begin{aligned} & \text{TagLift}_{uv}[f] \\ & : \{u\} \times \text{Dom}(f) \hookrightarrow \{v\} \times \text{Im}(f) \end{aligned}$$

by

$$\text{TagLift}_{uv}[f](u, x) := (v, f(x))$$

**Lemma 4.15** (Compatibility of Tagged Partial-Injection Lifts with Inverses). For any partial injection  $f$  and its inverse  $f^{-1}$  on its image,

$$(\text{TagLift}_{uv}[f])^{-1} = \text{TagLift}_{vu}[f^{-1}].$$

Consequently,

$$\begin{aligned} & \text{TagLift}_{vu}[f^{-1}] \circ \text{TagLift}_{uv}[f] \\ &= \text{id}_{\text{Dom}(\text{TagLift}_{uv}[f])}, \\ & \text{TagLift}_{uv}[f] \circ \text{TagLift}_{vu}[f^{-1}] \\ &= \text{id}_{\text{Im}(\text{TagLift}_{uv}[f])}. \end{aligned}$$

*Proof.* The domains of the two sides coincide with

$$\{v\} \times \text{Im}(f)$$

For arbitrary

$$y \in \text{Im}(f)$$

Definition 4.14 gives

$$\begin{aligned} & (\text{TagLift}_{uv}[f])^{-1}(v, y) \\ &= (u, f^{-1}(y)) \\ &= \text{TagLift}_{vu}[f^{-1}](v, y). \end{aligned}$$

The two maps therefore coincide. The two composition identities follow from the definition of an inverse map.  $\square$

**Definition 4.16** (Pure-Transport Terms and Formal Inverse Terms). Assume

$$\mathbf{L}_1(\mathcal{X})$$

and

$$\mathbf{L}_2[\mathbb{I}]$$

For arbitrary

$$\begin{aligned} & R \in \text{Rol}^{\text{cmp}}, \\ & (u, v) \in \text{Adm}_{\text{cmp}}[\mathbb{I}] \end{aligned}$$

define the forward pure-transport term by

$$\begin{aligned} & \text{EdgeTerm}_{uv}^{\text{tr}, R, +}[\mathbb{I}] \\ &:= \text{TagLift}_{uv}[\text{Tr}_{uv}^R[\mathbb{I}]] \end{aligned}$$

Define the formal inverse pure-transport term by

$$\begin{aligned} & \text{EdgeTerm}_{uv}^{\text{tr}, R, -}[\mathbb{I}] \\ &:= (\text{EdgeTerm}_{uv}^{\text{tr}, R, +}[\mathbb{I}])^{-1} \\ &= \text{TagLift}_{vu}[(\text{Tr}_{uv}^R[\mathbb{I}])^{-1}] \end{aligned}$$

**Lemma 4.17** (Mutual Inverseness of Pure-Transport Terms and Formal Inverse Terms). For each term in Definition 4.16, for the two choices of  $\pm$  and the corresponding opposite sign  $\mp$ ,

$$\begin{aligned} & \text{EdgeTerm}_{uv}^{\text{tr}, R, \mp}[\mathbb{I}] \circ \text{EdgeTerm}_{uv}^{\text{tr}, R, \pm}[\mathbb{I}] \\ &= \text{id}_{\text{Dom}(\text{EdgeTerm}_{uv}^{\text{tr}, R, \pm}[\mathbb{I}])}, \end{aligned}$$

*Proof.* Apply Lemma 4.15 to

$$f = \text{Tr}_{uv}^R [\text{I}]$$

□

**Definition 4.18** (Source-Matching Terms). Assume

$$\mathbf{L}_1(\mathcal{X})$$

For arbitrary

$$R \in \text{Rol}^{\text{cmp}}, \quad u, v \in U$$

define the source-matching term by

$$\text{EdgeTerm}_{uv}^{\text{src}, R} := \text{TagLift}_{uv} [\text{SrcMatch}_{uv}^R]$$

**Lemma 4.19** (Mutual Inverseness of Source-Matching Terms). For each term in Definition 4.18 and arbitrary

$$R \in \text{Rol}^{\text{cmp}}, \quad u, v \in U, \quad (u_0, u_1) \in \{(u, v), (v, u)\}$$

one has

$$\begin{aligned} \text{EdgeTerm}_{u_1 u_0}^{\text{src}, R} &= (\text{EdgeTerm}_{u_0 u_1}^{\text{src}, R})^{-1}, \\ \text{EdgeTerm}_{u_1 u_0}^{\text{src}, R} \circ \text{EdgeTerm}_{u_0 u_1}^{\text{src}, R} &= \text{id}_{\text{Dom}(\text{EdgeTerm}_{u_0 u_1}^{\text{src}, R})}. \end{aligned}$$

Because the domain of the inverse map is the image of the original map, the two ordered pairs in the second identity give the two composition identities.

*Proof.* Applying Lemma 4.9 to this ordered pair gives

$$\text{SrcMatch}_{u_1 u_0}^R = (\text{SrcMatch}_{u_0 u_1}^R)^{-1}.$$

Apply Lemma 4.15 to

$$f = \text{SrcMatch}_{u_0 u_1}^R$$

□

**Definition 4.20** (Typed Finite Partial-Injection Sequence Action). For sets  $Y_0, Y_1, Y_2$  and partial maps

$$\begin{aligned} f &: \text{Dom}(f) \subseteq Y_0 \longrightarrow Y_1, \\ g &: \text{Dom}(g) \subseteq Y_1 \longrightarrow Y_2 \end{aligned}$$

define their composition by

$$\begin{aligned} g \circ f &: \{x \in \text{Dom}(f) \mid f(x) \in \text{Dom}(g)\} \longrightarrow Y_2, \\ x &\longmapsto g(f(x)) \end{aligned}$$

For each  $m \in \mathbb{Z}_{\geq 0}$ , each sequence of sets  $(Y_j)_{j=0}^m$ , and each sequence of partial injections

$$f_j : \text{Dom}(f_j) \subseteq Y_{j-1} \longrightarrow Y_j, \quad j \in \{1, \dots, m\}$$

define the typed finite partial-injection sequence action by

$$\begin{aligned} \text{SeqAct}[(\cdot)_{Y_0}] &:= \text{id}_{Y_0}, \\ \text{SeqAct}[(f_j)_{j=1}^m] &:= f_m \circ \dots \circ f_1 \quad (m \geq 1) \end{aligned}$$

Its maximal domain and value satisfy

$$\begin{aligned} & \text{Dom}(\text{SeqAct}[(f_j)_{j=1}^m]) \\ &= \left\{ x_0 \in Y_0 \mid \exists (x_j)_{j=1}^m \in \prod_{j=1}^m Y_j, \right. \\ & \quad \left. \forall j \in \{1, \dots, m\}, \quad x_{j-1} \in \text{Dom}(f_j), \quad x_j = f_j(x_{j-1}) \right\}, \\ & \text{SeqAct}[(f_j)_{j=1}^m](x_0) = x_m \end{aligned}$$

**Lemma 4.21** (Action Laws for Typed Finite Partial-Injection Sequences). For each  $m, n \in \mathbb{Z}_{\geq 0}$  and two type-compatible sequences  $(f_j)_{j=1}^m$  and  $(g_k)_{k=1}^n$ , define the concatenated sequence and the inverse sequence by

$$\begin{aligned} h_\ell &:= \begin{cases} f_\ell, & 1 \leq \ell \leq m, \\ g_{\ell-m}, & m+1 \leq \ell \leq m+n, \end{cases} \\ f_j^{\text{rev}} &:= f_{m+1-j}^{-1}, \quad j \in \{1, \dots, m\} \end{aligned}$$

Then

$$\begin{aligned} & \text{Dom}(\text{SeqAct}[(h_\ell)_{\ell=1}^{m+n}]) \\ &= \{x \in \text{Dom}(\text{SeqAct}[(f_j)_{j=1}^m]) \mid \text{SeqAct}[(f_j)_{j=1}^m](x) \in \text{Dom}(\text{SeqAct}[(g_k)_{k=1}^n])\}, \\ & \text{SeqAct}[(h_\ell)_{\ell=1}^{m+n}] = \text{SeqAct}[(g_k)_{k=1}^n] \circ \text{SeqAct}[(f_j)_{j=1}^m], \\ & \text{SeqAct}[(f_j^{\text{rev}})_{j=1}^m] = (\text{SeqAct}[(f_j)_{j=1}^m])^{-1} \end{aligned}$$

hold. In addition,  $\text{SeqAct}[(f_j)_{j=1}^m]$  is a partial injection.

*Proof.* The partial-map composition and associativity in Definition 4.20 give the domain equality and the composition law. When  $m = 0$ , the empty-sequence action is the identity map, so the inverse-sequence law holds. When  $m \geq 1$ ,

$$(f_m \circ \dots \circ f_1)^{-1} = f_1^{-1} \circ \dots \circ f_m^{-1}$$

gives the inverse-sequence law. The existence of an inverse partial map gives partial injectivity.  $\square$

**Definition 4.22** (Abstract Typed Atomic-Action System). In this subsection, an atomic action is a one-stage partial action forming one component of a typed finite word. An atomic kind represents its action kind, and an admissible atom represents an admissible triple of atomic kind, initial position, and terminal position. Define an abstract typed atomic-action system by

$$\mathfrak{W} = (U, (Y_u)_{u \in U}, \text{Kind}, \text{AdmAtom}, \text{AtomAct}, \dagger)$$

Here,  $U$  is the position set,  $(Y_u)_{u \in U}$  is a family of underlying sets,  $\text{Kind}$  is the set of atomic kinds, and

$$\text{AdmAtom} \subseteq \text{Kind} \times U \times U$$

is the set of admissible atoms.

For each

$$(\kappa, u, v) \in \text{AdmAtom}$$

the map

$$\text{AtomAct}(\kappa; u, v) : \text{Dom}(\text{AtomAct}(\kappa; u, v)) \subseteq Y_u \longrightarrow Y_v$$

is a partial injection.

The inverse correspondence

$$\begin{aligned} \dagger : \text{AdmAtom} &\longrightarrow \text{AdmAtom}, \\ (\kappa, u, v) &\longmapsto (\kappa^\dagger, v, u) \end{aligned}$$

satisfies

$$\begin{aligned} (\kappa^\dagger)^\dagger &= \kappa, \\ \text{AtomAct}(\kappa^\dagger; v, u) &= \text{AtomAct}(\kappa; u, v)^{-1} \end{aligned}$$

**Definition 4.23** (Abstract Typed Finite Words and Induced Partial Actions). Take the abstract typed atomic-action system  $\mathfrak{W}$  of Definition 4.22. Define the family of typed finite words by

$$\begin{aligned} \text{Word}(\mathfrak{W}) := \left\{ \mathbf{w} = (m, (u_j)_{j=0}^m, (\kappa_j)_{j=1}^m) \mid \right. \\ m \in \mathbb{Z}_{\geq 0}, \quad (u_j)_{j=0}^m \in U^{m+1}, \quad (\kappa_j)_{j=1}^m \in \text{Kind}^m, \\ \left. \forall j \in \{1, \dots, m\}, \quad (\kappa_j, u_{j-1}, u_j) \in \text{AdmAtom} \right\} \end{aligned}$$

Two words are equal when their respective components—length, position sequence, and atomic-kind sequence—are equal.

For each

$$\mathbf{w} = (m, (u_j)_{j=0}^m, (\kappa_j)_{j=1}^m) \in \text{Word}(\mathfrak{W})$$

define the word length, initial type, terminal type, and atomic-kind support by

$$\begin{aligned} \text{len}(\mathbf{w}) &:= m, \\ \partial_-(\mathbf{w}) &:= u_0, \\ \partial_+(\mathbf{w}) &:= u_m, \\ \text{supp}_{\text{Kind}}(\mathbf{w}) &:= \{\kappa_j \mid 1 \leq j \leq m\} \end{aligned}$$

For each

$$\text{WordFam} \subseteq \text{Word}(\mathfrak{W})$$

define the closed-loop subfamily constructor by

$$\text{Loop}(\text{WordFam}) := \{\mathbf{w} \in \text{WordFam} \mid \partial_-(\mathbf{w}) = \partial_+(\mathbf{w})\}$$

For each  $\mathbf{w} = (m, (u_j)_{j=0}^m, (\kappa_j)_{j=1}^m) \in \text{Word}(\mathfrak{W})$  and each  $j \in \{1, \dots, m\}$ , define

$$\text{AtomAct}_j[\mathbf{w}] := \text{AtomAct}(\kappa_j; u_{j-1}, u_j)$$

and define the induced partial action by

$$\text{Act}_{\mathfrak{W}}[\mathbf{w}] := \text{SeqAct}[(\text{AtomAct}_j[\mathbf{w}])_{j=1}^m]$$

The type of this partial action is

$$\text{Act}_{\mathfrak{W}}[\mathbf{w}] : \text{Dom}(\text{Act}_{\mathfrak{W}}[\mathbf{w}]) \subseteq Y_{u_0} \longrightarrow Y_{u_m}$$



For an arbitrary partial self-map on a set  $\mathcal{Y}$ ,

$$f : \mathcal{Y} \rightharpoonup \mathcal{Y}$$

define the fixed-point set by

$$\text{Fix}(f) := \{y \in \text{Dom}(f) \mid f(y) = y\}$$

For each  $u \in U$ , define the empty word by

$$\mathbf{1}_u := (0, (u), ())$$

Write two words  $\mathbf{w}_0, \mathbf{w}_1 \in \text{Word}(\mathfrak{W})$  as

$$\begin{aligned} \mathbf{w}_0 &= \left( m, (u_j)_{j=0}^m, (\kappa_j^{(0)})_{j=1}^m \right), \\ \mathbf{w}_1 &= \left( n, (v_k)_{k=0}^n, (\kappa_k^{(1)})_{k=1}^n \right) \end{aligned}$$

When  $u_m = v_0$ , define the concatenated word that acts first by  $\mathbf{w}_0$  and then by  $\mathbf{w}_1$  by

$$\mathbf{w}_1 \star \mathbf{w}_0 := \left( m+n, (u_\ell^{(01)})_{\ell=0}^{m+n}, (\kappa_\ell^{(01)})_{\ell=1}^{m+n} \right)$$

where

$$\begin{aligned} u_\ell^{(01)} &:= \begin{cases} u_\ell, & 0 \leq \ell \leq m, \\ v_{\ell-m}, & m+1 \leq \ell \leq m+n, \end{cases} \\ \kappa_\ell^{(01)} &:= \begin{cases} \kappa_\ell^{(0)}, & 1 \leq \ell \leq m, \\ \kappa_{\ell-m}^{(1)}, & m+1 \leq \ell \leq m+n. \end{cases} \end{aligned}$$

For each  $\mathbf{w} = (m, (u_j)_{j=0}^m, (\kappa_j)_{j=1}^m) \in \text{Word}(\mathfrak{W})$ , define the inverse word by

$$\mathbf{w}^\dagger := \left( m, (u_{m-j})_{j=0}^m, (\kappa_{m+1-j}^\dagger)_{j=1}^m \right)$$

**Lemma 4.24** (Action Laws for Abstract Typed Finite Words). For each  $\mathbf{w} \in \text{Word}(\mathfrak{W})$ ,

$$\text{Act}_{\mathfrak{W}}[\mathbf{w}]$$

is a partial injection, and

$$\text{Act}_{\mathfrak{W}}[\mathbf{w}^\dagger] = (\text{Act}_{\mathfrak{W}}[\mathbf{w}])^{-1}$$

holds. For each  $u \in U$ ,

$$\text{Act}_{\mathfrak{W}}[\mathbf{1}_u] = \text{id}_{Y_u}$$

holds.

Let  $\mathbf{w}_0, \mathbf{w}_1 \in \text{Word}(\mathfrak{W})$ , and suppose that the terminal position of  $\mathbf{w}_0$  equals the initial position of  $\mathbf{w}_1$ . Then

$$\begin{aligned} &\text{Dom}(\text{Act}_{\mathfrak{W}}[\mathbf{w}_1 \star \mathbf{w}_0]) \\ &= \{x \in \text{Dom}(\text{Act}_{\mathfrak{W}}[\mathbf{w}_0]) \mid \text{Act}_{\mathfrak{W}}[\mathbf{w}_0](x) \in \text{Dom}(\text{Act}_{\mathfrak{W}}[\mathbf{w}_1])\}, \\ &\text{Act}_{\mathfrak{W}}[\mathbf{w}_1 \star \mathbf{w}_0] = \text{Act}_{\mathfrak{W}}[\mathbf{w}_1] \circ \text{Act}_{\mathfrak{W}}[\mathbf{w}_0] \end{aligned}$$

hold.

*Proof.* To the sequence of atomic partial injections in Definition 4.23, apply Lemma 4.21.  $\square$

**Definition 4.25** (Orbit Relation of Abstract Typed Word Actions). For the abstract typed atomic-action system  $\mathfrak{W}$  of Definition 4.22, define the orbit relation

$$\sim_{\mathfrak{W}}^{\text{orb}}$$

for arbitrary

$$x, y \in \bigcup_{u \in U} Y_u$$

by

$$x \sim_{\mathfrak{W}}^{\text{orb}} y \iff \exists \mathbf{w} \in \text{Word}(\mathfrak{W}), \quad x \in \text{Dom}(\text{Act}_{\mathfrak{W}}[\mathbf{w}]), \\ \text{Act}_{\mathfrak{W}}[\mathbf{w}](x) = y$$

**Lemma 4.26** (Equivalence-Relation Property of the Orbit Relation of Abstract Typed Word Actions). If

$$u \neq v \implies Y_u \cap Y_v = \emptyset$$

holds for arbitrary  $u, v \in U$ , then

$$\sim_{\mathfrak{W}}^{\text{orb}}$$

is an equivalence relation on

$$\bigcup_{u \in U} Y_u$$

*Proof.* Take arbitrary

$$x \in \bigcup_{u \in U} Y_u$$

and take  $x \in Y_u$  for some  $u \in U$ . The empty-word action law in Lemma 4.24 and Definition 4.25 give

$$x \sim_{\mathfrak{W}}^{\text{orb}} x.$$

Next, for

$$x \sim_{\mathfrak{W}}^{\text{orb}} y$$

let the witness be  $\mathbf{w}$ . The inverse-word action law in the same lemma makes  $\mathbf{w}^\dagger$  a witness of

$$y \sim_{\mathfrak{W}}^{\text{orb}} x$$

Finally, for

$$x \sim_{\mathfrak{W}}^{\text{orb}} y, \quad y \sim_{\mathfrak{W}}^{\text{orb}} z$$

let the respective witnesses be  $\mathbf{w}_0, \mathbf{w}_1$ . The action types give

$$y \in Y_{\partial_+(\mathbf{w}_0)} \cap Y_{\partial_-(\mathbf{w}_1)}.$$

Disjointness of the type domains therefore gives

$$\partial_+(\mathbf{w}_0) = \partial_-(\mathbf{w}_1).$$

The concatenated word  $\mathbf{w}_1 \star \mathbf{w}_0$  is defined, and the concatenated-action law in the same lemma gives

$$\text{Act}_{\mathfrak{W}}[\mathbf{w}_1 \star \mathbf{w}_0](x) = z.$$

Hence

$$x \sim_{\mathfrak{W}}^{\text{orb}} z.$$

The orbit relation is thus reflexive, symmetric, and transitive.  $\square$

**Definition 4.27** (Saturation of a Subset along an Equivalence Relation). For a set  $S$ , an equivalence relation on  $S$ , denoted by  $\sim$ , and a subset  $S_0 \subseteq S$ , define the saturation predicate by

$$\text{Sat}(S_0; \sim) \iff \left[ \forall p, q \in S, \quad p \sim q \implies (p \in S_0 \iff q \in S_0) \right]$$

**Lemma 4.28** (Complete Pullback and Minimality of the Image of a Saturated Set). Let  $S, T$  be sets, let an equivalence relation on  $S$  be denoted by  $\sim$ , and suppose that a map

$$\pi : S \longrightarrow T$$

satisfies, for arbitrary  $x, y \in S$ ,

$$\pi(x) = \pi(y) \iff x \sim y$$

If a subset  $S_0 \subseteq S$  satisfies

$$\text{Sat}(S_0; \sim)$$

then

$$\pi[S_0] \subseteq T$$

and, for arbitrary  $x \in S$ ,

$$\pi(x) \in \pi[S_0] \iff x \in S_0$$

hold.

Furthermore, for arbitrary  $A \subseteq T$ ,

$$[\forall x \in S_0, \quad \pi(x) \in A] \implies \pi[S_0] \subseteq A.$$

Thus  $\pi[S_0]$  is the unique minimal subset containing the entire image of  $S_0$ .

*Proof.* The definition of an image gives

$$\pi[S_0] \subseteq T.$$

If  $x \in S_0$ , the same definition gives

$$\pi(x) \in \pi[S_0].$$

Conversely, suppose

$$\pi(x) \in \pi[S_0]$$

By the definition of an image, some  $y \in S_0$  satisfies

$$\pi(x) = \pi(y).$$

The fiber condition gives

$$x \sim y.$$

From

$$\text{Sat}(S_0; \sim)$$

and  $y \in S_0$ , it follows that  $x \in S_0$ .

Finally, suppose that  $A \subseteq T$  satisfies

$$\forall x \in S_0, \quad \pi(x) \in A$$

For arbitrary  $z \in \pi[S_0]$ , some  $x \in S_0$  satisfies  $z = \pi(x)$ , and hence  $z \in A$ . Therefore

$$\pi[S_0] \subseteq A.$$

If there are two minimal subsets, mutual inclusion makes them equal. □

**Definition 4.29** (Identity-on-Domain Predicate). For a partial self-map on a set  $\mathcal{Y}$ ,

$$f : \mathcal{Y} \rightharpoonup \mathcal{Y}$$

define the identity-on-domain predicate by

$$\text{IdOnDom}(f) \iff \text{Fix}(f) = \text{Dom}(f)$$

**Definition 4.30** (Equivalence Classes and the Canonical Quotient Projection). For a set  $S$  and an equivalence relation  $\sim$  on it, define

$$\begin{aligned} [x]_\sim &:= \{y \in S \mid x \sim y\}, \\ S/\sim &:= \{[x]_\sim \mid x \in S\} \end{aligned}$$

Define the canonical quotient projection by

$$q_\sim : S \longrightarrow S/\sim, \quad q_\sim(x) := [x]_\sim$$

**Lemma 4.31** (Surjectivity and Equivalence-Class Criterion for the Canonical Quotient Projection). The canonical quotient projection  $q_\sim$  of Definition 4.30 is surjective and, for arbitrary  $x, y \in S$ ,

$$q_\sim(x) = q_\sim(y) \iff x \sim y.$$

*Proof.* Every equivalence class has the form  $[x]_\sim$  for some  $x \in S$ ; hence  $q_\sim$  is surjective.

Suppose  $[x]_\sim = [y]_\sim$ . Reflexivity gives  $x \in [x]_\sim$ , and therefore  $x \in [y]_\sim$ . Thus  $y \sim x$ , and symmetry gives  $x \sim y$ .

Conversely, suppose  $x \sim y$ . Symmetry and transitivity give, for arbitrary  $z \in S$ ,

$$x \sim z \iff y \sim z.$$

Extensionality gives  $[x]_\sim = [y]_\sim$ . □

**Definition 4.32** (Closed-Word Identity-Action Predicate). For an abstract typed atomic-action system  $\mathfrak{W}$ , define the closed-word identity-action predicate by

$$\begin{aligned} &\text{LoopId}(\mathfrak{W}) \\ &\iff \left[ \forall \mathbf{w} \in \text{Loop}(\text{Word}(\mathfrak{W})), \quad \text{IdOnDom}(\text{Act}_{\mathfrak{W}}[\mathbf{w}]) \right] \end{aligned}$$

**Theorem 4.33** (Closed-Word Identity Action and Injectivity of the Orbit Quotient Projection on Type Domains). Suppose that, for arbitrary  $u, v \in U$ ,

$$u \neq v \implies Y_u \cap Y_v = \emptyset$$

Then

$$\begin{aligned} &\text{LoopId}(\mathfrak{W}) \\ &\iff \left[ \forall u \in U, \quad q_{\sim_{\mathfrak{W}}^{\text{orb}}} \big|_{Y_u} : Y_u \longrightarrow \left( \bigcup_{v \in U} Y_v \right) / \sim_{\mathfrak{W}}^{\text{orb}} \text{ is injective} \right]. \end{aligned}$$

*Proof.* Assume

$$\text{LoopId}(\mathfrak{W})$$

and let  $u \in U$  and  $x, y \in Y_u$  satisfy

$$q_{\sim_{\mathfrak{W}}^{\text{orb}}}(x) = q_{\sim_{\mathfrak{W}}^{\text{orb}}}(y)$$

Lemma 4.31 gives

$$x \sim_{\mathfrak{W}}^{\text{orb}} y.$$

By Definition 4.25, there exists

$$\mathbf{w} \in \text{Word}(\mathfrak{W})$$

such that

$$\begin{aligned} x &\in \text{Dom}(\text{Act}_{\mathfrak{W}}[\mathbf{w}]), \\ \text{Act}_{\mathfrak{W}}[\mathbf{w}](x) &= y. \end{aligned}$$

The action type and disjointness of the type domains give

$$\partial_-(\mathbf{w}) = u = \partial_+(\mathbf{w}).$$

Therefore

$$\mathbf{w} \in \text{Loop}(\text{Word}(\mathfrak{W})).$$

Definitions 4.32 and 4.29 give

$$\text{Act}_{\mathfrak{W}}[\mathbf{w}](x) = x.$$

Hence  $x = y$ , so the restriction on each type domain is injective.

Conversely, suppose that for each  $u \in U$ ,

$$q_{\sim_{\mathfrak{W}}^{\text{orb}}} \big|_{Y_u}$$

is injective. Take arbitrary

$$\mathbf{w} \in \text{Loop}(\text{Word}(\mathfrak{W}))$$

and set

$$u := \partial_-(\mathbf{w}) = \partial_+(\mathbf{w})$$

For arbitrary

$$x \in \text{Dom}(\text{Act}_{\mathfrak{W}}[\mathbf{w}])$$

set

$$y := \text{Act}_{\mathfrak{W}}[\mathbf{w}](x)$$

The action type gives  $x, y \in Y_u$ , and Definition 4.25 gives

$$x \sim_{\mathfrak{W}}^{\text{orb}} y.$$

Lemma 4.31 gives

$$q_{\sim_{\mathfrak{W}}^{\text{orb}}}(x) = q_{\sim_{\mathfrak{W}}^{\text{orb}}}(y).$$

Injectivity on  $Y_u$  gives  $x = y$ . Therefore

$$\text{Fix}(\text{Act}_{\mathfrak{W}}[\mathbf{w}]) = \text{Dom}(\text{Act}_{\mathfrak{W}}[\mathbf{w}]).$$

Definitions 4.29 and 4.32 give

$$\text{LoopId}(\mathfrak{W})$$

□

**Definition 4.34** (Atomic-Kind Restrictions of an Abstract Typed Atomic-Action System). Take the abstract typed atomic-action system  $\mathfrak{W}$  of Definition 4.22. For an inverse-correspondence-invariant subset

$$\text{Kind}_0 \subseteq \text{Kind}$$

define the atomic-kind restriction by

$$\begin{aligned} \mathfrak{W}|_{\text{Kind}_0} &:= (U, (Y_u)_{u \in U}, \text{Kind}_0, \text{AdmAtom}_0, \text{AtomAct}|_{\text{AdmAtom}_0}, \dagger|_{\text{AdmAtom}_0}), \\ \text{AdmAtom}_0 &:= \text{AdmAtom} \cap (\text{Kind}_0 \times U \times U) \end{aligned}$$

**Definition 4.35** (Comparison-Stage Typed Atomic-Action System). Assume

$$\text{L}_1(\mathcal{X})$$

and

$$\text{L}_2(\mathcal{X}; \mathcal{J}_{\text{cmp}})$$

For each  $R \in \text{Rol}^{\text{cmp}}$ , define the underlying set at each  $u \in U$  by

$$Y_u^R := \{u\} \times \text{Im}(\text{Arr}_R^{\text{cmp}})$$

Define the atomic-kind set and the atomic-kind sets used for restrictions by

$$\begin{aligned} \text{Kind}^{\text{cmp}} &:= \{\text{tr}^+, \text{tr}^-, \text{src}\}, \\ \text{Kind}^{\text{tr}} &:= \{\text{tr}^+, \text{tr}^-\}, \quad \text{Kind}^{\text{src}} := \{\text{src}\} \end{aligned}$$

Define the set of admissible atoms by

$$\begin{aligned} \text{AdmAtom}_{\text{cmp}}^R &:= \left\{ (\kappa, u, v) \in \text{Kind}^{\text{cmp}} \times U \times U \mid \right. \\ &\quad [\kappa = \text{tr}^+ \wedge (u, v) \in \text{Adm}_{\text{cmp}}(\mathcal{X}; \mathcal{J}_{\text{cmp}})] \\ &\quad \vee [\kappa = \text{tr}^- \wedge (v, u) \in \text{Adm}_{\text{cmp}}(\mathcal{X}; \mathcal{J}_{\text{cmp}})] \\ &\quad \left. \vee [\kappa = \text{src}] \right\} \end{aligned}$$

For each  $(\kappa, u, v) \in \text{AdmAtom}_{\text{cmp}}^R$ , define the atomic partial injection by

$$\text{AtomAct}_{\text{cmp}}^R(\kappa; u, v) := \begin{cases} \text{EdgeTerm}_{uv}^{\text{tr}, R, +}(\mathcal{X}; \mathcal{J}_{\text{cmp}}), & \kappa = \text{tr}^+, \\ \text{EdgeTerm}_{vu}^{\text{tr}, R, -}(\mathcal{X}; \mathcal{J}_{\text{cmp}}), & \kappa = \text{tr}^-, \\ \text{EdgeTerm}_{uv}^{\text{src}, R}, & \kappa = \text{src} \end{cases}$$

Define the inverse correspondence by

$$(\text{tr}^+)^\dagger := \text{tr}^-, \quad (\text{tr}^-)^\dagger := \text{tr}^+, \quad (\text{src})^\dagger := \text{src}$$

Define the comparison-stage typed atomic-action system by

$$\begin{aligned} &\mathfrak{W}_{\text{cmp}}^R(\mathcal{X}; \mathcal{J}_{\text{cmp}}) \\ &:= (U, (Y_u^R)_{u \in U}, \text{Kind}^{\text{cmp}}, \text{AdmAtom}_{\text{cmp}}^R, \text{AtomAct}_{\text{cmp}}^R, \dagger) \end{aligned}$$

Lemmas 4.17 and 4.19 establish the conditions of Definition 4.22.

**Lemma 4.36** (Partition of the Comparison-Arrival Type-Domain Family). Under  $L_{\leq 2}[\mathbf{Ev}^{\text{cmp}}]$ , for each  $R \in \mathbf{Rol}^{\text{cmp}}$ ,

$$\bigcup_{u \in U} Y_u^R = \mathcal{R}_R^{\text{arr}},$$

$$u \neq v \implies Y_u^R \cap Y_v^R = \emptyset \quad (u, v \in U)$$

hold.

*Proof.* The result follows from Definitions 4.13 and 4.35.  $\square$

**Definition 4.37** (Displayed Typed Finite Pure-Transport Words). For each  $R \in \mathbf{Rol}^{\text{cmp}}$ , define the following objects. For a transport-direction-encoded word

$$\mathbf{w} = (m, (u_j)_{j=0}^m, (\varepsilon_j)_{j=1}^m)$$

define

$$\text{Enc}^{\text{tr}}[\mathbf{w}] := (m, (u_j)_{j=0}^m, (\kappa_j)_{j=1}^m), \quad \kappa_j := \begin{cases} \text{tr}^+, & \varepsilon_j = +, \\ \text{tr}^-, & \varepsilon_j = - \end{cases}$$

Define the family of all pure-transport words by

$$\begin{aligned} & \text{Word}_R^{\text{tr}}[\mathbf{l}] \\ & := \left\{ \mathbf{w} \mid \text{Enc}^{\text{tr}}[\mathbf{w}] \in \text{Word} \left( \mathfrak{W}_{\text{cmp}}^R[\mathbf{l}]|_{\text{Kind}^{\text{tr}}} \right) \right\} \end{aligned}$$

The transport-direction encoding is a bijection from the family of words on the left to the family on the right.

**Definition 4.38** (Induced Partial Action of a Pure-Transport Word). For each  $R \in \mathbf{Rol}^{\text{cmp}}$  and each  $\mathbf{w} \in \text{Word}_R^{\text{tr}}[\mathbf{l}]$ , define

$$\begin{aligned} & \text{Act}_R^{\text{tr}}[\mathbf{l}][\mathbf{w}] \\ & := \text{Act}_{\mathfrak{W}_{\text{cmp}}^R[\mathbf{l}]|_{\text{Kind}^{\text{tr}}}}[\text{Enc}^{\text{tr}}[\mathbf{w}]] \end{aligned}$$

**Definition 4.39** (Empty, Concatenated, and Inverse Pure-Transport Words). For each  $R \in \mathbf{Rol}^{\text{cmp}}$ , define the following operations. For each  $u \in U$ , define the empty word by

$$\text{Enc}^{\text{tr}}[\mathbf{1}_{R,u}^{\text{tr}}] := \mathbf{1}_u$$

For two words  $\mathbf{w}_0$  and  $\mathbf{w}_1$  whose terminal and initial positions agree,

$$\mathbf{w}_0, \mathbf{w}_1 \in \text{Word}_R^{\text{tr}}[\mathbf{l}]$$

define the concatenated word by

$$\text{Enc}^{\text{tr}}[\mathbf{w}_1 \star \mathbf{w}_0] := \text{Enc}^{\text{tr}}[\mathbf{w}_1] \star \text{Enc}^{\text{tr}}[\mathbf{w}_0]$$

For an arbitrary pure-transport word  $\mathbf{w}$ , define the inverse word by

$$\text{Enc}^{\text{tr}}[\mathbf{w}^\dagger] := (\text{Enc}^{\text{tr}}[\mathbf{w}])^\dagger$$

**Lemma 4.40** (Compatibility of Pure-Transport Word Operations with Induced Partial Actions). For each  $R \in \mathbf{Rol}^{\text{cmp}}$ , the action system for pure-transport words defined by Definitions 4.38 and 4.39 is a special case of the abstract typed finite-word action system of Lemma 4.24. Consequently, the conclusions of that lemma for inverse-word actions, empty-word actions, concatenated-word domains, and composed actions hold.

*Proof.* Through Definitions 4.38 and 4.39, apply Lemma 4.24.  $\square$

**Lemma 4.41** (Partial Injectivity of the Induced Partial Action of a Pure-Transport Word).

For each  $R \in \mathbf{Rol}^{\text{cmp}}$  and each  $\mathbf{w} \in \mathbf{Word}_R^{\text{tr}}[\mathbf{l}]$ ,

$$\mathbf{Act}_R^{\text{tr}}[\mathbf{l}][\mathbf{w}]$$

is a partial injection.

*Proof.* Apply Lemma 4.24.  $\square$

**Definition 4.42** (Family of All Pure-Transport Closed-Loop Words). For each  $R \in \mathbf{Rol}^{\text{cmp}}$ , define

$$\begin{aligned} \mathbf{LoopWord}_{\text{all}}^{\text{tr}, R}[\mathbf{l}] \\ := \{ \mathbf{w} = (m, (u_j)_{j=0}^m, (\varepsilon_j)_{j=1}^m) \in \mathbf{Word}_R^{\text{tr}}[\mathbf{l}] \mid u_m = u_0 \} \end{aligned}$$

**Definition 4.43** (Displayed Typed Finite Source-Matching Words). For each  $R \in \mathbf{Rol}^{\text{cmp}}$ , define the source-matching atomic-action system by

$$\mathfrak{W}_{\text{src}}^R := \mathfrak{W}_{\text{cmp}}^R[\mathbf{l}]|_{\mathbf{Kind}^{\text{src}}}.$$

For a source-encoded word

$$\mathbf{v} = (m, (u_j)_{j=0}^m)$$

define

$$\mathbf{Enc}^{\text{src}}[\mathbf{v}] := (m, (u_j)_{j=0}^m, (\text{src})_{j=1}^m)$$

Define the family of all source-matching words by

$$\begin{aligned} \mathbf{Word}_R^{\text{src}}(\mathcal{X}) \\ := \{ \mathbf{v} \mid \mathbf{Enc}^{\text{src}}[\mathbf{v}] \in \mathbf{Word}(\mathfrak{W}_{\text{src}}^R) \} \end{aligned}$$

The source encoding is a bijection from the family of words on the left to the family on the right.

**Definition 4.44** (Induced Partial Action of a Source-Matching Word). For each  $R \in \mathbf{Rol}^{\text{cmp}}$  and each  $\mathbf{v} \in \mathbf{Word}_R^{\text{src}}(\mathcal{X})$ , define

$$\mathbf{Act}_R^{\text{src}}[\mathbf{v}] := \mathbf{Act}_{\mathfrak{W}_{\text{src}}^R}[\mathbf{Enc}^{\text{src}}[\mathbf{v}]]$$

**Definition 4.45** (Empty, Concatenated, and Inverse Source-Matching Words). For each  $R \in \mathbf{Rol}^{\text{cmp}}$ , define the following operations. For each  $u \in U$ , define the empty word by

$$\mathbf{Enc}^{\text{src}}[\mathbf{1}_{R,u}^{\text{src}}] := \mathbf{1}_u$$

For two words  $\mathbf{v}_0$  and  $\mathbf{v}_1$  whose terminal and initial positions agree,

$$\mathbf{v}_0, \mathbf{v}_1 \in \mathbf{Word}_R^{\text{src}}(\mathcal{X})$$



define the concatenated word by

$$\text{Enc}^{\text{src}}[\mathbf{v}_1 \star \mathbf{v}_0] := \text{Enc}^{\text{src}}[\mathbf{v}_1] \star \text{Enc}^{\text{src}}[\mathbf{v}_0]$$

For an arbitrary source-matching word  $\mathbf{v}$ , define the inverse word by

$$\text{Enc}^{\text{src}}[\mathbf{v}^\dagger] := (\text{Enc}^{\text{src}}[\mathbf{v}])^\dagger$$

**Lemma 4.46** (Compatibility of Source-Matching Word Operations with Induced Partial Actions). For each  $R \in \text{Rol}^{\text{cmp}}$ , the action system of source-matching words determined by Definition 4.44 and Definition 4.45 is a special case of the abstract typed finite-word action system in Lemma 4.24. Consequently, the conclusions of that lemma concerning inverse-word actions, empty-word actions, concatenated-word domains, and composite actions hold.

*Proof.* Apply Definition 4.44 and Definition 4.45, and then Lemma 4.24.  $\square$

**Lemma 4.47** (Partial Injectivity of the Induced Partial Action of a Source-Matching Word).

For each  $R \in \text{Rol}^{\text{cmp}}$  and each  $\mathbf{v} \in \text{Word}_R^{\text{src}}(\mathcal{X})$ ,

$$\text{Act}_R^{\text{src}}[\mathbf{v}]$$

is a partial injection.

*Proof.* Apply Lemma 4.24.  $\square$

**Lemma 4.48** (Identity of a Source-Matching Word When Its Initial and Terminal Positions Coincide). For each  $R \in \text{Rol}^{\text{cmp}}$  and each

$$\mathbf{v} = (m, (u_j)_{j=0}^m) \in \text{Word}_R^{\text{src}}(\mathcal{X})$$

if  $u_m = u_0$ , then

$$\text{Act}_R^{\text{src}}[\mathbf{v}] = \text{id}_{\text{Dom}(\text{Act}_R^{\text{src}}[\mathbf{v}])}$$

holds.

*Proof.* Take any

$$x_0 = (u_0, \alpha_0) \in \text{Dom}(\text{Act}_R^{\text{src}}[\mathbf{v}])$$

and set

$$\text{SrcTok}_R := \text{Rec}_{R, u_0}^{\text{cmp}}(\alpha_0)$$

By Definition 4.5 and finite induction, the corresponding stage sequence satisfies

$$x_j = (u_j, \text{Arr}_{R, u_j}^{\text{cmp}}(\text{SrcTok}_R)), \quad j \in \{0, \dots, m\}$$

The equality  $u_m = u_0$  yields  $x_m = x_0$ .  $\square$

**Definition 4.49** (Displayed Typed Finite Extended Comparison Words). For each  $R \in \text{Rol}^{\text{cmp}}$ , define

$$\text{Word}_R^{\text{ext}}[\mathbb{I}] := \text{Word}(\mathfrak{W}_{\text{cmp}}^R[\mathbb{I}])$$

Each of its elements is displayed as

$$\mathbf{e} = (m, (u_j)_{j=0}^m, (\kappa_j(\mathbf{e}))_{j=1}^m)$$

**Definition 4.50** (Induced Partial Action of an Extended Comparison Word). For each  $R \in \text{Rol}^{\text{cmp}}$  and each  $\mathbf{e} \in \text{Word}_R^{\text{ext}}[\mathbb{I}]$ , define

$$\begin{aligned} \text{Act}_R^{\text{ext}}[\mathbb{I}][\mathbf{e}] \\ := \text{Act}_{\mathcal{W}_{\text{cmp}}^R[\mathbb{I}]}[\mathbf{e}] \end{aligned}$$

**Definition 4.51** (Empty, Concatenated, and Inverse Extended Comparison Words). For each  $R \in \text{Rol}^{\text{cmp}}$ , define the following objects. For each  $u \in U$ , define the empty word by

$$\mathbf{1}_{R,u}^{\text{ext}} := \mathbf{1}_u$$

For each  $\mathbf{e}_0$  whose terminal position coincides with the initial position of  $\mathbf{e}_1$ , where

$$\mathbf{e}_0, \mathbf{e}_1 \in \text{Word}_R^{\text{ext}}[\mathbb{I}],$$

define the concatenated word  $\mathbf{e}_1 \star \mathbf{e}_0$  by the concatenation in Definition 4.23. For each extended comparison word  $\mathbf{e}$ , define the inverse word  $\mathbf{e}^\dagger$  by the inverse-word construction in the same definition.

**Lemma 4.52** (Compatibility of Extended Comparison Word Operations with Induced Partial Actions). For each  $R \in \text{Rol}^{\text{cmp}}$ , the action system of extended comparison words determined by Definition 4.50 and Definition 4.51 is a special case of the abstract typed finite-word action system in Lemma 4.24. Consequently, the conclusions of that lemma concerning inverse-word actions, empty-word actions, concatenated-word domains, and composite actions hold.

*Proof.* Apply Definition 4.50 and Definition 4.51, and then Lemma 4.24.  $\square$

**Lemma 4.53** (Partial Injectivity of the Induced Partial Action of an Extended Comparison Word).

For each  $R \in \text{Rol}^{\text{cmp}}$  and each  $\mathbf{e} \in \text{Word}_R^{\text{ext}}[\mathbb{I}]$ ,

$$\text{Act}_R^{\text{ext}}[\mathbb{I}][\mathbf{e}]$$

is a partial injection.

*Proof.* Apply Lemma 4.24.  $\square$

**Definition 4.54** (Standard Embeddings of Pure-Transport Words and Source-Matching Words). For each  $R \in \text{Rol}^{\text{cmp}}$ , the encoding maps of Definition 4.37 and Definition 4.43 give the standard embeddings

$$\begin{aligned} \text{Enc}^{\text{tr}} : \text{Word}_R^{\text{tr}}[\mathbb{I}] &\hookrightarrow \text{Word}_R^{\text{ext}}[\mathbb{I}], \\ \text{Enc}^{\text{src}} : \text{Word}_R^{\text{src}}(\mathcal{X}) &\hookrightarrow \text{Word}_R^{\text{ext}}[\mathbb{I}] \end{aligned}$$

**Lemma 4.55** (Injectivity and Action Compatibility of the Standard Embeddings). For each  $R \in \text{Rol}^{\text{cmp}}$ , the two standard embeddings are respectively injections

$$\begin{aligned} \text{Word}_R^{\text{tr}}[\mathbb{I}] &\longrightarrow \text{Word}_R^{\text{ext}}[\mathbb{I}], \\ \text{Word}_R^{\text{src}}(\mathcal{X}) &\longrightarrow \text{Word}_R^{\text{ext}}[\mathbb{I}] \end{aligned}$$

For any  $\mathbf{w}$  and  $\mathbf{v}$  in Definition 4.54,

$$\begin{aligned} \text{Act}_R^{\text{ext}}[\mathbb{I}][\text{Enc}^{\text{tr}}[\mathbf{w}]] &= \text{Act}_R^{\text{tr}}[\mathbb{I}][\mathbf{w}], \\ \text{Act}_R^{\text{ext}}[\mathbb{I}][\text{Enc}^{\text{src}}[\mathbf{v}]] &= \text{Act}_R^{\text{src}}[\mathbb{I}][\mathbf{v}] \end{aligned}$$

hold as equalities of partial maps. For each  $u \in U$ ,

$$\text{Enc}^{\text{tr}}[\mathbf{1}_R^{\text{tr}}] = \mathbf{1}_R^{\text{ext}} = \text{Enc}^{\text{src}}[\mathbf{1}_R^{\text{src}}]$$

holds.

*Proof.* The claim follows from the bijectivity in Definition 4.37 and Definition 4.43, together with Definition 4.38 and Definition 4.44.  $\square$

**Definition 4.56** (Family of Mixed Words and Family of All Mixed Closed-Loop Words). For each  $R \in \text{Rol}^{\text{cmp}}$ , define the family of mixed words by

$$\begin{aligned} & \text{Word}_R^{\text{mix}}[\mathbb{I}] \\ & := \left\{ \mathbf{e} \in \text{Word}_R^{\text{ext}}[\mathbb{I}] \mid \begin{array}{l} \text{supp}_{\text{Kind}^{\text{cmp}}}(\mathbf{e}) \cap \text{Kind}^{\text{tr}} \neq \emptyset, \\ \text{supp}_{\text{Kind}^{\text{cmp}}}(\mathbf{e}) \cap \text{Kind}^{\text{src}} \neq \emptyset \end{array} \right\} \end{aligned}$$

Define the family of all mixed closed-loop words by

$$\text{LoopWord}_{\text{all}}^{\text{mix}, R}[\mathbb{I}] := \text{Loop}(\text{Word}_R^{\text{mix}}[\mathbb{I}])$$

**Definition 4.57** (One-Step Deletion of a Source-Matching Closed-Loop Subword). For each  $R \in \text{Rol}^{\text{cmp}}$  and each

$$\mathbf{e} = (m, (u_j)_{j=0}^m, (\kappa_j(\mathbf{e}))_{j=1}^m) \in \text{Word}_R^{\text{ext}}[\mathbb{I}]$$

define the following operation. When the indices  $0 \leq r_0 < r_1 \leq m$  satisfy

$$u_{r_0} = u_{r_1}, \quad \forall j \in \{r_0 + 1, \dots, r_1\}, \quad \kappa_j(\mathbf{e}) = \text{src}$$

define

$$\begin{aligned} & \text{Del}_{r_0, r_1}^{\text{src}}[\mathbf{e}] \\ & := (m - (r_1 - r_0), (u_0, \dots, u_{r_0}, u_{r_1+1}, \dots, u_m), \\ & \quad (\kappa_1(\mathbf{e}), \dots, \kappa_{r_0}(\mathbf{e}), \kappa_{r_1+1}(\mathbf{e}), \dots, \kappa_m(\mathbf{e}))) \end{aligned}$$

An initial or terminal sequence that becomes empty is taken to be the empty sequence. For each  $\mathbf{e}' \in \text{Word}_R^{\text{ext}}[\mathbb{I}]$ , define the one-step deletion relation by

$$\begin{aligned} \mathbf{e} \xrightarrow{\text{Del}^{\text{src}}} \mathbf{e}' & \iff \left[ \begin{array}{l} \exists r_0, r_1, \quad 0 \leq r_0 < r_1 \leq m, \quad u_{r_0} = u_{r_1}, \\ \forall j \in \{r_0 + 1, \dots, r_1\}, \quad \kappa_j(\mathbf{e}) = \text{src}, \\ \mathbf{e}' = \text{Del}_{r_0, r_1}^{\text{src}}[\mathbf{e}] \end{array} \right] \end{aligned}$$

**Definition 4.58** (Family of Source-Irreducible Mixed Closed-Loop Words). For each  $R \in \text{Rol}^{\text{cmp}}$ , define

$$\begin{aligned} & \text{LoopWord}_{\text{src-irr}}^{\text{mix}, R}[\mathbb{I}] \\ & := \left\{ \mathbf{e} \in \text{LoopWord}_{\text{all}}^{\text{mix}, R}[\mathbb{I}] \mid \nexists \mathbf{e}' \in \text{Word}_R^{\text{ext}}[\mathbb{I}], \quad \mathbf{e} \xrightarrow{\text{Del}^{\text{src}}} \mathbf{e}' \right\} \end{aligned}$$

**Lemma 4.59** (Exclusive Four-Case Decomposition of Extended Comparison Words). For each  $R \in \text{Rol}^{\text{cmp}}$  and each

$$\mathbf{e} \in \text{Word}_R^{\text{ext}}[\mathbb{I}]$$

exactly one of the following four cases holds:

- (i)  $\text{len}(\mathbf{e}) = 0$ ,
- (ii)  $\text{len}(\mathbf{e}) \geq 1$ ,  $\text{supp}_{\text{Kind}^{\text{cmp}}}(\mathbf{e}) \subseteq \text{Kind}^{\text{tr}}$ ,
- (iii)  $\text{len}(\mathbf{e}) \geq 1$ ,  $\text{supp}_{\text{Kind}^{\text{cmp}}}(\mathbf{e}) = \text{Kind}^{\text{src}}$ ,
- (iv)  $\mathbf{e} \in \text{Word}_R^{\text{mix}}[\mathbb{I}]$ .

In the second case, there exists a unique  $\mathbf{w} \in \text{Word}_R^{\text{tr}}[\mathbb{I}]$  such that

$$\mathbf{e} = \text{Enc}^{\text{tr}}[\mathbf{w}]$$

In the third case, there exists a unique  $\mathbf{v} \in \text{Word}_R^{\text{src}}(\mathcal{X})$  such that

$$\mathbf{e} = \text{Enc}^{\text{src}}[\mathbf{v}]$$

*Proof.* The identity

$$\text{Kind}^{\text{cmp}} = \text{Kind}^{\text{tr}} \sqcup \text{Kind}^{\text{src}}$$

makes the four cases exclusive and exhaustive. Existence and uniqueness in the second and third cases follow from Definition 4.37, Definition 4.43, and Lemma 4.55.  $\square$

**Lemma 4.60** (Extension of the Action under Deletion of a Source-Matching Closed-Loop Subword). For each  $R \in \text{Rol}^{\text{cmp}}$  and each  $\mathbf{e}, \mathbf{e}' \in \text{Word}_R^{\text{ext}}[\mathbb{I}]$ , suppose that

$$\mathbf{e} \xrightarrow{\text{Del}^{\text{src}}} \mathbf{e}'$$

Then

$$\begin{aligned} \text{Dom}(\text{Act}_R^{\text{ext}}[\mathbb{I}][\mathbf{e}]) &\subseteq \text{Dom}(\text{Act}_R^{\text{ext}}[\mathbb{I}][\mathbf{e}']), \\ \text{Act}_R^{\text{ext}}[\mathbb{I}][\mathbf{e}']|_{\text{Dom}(\text{Act}_R^{\text{ext}}[\mathbb{I}][\mathbf{e}])} &= \text{Act}_R^{\text{ext}}[\mathbb{I}][\mathbf{e}] \end{aligned}$$

*Proof.* For indices  $r_0, r_1$  that establish the deletion relation, define the source-matching word on the deleted interval by

$$\mathbf{v}_{r_0, r_1} := (r_1 - r_0, (u_{r_0+k})_{k=0}^{r_1-r_0}) \in \text{Word}_R^{\text{src}}(\mathcal{X})$$

By Lemma 4.55 and Lemma 4.48, the stage sequence of the original word satisfies  $x_{r_1} = x_{r_0}$ . Removing stages  $r_0 + 1, \dots, r_1$  yields the stage sequence of the word after deletion; hence the domain inclusion and equality of action values hold.  $\square$

**Lemma 4.61** (Finite Source Reduction of a Mixed Closed-Loop Word). For each  $R \in \text{Rol}^{\text{cmp}}$  and every

$$\mathbf{e} \in \text{LoopWord}_{\text{all}}^{\text{mix}, R}[\mathbb{I}]$$

there exists a finite sequence

$$\mathbf{e}_0 = \mathbf{e}, \quad \mathbf{e}_0 \xrightarrow{\text{Del}^{\text{src}}} \dots \xrightarrow{\text{Del}^{\text{src}}} \mathbf{e}_N$$

The terminal word satisfies

$$\begin{aligned} \mathbf{e}_N &\in \text{LoopWord}_{\text{src-irr}}^{\text{mix}, R}[\mathbb{I}] \\ \text{or } \exists \mathbf{w} &\in \text{LoopWord}_{\text{all}}^{\text{tr}, R}[\mathbb{I}], \quad \mathbf{e}_N = \text{Enc}^{\text{tr}}[\mathbf{w}] \end{aligned}$$

For each  $k \in \{0, \dots, N\}$ ,

$$\begin{aligned} \text{Dom}(\text{Act}_R^{\text{ext}}[\mathbb{I}][\mathbf{e}_0]) &\subseteq \text{Dom}(\text{Act}_R^{\text{ext}}[\mathbb{I}][\mathbf{e}_k]), \\ \text{Act}_R^{\text{ext}}[\mathbb{I}][\mathbf{e}_k]|_{\text{Dom}(\text{Act}_R^{\text{ext}}[\mathbb{I}][\mathbf{e}_0])} &= \text{Act}_R^{\text{ext}}[\mathbb{I}][\mathbf{e}_0] \end{aligned}$$

*Proof.* Use strong induction on word length. For a source-irreducible word, set  $N = 0$ . When a one-step deletion is possible, the word length decreases strictly, while the transport atoms and the closed-loop condition are preserved. If the word after deletion consists only of transport atoms, apply Lemma 4.59; otherwise apply the induction hypothesis. The two formulas concerning the action follow by applying Lemma 4.60 to each one-step deletion.  $\square$

**Definition 4.62** (Candidate Pure-Transport Word and Candidate Pure-Transport Action of a Local Closed-Loop Specification). Assume  $L_{\leq 2}[\text{Ev}^{\text{cmp}}]$ . For any  $R \in \text{Rol}^{\text{cmp}}$  and

$$\text{Spec} = (m, (u_j)_{j=0}^m, (\varepsilon_j)_{j=1}^m, \text{Dom}_{\text{Spec}}) \in \text{LoopSpec}_{\text{loc}}^R[\mathbb{I}]$$

define the candidate pure-transport word by

$$\text{Word}_R^{\text{tr}, \text{cand}}[\text{Spec}] := (m, (u_j)_{j=0}^m, (\varepsilon_j)_{j=1}^m)$$

Define the candidate pure-transport action by

$$\begin{aligned} \text{Act}_R^{\text{tr}, \text{cand}}[\text{Spec}] \\ := \text{Act}_R^{\text{tr}}[\mathbb{I}] \left[ \text{Word}_R^{\text{tr}, \text{cand}}[\text{Spec}] \right] \end{aligned}$$

**Lemma 4.63** (Admissibility of the Candidate Pure-Transport Word and Action from a Local Closed-Loop Specification). Under  $L_{\leq 2}[\text{Ev}^{\text{cmp}}]$ , for each  $R \in \text{Rol}^{\text{cmp}}$  and each  $\text{Spec} \in \text{LoopSpec}_{\text{loc}}^R[\mathbb{I}]$ ,

$$\text{Word}_R^{\text{tr}, \text{cand}}[\text{Spec}] \in \text{LoopWord}_{\text{all}}^{\text{tr}, R}[\mathbb{I}]$$

and

$$\text{Act}_R^{\text{tr}, \text{cand}}[\text{Spec}] : \text{Dom} \left( \text{Act}_R^{\text{tr}, \text{cand}}[\text{Spec}] \right) \longrightarrow Y_{u_0}^R$$

is a partial injection.

*Proof.* The type conditions in Definition 4.4 and the equality  $u_m = u_0$  place the candidate word in the family of all pure-transport closed-loop words. Partial injectivity follows from Lemma 4.41.  $\square$

**Definition 4.64** (Comparison-Stage Judgment Condition  $L_3$ : Realization of Local Closed-Loop Specifications). Define the local closed-loop specification realization condition by

$$\begin{aligned} L_3[\text{Ev}_3^{\text{cmp}}] &\iff \left[ L_{\leq 2}[\text{Ev}^{\text{cmp}}] \right. \\ &\implies \forall R \in \text{Rol}^{\text{cmp}}, \quad \forall \text{Spec} \in \text{LoopSpec}_{\text{loc}}^R[\mathbb{I}], \\ &\quad \left. \text{Dom}_{\text{Spec}} \subseteq \text{Dom} \left( \text{Act}_R^{\text{tr}, \text{cand}}[\text{Spec}] \right) \right] \end{aligned}$$

Define the third-stage establishment bundle by

$$L_{\leq 3}[\text{Ev}^{\text{cmp}}] := L_{\leq 2}[\text{Ev}^{\text{cmp}}] \wedge L_3[\text{Ev}_3^{\text{cmp}}]$$

**Definition 4.65** (Realized Local-Specification Closed-Loop Data). Assume  $L_{\leq 3}[\text{Ev}^{\text{cmp}}]$ . For any

$$R \in \text{Rol}^{\text{cmp}}$$

and

$$\text{Spec} \in \text{LoopSpec}_{\text{loc}}^R[\mathbb{I}]$$

define the realized restricted action by

$$\text{Act}_R^{\text{tr},\text{real}}[\text{Spec}] := \text{Act}_R^{\text{tr},\text{cand}}[\text{Spec}] \Big|_{\text{Dom}_{\text{Spec}}}$$

Define the realized local-specification closed-loop datum by

$$\text{LoopDat}_R^{\text{real}}[\text{Spec}] := \left( \text{Spec}, \text{Word}_R^{\text{tr},\text{cand}}[\text{Spec}], \text{Dom}_{\text{Spec}}, \text{Act}_R^{\text{tr},\text{real}}[\text{Spec}] \right)$$

Define the indexed family of data corresponding to all local specifications by

$$\text{LoopDat}_R^{\text{real},\text{loc}}[\mathbb{I}] := \left( \text{LoopDat}_R^{\text{real}}[\text{Spec}] \right)_{\text{Spec} \in \text{LoopSpec}_{\text{loc}}^R[\mathbb{I}]}$$

**Lemma 4.66** (Injectivity of a Realized Local Transport Action). Under  $\text{L}_{\leq 3}[\text{Ev}^{\text{cmp}}]$ , for any

$$R \in \text{Rol}^{\text{cmp}}$$

and

$$\text{Spec} \in \text{LoopSpec}_{\text{loc}}^R[\mathbb{I}]$$

the map

$$\text{Act}_R^{\text{tr},\text{real}}[\text{Spec}] : \text{Dom}_{\text{Spec}} \longrightarrow Y_{\text{Base}(\text{Spec})}^R$$

is injective.

*Proof.* By Definition 4.64,

$$\text{Dom}_{\text{Spec}} \subseteq \text{Dom} \left( \text{Act}_R^{\text{tr},\text{cand}}[\text{Spec}] \right).$$

By Lemma 4.63, the candidate pure-transport action is a partial injection on its domain. The domain restriction in Definition 4.65 therefore makes the restricted action stated in the lemma injective.  $\square$

**Definition 4.67** (Comparison-Stage Judgment Condition  $\text{L}_4$ : Identity Action of Realized Local-Specification Closed Loops). Define the identity-action condition for realized local-specification closed loops by

$$\begin{aligned} & \text{L}_4[\text{Ev}_4^{\text{cmp}}] \\ & \iff \left[ \text{L}_{\leq 3}[\text{Ev}^{\text{cmp}}] \right. \\ & \quad \implies \forall R \in \text{Rol}^{\text{cmp}}, \quad \forall \text{Spec} \in \text{LoopSpec}_{\text{loc}}^R[\mathbb{I}], \\ & \quad \left. \text{IdOnDom} \left( \text{Act}_R^{\text{tr},\text{real}}[\text{Spec}] \right) \right] \end{aligned}$$

Define the fourth-stage establishment bundle by

$$\text{L}_{\leq 4}[\text{Ev}^{\text{cmp}}] := \text{L}_{\leq 3}[\text{Ev}^{\text{cmp}}] \wedge \text{L}_4[\text{Ev}_4^{\text{cmp}}]$$

**Definition 4.68** (Extended Comparison-Reachability Equivalence Relation). Assume

$$\text{L}_{\leq 2}[\text{Ev}^{\text{cmp}}]$$

For each

$$R \in \text{Rol}^{\text{cmp}}$$

define the extended comparison-reachability equivalence relation by

$$\sim_R^{\text{cmp}}[\mathbb{I}] := \sim_{\mathfrak{M}_{\text{cmp}}^R[\mathbb{I}]}^{\text{orb}}$$

**Lemma 4.69** (Equivalence-Relation Property of the Extended Comparison-Reachability Relation). Under

$$L_{\leq 2}[\text{Ev}^{\text{cmp}}]$$

for each

$$R \in \text{Rol}^{\text{cmp}}$$

the relation

$$\sim_R^{\text{cmp}} [I]$$

is an equivalence relation on

$$\mathcal{R}_R^{\text{arr}}$$

*Proof.* Lemma 4.36 establishes the union and pairwise-disjointness conditions on the underlying family required by Lemma 4.26. Apply that lemma to the comparison-stage typed atomic-action system  $\mathfrak{M}_{\text{cmp}}^R[I]$  of Definition 4.35, and then use Definition 4.68.  $\square$

**Definition 4.70** (Comparison Quotient Sets and Canonical Projections). Assume

$$L_{\leq 2}[\text{Ev}^{\text{cmp}}]$$

For each

$$R \in \text{Rol}^{\text{cmp}}$$

define the comparison quotient set by

$$\text{Qu}_R^{\text{can}}[I] := \mathcal{R}_R^{\text{arr}} / \sim_R^{\text{cmp}} [I]$$

Define the canonical projection by

$$\text{Prj}_R^{\text{can}}[I] : \mathcal{R}_R^{\text{arr}} \longrightarrow \text{Qu}_R^{\text{can}}[I]$$

and

$$\text{Prj}_R^{\text{can}}[I](x) := [x]_{\sim_R^{\text{cmp}} [I]}$$

**Lemma 4.71** (Surjectivity and Equivalence-Class Criterion for the Canonical Projection). For each

$$R \in \text{Rol}^{\text{cmp}}$$

the map

$$\text{Prj}_R^{\text{can}}[I] : \mathcal{R}_R^{\text{arr}} \twoheadrightarrow \text{Qu}_R^{\text{can}}[I]$$

is surjective. Moreover, for any

$$x, y \in \mathcal{R}_R^{\text{arr}}$$

the equivalence

$$\text{Prj}_R^{\text{can}}[I](x) = \text{Prj}_R^{\text{can}}[I](y) \iff x \sim_R^{\text{cmp}} [I] y.$$

holds.

*Proof.* Apply Lemma 4.69 and Lemma 4.31 to

$$S = \mathcal{R}_R^{\text{arr}}, \quad \sim = \sim_R^{\text{cmp}} [I]$$

$\square$

**Definition 4.72** (Local Source-Compatible Two-Role Comparison Relation). For each  $u \in U$ , define the local source-compatible two-role comparison relation by

$$\begin{aligned} & \text{Rel}_u^{\text{cmp}, \text{src}}[\mathbb{I}] \\ & := \left\{ (\alpha_R)_{R \in \text{Rol}^{\text{cmp}}} \mid \right. \\ & \quad (\alpha_R)_{R \in \text{Rol}^{\text{cmp}}} \in \prod_{R \in \text{Rol}^{\text{cmp}}} \text{Im}(\text{Arr}_{R,u}^{\text{cmp}}), \\ & \quad \left. (\text{Rec}_{R,u}^{\text{cmp}}(\alpha_R))_{R \in \text{Rol}^{\text{cmp}}} \in \text{Rel}^{\text{src}} \right\} \end{aligned}$$

**Definition 4.73** (Comparison-Stage Judgment Condition  $L_5$ : Source Compatibility of the Local Two-Role Comparison Relation). Define the source-compatibility condition for the local two-role comparison relation by

$$\begin{aligned} & L_5[\text{Ev}_5^{\text{cmp}}] \\ & \iff \left[ L_{\leq 4}[\text{Ev}^{\text{cmp}}] \implies \forall u \in U, \right. \\ & \quad \left. \text{Rel}_u^{\text{cmp}} = \text{Rel}_u^{\text{cmp}, \text{src}}[\mathbb{I}] \right] \end{aligned}$$

Define the fifth-stage establishment bundle by

$$L_{\leq 5}[\text{Ev}^{\text{cmp}}] := L_{\leq 4}[\text{Ev}^{\text{cmp}}] \wedge L_5[\text{Ev}_5^{\text{cmp}}]$$

**Definition 4.74** (Synchronized Two-Role Comparison-Arrival Space and Local Relation Bundle). Define the synchronized two-role comparison-arrival space by

$$\mathcal{R}^{\text{pair}} := \left\{ (u, (\alpha_R)_{R \in \text{Rol}^{\text{cmp}}}) \mid \begin{array}{l} u \in U, \\ (\alpha_R)_{R \in \text{Rol}^{\text{cmp}}} \in \prod_{R \in \text{Rol}^{\text{cmp}}} \text{Im}(\text{Arr}_{R,u}^{\text{cmp}}) \end{array} \right\}$$

Define the local two-role comparison-relation bundle by

$$\text{Rel}^{\text{pair}} := \{(u, \alpha) \in \mathcal{R}^{\text{pair}} \mid \alpha \in \text{Rel}_u^{\text{cmp}}\}$$

where

$$\alpha = (\alpha_R)_{R \in \text{Rol}^{\text{cmp}}}$$

**Definition 4.75** (Synchronized Two-Role Comparison-Pair Equivalence Relation). Assume

$$L_{\leq 2}[\text{Ev}^{\text{cmp}}]$$

Define the synchronized two-role comparison-pair equivalence relation

$$\sim^{\text{pair}}[\mathbb{I}]$$

on

$$\mathcal{R}^{\text{pair}}$$



as follows. For any

$$p, q \in \mathcal{R}^{\text{pair}}$$

written as

$$\begin{aligned} p &= (u, (\alpha_R)_{R \in \text{Rol}^{\text{cmp}}}), \\ q &= (v, (\beta_R)_{R \in \text{Rol}^{\text{cmp}}}) \end{aligned}$$

set

$$\begin{aligned} p &\sim^{\text{pair}} [\mathbb{I}] q \\ &\iff [\forall R \in \text{Rol}^{\text{cmp}}, \quad (u, \alpha_R) \sim_R^{\text{cmp}} [\mathbb{I}] (v, \beta_R)]. \end{aligned}$$

**Lemma 4.76** (Equivalence-Relation Property of the Synchronized Two-Role Comparison-Pair Relation). Under

$$\mathbf{L}_{\leq 2}[\text{Ev}^{\text{cmp}}]$$

the relation

$$\sim^{\text{pair}} [\mathbb{I}]$$

is an equivalence relation on

$$\mathcal{R}^{\text{pair}}$$

*Proof.* By Lemma 4.69, for each

$$R \in \text{Rol}^{\text{cmp}}$$

the relation

$$\sim_R^{\text{cmp}} [\mathbb{I}]$$

is an equivalence relation on the corresponding comparison-arrival space. The right-hand side of Definition 4.75 is the universal conjunction of the equivalence relations for the respective role types; it therefore satisfies reflexivity, symmetry, and transitivity componentwise.  $\square$

**Definition 4.77** (Comparison-Stage Judgment Condition  $\mathbf{L}_6$ : Consistency of Local Two-Role Comparison-Relation Values across Equivalent Representatives). Define the consistency condition for local two-role comparison-relation values across equivalent representatives by

$$\begin{aligned} &\mathbf{L}_6[\text{Ev}_6^{\text{cmp}}] \\ &\iff \left[ \mathbf{L}_{\leq 5}[\text{Ev}^{\text{cmp}}] \implies \text{Sat}(\text{Rel}^{\text{pair}}, \sim^{\text{pair}} [\mathbb{I}]) \right] \end{aligned}$$

Define the sixth-stage establishment bundle by

$$\mathbf{L}_{\leq 6}[\text{Ev}^{\text{cmp}}] := \mathbf{L}_{\leq 5}[\text{Ev}^{\text{cmp}}] \wedge \mathbf{L}_6[\text{Ev}_6^{\text{cmp}}]$$

**Definition 4.78** (Comparison-Stage Judgment Condition  $\mathbf{L}_7$ : Identity Action of Mixed Closed Loops). Define the mixed closed-loop identity-action condition by

$$\begin{aligned} &\mathbf{L}_7[\text{Ev}_7^{\text{cmp}}] \\ &\iff \left[ \mathbf{L}_{\leq 6}[\text{Ev}^{\text{cmp}}] \implies \forall R \in \text{Rol}^{\text{cmp}}, \right. \\ &\quad \forall \mathbf{e} \in \text{LoopWord}_{\text{src-irr}}^{\text{mix}, R}[\mathbb{I}], \\ &\quad \left. \text{IdOnDom}(\text{Act}_R^{\text{ext}}[\mathbb{I}][\mathbf{e}]) \right]. \end{aligned}$$

Define the seventh-stage establishment bundle by

$$\mathbf{L}_{\leq 7}[\text{Ev}^{\text{cmp}}] := \mathbf{L}_{\leq 6}[\text{Ev}^{\text{cmp}}] \wedge \mathbf{L}_7[\text{Ev}_7^{\text{cmp}}]$$

**Definition 4.79** (Comparison-Stage Judgment Condition G: Identity Action of All Pure-Transport Closed Loops). Define the identity-action condition for all pure-transport closed loops by

$$\begin{aligned} & \mathbf{G}[\mathbf{Ev}^{\text{cmp}}] \\ & \iff \left[ \mathbf{L}_{\leq 7}[\mathbf{Ev}^{\text{cmp}}] \implies \forall R \in \mathbf{Rol}^{\text{cmp}}, \right. \\ & \quad \forall \mathbf{w} \in \mathbf{LoopWord}_{\text{all}}^{\text{tr}, R}[\mathbf{l}], \\ & \quad \left. \text{IdOnDom}(\text{Act}_R^{\text{tr}}[\mathbf{l}][\mathbf{w}]) \right]. \end{aligned}$$

**Theorem 4.80** (Global Gluing Condition and Identity Action of Extended Comparison Closed Words).

$$\begin{aligned} & \mathbf{L}_{\leq 7}[\mathbf{Ev}^{\text{cmp}}] \\ & \implies \left[ \mathbf{G}[\mathbf{Ev}^{\text{cmp}}] \iff \forall R \in \mathbf{Rol}^{\text{cmp}}, \quad \text{LoopId}(\mathfrak{W}_{\text{cmp}}^R[\mathbf{l}]) \right]. \end{aligned}$$

*Proof.* Assume  $\mathbf{L}_{\leq 7}[\mathbf{Ev}^{\text{cmp}}]$ .

Assume  $\mathbf{G}[\mathbf{Ev}^{\text{cmp}}]$ . For each  $R \in \mathbf{Rol}^{\text{cmp}}$ , take any

$$\mathbf{e} \in \text{Loop}(\mathbf{Word}_R^{\text{ext}}[\mathbf{l}])$$

Apply Lemma 4.59. For the empty-word case, use Lemma 4.52. For the standard embedding of a pure-transport word, use Definition 4.79 and Lemma 4.55. For the standard embedding of a source-matching word, use endpoint preservation by the standard embedding, Lemma 4.48, and the same embedding compatibility. For the mixed-word case, apply Lemma 4.61; use Definition 4.78 for a source-irreducible terminal word and Definition 4.79 for a pure-transport terminal word. The domain inclusion and equality of action restrictions in that lemma yield, in every case,

$$\text{IdOnDom}(\text{Act}_R^{\text{ext}}[\mathbf{l}][\mathbf{e}])$$

Definition 4.32 then gives

$$\text{LoopId}(\mathfrak{W}_{\text{cmp}}^R[\mathbf{l}])$$

Conversely, assume the same closed-word identity-action condition for each  $R \in \mathbf{Rol}^{\text{cmp}}$ . Map any pure-transport closed-loop word to an extended comparison closed word by the standard embedding. Definition 4.32 and Lemma 4.55 show that its pure-transport action is the identity on its domain. Definition 4.79 yields  $\mathbf{G}[\mathbf{Ev}^{\text{cmp}}]$ .  $\square$

**Lemma 4.81** (Identity Action of All Pure-Transport Closed Loops and Local Injectivity of the Canonical Projection). Assume

$$\mathbf{L}_{\leq 7}[\mathbf{Ev}^{\text{cmp}}]$$

Then

$$\begin{aligned} & \mathbf{G}[\mathbf{Ev}^{\text{cmp}}] \\ & \iff \left[ \forall R \in \mathbf{Rol}^{\text{cmp}}, \quad \forall u \in U, \right. \\ & \quad \left. \text{Prj}_R^{\text{can}}[\mathbf{l}]|_{Y_u^R} : Y_u^R \longrightarrow \text{Qu}_R^{\text{can}}[\mathbf{l}] \text{ is injective} \right]. \end{aligned}$$

*Proof.* For each  $R \in \mathbf{Rol}^{\text{cmp}}$ , Lemma 4.36 establishes the union and pairwise-disjointness conditions on the underlying family required by Theorem 4.33. Apply that theorem to the comparison-stage typed atomic-action system  $\mathfrak{M}_{\text{cmp}}^R[\mathbb{I}]$  of Definition 4.35. By Definition 4.68 and Definition 4.70, the orbit relation, orbit quotient, and canonical quotient projection in that theorem coincide respectively with

$$\sim_R^{\text{cmp}}[\mathbb{I}], \quad \text{Qu}_R^{\text{can}}[\mathbb{I}], \quad \text{Prj}_R^{\text{can}}[\mathbb{I}]$$

Thus, for each  $R \in \mathbf{Rol}^{\text{cmp}}$ , that theorem identifies the closed-word identity-action condition with the local-injectivity condition on the canonical quotient projection stated in this lemma. Composing this equivalence with the equivalence in Theorem 4.80 identifies

$$\mathbf{G}[\mathbf{Ev}^{\text{cmp}}]$$

with the injectivity of every local restriction of a canonical projection.  $\square$

**Definition 4.82** (Canonical Projection of Synchronized Two-Role Comparison Pairs). Assume

$$\mathbf{L}_{\leq 2}[\mathbf{Ev}^{\text{cmp}}]$$

Define the canonical projection of synchronized two-role comparison pairs by

$$\begin{aligned} \text{Prj}_{\mathfrak{C}\mathfrak{D}}^{\text{can,pair}}[\mathbb{I}] : \mathcal{R}^{\text{pair}} \\ \longrightarrow \prod_{R \in \mathbf{Rol}^{\text{cmp}}} \text{Qu}_R^{\text{can}}[\mathbb{I}] \end{aligned}$$

and

$$\begin{aligned} \text{Prj}_{\mathfrak{C}\mathfrak{D}}^{\text{can,pair}}[\mathbb{I}](u, (\alpha_R)_{R \in \mathbf{Rol}^{\text{cmp}}}) \\ := (\text{Prj}_R^{\text{can}}[\mathbb{I}](u, \alpha_R))_{R \in \mathbf{Rol}^{\text{cmp}}} \end{aligned}$$

**Lemma 4.83** (Fiber Criterion for the Canonical Projection of Synchronized Two-Role Comparison Pairs). Under

$$\mathbf{L}_{\leq 2}[\mathbf{Ev}^{\text{cmp}}]$$

for any

$$p, q \in \mathcal{R}^{\text{pair}}$$

the equivalence

$$\text{Prj}_{\mathfrak{C}\mathfrak{D}}^{\text{can,pair}}[\mathbb{I}](p) = \text{Prj}_{\mathfrak{C}\mathfrak{D}}^{\text{can,pair}}[\mathbb{I}](q) \iff p \sim^{\text{pair}}[\mathbb{I}] q.$$

holds.

*Proof.* Write

$$\begin{aligned} p &= (u, (\alpha_R)_{R \in \mathbf{Rol}^{\text{cmp}}}), \\ q &= (v, (\beta_R)_{R \in \mathbf{Rol}^{\text{cmp}}}) \end{aligned}$$

By Definition 4.82, equality of the two families of canonical images is equivalent to equality of the corresponding canonical images for each

$$R \in \mathbf{Rol}^{\text{cmp}}$$

Apply Lemma 4.71 to each component and combine the resulting equivalences with Definition 4.75. Equality of the families of canonical images is then equivalent to comparison equivalence of the synchronized pairs.  $\square$

**Lemma 4.84** (Descent, Local Pullback, and Minimality of the Two-Role Comparison Relation on the Quotient). Assume

$$L_{\leq 6}[\text{Ev}^{\text{cmp}}]$$

Define the two-role comparison relation on the quotient by

$$\text{Rel}^{\text{can}}[\mathbb{I}] := \text{Prj}_{\mathcal{CD}}^{\text{can,pair}}[\mathbb{I}] [\text{Rel}^{\text{pair}}]$$

Then

$$\text{Rel}^{\text{can}}[\mathbb{I}] \subseteq \prod_{R \in \text{Rol}^{\text{cmp}}} \text{Qu}_R^{\text{can}}[\mathbb{I}].$$

Moreover, for any

$$(u, \alpha) \in \mathcal{R}^{\text{pair}}, \\ \alpha = (\alpha_R)_{R \in \text{Rol}^{\text{cmp}}}$$

the equivalences

$$\begin{aligned} \text{Prj}_{\mathcal{CD}}^{\text{can,pair}}[\mathbb{I}](u, \alpha) \in \text{Rel}^{\text{can}}[\mathbb{I}] \\ \iff \alpha \in \text{Rel}_u^{\text{cmp}} \\ \iff (\text{Rec}_{R,u}^{\text{cmp}}(\alpha_R))_{R \in \text{Rol}^{\text{cmp}}} \in \text{Rel}^{\text{src}}. \end{aligned}$$

hold.

For any relation

$$\text{Rel}_*^{\text{can}} \subseteq \prod_{R \in \text{Rol}^{\text{cmp}}} \text{Qu}_R^{\text{can}}[\mathbb{I}]$$

the implication

$$\text{Prj}_{\mathcal{CD}}^{\text{can,pair}}[\mathbb{I}] [\text{Rel}^{\text{pair}}] \subseteq \text{Rel}_*^{\text{can}} \implies \text{Rel}^{\text{can}}[\mathbb{I}] \subseteq \text{Rel}_*^{\text{can}}.$$

holds. Consequently,

$$\text{Rel}^{\text{can}}[\mathbb{I}]$$

is the unique minimal relation on the quotient that contains the canonical images of all local two-role comparison relations.

*Proof.* By Lemma 4.83, the fibers of

$$\text{Prj}_{\mathcal{CD}}^{\text{can,pair}}[\mathbb{I}]$$

are the equivalence classes of

$$\sim^{\text{pair}}[\mathbb{I}]$$

By Definition 4.77 and the assumption

$$L_{\leq 6}[\text{Ev}^{\text{cmp}}]$$

the relation

$$\text{Sat}(\text{Rel}^{\text{pair}}; \sim^{\text{pair}}[\mathbb{I}]).$$

holds. Applying Lemma 4.28 gives the type of the relation on the quotient, the local-pullback equivalence, and minimality. By Definition 4.74, the local pullback is equivalent to

$$\alpha \in \text{Rel}_u^{\text{cmp}}$$

By Definition 4.73, this is equivalent to

$$(\text{Rec}_{R,u}^{\text{cmp}}(\alpha_R))_{R \in \text{Rol}^{\text{cmp}}} \in \text{Rel}^{\text{src}}$$

□

**Definition 4.85** (Comparison-Stage Preceding-Condition Establishment Bundles). Define the establishment bundle preceding the first condition by

$$L_{<1} [\mathbf{Ev}^{\text{cmp}}] := \top$$

For each

$$i \in \{2, \dots, 7\}$$

define the establishment bundle preceding condition  $i$  by

$$L_{<i} [\mathbf{Ev}^{\text{cmp}}] := L_{\leq i-1} [\mathbf{Ev}^{\text{cmp}}]$$

For each

$$i \in \{1, \dots, 7\}$$

the identity

$$L_{\leq i} [\mathbf{Ev}^{\text{cmp}}] = L_{<i} [\mathbf{Ev}^{\text{cmp}}] \wedge L_i [\mathbf{Ev}_i^{\text{cmp}}]$$

holds.

**Definition 4.86** (Comparison-Stage Operability). Let

$$\text{Operative}(\mathcal{X})$$

denote comparison-stage operability, defined by

$$\text{Operative}(\mathcal{X}) \iff L_{\leq 7} [\mathbf{Ev}^{\text{cmp}}] \wedge G [\mathbf{Ev}^{\text{cmp}}]$$

**Definition 4.87** (Family of Comparison Index Domains). Define the comparison index domains for the family of local objects and the family of local transitions by

$$\text{Idx}_m^{\text{cmp}} := \text{Rol}^{\text{cmp}} \times U^m, \quad m \in \{1, 2\}$$

**Definition 4.88** (Comparison Gluing Datum). Assume  $\text{Operative}(\mathcal{X})$ . Define the comparison gluing datum by

$$\begin{aligned} \text{Dat}^{\text{glue}}[\mathbb{I}] &:= \left( \text{Dat}^{\text{tr}}[\mathbb{I}], \right. \\ &\quad (\text{LoopSpec}_{\text{loc}}^R[\mathbb{I}])_{R \in \text{Rol}^{\text{cmp}}}, \\ &\quad (\text{LoopDat}_R^{\text{real, loc}}[\mathbb{I}])_{R \in \text{Rol}^{\text{cmp}}}, \\ &\quad (\sim_R^{\text{cmp}}[\mathbb{I}])_{R \in \text{Rol}^{\text{cmp}}}, \\ &\quad (\text{Qu}_R^{\text{can}}[\mathbb{I}])_{R \in \text{Rol}^{\text{cmp}}}, \\ &\quad (\text{Prj}_R^{\text{can}}[\mathbb{I}])_{R \in \text{Rol}^{\text{cmp}}}, \\ &\quad \left( \text{Prj}_R^{\text{can}}[\mathbb{I}]|_{Y_u^R} \right)_{(R, u) \in \text{Idx}_1^{\text{cmp}}}, \\ &\quad \left. \text{Rel}^{\text{can}}[\mathbb{I}] \right) \end{aligned}$$

By Lemma 4.81, for each

$$(R, u) \in \text{Idx}_1^{\text{cmp}}$$

the restriction

$$\text{Prj}_R^{\text{can}}[\mathbb{I}]|_{Y_u^R}$$

is injective.

**Definition 4.89** (Canonical Comparison Quotient). Assume  $\text{Operative}(\mathcal{X})$ . Define the canonical comparison quotient by

$$\begin{aligned} \text{Qu}^{\text{can}}[\mathbb{I}] \\ := & \left( (\text{Qu}_R^{\text{can}}[\mathbb{I}])_{R \in \text{Rol}^{\text{cmp}}}, \right. \\ & \left. (\text{Prj}_R^{\text{can}}[\mathbb{I}])_{R \in \text{Rol}^{\text{cmp}}}, \text{Rel}^{\text{can}}[\mathbb{I}] \right) \end{aligned}$$

**Theorem 4.90** (Comparison-Stage Operability and Generation of the Canonical Comparison Quotient). For any realized operational configuration  $\mathcal{X}$  and any family of typed comparison rules  $\mathcal{J}_{\text{cmp}}$ ,

$$\text{Operative}(\mathcal{X}) \iff \text{L}_{\leq 7}[\text{Ev}^{\text{cmp}}] \wedge \text{G}[\text{Ev}^{\text{cmp}}].$$

If  $\text{Operative}(\mathcal{X})$  holds, then the following statements hold.

1. The comparison gluing datum

$$\text{Dat}^{\text{glue}}[\mathbb{I}]$$

and the canonical comparison quotient

$$\text{Qu}^{\text{can}}[\mathbb{I}]$$

are uniquely determined in dependency order.

2. For any

$$(R, u) \in \text{Idx}_1^{\text{cmp}}$$

the restriction

$$\text{Prj}_R^{\text{can}}[\mathbb{I}]|_{Y_u^R}$$

is injective.

3. The local-pullback equivalence in Lemma 4.84 holds.

4. By the minimality in Lemma 4.84,

$$\text{Rel}^{\text{can}}[\mathbb{I}]$$

is the unique minimal relation on the quotient that contains the canonical images of all local two-role comparison relations.

*Proof.* Definition 4.86 gives the first equivalence.

Assume  $\text{Operative}(\mathcal{X})$ . Definition 4.88 and Definition 4.89 give the first item.

Lemma 4.81 gives the second item.

Lemma 4.84 gives the third and fourth items and the uniqueness of the minimal relation on the quotient.  $\square$

### Comparison reachability and equivalence classes

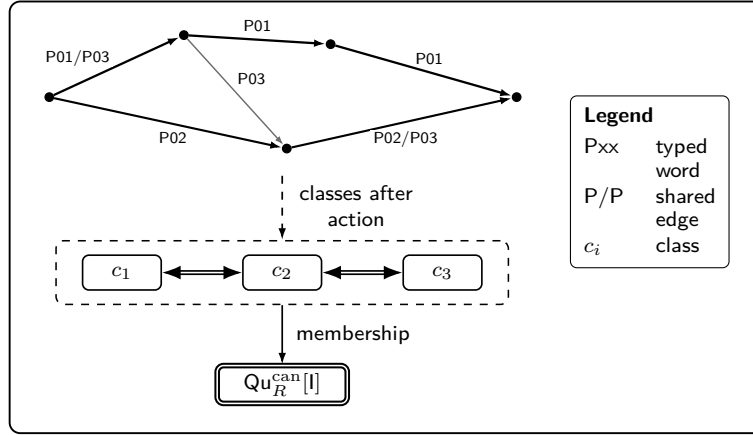


Figure 8: Actions of extended comparison words and comparison-reachability classes. Assume  $L_{\leq 2}[Ev^{cmp}]$  and fix  $R \in RoI^{cmp}$ . The dots represent reception-phase types and the edges represent admissible atomic actions. P01, P02, and P03 denote three extended comparison words  $e_1, e_2, e_3$  with the same source and target types. Take a point  $x$  in the common domain of the three actions and set  $y_i := Act_R^{ext}[l][e_i](x)$  and  $c_i := Prj_R^{can}[l](y_i)$ . Definition 4.25 and Definition 4.68 give  $x \sim_R^{cmp}[l]y_i$ . Lemma 4.71 then gives  $c_1 = c_2 = c_3 = Prj_R^{can}[l](x)$ . The lower part shows that this common class belongs to  $Qu_R^{can}[l]$ .

Figure 6 shows the dependency structure of the comparison stage from the comparison-rule-dependent common input to the canonical comparison quotient. Figure 8 shows that the images of a common domain point under three extended comparison words project to the same comparison-reachability class.

Theorem 4.80 states the equivalence between the judgment of closed-loop actions, including finite source-word deletion, and the identity-action condition for the family of extended comparison closed words. Theorem 4.90 connects the comparison-stage establishment conditions to the eight-component datum of Definition 4.88 and the three-component canonical comparison quotient of Definition 4.89. The injectivity of local restrictions of the canonical projections, the equivalence between local two-role comparison relations and source-side two-role comparison relations, and the minimality of the two-role comparison relation on the quotient are contained in items two through four of that theorem.

The canonical comparison quotient is an input to the local quotient charts, representational coordinates, and canonical orders of the subsequent representation stage.

### 4.1.3 Operational Skeleton

The operational skeleton of the comparison stage consists of four layers: input, action, quotient, and canonical construction.

$$\begin{aligned}
I &\xrightarrow{\text{gen}} \text{Dat}^{\text{mat}}[I] \\
&\xrightarrow{\text{gen}} \begin{array}{c} \text{Dat}^{\text{tr}}[I], \\ (\text{LoopSpec}_{\text{loc}}^R[I])_{R \in \text{Rol}^{\text{cmp}}} \end{array} \\
&\xrightarrow{\text{gen}} \left( \mathcal{R}_R^{\text{arr}}, \mathfrak{M}_{\text{cmp}}^R[I], \text{LoopDat}_R^{\text{real,loc}}[I] \right)_{R \in \text{Rol}^{\text{cmp}}} \\
&\xrightarrow{\text{gen}} (\sim_R^{\text{cmp}}[I], \text{Qu}_R^{\text{can}}[I], \text{Prj}_R^{\text{can}}[I])_{R \in \text{Rol}^{\text{cmp}}}.
\end{aligned}$$

Here, Definition 4.68 gives

$$\sim_R^{\text{cmp}}[I] = \sim_{\mathfrak{M}_{\text{cmp}}^R[I]}^{\text{orb}}$$

The establishment conditions connect to the respective layers as follows:

$$\begin{aligned}
&\left\{ L_{\leq 3}[\text{Ev}^{\text{cmp}}], (\text{LoopSpec}_{\text{loc}}^R[I])_{R \in \text{Rol}^{\text{cmp}}} \right\} \\
&\xrightarrow{\text{gen}} \left( \text{LoopDat}_R^{\text{real,loc}}[I] \right)_{R \in \text{Rol}^{\text{cmp}}}, \\
&\left\{ L_{\leq 6}[\text{Ev}^{\text{cmp}}], (\text{Rel}_u^{\text{cmp}})_{u \in U}, (\text{Qu}_R^{\text{can}}[I], \text{Prj}_R^{\text{can}}[I])_{R \in \text{Rol}^{\text{cmp}}} \right\} \\
&\xrightarrow{\text{gen}} \text{Rel}^{\text{can}}[I], \\
&L_{\leq 7}[\text{Ev}^{\text{cmp}}] \wedge G[\text{Ev}^{\text{cmp}}] \iff \text{Operative}(\mathcal{X}).
\end{aligned}$$

By Definition 4.67, the fourth condition  $L_4[\text{Ev}_4^{\text{cmp}}]$  requires

$$\text{IdOnDom} \left( \text{Act}_R^{\text{tr,real}}[\text{Spec}] \right)$$

Apply Theorem 4.80 to the judgment of finite source-word deletion, source-irreducible terminal words, and pure-transport terminal words.

By Theorem 4.90, the canonical-construction layer consists of the uniquely determined objects

$$\text{Dat}^{\text{glue}}[I], \quad \text{Qu}^{\text{can}}[I]$$

The eight components of the former are given in Definition 4.88; the three components of the latter are given in Definition 4.89.



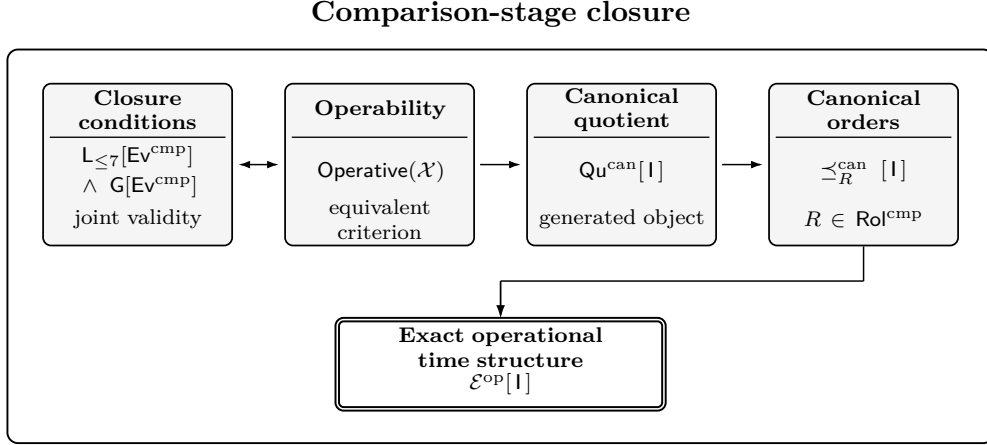


Figure 9: Separation between comparison-stage closure and the subsequent object sequence. Comparison-stage closure yields the operability judgment; the canonical comparison quotient derived from the C and D roles, the descent of canonical order data, and the exact operational time structure are determined as subsequent objects.

## 4.2 Exact Operational Time Structure and Representational Atlas

**Definition 4.91** (Representation-Stage Input Interface). Every object at the representation stage takes as input the comparison-rule-dependent common input

$$I = (\mathcal{X}, \mathcal{J}_{\text{cmp}})$$

of Definition 4.1. By the expansion rule in that definition, evaluation of an object  $F$  that depends on this input satisfies

$$F[I] = F(\mathcal{X}; \mathcal{J}_{\text{cmp}})$$

Throughout this subsection, each symbol root is first introduced in its full form with both a role type and a reception phase. Thereafter, only the reception-phase subscript is omitted, and only when the reception phase is fixed within the same definition, proposition, or proof and the full form is uniquely recoverable. The reception-phase index  $u$  of the local incomparability relation  $\asymp_{R,u}[I]$  is retained to distinguish it from the global relation  $\asymp_R[I]$ . For a binary relation  $r \subseteq S \times S$  and a subset  $A \subseteq S$ , its restriction is  $r|_A := r \cap (A \times A)$ . The role type, any varying family index, distinct reception phases, and any residual modifier that selects a genuine branch remain explicit in the full form.

**Definition 4.92** (Representation-Stage Partial Establishment Bundles). Recursively define the representation-stage partial establishment bundles by

$$\begin{aligned} \text{Repr}_{\leq 1}[I] &\iff K_{\text{CanOrdDesc}}^{\text{repr}}[I], \\ \text{Repr}_{\leq 2}[I] &\iff \text{Repr}_{\leq 1}[I] \wedge K_{\text{LocIncTrans}}^{\text{repr}}[I], \\ \text{Repr}_{\leq 3}[I] &\iff \text{Repr}_{\leq 2}[I] \wedge K_{\text{GlobIncTrans}}^{\text{repr}}[I], \\ \text{ReprCtx}_{\leq k}[I] &\iff \text{Operative}(\mathcal{X}) \wedge \text{Repr}_{\leq k}[I], \quad k \in \{1, 2, 3\}. \end{aligned}$$

The representation stage takes as input  $\text{Operative}(\mathcal{X})$  and, from Definition 4.89, the  $\mathfrak{C}/\mathfrak{D}$ -derived canonical comparison quotient  $\text{Qu}^{\text{can}}[I]$ . In dependency order, its structural generation sequence constructs the  $\mathfrak{C}/\mathfrak{D}$ -source-induced arrival preorders and the exact

operational time structure; the local order-coordinate structures and the comparison-order atlas; and the global order-coordinate structures. The comparison representational coordinates, quantitative representations, and continuum-approximation evaluation input are constructed by additionally specifying a selected ordered value range and embedding, a quantitative affine realization, a finite operational granularity datum, and componentwise evaluation inputs. Definition 4.92 uses  $\text{Repr}_{\leq k}[\mathbb{I}]$ , ( $k \in \{1, 2, 3\}$ ), for the cumulative establishment conditions of the three representation stages.

#### 4.2.1 Source-Induced Arrival Preorders

Assume  $\text{Operative}(\mathcal{X})$ . The representation stage takes as input the  $\mathfrak{C}$ -derived direct-sum order on source families from Definition 3.17, the  $\mathfrak{D}$ -derived record-index order from Definition 3.12, the comparison-side recovery maps from Definition 3.21, and the tagged comparison-arrival reception spaces from Definition 4.13.

**Definition 4.93** (Pullback of a Binary Relation along a Map). Let  $A, T$  be sets. On  $T$ , let  $\rho \subseteq T^2$  be a binary relation, and let  $r : A \rightarrow T$  be a map. Define the pullback relation by

$$r^*(\rho) := \{(a_i, a_j) \in A^2 \mid (r(a_i), r(a_j)) \in \rho\}$$

**Lemma 4.94** (Preorder Property under Pullback of a Binary Relation). If  $\rho$  in Definition 4.93 is a preorder, then  $r^*(\rho)$  is a preorder on  $A$ .

*Proof.* Reflexivity of  $\rho$  gives  $(r(a_i), r(a_i)) \in \rho$ . Applying transitivity to  $(r(a_i), r(a_j)) \in \rho$  and  $(r(a_j), r(a_k)) \in \rho$  gives  $(r(a_i), r(a_k)) \in \rho$ .  $\square$

**Definition 4.95** (Source-Induced Arrival Preorders). For each  $R \in \text{Rol}^{\text{cmp}}$ , define the tagged comparison-arrival recovery map and the source-induced arrival preorder by

$$\begin{aligned} \text{Rec}_R^{\text{arr}} : \mathcal{R}_R^{\text{arr}} &\longrightarrow \text{Tok}_R^*, \\ \text{Rec}_R^{\text{arr}}(u, \text{ArrVal}_{R,u}) &:= \text{Rec}_{R,u}^{\text{cmp}}(\text{ArrVal}_R), \\ \preceq_R^{\text{arr}} &:= (\text{Rec}_R^{\text{arr}})^*(\preceq_*^R) \end{aligned}$$

Expanding Lemma 4.94 yields the source-side order judgment on the recovered images of two arrival points.

**Proposition 4.96** (Preorder Property of Source-Induced Arrival Preorders). For each  $R \in \text{Rol}^{\text{cmp}}$ ,

$$(\mathcal{R}_R^{\text{arr}}, \preceq_R^{\text{arr}})$$

is a preordered set.

*Proof.* By Proposition 3.18 and Proposition 3.13, for each  $R \in \text{Rol}^{\text{cmp}}$ , the relation  $\preceq_*^R$  is a preorder on  $\text{Tok}_R^*$ . By Definition 4.95, the relation  $\preceq_R^{\text{arr}}$  is the pullback of  $\preceq_*^R$  along  $\text{Rec}_R^{\text{arr}}$ . Apply Lemma 4.94.  $\square$

**Definition 4.97** (Descent of a Preorder to a Quotient). For a set  $A$ , an equivalence relation  $\sim$  on that set, and a preorder  $\preceq$  on that set, define the order-descent condition

and the order-separation condition by

$$\begin{aligned} \text{OrdDesc}(A, \sim, \preceq) &\iff \left[ \begin{array}{l} \forall a_0, a'_0, a_1, a'_1 \in A, \\ (a_0 \sim a'_0) \wedge (a_1 \sim a'_1) \\ \implies (a_0 \preceq a_1 \iff a'_0 \preceq a'_1) \end{array} \right], \\ \text{OrdSep}(A, \sim, \preceq) &\iff \left[ \begin{array}{l} \forall a_0, a_1 \in A, \\ (a_0 \preceq a_1) \wedge (a_1 \preceq a_0) \\ \implies a_0 \sim a_1 \end{array} \right] \end{aligned}$$

When  $\text{OrdDesc}(A, \sim, \preceq)$  holds, define the quotient relation by

$$[a_0]_{\sim} \preceq / \sim [a_1]_{\sim} \iff a_0 \preceq a_1$$

**Lemma 4.98** (Partial Order on a Preorder Quotient). If  $\text{OrdDesc}(A, \sim, \preceq)$  and  $\text{OrdSep}(A, \sim, \preceq)$  in Definition 4.97 hold, then  $\preceq / \sim$  is a partial order on  $A/\sim$ .

*Proof.* The condition **OrdDesc** makes the quotient relation independent of representatives. Reflexivity and transitivity of  $\preceq$  descend to the quotient relation. If the relation holds in both directions on the quotient, the representatives satisfy  $\preceq$  in both directions. By **OrdSep**, the representatives are  $\sim$ -equivalent, and the corresponding quotient elements are equal.  $\square$

**Definition 4.99** (Canonical Order-Descent Condition at the Representation Stage). Assume **Operative**( $\mathcal{X}$ ). Define the representative-invariant descent condition for the source-induced arrival preorders by

$$K_{\text{CanOrdDesc}}^{\text{repr}}[\mathbb{I}] \iff [\forall R \in \text{Rol}^{\text{cmp}}, \quad \text{OrdDesc}(\mathcal{R}_R^{\text{arr}}, \sim_R^{\text{cmp}}[\mathbb{I}], \preceq_R^{\text{arr}})].$$

Expanding Definition 4.97 gives the representative-invariance condition on four arrival points.

**Lemma 4.100** (Comparison Equivalence Obtained from Mutual Arrival Order). Assume **Operative**( $\mathcal{X}$ ). For each  $R \in \text{Rol}^{\text{cmp}}$ ,

$$\text{OrdSep}(\mathcal{R}_R^{\text{arr}}, \sim_R^{\text{cmp}}[\mathbb{I}], \preceq_R^{\text{arr}})$$

holds.

*Proof.* To prove **OrdSep** in Definition 4.97, take arbitrary

$$\text{ArrPt}_{R,0}, \text{ArrPt}_{R,1} \in \mathcal{R}_R^{\text{arr}}$$

and assume

$$\text{ArrPt}_{R,0} \preceq_R^{\text{arr}} \text{ArrPt}_{R,1}, \quad \text{ArrPt}_{R,1} \preceq_R^{\text{arr}} \text{ArrPt}_{R,0}$$

By Definition 4.13, there exist  $u, v \in U$  and

$$\text{ArrVal}_{R,u}^{(0)} \in \text{Im}(\text{Arr}_{R,u}^{\text{cmp}}), \quad \text{ArrVal}_{R,v}^{(1)} \in \text{Im}(\text{Arr}_{R,v}^{\text{cmp}})$$

such that

$$\begin{aligned} \text{ArrPt}_{R,0} &= \left( u, \text{ArrVal}_{R,u}^{(0)} \right), \\ \text{ArrPt}_{R,1} &= \left( v, \text{ArrVal}_{R,v}^{(1)} \right), \\ \text{SrcTok}_{R,i} &:= \text{Rec}_R^{\text{arr}}(\text{ArrPt}_{R,i}), \quad i \in \{0, 1\}. \end{aligned}$$

Expand the preceding two assumptions with Definition 4.95, and use antisymmetry from Proposition 3.18 and Proposition 3.13 to obtain

$$\begin{aligned}\text{SrcTok}_{R,0} &\preceq_*^R \text{SrcTok}_{R,1}, \\ \text{SrcTok}_{R,1} &\preceq_*^R \text{SrcTok}_{R,0} \\ &\implies \text{SrcTok}_{R,0} = \text{SrcTok}_{R,1}.\end{aligned}$$

Definition 4.86, Definition 4.85, Definition 4.5, and Definition 4.8 give

$$\begin{aligned}\text{SrcTok}_{R,0} &\in \text{Dom}(\text{Arr}_{R,v}^{\text{cmp}}), \\ \text{Arr}_{R,v}^{\text{cmp}}(\text{SrcTok}_{R,1}) &= \text{ArrVal}_{R,v}^{(1)}, \\ \text{ArrVal}_{R,u}^{(0)} &\in \text{Dom}(\text{SrcMatch}_{uv}^R), \\ \text{SrcMatch}_{uv}^R(\text{ArrVal}_{R,u}^{(0)}) &= \text{Arr}_{R,v}^{\text{cmp}}(\text{SrcTok}_{R,0}) \\ &= \text{Arr}_{R,v}^{\text{cmp}}(\text{SrcTok}_{R,1}) \\ &= \text{ArrVal}_{R,v}^{(1)}.\end{aligned}$$

By Definition 4.49, Definition 4.18, and Definition 4.50, the length-one source-matching word and its action satisfy

$$\begin{aligned}\text{SrcldWord}_{R,01}^{(1)} &:= (1, (u, v), (\text{src})) \in \text{Word}_R^{\text{ext}}[\mathbb{I}], \\ \text{ArrPt}_{R,0} &\in \text{Dom}\left(\text{Act}_R^{\text{ext}}[\mathbb{I}] \left[ \text{SrcldWord}_{R,01}^{(1)} \right]\right), \\ \text{Act}_R^{\text{ext}}[\mathbb{I}] \left[ \text{SrcldWord}_{R,01}^{(1)} \right] (\text{ArrPt}_{R,0}) &= \text{ArrPt}_{R,1}\end{aligned}$$

Definition 4.68 then makes the two arrival points comparison-equivalent. This proves **OrdSep** in Definition 4.97.  $\square$

**Definition 4.101** (Canonical Partial Orders for the  $\mathfrak{C}/\mathfrak{D}$  Roles). Assume  $\text{ReprCtx}_{\leq 1}[\mathbb{I}]$ . For each  $R \in \text{Rol}^{\text{cmp}}$ , use the comparison quotient set and canonical projection of Definition 4.70 to define the canonical partial order for the  $\mathfrak{C}/\mathfrak{D}$  roles by

$$\preceq_R^{\text{can}}[\mathbb{I}] := \preceq_R^{\text{arr}} / \sim_R^{\text{cmp}}[\mathbb{I}]$$

Expanding Definition 4.97 gives the representative order judgment on canonical-projection images.

**Lemma 4.102** (Partial-Order Property of the Canonical Orders for the  $\mathfrak{C}/\mathfrak{D}$  Roles). Under  $\text{ReprCtx}_{\leq 1}[\mathbb{I}]$ , for each  $R \in \text{Rol}^{\text{cmp}}$ ,

$$(\text{Qu}_R^{\text{can}}[\mathbb{I}], \preceq_R^{\text{can}}[\mathbb{I}])$$

is a partially ordered set.

*Proof.* The preorder property of  $\preceq_R^{\text{arr}}$  follows from Proposition 4.96, and the equivalence-relation property of comparison equivalence follows from Lemma 4.69. The order-descent condition follows from Definition 4.99, and the order-separation condition follows from Lemma 4.100. For the quotient relation of Definition 4.101, apply Lemma 4.98.  $\square$

**Definition 4.103** (Exact Operational Time Structure). Assume  $\text{ReprCtx}_{\leq 1}[\mathbb{I}]$ . Define the exact operational time structure by

$$\begin{aligned} \mathcal{E}^{\text{op}}[\mathbb{I}] &:= (\text{Qu}^{\text{can}}[\mathbb{I}], (\preceq_R^{\text{can}}[\mathbb{I}])_{R \in \text{Rol}^{\text{cmp}}}), \\ (\preceq_R^{\text{can}}[\mathbb{I}])_{R \in \text{Rol}^{\text{cmp}}} &\longleftrightarrow (\preceq_{\mathcal{C}}^{\text{can}}[\mathbb{I}], \preceq_{\mathcal{D}}^{\text{can}}[\mathbb{I}]). \end{aligned}$$

The first component is the canonical comparison quotient whose components are the family of comparison quotient sets, the family of canonical projections, and the two-role comparison relation on the quotient. The second component is the role-indexed family of canonical partial orders on the comparison quotient sets of the respective comparison roles.

**Definition 4.104** (Local Canonical Quotient Regions). Assume  $\text{Operative}(\mathcal{X})$ . For each  $R \in \text{Rol}^{\text{cmp}}$  and  $u \in U$ , define the local canonical quotient region by

$$\begin{aligned} \mathcal{U}_{R,u}[\mathbb{I}] &:= \text{Prj}_R^{\text{can}}[\mathbb{I}] (\{u\} \times \text{Im}(\text{Arr}_R^{\text{cmp}})) \\ &\subseteq \text{Qu}_R^{\text{can}}[\mathbb{I}]. \end{aligned}$$

**Lemma 4.105** (Strict Part of a Partial Order). For a set  $S$  and a partial order  $\preceq$  on that set, define a binary relation  $\prec$  by

$$x \prec y \iff (x \preceq y) \wedge (x \neq y) \quad (x, y \in S)$$

Then  $\prec$  is a strict partial order on  $S$ .

*Proof.* For any  $x \in S$ , the relation  $x \prec x$  does not hold because it contains  $x \neq x$ . Suppose  $x \prec y$  and  $y \prec z$ . Transitivity of the partial order  $\preceq$  gives  $x \preceq z$ . If  $x = z$ , antisymmetry applied to  $y \preceq x$  and  $x \preceq y$  gives  $x = y$ . This is incompatible with  $x \prec y$ , which contains  $x \neq y$ . Hence  $x \neq z$ , and  $x \prec z$  holds.  $\square$

**Definition 4.106** (Linearization of Incomparability in a Strict Partial Order). For a strict-partial-order datum  $\mathbf{P} = (S, \prec)$ , define the incomparability relation and its transitivity condition by

$$\begin{aligned} \text{Inc}[\mathbf{P}] &:= \{(x, y) \in S^2 \mid \neg(x \prec y) \wedge \neg(y \prec x)\}, \\ \text{IncTrans}[\mathbf{P}] &\iff \left[ \begin{array}{c} \forall x_0, x_1, x_2 \in S, \\ (x_0 \text{ Inc}[\mathbf{P}] x_1) \wedge (x_1 \text{ Inc}[\mathbf{P}] x_2) \\ \implies x_0 \text{ Inc}[\mathbf{P}] x_2 \end{array} \right] \end{aligned}$$

When  $\text{IncTrans}[\mathbf{P}]$  holds, define the incomparability linearization by

$$\begin{aligned} \text{LinQuot}[\mathbf{P}] &:= (\text{Qu}^{\text{inc}}[\mathbf{P}], \text{Prj}_{\mathbf{P}}^{\text{inc}}, <_{\mathbf{P}}^{\text{inc}}), \\ \text{Qu}^{\text{inc}}[\mathbf{P}] &:= S / \text{Inc}[\mathbf{P}], \\ \text{Prj}_{\mathbf{P}}^{\text{inc}} : S &\twoheadrightarrow \text{Qu}^{\text{inc}}[\mathbf{P}], \quad \text{Prj}_{\mathbf{P}}^{\text{inc}}(x) := [x]_{\text{Inc}[\mathbf{P}]}, \\ \text{Prj}_{\mathbf{P}}^{\text{inc}}(x) <_{\mathbf{P}}^{\text{inc}} \text{Prj}_{\mathbf{P}}^{\text{inc}}(y) &\iff x \prec y \end{aligned}$$

**Definition 4.107** (Equality Kernel of a Map). For a map

$$\Phi : S \longrightarrow T$$

define the equality kernel by

$$\text{EqKer}(\Phi) := \{(x_0, x_1) \in S^2 \mid \Phi(x_0) = \Phi(x_1)\}$$

**Definition 4.108** (Ordered-Representation Judgment Condition). For a set  $S$ , binary relations  $\sim$  and  $\prec$  on that set, a linearly ordered set  $(T, <)$ , and a map  $\Phi : S \rightarrow T$ , define the ordered-representation judgment condition using the relation pullback of Definition 4.93 by

$$\begin{aligned} \text{OrdRepCond}(\Phi; \sim, \prec, <) \\ \iff [\text{EqKer}(\Phi) = \sim] \wedge [\Phi^*(<) = \prec] \end{aligned}$$

**Lemma 4.109** (Strict Linear-Order Quotient by Transitive Incomparability). Suppose that the strict-partial-order datum  $\mathbf{P} = (S, \prec)$  satisfies  $\text{IncTrans}[\mathbf{P}]$ . Then  $\text{Inc}[\mathbf{P}]$  is an equivalence relation, and  $\text{LinQuot}[\mathbf{P}]$  has a third component  $<_{\mathbf{P}}^{\text{inc}}$  that is independent of representatives and is a strict linear order on  $\text{Qu}^{\text{inc}}[\mathbf{P}]$ . Moreover, for any  $x, x', y, y' \in S$ ,

$$(x \text{ Inc}[\mathbf{P}] x') \wedge (y \text{ Inc}[\mathbf{P}] y') \implies (x \prec y \iff x' \prec y').$$

In addition,

$$\text{OrdRepCond}(\text{Prj}_{\mathbf{P}}^{\text{inc}}; \text{Inc}[\mathbf{P}], \prec, <_{\mathbf{P}}^{\text{inc}})$$

holds.

*Proof.* Irreflexivity of  $\prec$  gives  $\neg(x \prec x)$ , and the definition then gives  $x \text{ Inc}[\mathbf{P}] x$ . Exchanging the two negated components in the definition shows that  $x \text{ Inc}[\mathbf{P}] y$  implies  $y \text{ Inc}[\mathbf{P}] x$ . Together with  $\text{IncTrans}[\mathbf{P}]$ , these properties make  $\text{Inc}[\mathbf{P}]$  an equivalence relation.

Assume  $x \text{ Inc}[\mathbf{P}] x'$ ,  $y \text{ Inc}[\mathbf{P}] y'$ , and  $x \prec y$ . Suppose that  $x' \prec y$  does not hold. If  $y \prec x'$ , then  $x \prec y \prec x'$  gives  $x \prec x'$ , which is incompatible with  $x \text{ Inc}[\mathbf{P}] x'$ . If  $y \prec x'$  also does not hold, then  $x' \text{ Inc}[\mathbf{P}] y$ ; transitivity with  $x \text{ Inc}[\mathbf{P}] x'$  gives  $x \text{ Inc}[\mathbf{P}] y$ , which is incompatible with  $x \prec y$ . Therefore  $x' \prec y$ .

Next suppose that  $x' \prec y'$  does not hold. If  $y' \prec x'$ , then  $y' \prec x' \prec y$  gives  $y' \prec y$ , which is incompatible with  $y \text{ Inc}[\mathbf{P}] y'$ . If  $y' \prec x'$  also does not hold, then  $x' \text{ Inc}[\mathbf{P}] y'$ ; transitivity with  $y' \text{ Inc}[\mathbf{P}] y$  gives  $x' \text{ Inc}[\mathbf{P}] y$ , which is incompatible with  $x' \prec y$ . Hence  $x' \prec y'$ . Symmetry gives  $x' \text{ Inc}[\mathbf{P}] x$  and  $y' \text{ Inc}[\mathbf{P}] y$ . Applying the implication just proved to  $(x', x, y', y)$  gives the reverse implication and establishes representative invariance.

The equivalence-class criterion follows from the standard equivalence-class criterion for the quotient projection of the equivalence relation  $\text{Inc}[\mathbf{P}]$ . Every  $A \in \text{Qu}^{\text{inc}}[\mathbf{P}]$  has some  $x \in S$  for which  $A = \text{Prj}_{\mathbf{P}}^{\text{inc}}(x)$ ; hence  $\text{Prj}_{\mathbf{P}}^{\text{inc}}$  is surjective.

By representative invariance, the truth of  $\text{Prj}_{\mathbf{P}}^{\text{inc}}(x) <_{\mathbf{P}}^{\text{inc}} \text{Prj}_{\mathbf{P}}^{\text{inc}}(y)$  is independent of the representatives. The relation with the same element on both sides is equivalent to  $x \prec x$ , and therefore does not hold.

Suppose that  $\text{Prj}_{\mathbf{P}}^{\text{inc}}(x) <_{\mathbf{P}}^{\text{inc}} \text{Prj}_{\mathbf{P}}^{\text{inc}}(y)$  and  $\text{Prj}_{\mathbf{P}}^{\text{inc}}(y) <_{\mathbf{P}}^{\text{inc}} \text{Prj}_{\mathbf{P}}^{\text{inc}}(z)$ . Let  $y_0, y_1$  be the respective intermediate representatives used in the two relations; then  $y_0 \text{ Inc}[\mathbf{P}] y_1$ . Representative invariance turns  $x \prec y_0$  into  $x \prec y_1$ , and transitivity of  $x \prec y_1 \prec z$  gives  $x \prec z$ . Therefore  $\text{Prj}_{\mathbf{P}}^{\text{inc}}(x) <_{\mathbf{P}}^{\text{inc}} \text{Prj}_{\mathbf{P}}^{\text{inc}}(z)$ .

If  $\text{Prj}_{\mathbf{P}}^{\text{inc}}(x) \neq \text{Prj}_{\mathbf{P}}^{\text{inc}}(y)$ , the equivalence-class criterion shows that  $x \text{ Inc}[\mathbf{P}] y$  does not hold. By the definition of incomparability, either  $x \prec y$  or  $y \prec x$  holds. If both held, they would give  $x \prec x$ ; hence exactly one holds. Thus  $<_{\mathbf{P}}^{\text{inc}}$  is a strict linear order that compares any two distinct elements in a unique direction. The class criterion and order criterion in Definition 4.106 coincide with the two components in Definition 4.107 and 4.108.  $\square$

**Definition 4.110** (Canonical strict orders on  $\mathfrak{C}/\mathfrak{D}$ ). Assume  $\text{ReprCtx}_{\leq 1}[\mathbb{I}]$ . For each  $R \in \text{Rol}^{\text{cmp}}$  and any

$$(x_0, x_1) \in (\text{Qu}_R^{\text{can}}[\mathbb{I}])^2$$

define the canonical strict order by

$$\begin{aligned} \prec_R^{\text{can}} [\mathbb{I}] &\subseteq (\text{Qu}_R^{\text{can}} [\mathbb{I}])^2, \\ x_0 \prec_R^{\text{can}} [\mathbb{I}] x_1 &\iff (x_0 \preceq_R^{\text{can}} [\mathbb{I}] x_1) \\ &\quad \wedge (x_0 \neq x_1) \end{aligned}$$

**Definition 4.111** (Local incomparability relations). Assume  $\text{ReprCtx}_{\leq 1} [\mathbb{I}]$ . For each  $R \in \text{Rol}^{\text{cmp}}$  and  $u \in U$ , define the local strict-order datum and the local incomparability relation by

$$\begin{aligned} \text{OrdDat}_{R,u}^{\text{loc}} [\mathbb{I}] &:= (\mathcal{U}_R [\mathbb{I}], \prec_R^{\text{can}} [\mathbb{I}]|_{\mathcal{U}_R [\mathbb{I}]}) , \\ \succ_{R,u} [\mathbb{I}] &:= \text{Inc} [\text{OrdDat}_{R,u}^{\text{loc}} [\mathbb{I}]] \end{aligned}$$

Expanding Definition 4.106 gives the bidirectional incomparability condition for any two local canonical comparison quotient classes.

**Definition 4.112** (Representation-stage local incomparability-transitivity condition). Assume  $\text{ReprCtx}_{\leq 1} [\mathbb{I}]$ . Define the representation-stage local incomparability-transitivity condition by

$$K_{\text{LocIncTrans}}^{\text{repr}} [\mathbb{I}] \iff \left[ \begin{array}{c} \forall (R, u) \in \text{Idx}_1^{\text{cmp}}, \\ \text{IncTrans} [\text{OrdDat}_{R,u}^{\text{loc}} [\mathbb{I}]] \end{array} \right].$$

Expanding Definition 4.106 gives the transitivity condition quantified over three points.

**Lemma 4.113** (Strict-partial-order property of the canonical strict orders on  $\mathfrak{C}/\mathfrak{D}$ ). Under  $\text{ReprCtx}_{\leq 1} [\mathbb{I}]$ , for each  $R \in \text{Rol}^{\text{cmp}}$ ,

$$(\text{Qu}_R^{\text{can}} [\mathbb{I}], \prec_R^{\text{can}} [\mathbb{I}])$$

is a strictly partially ordered set.

*Proof.* By Lemma 4.102, the canonical partial order is a partial order. Definition 4.110 is its strictification, so Lemma 4.105 applies.  $\square$

**Lemma 4.114** (Equivalence-relation property of the local incomparability relations). Under  $\text{ReprCtx}_{\leq 2} [\mathbb{I}]$ , for each  $R \in \text{Rol}^{\text{cmp}}$  and  $u \in U$ ,

$$(\mathcal{U}_R [\mathbb{I}], \succ_{R,u} [\mathbb{I}])$$

is a set equipped with an equivalence relation.

*Proof.* By Lemma 4.113, the restriction of the canonical strict order to the local canonical quotient region is a strict partial order. Definition 4.111 gives its incomparability relation, and Definition 4.112 gives transitivity. Apply Lemma 4.109.  $\square$

**Lemma 4.115** (Invariance of the canonical strict order along local incomparability classes).

Assume  $\text{ReprCtx}_{\leq 2} [\mathbb{I}]$ . For each  $R \in \text{Rol}^{\text{cmp}}$  and  $u \in U$ , the representative invariance of Lemma 4.109 holds under

$$P = \text{OrdDat}_R^{\text{loc}} [\mathbb{I}]$$

*Proof.* Lemma 4.113, Definition 4.111, and Definition 4.112 permit application of the representative invariance in Lemma 4.109.  $\square$

**Definition 4.116** (Local order-coordinate quotients). Assume  $\text{ReprCtx}_{\leq 2}[I]$ . For each  $R \in \text{Rol}^{\text{cmp}}$  and  $u \in U$ , from this section onward the superscript  $(\text{ord}, \text{loc})$  on quotients, projections, orders, atlases, and chart changes belonging to the local order-coordinate family is abbreviated to  $(\text{loc})$ . The full form  $(\text{ord}, \text{can})$  is retained for the global order-coordinate family. Under this abbreviation, define the local order-coordinate quotient, the local order-coordinate projection, and the strict order on the quotient by

$$\begin{aligned}\text{LinQuot}_{R,u}^{\text{loc}}[I] &:= \text{LinQuot}[\text{OrdDat}_R^{\text{loc}}[I]], \\ \text{LinQuot}_R^{\text{loc}}[I] &= (\text{Qu}_{R,u}^{\text{loc}}[I], \text{Prj}_{R,u}^{\text{loc}}[I], <_{R,u}^{\text{loc}}[I]), \\ \text{OrdCls}_{R,i}^{\text{loc}} &:= \text{Prj}_R^{\text{loc}}[I](\text{CmpCls}_{R,i}), \quad i \in \{0, 1\}.\end{aligned}$$

Expanding Definition 4.106 gives the local incomparability classes, quotient set, quotient projection, and representative-based strict-order judgment.

**Lemma 4.117** (Strict linearity of the local order-coordinate quotients). Under  $\text{ReprCtx}_{\leq 2}[I]$ , for each  $R \in \text{Rol}^{\text{cmp}}$  and  $u \in U$ ,  $<_R^{\text{loc}}[I]$  is a strict linear order on  $\text{Qu}_R^{\text{loc}}[I]$ .

*Proof.* Lemma 4.113, Definition 4.111, and Definition 4.112 permit application of Lemma 4.109. The three components of Definition 4.116 coincide with  $\text{LinQuot}$  in that lemma.  $\square$

**Definition 4.118** (Surjective quotient-atlas datum). Let the datum whose components are a set  $S$ , a strict partial order  $\prec$  on that set, an index set  $I$ , and, for each  $i \in I$ , sets  $A_i, S_i$  and a map  $\vartheta_i$  be

$$\mathfrak{A} := (S, \prec, I, (A_i, S_i, \vartheta_i)_{i \in I})$$

and define each local strict-order datum by

$$\text{OrdDat}_i[\mathfrak{A}] := (S_i, \prec|_{S_i}), \quad i \in I$$

Define the surjective quotient-atlas condition by

$$\text{QuotAtlasDat}(\mathfrak{A}) \iff \left[ \forall i \in I, \quad S_i \subseteq S, \quad \vartheta_i : A_i \twoheadrightarrow S_i, \quad \text{IncTrans}[\text{OrdDat}_i[\mathfrak{A}]] \right]$$

**Definition 4.119** (Local order-coordinate projection system and local quotient-atlas datum). Assume  $\text{ReprCtx}_{\leq 2}[I]$ . For each  $R \in \text{Rol}^{\text{cmp}}$  and  $u \in U$ , define the surjection from the arrival-value image to the local canonical quotient region by

$$\text{ArrCan}_{R,u}[I] : \text{Im}(\text{Arr}_R^{\text{cmp}}) \twoheadrightarrow \mathcal{U}_R[I], \quad \text{ArrVal}_R^{(0)} \longmapsto \text{Prj}_R^{\text{can}}[I](u, \text{ArrVal}_R^{(0)})$$

Using the local order-coordinate projection system of Definition 4.116 as its components, define the local quotient-atlas datum by

$$\mathfrak{A}_R^{\text{loc}}[I] := \left( \text{Qu}_R^{\text{can}}[I], <_R^{\text{can}}[I], U, \left( \text{Im}(\text{Arr}_{R,u}^{\text{cmp}}), \mathcal{U}_{R,u}[I], \text{ArrCan}_{R,u}[I] \right)_{u \in U} \right)$$

**Lemma 4.120** (Local quotient-atlas datum and local order-coordinate projections). Under  $\text{ReprCtx}_{\leq 2}[I]$ , for each  $R \in \text{Rol}^{\text{cmp}}$  and  $u \in U$ , the local order-coordinate projection  $\text{Prj}_R^{\text{loc}}[I] : \mathcal{U}_R[I] \twoheadrightarrow \text{Qu}_R^{\text{loc}}[I]$  is surjective and satisfies, on the local canonical quotient region, the class judgment and order judgment of Lemma 4.109. Moreover,

$$\text{QuotAtlasDat}(\mathfrak{A}_R^{\text{loc}}[I])$$

holds, and its linearized-quotient component coincides with

$$\text{LinQuot}_u[\mathfrak{A}_R^{\text{loc}}[I]] = \text{LinQuot}_R^{\text{loc}}[I]$$



*Proof.* By Definition 4.116, the local order-coordinate quotient, the local order-coordinate projection, and the strict order on the quotient are the three components of  $\text{LinQuot}$ . Apply the surjectivity, class judgment, and order judgment of Lemma 4.109. By Definition 4.104,  $\text{ArrCan}_R[l]$  is surjective. By Definition 4.111,

$$\asymp_{R,u} [l] = \text{Inc} [\text{OrdDat}_R^{\text{loc}} [l]]$$

holds, and Definition 4.112 supplies  $\text{IncTrans}$ . Hence every condition in Definition 4.118 holds.  $\square$

**Definition 4.121** (Surjective quotient-atlas operations). Under  $\text{QuotAtlasDat}(\mathfrak{A})$  of Definition 4.118, define the composite charts, linearized quotients, double and triple overlaps, and quotient-chart changes by

$$\begin{aligned} \text{LinQuot}_i[\mathfrak{A}] &:= \text{LinQuot}[\text{OrdDat}_i[\mathfrak{A}]] \\ &= (Q_i[\mathfrak{A}], \pi_i[\mathfrak{A}], <_i[\mathfrak{A}]), \\ \text{QuotChart}_i[\mathfrak{A}] &:= \pi_i[\mathfrak{A}] \circ \vartheta_i : A_i \rightarrow Q_i[\mathfrak{A}], \\ S_{ij}^\cap[\mathfrak{A}] &:= S_i \cap S_j, \\ S_{ijk}^\cap[\mathfrak{A}] &:= S_i \cap S_j \cap S_k, \\ \text{QuotChg}_{ij}[\mathfrak{A}] &: \pi_i[\mathfrak{A}] (S_{ij}^\cap[\mathfrak{A}]) \\ &\quad \longrightarrow \pi_j[\mathfrak{A}] (S_{ij}^\cap[\mathfrak{A}]), \\ \text{QuotChg}_{ij}[\mathfrak{A}] (\pi_i[\mathfrak{A}](x)) &:= \pi_j[\mathfrak{A}](x) \\ &\quad (x \in S_{ij}^\cap[\mathfrak{A}]) \end{aligned}$$

If  $S_{ij}^\cap[\mathfrak{A}] = \emptyset$ , then  $\text{QuotChg}_{ij}[\mathfrak{A}]$  is the empty map  $\emptyset \rightarrow \emptyset$ .

**Lemma 4.122** (Composition of a surjection and a quotient projection). Under  $\text{QuotAtlasDat}(\mathfrak{A})$ , for every  $i \in I$ ,  $\text{QuotChart}_i[\mathfrak{A}]$  is surjective, and

$$\text{OrdRepCond} (\text{QuotChart}_i[\mathfrak{A}]; \vartheta_i^* (\text{Inc}[\text{OrdDat}_i[\mathfrak{A}]]), \vartheta_i^* (<_{|S_i}), <_i[\mathfrak{A}])$$

holds.

*Proof.* By Lemma 4.109,  $\pi_i[\mathfrak{A}]$  is surjective and satisfies the class judgment and order judgment. Its composition with the surjection  $\vartheta_i$  is surjective. By Definitions 4.107 and 4.93, precomposition with  $\vartheta_i$  pulls the class-judgment relation and the order-judgment relation back to  $A_i$ , respectively. Apply Definition 4.108.  $\square$

**Lemma 4.123** (Overlap chart changes between subregion quotients). Under  $\text{QuotAtlasDat}(\mathfrak{A})$ , for every  $i, j \in I$ ,  $\text{QuotChg}_{ij}[\mathfrak{A}]$  is a representative-independent bijection and a strict linear-order isomorphism, with inverse  $\text{QuotChg}_{ji}[\mathfrak{A}]$ . Moreover,

$$\begin{aligned} \text{QuotChg}_{ii}[\mathfrak{A}] &= \text{id}_{Q_i[\mathfrak{A}]}, \\ \text{QuotChg}_{ji}[\mathfrak{A}] \circ \text{QuotChg}_{ij}[\mathfrak{A}] &= \text{id}_{\pi_i[\mathfrak{A}](S_{ij}^\cap[\mathfrak{A}])}, \\ \text{QuotChg}_{jk}[\mathfrak{A}] (\text{QuotChg}_{ij}[\mathfrak{A}] (\pi_i[\mathfrak{A}](x))) &= \text{QuotChg}_{ik}[\mathfrak{A}] (\pi_i[\mathfrak{A}](x)) \\ &\quad (x \in S_{ijk}^\cap[\mathfrak{A}]) \end{aligned}$$

hold.

*Proof.* Let  $x, y \in S_{ij}^\cap[\mathfrak{A}]$ . The class judgment for  $\pi_i[\mathfrak{A}](x) = \pi_i[\mathfrak{A}](y)$ , together with the two restricted orders that agree on the overlap, gives  $\pi_j[\mathfrak{A}](x) = \pi_j[\mathfrak{A}](y)$ . Using this implication after exchanging the indices gives injectivity, and the pointwise definition gives surjectivity. Since the two order judgments are both equivalent to  $x \prec y$ , the map preserves and reflects order. Composing the pointwise definition twice gives the inverse condition and the triple-overlap condition. The identities  $S_{ii}^\cap[\mathfrak{A}] = S_i$  and  $\pi_i[\mathfrak{A}](S_i) = Q_i[\mathfrak{A}]$  give the identity condition.  $\square$

**Definition 4.124** (Local comparison charts). Assume  $\text{ReprCtx}_{\leq 2}[\mathbb{I}]$ . For each  $R \in \text{Rol}^{\text{cmp}}$  and  $u \in U$ , define the local comparison chart by

$$\begin{aligned} \text{Chart}_{R,u}^{\text{loc}}[\mathbb{I}] &:= \text{QuotChart}_u[\mathfrak{A}_R^{\text{loc}}[\mathbb{I}]] \\ &= \text{Prj}_R^{\text{loc}}[\mathbb{I}] \circ \text{ArrCan}_R[\mathbb{I}] \end{aligned}$$

This family of surjections is the local comparison-chart system.

**Proposition 4.125** (Canonical compositionality of local comparison charts). Under  $\text{ReprCtx}_{\leq 2}[\mathbb{I}]$ , the local comparison chart is the surjection uniquely determined by  $\text{ArrCan}_{R,u}[\mathbb{I}]$  and  $\text{Prj}_{R,u}^{\text{loc}}[\mathbb{I}]$ , and it satisfies the class judgment and order judgment of Lemma 4.122.

*Proof.* By Lemma 4.120,  $\text{QuotAtlasDat}(\mathfrak{A}_R^{\text{loc}}[\mathbb{I}])$  holds. Apply Lemma 4.122 to the local datum of Definition 4.124.  $\square$

**Definition 4.126** (Local order-coordinate chart-change maps). Assume  $\text{ReprCtx}_{\leq 2}[\mathbb{I}]$ . For each  $R \in \text{Rol}^{\text{cmp}}$  and  $u, v, w \in U$ , define the double and triple overlaps and the local order-coordinate chart-change maps by

$$\begin{aligned} \mathcal{U}_{R,uv}^\cap[\mathbb{I}] &:= S_{uv}^\cap[\mathfrak{A}_R^{\text{loc}}[\mathbb{I}]] , \\ \mathcal{U}_{R,uvw}^\cap[\mathbb{I}] &:= S_{uvw}^\cap[\mathfrak{A}_R^{\text{loc}}[\mathbb{I}]] , \\ \text{ChartChg}_{R,uv}^{\text{loc}}[\mathbb{I}] &:= \text{QuotChg}_{uv}[\mathfrak{A}_R^{\text{loc}}[\mathbb{I}]] . \end{aligned}$$

On an empty overlap,  $\text{ChartChg}_{R,uv}^{\text{loc}}[\mathbb{I}]$  is the empty map.

**Lemma 4.127** (Well-definedness and order-isomorphism property of local order-coordinate chart-change maps). Under  $\text{ReprCtx}_{\leq 2}[\mathbb{I}]$ , for each  $R \in \text{Rol}^{\text{cmp}}$  and  $u, v \in U$ , the local order-coordinate chart-change map is a representative-independent bijection and a strict linear-order isomorphism. Its inverse is

$$\text{ChartChg}_{R,vu}^{\text{loc}}[\mathbb{I}]$$

and it satisfies the order-preservation and order-reflection formulas of Lemma 4.123.

*Proof.* By Lemma 4.120,  $\text{QuotAtlasDat}(\mathfrak{A}_R^{\text{loc}}[\mathbb{I}])$  holds. Apply Lemma 4.123 to the local datum of Definition 4.126.  $\square$

**Definition 4.128** (Comparison-order atlas). Assume  $\text{ReprCtx}_{\leq 2}[\mathbb{I}]$ . Define the comparison-order atlas by

$$\begin{aligned} \text{Atlas}^{\text{ord}}[\mathbb{I}] &:= \left( \left( Q_u[\mathfrak{A}_R^{\text{loc}}[\mathbb{I}]] , <_u[\mathfrak{A}_R^{\text{loc}}[\mathbb{I}]] , \right. \right. \\ &\quad \left. \left. \text{QuotChart}_u[\mathfrak{A}_R^{\text{loc}}[\mathbb{I}]] \right)_{(R,u) \in \text{Idx}_1^{\text{cmp}}} , \right. \\ &\quad \left. \left( \text{QuotChg}_{uv}[\mathfrak{A}_R^{\text{loc}}[\mathbb{I}]] \right)_{(R,u,v) \in \text{Idx}_2^{\text{cmp}}} \right) \end{aligned}$$

The first component is the family of quotients, orders, and composite charts on the respective  $\mathfrak{C}/\mathfrak{D}$  branches; the second is the family of chart-change maps on the respective  $\mathfrak{C}/\mathfrak{D}$  branches.

**Proposition 4.129** (Cocycle condition for local order-coordinate chart-change maps). Under  $\text{ReprCtx}_{\leq 2}[\mathbb{I}]$ , for each  $R \in \text{Rol}^{\text{cmp}}$  and  $u, v, w \in U$ , the local order-coordinate chart-change maps satisfy the identity condition, the inverse condition, and the triple-overlap cocycle condition. The domains, codomains, and pointwise equalities of these three conditions are given by the formulas of Lemma 4.123 for  $\mathfrak{A}_R^{\text{loc}}[\mathbb{I}]$ .

*Proof.* Apply Lemma 4.120 and Lemma 4.123 to  $\mathfrak{A}_R^{\text{loc}}[\mathbb{I}]$ . □

**Definition 4.130** (Global incomparability relations). Assume  $\text{ReprCtx}_{\leq 1}[\mathbb{I}]$ . For each  $R \in \text{Rol}^{\text{cmp}}$ , define the global strict-order datum and the global incomparability relation by

$$\begin{aligned} \text{OrdDat}_R^{\text{can}}[\mathbb{I}] &:= (\text{Qu}_R^{\text{can}}[\mathbb{I}], \prec_R^{\text{can}}[\mathbb{I}]), \\ \succsim_R[\mathbb{I}] &:= \text{Inc}[\text{OrdDat}_R^{\text{can}}[\mathbb{I}]] \end{aligned}$$

Expanding Definition 4.106 gives the bidirectional incomparability condition for any two canonical comparison quotient classes.

**Lemma 4.131** (Agreement under restriction between the local and global incomparability relations). Under  $\text{ReprCtx}_{\leq 1}[\mathbb{I}]$ , for each  $R \in \text{Rol}^{\text{cmp}}$ , each  $u \in U$ , and any  $(x_0, x_1) \in (\mathcal{U}_R[\mathbb{I}])^2$ ,

$$x_0 \succsim_{R,u}[\mathbb{I}] x_1 \iff x_0 \succsim_R[\mathbb{I}] x_1.$$

*Proof.* Definitions 4.111 and 4.130 are determined on the local canonical quotient region by the same bidirectional incomparability condition for the canonical strict order. □

**Definition 4.132** (Representation-stage global incomparability-transitivity condition). Assume  $\text{ReprCtx}_{\leq 2}[\mathbb{I}]$ . Define the representation-stage global incomparability-transitivity condition by

$$K_{\text{GlobIncTrans}}^{\text{repr}}[\mathbb{I}] \iff [\forall R \in \text{Rol}^{\text{cmp}}, \quad \text{IncTrans}[\text{OrdDat}_R^{\text{can}}[\mathbb{I}]]].$$

Expanding Definition 4.106 gives the transitivity condition quantified over three points.

**Lemma 4.133** (Equivalence-relation property of the global incomparability relations). Under  $\text{ReprCtx}_{\leq 3}[\mathbb{I}]$ , for each  $R \in \text{Rol}^{\text{cmp}}$ ,  $\succsim_R[\mathbb{I}]$  is an equivalence relation on  $\text{Qu}_R^{\text{can}}[\mathbb{I}]$ .

*Proof.* Lemma 4.113, Definition 4.130, and Definition 4.132 permit application of Lemma 4.109 to the entire canonical comparison quotient. □

**Lemma 4.134** (Invariance of the canonical strict order along global incomparability classes). Assume

$$\text{ReprCtx}_{\leq 3}[\mathbb{I}]$$

For each

$$R \in \text{Rol}^{\text{cmp}}$$

the representative invariance of Lemma 4.109 holds under

$$P = \text{OrdDat}_R^{\text{can}}[\mathbb{I}]$$

*Proof.* Lemma 4.113, Definition 4.130, and Definition 4.132 permit application of the representative invariance in Lemma 4.109 to the entire canonical comparison quotient.  $\square$

**Definition 4.135** (Global order-coordinate quotients on  $\mathfrak{C}/\mathfrak{D}$  and canonical global order-coordinate projections).

Assume  $\text{ReprCtx}_{\leq 3}[\mathbb{I}]$ . For each  $R \in \text{Rol}^{\text{cmp}}$ , define the global order-coordinate quotient on  $\mathfrak{C}/\mathfrak{D}$ , the canonical global order-coordinate projection, and the strict order on the quotient by

$$\begin{aligned}\text{LinQuot}_R^{\text{can}}[\mathbb{I}] &:= \text{LinQuot}[\text{OrdDat}_R^{\text{can}}[\mathbb{I}]], \\ \text{LinQuot}_R^{\text{can}}[\mathbb{I}] &= \left( \text{Qu}_R^{\text{ord,can}}[\mathbb{I}], \text{Prj}_R^{\text{ord,can}}[\mathbb{I}], <_R^{\text{ord,can}}[\mathbb{I}] \right), \\ \text{OrdCls}_{R,i}^{\text{can}} &:= \text{Prj}_R^{\text{ord,can}}[\mathbb{I}](\text{CmpCls}_{R,i}), \quad i \in \{0, 1\}.\end{aligned}$$

Expanding Definition 4.106 gives the global incomparability classes, quotient set, quotient projection, and representative-based strict-order judgment.

**Lemma 4.136** (Strict linearity of global order-coordinate quotients and canonical projections). Under

$$\text{ReprCtx}_{\leq 3}[\mathbb{I}]$$

for each

$$\begin{aligned}R &\in \text{Rol}^{\text{cmp}} \\ &<_R^{\text{ord,can}}[\mathbb{I}]\end{aligned}$$

is a strict linear order on

$$\text{Qu}_R^{\text{ord,can}}[\mathbb{I}]$$

The canonical projection

$$\text{Prj}_R^{\text{ord,can}}[\mathbb{I}] : \text{Qu}_R^{\text{can}}[\mathbb{I}] \twoheadrightarrow \text{Qu}_R^{\text{ord,can}}[\mathbb{I}]$$

is surjective and satisfies the class judgment and order judgment of Lemma 4.109 on the entire canonical comparison quotient.

*Proof.* Lemma 4.113, Definition 4.130, and Definition 4.132 permit application of Lemma 4.109 to the entire canonical comparison quotient. The three components of Definition 4.135 coincide with  $\text{LinQuot}$  in that lemma.  $\square$

#### 4.2.2 Representational coordinates obtained from global order-coordinate quotients

The global order-coordinate quotient sends each incomparability class in the entire canonical comparison quotient to a single order-coordinate class. The comparison representational coordinate derived from  $\mathfrak{C}/\mathfrak{D}$  is determined by composing the canonical global order-coordinate projection with an order embedding into the selected linearly ordered codomain.

**Definition 4.137** (Strict order embedding). For strictly ordered sets  $(S, <_S)$  and  $(T, <_T)$ , and an injection  $f : S \hookrightarrow T$ , define the strict-order-embedding condition by

$$\text{OrdEmb}(f; S, <_S; T, <_T) \iff f^*(<_T) = <_S$$

**Lemma 4.138** (Composition of a quotient projection and an order embedding). Take sets  $S, Q, V$ ; on  $S$ , take relations  $\succsim, \prec$ ; on  $Q, V$ , take strict orders  $<_Q, <_V$ , a map  $\pi$ , and an injection  $\iota : Q \hookrightarrow V$ . Then the following implication holds:

$$\begin{aligned} \pi : S \twoheadrightarrow Q, \quad & \text{OrdRepCond}(\pi; \succsim, \prec, <_Q), \\ & \text{OrdEmb}(\iota; Q, <_Q; V, <_V), \\ & \implies \\ f := \iota \circ \pi : S \longrightarrow V, \\ & \text{OrdRepCond}(f; \succsim, \prec, <_V) \end{aligned}$$

*Proof.* Injectivity of  $\iota$  gives  $\text{EqKer}(f) = \text{EqKer}(\pi)$ . Preservation and reflection of strict order give  $f^*(<_V) = \pi^*(<_Q)$ . Apply the assumed  $\text{OrdRepCond}(\pi; \succsim, \prec, <_Q)$  and Definition 4.108.  $\square$

**Lemma 4.139** (Quotient-order embedding for nested domains). Let  $A \subseteq S$ , let  $Q_A, Q_S$  carry strict orders  $<_A, <_S$ , and let  $\pi_A : A \twoheadrightarrow Q_A$  and  $\pi_S : S \twoheadrightarrow Q_S$  be surjections. Assume that both of the following equivalences hold for every  $x, y \in A$ :

$$\begin{aligned} \pi_A(x) = \pi_A(y) &\iff \pi_S(x) = \pi_S(y), \\ \pi_A(x) <_A \pi_A(y) &\iff \pi_S(x) <_S \pi_S(y). \end{aligned}$$

Then the rule

$$\iota_{A,S} : Q_A \hookrightarrow Q_S, \quad \iota_{A,S}(\pi_A(x)) := \pi_S(x) \quad (x \in A)$$

defines an injection satisfying

$$\text{OrdEmb}(\iota_{A,S}; Q_A, <_A; Q_S, <_S), \quad \iota_{A,S} \circ \pi_A = \pi_S|_A.$$

*Proof.* Since  $\pi_A$  is surjective, every element of  $Q_A$  has the form  $\pi_A(x)$  for some  $x \in A$ . By the class-judgment equivalence,  $\pi_A(x) = \pi_A(y)$  implies  $\pi_S(x) = \pi_S(y)$ , so the value of the map is independent of the representative. By the converse class judgment, two elements with the same image satisfy  $\pi_A(x) = \pi_A(y)$ ; hence  $\iota_{A,S}$  is injective. The order-judgment equivalence and the definition give

$$\begin{aligned} \pi_A(x) <_A \pi_A(y) &\iff \iota_{A,S}(\pi_A(x)) <_S \iota_{A,S}(\pi_A(y)), \\ \iota_{A,S}(\pi_A(x)) &= \pi_S(x) \quad (x \in A) \end{aligned}$$

and thus give preservation and reflection of strict order and the commutative identity.  $\square$

**Definition 4.140** (Selected ordered codomains and representational coordinates). Assume  $\text{ReprCtx}_{\leq 3}[I]$ . For each  $R \in \text{Rol}^{\text{cmp}}$ , specify a selected representational codomain derived from  $\mathfrak{C}/\mathfrak{D}$ ,  $(\text{RepVal}_R, \leq_R^{\text{repr}})$ , and its strict-order part  $<_R^{\text{repr}}$ . Define the order embedding into the representational codomain derived from  $\mathfrak{C}/\mathfrak{D}$  and the comparison representational coordinate derived from  $\mathfrak{C}/\mathfrak{D}$  by

$$\begin{aligned} \text{RepEmb}_R^{\text{ord}}[I] : \text{Qu}_R^{\text{ord}, \text{can}}[I] &\hookrightarrow \text{RepVal}_R, \\ \text{OrdEmb}(\text{RepEmb}_R^{\text{ord}}[I]; \text{Qu}_R^{\text{ord}, \text{can}}[I], <_R^{\text{ord}, \text{can}}[I]; \text{RepVal}_R, <_R^{\text{repr}}), \\ \text{Rep}_R^{\text{cmp}}[I] : \text{Qu}_R^{\text{can}}[I] &\longrightarrow \text{RepVal}_R, \\ \text{Rep}_R^{\text{cmp}}[I] &:= \text{RepEmb}_R^{\text{ord}}[I] \circ \text{Prj}_R^{\text{ord}, \text{can}}[I] \end{aligned}$$

**Proposition 4.141** (Canonical factorization and order judgment of representational coordinates). Under the assumptions of Definition 4.140, for each  $R \in \text{Rol}^{\text{cmp}}$ ,

$$\text{OrdRepCond}(\text{Rep}_R^{\text{cmp}}[\mathbb{I}]; \asymp_R[\mathbb{I}], \prec_R^{\text{can}}[\mathbb{I}], \prec_R^{\text{repr}})$$

holds.

*Proof.* In Lemma 4.138, substitute

$$\begin{aligned} \pi &= \text{Prj}_R^{\text{ord,can}}[\mathbb{I}], & \iota &= \text{RepEmb}_R^{\text{ord}}[\mathbb{I}], \\ f &= \text{Rep}_R^{\text{cmp}}[\mathbb{I}], & \asymp &= \asymp_R[\mathbb{I}], \\ \prec &= \prec_R^{\text{can}}[\mathbb{I}] \end{aligned}$$

The class judgment and order judgment of the quotient projection follow from Lemma 4.136; the order-embedding assumption and the composition identity follow from Definition 4.140.  $\square$

**Definition 4.142** (Local-to-global order-coordinate embeddings). Assume  $\text{ReprCtx}_{\leq 3}[\mathbb{I}]$ . For each  $R \in \text{Rol}^{\text{cmp}}$ , each  $u \in U$ , and any  $x \in \mathcal{U}_R[\mathbb{I}]$ , define the local-to-global order-coordinate embedding by

$$\begin{aligned} \text{LocEmb}_{R,u}^{\text{ord}}[\mathbb{I}] : \text{Qu}_R^{\text{loc}}[\mathbb{I}] &\hookrightarrow \text{Qu}_R^{\text{ord,can}}[\mathbb{I}], \\ \text{LocEmb}_R^{\text{ord}}[\mathbb{I}] (\text{Prj}_R^{\text{loc}}[\mathbb{I}](x)) &:= \text{Prj}_R^{\text{ord,can}}[\mathbb{I}](x) \end{aligned}$$

**Proposition 4.143** (Local-to-global order-coordinate compatibility). Under the assumptions of Definition 4.140, for each

$$R \in \text{Rol}^{\text{cmp}}, \quad u \in U$$

$$\text{LocEmb}_R^{\text{ord}}[\mathbb{I}]$$

is a well-defined injection and strict order embedding. For any

$$(x_0, x_1) \in (\text{Qu}_R^{\text{loc}}[\mathbb{I}])^2$$

and

$$\begin{aligned} a &\in \text{Im}(\text{Arr}_R^{\text{cmp}}) \\ x_0 &\prec_R^{\text{loc}}[\mathbb{I}] x_1 \iff \text{LocEmb}_R^{\text{ord}}[\mathbb{I}](x_0) <_R^{\text{ord,can}}[\mathbb{I}] \text{LocEmb}_R^{\text{ord}}[\mathbb{I}](x_1), \\ y &:= \text{Prj}_R^{\text{can}}[\mathbb{I}](u, a), \\ \text{Rep}_R^{\text{cmp}}[\mathbb{I}](y) &= \text{RepEmb}_R^{\text{ord}}[\mathbb{I}] \left( \text{LocEmb}_R^{\text{ord}}[\mathbb{I}](\text{Chart}_R^{\text{loc}}[\mathbb{I}](a)) \right) \end{aligned}$$

hold.

*Proof.* In Lemma 4.139, substitute

$$\begin{aligned} A &= \mathcal{U}_R[\mathbb{I}], & S &= \text{Qu}_R^{\text{can}}[\mathbb{I}], \\ \pi_A &= \text{Prj}_R^{\text{loc}}[\mathbb{I}], & \pi_S &= \text{Prj}_R^{\text{ord,can}}[\mathbb{I}]. \end{aligned}$$

The lemma gives the commutative identity

$$\begin{aligned} \text{LocEmb}_R^{\text{ord}}[\mathbb{I}] \circ \text{Prj}_R^{\text{loc}}[\mathbb{I}] \\ = \text{Prj}_R^{\text{ord,can}}[\mathbb{I}] \Big|_{\mathcal{U}_R[\mathbb{I}]} \end{aligned}$$

On the local comparison chart,

$$\begin{aligned} \text{LocEmb}_R^{\text{ord}}[\mathbb{I}] (\text{Chart}_R^{\text{loc}}[\mathbb{I}](a)) \\ = \text{Prj}_R^{\text{ord,can}}[\mathbb{I}](y) \end{aligned}$$

holds. Lemma 4.131, together with the class judgments and order judgments of Lemma 4.120 and Lemma 4.136, establishes the two equivalences required by the lemma. The commutative identity follows from its conclusion. Definition 4.124 and the commutative identity give the identity concerning the local comparison chart. Compose both sides of that identity with  $\text{RepEmb}_R^{\text{ord}}[\mathbb{I}]$  and apply Definition 4.140.  $\square$

### 4.2.3 Partial-establishment bundles and structural completion at the representation stage

**Definition 4.144** (Generated-structure bundles at the representation stage). For any object  $a$  and finite tuple  $(b_1, \dots, b_n)$ , define the front-preserving extension by

$$\text{TupExt}(a; (b_1, \dots, b_n)) := (a, b_1, \dots, b_n)$$

Evaluate each component at  $\mathbb{I}$ , and define the generated-structure bundle at each stage by

$$\begin{aligned} \text{ReprStr}_{=1}[\mathbb{I}] &:= \left( (\preceq_R^{\text{can}})_{R \in \text{Rol}^{\text{cmp}}}, \mathcal{E}^{\text{op}}, (\mathcal{U}_{R,u})_{(R,u) \in \text{IdX}_1^{\text{cmp}}}, \right. \\ &\quad \left. (\prec_R^{\text{can}})_{R \in \text{Rol}^{\text{cmp}}}, (\succ_{R,u})_{(R,u) \in \text{IdX}_1^{\text{cmp}}}, (\succ_R)_{R \in \text{Rol}^{\text{cmp}}} \right), \\ \text{ReprStr}_{=2}[\mathbb{I}] &:= \left( (\text{LinQuot}_{R,u}^{\text{loc}}, \text{Chart}_{R,u}^{\text{loc}})_{(R,u) \in \text{IdX}_1^{\text{cmp}}}, \right. \\ &\quad \left. (\text{ChartChg}_{R,uv}^{\text{loc}})_{(R,u,v) \in \text{IdX}_2^{\text{cmp}}}, \text{Atlas}^{\text{ord}} \right), \\ \text{ReprStr}_{=3}[\mathbb{I}] &:= \left( (\text{LinQuot}_R^{\text{can}})_{R \in \text{Rol}^{\text{cmp}}}, \right. \\ &\quad \left. (\text{LocEmb}_{R,u}^{\text{ord}})_{(R,u) \in \text{IdX}_1^{\text{cmp}}} \right), \\ \text{ReprStr}_{\leq 1}[\mathbb{I}] &:= \text{ReprStr}_{=1}[\mathbb{I}], \\ \text{ReprStr}_{\leq k}[\mathbb{I}] &:= \text{TupExt}(\text{ReprStr}_{\leq k-1}[\mathbb{I}], ; \text{ReprStr}_{=k}[\mathbb{I}]), \quad k \in \{2, 3\}. \end{aligned}$$

**Proposition 4.145** (Implication chain for the partial-establishment bundles at the representation stage).

$$\text{Repr}_{\leq 3}[\mathbb{I}] \implies \text{Repr}_{\leq 2}[\mathbb{I}] \implies \text{Repr}_{\leq 1}[\mathbb{I}].$$

*Proof.* In the recursive formula of Definition 4.92,  $\text{Repr}_{\leq k}$  contains  $\text{Repr}_{\leq k-1}$  as a conjunctive component.  $\square$

**Proposition 4.146** (Structural distinction among exact operational time structures, comparison-order atlases, global order-coordinate quotients, and representational coordinates). Assume  $\text{Operative}(\mathcal{X})$ . For each  $k \in \{1, 2, 3\}$ , under  $\text{Repr}_{\leq k}[\mathbb{I}]$ ,  $\text{ReprStr}_{\leq k}[\mathbb{I}]$  is generated in dependency order.

The exact operational time structure is a component of  $\text{ReprStr}_{=1}[\mathbb{I}]$ ; the comparison-order atlas is a component of  $\text{ReprStr}_{=2}[\mathbb{I}]$ ; and the global order-coordinate quotient on  $\mathfrak{C}/\mathfrak{D}$  and the canonical global order-coordinate projection are components of  $\text{ReprStr}_{=3}[\mathbb{I}]$ .

For each  $R \in \text{Rol}^{\text{cmp}}$ , once the selected representational codomain derived from  $\mathfrak{C}/\mathfrak{D}$  and the order embedding into the representational codomain derived from  $\mathfrak{C}/\mathfrak{D}$  in Definition 4.140 are given, the composition formula of Definition 4.140 determines the comparison representational coordinate derived from  $\mathfrak{C}/\mathfrak{D}$ . Thus the exact operational time structure, comparison-order atlas, global order-coordinate quotient, and representational coordinate have distinct types and generation formulas.

*Proof.* For  $k = 1$ , Definition 4.101, Lemma 4.102, Definition 4.103, Definition 4.104, Definition 4.110, Definition 4.111, and Definition 4.130 generate  $\text{ReprStr}_{=1}[\mathbb{I}]$  in dependency order.

For  $k = 2$ , Proposition 4.145, Lemma 4.114, Definition 4.116, Lemma 4.117, Lemma 4.120, Definition 4.124, Definition 4.126, Lemma 4.127, and Definition 4.128 generate  $\text{ReprStr}_{=2}[\mathbb{I}]$  in dependency order. Proposition 4.129 gives the cocycle condition for the local order-coordinate chart-change maps.

For  $k = 3$ , the implication chain of Proposition 4.145, Lemma 4.133, Definition 4.135, Lemma 4.136, Definition 4.142, and Lemma 4.139 generate  $\text{ReprStr}_{=3}[\mathbb{I}]$  in dependency order.

The recursive formula of Definition 4.144 gives each cumulative bundle. Definition 4.140 and Proposition 4.141 give the composition formula, equivalence-class judgment, and strict-order judgment for the representational coordinate. Proposition 4.143 gives the local-to-global compatibility identity with the local comparison chart.  $\square$

**Theorem 4.147** (Representation-stage generation theorem). Assume

$$\text{Operative}(\mathcal{X})$$

For each

$$k \in \{1, 2, 3\}$$

whenever

$$\text{Repr}_{\leq k}[\mathbb{I}]$$

holds,

$$\text{ReprStr}_{\leq k}[\mathbb{I}]$$

is generated in dependency order.

The local order-coordinate chart-change maps in

$$\text{ReprStr}_{=2}[\mathbb{I}]$$

satisfy the cocycle condition. Under

$$\text{Repr}_{\leq 3}[\mathbb{I}]$$

once the selected representational codomain derived from  $\mathfrak{C}/\mathfrak{D}$  and the order embedding into the representational codomain derived from  $\mathfrak{C}/\mathfrak{D}$  in Proposition 4.146 are given, the representational-coordinate factorization and the local-to-global compatibility identity with the local comparison chart hold.

*Proof.* Proposition 4.146 generates the three generated-structure bundles in dependency order, and Proposition 4.129 gives the cocycle condition. Definition 4.140, Proposition 4.141, and Proposition 4.143 give the factorization, equivalence-class judgment, and strict-order judgment of the representational coordinate and the local-to-global compatibility identity with the local comparison chart.  $\square$



#### 4.2.4 Conditional quantitative affine realization and frequency representation

**Definition 4.148** (Conditional quantitative affine realization derived from  $\mathfrak{C}/\mathfrak{D}$ ). Assume  $\text{ReprCtx}_{\leq 3}[\mathbb{I}]$  and the selected representational codomain derived from  $\mathfrak{C}/\mathfrak{D}$  in Definition 4.140. For each  $R \in \text{Rol}^{\text{cmp}}$ , let  $\text{AffAct}_R^{\text{qty}}$  be, on  $\text{RepVal}_R$ , a free and transitive action of  $\text{QtyVal}_R$ . For arbitrary

$$(a_0, a_1) \in (\text{RepVal}_R)^2, \quad \delta \in \text{QtyVal}_R$$

and

$$(x_0, x_1) \in (\text{Qu}_R^{\text{can}}[\mathbb{I}])^2$$

define the quantitatively linearly ordered field derived from  $\mathfrak{C}/\mathfrak{D}$ , the quantitative affine action, the quantitative affine difference, the conditional quantitative affine realization, and the quantitative difference of representational coordinates by

$$\begin{aligned} \text{QtyFld}_R &:= (\text{QtyVal}_R, +_R^{\text{qty}}, \cdot_R^{\text{qty}}, 0_R^{\text{qty}}, 1_R^{\text{qty}}, \leq_R^{\text{qty}}), \\ \text{AffAct}_R^{\text{qty}} : \text{RepVal}_R \times \text{QtyVal}_R &\longrightarrow \text{RepVal}_R, \\ \text{AffAct}_R^{\text{qty}}(a_0, \delta) = a_1 &\iff \delta = \text{AffDiff}_R^{\text{qty}}(a_1, a_0), \\ a_0 <_R^{\text{repr}} a_1 &\iff 0_R^{\text{qty}} <_R^{\text{qty}} \text{AffDiff}_R^{\text{qty}}(a_1, a_0), \\ \text{QtyAffReal}_R[\mathbb{I}] &:= (\text{QtyFld}_R, \text{RepVal}_R, \text{AffAct}_R^{\text{qty}}, \text{AffDiff}_R^{\text{qty}}, \text{RepEmb}_R^{\text{ord}}[\mathbb{I}]). \end{aligned}$$

Define the quantitative difference of representational coordinates derived from  $\mathfrak{C}/\mathfrak{D}$  by

$$\begin{aligned} &\text{CoordDiff}_R^{\text{qty}}[\mathbb{I}](x_1, x_0) \\ &:= \text{AffDiff}_R^{\text{qty}} \left( \text{Rep}_R^{\text{cmp}}[\mathbb{I}](x_1), \text{Rep}_R^{\text{cmp}}[\mathbb{I}](x_0) \right) \in \text{QtyVal}_R \end{aligned}$$

**Lemma 4.149** (Nonnegativity judgment for ordered affine differences). Under Definition 4.148, for each  $R \in \text{Rol}^{\text{cmp}}$  and arbitrary  $a_0, a_1 \in \text{RepVal}_R$ ,

$$a_0 \leq_R^{\text{repr}} a_1 \iff 0_R^{\text{qty}} \leq_R^{\text{qty}} \text{AffDiff}_R^{\text{qty}}(a_1, a_0).$$

*Proof.* If  $a_0 = a_1$ , the zero-element action and uniqueness of the quantitative affine difference give  $\text{AffDiff}_R^{\text{qty}}(a_1, a_0) = 0_R^{\text{qty}}$ . By linearity of the order on the quantitative codomain, a nonnegative quantitative affine difference is either zero or positive. In the zero case,  $a_1 = \text{AffAct}_R^{\text{qty}}(a_0, 0_R^{\text{qty}}) = a_0$ ; in the positive case, order compatibility gives  $a_0 <_R^{\text{repr}} a_1$ .  $\square$

**Definition 4.150** (Frequency representation). Assume the conditional quantitative affine realization derived from  $\mathfrak{C}$  in Definition 4.148,

$$\text{QtyAffReal}_{\mathfrak{C}}[\mathbb{I}]$$

Define the counting-window space, the comparison quotient classes at the counting-window endpoints, the set of admissible counting windows, the quantitative counting difference, the quantitative difference of representational coordinates, the counting-window frequency

representation derived from  $\mathfrak{C}$ , and the period representation by

$$\begin{aligned}
\text{CntWin}_{\mathfrak{C}} &:= U \times \text{Idx}_{\mathfrak{C}}^{\text{src}} \times \mathbb{N}^2, \\
\text{CntCls}_{\mathfrak{C},i}[\mathbf{l}; \text{win}] &:= \text{Prj}_{\mathfrak{C}}^{\text{can}}[\mathbf{l}] \left( u, \text{Arr}_{\mathfrak{C},u}^{\text{cmp}}(\text{tok}_{\mathfrak{C}}(j_{\mathfrak{C}}^{\text{src}}, k_i)) \right), \quad i \in \{0, 1\}, \\
\text{CntWin}_{\mathfrak{C}}^{\text{adm}}[\mathbf{l}] &:= \left\{ \text{win} = (u, j_{\mathfrak{C}}^{\text{src}}, k_0, k_1) \in \text{CntWin}_{\mathfrak{C}} \mid \begin{array}{l} k_0 < k_1, \\ \text{tok}_{\mathfrak{C}}(j_{\mathfrak{C}}^{\text{src}}, k_i) \in \text{Dom}(\text{Arr}_{\mathfrak{C},u}^{\text{cmp}}) \ (i = 0, 1), \\ \text{CntCls}_{\mathfrak{C},0}[\mathbf{l}; \text{win}] \prec_{\mathfrak{C}}^{\text{can}}[\mathbf{l}] \text{CntCls}_{\mathfrak{C},1}[\mathbf{l}; \text{win}] \end{array} \right\}, \\
\text{CntDiff}_{\mathfrak{C}}^{\text{qty}}[\text{win}] &:= (k_1 - k_0) \cdot_{\mathfrak{C}}^{\text{qty}} 1_{\mathfrak{C}}^{\text{qty}}, \\
\text{RepDiff}_{\mathfrak{C}}^{\text{qty}}[\mathbf{l}; \text{win}] &:= \text{CoordDiff}_{\mathfrak{C}}^{\text{qty}}[\mathbf{l}] (\text{CntCls}_{\mathfrak{C},1}[\mathbf{l}; \text{win}], \text{CntCls}_{\mathfrak{C},0}[\mathbf{l}; \text{win}]), \\
\nu_{\mathfrak{C}}^{\text{win}}[\mathbf{l}; \text{win}] &:= \frac{\text{CntDiff}_{\mathfrak{C}}^{\text{qty}}[\text{win}]}{\text{RepDiff}_{\mathfrak{C}}^{\text{qty}}[\mathbf{l}; \text{win}]}, \\
\text{Per}_{\mathfrak{C}}^{\text{win}}[\mathbf{l}; \text{win}] &:= (\nu_{\mathfrak{C}}^{\text{win}}[\mathbf{l}; \text{win}])^{-1}.
\end{aligned}$$

$\text{CntDiff}_{\mathfrak{C}}^{\text{qty}}$ ,  $\text{RepDiff}_{\mathfrak{C}}^{\text{qty}}$ ,  $\nu_{\mathfrak{C}}^{\text{win}}$ , and  $\text{Per}_{\mathfrak{C}}^{\text{win}}$  are defined for every  $\text{win} \in \text{CntWin}_{\mathfrak{C}}^{\text{adm}}[\mathbf{l}]$  and belong to  $\text{QtyVal}_{\mathfrak{C}}$ .

For each  $j_{\mathfrak{C}}^{\text{src}} \in \text{Idx}_{\mathfrak{C}}^{\text{src}}$ , define a nonempty family of admissible counting windows belonging to the same source by

$$\begin{aligned}
\emptyset \neq \mathcal{W}_{j_{\mathfrak{C}}^{\text{src}}} &\subseteq \{(u, j_{\mathfrak{C}}^{\text{src}}, k_0, k_1) \in \text{CntWin}_{\mathfrak{C}}^{\text{adm}}[\mathbf{l}]\}, \\
\nu_{\mathfrak{C}}^{\text{win}}[\mathbf{l}; \text{win}] &= \nu_{\mathfrak{C}}^{\text{win}}[\mathbf{l}; \text{win}'], \\
\text{Per}_{\mathfrak{C}}^{\text{win}}[\mathbf{l}; \text{win}] &= \text{Per}_{\mathfrak{C}}^{\text{win}}[\mathbf{l}; \text{win}'] \quad (\text{win}, \text{win}' \in \mathcal{W}_{j_{\mathfrak{C}}^{\text{src}}}), \\
\nu_{\mathfrak{C}}^{\text{src}}[\mathbf{l}; j_{\mathfrak{C}}^{\text{src}}] &:= \nu_{\mathfrak{C}}^{\text{win}}[\mathbf{l}; \text{win}], \\
\text{Per}_{\mathfrak{C}}^{\text{src}}[\mathbf{l}; j_{\mathfrak{C}}^{\text{src}}] &:= \text{Per}_{\mathfrak{C}}^{\text{win}}[\mathbf{l}; \text{win}] \quad (\text{win} \in \mathcal{W}_{j_{\mathfrak{C}}^{\text{src}}})
\end{aligned}$$

For  $\text{Src}_{\mathfrak{C}}(j_{\mathfrak{C}}^{\text{src}})$ , these are the source-relative frequency representation and period representation derived from  $\mathfrak{C}$ .

**Definition 4.151** (Frequency, period, and Hz representations). Specify a frequency-unit scalar derived from  $\mathfrak{C}$ ,  $\nu_{\mathfrak{C}}^{\text{unit}} \in \text{QtyVal}_{\mathfrak{C}}$ , such that  $0_{\mathfrak{C}}^{\text{qty}} <_{\mathfrak{C}}^{\text{qty}} \nu_{\mathfrak{C}}^{\text{unit}}$ . For a positive frequency representation  $\nu \in \text{QtyVal}_{\mathfrak{C}}$ , define its Hz numerical value relative to the frequency unit by

$$\text{HzNum}_{\nu_{\mathfrak{C}}^{\text{unit}}}(\nu) := \frac{\nu}{\nu_{\mathfrak{C}}^{\text{unit}}}$$

If the unit name Hz is assigned to  $\nu_{\mathfrak{C}}^{\text{unit}}$ , this value is the Hz numerical value of the frequency representation.

**Definition 4.152** (Standard frequency value as a representation). For the reference source derived from  $\mathfrak{C}$ ,  $\text{Src}_{\mathfrak{C}}(j_{\text{std}}^{\text{src}})$ , assume its source-relative frequency representation derived from  $\mathfrak{C}$ . For a normalization convention  $N_{\text{std}} \in \mathbb{N}_{>0}$ , define the standard frequency representation derived from  $\mathfrak{C}$  and the Hz unit scalar derived from  $\mathfrak{C}$  by

$$\begin{aligned}
\nu_{\mathfrak{C}}^{\text{std}}[\mathbf{l}] &:= \nu_{\mathfrak{C}}^{\text{src}}[\mathbf{l}; j_{\text{std}}^{\text{src}}], \\
\nu_{\mathfrak{C}}^{\text{Hz}}[\mathbf{l}] &:= \frac{\nu_{\mathfrak{C}}^{\text{std}}[\mathbf{l}]}{N_{\text{std}} \cdot_{\mathfrak{C}}^{\text{qty}} 1_{\mathfrak{C}}^{\text{qty}}}, \\
\text{HzNum}_{\nu_{\mathfrak{C}}^{\text{Hz}}[\mathbf{l}]}(\nu_{\mathfrak{C}}^{\text{std}}[\mathbf{l}]) &= N_{\text{std}} \cdot_{\mathfrak{C}}^{\text{qty}} 1_{\mathfrak{C}}^{\text{qty}}
\end{aligned}$$

**Proposition 4.153** (Well-definedness and generative position of frequency and period representations).

Under  $\text{ReprCtx}_{\leq 3}[l]$  and  $\text{QtyAffReal}_{\mathfrak{C}}[l]$ , for any  $\text{win} \in \text{CntWin}_{\mathfrak{C}}^{\text{adm}}[l]$ ,

$$\begin{aligned} 0_{\mathfrak{C}}^{\text{qty}} &<_{\mathfrak{C}}^{\text{qty}} \text{RepDiff}_{\mathfrak{C}}^{\text{qty}}[l; \text{win}], \\ \nu_{\mathfrak{C}}^{\text{win}}[l; \text{win}] \cdot_{\mathfrak{C}}^{\text{qty}} \text{Per}_{\mathfrak{C}}^{\text{win}}[l; \text{win}] &= 1_{\mathfrak{C}}^{\text{qty}}. \end{aligned}$$

Frequency, period, Hz numerical values, and the standard frequency representation are determined from the structural representation stage, the quantitative affine realization, admissible counting windows, window invariance, and a unit-normalization convention as inputs.

*Proof.* Admissibility of

$$\text{win} = (u, j_{\mathfrak{C}}^{\text{src}}, k_0, k_1)$$

and

$$\text{CntCls}_{\mathfrak{C},0}[l; \text{win}] \prec_{\mathfrak{C}}^{\text{can}} [l] \text{CntCls}_{\mathfrak{C},1}[l; \text{win}]$$

hold. By the strict-order judgment of Proposition 4.141 and the order-compatibility condition of Definition 4.148, the quantitative difference of representational coordinates is positive; and  $k_0 < k_1$  gives  $\text{CntDiff}_{\mathfrak{C}}^{\text{qty}}[\text{win}] \neq 0_{\mathfrak{C}}^{\text{qty}}$ . Thus the frequency quotient and its multiplicative inverse in Definition 4.150 are defined, and the multiplication and inverse axioms of the quantitatively linearly ordered field make their product  $1_{\mathfrak{C}}^{\text{qty}}$ .

The domains and dependency order of the source-relative frequency representation derived from  $\mathfrak{C}$ , the Hz numerical value, and the standard frequency representation derived from  $\mathfrak{C}$  follow from Definition 4.150, Definition 4.151, and Definition 4.152.  $\square$

*Remark 4.154* (Generation order of the frequency standard). For a source derived from  $\mathfrak{C}$ , the internal counter position  $k \in \mathbb{N}$  and update  $k \mapsto k + 1$  belong to the primitive input layer. Once the image of a token derived from  $\mathfrak{C}$  on the canonical comparison quotient, the representational coordinate, the quantitative affine difference, and an admissible counting window have been determined in dependency order, the ratio of the cycle-count difference to the representational-coordinate difference gives the frequency representation. The Hz unit is determined from the frequency representation of the reference source derived from  $\mathfrak{C}$  and the normalization convention. The cycle-count input, representational coordinate, frequency representation, and unit normalization belong to distinct generation stages.

#### 4.2.5 Finite operational granularity input

**Definition 4.155** (Quantitative coordinate width of a counting step derived from  $\mathfrak{C}$ ). Assume

$$\begin{aligned} &\text{Operative}(\mathcal{X}), \\ &\text{Repr}_{\leq 3}[l], \\ &\text{QtyAffReal}_{\mathfrak{C}}[l] \end{aligned}$$

Define the set of admissible single-step counting windows derived from  $\mathfrak{C}$  and its quantitative coordinate width by

$$\begin{aligned} \text{CntWin}_{\mathfrak{C}}^{\text{step}}[l] &:= \{ \text{win} = (u, j_{\mathfrak{C}}^{\text{src}}, k_0, k_1) \in \text{CntWin}_{\mathfrak{C}}^{\text{adm}}[l] \mid k_1 = k_0 + 1 \}, \\ \text{CntStepWid}_{\mathfrak{C}}^{\text{qty}}[l; \text{win}] &:= \text{RepDiff}_{\mathfrak{C}}^{\text{qty}}[l; \text{win}] \in \text{QtyVal}_{\mathfrak{C}}, \\ 0_{\mathfrak{C}}^{\text{qty}} &<_{\mathfrak{C}}^{\text{qty}} \text{CntStepWid}_{\mathfrak{C}}^{\text{qty}}[l; \text{win}], \quad \text{win} \in \text{CntWin}_{\mathfrak{C}}^{\text{step}}[l] \end{aligned}$$

The final inequality follows from the strict-order condition for admissible counting windows and the order-compatibility condition of Definition 4.148.

**Definition 4.156** (Finite operational granularity input). Assume  $\text{ReprCtx}_{\leq 3}[I]$  and  $\text{QtyAffReal}_R[I]$  ( $R \in \text{Rol}^{\text{cmp}}$ ). Specify the three nonempty auxiliary component codomains and the five component values of finite operational granularity. Define the component codomains and the product-type datum consisting of the specified values by

$$\begin{aligned} (\text{QtyVal}_R)_{\geq 0} &:= \{\delta \in \text{QtyVal}_R \mid 0_R^{\text{qty}} \leq_R^{\text{qty}} \delta\}, & R \in \text{Rol}^{\text{cmp}}, \\ \text{StabVal}_{\mathfrak{D}} &\neq \emptyset, & \text{DiscVal}_{\mathfrak{B}} \neq \emptyset, & \text{CouplVal}_{\mathfrak{D}} \neq \emptyset, \\ \text{GrnVal}^{\text{op}} &:= (\text{QtyVal}_{\mathfrak{C}})_{\geq 0} \times \text{StabVal}_{\mathfrak{D}} \times (\text{QtyVal}_{\mathfrak{D}})_{\geq 0} \times \text{DiscVal}_{\mathfrak{B}} \times \text{CouplVal}_{\mathfrak{D}}, \\ \text{Grn}^{\text{op}}(\mathcal{X}) &:= \left( \text{Grn}_{\mathfrak{C}, \text{cntstep}}, \text{Grn}_{\mathfrak{D}, \text{stab}}, \text{Grn}_{\mathfrak{D}, \text{cellwid}}, \text{Grn}_{\mathfrak{B}, \text{disc}}, \text{Grn}_{\mathfrak{D}, \text{coupl}} \right) \in \text{GrnVal}^{\text{op}} \end{aligned}$$

In order, the five components are a finite admissible scale for the quantitative coordinate width of an admissible single-step counting window derived from  $\mathfrak{C}$ ; a finite stability value of a record value derived from  $\mathfrak{D}$ ; a record-cell-width scale on quantitative coordinates derived from  $\mathfrak{D}$ ; a finite discrimination scale for an object-state difference derived from  $\mathfrak{B}$ ; and a finite limit on receptive coupling derived from  $\mathfrak{D}$ .

**Proposition 4.157** (Structural separation of finite operational granularity input). Under  $\text{ReprCtx}_{\leq 3}[I]$ , Theorem 4.147 generates  $\text{ReprStr}_{\leq 3}[I]$  in dependency order. Once the selected representational codomain derived from  $\mathfrak{C}/\mathfrak{D}$  and the order embedding into the representational codomain derived from  $\mathfrak{C}/\mathfrak{D}$  are given, the comparison representational coordinate derived from  $\mathfrak{C}/\mathfrak{D}$  is also determined.

The finite operational granularity datum has the five-component product type

$$\text{Grn}^{\text{op}}(\mathcal{X}) \in \text{GrnVal}^{\text{op}}$$

and supplies inputs for evaluating the counting-step width derived from  $\mathfrak{C}$ , judging the admissibility of the record-cell width derived from  $\mathfrak{D}$ , and evaluating the continuum approximation.

*Proof.* Theorem 4.147 generates the components of  $\text{ReprStr}_{\leq 3}[I]$ , and Definition 4.140 generates the comparison representational coordinate derived from  $\mathfrak{C}/\mathfrak{D}$ . Definition 4.148 equips the selected representational codomain derived from  $\mathfrak{C}/\mathfrak{D}$  with a quantitative affine structure, and Definition 4.156 makes the five components a product-type datum on  $\text{GrnVal}^{\text{op}}$ . These results establish the generation dependencies of the representation-stage structure, the type of the finite operational granularity datum, and the three constructions that use that datum.  $\square$

#### 4.2.6 Record-cell representational images derived from $\mathfrak{D}$ and $\mathfrak{C}$ -token record separation

**Definition 4.158** (Record-cell representational image derived from  $\mathfrak{D}$ ). Assume

$$\text{ReprCtx}_{\leq 3}[I]$$

and

$$\text{QtyAffReal}_{\mathfrak{D}}[I]$$

For each  $u \in U$ , each  $x_{\mathfrak{D}} \in \text{Tok}_{\mathfrak{D}}^*$ , and each  $y_{\mathfrak{D}} \in \text{Dom}(\text{Arr}_{\mathfrak{D},u}^{\text{cmp}})$ , define the set of arrival-capable  $\mathfrak{D}$ -record-cell tokens, the map from  $\mathfrak{D}$ -record tokens to representational values, and the record-cell representational image derived from  $\mathfrak{D}$  by

$$\begin{aligned}\text{CellTok}_{\mathfrak{D},u}^{\text{arr}}(x_{\mathfrak{D}}) &:= W_{\mathfrak{D}}(x_{\mathfrak{D}}) \cap \text{Dom}(\text{Arr}_{\mathfrak{D},u}^{\text{cmp}}), \\ \text{RecordVal}_{\mathfrak{D},u}^{\text{repr}}[\mathbb{I}](y_{\mathfrak{D}}) &:= \text{Rep}_{\mathfrak{D}}^{\text{cmp}}[\mathbb{I}]\left(\text{Prj}_{\mathfrak{D}}^{\text{can}}[\mathbb{I}](u, \text{Arr}_{\mathfrak{D},u}^{\text{cmp}}(y_{\mathfrak{D}}))\right) \in \text{RepVal}_{\mathfrak{D}}, \\ \text{CellImg}_{\mathfrak{D},u}^{\text{repr}}[\mathbb{I}](x_{\mathfrak{D}}) &:= \text{RecordVal}_{\mathfrak{D},u}^{\text{repr}}[\mathbb{I}][\text{CellTok}_{\mathfrak{D},u}^{\text{arr}}(x_{\mathfrak{D}})] \subseteq \text{RepVal}_{\mathfrak{D}}\end{aligned}$$

**Definition 4.159** (Extrema and quantitative coordinate width of a record-cell representational image derived from  $\mathfrak{D}$ ). Let  $\text{Bool} := \{\perp, \top\}$ . For each  $S \subseteq \text{RepVal}_{\mathfrak{D}}$ ,  $u \in U$ , and  $x_{\mathfrak{D}} \in \text{Tok}_{\mathfrak{D}}^*$ , define the extremum-existence predicate and ordered affine width on a set of  $\mathfrak{D}$ -representational values, together with the extrema and quantitative coordinate width of the record-cell representational image derived from  $\mathfrak{D}$ , as follows:

$$\begin{aligned}\text{HasExt}_{\mathfrak{D}}^{\text{repr}}(S) &\iff S \neq \emptyset \\ &\quad \wedge [\exists a_-, a_+ \in S \forall a \in S, \quad a_- \leq_{\mathfrak{D}}^{\text{repr}} a \leq_{\mathfrak{D}}^{\text{repr}} a_+].\end{aligned}$$

The value of this predicate satisfies

$$\text{HasExt}_{\mathfrak{D}}^{\text{repr}}(S) \in \text{Bool}$$

Further, set

$$\begin{aligned}\text{AffSpan}_{\mathfrak{D}}^{\text{qty}}(S) &:= \text{AffDiff}_{\mathfrak{D}}^{\text{qty}}(\max S, \min S), \\ S_{u,x_{\mathfrak{D}}} &:= \text{CellImg}_{\mathfrak{D},u}^{\text{repr}}[\mathbb{I}](x_{\mathfrak{D}}).\end{aligned}$$

Define the existence of extrema for the record-cell representational image derived from  $\mathfrak{D}$  by

$$\text{CellExtrema}_{\mathfrak{D},u}^{\text{repr}}[\mathbb{I}](x_{\mathfrak{D}}) \iff \text{HasExt}_{\mathfrak{D}}^{\text{repr}}(S_{u,x_{\mathfrak{D}}})$$

and let

$$\text{CellExtrema}_{\mathfrak{D},u}^{\text{repr}}[\mathbb{I}](x_{\mathfrak{D}}) \in \text{Bool}$$

Define the extrema and quantitative coordinate width by

$$\begin{aligned}\text{CellMin}_{\mathfrak{D},u}^{\text{repr}}[\mathbb{I}](x_{\mathfrak{D}}) &:= \min S_{u,x_{\mathfrak{D}}}, \\ \text{CellMax}_{\mathfrak{D},u}^{\text{repr}}[\mathbb{I}](x_{\mathfrak{D}}) &:= \max S_{u,x_{\mathfrak{D}}}, \\ \text{CellWid}_{\mathfrak{D},u}^{\text{qty}}[\mathbb{I}](x_{\mathfrak{D}}) &:= \text{AffSpan}_{\mathfrak{D}}^{\text{qty}}(S_{u,x_{\mathfrak{D}}})\end{aligned}$$

The minimum, maximum, and ordered affine width are defined on the domain where  $\text{HasExt}_{\mathfrak{D}}^{\text{repr}}$  holds.

**Definition 4.160** (Admissibility of record-cell width derived from  $\mathfrak{D}$ ). Define the predicate for admissibility of record-cell width derived from  $\mathfrak{D}$  by

$$\begin{aligned}\text{CellWidAdm}_{\mathfrak{D}}[\mathbb{I}] &\iff \left[ \begin{aligned} &\forall (u, x_{\mathfrak{D}}) \in U \times \text{Tok}_{\mathfrak{D}}^*, \\ &(\text{CellTok}_{\mathfrak{D},u}^{\text{arr}}(x_{\mathfrak{D}}) \neq \emptyset \implies \text{CellExtrema}_{\mathfrak{D},u}^{\text{repr}}[\mathbb{I}](x_{\mathfrak{D}})) \\ &\wedge (\text{CellExtrema}_{\mathfrak{D},u}^{\text{repr}}[\mathbb{I}](x_{\mathfrak{D}}) \implies \text{CellWid}_{\mathfrak{D},u}^{\text{qty}}[\mathbb{I}](x_{\mathfrak{D}}) \leq_{\mathfrak{D}}^{\text{qty}} \text{Grn}_{\mathfrak{D},\text{cellwid}}) \end{aligned} \right].\end{aligned}$$

The value of this predicate satisfies

$$\text{CellWidAdm}_{\mathfrak{D}}[\mathbb{I}] \in \text{Bool}$$

**Definition 4.161** ( $\mathfrak{C}$ -token-associated representational image of  $\mathfrak{D}$ -records). For each  $u \in U$  and  $x_{\mathfrak{C}} \in \text{Dom}(\text{Arr}_{\mathfrak{C},u}^{\text{cmp}})$ , define the  $\mathfrak{C}$ -token-associated representational image of  $\mathfrak{D}$ -records and the global order-coordinate class of a  $\mathfrak{C}$ -token by

$$\begin{aligned} \text{RecordImg}_{\mathfrak{C}\mathfrak{D},u}^{\text{repr}}[\mathbb{I}](x_{\mathfrak{C}}) &:= \bigcup_{\substack{x_{\mathfrak{D}} \in \text{Dom}(\text{Arr}_{\mathfrak{D},u}^{\text{cmp}}) \\ (x_{\mathfrak{C}}, x_{\mathfrak{D}}) \in \text{Rel}^{\text{src}}}} \text{CellImg}_{\mathfrak{D},u}^{\text{repr}}[\mathbb{I}](x_{\mathfrak{D}}) \subseteq \text{RepVal}_{\mathfrak{D}}, \\ \text{OrdCls}_{\mathfrak{C},u}^{\text{can}}[\mathbb{I}](x_{\mathfrak{C}}) &:= \text{Prj}_{\mathfrak{C}}^{\text{ord,can}}[\mathbb{I}]\left(\text{Prj}_{\mathfrak{C}}^{\text{can}}[\mathbb{I}](u, \text{Arr}_{\mathfrak{C},u}^{\text{cmp}}(x_{\mathfrak{C}}))\right) \end{aligned}$$

The latter satisfies

$$\text{OrdCls}_{\mathfrak{C},u}^{\text{can}}[\mathbb{I}](x_{\mathfrak{C}}) \in \text{Qu}_{\mathfrak{C}}^{\text{ord,can}}[\mathbb{I}]$$

**Definition 4.162** (Record separation of  $\mathfrak{C}$ -tokens). Define the predicate for record separation of  $\mathfrak{C}$ -tokens by

$$\begin{aligned} \text{TokSep}_{\mathfrak{C}\mathfrak{D}}^{\text{rec}}[\mathbb{I}] &\iff \forall u \in U \forall (x_0, x_1) \in (\text{Dom}(\text{Arr}_{\mathfrak{C},u}^{\text{cmp}}))^2, \\ &\quad \text{OrdCls}_{\mathfrak{C},u}^{\text{can}}[\mathbb{I}](x_0) \neq \text{OrdCls}_{\mathfrak{C},u}^{\text{can}}[\mathbb{I}](x_1) \\ &\implies \left\{ \left[ \bigwedge_{i \in \{0,1\}} \text{RecordImg}_{\mathfrak{C}\mathfrak{D},u}^{\text{repr}}[\mathbb{I}](x_i) \neq \emptyset \right] \right. \\ &\quad \left. \wedge [\text{RecordImg}_{\mathfrak{C}\mathfrak{D},u}^{\text{repr}}[\mathbb{I}](x_0) \cap \text{RecordImg}_{\mathfrak{C}\mathfrak{D},u}^{\text{repr}}[\mathbb{I}](x_1) = \emptyset] \right\}, \\ \text{TokSep}_{\mathfrak{C}\mathfrak{D}}^{\text{rec}}[\mathbb{I}] &\in \text{Bool} \end{aligned}$$

**Proposition 4.163** (Generation dependencies of  $\mathfrak{D}$ -record-cell width and  $\mathfrak{C}$ -token record separation). Assume  $\text{ReprCtx}_{\leq 3}[\mathbb{I}]$ ,  $\text{QtyAffReal}_{\mathfrak{D}}[\mathbb{I}]$ , and  $\text{Grn}^{\text{op}}(\mathcal{X})$ . For arbitrary  $u \in U$  and  $x_{\mathfrak{D}} \in \text{Tok}_{\mathfrak{D}}^*$ ,

$$\text{CellExtrema}_{\mathfrak{D},u}^{\text{repr}}[\mathbb{I}](x_{\mathfrak{D}}) \implies \text{CellWid}_{\mathfrak{D},u}^{\text{qty}}[\mathbb{I}](x_{\mathfrak{D}}) \in (\text{QtyVal}_{\mathfrak{D}})_{\geq 0}.$$

$\text{CellWidAdm}_{\mathfrak{D}}[\mathbb{I}]$  requires the existence of extrema of the representational image in each record cell derived from  $\mathfrak{D}$  whose arrival-capable part is nonempty, and, when extrema exist, comparison of its quantitative coordinate width with  $\text{Grn}_{\mathfrak{D},\text{cellwid}}$ .  $\text{TokSep}_{\mathfrak{C}\mathfrak{D}}^{\text{rec}}[\mathbb{I}]$  requires that, for distinct global order-coordinate classes of  $\mathfrak{C}$ -tokens, the corresponding two  $\mathfrak{D}$ -record representational images are both nonempty and have empty intersection.

The record-cell representational image derived from  $\mathfrak{D}$ , its extrema and quantitative coordinate width, record-cell-width admissibility, the  $\mathfrak{C}$ -token-associated representational image of  $\mathfrak{D}$ -records, the global order-coordinate class of a  $\mathfrak{C}$ -token, and record separation of  $\mathfrak{C}$ -tokens are determined in dependency order from record cells derived from  $\mathfrak{D}$ , the source-side relation derived from  $\mathfrak{C}\mathfrak{D}$ , the canonical comparison quotient, the global order-coordinate quotient, the representational coordinate, the quantitative affine realization, and finite operational granularity.

*Proof.* Assume the antecedent of the proposition's implication. Then

$$\text{CellMin}_{\mathfrak{D},u}^{\text{repr}}[\mathbb{I}](x_{\mathfrak{D}}) \leq_{\mathfrak{D}}^{\text{repr}} \text{CellMax}_{\mathfrak{D},u}^{\text{repr}}[\mathbb{I}](x_{\mathfrak{D}})$$

holds. By Definition 4.159 and Lemma 4.149, the record-cell width, which is the quantitative affine difference between the two extrema, is nonnegative.

The two implications in Definition 4.160 give nonemptiness of the arrival-capable part, existence of extrema, and comparison with the granularity component. Definition 4.162 gives nonemptiness and empty intersection for the two  $\mathfrak{D}$ -record representational images corresponding to distinct global order-coordinate classes. The defining formulas of the respective objects and the record-cell-width granularity component derived from  $\mathfrak{D}$  in Definition 4.156 determine, in dependency order from the inputs listed in the proposition, admissibility of the record-cell width derived from  $\mathfrak{D}$ , the  $\mathfrak{C}$ -token-associated representational image of  $\mathfrak{D}$ -records, and record separation of  $\mathfrak{C}$ -tokens.  $\square$

#### 4.2.7 Continuum-approximation evaluation input

**Definition 4.164** (Componentwise evaluation input for continuum approximation). Define the index set and component types for continuum-approximation evaluation, and the evaluation source for the counting-step-width component derived from  $\mathfrak{C}$ , by

$$\begin{aligned} \text{EvalIdx}^{\text{cont}} &:= \left\{ (\mathfrak{C}, \text{cntstep}), (\mathfrak{D}, \text{stab}), (\mathfrak{D}, \text{cellwid}), \right. \\ &\quad \left. (\mathfrak{B}, \text{disc}), (\mathfrak{D}, \text{coupl}), \text{OrdStab} \right\}, \\ \text{EvalVal}_{\kappa}^{\text{cont}} &\neq \emptyset, \quad \preceq_{\kappa}^{\text{cont,eval}} \subseteq (\text{EvalVal}_{\kappa}^{\text{cont}})^2, \\ 0_{\kappa}^{\text{cont,eval}} &\in \text{EvalVal}_{\kappa}^{\text{cont}}, \\ \forall z \in \text{EvalVal}_{\kappa}^{\text{cont}}, \quad 0_{\kappa}^{\text{cont,eval}} &\preceq_{\kappa}^{\text{cont,eval}} z, \\ \text{Eval}_{\kappa}^{\text{cont}}[\mathbb{I}], \text{Tol}_{\kappa}^{\text{cont}} &\in \text{EvalVal}_{\kappa}^{\text{cont}}, \\ \text{EvalComp}_{\kappa}^{\text{cont}}[\mathbb{I}] &:= \left( \begin{array}{c} \text{EvalVal}_{\kappa}^{\text{cont}}, \\ \preceq_{\kappa}^{\text{cont,eval}}, \\ 0_{\kappa}^{\text{cont,eval}}, \text{Eval}_{\kappa}^{\text{cont}}[\mathbb{I}], \\ \text{Tol}_{\kappa}^{\text{cont}} \end{array} \right), \\ \kappa &\in \text{EvalIdx}^{\text{cont}}. \end{aligned}$$

Define the evaluation source for the counting-step-width component derived from  $\mathfrak{C}$  by

$$\begin{aligned} \text{EvalSrc}_{(\mathfrak{C}, \text{cntstep})}^{\text{cont}}[\mathbb{I}] &:= \left( (\text{CntStepWid}_{\mathfrak{C}}^{\text{qty}}[\mathbb{I}; \text{win}])_{\text{win} \in \text{CntWin}_{\mathfrak{C}}^{\text{step}}[\mathbb{I}]}, \right. \\ &\quad \left. \text{Grn}_{\mathfrak{C}, \text{cntstep}} \right) \end{aligned}$$

The relation  $\preceq_{\kappa}^{\text{cont,eval}}$  is a partial order, and  $0_{\kappa}^{\text{cont,eval}}$  is its least element. The counting-step-width component derived from  $\mathfrak{C}$ ,  $\text{Eval}_{(\mathfrak{C}, \text{cntstep})}^{\text{cont}}[\mathbb{I}]$ , takes  $\text{EvalSrc}_{(\mathfrak{C}, \text{cntstep})}^{\text{cont}}[\mathbb{I}]$  as input.

**Definition 4.165** (Order-preserving representational-stability evaluation input). The order-preserving representational-stability evaluation input is

$$\text{EvalComp}_{\text{OrdStab}}^{\text{cont}}[\mathbb{I}]$$

and its componentwise evaluation formula is defined by

$$\text{Eval}_{\text{OrdStab}}^{\text{cont}}[\mathbb{I}] \preceq_{\text{OrdStab}}^{\text{cont,eval}} \text{Tol}_{\text{OrdStab}}^{\text{cont}}$$

**Definition 4.166** (Boolean truth-value evaluation inputs for record-cell width and record separation). Let the predicate for admissibility of record-cell width derived from  $\mathfrak{D}$  and the predicate for record separation of  $\mathfrak{C}$ -tokens be the Boolean truth-value components of the continuum-approximation evaluation-input datum by

$$(\text{CellWidAdm}_{\mathfrak{D}}[\mathbb{I}], \text{TokSep}_{\mathfrak{C}\mathfrak{D}}^{\text{rec}}[\mathbb{I}]) \in \text{Bool}^2$$

**Definition 4.167** (Order-topological coordinate candidates). For a subbasis  $\mathcal{S}$ , write  $\text{TopSub}(\mathcal{S})$  for the topology it generates. For each  $R \in \text{Rol}^{\text{cmp}}$  and  $a_0 \in \text{RepVal}_R$ , define the open rays, order topology, and candidate datum for a continuous coordinate representation by

$$\begin{aligned} \text{Ray}_R^\sigma(a_0) &:= \begin{cases} \{a \in \text{RepVal}_R \mid a <_R^{\text{repr}} a_0\}, & \sigma = -, \\ \{a \in \text{RepVal}_R \mid a_0 <_R^{\text{repr}} a\}, & \sigma = +. \end{cases} \\ \text{OrdTop}_R^{\text{repr}} &:= \text{TopSub}(\{\text{Ray}_R^\sigma(a_0) \mid \sigma \in \{-, +\}, a_0 \in \text{RepVal}_R\}), \\ \text{CoordCand}_R^{\text{cont}}[\mathbb{I}] &:= (\text{RepVal}_R, \leq_R^{\text{repr}}, \text{OrdTop}_R^{\text{repr}}, \text{Rep}_R^{\text{cmp}}[\mathbb{I}]) \end{aligned}$$

**Definition 4.168** (Continuum-approximation evaluation-input datum). Assume

$$\text{ReprCtx}_{\leq 3}[\mathbb{I}]$$

and

$$\text{QtyAffReal}_R[\mathbb{I}], \quad R \in \text{Rol}^{\text{cmp}}$$

Specify the finite operational granularity datum of Definition 4.156 and the six componentwise evaluation inputs of Definition 4.164. Construct record-cell-width admissibility, record separation, and the order-topological coordinate candidates for these specifications and the generated representation-stage structures according to their defining clauses. Define the continuum-approximation evaluation-input datum by

$$\begin{aligned} \text{EvalInput}^{\text{cont}}[\mathbb{I}] &:= \left( (\text{QtyAffReal}_R[\mathbb{I}])_{R \in \text{Rol}^{\text{cmp}}}, \text{Grn}^{\text{op}}(\mathcal{X}), \right. \\ &\quad (\text{EvalComp}_\kappa^{\text{cont}}[\mathbb{I}])_{\kappa \in \text{EvalIdx}^{\text{cont}}}, \\ &\quad \text{CellWidAdm}_{\mathfrak{D}}[\mathbb{I}], \text{TokSep}_{\mathfrak{C}/\mathfrak{D}}^{\text{rec}}[\mathbb{I}], \\ &\quad \left. (\text{CoordCand}_R^{\text{cont}}[\mathbb{I}])_{R \in \text{Rol}^{\text{cmp}}} \right) \end{aligned}$$

**Proposition 4.169** (Component types of continuum-approximation evaluation input). The componentwise establishment-judgment formulas and the types of the Boolean truth-value components of the continuum-approximation evaluation-input datum are

$$\begin{aligned} \text{Eval}_\kappa^{\text{cont}}[\mathbb{I}] &\preceq_{\kappa}^{\text{cont,eval}} \text{Tok}_\kappa^{\text{cont}}, \quad \kappa \in \text{EvalIdx}^{\text{cont}}, \\ (\text{CellWidAdm}_{\mathfrak{D}}[\mathbb{I}], \text{TokSep}_{\mathfrak{C}/\mathfrak{D}}^{\text{rec}}[\mathbb{I}]) &\in \text{Bool}^2 \end{aligned}$$

The candidate datum for a continuous coordinate representation derived from  $\mathfrak{C}/\mathfrak{D}$  has as components a coordinate carrier, a linear order, an order topology, and a comparison representational coordinate.

*Proof.* Definition 4.164 gives the six  $\text{EvalComp}_\kappa^{\text{cont}}$  an evaluation-value set, a partial order, a least element, an evaluation value, and an admissible threshold. Definition 4.165 gives order-preserving representational stability as the  $\kappa = \text{OrdStab}$  component of the same component family. Definition 4.166 gives the two truth-value components, and Definition 4.167 gives the candidate for a continuous coordinate representation derived from  $\mathfrak{C}/\mathfrak{D}$ .

Thus the tuple of Definition 4.168 has as components conditional quantitative affine realizations, finite operational granularity, six componentwise evaluation types, two Boolean truth values, and candidates for continuous coordinate representations derived from  $\mathfrak{C}/\mathfrak{D}$ . The componentwise establishment judgment and continuous-representation evaluation apply the evaluation relation to the corresponding components, and the complete evaluation specification determines the establishment value of the continuum approximation.  $\square$



### 4.2.8 Consequences of the Representation Stage

**Proposition 4.170** (Separation of representation-stage structures from quantitative-evaluation inputs). Assume  $\text{ReprCtx}_{\leq 3}[\mathbb{I}]$ . For each  $R \in \text{Rol}^{\text{cmp}}$ , specify the selected  $\mathfrak{C}/\mathfrak{D}$ -derived representational value range and its  $\mathfrak{C}/\mathfrak{D}$ -derived representational-value-range order embedding from Definition 4.140. Then

$$\begin{aligned} \text{Repr}_{\leq 3}[\mathbb{I}] &\implies \text{ReprStr}_{\leq 3}[\mathbb{I}], \\ \text{Rep}_R^{\text{cmp}}[\mathbb{I}] &= \text{RepEmb}_R^{\text{ord}}[\mathbb{I}] \circ \text{Prj}_R^{\text{ord,can}}[\mathbb{I}] \end{aligned}$$

are determined in dependency order. The components of  $\text{ReprStr}_{\leq 3}[\mathbb{I}]$  are given in Definition 4.144.

Further specify the  $\mathfrak{C}/\mathfrak{D}$ -derived conditional quantitative affine realizations of Definition 4.148, the finite operational granularity datum of Definition 4.156, and the six componentwise evaluation inputs of Definition 4.164. Under these specifications, the  $\mathfrak{D}$ -derived record-cell-width admissibility predicate, the  $\mathfrak{C}$ -token record-separation predicate, and the order-topological coordinate candidates are determined and form the continuum-approximation evaluation-input datum of Definition 4.168. The bundles of structures generated at the representation stage and the quantitative-evaluation inputs have distinct types and dependency orders.

*Proof.* Theorem 4.147 generates  $\text{ReprStr}_{\leq 3}[\mathbb{I}]$ , and Definition 4.140 gives the factorization of the  $\mathfrak{C}/\mathfrak{D}$ -derived comparison representational coordinates. The specified quantitative affine realizations satisfy the compatibility conditions of Definition 4.148. Definition 4.156 forms a product-type datum from the five specified granularity components. Definitions 4.160 and 4.162 determine record-cell-width admissibility and record separation from the generated record images and the specified inputs. Definition 4.167 determines the order-topological coordinate candidates from the order on the selected representational value ranges and the comparison representational coordinates. Definition 4.168 collects these objects, together with the six specified componentwise evaluation inputs, into one tuple. Hence, the bundles of structures generated at the representation stage, the  $\mathfrak{C}/\mathfrak{D}$ -derived comparison representational coordinates, and the quantitative-evaluation inputs satisfy the types and dependency orders stated in the proposition.  $\square$

**Definition 4.171** (Typed generation closure). Let

$\text{Expr}$

denote the collection of all typed formal expressions whose components are object expressions or truth-valued expressions. For a finite tuple  $T$ , let  $\text{Idx}(T)$  denote its component-index set, and define the component relation by

$$A \in_{\text{comp}} T \iff \exists i \in \text{Idx}(T), \text{pr}_i(T) = A$$

For an object set  $\text{In}$  and a family of typed generative-dependency rules

$$\text{Dep} \subseteq \mathcal{P}(\text{Expr}) \times \text{Expr}$$

define the generation closure by

$$\begin{aligned} \text{Cl}_{\text{Dep}}(\text{In}) := \bigcap \left\{ S \subseteq \text{Expr} \mid \text{In} \subseteq S, \right. \\ \left. \forall (A, Y) \in \text{Dep}, \quad A \subseteq S \implies Y \in S \right\} \end{aligned}$$

and define generability by

$$\text{GenDef}_{\text{Dep}}(Y \mid \text{In}) \iff Y \in \text{Cl}_{\text{Dep}}(\text{In})$$

**Proposition 4.172** (Generation closure of continuum-approximation evaluation inputs). Define the set of direct inputs to continuum-approximation evaluation by

$$\begin{aligned} \text{In}^{\text{cont}} &:= \{A \in \text{Expr} \mid A \in_{\text{comp}} \text{EvalInput}^{\text{cont}}[\mathbb{I}]\}, \\ \text{Dep}^{\text{cont}} &:= \{(\text{In}^{\text{cont}}, \text{EvalInput}^{\text{cont}}[\mathbb{I}])\} \end{aligned}$$

Then

$$\text{GenDef}_{\text{Dep}^{\text{cont}}}(\text{EvalInput}^{\text{cont}}[\mathbb{I}] \mid \text{In}^{\text{cont}})$$

holds.

*Proof.* The right-hand side of Definition 4.168 is a tuple containing every element of  $\text{In}^{\text{cont}}$  as a component. Therefore,

$$(\text{In}^{\text{cont}}, \text{EvalInput}^{\text{cont}}[\mathbb{I}]) \in \text{Dep}^{\text{cont}}.$$

Apply, in order, the generation closure and the generability criterion of Definition 4.171.  $\square$

By Definition 4.144, Theorem 4.147, Proposition 4.170, and Proposition 4.172, the structural generation sequence of the representation stage is

$$\begin{aligned} \text{Operative}(\mathcal{X}) &\longmapsto (\preceq_R^{\text{arr}})_{R \in \text{Rol}^{\text{cmp}}}, \\ \text{Repr}_{\leq k}[\mathbb{I}] &\longmapsto \text{ReprStr}_{\leq k}[\mathbb{I}], \quad k \in \{1, 2, 3\}, \\ (\text{ReprStr}_{\leq 3}[\mathbb{I}], (\text{RepEmb}_R^{\text{ord}}[\mathbb{I}])_{R \in \text{Rol}^{\text{cmp}}}) &\longmapsto (\text{Rep}_R^{\text{cmp}}[\mathbb{I}])_{R \in \text{Rol}^{\text{cmp}}}. \end{aligned}$$

The object  $\text{Repr}_{\leq 3}[\mathbb{I}]$  of Definition 4.92 expresses the conjunction of the three representation-stage conditions. Specifying the  $\mathfrak{C}/\mathfrak{D}$ -derived conditional quantitative affine realizations determines quantitative affine differences and frequency and period indexed by admissible counting windows. Source-relative frequency requires window invariance on a nonempty family of windows; a hertz numerical value requires a positive unit scalar. Specifying a unit through a standard frequency uses a reference source and a normalization convention. Further specifying finite operational granularity and the six componentwise evaluation inputs determines the continuum-approximation evaluation-input datum, together with record-cell-width admissibility, record separation, and coordinate candidates. Proposition 4.172 expresses tuple construction when these direct inputs are given.

Table 2: Hierarchy of generated representation-stage objects

| Stage                                          | Condition or generated object                                                                                                                                                                                                                                       |
|------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Comparison-input layer                         | $\text{Operative}(\mathcal{X}) \mapsto (\preceq_R^{\text{arr}})_{R \in \text{Rol}^{\text{cmp}}}$                                                                                                                                                                    |
| Canonical-order layer                          | $\text{Repr}_{\leq 1}[I] \mapsto \text{ReprStr}_{\leq 1}[I]$                                                                                                                                                                                                        |
| Local order-coordinate layer                   | $\text{Repr}_{\leq 2}[I] \mapsto \text{ReprStr}_{\leq 2}[I]$<br>$\text{Repr}_{\leq 3}[I] \mapsto \text{ReprStr}_{\leq 3}[I]$                                                                                                                                        |
| Global order-coordinate layer                  | $\xrightarrow{\text{RepEmb}_R^{\text{ord}}} \text{Rep}_R^{\text{cmp}}[I]$                                                                                                                                                                                           |
| Quantitative-representation layer              | $\text{Repr}_{\leq 3}[I], \text{QtyAffReal}_R[I] \ (R \in \text{Rol}^{\text{cmp}})$<br>$\mapsto$ quantitative affine differences, frequency and period for each admissible counting window<br>A hertz numerical value additionally requires a positive unit scalar. |
| Continuum-approximation evaluation-input layer | $\text{In}^{\text{cont}} \mapsto \text{EvalInput}^{\text{cont}}[I]$                                                                                                                                                                                                 |

The structures generated at each stage of Table 2 are as follows.

**Comparison-input layer** The  $\mathfrak{C}/\mathfrak{D}$ -source-induced arrival preorders are determined from the  $\mathfrak{C}/\mathfrak{D}$  source-side partial orders, the comparison-side recovery maps, and the tagged comparison-arrival reception spaces.

**Canonical-order layer** When  $K_{\text{CanOrdDesc}}^{\text{repr}}[I]$  holds, the source-induced arrival preorders descend to the comparison equivalence classes and determine the canonical  $\mathfrak{C}/\mathfrak{D}$  partial orders and the exact operational time structures.

**Local order-coordinate layer** When  $K_{\text{LocIncTrans}}^{\text{repr}}[I]$  holds, it determines the quotients by local incomparability classes, the local strict linear orders, the local order-coordinate projections, and the local comparison charts. The local order-coordinate chart-transition maps yield the comparison-order atlas.

**Global order-coordinate layer** When  $K_{\text{GlobIncTrans}}^{\text{repr}}[I]$  holds, it determines the quotients by global incomparability classes, the global strict linear orders, and the canonical global order-coordinate projections. Their composition with the  $\mathfrak{C}/\mathfrak{D}$ -derived representational-value-range order embeddings yields the  $\mathfrak{C}/\mathfrak{D}$ -derived comparison representational coordinates.

**Quantitative-representation layer** Specifying the  $\mathfrak{C}/\mathfrak{D}$ -derived conditional quantitative affine realizations determines quantitative affine differences and frequency and period indexed by admissible counting windows. Source-relative frequency and hertz numerical values use the window-family, window-invariance, and unit-normalization inputs specified in their respective definitions.

**Continuum-approximation evaluation-input layer** Further specify the finite operational granularity datum and the six componentwise evaluation inputs. Each evaluation input includes an evaluation-value carrier, a partial order, a least element, an evaluation value, and an admissible threshold. Construct record-cell-width admissibility, record separation, and the order-topological coordinate candidates, and collect them into the tuple of Definition 4.168.

The  $\mathfrak{C}/\mathfrak{D}$  source-side partial orders, the comparison-side recovery maps, and the tagged comparison-arrival reception spaces are the generative inputs to  $(\preceq_R^{\text{arr}})_{R \in \text{Rol}^{\text{cmp}}}$ .

**Definition 4.173** (Representation-stage dependency layers). Fix the input  $\mathbb{I}$ . Define the outputs of the canonical quotient and order constructions by dependency layer as

$$(\text{Out}_k^{\text{repr}})_{0 \leq k \leq 3} := \left( \begin{array}{c} \{\text{Qu}^{\text{can}}[\mathbb{I}]\}, \\ \{\mathcal{E}^{\text{op}}[\mathbb{I}]\} \cup \bigcup_{R \in \text{Rol}^{\text{cmp}}} \{\preceq_R^{\text{can}}[\mathbb{I}]\}, \\ \{\text{Atlas}^{\text{ord}}[\mathbb{I}]\} \cup \bigcup_{\substack{R \in \text{Rol}^{\text{cmp}} \\ u \in U}} \{\text{Qu}_{R,u}^{\text{loc}}[\mathbb{I}], \text{Prj}_{R,u}^{\text{loc}}[\mathbb{I}], \text{Chart}_{R,u}^{\text{loc}}[\mathbb{I}]\}, \\ \bigcup_{R \in \text{Rol}^{\text{cmp}}} \{\text{Qu}_R^{\text{ord,can}}[\mathbb{I}], \text{Prj}_R^{\text{ord,can}}[\mathbb{I}]\} \end{array} \right).$$

The selected representational value ranges and embeddings, and the quantitative-evaluation inputs, are not members of this canonical structural output family. Define the premise set for each layer's generation rules and the family of generative-dependency rules for canonical structures by

$$\begin{aligned} \text{Req}_0^{\text{repr}} &:= \{A \in \text{Expr} \mid A \in_{\text{comp}} \mathbb{I}\} \cup \{\text{Operative}(\mathcal{X})\}, \\ \text{Req}_k^{\text{repr}} &:= \text{Req}_0^{\text{repr}} \cup \{\text{Repr}_{\leq k}[\mathbb{I}]\} \cup \bigcup_{0 \leq j < k} \text{Out}_j^{\text{repr}}, \quad k \in \{1, 2, 3\}, \\ \text{Dep}^{\text{repr}} &:= \bigcup_{k=0}^3 \{(\text{Req}_k^{\text{repr}}, Y) \mid Y \in \text{Out}_k^{\text{repr}}\}, \\ \text{Out}_{\neg G}^{\text{repr}} &:= \bigcup_{k=0}^3 \text{Out}_k^{\text{repr}} \end{aligned}$$

and let the context in which global gluing fails be

$$\begin{aligned} \text{Ctx}_{\neg G}^{\text{repr}} &:= \{A \in \text{Expr} \mid A \in_{\text{comp}} \mathbb{I}\} \\ &\cup \{\text{L}_{\leq 7}[\text{Ev}^{\text{cmp}}], \neg G[\text{Ev}^{\text{cmp}}], \neg \text{Operative}(\mathcal{X})\} \end{aligned}$$

**Proposition 4.174** (Failure of global gluing prevents generation of canonical representation structures).

$$\begin{aligned} &\text{L}_{\leq 7}[\text{Ev}^{\text{cmp}}] \wedge \neg G[\text{Ev}^{\text{cmp}}] \\ &\implies \neg \text{Operative}(\mathcal{X}) \wedge \bigwedge_{Y \in \text{Out}_{\neg G}^{\text{repr}}} \neg \text{GenDef}_{\text{Dep}^{\text{repr}}}(Y \mid \text{Ctx}_{\neg G}^{\text{repr}}). \end{aligned}$$

*Proof.* The equivalence in Definition 4.86 yields  $\neg \text{Operative}(\mathcal{X})$  from the antecedent. Take the set

$$S_{\neg G} := \text{Expr} \setminus (\text{Out}_{\neg G}^{\text{repr}} \cup \{\text{Operative}(\mathcal{X})\})$$

By type,  $\text{Ctx}_{\neg G}^{\text{repr}} \subseteq S_{\neg G}$ . For each  $k \in \{0, 1, 2, 3\}$ ,

$$\begin{aligned} \text{Operative}(\mathcal{X}) &\in \text{Req}_k^{\text{repr}}, \\ \text{Operative}(\mathcal{X}) &\notin S_{\neg G} \end{aligned}$$

and hence  $\text{Req}_k^{\text{repr}} \not\subseteq S_{-G}$ . Thus,  $S_{-G}$  is closed under every rule in  $\text{Dep}^{\text{repr}}$ . By the minimality of the generation closure,

$$\text{Cl}_{\text{Dep}^{\text{repr}}}(\text{Ctx}_{-G}^{\text{repr}}) \subseteq S_{-G}.$$

Therefore, no  $Y \in \text{Out}_{-G}^{\text{repr}}$  belongs to the generation closure. The second conjunct of the proposition follows from the definition of generability.  $\square$

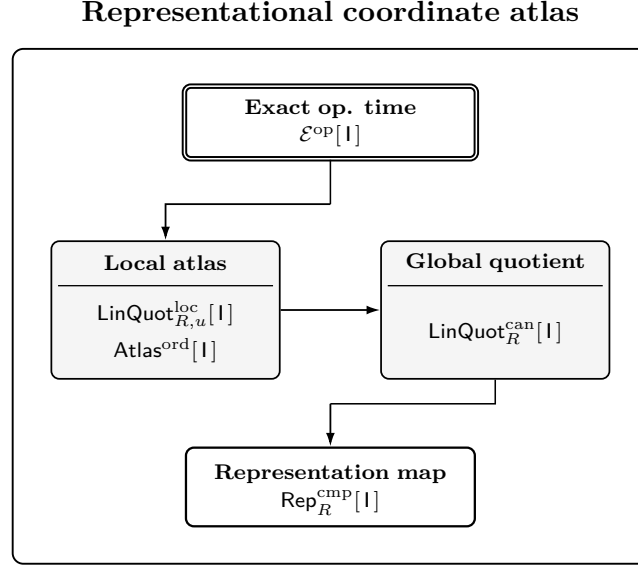


Figure 10: Generation order under  $\text{ReprCtx}_{\leq 3}[l]$ , from exact operational time structures through local order coordinates and the comparison-order atlas to global order-coordinate quotients. The final comparison representational coordinate is the composition of the order embedding into the selected representational value range with the canonical global order-coordinate projection.

### 4.3 Exact Operability, Canonical Uniqueness, and Universal Factorization

For each symbol root, one pair consisting of a role type and a reception phase first appears in complete form. Thereafter, only the reception-phase subscript is omitted within the same definition, proposition, or proof when the reception phase is fixed and the complete form is uniquely recoverable. The complete forms are retained for the role type, a varying family index, distinct reception phases, and residual modifiers that select genuine branches.

When the structural completion of the comparison stage  $\text{Operative}(\mathcal{X})$  and the structural completion of the representation stage  $\text{Repr}_{\leq 3}[l]$  from Definition 4.92 both hold, the dependency sequence from comparison-stage material to the global order-coordinate structure is generated. The selected  $\mathfrak{C}/\mathfrak{D}$ -derived representational value ranges and the  $\mathfrak{C}/\mathfrak{D}$ -derived representational-value-range order embeddings determine the  $\mathfrak{C}/\mathfrak{D}$ -derived comparison representational coordinates through composition with the canonical global order-coordinate projections. Specifying the finite operational granularity datum and the six componentwise evaluation inputs in addition to the  $\mathfrak{C}/\mathfrak{D}$ -derived conditional quantitative affine realizations determines the continuum-approximation evaluation-input datum.

**Definition 4.175** (Exact-operability object sequence and exact operability). For a realized operational configuration  $\mathcal{X}$  and a typed family of comparison rules  $\mathcal{J}_{\text{cmp}}$ , define exact operability by

$$\text{ExactOperable}[\mathbb{I}] \iff \text{ReprCtx}_{\leq 3}[\mathbb{I}]$$

When  $\text{ExactOperable}[\mathbb{I}]$  holds, define the exact-operability object sequence by

$$\text{EOSeq}[\mathbb{I}] := (\text{Dat}^{\text{mat}}[\mathbb{I}], \text{Qu}^{\text{can}}[\mathbb{I}], (\preceq_R^{\text{arr}})_{R \in \text{Rol}^{\text{cmp}}}, \text{ReprStr}_{\leq 3}[\mathbb{I}])$$

An object in the transitive closure of the component relation of a finite tuple  $T$  is called a recursive component of  $T$ . The ordering of the recursive components is determined by the dependencies among the comparison-stage material datum, the  $\mathfrak{C}\mathfrak{D}$ -derived canonical comparison quotient, the source-induced arrival preorders, the canonical  $\mathfrak{C}/\mathfrak{D}$  partial orders, the exact operational time structures, the local order-coordinate structures, the comparison-order atlas, and the global order-coordinate structures.

**Proposition 4.176** (Expansion of the exact-operability conditions). For a realized operational configuration  $\mathcal{X}$  and a typed family of comparison rules  $\mathcal{J}_{\text{cmp}}$ ,

$$\begin{aligned} &\text{ExactOperable}[\mathbb{I}] \\ &\iff \text{Operative}(\mathcal{X}) \\ &\quad \wedge K_{\text{CanOrdDesc}}^{\text{repr}}[\mathbb{I}] \\ &\quad \wedge K_{\text{LocIncTrans}}^{\text{repr}}[\mathbb{I}] \\ &\quad \wedge K_{\text{GlobIncTrans}}^{\text{repr}}[\mathbb{I}]. \end{aligned}$$

holds.

When every condition in this equivalence holds, every recursive component of

$$\text{EOSeq}[\mathbb{I}]$$

from Definition 4.175 is generated according to the dependencies among those components.

*Proof.* Definitions 4.175 and 4.92 give

$$\begin{aligned} &\text{ExactOperable}[\mathbb{I}] \\ &\iff \text{Operative}(\mathcal{X}) \\ &\quad \wedge \text{Repr}_{\leq 3}[\mathbb{I}]. \end{aligned}$$

Substituting the equivalence from Definition 4.92 into the preceding equivalence yields the equivalence in the proposition.

By Definition 4.2, the realized operational configuration and the typed family of comparison rules determine the comparison-stage material datum

$$\text{Dat}^{\text{mat}}[\mathbb{I}]$$

By Theorem 4.90, under  $\text{Operative}(\mathcal{X})$ , the  $\mathfrak{C}\mathfrak{D}$ -derived canonical comparison quotient

$$\text{Qu}^{\text{can}}[\mathbb{I}]$$

is generated.

By Theorem 4.147, under

$$\text{Operative}(\mathcal{X}), \quad \text{Repr}_{\leq 3}[\mathbb{I}]$$

the  $\mathfrak{C}/\mathfrak{D}$  source-induced arrival preorders, the canonical  $\mathfrak{C}/\mathfrak{D}$  partial orders, the exact operational time structures, the local order-coordinate structures, the comparison-order atlas, the global order-coordinate structures, and the local-to-global order-coordinate embeddings are generated in dependency order.

Therefore, every recursive component of Equation (4.176) is generated according to the dependencies among those components.  $\square$

Let a realized operational configuration  $\mathcal{X}$  from Definition 3.25 and a typed family of comparison rules  $\mathcal{J}_{\text{cmp}}$  be given. The structural completion of the comparison stage is expressed by  $\text{Operative}(\mathcal{X})$ .

The structural completion of the representation stage is expressed by  $\text{Repr}_{\leq 3}[\mathbb{I}]$  from Definition 4.92 and is equivalent to the simultaneous satisfaction of the three conditions in that definition.

Structural completion of the comparison and representation stages determines, in dependency order, the  $\mathfrak{C}\mathfrak{D}$ -derived canonical comparison quotient, the canonical  $\mathfrak{C}/\mathfrak{D}$  partial orders, the exact operational time structures, the local order-coordinate structures, the comparison-order atlas, and the global order-coordinate structures. This structural sequence is expressed by the exact-operability object sequence  $\text{EOSeq}[\mathbb{I}]$  from Definition 4.175.

Specifying the selected  $\mathfrak{C}/\mathfrak{D}$ -derived representational value ranges and the  $\mathfrak{C}/\mathfrak{D}$ -derived representational-value-range order embeddings determines the factorization of the  $\mathfrak{C}/\mathfrak{D}$ -derived comparison representational coordinates in Proposition 4.170.

Specifying the  $\mathfrak{C}/\mathfrak{D}$ -derived conditional quantitative affine realizations

$$(\text{QtyAffReal}_R[\mathbb{I}])_{R \in \text{Rol}^{\text{cmp}}}$$

together with the finite operational granularity datum of Definition 4.156 and the six componentwise evaluation inputs of Definition 4.164 determines the continuum-approximation evaluation-input datum of Definition 4.168,

$$\text{EvalInput}^{\text{cont}}[\mathbb{I}].$$

### 4.3.1 Operability Theorem

Comparison-stage operability is equivalent to the conjunction of the stage-seven satisfaction bundle and the identity-action condition for every pure transport loop.

**Theorem 4.177** (Operability theorem). For every realized operational configuration  $\mathcal{X}$  and typed family of comparison rules  $\mathcal{J}_{\text{cmp}}$ ,

$$\begin{aligned} & \text{Operative}(\mathcal{X}) \\ & \iff \text{L}_{\leq 7}[\text{Ev}^{\text{cmp}}] \wedge \text{G}[\text{Ev}^{\text{cmp}}]. \end{aligned}$$

*Proof.* By Definition 4.78,  $\text{L}_{\leq 7}[\text{Ev}^{\text{cmp}}]$  is the stage-seven satisfaction bundle.

By Definition 4.86, comparison-stage operability is equivalent to the conjunction of the stage-seven satisfaction bundle and the identity-action condition for every pure transport loop. The claim follows.  $\square$

**Proposition 4.178** (Exact operability implies comparison-stage operability). For every realized operational configuration  $\mathcal{X}$  and typed family of comparison rules  $\mathcal{J}_{\text{cmp}}$ ,

$$\text{ExactOperable}[\mathbb{I}] \implies \text{Operative}(\mathcal{X})$$

holds.

*Proof.* The first component on the right-hand side of the equivalence in Proposition 4.176 yields  $\text{Operative}(\mathcal{X})$ .  $\square$

### 4.3.2 Canonical Uniqueness and Universality

Under exact operability, the canonical objects of the comparison and representation stages are uniquely determined by the realized operational configuration and the typed family of comparison rules. Admissible comparison representations, admissible local order-atlas representations, and admissible global order representations factor, respectively, as compositions with the canonical projection, the local order-coordinate projections, and the canonical global order-coordinate projections.

**Theorem 4.179** (Canonical-structure uniqueness theorem). Assume  $\text{ExactOperable}[\mathbb{I}]$ . Then

$$(\text{Qu}^{\text{can}}[\mathbb{I}], \text{ReprStr}_{\leq 3}[\mathbb{I}])$$

is uniquely determined by  $\mathbb{I}$ .

*Proof.* By Proposition 4.176, comparison-stage operability, the canonical-order descent condition, the local incomparability transitivity condition, and the global incomparability transitivity condition that constitute the right-hand side of its equivalence all hold.

By Theorem 4.90, the extended comparison-reachability equivalence relation, the  $\mathfrak{C}/\mathfrak{D}$  comparison quotient set, the canonical projection, the quotient relation derived from  $\mathfrak{C}\mathfrak{D}$ , and the canonical comparison quotient derived from  $\mathfrak{C}\mathfrak{D}$  are uniquely determined in dependency order.

By the representative invariance in Definition 4.99, the right-hand side of Definition 4.101 is invariant under the choice of representative of a comparison equivalence class. Hence, the canonical  $\mathfrak{C}/\mathfrak{D}$  partial orders are uniquely determined as binary relations on the  $\mathfrak{C}/\mathfrak{D}$  comparison quotient sets. By Definition 4.103,

$$\mathcal{E}^{\text{op}}[\mathbb{I}] = (\text{Qu}^{\text{can}}[\mathbb{I}], (\preceq_R^{\text{can}}[\mathbb{I}])_{R \in \text{Rol}^{\text{cmp}}})$$

is uniquely determined.

Definition 4.111 determines the local incomparability relations on each local canonical quotient domain from the canonical strict  $\mathfrak{C}/\mathfrak{D}$  orders. By Lemma 4.114, the local incomparability relations are equivalence relations. Definition 4.116 uniquely determines the local order-coordinate quotients and the local strict linear orders. Definition 4.119 uniquely determines the local order-coordinate projections, and Definition 4.124 uniquely determines the local comparison charts as compositions of the canonical projection with the local order-coordinate projections.

By Lemma 4.127, the local order-coordinate chart-transition maps of Definition 4.126 are representative-invariant and uniquely determined. By Definition 4.128, the family of local order-coordinate projections, local comparison charts, and local order-coordinate chart-transition maps uniquely determines the comparison-order atlas.



Definition 4.130 determines the global incomparability relations from the canonical strict  $\mathfrak{C}/\mathfrak{D}$  orders. By Lemma 4.133, the global incomparability relations are equivalence relations. Definition 4.135 uniquely determines the  $\mathfrak{C}/\mathfrak{D}$  global order-coordinate quotients, the global strict linear orders, and the canonical global order-coordinate projections.

By Definition 4.142, each local-to-global order-coordinate embedding is uniquely determined as the map that sends a local incomparability class to the global incomparability class of the same representative.  $\square$

**Lemma 4.180** (Unique factorization through a surjection under equality-kernel inclusion).  
Let sets

$$S, \quad Q, \quad T$$

be given, together with a surjection

$$\pi : S \twoheadrightarrow Q$$

and a map

$$\Phi : S \longrightarrow T$$

If

$$\text{EqKer}(\pi) \subseteq \text{EqKer}(\Phi)$$

holds, there exists a unique map

$$\overline{\Phi} : Q \longrightarrow T$$

such that

$$\Phi = \overline{\Phi} \circ \pi$$

*Proof.* For each

$$x \in S$$

define

$$\overline{\Phi}(\pi(x)) := \Phi(x)$$

By the equality-kernel inclusion, any two elements with equal values under  $\pi$  have equal images under  $\Phi$ ; hence this map is well-defined. Surjectivity of  $\pi$  makes its domain equal to  $Q$ , and the defining equation gives the factorization equation. For each  $x \in S$ , the value of any map satisfying the factorization equation is fixed by  $\pi(x) \mapsto \Phi(x)$ . Since  $\pi$  is surjective, this map is unique.  $\square$

**Lemma 4.181** (Cancellation of precomposition by a surjection). If a surjection

$$\pi : S \twoheadrightarrow Q$$

and two maps

$$F, G : Q \longrightarrow T$$

satisfy

$$F \circ \pi = G \circ \pi$$

then

$$F = G$$

holds.

*Proof.* Take any  $q \in Q$ . By surjectivity of  $\pi$ , the equation  $\pi(x) = q$  holds for some  $x \in S$ . Evaluating the equality after precomposition at  $x$  gives  $F(q) = G(q)$ .  $\square$

**Lemma 4.182** (Common core of the order isomorphism between representational images).  
Let a set  $S$  and binary relations

$$\sim_S, \quad \prec_S \subseteq S \times S$$

be given. For each

$$a \in \{0, 1\}$$

let a linearly ordered set

$$\left( \text{RepVal}^{(a)}, <^{(a)} \right)$$

and a map

$$\text{Rep}^{(a)} : S \longrightarrow \text{RepVal}^{(a)}$$

be given. Suppose that the map satisfies

$$\text{OrdRepCond} \left( \text{Rep}^{(a)} ; \sim_S, \prec_S, <^{(a)} \right)$$

Then there exists a unique strict-order isomorphism

$$\text{Replso}^{01} : \text{Im} \left( \text{Rep}^{(0)} \right) \longrightarrow \text{Im} \left( \text{Rep}^{(1)} \right)$$

such that

$$\text{Rep}^{(1)} = \text{Replso}^{01} \circ \text{Rep}^{(0)}$$

*Proof.* Define

$$\text{Replso}^{01} \left( \text{Rep}^{(0)}(x) \right) := \text{Rep}^{(1)}(x)$$

By Definition 4.107 and the equality-kernel condition for  $a = 0$ , any two elements having equal values under  $\text{Rep}^{(0)}$  are related by  $\sim_S$ . Applying the equality-kernel condition for  $a = 1$ , their images under  $\text{Rep}^{(1)}$  are equal. Thus,  $\text{Replso}^{01}$  is well-defined as a map. Its definition with codomain  $\text{Im} \left( \text{Rep}^{(1)} \right)$  gives surjectivity.

Suppose that two values of  $\text{Replso}^{01}$  are equal. By definition, the corresponding values of  $\text{Rep}^{(1)}$  are equal. Applying the equality-kernel condition for  $a = 1$ , followed by that for  $a = 0$ , shows that the corresponding values of  $\text{Rep}^{(0)}$  are equal. Thus, injectivity holds.

For any

$$x_0, x_1 \in S$$

the two order-pullback conditions give

$$\begin{aligned} & \text{Rep}^{(0)}(x_0) <^{(0)} \text{Rep}^{(0)}(x_1) \\ & \iff x_0 \prec_S x_1 \\ & \iff \text{Rep}^{(1)}(x_0) <^{(1)} \text{Rep}^{(1)}(x_1) \end{aligned}$$

Therefore,  $\text{Replso}^{01}$  preserves and reflects the strict order.

For any

$$x \in S$$

the value of a map satisfying the commutative equation is fixed by

$$\text{Rep}^{(0)}(x) \longmapsto \text{Rep}^{(1)}(x)$$

Every element of the domain is in the image of  $\text{Rep}^{(0)}$ ; hence  $\text{Replso}^{01}$  is unique.  $\square$

**Theorem 4.183** (Uniqueness of the order type of representational images). Assume  $\text{ExactOperable}[\mathbb{I}]$  and fix

$$R \in \text{Rol}^{\text{cmp}}$$

For each

$$a \in \{0, 1\}$$

suppose that a map into the linearly ordered set

$$\left( \text{RepVal}_R^{(a)}, <_R^{(a)} \right)$$

is given by

$$\text{Rep}_R^{(a)} : \text{Qu}_R^{\text{can}}[\mathbb{I}] \longrightarrow \text{RepVal}_R^{(a)}$$

and satisfies

$$\text{OrdRepCond} \left( \text{Rep}_R^{(a)} ; \asymp_R[\mathbb{I}], \prec_R^{\text{can}}[\mathbb{I}], <_R^{(a)} \right)$$

Then there exists a unique strict-order isomorphism

$$\text{Replso}_R^{\text{ord},01} : \text{Im} \left( \text{Rep}_R^{(0)} \right) \longrightarrow \text{Im} \left( \text{Rep}_R^{(1)} \right)$$

such that

$$\text{Rep}_R^{(1)} = \text{Replso}_R^{\text{ord},01} \circ \text{Rep}_R^{(0)}$$

holds.

Further, fix

$$u \in U$$

For each

$$a \in \{0, 1\}$$

suppose that a map into the linearly ordered set

$$\left( \text{LocRepVal}_{R,u}^{(a)}, <_{R,u}^{\text{loc},(a)} \right)$$

is given by

$$\text{LocRep}_{R,u}^{(a)} : \mathcal{U}_{R,u}[\mathbb{I}] \longrightarrow \text{LocRepVal}_{R,u}^{(a)}$$

and satisfies

$$\text{OrdRepCond} \left( \text{LocRep}_{R,u}^{(a)} ; \asymp_{R,u}[\mathbb{I}], \prec_R^{\text{can}}[\mathbb{I}]|_{\mathcal{U}_{R,u}[\mathbb{I}]}, <_{R,u}^{\text{loc},(a)} \right)$$

Then there exists a unique strict-order isomorphism

$$\text{Replso}_{R,u}^{\text{loc},01} : \text{Im} \left( \text{LocRep}_{R,u}^{(0)} \right) \longrightarrow \text{Im} \left( \text{LocRep}_{R,u}^{(1)} \right)$$

such that

$$\text{LocRep}_{R,u}^{(1)} = \text{Replso}_{R,u}^{\text{loc},01} \circ \text{LocRep}_{R,u}^{(0)}$$

holds.

*Proof.* For the global representations, apply Lemma 4.182 to

$$\begin{aligned} & \left( S, \sim_S, \prec_S; \text{Rep}^{(a)}, \text{RepVal}^{(a)}, <^{(a)} \right) \\ &= \left( \text{Qu}_R^{\text{can}}[\mathbb{I}], \asymp_R[\mathbb{I}], \prec_R^{\text{can}}[\mathbb{I}]; \text{Rep}_R^{(a)}, \text{RepVal}_R^{(a)}, <_R^{(a)} \right) \end{aligned}$$

The assumed equality criterion and strict-order criterion yield the unique strict-order isomorphism  $\text{Replso}_R^{\text{ord},01}$  and the global commutative equation.

For the local representations, apply the same lemma to

$$\begin{aligned} & \left( S, \sim_S, \prec_S; \text{Rep}^{(a)}, \text{RepVal}^{(a)}, <^{(a)} \right) \\ &= \left( \mathcal{U}_{R,u}[\mathbb{I}], \asymp_{R,u}[\mathbb{I}], \prec_R^{\text{can}}[\mathbb{I}]|_{\mathcal{U}_{R,u}[\mathbb{I}]}; \text{LocRep}_R^{(a)}, \text{LocRepVal}_R^{(a)}, <_R^{\text{loc},(a)} \right) \end{aligned}$$

The assumed local equality criterion and local strict-order criterion yield the unique strict-order isomorphism  $\text{Replso}_R^{\text{loc},01}$  and the local commutative equation.  $\square$

**Definition 4.184** (Admissible comparison representation). Let a realized operational configuration  $\mathcal{X}$  and a typed family of comparison rules  $\mathcal{J}_{\text{cmp}}$  be given. Assume  $\mathbf{L}_{\leq 2}[\text{Ev}^{\text{cmp}}]$ . Let the family of comparison-representation target sets be

$$(\text{CmpTgt}_R)_{R \in \text{Rol}^{\text{cmp}}}$$

and let a relation on their product be

$$\text{Rel}^{\text{tgt}} \subseteq \prod_{R \in \text{Rol}^{\text{cmp}}} \text{CmpTgt}_R$$

For each  $R \in \text{Rol}^{\text{cmp}}$ , take a map

$$\Phi_R^{\text{cmp}} : \mathcal{R}_R^{\text{arr}} \longrightarrow \text{CmpTgt}_R$$

First, suppose that for every  $R \in \text{Rol}^{\text{cmp}}$ ,

$$\text{EqKer}(\text{Prj}_R^{\text{can}}[\mathbb{I}]) \subseteq \text{EqKer}(\Phi_R^{\text{cmp}})$$

holds.

Second, suppose that for every

$$\begin{aligned} (u, \alpha) &\in \mathcal{R}^{\text{pair}}, \\ \alpha &= (\alpha_R)_{R \in \text{Rol}^{\text{cmp}}} \end{aligned}$$

the equivalence

$$(\Phi_R^{\text{cmp}}(u, \alpha_R))_{R \in \text{Rol}^{\text{cmp}}} \in \text{Rel}^{\text{tgt}} \iff \alpha \in \text{Rel}_u^{\text{cmp}}$$

holds.

A pair

$$((\Phi_R^{\text{cmp}})_{R \in \text{Rol}^{\text{cmp}}}, \text{Rel}^{\text{tgt}})$$

satisfying these two conditions is called an admissible comparison representation.

**Definition 4.185** (Admissible global order representation). Let a realized operational configuration  $\mathcal{X}$  and a typed family of comparison rules  $\mathcal{J}_{\text{cmp}}$  be given. Assume  $\text{ReprCtx}_{\leq 1}[\mathbb{I}]$ . For each

$$R \in \text{Rol}^{\text{cmp}}$$

take a map into the linearly ordered set

$$\left( \text{OrdTgt}_R, <_R^{\text{ord,tgt}} \right)$$

given by

$$\Phi_R^{\text{ord}} : \text{Qu}_R^{\text{can}}[\mathbb{I}] \longrightarrow \text{OrdTgt}_R$$

When

$$\text{OrdRepCond} \left( \Phi_R^{\text{ord}} ; \asymp_R [\mathbb{I}], \prec_R^{\text{can}} [\mathbb{I}], <_R^{\text{ord,tgt}} \right)$$

holds,  $\Phi_R^{\text{ord}}$  is called an admissible global order representation.

**Definition 4.186** (Admissible local order-atlas representation). Let a realized operational configuration  $\mathcal{X}$  and a typed family of comparison rules  $\mathcal{J}_{\text{cmp}}$  be given. Assume  $\text{ReprCtx}_{\leq 1}[\mathbb{I}]$ . For each

$$R \in \text{Rol}^{\text{cmp}}, \quad u \in U$$

take a map into the linearly ordered set

$$\left( \text{LocOrdTgt}_{R,u}, <_{R,u}^{\text{loc,tgt}} \right)$$

given by

$$\Phi_{R,u}^{\text{loc}} : \mathcal{U}_{R,u}[\mathbb{I}] \longrightarrow \text{LocOrdTgt}_R$$

Suppose that

$$\text{OrdRepCond} \left( \Phi_R^{\text{loc}} ; \asymp_{R,u} [\mathbb{I}], \prec_R^{\text{can}} [\mathbb{I}]|_{\mathcal{U}_{R,u}[\mathbb{I}]}, <_R^{\text{loc,tgt}} \right)$$

holds.

For each

$$u, v \in U$$

let the strict-order isomorphism between the overlap images be

$$\begin{aligned} \text{TgtChg}_{R,uv}^{\text{loc}} : \Phi_{R,u}^{\text{loc}} (\mathcal{U}_{R,uv}^{\cap}[\mathbb{I}]) \\ \longrightarrow \Phi_{R,v}^{\text{loc}} (\mathcal{U}_{R,uv}^{\cap}[\mathbb{I}]) \end{aligned}$$

Suppose that, for every

$$\text{CmpCls}_{R,0} \in \mathcal{U}_{R,uv}^{\cap}[\mathbb{I}]$$

the equality

$$\begin{aligned} \text{TgtChg}_{R,uv}^{\text{loc}} (\Phi_{R,u}^{\text{loc}} (\text{CmpCls}_{R,0})) \\ = \Phi_{R,v}^{\text{loc}} (\text{CmpCls}_{R,0}) \end{aligned}$$

holds.

Further, suppose that for every

$$u, v, w \in U$$

the equality

$$\mathsf{TgtChg}_{R,uu}^{\text{loc}} = \text{id}$$

holds on

$$\Phi_{R,u}^{\text{loc}}(\mathcal{U}_{R,u}[\mathbb{I}])$$

and that

$$\mathsf{TgtChg}_{R,vw}^{\text{loc}} \circ \mathsf{TgtChg}_{R,uv}^{\text{loc}} = \mathsf{TgtChg}_{R,uw}^{\text{loc}}$$

holds on

$$\Phi_{R,u}^{\text{loc}}(\mathcal{U}_{R,uvw}^{\cap}[\mathbb{I}])$$

The family of maps

$$\left( (\Phi_{R,u}^{\text{loc}})_{(R,u) \in \text{Id}\mathbf{x}_1^{\text{cmp}}}, (\mathsf{TgtChg}_{R,uv}^{\text{loc}})_{(R,u,v) \in \text{Id}\mathbf{x}_2^{\text{cmp}}} \right)$$

is called an admissible local order-atlas representation.

**Theorem 4.187** (Three-layer universal factorization theorem). Assume  $\text{ExactOperable}[\mathbb{I}]$ .

First, for an admissible comparison representation

$$((\Phi_R^{\text{cmp}})_{R \in \text{Rol}^{\text{cmp}}}, \text{Rel}^{\text{tgt}})$$

from Definition 4.184, there exists, for each  $R \in \text{Rol}^{\text{cmp}}$ , a unique map

$$\overline{\Phi}_R^{\text{cmp}} : \text{Qu}_R^{\text{can}}[\mathbb{I}] \longrightarrow \text{CmpTgt}_R$$

such that

$$\Phi_R^{\text{cmp}} = \overline{\Phi}_R^{\text{cmp}} \circ \text{Prj}_R^{\text{can}}[\mathbb{I}]$$

holds.

Moreover, for every

$$(u, \alpha) \in \mathcal{R}^{\text{pair}}, \\ \alpha = (\alpha_R)_{R \in \text{Rol}^{\text{cmp}}}$$

the equivalence

$$\left( \overline{\Phi}_R^{\text{cmp}}(\text{Prj}_R^{\text{can}}[\mathbb{I}](u, \alpha_R)) \right)_{R \in \text{Rol}^{\text{cmp}}} \in \text{Rel}^{\text{tgt}} \\ \iff \text{Prj}_{\mathfrak{E}\mathfrak{D}}^{\text{can,pair}}[\mathbb{I}](u, \alpha) \in \text{Rel}^{\text{can}}[\mathbb{I}]$$

holds.

Second, for an admissible local order-atlas representation from Definition 4.186, there exists, for each

$$R \in \text{Rol}^{\text{cmp}}, \quad u \in U$$

a unique strict-order isomorphism

$$\overline{\Phi}_{R,u}^{\text{loc}} : \text{Qu}_R^{\text{loc}}[\mathbb{I}] \longrightarrow \text{Im}(\Phi_R^{\text{loc}})$$

such that

$$\Phi_R^{\text{loc}} = \overline{\Phi}_R^{\text{loc}} \circ \text{Prj}_R^{\text{loc}}[\mathbb{I}]$$

holds.

For every

$$u, v \in U$$

the following equality holds on the overlap image:

$$\begin{aligned} & \overline{\Phi}_{R,v}^{\text{loc}} \circ \text{ChartChg}_{R,uv}^{\text{loc}} [I] \\ &= \text{TgtChg}_{R,uv}^{\text{loc}} \circ \overline{\Phi}_{R,u}^{\text{loc}} \end{aligned}$$

Third, for an admissible global order representation from Definition 4.185, there exists, for each

$$R \in \text{Rol}^{\text{cmp}}$$

a unique strict-order isomorphism

$$\overline{\Phi}_R^{\text{ord}} : \text{Qu}_R^{\text{ord},\text{can}} [I] \longrightarrow \text{Im} (\Phi_R^{\text{ord}})$$

such that

$$\Phi_R^{\text{ord}} = \overline{\Phi}_R^{\text{ord}} \circ \text{Prj}_R^{\text{ord},\text{can}} [I]$$

holds.

*Proof.* Take an admissible comparison representation. For each  $R \in \text{Rol}^{\text{cmp}}$ , apply Lemma 4.180 to

$$\begin{aligned} & (S, Q, T; \pi, \Phi) \\ &= (\mathcal{R}_R^{\text{arr}}, \text{Qu}_R^{\text{can}} [I], \text{CmpTgt}_R; \text{Prj}_R^{\text{can}} [I], \Phi_R^{\text{cmp}}) \end{aligned}$$

The surjectivity of the canonical projection from Definition 4.70 and the equality-kernel inclusion condition in Definition 4.184 yield the unique comparison factor map and the comparison factorization equation.

Take any  $(u, \alpha)$  quantified in the first claim. Substituting the comparison factorization equation into the relation-preservation condition of the admissible comparison representation makes the left-hand side of the theorem equivalent to  $\alpha \in \text{Rel}_u^{\text{cmp}}$ . By the local pullback agreement in Lemma 4.84, this is equivalent to the right-hand side of the theorem.

Take an admissible local order-atlas representation. Apply Lemma 4.182 to

$$\begin{aligned} & \left( S, \sim_S, \prec_S; \left( \text{Rep}^{(a)}, \text{RepVal}^{(a)}, <^{(a)} \right)_{a \in \{0,1\}} \right) \\ &= \left( \mathcal{U}_{R,u} [I], \asymp_{R,u} [I], \prec_R^{\text{can}} [I]|_{\mathcal{U}_{R,u} [I]}; \left( \text{Prj}_R^{\text{loc}} [I], \text{Qu}_R^{\text{loc}} [I], <_R^{\text{loc}} [I] \right), \left( \Phi_R^{\text{loc}}, \text{LocOrdTgt}_R, <_R^{\text{loc},\text{tgt}} \right) \right) \end{aligned}$$

The class criterion, order criterion, and surjectivity in Lemma 4.120, together with the order-representation criterion in Definition 4.186, yield the unique local strict-order isomorphism and factorization equation stated in the theorem.

Apply Lemma 4.181 to the local order-coordinate projection restricted to  $\mathcal{U}_{R,uv}^\cap [I]$ , and to the composite maps on the left- and right-hand sides of the local chart-transition commutative equation in the theorem. The restricted projection is surjective onto its image. By the local chart-transition equation in Definition 4.126, the two local factorization equations, and the overlap-compatibility equation in Definition 4.186, the two composite maps agree after precomposition with the same projection. Therefore, the local chart-transition commutative equation holds.

Take an admissible global order representation. Apply Lemma 4.182 to

$$\begin{aligned} & \left( S, \sim_S, \prec_S; \left( \text{Rep}^{(a)}, \text{RepVal}^{(a)}, <^{(a)} \right)_{a \in \{0,1\}} \right) \\ &= \left( \text{Qu}_R^{\text{can}} [I], \asymp_R [I], \prec_R^{\text{can}} [I]; \left( \text{Prj}_R^{\text{ord},\text{can}} [I], \text{Qu}_R^{\text{ord},\text{can}} [I], <_R^{\text{ord},\text{can}} [I] \right), \left( \Phi_R^{\text{ord}}, \text{OrdTgt}_R, <_R^{\text{ord},\text{tgt}} \right) \right) \end{aligned}$$

The class criterion, order criterion, and surjectivity in Lemma 4.136, together with the order-representation criterion in Definition 4.185, yield the unique global strict-order isomorphism and factorization equation stated in the theorem.  $\square$

**Corollary 4.188** (Uniqueness of reparameterization on representational images). Assume `ExactOperable` [1] and fix

$$R \in \mathbf{Rol}^{\text{cmp}}$$

For each

$$a \in \{0, 1\}$$

let

$$\Phi_R^{\text{ord},(a)} : \mathbf{Qu}_R^{\text{can}}[\mathbf{I}] \longrightarrow \mathbf{OrdTgt}_R^{(a)}$$

be an admissible global order representation. Then there exists a unique strict-order isomorphism

$$\mathbf{Replso}_R^{\text{ord},01} : \mathbf{Im} \left( \Phi_R^{\text{ord},(0)} \right) \longrightarrow \mathbf{Im} \left( \Phi_R^{\text{ord},(1)} \right)$$

such that

$$\Phi_R^{\text{ord},(1)} = \mathbf{Replso}_R^{\text{ord},01} \circ \Phi_R^{\text{ord},(0)}$$

holds.

Further, for each

$$a \in \{0, 1\}$$

let

$$\left( \left( \Phi_{R,u}^{\text{loc},(a)} \right)_{u \in U}, \left( \mathbf{TgtChg}_{R,uv}^{\text{loc},(a)} \right)_{(u,v) \in U \times U} \right)$$

be an admissible local order-atlas representation. Then, for each

$$u \in U$$

there exists a unique strict-order isomorphism

$$\mathbf{Replso}_R^{\text{loc},01} : \mathbf{Im} \left( \Phi_R^{\text{loc},(0)} \right) \longrightarrow \mathbf{Im} \left( \Phi_R^{\text{loc},(1)} \right)$$

such that

$$\Phi_R^{\text{loc},(1)} = \mathbf{Replso}_R^{\text{loc},01} \circ \Phi_R^{\text{loc},(0)}$$

holds.

For every

$$u, v \in U$$

the following equality holds on the overlap image:

$$\begin{aligned} & \mathbf{Replso}_{R,v}^{\text{loc},01} \circ \mathbf{TgtChg}_{R,uv}^{\text{loc},(0)} \\ &= \mathbf{TgtChg}_{R,uv}^{\text{loc},(1)} \circ \mathbf{Replso}_{R,u}^{\text{loc},01} \end{aligned}$$

*Proof.* For the global representations, apply Theorem 4.183 to

$$\mathbf{Rep}_R^{(a)} = \Phi_R^{\text{ord},(a)}$$

The equality criterion and strict-order criterion of an admissible global order representation satisfy the assumptions of that theorem. Hence, a unique strict-order isomorphism  $\mathbf{Replso}_R^{\text{ord},01}$  exists and the global commutative equation holds.



For the local representations, at each

$$u \in U$$

apply the same theorem to

$$\text{LocRep}_R^{(a)} = \Phi_R^{\text{loc},(a)}$$

The equality criterion and strict-order criterion of an admissible local order-atlas representation satisfy the local assumptions of that theorem. Hence, a unique strict-order isomorphism  $\text{Replso}_R^{\text{loc},01}$  exists and the local commutative equation holds.

For commutativity with the chart-transition maps, apply Lemma 4.181 to the restriction to  $\mathcal{U}_{R,uv}^\cap[\mathbb{I}]$  of  $\Phi_{R,u}^{\text{loc},(0)}$ , and to the composite maps on the left- and right-hand sides of the chart-transition commutative equation in the corollary. The restricted map is surjective onto its image. By the overlap-compatibility equations of the two admissible local order-atlas representations and the two local commutative equations, the two composite maps agree after precomposition with the same map. Therefore, the chart-transition commutative equation holds. Uniqueness of the strict-order isomorphism at each phase also makes the reparameterization family satisfying this commutative equation unique.  $\square$

**Corollary 4.189** (Comparison-stage conditions imply operability). If

$$\mathbf{L}_{\leq 7} [\mathbf{Ev}^{\text{cmp}}] \wedge \mathbf{G} [\mathbf{Ev}^{\text{cmp}}]$$

holds, then  $\text{Operative}(\mathcal{X})$  holds.

*Proof.* The conclusion follows from the  $\Leftarrow$  direction of the equivalence in Theorem 4.177.  $\square$

**Corollary 4.190** (Dependence of the exact-operability object sequence on generative inputs). When  $\text{ExactOperable}[\mathbb{I}]$  holds, every recursive component of  $\text{EOSeq}[\mathbb{I}]$  is generated in dependency order from

$$\mathcal{I}^{\text{raw}}, \quad \text{Dat}^{\text{src}}, \quad \mathcal{J}_{\text{cmp}}$$

*Proof.* By Definition 3.25,

$$\mathcal{X} = (\mathcal{I}^{\text{raw}}, \text{Dat}^{\text{src}}).$$

By Definition 4.175, satisfaction of  $\text{ExactOperable}[\mathbb{I}]$  gives satisfaction of  $\text{Operative}(\mathcal{X})$  and  $\text{Repr}_{\leq 3}[\mathbb{I}]$ , and types every recursive component of  $\text{EOSeq}[\mathbb{I}]$ .

The comparison-stage material datum

$$\text{Dat}^{\text{mat}}[\mathbb{I}]$$

is determined from

$$\mathcal{X}, \quad \mathcal{J}_{\text{cmp}}$$

The canonical comparison quotient derived from  $\mathfrak{C}\mathfrak{D}$ ,

$$\text{Qu}^{\text{can}}[\mathbb{I}]$$

is determined from the comparison-stage material datum, the comparison-stage seven-condition satisfaction bundle, and the global gluing condition.

Definitions 4.95 through 4.103 determine, in dependency order, the source-induced arrival preorders, the canonical  $\mathfrak{C}/\mathfrak{D}$  partial orders, and the exact operational time structures.

Definitions 4.104 through 4.128 determine, in dependency order, the local order-coordinate structures, the local order-coordinate chart-transition maps, and the comparison-order atlas.

Definitions 4.130 through 4.142 determine, in dependency order, the global order-coordinate structures and the local-to-global order-coordinate embeddings.

Each generative input to the comparison-stage material datum, the canonical comparison quotient derived from  $\mathfrak{C}\mathfrak{D}$ , the source-induced arrival preorders, the canonical  $\mathfrak{C}/\mathfrak{D}$  partial orders, the exact operational time structures, the local order-coordinate structures, the comparison-order atlas, the global order-coordinate structures, and the local-to-global order-coordinate embeddings is either

$$\mathcal{X}, \quad \mathcal{J}_{\text{cmp}}$$

or a previously generated recursive component. Substituting the construction equation for  $\mathcal{X}$  from Definition 3.25 shows that every recursive component of  $\text{EOSeq}[\mathbb{I}]$  is generated in dependency order from the three inputs in Equation (4.190).  $\square$

**Corollary 4.191** (Factorization of a conventional time parameter through the canonical global order coordinate). Assume **ExactOperable**  $[\mathbb{I}]$  and fix

$$R \in \text{Rol}^{\text{cmp}}$$

Suppose that a map into the linearly ordered set

$$(\text{TimeVal}_R, <_R^{\text{time}})$$

given by

$$t_R : \text{Qu}_R^{\text{can}}[\mathbb{I}] \longrightarrow \text{TimeVal}_R$$

satisfies

$$\text{OrdRepCond} (t_R ; \asymp_R [\mathbb{I}], \prec_R^{\text{can}} [\mathbb{I}], <_R^{\text{time}})$$

Then there exists a unique strict-order isomorphism

$$\bar{t}_R : \text{Qu}_R^{\text{ord,can}}[\mathbb{I}] \longrightarrow \text{Im} (t_R)$$

such that

$$t_R = \bar{t}_R \circ \text{Prj}_R^{\text{ord,can}}[\mathbb{I}]$$

holds.

*Proof.* By Definition 4.108, the order-representation criterion for  $t_R$  is the condition for an admissible global order representation in Definition 4.185.

Apply the global order factorization in Theorem 4.187 to

$$(\Phi_R^{\text{ord}}, \text{OrdTgt}_R, <_R^{\text{ord,tgt}}) = (t_R, \text{TimeVal}_R, <_R^{\text{time}})$$

This yields the unique strict-order isomorphism and factorization equation stated in the corollary.

Therefore, a conventional time parameter satisfying the equality criterion and the strict-order criterion is determined as the composition of the canonical global order-coordinate projection with a strict-order isomorphism into a linearly ordered image.  $\square$

## 5 Generation of Observer Time at the Dynamical-Description Stage

Take as input the exact operability

$$\text{ExactOperable}[\mathbb{I}]$$

of Definition 4.175. For an observer-side information-coupling position and an object-specification index, the dynamical-description stage specifies the conditions that generate the  $\mathfrak{C}$ -derived observer-time quotient, the  $\mathfrak{B}$ -derived object-state quotient, the  $\mathfrak{C}\mathfrak{D}\mathfrak{B}$ -derived canonical trajectory relation, graph trajectories, and the family of canonical observable functions. In dependency order, these conditions provide the construction conditions for dynamical-description operability, law-description operability, differential-stage operability, and multi-object relations.

### 5.1 Generation of Dynamical Descriptions from the Canonical Comparison Quotient

**Definition 5.1** (Dynamical-description quotient-role family). Within the previously defined source-role family  $\text{Rol}^{\text{src}}$ , define the family of roles that carry quotients at the dynamical-description stage by

$$\text{Rol}^{\text{dyn,quot}} := \{\mathfrak{C}, \mathfrak{B}\}$$

Throughout this chapter, the compound superscript  $(\text{dyn}, \sigma)$  on an object specific to the dynamical-description stage is abbreviated to  $\sigma$  only when the symbol root and  $\sigma$  uniquely determine that the object belongs to the dynamical-description stage; a lone  $(\text{dyn})$  is omitted. To distinguish it from the source-side relation  $\text{Rel}^{\text{src}}$ , the complete form  $\text{Rel}^{\text{dyn,src}}$  is retained.

**Definition 5.2** (Complete evaluation specification). For the comparison-rule-dependent common input  $\mathbb{I}$  of Definition 4.1, an observer-side information-coupling position  $o \in U$ , and an object-specification index  $\zeta \in Z_0$  from Definition 2.10, define the complete-evaluation-specification set by

$$\text{EvSpec} := \{\mathbb{I}\} \times U \times Z_0$$

Define a complete evaluation specification by

$$\text{Ev} := (\mathbb{I}, o, \zeta) \in \text{EvSpec}$$

Define the exactly operable complete-evaluation-specification set by

$$\text{EvSpec}^{\text{exact}} := \{(\mathbb{I}, o, \zeta) \in \text{EvSpec} \mid \text{ExactOperable}[\mathbb{I}]\}$$

**Definition 5.3** (Common evaluation rule for dynamical-description conditions). The eight dynamical-description conditions below use the following common rule. For each condition, consider the subset of  $\text{EvSpec}^{\text{exact}}$  on which all predecessor conditions explicitly assumed in its definition hold. Denote this subset by  $D$ , and let  $P$  be the test stated in that definition on this domain. Define the condition  $K$ , for every  $\text{Ev} \in \text{EvSpec}$ , by

$$K(\text{Ev}) \iff \exists e \in D, \quad \text{Ev} = e \wedge P(e).$$

The test formula in each individual definition is its expression on  $D$ ; outside  $D$ ,  $K$  is false. This rule makes the conditions total without extending the domains on which their quotients or maps are constructed.

**Proposition 5.4** (Foundational structure for dynamical description). For a complete evaluation specification  $\mathbf{Ev} \in \mathbf{EvSpec}^{\text{exact}}$ , the family of source-token sets

$$(\mathbf{Tok}_R^*)_{R \in \mathbf{Rol}^{\text{src}}}$$

is determined, and the exact-operability object sequence

$$\mathbf{EOSeq}[\mathbf{I}]$$

is generated.

Consequently, the canonical comparison quotient derived from  $\mathfrak{CD}$ ,  $\mathbf{Qu}^{\text{cmp}, \text{can}}[\mathbf{I}]$ , the exact operational time structure  $\mathcal{E}^{\text{op}}[\mathbf{I}]$ , the local order-coordinate structures

$$(\mathbf{Qu}_{R,u}^{\text{loc}}[\mathbf{I}], <_{R,u}^{\text{loc}}[\mathbf{I}], \mathbf{Prj}_{R,u}^{\text{loc}}[\mathbf{I}])_{(R,u) \in \mathbf{Rol}^{\text{cmp}} \times U}$$

and the global order-coordinate structures

$$(\mathbf{Qu}_R^{\text{ord}, \text{can}}[\mathbf{I}], <_R^{\text{ord}, \text{can}}[\mathbf{I}], \mathbf{Prj}_R^{\text{ord}, \text{can}}[\mathbf{I}])_{R \in \mathbf{Rol}^{\text{cmp}}}$$

are determined.

*Proof.* By Definition 3.25,  $\mathcal{X} = (\mathcal{I}^{\text{raw}}, \mathbf{Dat}^{\text{src}})$ . By Definition 3.24,  $(\mathbf{Tok}_R^*)_{R \in \mathbf{Rol}^{\text{src}}}$  is a component of  $\mathbf{Dat}^{\text{src}}$ .

By Definition 4.175, under  $\mathbf{ExactOperable}[\mathbf{I}]$ ,  $\mathbf{EOSeq}[\mathbf{I}]$  is generated. The structures listed in the proposition are recursive components of  $\mathbf{EOSeq}[\mathbf{I}]$ .  $\square$

**Definition 5.5** (Direct arrival maps at the dynamical-description stage). For a complete evaluation specification  $\mathbf{Ev} = (\mathbf{I}, o, \zeta)$ , define the direct arrival map at the dynamical-description stage for each  $R \in \mathbf{Rol}^{\text{src}}$  by

$$\mathbf{Arr}_{R,o} : \text{Dom}(\mathbf{Arr}_{R,o}) \subseteq \mathbf{Tok}_R^* \longrightarrow \mathbf{Rcv}_{R,o}$$

Define the corresponding arrival set by

$$\mathcal{R}_{R,o}^{\text{arr}} := \text{Im}(\mathbf{Arr}_{R,o}) \quad (R \in \mathbf{Rol}^{\text{src}})$$

**Definition 5.6** (Source-role relation at the dynamical-description stage). For a complete evaluation specification  $\mathbf{Ev} = (\mathbf{I}, o, \zeta)$ , define the source-role relation at the dynamical-description stage by

$$\mathbf{Rel}_o \subseteq \prod_{R \in \mathbf{Rol}^{\text{src}}} \mathcal{R}_{R,o}^{\text{arr}}$$

**Definition 5.7** (Dynamical-description source-role relation). Define the dynamical-description source-role relation by

$$\mathbf{Rel}^{\text{dyn}, \text{src}} \subseteq \prod_{R \in \mathbf{Rol}^{\text{src}}} \mathbf{Tok}_R^*$$

**Definition 5.8** (Exact four-party descent). Define exact four-party descent at an observer-side information-coupling position  $o \in U$  by satisfaction of

$$\mathbf{Rel}_o = \left\{ \left( \mathbf{Arr}_{R,o}(x_R) \right)_{R \in \mathbf{Rol}^{\text{src}}} \mid (x_R)_{R \in \mathbf{Rol}^{\text{src}}} \in \mathbf{Rel}^{\text{dyn}, \text{src}}, \right. \\ \left. \forall R \in \mathbf{Rol}^{\text{src}}, \quad x_R \in \text{Dom}(\mathbf{Arr}_{R,o}) \right\}$$

**Definition 5.9** (Four-party reception structure at the dynamical-description stage). Define the four-party reception structure at the dynamical-description stage at an observer-side information-coupling position  $o \in U$  by

$$\text{RcvStr}_o := \left( (\mathcal{R}_{R,o}^{\text{arr}})_{R \in \text{Rol}^{\text{src}}}, (\text{Arr}_{R,o})_{R \in \text{Rol}^{\text{src}}}, \text{Rel}_o, \text{Rel}^{\text{dyn,src}} \right)$$

**Definition 5.10** (Recovery maps at the dynamical-description stage). For an observer-side information-coupling position  $o \in U$  and each  $R \in \text{Rol}^{\text{src}}$ , define the recovery map at the dynamical-description stage by

$$\text{Rec}_{R,o} : \mathcal{R}_{R,o}^{\text{arr}} \longrightarrow \text{Tok}_R^*$$

For every  $a_R \in \mathcal{R}_{R,o}^{\text{arr}}$ ,

$$\begin{aligned} \text{Rec}_{R,o}(a_R) &\in \text{Dom}(\text{Arr}_{R,o}), \\ \text{Arr}_{R,o}(\text{Rec}_{R,o}(a_R)) &= a_R \end{aligned}$$

holds.

**Definition 5.11** (Four-party binding relation at the dynamical-description stage). Define the four-party binding relation at the dynamical-description stage at an observer-side information-coupling position  $o \in U$  by

$$\text{Bind}_o \subseteq (\text{Rel}_o)^2$$

**Definition 5.12** (Source-side four-party binding relation). For an observer-side information-coupling position  $o \in U$ , define the source-side four-party binding relation

$$\text{Bind}_o^{\text{src}} \subseteq (\text{Rel}^{\text{dyn,src}})^2$$

For every  $\mathbf{x}_i \in \text{Rel}^{\text{dyn,src}}$ , ( $i \in \{0, 1\}$ ),

$$\begin{aligned} &\mathbf{x}_0 \text{ Bind}_o^{\text{src}} \mathbf{x}_1 \\ &\iff \left[ \exists \mathbf{a}_0, \mathbf{a}_1 \in \text{Rel}_o, \quad (\mathbf{a}_0, \mathbf{a}_1) \in \text{Bind}_o, \quad \forall (i, R) \in \{0, 1\} \times \text{Rol}^{\text{src}}, \right. \\ &\quad \left. \text{Rec}_{R,o}((\mathbf{a}_i)_R) = (\mathbf{x}_i)_R \right]. \end{aligned}$$

**Definition 5.13** (Generated relations and least equivalence closures). For each  $R \in \text{Rol}^{\text{quot}}$ , define the generated relation

$$\rightsquigarrow_{R,o} \subseteq (\text{Tok}_R^*)^2$$

by, for every  $x_{R,0}, x_{R,1} \in \text{Tok}_R^*$ ,

$$\begin{aligned} &x_{R,0} \rightsquigarrow_{R,o} x_{R,1} \\ &\iff \left[ \exists \mathbf{x}_0, \mathbf{x}_1 \in \text{Rel}^{\text{dyn,src}}, \quad \forall i \in \{0, 1\}, \quad (\mathbf{x}_i)_R = x_{R,i}, \quad \mathbf{x}_0 \text{ Bind}_o^{\text{src}} \mathbf{x}_1 \right] \end{aligned}$$

For each  $R \in \text{Rol}^{\text{quot}}$ , let  $\sim_{R,o}$  be the least equivalence relation that contains  $\rightsquigarrow_{R,o}$  and is defined on  $\text{Tok}_R^*$ . That is,

$$\rightsquigarrow_{R,o} \subseteq \sim_{R,o},$$

and, for every equivalence relation  $E_R \subseteq (\text{Tok}_R^*)^2$ ,

$$\rightsquigarrow_{R,o} \subseteq E_R \implies \sim_{R,o} \subseteq E_R$$

holds.

**Definition 5.14** (Base datum for the dynamical-description extension). For a complete evaluation specification  $\text{Ev} \in \text{EvSpec}^{\text{exact}}$ , define the base datum for the dynamical-description extension by

$$\text{Dat}_{o,\zeta}^0 := \left( \text{RcvStr}_o, (\text{Rec}_{R,o})_{R \in \text{Rol}^{\text{src}}}, \right. \\ \left. \text{Bind}_o, \text{Bind}_o^{\text{src}}, (\rightsquigarrow_{R,o}, \sim_{R,o})_{R \in \text{Rol}^{\text{quot}}} \right)$$

**Definition 5.15** (Specified local-observable datum). For a complete evaluation specification  $\text{Ev} = (\text{I}, o, \zeta)$ , let  $\text{ObsIdx}^{\text{loc}}$  be the local-observable index set, let  $\text{ObsIdx}_{o,\zeta}^{\text{quot}}$  be the quotient-stage observable index set, and let the local-index correspondence map be

$$\text{ObsIdxMap}_{o,\zeta} : \text{ObsIdx}^{\text{loc}} \longrightarrow \text{ObsIdx}^{\text{quot}}$$

For each  $u, v \in U$  and  $\ell \in \text{ObsIdx}^{\text{loc}}$ , let the local-observable value range, local-observable map, local-observable-value transition partial map, common comparison value range, and common comparison-value map, respectively, be

$$\begin{aligned} & \text{ObsVal}_{u,\ell}^{\text{loc}}, \\ & \text{Obs}_{u,\ell}^{\text{loc}} : \mathcal{R}_{\mathfrak{B},u}^{\text{arr}} \longrightarrow \text{ObsVal}_{u,\ell}^{\text{loc}}, \\ & \text{ObsValChg}_{uv,\ell}^{\text{loc}} : \text{ObsVal}_{u,\ell}^{\text{loc}} \rightharpoonup \text{ObsVal}_{v,\ell}^{\text{loc}}, \\ & \text{ObsVal}_{\ell}^{\text{cmp}}, \\ & \text{ObsValMap}_{u,\ell}^{\text{cmp}} : \text{ObsVal}_{u,\ell}^{\text{loc}} \longrightarrow \text{ObsVal}_{\ell}^{\text{cmp}} \end{aligned}$$

For each  $\ell \in \text{ObsIdx}^{\text{loc}}$ , define the local-observable family by

$$\text{ObsFam}_{\ell}^{\text{loc}} := \left( \text{ObsVal}_{\ell}^{\text{cmp}}, \right. \\ \left( \text{ObsVal}_{u,\ell}^{\text{loc}}, \text{Obs}_{u,\ell}^{\text{loc}}, \text{ObsValMap}_{u,\ell}^{\text{cmp}} \right)_{u \in U}, \\ \left. \left( \text{ObsValChg}_{uv,\ell}^{\text{loc}} \right)_{(u,v) \in U^2} \right)$$

Define the specified local-observable datum by

$$\text{Dat}^{\text{obs,loc}} := \left( \text{ObsIdx}^{\text{loc}}, \text{ObsIdx}^{\text{quot}}, \right. \\ \left. \text{ObsIdxMap}, (\text{ObsFam}_{\ell}^{\text{loc}})_{\ell \in \text{ObsIdx}^{\text{loc}}} \right)$$

## 5.2 Construction of the Observer-Time Quotient

Take a complete evaluation specification

$$\text{Ev} \in \text{EvSpec}^{\text{exact}}$$

from Definition 5.2.

The base datum for the dynamical-description extension

$$\text{Dat}^0,$$

from Definition 5.14 and the specified local-observable datum

$$\text{Dat}^{\text{obs,loc}}$$

from Definition 5.15 are determined.

From the  $\mathfrak{C}/\mathfrak{B}$ -derived generated equivalence relations

$$\sim_{\mathfrak{C},o}, \quad \sim_{\mathfrak{B},o},$$

of Definition 5.13, the  $\mathfrak{C}$ -derived source-family direct-sum order

$$\preceq_{*}^{\mathfrak{C}},$$

of Definition 3.17, and the source-side  $\mathfrak{C}\mathfrak{O}\mathfrak{B}$ -derived relation

$$\text{Rel}^{\text{dyn,src}}$$

of Definition 5.7, the  $\mathfrak{C}$ -derived observer-time quotient, the  $\mathfrak{B}$ -derived object-state quotient, the canonical order on the  $\mathfrak{C}$ -derived observer-time quotient, the  $\mathfrak{C}\mathfrak{O}\mathfrak{B}$ -derived canonical trajectory relation, and the family of canonical observable functions are generated in sequence under the stage-specific establishment conditions defined below.

**Definition 5.16** (The  $\mathfrak{C}$ -derived observer-time quotient and the  $\mathfrak{B}$ -derived object-state quotient). For a complete evaluation specification

$$\text{Ev} \in \text{EvSpec}^{\text{exact}}$$

define, for each

$$R \in \text{Rol}^{\text{quot}} = \{\mathfrak{C}, \mathfrak{B}\}$$

the canonical dynamical-description quotient and natural projection by

$$\text{Qu}_{R,o}^{\text{can}} := \text{Tok}_R^* / \sim_{R,o},$$

$$\Theta_{R,o}^{\text{can}} : \text{Tok}_R^* \twoheadrightarrow \text{Qu}_{R,o}^{\text{can}},$$

$$\Theta_{R,o}^{\text{can}}(x_R) := [x_R]_{\sim_{R,o}}$$

The component with  $R = \mathfrak{C}$  is called the  $\mathfrak{C}$ -derived observer-time quotient, and the component with  $R = \mathfrak{B}$  is called the  $\mathfrak{B}$ -derived object-state quotient.

**Proposition 5.17** (Minimality of the generated closure). For each

$$R \in \text{Rol}^{\text{quot}}$$

the relation  $\sim_{R,o}$  is the least equivalence relation that contains  $\rightsquigarrow_{R,o}$  and is defined on  $\text{Tok}_R^*$ .

*Proof.* The conclusion follows from the inclusion condition and minimality condition in Definition 5.13.  $\square$

**Definition 5.18** (Generated time preorder and its antisymmetry condition). Fix a complete evaluation specification  $\text{Ev} \in \text{EvSpec}^{\text{exact}}$ . Define the image of the source order in the canonical time quotient and the preorder it generates by

$$\begin{aligned} \text{TimeOrdEdge}_o &:= \left\{ (\Theta_{\mathfrak{C},o}^{\text{can}}(x), \Theta_{\mathfrak{C},o}^{\text{can}}(y)) \mid \right. \\ &\quad \left. x, y \in \text{Tok}_{\mathfrak{C}}^*, \quad x \preceq_*^{\mathfrak{C}} y \right\}, \\ \preceq_o^{\text{time,gen}} &:= (\text{TimeOrdEdge}_o)^*. \end{aligned}$$

Here the superscript  $*$  on a relation denotes its reflexive transitive closure, defined by finite paths, including paths of length zero. Define the antisymmetry condition for the generated time preorder by

$$\begin{aligned} K_{\text{GenTimeOrdAntiSym}}(\text{Ev}) &\iff \left[ \forall T_0, T_1 \in \text{Qu}_{\mathfrak{C},o}^{\text{can}}, \right. \\ &\quad \left. (T_0 \preceq_o^{\text{time,gen}} T_1 \wedge T_1 \preceq_o^{\text{time,gen}} T_0) \implies T_0 = T_1 \right]. \end{aligned}$$

**Definition 5.19** (Canonical observer-time order). Assume  $K_{\text{GenTimeOrdAntiSym}}(\text{Ev})$ . Define the canonical observer-time order by

$$\preceq_o^{\text{time,can}} := \preceq_o^{\text{time,gen}} \subseteq (\text{Qu}_{\mathfrak{C},o}^{\text{can}})^2.$$

Throughout this subsection, the time-type superscript is omitted from the symbol for the canonical observer-time order.

**Proposition 5.20** (Partial-order and generation properties of the canonical observer-time order). Under

$$\text{ExactOperable}[\mathbb{I}], \quad K_{\text{GenTimeOrdAntiSym}}(\text{Ev}),$$

the relation  $\preceq_o^{\text{can}}$  is a partial order on  $\text{Qu}_{\mathfrak{C},o}^{\text{can}}$ . The natural projection  $\Theta_{\mathfrak{C},o}^{\text{can}}$  preserves the source order. Moreover,  $\preceq_o^{\text{can}}$  is the least preorder on the canonical time quotient for which this natural projection preserves the source order.

*Proof.* Paths of length zero give reflexivity of the generated preorder, and concatenation of finite paths gives transitivity. Antisymmetry is the condition  $K_{\text{GenTimeOrdAntiSym}}(\text{Ev})$ . The canonical order is therefore a partial order.

If  $x \preceq_*^{\mathfrak{C}} y$ , the pair of natural-projection images belongs to  $\text{TimeOrdEdge}_o$ . The resulting path of length one shows that the natural projection preserves order.

Take any preorder on the canonical time quotient for which the natural projection preserves the source order. Its relation contains  $\text{TimeOrdEdge}_o$ . By reflexivity and transitivity, it also contains the reflexive transitive closure of that relation. This proves the stated minimality.  $\square$

**Definition 5.21** ( $\mathfrak{C}/\mathfrak{B}$  source trajectory relation). Define the  $\mathfrak{C}/\mathfrak{B}$  source trajectory relation by

$$\begin{aligned} \text{TrjRel}^{\text{src}} &:= \left\{ (x_{\mathfrak{C}}, x_{\mathfrak{B}}) \in \text{Tok}_{\mathfrak{C}}^* \times \text{Tok}_{\mathfrak{B}}^* \mid \right. \\ &\quad \left. \exists x_{\mathfrak{D}} \in \text{Tok}_{\mathfrak{D}}^*, \quad (x_{\mathfrak{C}}, x_{\mathfrak{D}}, x_{\mathfrak{B}}) \in \text{Rel}^{\text{dyn,src}} \right\} \end{aligned}$$



**Definition 5.22** (Canonical trajectory-relation descent condition at the dynamical-description stage). For a complete evaluation specification  $\text{Ev} \in \text{EvSpec}^{\text{exact}}$ , define representative invariance of the  $\mathfrak{CDB}$ -derived trajectory relation by

$$K_{\text{CanTrjDesc}}(\text{Ev}) \iff \left[ \forall (x_R, x'_R)_{R \in \text{Rol}^{\text{quot}}} \in \prod_{R \in \text{Rol}^{\text{quot}}} (\text{Tok}_R^*)^2, \right. \\ \left. \left( \bigwedge_{R \in \text{Rol}^{\text{quot}}} x_R \sim_{R,o} x'_R \right) \implies \right. \\ \left. \left( (x_{\mathfrak{C}}, x_{\mathfrak{B}}) \in \text{TrjRel}^{\text{src}} \right. \right. \\ \left. \left. \iff (x'_{\mathfrak{C}}, x'_{\mathfrak{B}}) \in \text{TrjRel}^{\text{src}} \right) \right]$$

**Definition 5.23** (Canonical trajectory relation derived from  $\mathfrak{CDB}$ ). Assume

$$K_{\text{CanTrjDesc}}(\text{Ev})$$

Define the canonical trajectory relation derived from  $\mathfrak{CDB}$  by

$$\text{TrjRel}_o^{\text{can}} := (\Theta_{\mathfrak{C},o}^{\text{can}} \times \Theta_{\mathfrak{B},o}^{\text{can}}) [\text{TrjRel}^{\text{src}}]$$

This relation satisfies

$$\text{TrjRel}_o^{\text{can}} \subseteq \text{Qu}_{\mathfrak{C},o}^{\text{can}} \\ \times \text{Qu}_{\mathfrak{B},o}^{\text{can}}$$

**Lemma 5.24** (Descent of the  $\mathfrak{CDB}$ -derived trajectory relation along the generated closure). Suppose that  $K_{\text{CanTrjDesc}}(\text{Ev})$  holds. For each  $R \in \text{Rol}^{\text{quot}}$ , if  $x_R \sim_{R,o} x'_R$ , then

$$(\Theta_{\mathfrak{C},o}^{\text{can}}(x_{\mathfrak{C}}), \Theta_{\mathfrak{B},o}^{\text{can}}(x_{\mathfrak{B}})) \in \text{TrjRel}_o^{\text{can}} \\ \iff (\Theta_{\mathfrak{C},o}^{\text{can}}(x'_{\mathfrak{C}}), \Theta_{\mathfrak{B},o}^{\text{can}}(x'_{\mathfrak{B}})) \in \text{TrjRel}_o^{\text{can}}.$$

*Proof.* This follows from the equivalence of the source-trajectory conditions supplied by Definition 5.22 and the membership condition for the canonical trajectory relation in Definition 5.23.  $\square$

**Definition 5.25** (Canonical trajectory domain). Assume  $K_{\text{CanTrjDesc}}(\text{Ev})$ . Define the canonical trajectory domain by

$$\text{TrjDom}_o^{\text{can}} := \text{pr}_{\mathfrak{C}}(\text{TrjRel}_o^{\text{can}}) \\ = \left\{ t \in \text{Qu}_{\mathfrak{C},o}^{\text{can}} \mid \exists s \in \text{Qu}_{\mathfrak{B},o}^{\text{can}}, (t, s) \in \text{TrjRel}_o^{\text{can}} \right\}.$$

**Definition 5.26** (Canonical trajectory-function condition at the dynamical-description stage). Assume  $K_{\text{CanTrjDesc}}(\text{Ev})$ . Define the single-valuedness condition for the canonical trajectory relation derived from  $\mathfrak{CDB}$  by

$$K_{\text{CanTrjFunc}}(\text{Ev}) \iff \left[ \forall t \in \text{TrjDom}_o^{\text{can}}, \forall s_0, s_1 \in \text{Qu}_{\mathfrak{B},o}^{\text{can}}, \right. \\ \left. \left( \bigwedge_{k=0}^1 (t, s_k) \in \text{TrjRel}_o^{\text{can}} \right) \implies s_0 = s_1 \right]$$

**Proposition 5.27** (Existence and uniqueness of the graph trajectory). When

$$K_{\text{CanTrjDesc}}(\text{Ev}), \quad K_{\text{CanTrjFunc}}(\text{Ev})$$

hold, there exists a unique map

$$\text{TrjMap}_o^{\text{can}} : \text{TrjDom}_o^{\text{can}} \longrightarrow \text{Qu}_{\mathfrak{B},o}^{\text{can}}$$

satisfying

$$\text{TrjRel}_o^{\text{can}} = \{(t, \text{TrjMap}_o^{\text{can}}(t)) \mid t \in \text{TrjDom}_o^{\text{can}}\}$$

*Proof.* For any  $t \in \text{TrjDom}_o^{\text{can}}$ , Definition 5.25 gives

$$(t, s) \in \text{TrjRel}_o^{\text{can}}$$

for some  $s \in \text{Qu}_{\mathfrak{B},o}^{\text{can}}$ . Definition 5.26 makes this  $s$  unique, so one may define

$$\text{TrjMap}_o^{\text{can}}(t) := s$$

By this definition, for each  $t$ ,  $(t, \text{TrjMap}_o^{\text{can}}(t))$  belongs to the trajectory relation. Conversely, if  $(t, s) \in \text{TrjRel}_o^{\text{can}}$ , then  $t \in \text{TrjDom}_o^{\text{can}}$ , and single-valuedness gives  $s = \text{TrjMap}_o^{\text{can}}(t)$ . Therefore, the stated graph equality holds.

At each  $t$ , any map satisfying the same graph equality has the same unique  $s$  as its value, so the map is unique.  $\square$

**Definition 5.28** (Graph trajectory). The unique map

$$\begin{aligned} &\text{TrjMap}_o^{\text{can}} : \\ &\text{TrjDom}_o^{\text{can}} \longrightarrow \\ &\text{Qu}_{\mathfrak{B},o}^{\text{can}} \end{aligned}$$

of Proposition 5.27 is called the graph trajectory.

Take as input the specified local-observable datum  $\text{Dat}^{\text{obs,loc}}$  from Definition 5.15.

**Definition 5.29** (Common-comparison-value composites of local observables). For each  $u, v \in U$  and  $\ell \in \text{ObsIdx}^{\text{loc}}$ , define the common-comparison-value map and the post-transition common-comparison-value partial map of the local observable by

$$\text{ObsCmp}_{u,\ell} := \text{ObsValMap}_{u,\ell}^{\text{cmp}} \circ \text{Obs}_{u,\ell}^{\text{loc}}$$

$$\text{ObsCmpChg}_{uv,\ell} := \text{ObsValMap}_{v,\ell}^{\text{cmp}} \circ \text{ObsValChg}_{uv,\ell}^{\text{loc}} \circ \text{Obs}_{u,\ell}^{\text{loc}}$$

**Definition 5.30** ( $\mathfrak{B}$ -derived local-reception canonical-state maps). For each  $u \in U$ , define the  $\mathfrak{B}$ -derived local-reception canonical-state map by

$$\begin{aligned} \Theta_{\mathfrak{B},o \leftarrow u}^{\text{can}} &:= \Theta_{\mathfrak{B},o}^{\text{can}} \circ \text{Rec}_{\mathfrak{B},u} \\ &: \mathcal{R}_{\mathfrak{B},u}^{\text{arr}} \longrightarrow \text{Qu}_{\mathfrak{B},o}^{\text{can}} \end{aligned}$$

**Definition 5.31** (Descent condition for the canonical observable-function family at the dynamical-description stage). For a complete evaluation specification  $\text{Ev} \in \text{EvSpec}^{\text{exact}}$ , define  $K_{\text{CanObsDesc}}(\text{Ev})$  as the conjunction of the following four conditions.

First,  $\text{ObsIdxMap}$  is a surjection onto  $\text{ObsIdx}^{\text{quot}}$ . Within this definition, write this map as  $\pi_{\text{obs}}$ .

Second, for any  $\ell_0, \ell_1 \in \text{ObsIdx}^{\text{loc}}$ ,

$$\pi_{\text{obs}}(\ell_0) = \pi_{\text{obs}}(\ell_1) \implies \text{ObsVal}_{\ell_0}^{\text{cmp}} = \text{ObsVal}_{\ell_1}^{\text{cmp}}.$$

Third, for any

$$q \in \text{ObsIdx}^{\text{quot}}, \quad s \in \text{Qu}_{\mathfrak{B},o}^{\text{can}}$$

there exist  $u \in U$ ,  $\ell \in \text{ObsIdx}^{\text{loc}}$ , and  $a \in \mathcal{R}_{\mathfrak{B},u}^{\text{arr}}$  such that

$$\begin{aligned} \pi_{\text{obs}}(\ell) &= q, \\ \Theta_{\mathfrak{B},o \leftarrow u}^{\text{can}}(a) &= s. \end{aligned}$$

Fourth, for  $k \in \{0, 1\}$ , let

$$u_k \in U, \quad \ell_k \in \text{ObsIdx}^{\text{loc}}, \quad a_k \in \mathcal{R}_{\mathfrak{B},u_k}^{\text{arr}}$$

If

$$\begin{aligned} \pi_{\text{obs}}(\ell_0) &= \pi_{\text{obs}}(\ell_1), \\ \Theta_{\mathfrak{B},o \leftarrow u_0}^{\text{can}}(a_0) &= \Theta_{\mathfrak{B},o \leftarrow u_1}^{\text{can}}(a_1) \end{aligned}$$

then

$$\text{ObsCmp}_{u_0,\ell_0}(a_0) = \text{ObsCmp}_{u_1,\ell_1}(a_1).$$

**Definition 5.32** (Canonical observable-function family). Assume  $K_{\text{CanObsDesc}}(\text{Ev})$ . For each  $q \in \text{ObsIdx}^{\text{quot}}$ , take a solution of  $\text{ObsIdxMap}(\ell) = q$  with  $\ell \in \text{ObsIdx}^{\text{loc}}$ , and define

$$\text{ObsVal}_{o,\zeta,q}^{\text{can}} := \text{ObsVal}_{\ell}^{\text{cmp}}$$

For each  $s \in \text{Qu}_{\mathfrak{B},o}^{\text{can}}$ , take a solution of

$$\Theta_{\mathfrak{B},o \leftarrow u}^{\text{can}}(a) = s$$

with  $u \in U$  and  $a \in \mathcal{R}_{\mathfrak{B},u}^{\text{arr}}$ , and define

$$\text{Obs}_{o,\zeta,q}^{\text{can}}(s) := \text{ObsCmp}_{u,\ell}(a)$$

Define the canonical observable-function family by

$$\mathcal{O}_{o,\zeta}^{\text{can}} := \left( \text{ObsIdx}^{\text{quot}}, \left( \text{ObsVal}_q^{\text{can}}, \text{Obs}_q^{\text{can}} \right)_{q \in \text{ObsIdx}^{\text{quot}}} \right)$$

**Proposition 5.33** (Representative independence of the canonical observable-function family). The canonical observable value ranges and canonical observable functions do not depend on the choice of local-observable index or local  $\mathfrak{B}$ -derived reception.

*Proof.* The first condition in Definition 5.31 gives the existence of a local-observable index; the second gives representative independence of the value range; and the third gives the existence of a local  $\mathfrak{B}$ -derived reception. For the same quotient-stage observable index and the same  $\mathfrak{B}$ -derived object-state quotient class, the fourth condition gives, for any two representing triples  $(u_0, \ell_0, a_0)$  and  $(u_1, \ell_1, a_1)$ ,

$$\text{ObsCmp}_{u_0,\ell_0}(a_0) = \text{ObsCmp}_{u_1,\ell_1}(a_1)$$

Therefore, the value ranges and function values in Definition 5.32 do not depend on any of these choices.  $\square$

**Definition 5.34** (Common-comparison-value commutativity condition for local-observable value changes at the dynamical-description stage). For a complete evaluation specification  $\text{Ev} \in \text{EvSpec}^{\text{exact}}$ , define  $K_{\text{LocObsChgComm}}(\text{Ev})$  by requiring, for every

$$\begin{aligned} u, v &\in U, \\ \ell &\in \text{ObsIdx}^{\text{loc}}, \\ a &\in \mathcal{R}_{\mathfrak{B},u}^{\text{arr}} \end{aligned}$$

that

$$\text{Obs}_{u,\ell}^{\text{loc}}(a) \in \text{Dom}(\text{ObsValChg}_{uv,\ell}^{\text{loc}}) \implies \text{ObsCmpChg}_{uv,\ell}(a) = \text{ObsCmp}_{u,\ell}(a)$$

holds.

**Proposition 5.35** (Compatibility of the canonical observable-function family with local-observable value changes). Suppose that

$$K_{\text{CanObsDesc}}(\text{Ev}), \quad K_{\text{LocObsChgComm}}(\text{Ev})$$

hold. For any  $u, v \in U$ ,  $\ell \in \text{ObsIdx}^{\text{loc}}$ , and  $a \in \mathcal{R}_{\mathfrak{B},u}^{\text{arr}}$ , if  $\text{Obs}_{u,\ell}^{\text{loc}}(a)$  belongs to the domain of  $\text{ObsValChg}_{uv,\ell}^{\text{loc}}$ , then

$$\text{Obs}_{\text{ObsIdxMap}(\ell)}^{\text{can}}(\Theta_{\mathfrak{B},o \leftarrow u}^{\text{can}}(a)) = \text{ObsCmpChg}_{uv,\ell}(a).$$

*Proof.* By Definition 5.32,

$$\text{Obs}_{\text{ObsIdxMap}(\ell)}^{\text{can}}(\Theta_{\mathfrak{B},o \leftarrow u}^{\text{can}}(a)) = \text{ObsCmp}_{u,\ell}(a).$$

By Definition 5.34, the right-hand side equals  $\text{ObsCmpChg}_{uv,\ell}(a)$ .  $\square$

**Definition 5.36** ( $\mathfrak{C}\mathfrak{D}\mathfrak{B}$ -derived dynamical-description quotient). Assume

$$K_{\text{GenTimeOrdAntiSym}}(\text{Ev}), \quad K_{\text{CanTrjDesc}}(\text{Ev})$$

Define the  $\mathfrak{C}\mathfrak{D}\mathfrak{B}$ -derived dynamical-description quotient by

$$\text{Qu}_{o,\zeta}^{\text{dyn,can}} := \left( (\text{Qu}_{R,o}^{\text{can}}, \Theta_{R,o}^{\text{can}})_{R \in \text{Rel}^{\text{quot}}}, \preceq_o^{\text{can}}, \text{TrjRel}_o^{\text{can}} \right).$$

When, in addition,

$$K_{\text{CanTrjFunc}}(\text{Ev})$$

holds, the graph trajectory of Definition 5.28 satisfies

$$\text{TrjRel}_o^{\text{dyn,can}} = \left\{ \begin{aligned} &(\text{DynCls}_{\mathfrak{C},o,0}, \text{TrjMap}_o^{\text{dyn,can}}(\text{DynCls}_{\mathfrak{C},o,0})) \\ &\left| \text{DynCls}_{\mathfrak{C},o,0} \in \text{TrjDom}_o^{\text{dyn,can}} \right. \end{aligned} \right\}.$$

**Definition 5.37** ( $\mathfrak{D}$ -derived record code). Define the  $\mathfrak{D}$ -derived record-code map, for each  $x_{\mathfrak{D}} \in \text{Tok}_{\mathfrak{D}}^*$ , by

$$\text{RecordCode}_{\mathfrak{D}}^{\text{dyn}}(x_{\mathfrak{D}}) := (W_{\mathfrak{D}}(x_{\mathfrak{D}}), \phi_{\mathfrak{D}}(x_{\mathfrak{D}}), \text{seq}_{\mathfrak{D}}(x_{\mathfrak{D}})).$$

Define the  $\mathfrak{D}$ -derived record-code value range and the type of the map by

$$\begin{aligned} \text{RecordCodeVal}_{\mathfrak{D}}^{\text{dyn}} &:= \text{RecordCode}_{\mathfrak{D}}[\text{Tok}_{\mathfrak{D}}^*], \\ \text{RecordCode}_{\mathfrak{D}} &: \text{Tok}_{\mathfrak{D}}^* \longrightarrow \text{RecordCodeVal}_{\mathfrak{D}} \end{aligned}$$

**Definition 5.38** (Typed  $\mathfrak{D}$ -record-code relations). Assume

$$K_{\text{CanTrjDesc}}(\mathbf{Ev})$$

Define the  $\mathfrak{C}/\mathfrak{B}$ -correspondence-typed  $\mathfrak{D}$ -record-code relation by

$$\text{RecordCodeRel}_o^{\text{dyn,pair}} := \left\{ \left( \left( \Theta_{\mathfrak{C},o}^{\text{can}}(x_{\mathfrak{C}}), \Theta_{\mathfrak{B},o}^{\text{can}}(x_{\mathfrak{B}}) \right), \text{RecordCode}_{\mathfrak{D}}^{\text{dyn}}(x_{\mathfrak{D}}) \right) \mid \right. \\ \left. (x_{\mathfrak{C}}, x_{\mathfrak{D}}, x_{\mathfrak{B}}) \in \text{Rel}^{\text{dyn,src}} \right\}.$$

This relation satisfies

$$\text{RecordCodeRel}_o^{\text{pair}} \subseteq \text{TrjRel}_o^{\text{can}} \times \text{RecordCodeVal}_{\mathfrak{D}}$$

For each  $R \in \text{Rol}^{\text{quot}}$ , define the typed  $\mathfrak{D}$ -record-code relation by

$$\text{RecordCodeRel}_{R,o}^{\text{dyn}} := (\text{pr}_R \times \text{id}) [\text{RecordCodeRel}_o^{\text{pair}}] \\ \subseteq \text{Qu}_{R,o}^{\text{can}} \times \text{RecordCodeVal}_{\mathfrak{D}}.$$

**Definition 5.39** ( $\mathfrak{D}$ -derived record-code datum). Define the record-code datum by

$$\text{RecordDat}_o^{\text{dyn}} := \left( \text{RecordCodeVal}_{\mathfrak{D}}^{\text{dyn}}, \text{RecordCode}_{\mathfrak{D}}^{\text{dyn}}, \right. \\ \left. \left( \text{RecordCodeRel}_{R,o}^{\text{dyn}} \right)_{R \in \text{Rol}^{\text{dyn,quot}}}, \text{RecordCodeRel}_o^{\text{dyn,pair}} \right).$$

**Proposition 5.40** (Projection compatibility of the typed  $\mathfrak{D}$ -record-code relations). For each

$$R \in \text{Rol}^{\text{quot}}$$

the equality

$$\text{RecordCodeRel}_{R,o} = (\text{pr}_R \times \text{id}) [\text{RecordCodeRel}_o^{\text{pair}}]$$

holds.

*Proof.* Apply Definition 5.38 to each  $R \in \text{Rol}^{\text{quot}}$ . □

**Definition 5.41** (Dynamical-description admissible quotients). Write the dynamical-description admissible quotients as

$$\text{AdmQuPrs}_{o,\zeta,i}^{\text{dyn}} := \left( \left( \text{Qu}_{R,o,i}^{\text{dyn,adm}}, \Theta_{R,o,i}^{\text{dyn,adm}} \right)_{R \in \text{Rol}^{\text{quot}}}, \preceq_{o,i}^{\text{dyn,time,adm}}, \text{TrjRel}_{o,i}^{\text{dyn,adm}} \right)$$

and define them by the following conditions.

(DQ1) For each  $R \in \text{Rol}^{\text{quot}}$ ,

$$\Theta_{R,o,i}^{\text{adm}} : \text{Tok}_R^* \rightarrow \text{Qu}_{R,o,i}^{\text{adm}}$$

is surjective.

(DQ2) For each  $R \in \text{Rol}^{\text{quot}}$  and all  $x_R, x'_R \in \text{Tok}_R^*$ ,

$$x_R \sim_{R,o}^{\text{dyn}} x'_R \implies \Theta_{R,o,i}^{\text{adm}}(x_R) = \Theta_{R,o,i}^{\text{adm}}(x'_R)$$

holds.

(DQ3)  $\preceq_{o,i}^{\text{time,adm}}$  is a partial order on  $\text{Qu}_{\mathfrak{C},o,i}^{\text{adm}}$ , and, for all  $x_{\mathfrak{C}}, y_{\mathfrak{C}} \in \text{Tok}_{\mathfrak{C}}^*$ ,

$$x_{\mathfrak{C}} \preceq_*^{\mathfrak{C}} y_{\mathfrak{C}} \implies \Theta_{\mathfrak{C},o,i}^{\text{adm}}(x_{\mathfrak{C}}) \preceq_{o,i}^{\text{time,adm}} \Theta_{\mathfrak{C},o,i}^{\text{adm}}(y_{\mathfrak{C}})$$

holds.

(DQ4)

$$\text{TrjRel}_{o,i}^{\text{adm}} = (\Theta_{\mathfrak{C},o,i}^{\text{adm}} \times \Theta_{\mathfrak{B},o,i}^{\text{adm}}) [\text{TrjRel}^{\text{dyn,src}}].$$

**Definition 5.42** (Refinement relation for dynamical-description admissible quotients).  
For two dynamical-description admissible quotients

$$\text{AdmQuPrs}_i, \quad \text{AdmQuPrs}_j$$

define

$$\text{AdmQuPrs}_i \sqsubseteq_{\text{adm}} \text{AdmQuPrs}_j$$

by the existence, for each  $R \in \text{Rol}^{\text{quot}}$ , of a surjection

$$\text{RefMap}_{R,o;j \rightarrow i} : \text{Qu}_{R,o,j}^{\text{adm}} \twoheadrightarrow \text{Qu}_{R,o,i}^{\text{adm}}$$

satisfying

$$\Theta_{R,o,i}^{\text{adm}} = \text{RefMap}_{R,o;j \rightarrow i} \circ \Theta_{R,o,j}^{\text{adm}} \quad (R \in \text{Rol}^{\text{quot}}),$$

$$\text{TrjRel}_{o,i}^{\text{adm}} = (\text{RefMap}_{\mathfrak{C},o;j \rightarrow i} \times \text{RefMap}_{\mathfrak{B},o;j \rightarrow i}) [\text{TrjRel}_{o,j}^{\text{adm}}]$$

and by requiring  $\text{RefMap}_{\mathfrak{C},o;j \rightarrow i}$  to be order-preserving from  $\preceq_{o,j}^{\text{time,adm}}$  to  $\preceq_{o,i}^{\text{time,adm}}$ .

When there exists, between the two presentations, a family of bijections that commutes with the natural projections and the trajectory relations and whose  $\mathfrak{C}$ -component is an order isomorphism between the observer-time partial orders,

$$(\text{RefMap}_{R,o;j \rightarrow i})_{R \in \text{Rol}^{\text{quot}}}$$

define

$$\text{AdmQuPrs}_i \cong_{\text{adm}} \text{AdmQuPrs}_j$$

**Proposition 5.43** (Preorder property of the refinement relation). The relation  $\sqsubseteq_{\text{adm}}$  is a preorder on presentations of dynamical-description admissible quotients. Two mutually refining presentations are isomorphic under  $\cong_{\text{adm}}$ , and  $\sqsubseteq_{\text{adm}}$  induces a partial order on the isomorphism classes.

*Proof.* The family of identity maps satisfies every condition in Definition 5.42; hence reflexivity holds.

Suppose that

$$\text{AdmQuPrs}_0 \sqsubseteq_{\text{adm}} \text{AdmQuPrs}_1 \sqsubseteq_{\text{adm}} \text{AdmQuPrs}_2$$

For each  $R \in \text{Rol}^{\text{quot}}$ , define

$$\text{RefMap}_{R,o;2 \rightarrow 0} := \text{RefMap}_{R,o;1 \rightarrow 0} \circ \text{RefMap}_{R,o;2 \rightarrow 1}$$

Surjectivity, factorization through the natural projection, equality of the images of the trajectory relations, and order preservation by the  $\mathfrak{C}$ -component are all preserved under composition; hence transitivity holds.

Suppose that two presentations mutually refine one another. For each  $R \in \mathbf{Rol}^{\text{quot}}$ , composing the two factorization identities shows that each of the two composites of the oppositely directed refinement maps agrees with the corresponding natural projection after precomposition by that projection. Since the natural projections are surjective, Lemma 4.181 implies that both composites are identity maps. The two families of maps are therefore mutually inverse families of bijections preserving the order and the trajectory relation. Thus mutual refinement gives an isomorphism, and antisymmetry holds on the isomorphism classes.  $\square$

**Theorem 5.44** (Maximal admissibility of the  $\mathfrak{CDB}$ -derived dynamical-description quotient). Assume

$$\begin{aligned} &\text{ExactOperable}[\mathbb{I}], \\ &K_{\text{GenTimeOrdAntiSym}}(\text{Ev}), \\ &K_{\text{CanTrjDesc}}(\text{Ev}) \end{aligned}$$

The  $\mathfrak{CDB}$ -derived dynamical-description quotient  $\mathbf{Qu}^{\text{can}}$  is a dynamical-description admissible quotient.

For every dynamical-description admissible quotient  $\mathbf{AdmQuPrs}_i$ , there exists a unique family of surjections

$$\text{FactorMap}_{R,o;i} : \mathbf{Qu}_{R,o}^{\text{can}} \twoheadrightarrow \mathbf{Qu}_{R,o,i}^{\text{adm}} \quad (R \in \mathbf{Rol}^{\text{quot}})$$

such that

$$\begin{aligned} \Theta_{R,o,i}^{\text{adm}} &= \text{FactorMap}_{R,o,i} \circ \Theta_{R,o}^{\text{can}} \quad (R \in \mathbf{Rol}^{\text{quot}}), \\ \text{TrjRel}_{o,i}^{\text{adm}} &= (\text{FactorMap}_{\mathfrak{C},o,i} \times \text{FactorMap}_{\mathfrak{B},o,i}) [\text{TrjRel}_o^{\text{can}}] \end{aligned}$$

hold. Moreover,  $\text{FactorMap}_{\mathfrak{C},o,i}$  is order-preserving from the canonical observer-time order to the admissible time order.

Therefore,

$$[\mathbf{Qu}^{\text{can}}]_{\cong_{\text{adm}}}$$

is the greatest element in the refinement order on the isomorphism classes under  $\cong_{\text{adm}}$ .

*Proof.* By Definition 5.16, for each  $R \in \mathbf{Rol}^{\text{quot}}$ , the map  $\Theta_{R,o}^{\text{can}}$  is surjective and takes generated equivalence classes as its values, so it satisfies (DQ1) and (DQ2). Proposition 5.20 and Definition 5.19 give (DQ3), while Definition 5.23 gives (DQ4). Hence  $\mathbf{Qu}^{\text{can}}$  is a dynamical-description admissible quotient.

For each  $R \in \mathbf{Rol}^{\text{quot}}$  and  $\text{DynCls}_{R,o,0} \in \mathbf{Qu}_{R,o}^{\text{can}}$ , choose a representative through

$$\text{DynCls}_{R,o,0} = \Theta_{R,o}^{\text{can}}(x_R)$$

with  $x_R \in \mathbf{Tok}_R^*$ , and define

$$\text{FactorMap}_{R,o,i}(\text{DynCls}_{R,o,0}) := \Theta_{R,o,i}^{\text{adm}}(x_R)$$

Replacing the representative gives  $x_R \sim_{R,o} x'_R$ , and (DQ2) leaves the right-hand side unchanged. The family of maps is therefore well-defined. By (DQ1), every map is surjective, and the natural-projection factorization identity follows from the definition.

Take any edge in  $\text{TimeOrdEdge}_o$ . It has source representatives satisfying  $x \preceq_*^{\mathfrak{C}} y$ . By (DQ3), their images under the admissible natural projection are ordered in the same direction. The natural-projection factorization identity therefore shows that  $\text{FactorMap}_{\mathfrak{C},o,i}$

preserves each generating edge. Reflexivity and transitivity of the admissible time order extend this preservation to all finite paths, including paths of length zero. By Definition 5.19, this finite-path relation is the canonical observer-time order. Thus the  $\mathfrak{C}$ -component preserves order. Equality of the images of the trajectory relations follows from Definition 5.23 and (DQ4).

Finally, take any family of maps satisfying the natural-projection factorization identity. For each  $R \in \text{Rol}^{\text{quot}}$ , if

$$\text{DynCls}_{R,o,0} = \Theta_{R,o}^{\text{can}}(x_R)$$

then the factorization identity makes the value of that map equal to  $\Theta_{R,o,i}^{\text{adm}}(x_R)$ . Because the canonical natural projection is surjective, the family of maps is unique.  $\square$

**Definition 5.45** (Canonical time-and-trajectory condition bundle). Define the canonical time-and-trajectory condition bundle by

$$\begin{aligned} \text{DynCanTimeTrjCondAll}[\text{Ev}] \iff & K_{\text{GenTimeOrdAntiSym}}(\text{Ev}) \\ & \wedge K_{\text{CanTrjDesc}}(\text{Ev}) \\ & \wedge K_{\text{CanTrjFunc}}(\text{Ev}). \end{aligned}$$

**Definition 5.46** (Observer-time law-input condition bundle).

Define the observer-time law-input condition bundle by

$$\begin{aligned} \text{DynLawInputCondAll}[\text{Ev}] \iff & \text{DynCanTimeTrjCondAll}[\text{Ev}] \\ & \wedge K_{\text{CanObsDesc}}(\text{Ev}) \\ & \wedge K_{\text{LocObsChgComm}}(\text{Ev}). \end{aligned}$$

**Definition 5.47** (Observer-time law-input datum). Assume

$$\text{ExactOperable}[\text{I}], \quad \text{DynLawInputCondAll}[\text{Ev}]$$

Define the observer-time law-input datum by

$$\text{LawInput}_{o,\zeta}^{\text{dyn}} := (\text{Qu}^{\text{can}}, \text{TrjDom}_o^{\text{can}}, \text{TrjMap}_o^{\text{can}}, \mathcal{O}^{\text{can}}, \text{RecordDat}_o).$$

The type of each component is given, respectively, by Definition 5.36, Definition 5.25, Definition 5.28, Definition 5.32, and Definition 5.39.

**Proposition 5.48** (Existence and uniqueness of the observer-time law-input datum). Under

$$\text{ExactOperable}[\text{I}], \quad \text{DynLawInputCondAll}[\text{Ev}]$$

$\text{LawInput}$  exists uniquely.

*Proof.* Definition 5.36, Definition 5.25, Proposition 5.27, Definition 5.32, and Definition 5.39 uniquely determine every component and the component order in Definition 5.47.  $\square$

### 5.3 Observer-Time Law Core and Dynamical-Description Operability

Fix a complete evaluation specification  $\text{Ev} = (\text{I}, o, \zeta)$  and assume

$$\text{ExactOperable}[\text{I}], \quad K_{\text{GenTimeOrdAntiSym}}(\text{Ev})$$



Use the observer-time quotient and canonical order from Definition 5.19:

$$\mathbf{Qu}_{\mathfrak{E},o}^{\text{can}}, \quad \preceq_o^{\text{time,can}}$$

The constructions involving trajectories and observables below carry their respective existence conditions.

From here through the end of Chapter 6, each time object is first introduced in full form. Its time-type superscript is then omitted only when that type is uniquely determined by the symbol root and the remaining modifiers.

By Proposition 5.20,

$$\preceq_o^{\text{can}}$$

is a partial order on

$$\mathbf{Qu}_{\mathfrak{E},o}^{\text{can}}$$

**Definition 5.49** (Observer-time strict order). For any  $T_0, T_1 \in \mathbf{Qu}_{\mathfrak{E},o}^{\text{can}}$ , define the observer-time strict order by

$$T_0 \prec_o^{\text{time,can}} T_1 \iff (T_0 \preceq_o^{\text{can}} T_1) \wedge (T_0 \neq T_1)$$

Let the resulting strict-order datum be

$$\mathbf{P}_o^{\text{dyn,time}} := (\mathbf{Qu}_{\mathfrak{E},o}^{\text{can}}, \prec_o^{\text{can}})$$

**Lemma 5.50** (Strict-partial-order property of the observer-time strict order). The relation  $\prec_o^{\text{can}}$  is a strict partial order on  $\mathbf{Qu}_{\mathfrak{E},o}^{\text{can}}$ .

*Proof.* Apply Proposition 5.20 and Lemma 4.105 to

$$S = \mathbf{Qu}_{\mathfrak{E},o}^{\text{can}}, \quad \preceq = \preceq_o^{\text{can}}$$

□

**Definition 5.51** (Observer-time incomparability relation). Define the observer-time incomparability relation by

$$\asymp_o^{\text{dyn,time,can}} := \text{Inc}[\mathbf{P}_o] \subseteq (\mathbf{Qu}_{\mathfrak{E},o}^{\text{can}})^2$$

Consequently, for any  $T_0, T_1 \in \mathbf{Qu}_{\mathfrak{E},o}^{\text{can}}$ ,

$$T_0 \asymp_o^{\text{can}} T_1 \iff \neg (T_0 \prec_o^{\text{can}} T_1) \wedge \neg (T_1 \prec_o^{\text{can}} T_0).$$

**Proposition 5.52** (Reflexivity and symmetry of the observer-time incomparability relation). The relation  $\asymp_o^{\text{can}}$  is reflexive and symmetric.

*Proof.* By the irreflexivity in Lemma 5.50 and Definition 5.51, for every

$$T \in \mathbf{Qu}_{\mathfrak{E},o}^{\text{can}}$$

one has  $T \asymp_o^{\text{can}} T$ . The two negated components in the same definition interchange when the arguments are exchanged, so the relation is symmetric. □

**Definition 5.53** (Observer-time incomparability transitivity condition). Assume  $K_{\text{GenTimeOrdAntiSym}}(\text{Ev})$ . Define the observer-time incomparability transitivity condition by

$$K_{\text{ObsTimeIncTrans}}^{\text{dyn}}(\text{Ev}) \iff \text{IncTrans}[\mathbf{P}_o]$$

**Lemma 5.54** (Equivalence-relation property of the observer-time incomparability relation). Under  $K_{\text{ObsTimeIncTrans}}(\text{Ev})$ ,  $\succsim_o^{\text{can}}$  is an equivalence relation.

*Proof.* Apply Definition 5.53 and Lemma 4.109 to  $P_o$ .  $\square$

**Lemma 5.55** (Invariance of the strict order along observer-time incomparability classes). Under  $K_{\text{ObsTimeIncTrans}}(\text{Ev})$ , for any  $T_0, T'_0, T_1, T'_1 \in \text{Qu}_{\mathcal{E},o}^{\text{can}}$ ,

$$\begin{aligned} (T_0 \succsim_o^{\text{can}} T'_0) \wedge (T_1 \succsim_o^{\text{can}} T'_1) \\ \implies (T_0 \prec_o^{\text{can}} T_1 \iff T'_0 \prec_o^{\text{can}} T'_1). \end{aligned}$$

*Proof.* Apply the strict-order invariance of Lemma 4.109 to  $P_o$ .  $\square$

**Definition 5.56** (Observer-time order-coordinate quotient). Assume  $K_{\text{ObsTimeIncTrans}}(\text{Ev})$ . Define the observer-time order-coordinate quotient by

$$\begin{aligned} (\text{Qu}_o^{\text{dyn,time,ord}}, \text{Prj}_o^{\text{dyn,time,ord}}, \prec_o^{\text{dyn,time,ord}}) \\ := \text{LinQuot}[P_o] \end{aligned}$$

Consequently,

$$\begin{aligned} \text{Qu}_o^{\text{ord}} &= \text{Qu}_{\mathcal{E},o}^{\text{can}} / \succsim_o^{\text{can}}, \\ \text{Prj}_o^{\text{ord}} : \text{Qu}_{\mathcal{E},o}^{\text{can}} &\rightarrow \text{Qu}_o^{\text{ord}}, \quad \text{Prj}_o^{\text{ord}}(T) = [T]_{\succsim_o^{\text{can}}}, \\ \text{Prj}_o^{\text{ord}}(T_0) &\prec_o^{\text{ord}} \text{Prj}_o^{\text{ord}}(T_1) \iff T_0 \prec_o^{\text{can}} T_1. \end{aligned}$$

**Lemma 5.57** (Strict-linear-order property of the observer-time order-coordinate quotient). Under  $K_{\text{ObsTimeIncTrans}}(\text{Ev})$ ,  $\prec_o^{\text{ord}}$  is a strict linear order on  $\text{Qu}_o^{\text{ord}}$ .

*Proof.* Apply Lemma 4.109 to  $P_o$ .  $\square$

**Proposition 5.58** (Surjectivity, class determination, and order determination of the canonical observer-time order-coordinate projection). Under  $K_{\text{ObsTimeIncTrans}}(\text{Ev})$ ,  $\text{Prj}_o^{\text{ord}}$  is surjective, and

$$\text{OrdRepCond}(\text{Prj}_o^{\text{ord}}; \succsim_o^{\text{can}}, \prec_o^{\text{can}}, \prec_o^{\text{ord}})$$

holds.

*Proof.* Apply Definition 5.56 and Lemma 4.109 to  $P_o$ , and use Definition 4.108.  $\square$

**Definition 5.59** (Real-valued time-order embedding condition).

Assume  $K_{\text{ObsTimeIncTrans}}(\text{Ev})$ . Let the set of all order embeddings from the observer-time order-coordinate quotient into the real numbers be

$$\text{OrdEmbSet}_o^{\text{time}}(\mathbb{R}) := \left\{ \iota_o \left| \begin{array}{l} \iota_o : \text{Qu}_o^{\text{ord}} \rightarrow \mathbb{R}, \\ \forall \bar{T}_0, \bar{T}_1, \\ \bar{T}_0 \prec_o^{\text{ord}} \bar{T}_1 \iff \iota_o(\bar{T}_0) < \iota_o(\bar{T}_1) \end{array} \right. \right\}.$$

Define the real-valued time-order embedding condition by

$$K_{\text{ObsTimeRealOrdEmb}}(\text{Ev}) \iff \text{OrdEmbSet}_o^{\text{time}}(\mathbb{R}) \neq \emptyset$$

Under this condition, let

$$\text{RepEmb}_o^{\text{time}} \in \text{OrdEmbSet}_o^{\text{time}}(\mathbb{R})$$

be a real-valued time-order embedding.

**Proposition 5.60** (Injectivity of a real-valued time-order embedding). Under

$$K_{\text{ObsTimeIncTrans}}(\mathbf{Ev}), \quad K_{\text{ObsTimeRealOrdEmb}}(\mathbf{Ev})$$

every  $\rho_o \in \text{OrdEmbSet}_o(\mathbb{R})$  is injective.

*Proof.* If  $\bar{T}_0 \neq \bar{T}_1$ , Lemma 5.57 gives  $\bar{T}_0 <_o^{\text{ord}} \bar{T}_1$  or its converse. Definition 5.59 gives the corresponding strict inequality between their images, so the images are unequal.  $\square$

**Definition 5.61** (Inverse on the image of a real-valued time-order embedding). Under Proposition 5.60, for any  $\rho_o \in \text{OrdEmbSet}_o(\mathbb{R})$ , define its inverse on its image by

$$\begin{aligned} \rho_o^{-1} : \text{Im}(\rho_o) &\longrightarrow \text{Qu}_o^{\text{ord}}, \\ \rho_o^{-1}(t) = \bar{T} &\iff \rho_o(\bar{T}) = t \end{aligned}$$

Here  $t \in \text{Im}(\rho_o)$  and  $\bar{T} \in \text{Qu}_o^{\text{ord}}$ .

**Proposition 5.62** (Composition of a real-valued time-order embedding with its inverse on the image). Assume  $K_{\text{ObsTimeIncTrans}}(\mathbf{Ev})$ . For every

$$\begin{aligned} \rho_o &\in \text{OrdEmbSet}_o(\mathbb{R}), \\ \bar{T} &\in \text{Qu}_o^{\text{ord}}, \\ t &\in \text{Im}(\rho_o) \end{aligned}$$

one has

$$\rho_o^{-1}(\rho_o(\bar{T})) = \bar{T}, \quad \rho_o(\rho_o^{-1}(t)) = t.$$

*Proof.* This follows from Definition 5.61 and Proposition 5.60.  $\square$

**Definition 5.63** (Law-admissible time representation). Assume  $K_{\text{ObsTimeIncTrans}}(\mathbf{Ev})$ . For any  $\rho_o \in \text{OrdEmbSet}_o(\mathbb{R})$ , define the law-admissible time representation by

$$\text{TimeRep}_o[\rho_o] := \rho_o \circ \text{Prj}_o^{\text{ord}} : \text{Qu}_{\mathfrak{E},o}^{\text{can}} \longrightarrow \mathbb{R}$$

When  $K_{\text{ObsTimeRealOrdEmb}}(\mathbf{Ev})$  holds, let the time representation corresponding to  $\text{RepEmb}_o$  be

$$\text{TimeRep}_o := \text{TimeRep}_o[\text{RepEmb}_o]$$

**Proposition 5.64** (Class determination and order determination by a law-admissible time representation). Assume  $K_{\text{ObsTimeIncTrans}}(\mathbf{Ev})$ . For every  $\rho_o \in \text{OrdEmbSet}_o(\mathbb{R})$ ,

$$\text{OrdRepCond}(\text{TimeRep}_o[\rho_o]; \asymp_o^{\text{can}}, \prec_o^{\text{can}}, <)$$

holds.

*Proof.* Proposition 5.60, Definition 5.59, and Proposition 5.58 give the two components of Definition 4.108.  $\square$

**Definition 5.65** (Scalar-time trajectory factorization condition).

Assume  $\text{DynCanTimeTrjCondAll}[\mathbf{Ev}]$  and  $K_{\text{ObsTimeIncTrans}}(\mathbf{Ev})$ . Define the projected trajectory-time domain by

$$\text{TrjDom}_o^{\text{time,ord}} := \text{Prj}_o^{\text{ord}}(\text{TrjDom}_o^{\text{can}}) \subseteq \text{Qu}_o^{\text{ord}}$$

Let the restricted projection be

$$\pi_o := \text{Prj}_o^{\text{ord}}|_{\text{TrjDom}_o^{\text{can}}}$$

Define the scalar-time trajectory factorization condition by

$$K_{\text{ObsTimeTrjFact}}(\mathbf{Ev}) \iff \text{EqKer}(\pi_o) \subseteq \text{EqKer}(\text{TrjMap}_o^{\text{can}}).$$

**Proposition 5.66** (Existence and uniqueness of the scalar-time trajectory). Assume  $\text{DynCanTimeTrjCondAll}[\text{Ev}]$  and  $K_{\text{ObsTimeIncTrans}}(\text{Ev})$ . Under  $K_{\text{ObsTimeTrjFact}}(\text{Ev})$ , there exists a unique map

$$\text{TrjMap}_o^{\text{time,ord}} : \text{TrjDom}_o^{\text{ord}} \longrightarrow \text{Qu}_{\mathfrak{B},o}^{\text{can}}$$

satisfying

$$\text{TrjMap}_o^{\text{can}} = \text{TrjMap}_o^{\text{ord}} \circ \pi_o$$

*Proof.* The map  $\pi_o$  is surjective onto the image in Definition 5.65. Using the equality-kernel inclusion in that definition, apply Lemma 4.180 to

$$\pi = \pi_o, \quad \Phi = \text{TrjMap}_o^{\text{can}}$$

□

**Definition 5.67** (Trajectory-factorization condition bundle and all-dynamical-description-conditions bundle).

$$\begin{aligned} \text{DynTrjFactCondAll}[\text{Ev}] &\iff \text{DynCanTimeTrjCondAll}[\text{Ev}] \\ &\quad \wedge K_{\text{ObsTimeIncTrans}}(\text{Ev}) \\ &\quad \wedge K_{\text{ObsTimeTrjFact}}(\text{Ev}), \\ \text{DynCondAll}[\text{Ev}] &\iff \text{DynLawInputCondAll}[\text{Ev}] \\ &\quad \wedge K_{\text{ObsTimeIncTrans}}(\text{Ev}) \\ &\quad \wedge K_{\text{ObsTimeRealOrdEmb}}(\text{Ev}) \\ &\quad \wedge K_{\text{ObsTimeTrjFact}}(\text{Ev}). \end{aligned}$$

**Definition 5.68** (Canonical observable time curves). Assume  $\text{DynTrjFactCondAll}[\text{Ev}]$  and  $K_{\text{CanObsDesc}}(\text{Ev})$ . For any  $\rho_o \in \text{OrdEmbSet}_o(\mathbb{R})$ , define the real-valued trajectory-time domain by

$$\begin{aligned} \text{TrjDom}_o^{\text{time,real}}[\rho_o] &:= \rho_o[\text{TrjDom}_o^{\text{ord}}] \\ &= \text{TimeRep}_o[\rho_o][\text{TrjDom}_o^{\text{can}}] \subseteq \mathbb{R}. \end{aligned}$$

For each  $\ell \in \text{ObsIdx}^{\text{quot}}$ , define

$$\begin{aligned} \text{ObsCurve}_{o,\zeta,\ell}^{\text{time,ord}} &:= \text{Obs}_\ell^{\text{can}} \circ \text{TrjMap}_o^{\text{ord}}, \\ \text{ObsCurve}_{o,\zeta,\ell}^{\text{time,real}}[\rho_o] &:= \text{ObsCurve}_{o,\zeta,\ell}^{\text{ord}} \circ \rho_o^{-1} \big|_{\text{TrjDom}_o^{\text{real}}[\rho_o]}. \end{aligned}$$

When  $K_{\text{ObsTimeRealOrdEmb}}(\text{Ev})$  holds, let the real trajectory-time domain and the real-valued observable curve corresponding to  $\text{RepEmb}_o$  be

$$\begin{aligned} \text{TrjDom}_o^{\text{real}} &:= \text{TrjDom}_o^{\text{real}}[\text{RepEmb}_o], \\ \text{ObsCurve}_\ell^{\text{real}} &:= \text{ObsCurve}_\ell^{\text{real}}[\text{RepEmb}_o]. \end{aligned}$$

**Proposition 5.69** (Canonical-time factorization of trajectories and observables). Assume  $\text{DynTrjFactCondAll}[\text{Ev}]$  and  $K_{\text{CanObsDesc}}(\text{Ev})$ . For every  $\rho_o \in \text{OrdEmbSet}_o(\mathbb{R})$ ,  $T \in \text{TrjDom}_o^{\text{can}}$ , and  $\ell \in \text{ObsIdx}^{\text{quot}}$ ,

$$\begin{aligned} \text{TrjMap}_o^{\text{can}}(T) &= \text{TrjMap}_o^{\text{ord}}(\text{Prj}_o^{\text{ord}}(T)), \\ \text{Obs}_\ell^{\text{can}}(\text{TrjMap}_o^{\text{can}}(T)) &= \text{ObsCurve}_\ell^{\text{ord}}(\text{Prj}_o^{\text{ord}}(T)) \\ &= \text{ObsCurve}_\ell^{\text{real}}[\rho_o](\text{TimeRep}_o[\rho_o](T)). \end{aligned}$$

Moreover, the real-valued time curve is the unique map on  $\text{TrjDom}_o^{\text{real}}[\rho_o]$  satisfying this factorization identity.

*Proof.* The first identity follows from Proposition 5.66, the second from Definition 5.68, and the third from Proposition 5.62. The real-valued time domain is the image of the order-time domain under  $\rho_o$ , and injectivity gives each element a unique preimage. Since the factorization identity determines the value at that preimage, the real-valued time curve is unique.  $\square$

**Definition 5.70** (Dynamical-description operability). Define dynamical-description operability by

$$\text{DynOperative}[\text{Ev}] \iff \text{ExactOperable}[\text{I}] \wedge \text{DynCondAll}[\text{Ev}].$$

**Definition 5.71** (Dynamical-description object). Assume  $\text{DynOperative}[\text{Ev}]$ , and let  $\rho_o \in \text{OrdEmbSet}_o(\mathbb{R})$ . Define the dynamical-description time structure and the trajectory-and-observation structure relative to this embedding by

$$\begin{aligned} \text{DynTimeStr}_o[\rho_o] &:= \left( \text{Qu}_o^{\text{ord}}, <_o^{\text{ord}}, \text{Prj}_o^{\text{ord}}, \right. \\ &\quad \left. \rho_o, \text{TimeRep}_o[\rho_o] \right), \\ \text{DynTrjObsStr}_{o,\zeta}[\rho_o] &:= \left( \begin{array}{c} \text{TrjDom}_o^{\text{ord}}, \text{TrjMap}_o^{\text{ord}}, \\ \text{TrjDom}_o^{\text{real}}[\rho_o], \\ \left( \begin{array}{c} \text{ObsCurve}_\ell^{\text{ord}}, \\ \text{ObsCurve}_\ell^{\text{real}}[\rho_o] \end{array} \right)_{\ell \in \text{ObsIdx}^{\text{quot}}} \end{array} \right). \end{aligned}$$

Define the dynamical-description object by

$$\text{DynDesc}_{o,\zeta}[\rho_o] := (\text{LawInput}, \text{DynTimeStr}_o[\rho_o], \text{DynTrjObsStr}_{o,\zeta}[\rho_o]).$$

**Definition 5.72** (Dynamical-description witness object). For any  $\rho_o \in \text{OrdEmbSet}_o(\mathbb{R})$ , define the dynamical-description witness object by

$$\text{DynWit}_{o,\zeta}[\rho_o] := (\text{LawInput}, \text{DynDesc}[\rho_o]).$$

**Proposition 5.73** (Existence and uniqueness of the dynamical-description object and witness object). Under  $\text{DynOperative}[\text{Ev}]$ , for every  $\rho_o \in \text{OrdEmbSet}_o(\mathbb{R})$ ,  $\text{DynDesc}[\rho_o]$  and  $\text{DynWit}[\rho_o]$  exist uniquely.

*Proof.* Proposition 5.48, Definition 5.56, Definition 5.63, Proposition 5.66, and Definition 5.68 uniquely determine every component of the two inner structures depending on the same  $\rho_o$ . The conclusion follows from the two outer ordered tuples.  $\square$

**Definition 5.74** ( $\mathfrak{D}$ -record-cell code image). For any  $x_{\mathfrak{D}} \in \text{Tok}_{\mathfrak{D}}^*$ , define the  $\mathfrak{D}$ -record-cell code image by

$$\text{RecordImg}_{\mathfrak{D}}^{\text{code}}(x_{\mathfrak{D}}) := \text{RecordCode}_{\mathfrak{D}}[W_{\mathfrak{D}}(x_{\mathfrak{D}})] \subseteq \text{RecordCodeVal}_{\mathfrak{D}}.$$

**Definition 5.75** (Observer-time-relative  $\mathfrak{D}$ -record images). For a relation  $R \subseteq X \times Y$  and  $A \subseteq Y$ , let the relational inverse image be

$$R^{-1}[A] := \{x \in X \mid \exists y \in A, (x, y) \in R\}$$

For any  $x_{\mathfrak{D}} \in \text{Tok}_{\mathfrak{D}}^*$ , define the canonical-time image, order-time image, and real-time image of the  $\mathfrak{D}$ -record cell by

$$\begin{aligned} \text{RecordImg}_{\mathfrak{D},o}^{\text{time,can}}(x_{\mathfrak{D}}) &:= (\text{RecordCodeRel}_{\mathfrak{C},o})^{-1}[\text{RecordImg}_{\mathfrak{D}}^{\text{code}}(x_{\mathfrak{D}})], \\ \text{RecordImg}_{\mathfrak{D},o}^{\text{time,ord}}(x_{\mathfrak{D}}) &:= \text{Prj}_o^{\text{ord}}[\text{RecordImg}_{\mathfrak{D},o}^{\text{can}}(x_{\mathfrak{D}})], \\ \text{RecordImg}_{\mathfrak{D},o}^{\text{time,real}}(x_{\mathfrak{D}}) &:= \text{TimeRep}_o[\text{RecordImg}_{\mathfrak{D},o}^{\text{can}}(x_{\mathfrak{D}})]. \end{aligned}$$

These images are subsets of  $\text{Qu}_{\mathfrak{C},o}^{\text{can}}$ ,  $\text{Qu}_o^{\text{ord}}$ , and  $\mathbb{R}$ , respectively.

**Proposition 5.76** (Factorization of observer-time-relative  $\mathfrak{D}$ -record images). For every  $x_{\mathfrak{D}} \in \text{Tok}_{\mathfrak{D}}^*$ ,

$$\text{RecordImg}_{\mathfrak{D},o}^{\text{real}}(x_{\mathfrak{D}}) = \text{RepEmb}_o [\text{RecordImg}_{\mathfrak{D},o}^{\text{ord}}(x_{\mathfrak{D}})]$$

holds, and the four images from the code image through the real-time image exist uniquely.

*Proof.* The uniqueness of the first three images follows from the map images and relational inverse images in Definition 5.74 and Definition 5.75. Applying the factorization identity in Definition 5.63 to the canonical-time image gives the displayed factorization identity and uniqueness of the real-time image.  $\square$

**Definition 5.77** (Disjoint union of single-object and joint-object specifications). Define the joint-object specification set and the general evaluation-specification space by

$$\begin{aligned} \text{JointEvSpec} &:= \{(l, o', J) \mid o' \in U, J \subseteq Z_0, 2 \leq |J| < \infty\}, \\ \text{EvalSpecSp} &:= \text{EvSpec} \sqcup \text{JointEvSpec}. \end{aligned}$$

Write the two injections into the disjoint union as

$$\begin{aligned} \iota_{\text{single}} &: \text{EvSpec} \longrightarrow \text{EvalSpecSp}, \\ \iota_{\text{joint}} &: \text{JointEvSpec} \longrightarrow \text{EvalSpecSp}. \end{aligned}$$

The disjoint-union tags distinguish the two branches.

**Definition 5.78** (Agreement of the joint  $\mathfrak{C}$ -time bases). Define the set of exactly operable joint specifications by

$$\text{JointEvSpec}^{\text{exact}} := \{(l, o, J) \in \text{JointEvSpec} \mid \text{ExactOperable}[l]\}.$$

Fix an element  $(l, o, J)$  of this set. For each  $\zeta \in J$ , write the generated equivalence relation relative to  $\text{Ev}_{\zeta} = (l, o, \zeta)$  as  $\sim_{\mathfrak{C},o;\zeta}$ . The source-token set and source order are fixed by the common input  $l$ . Define agreement of the joint-time bases by

$$\begin{aligned} K_{\text{JointCTime}}^{\text{dyn}}(\iota_{\text{joint}}(l, o, J)) \\ \iff \forall \zeta, \eta \in J, \quad \sim_{\mathfrak{C},o;\zeta} = \sim_{\mathfrak{C},o;\eta}. \end{aligned}$$

The domain of this predicate is  $\iota_{\text{joint}}[\text{JointEvSpec}^{\text{exact}}]$ .

**Definition 5.79** (Dynamical readiness of each evaluation-specification branch). Define the set of ready evaluation specifications by

$$\begin{aligned} \text{EvalSpecReady} \\ &:= \iota_{\text{single}}[\{\text{Ev} \in \text{EvSpec} \mid \text{DynOperative}[\text{Ev}]\}] \\ &\cup \iota_{\text{joint}} \left[ \left\{ (l, o, J) \in \text{JointEvSpec}^{\text{exact}} \mid \begin{array}{l} \forall \zeta \in J, \text{DynOperative}[(l, o, \zeta)], \\ K_{\text{JointCTime}}^{\text{dyn}}(\iota_{\text{joint}}(l, o, J)) \end{array} \right\} \right]. \end{aligned}$$

For every  $e \in \text{EvalSpecSp}$ , set

$$\text{DynReady}[e] \iff e \in \text{EvalSpecReady}.$$

**Definition 5.80** (Generated time base of a ready evaluation specification). Take  $e \in \mathbf{EvalSpecReady}$ . In the single-object branch  $e = \iota_{\text{single}}(\mathbf{l}, o, \zeta)$ , use the order-time quotient, strict linear order and trajectory domain for that complete evaluation specification:

$$Q_e := \mathbf{Qu}_o^{\text{ord}}, \quad <_e := <_o^{\text{ord}}, \quad D_e := \mathbf{TrjDom}_o^{\text{ord}}.$$

In the joint branch  $e = \iota_{\text{joint}}(\mathbf{l}, o, J)$ , the object-specific order-time quotients and orders agree. Write their common quotient and order as  $Q_e, <_e$ , and set

$$D_e := \bigcap_{\zeta \in J} \mathbf{TrjDom}_{o;\zeta}^{\text{ord}} \subseteq Q_e.$$

Define the generated time base by

$$\mathbf{EvalTimeBase}[e] := (Q_e, <_e, D_e).$$

This definition does not require the joint domain to be nonempty.

**Proposition 5.81** (Uniqueness of the generated time base and existence of a real representation). For each  $e \in \mathbf{EvalSpecReady}$ , the datum  $\mathbf{EvalTimeBase}[e]$  is uniquely determined, and  $<_e$  is a strict linear order on  $Q_e$ . Moreover,

$$\Omega[e] := \{\rho : Q_e \rightarrow \mathbb{R} \mid \forall q, r \in Q_e, q <_e r \iff \rho(q) < \rho(r)\}$$

is nonempty.

*Proof.* For the single-object branch, use dynamical-description operability and Definition 5.56, Definition 5.65, and Definition 5.59. For the joint branch, the same source set, order and generated equivalence relation determine the same canonical time quotient, natural projection, generating edges and reflexive transitive closure. The antisymmetry and incomparability-transitivity conditions contained in each object's dynamical-description operability therefore give the same order-time quotient and strict linear order. Each individual trajectory domain is a subset of this common quotient, and their intersection is unique. Since the object set is nonempty, a real order embedding for one object gives an embedding of the common quotient. Agreement of the quotient and order makes this existence condition independent of the object used.  $\square$

**Definition 5.82** (General law-candidate relation-structure space). For a set  $\mathbb{T}$ , define the law-component space by

$$\begin{aligned} \mathbf{LawCompSp}[\mathbb{T}] := \Big\{ \mathbf{L} = (X, Y, f, g, E, R, H, V, h) \mid \\ & H \subseteq \mathbb{T}, \quad f : H \rightarrow X, \quad g : H \rightarrow Y, \quad E \subseteq \mathbb{T} \times X, \\ & R \subseteq E \times Y, \\ & V = (\mathbb{T} \times X) \times Y, \quad h : H \rightarrow V, \quad h(t) = ((t, f(t)), g(t)) \Big\}. \end{aligned}$$

For each  $\mathbf{L} \in \mathbf{LawCompSp}[\mathbb{T}]$ , let its component accessors satisfy

$$\begin{aligned} & \left( \mathbf{InValSp}, \mathbf{OutValSp}, \mathbf{InVal}, \mathbf{OutVal}, \right. \\ & \quad \mathbf{EvalDom}, \mathbf{CandRel}, \mathbf{EvalTimeDom}, \\ & \quad \left. \mathbf{EvalTupleSp}, \mathbf{EvalTuple} \right) [\mathbf{L}] := \mathbf{L} \end{aligned}$$

Also let

$$\text{Bij}(S) := \{f : S \rightarrow S \mid f \text{ is a bijection}\}$$

Define the general law-candidate relation-structure space by

$$\begin{aligned} \text{CandRelStrSp}^{\text{law}} := & \left\{ (\mathbb{T}, A, (L_a)_{a \in A}, \mathcal{U}, \mathcal{F}) \mid A \neq \emptyset, \right. \\ & \forall a \in A, L_a \in \text{LawCompSp}[\mathbb{T}], \quad \mathcal{U} \subseteq \mathcal{P}(\mathbb{T}), \\ & \left. \mathcal{F} \subseteq \prod_{a \in A} \text{Bij}(\text{EvalTupleSp}[L_a]) \right\}. \end{aligned}$$

**Definition 5.83** (Single-object law-evaluation components). Let  $\text{LawIdx}$  be a nonempty law-index set, and, for each  $a \in \text{LawIdx}$ , let

$$\text{InIdx}_a^{\text{law}}, \text{OutIdx}_a^{\text{law}} \subseteq \text{ObsIdx}^{\text{quot}}$$

Let the side-index set be  $\text{IOSide} := \{\text{in}, \text{out}\}$ , and set

$$\begin{aligned} \text{ObsIdx}_{a,\text{in}}^{\text{law}} &:= \text{InIdx}_a^{\text{law}}, \\ \text{ObsIdx}_{a,\text{out}}^{\text{law}} &:= \text{OutIdx}_a^{\text{law}} \end{aligned}$$

Let the individual law-evaluation time domain satisfy

$$\text{EvalTimeDom}_a^{\text{law}} \subseteq \text{TrjDom}_o^{\text{real}}$$

For each  $s \in \text{IOSide}$ , define

$$\begin{aligned} \text{ValSp}_{a,s}^{\text{law}} &:= \prod_{\ell \in \text{ObsIdx}_{a,s}^{\text{law}}} \text{ObsVal}_\ell^{\text{can}}, \\ \text{Val}_{a,s}^{\text{law}} : \text{EvalTimeDom}_a^{\text{law}} &\longrightarrow \text{ValSp}_{a,s}^{\text{law}}, \\ \text{Val}_{a,s}^{\text{law}}(t) &:= (\text{ObsCurve}_\ell^{\text{real}}(t))_{\ell \in \text{ObsIdx}_{a,s}^{\text{law}}} \end{aligned}$$

and let the component projections be

$$\begin{aligned} (\text{InValSp}_a^{\text{law}}, \text{OutValSp}_a^{\text{law}}) &:= (\text{ValSp}_{a,\text{in}}^{\text{law}}, \text{ValSp}_{a,\text{out}}^{\text{law}}), \\ (\text{InVal}_a^{\text{law}}, \text{OutVal}_a^{\text{law}}) &:= (\text{Val}_{a,\text{in}}^{\text{law}}, \text{Val}_{a,\text{out}}^{\text{law}}). \end{aligned}$$

Let the individual law-application domain and candidate relation satisfy

$$\begin{aligned} \text{EvalDom}_a^{\text{law}} &\subseteq \text{TrjDom}_o^{\text{real}} \times \text{InValSp}_a^{\text{law}}, \\ \text{CandRel}_a^{\text{law}} &\subseteq \text{EvalDom}_a^{\text{law}} \times \text{OutValSp}_a^{\text{law}}. \end{aligned}$$

Define the law-evaluation-tuple space and evaluation tuple by

$$\begin{aligned} \text{EvalTupleSp}_a^{\text{law}} &:= (\text{TrjDom}_o^{\text{real}} \times \text{InValSp}_a^{\text{law}}) \times \text{OutValSp}_a^{\text{law}}, \\ \text{EvalTuple}_a^{\text{law}} : \text{EvalTimeDom}_a^{\text{law}} &\longrightarrow \text{EvalTupleSp}_a^{\text{law}}, \\ \text{EvalTuple}_a^{\text{law}}(t) &:= ((t, \text{InVal}_a^{\text{law}}(t)), \text{OutVal}_a^{\text{law}}(t)). \end{aligned}$$

Let these nine components be

$$\begin{aligned} \text{LawEvalComp}_a &:= (\text{InValSp}_a^{\text{law}}, \text{OutValSp}_a^{\text{law}}, \text{InVal}_a^{\text{law}}, \text{OutVal}_a^{\text{law}}, \text{EvalDom}_a^{\text{law}}, \\ &\quad \text{CandRel}_a^{\text{law}}, \text{EvalTimeDom}_a^{\text{law}}, \text{EvalTupleSp}_a^{\text{law}}, \text{EvalTuple}_a^{\text{law}}). \end{aligned}$$

The family of admissible common law-time domains satisfies

$$\text{AdmCommonTimeDomSet}^{\text{law}} \subseteq \mathcal{P}(\text{TrjDom}_o^{\text{real}})$$



**Definition 5.84** (Family of admissible faithful law-representation transformations). For each  $a \in \text{LawIdx}$ , let

$$\begin{aligned} \text{EvalTupleBijSet}_a^{\text{law}} &:= \text{Bij}(\text{EvalTupleSp}_a^{\text{law}}), \\ \text{FaithTransSet}^{\text{law}} &\subseteq \prod_{a \in \text{LawIdx}} \text{EvalTupleBijSet}_a^{\text{law}}. \end{aligned}$$

**Definition 5.85** (Law-evaluation datum). Define the single-object law-candidate relation structure by

$$\begin{aligned} &\text{CandRelStr}_{o, \zeta}^{\text{law}}, \\ &\text{TrjDom}_o^{\text{real}}, \\ &:= \left( \begin{array}{c} \text{LawIdx}, (\text{LawEvalComp}_a)_{a \in \text{LawIdx}}, \\ \text{AdmCommonTimeDomSet}^{\text{law}}, \text{FaithTransSet}^{\text{law}} \end{array} \right) \\ &\in \text{CandRelStrSp}^{\text{law}}. \end{aligned}$$

Define the law-evaluation-datum space and the single-object law-evaluation datum by

$$\begin{aligned} \text{LawEvalDatSp} &:= \text{EvalSpecSp} \times \text{CandRelStrSp}^{\text{law}}, \\ \text{LawEvalDat} &:= (\iota_{\text{single}}(\text{Ev}), \text{CandRelStr}^{\text{law}}) \in \text{LawEvalDatSp}. \end{aligned}$$

For any  $\text{LawDat} \in \text{LawEvalDatSp}$ , let

$$\begin{aligned} \text{EvalSpec}[\text{LawDat}] &:= \text{pr}_1(\text{LawDat}), \\ \text{CandRelStr}[\text{LawDat}] &:= \text{pr}_2(\text{LawDat}), \\ \text{EvalCouplLoc}[\text{LawDat}] &:= \begin{cases} o', & \text{EvalSpec}[\text{LawDat}] = \iota_{\text{single}}(\text{I}, o', \zeta'), \\ o', & \text{EvalSpec}[\text{LawDat}] = \iota_{\text{joint}}(\text{I}, o', J) \end{cases} \in U. \end{aligned}$$

Write the second component as

$$\begin{aligned} \text{CandRelStr}[\text{LawDat}] &= \\ &\left( \begin{array}{c} \text{LawTimeDom}[\text{LawDat}], \text{LawIdx}[\text{LawDat}], \\ (\text{LawEvalComp}_a[\text{LawDat}])_{a \in \text{LawIdx}[\text{LawDat}]}, \\ \text{AdmCommonTimeDomSet}^{\text{law}}[\text{LawDat}], \text{FaithTransSet}^{\text{law}}[\text{LawDat}] \end{array} \right). \end{aligned}$$

For any  $\text{LawDat} \in \text{LawEvalDatSp}$  and  $a \in \text{LawIdx}[\text{LawDat}]$ , define dot notation for an outer-component accessor  $F$  and a law-component accessor  $G$  by

$$\begin{aligned} \text{LawDat}.F &:= F[\text{LawDat}], \\ \text{LawDat}[a] &:= \text{LawEvalComp}_a[\text{LawDat}], \\ \text{LawDat}[a].G &:= G[\text{LawDat}[a]]. \end{aligned}$$

Here  $F$  ranges over the outer-component accessors of this definition, and  $G$  ranges over the law-component accessors of Definition 5.82.

**Proposition 5.86** (Type consistency of the single-object law-evaluation datum). Hereafter set  $\text{LawDat} := \text{LawEvalDat}$ . Then

$$\begin{aligned} \iota_{\text{single}}(\text{Ev}) &\in \text{EvalSpecSp}, \\ \text{CandRelStr}^{\text{law}} &\in \text{CandRelStrSp}^{\text{law}}, \\ \text{LawDat} &\in \text{LawEvalDatSp}. \end{aligned}$$

Moreover,

$$\begin{aligned}\text{LawDat.EvalSpec} &= \iota_{\text{single}}(\text{Ev}), \\ \text{LawDat.EvalCouplLoc} &= o, \\ \text{LawDat.CandRelStr} &= \text{CandRelStr}^{\text{law}}, \\ \text{LawDat}[a] &= \text{LawEvalComp}_a \quad (a \in \text{LawIdx}).\end{aligned}$$

All projected values of the outer and law components are obtained by applying the corresponding component accessors to these record equalities.

*Proof.* Definition 5.2 and Definition 5.77 give  $\iota_{\text{single}}(\text{Ev}) \in \text{EvalSpecSp}$ . By Definition 5.83 and Definition 5.84, every law component and the two family components have the types in Definition 5.82. Thus the ordered pair of the evaluation specification and the candidate relation structure belongs to the displayed space. The displayed equalities follow by applying the dot projections of Definition 5.85 to **LawDat**.  $\square$

**Definition 5.87** (Individual law-application-domain condition). For any  $\text{LawDat} \in \text{LawEvalDatSp}$ , define the individual law-application-domain condition by

$$\begin{aligned}K_{\text{IndivEvalDomAdm}}^{\text{law}}[\text{LawDat}] &\iff \forall a \in \text{LawDat.LawIdx}, \\ &\quad \left[ \text{LawDat}[a].\text{EvalTimeDom} \neq \emptyset \right. \\ &\quad \wedge \forall t \in \text{LawDat}[a].\text{EvalTimeDom}, \\ &\quad \left. (t, \text{LawDat}[a].\text{InVal}(t)) \in \text{LawDat}[a].\text{EvalDom} \right].\end{aligned}$$

**Proposition 5.88** (Type consistency of the individual law-application-domain condition). For any  $\text{LawDat} \in \text{LawEvalDatSp}$ ,  $a \in \text{LawDat.LawIdx}$ , and  $t \in \text{LawDat}[a].\text{EvalTimeDom}$ ,

$$(t, \text{LawDat}[a].\text{InVal}(t)) \in \text{LawDat.LawTimeDom} \times \text{LawDat}[a].\text{InValSp}.$$

*Proof.* The law-component type in Definition 5.82 and the dot projections in Definition 5.85 give

$$\begin{aligned}\text{LawDat}[a].\text{EvalTimeDom} &\subseteq \text{LawDat.LawTimeDom}, \\ \text{LawDat}[a].\text{InVal} : \text{LawDat}[a].\text{EvalTimeDom} &\rightarrow \text{LawDat}[a].\text{InValSp}.\end{aligned}$$

Hence the product membership holds.  $\square$

**Definition 5.89** (Law-output determinacy condition). For sets  $X, Y$  and a relation  $R \subseteq X \times Y$ , define right uniqueness by

$$\text{RightUnique}(R) \iff \forall x \in X \forall y_0, y_1 \in Y, \quad [(x, y_0) \in R \wedge (x, y_1) \in R] \implies y_0 = y_1.$$

For any  $\text{LawDat} \in \text{LawEvalDatSp}$ , define the law-output determinacy condition by

$$\begin{aligned}K_{\text{CandRelRightUnique}}^{\text{law}}[\text{LawDat}] &\iff \forall a \in \text{LawDat.LawIdx}, \\ &\quad \text{RightUnique}(\text{LawDat}[a].\text{CandRel}).\end{aligned}$$

**Definition 5.90** (Domains of law-candidate relations). For any  $\text{LawDat} \in \text{LawEvalDatSp}$  and  $a \in \text{LawDat.LawIdx}$ , define the domain of the law-candidate relation by

$$\text{CandRelDom}_a^{\text{law}}[\text{LawDat}] := \text{Dom}(\text{LawDat}[a].\text{CandRel}).$$

**Proposition 5.91** (Partial-output-map representation of a law-candidate relation). For any  $\text{LawDat} \in \text{LawEvalDatSp}$ , assume

$$K_{\text{CandRelRightUnique}}^{\text{law}}[\text{LawDat}]$$

For each  $a \in \text{LawDat.LawIdx}$ , there exists a unique map

$$\text{CandRelOutMap}_a^{\text{law}}[\text{LawDat}] : \text{CandRelDom}_a^{\text{law}}[\text{LawDat}] \longrightarrow \text{LawDat}[a].\text{OutValSp}$$

satisfying

$$\text{LawDat}[a].\text{CandRel} = \left\{ \begin{array}{l} (x, \text{CandRelOutMap}_a^{\text{law}}[\text{LawDat}](x)) \\ \mid x \in \text{CandRelDom}_a^{\text{law}}[\text{LawDat}] \end{array} \right\}.$$

*Proof.* By Definition 5.90, at least one output value corresponds to each element of the relation domain. Right uniqueness in Definition 5.89 makes that output value unique. Hence there exists a map taking each domain element to its unique corresponding output value, and its graph equals the law-candidate relation. A map with the same graph takes the same value at every domain element and is therefore unique.  $\square$

**Definition 5.92** (Generated law-evaluation relation-membership condition). For any  $\text{LawDat} \in \text{LawEvalDatSp}$ , define the generated law-evaluation relation-membership condition by

$$\begin{aligned} & K_{\text{GenEvalCandRelMem}}^{\text{law}}[\text{LawDat}] \\ & \iff K_{\text{IndivEvalDomAdm}}^{\text{law}}[\text{LawDat}] \\ & \wedge \forall a \in \text{LawDat.LawIdx}, \quad \forall t \in \text{LawDat}[a].\text{EvalTimeDom}, \\ & \quad \text{LawDat}[a].\text{EvalTuple}(t) \in \text{LawDat}[a].\text{CandRel}. \end{aligned}$$

**Proposition 5.93** (Type consistency of the generated law-evaluation relation-membership condition). For any  $\text{LawDat} \in \text{LawEvalDatSp}$ , assume

$$K_{\text{IndivEvalDomAdm}}^{\text{law}}[\text{LawDat}].$$

For every

$$a \in \text{LawDat.LawIdx}, \quad t \in \text{LawDat}[a].\text{EvalTimeDom}$$

one has

$$\text{LawDat}[a].\text{EvalTuple}(t) \in \text{LawDat}[a].\text{EvalDom} \times \text{LawDat}[a].\text{OutValSp}.$$

*Proof.* Definition 5.87 gives  $(t, \text{LawDat}[a].\text{InVal}(t)) \in \text{LawDat}[a].\text{EvalDom}$ . The law-component type in Definition 5.82 gives  $\text{LawDat}[a].\text{OutVal}(t) \in \text{LawDat}[a].\text{OutValSp}$ , and expanding the definition of the evaluation-tuple projection yields the product membership.  $\square$

**Definition 5.94** (Faithful-representation-transformation invariance condition for law-candidate relations). For any  $\text{LawDat} \in \text{LawEvalDatSp}$ , define the faithful-representation-transformation invariance condition for law-candidate relations by

$$\begin{aligned} & K_{\text{CandRelFaithTransInv}}^{\text{law}}[\text{LawDat}] \\ & \iff \left[ \forall (\phi_a)_{a \in \text{LawDat.LawIdx}} \in \text{LawDat.FaithTransSet}, \right. \\ & \quad \left. \forall a \in \text{LawDat.LawIdx}, \quad \phi_a[\text{LawDat}[a].\text{CandRel}] = \text{LawDat}[a].\text{CandRel} \right]. \end{aligned}$$

**Proposition 5.95** (Law-candidate-relation invariance under inverses of faithful representation transformations).

For any  $\text{LawDat} \in \text{LawEvalDatSp}$ , assume

$$K_{\text{CandRelFaithTransInv}}^{\text{law}}[\text{LawDat}].$$

For every

$$(\phi_a)_{a \in \text{LawDat.LawIdx}} \in \text{LawDat.FaithTransSet}$$

and  $a \in \text{LawDat.LawIdx}$ , the inverse map

$$\phi_a^{-1} : \text{LawDat}[a].\text{EvalTupleSp} \longrightarrow \text{LawDat}[a].\text{EvalTupleSp}$$

exists and satisfies

$$\phi_a^{-1}[\text{LawDat}[a].\text{CandRel}] = \text{LawDat}[a].\text{CandRel}.$$

Consequently, for every  $z \in \text{LawDat}[a].\text{EvalTupleSp}$ ,

$$z \in \text{LawDat}[a].\text{CandRel} \iff \phi_a^{-1}(z) \in \text{LawDat}[a].\text{CandRel}$$

holds.

*Proof.* By Definition 5.84, every  $\phi_a$  is a bijection on the law-evaluation-tuple space. Apply  $\phi_a^{-1}$  to both sides of the image-invariance identity in Definition 5.94, and use  $\phi_a^{-1} \circ \phi_a = \text{id}$  to obtain the set-image identity under the inverse map. Expanding this identity as element membership gives the stated equivalence.  $\square$

**Definition 5.96** (Individual and common valid time domains for laws). For any  $\text{LawDat} \in \text{LawEvalDatSp}$  and  $a \in \text{LawDat.LawIdx}$ , define the individual valid time domain for the law by

$$\begin{aligned} \text{ValidTimeDom}_a^{\text{law}}[\text{LawDat}] &:= \text{LawDat}[a].\text{EvalTimeDom} \\ &\quad \cap (\text{LawDat}[a].\text{EvalTuple})^{-1}[\text{LawDat}[a].\text{CandRel}]. \end{aligned}$$

Define the common valid time domain by

$$\text{CommonValidTimeDom}^{\text{law}}[\text{LawDat}] := \bigcap_{a \in \text{LawDat.LawIdx}} \text{ValidTimeDom}_a^{\text{law}}[\text{LawDat}].$$

**Definition 5.97** (Common valid time-domain condition for laws). For any  $\text{LawDat} \in \text{LawEvalDatSp}$ , define the common valid time-domain condition for laws by

$$\begin{aligned} K_{\text{CommonValidTimeDomAdm}}^{\text{law}}[\text{LawDat}] &\iff \left[ K_{\text{GenEvalCandRelMem}}^{\text{law}}[\text{LawDat}] \right. \\ &\quad \wedge \text{CommonValidTimeDom}^{\text{law}}[\text{LawDat}] \neq \emptyset \\ &\quad \wedge \text{CommonValidTimeDom}^{\text{law}}[\text{LawDat}] \\ &\quad \left. \in \text{LawDat.AdmCommonTimeDomSet} \right]. \end{aligned}$$

**Proposition 5.98** (Type of the common valid time domain and simultaneous satisfaction of all laws). For any  $\text{LawDat} \in \text{LawEvalDatSp}$ , assume  $K_{\text{CommonValidTimeDomAdm}}^{\text{law}}[\text{LawDat}]$ . Then

$$\emptyset \neq \text{CommonValidTimeDom}^{\text{law}}[\text{LawDat}] \subseteq \text{LawDat.LawTimeDom}.$$

Moreover, for every  $t \in \text{CommonValidTimeDom}^{\text{law}}[\text{LawDat}]$  and  $a \in \text{LawDat.LawIdx}$ ,

$$\begin{aligned} t &\in \text{LawDat}[a].\text{EvalTimeDom}, \\ \text{LawDat}[a].\text{EvalTuple}(t) &\in \text{LawDat}[a].\text{CandRel} \end{aligned}$$

holds.

*Proof.* Definition 5.97 and

$$\text{LawDat.AdmCommonTimeDomSet} \subseteq \mathcal{P}(\text{LawDat.LawTimeDom})$$

give nonemptiness and the subset relation. By the definition of the intersection, for every  $a$ ,  $t \in \text{ValidTimeDom}_a^{\text{law}}[\text{LawDat}]$ . Expanding the inverse-image expression in Definition 5.96 gives the two memberships.  $\square$

**Definition 5.99** (Directed acyclic graph of law-condition dependencies). Define the law-condition node set and the five-condition vector by

$$\text{LawCondNodeSet} := \{1, 2, 3, 4, 5\},$$

$$\begin{aligned} \mathbf{K}^{\text{law}}[\text{LawDat}] := & \left( K_{\text{IndivEvalDomAdm}}^{\text{law}}[\text{LawDat}], K_{\text{CandRelRightUnique}}^{\text{law}}[\text{LawDat}], \right. \\ & K_{\text{GenEvalCandRelMem}}^{\text{law}}[\text{LawDat}], \\ & \left. K_{\text{CandRelFaithTransInv}}^{\text{law}}[\text{LawDat}], K_{\text{CommonValidTimeDomAdm}}^{\text{law}}[\text{LawDat}] \right). \end{aligned}$$

For every  $i \in \text{LawCondNodeSet}$ , let

$$\text{LawCondSat}[\text{LawDat}, i] \iff \mathbf{K}_i^{\text{law}}[\text{LawDat}].$$

Define the direct-dependency edge set and the law-condition dependency graph by

$$\begin{aligned} \text{LawCondEdgeSet} &:= \{(1, 3), (3, 5)\}, \\ \text{LawCondDAG} &:= (\text{LawCondNodeSet}, \text{LawCondEdgeSet}). \end{aligned}$$

**Proposition 5.100** (Acyclicity of the law-condition dependency graph). The graph  $\text{LawCondDAG}$  is a directed acyclic graph. Its root-node set is  $\{1, 2, 4\}$ , and the strict-ancestor sets of its nodes are

$$\begin{aligned} \text{Anc}(1) &= \text{Anc}(2) = \text{Anc}(4) = \emptyset, \\ \text{Anc}(3) &= \{1\}, \quad \text{Anc}(5) = \{1, 3\}. \end{aligned}$$

*Proof.* Take the rank map

$$\begin{aligned} \text{LawCondRank} : \text{LawCondNodeSet} &\longrightarrow \{0, 1, 2\}, \\ (\text{LawCondRank}(i))_{i=1}^5 &:= (0, 0, 1, 0, 2). \end{aligned}$$

For every  $(i, j) \in \text{LawCondEdgeSet}$ , one has  $\text{LawCondRank}(i) < \text{LawCondRank}(j)$ ; hence no directed cycle exists. Expanding the edge set gives the root-node set and strict-ancestor sets.  $\square$

**Definition 5.101** (All-law-conditions bundle). For any  $\text{LawDat} \in \text{LawEvalDatSp}$ , define the all-law-conditions bundle by

$$\text{LawCondAll}[\text{LawDat}] \iff \forall i \in \text{LawCondNodeSet}, \quad \mathbf{K}_i^{\text{law}}[\text{LawDat}].$$

**Definition 5.102** (Law-description operability). For any  $\text{LawDat} \in \text{LawEvalDatSp}$ , define law-description operability by

$$\text{LawOperative}[\text{LawDat}] \iff \text{DynReady}[\text{LawDat.EvalSpec}] \wedge \text{LawCondAll}[\text{LawDat}].$$

**Definition 5.103** (Transport of a law datum along a bijection of time domains). Take a law-evaluation datum and a bijection

$$D \in \text{LawEvalDatSp}, \quad \tau : D.\text{LawTimeDom} \longrightarrow \mathbb{T}'.$$

Use the following abbreviations.

$$\mathbb{T} := D.\text{LawTimeDom}, \quad A := D.\text{LawIdx}, \quad (X_a, Y_a, f_a, g_a, E_a, R_a, H_a, V_a, h_a) := D[a].$$

For each  $a \in A$ , define the change of time component on evaluation tuples by

$$\begin{aligned} \tau_a^{\text{eval}} : V_a &\longrightarrow (\mathbb{T}' \times X_a) \times Y_a, \\ \tau_a^{\text{eval}}((t, x), y) &:= ((\tau(t), x), y). \end{aligned}$$

Define the nine transported components by

$$\begin{aligned} X'_a &:= X_a, & Y'_a &:= Y_a, \\ H'_a &:= \tau[H_a], \\ f'_a &:= f_a \circ (\tau^{-1}|_{H'_a}), & g'_a &:= g_a \circ (\tau^{-1}|_{H'_a}), \\ E'_a &:= (\tau \times \text{id}_{X_a})[E_a], \\ R'_a &:= \tau_a^{\text{eval}}[R_a], \\ V'_a &:= (\mathbb{T}' \times X_a) \times Y_a, \\ h'_a &:= \tau_a^{\text{eval}} \circ h_a \circ (\tau^{-1}|_{H'_a}), \end{aligned}$$

and transport the families of admissible common time domains and faithful representation changes by

$$\begin{aligned} \mathcal{U}' &:= \{\tau[U] \mid U \in D.\text{AdmCommonTimeDomSet}\}, \\ \mathcal{F}' &:= \left\{ (\tau_a^{\text{eval}} \circ \phi_a \circ (\tau_a^{\text{eval}})^{-1})_{a \in A} \mid (\phi_a)_{a \in A} \in D.\text{FaithTransSet} \right\}. \end{aligned}$$

Define transport of the law datum along the bijection of time domains by

$$\begin{aligned} L'_a &:= (X'_a, Y'_a, f'_a, g'_a, E'_a, R'_a, H'_a, V'_a, h'_a), \\ \text{LawTimeTransport}[D; \tau] &:= (D.\text{EvalSpec}, (\mathbb{T}', A, (L'_a)_{a \in A}, \mathcal{U}', \mathcal{F}')). \end{aligned}$$

**Proposition 5.104** (Typing and five-condition preservation under law-datum transport). Under the preceding definition,

$$D' := \text{LawTimeTransport}[D; \tau] \in \text{LawEvalDatSp}.$$

Moreover, the five law conditions and the common valid-time domain satisfy

$$\begin{aligned} \forall i \in \text{LawCondNodeSet}, \quad \mathbf{K}_i^{\text{law}}[D'] &\iff \mathbf{K}_i^{\text{law}}[D], \\ \text{CommonValidTimeDom}^{\text{law}}[D'] &= \tau[\text{CommonValidTimeDom}^{\text{law}}[D]]. \end{aligned}$$

*Proof.* Images under the bijection and restrictions of its inverse give the nine components the types required by Definition 5.82. The admissible domains form a family of subsets of  $\mathbb{T}'$ , and each conjugated change is a bijection. Thus the transported outer datum belongs to the same law-evaluation space.

For the first condition, bijective images preserve nonemptiness and membership in the applicability domain. For the second condition, replacing the time component by a bijection while retaining the input and output values preserves and reflects right uniqueness of the output. For the third condition, evaluation tuples are transported by  $\tau_a^{\text{eval}}$ , which preserves and reflects membership in the candidate relation. For the fourth condition, the definitions of the relation image and conjugated changes make image invariance equivalent before and after transport. Each individual valid-time domain is the image of its original domain under  $\tau$ . Since bijective images commute with intersections, the displayed equality of common valid-time domains follows. Nonemptiness and membership in the admissible-domain family are also preserved, giving equivalence of the fifth condition.  $\square$

**Definition 5.105** (Law-evaluation data compatible with a generated time domain). For a map  $\rho$ , use the time base of Definition 5.80 to define

$$\begin{aligned} & \text{LawEvalDatSp}^{\text{time, gen}}[\rho] \\ & := \left\{ \text{LawDat} \in \text{LawEvalDatSp} \mid \begin{array}{l} \exists e \in \text{EvalSpecReady}, \\ \text{LawDat.EvalSpec} = e, \quad \rho \in \Omega[e], \\ \text{LawDat.LawTimeDom} \subseteq \rho[D_e] \end{array} \right\}. \end{aligned}$$

Here  $e$  is uniquely determined by  $\text{LawDat.EvalSpec}$ . Below, the superscript  $\text{time, gen}$  is abbreviated to  $\text{gen}$ .

**Definition 5.106** (Law-time admissibility). For any  $\text{LawDat} \in \text{LawEvalDatSp}$  and map  $\rho$ , define law-time admissibility by

$$\begin{aligned} & \text{LawTimeAdm}[\text{LawDat}; \rho] \\ & \iff \text{LawDat} \in \text{LawEvalDatSp}^{\text{gen}}[\rho] \wedge \text{LawCondAll}[\text{LawDat}]. \end{aligned}$$

**Definition 5.107** (Law data relative to a time representation). Fix a base representation  $\rho^0$  and a law datum satisfying

$$\text{LawDat} \in \text{LawEvalDatSp}^{\text{gen}}[\rho^0].$$

Set

$$\begin{aligned} e &:= \text{LawDat.EvalSpec}, \\ D^0 &:= (\rho^0)^{-1}[\text{LawDat.LawTimeDom}] \subseteq D_e, \\ \tau_\rho &:= \rho \circ (\rho^0|_{D^0})^{-1} : \text{LawDat.LawTimeDom} \longrightarrow \rho[D^0], \\ \text{LawDat}[\rho] &:= \text{LawTimeTransport}[\text{LawDat}; \tau_\rho] \quad (\rho \in \Omega[e]). \end{aligned}$$

The map  $\tau_\rho$  is the bijection between two representations of the same generated time subdomain  $D^0$ . The family is uniquely determined by the base representation and base law datum, and satisfies

$$\begin{aligned} & \text{LawDat}[\rho^0] = \text{LawDat}, \\ & \text{LawDat}[\rho] \in \text{LawEvalDatSp}^{\text{gen}}[\rho]. \end{aligned}$$

For the single-object datum  $\text{LawEvalDat}$  of Definitions 5.83 and 5.85,  $\rho^0$  is the representation used in its construction,  $e = \iota_{\text{single}}(\text{Ev})$ , and  $D^0 = D_e$ . Hereafter  $\text{LawEvalDat}[\rho]$  denotes this specialization.

**Definition 5.108** (Observer-time law core). Assume

$$\text{LawTimeAdm}[\text{LawEvalDat}[\text{RepEmb}_o]; \text{RepEmb}_o].$$

Define the observer-time law core relative to this real-valued time-order embedding by

$$\text{LawCore}_{o,\zeta}^{\text{obs,time}}(\text{RepEmb}_o) := \left( \text{DynDesc}[\text{RepEmb}_o], \right. \\ \left. \text{LawEvalDat}[\text{RepEmb}_o].\text{CandRelStr} \right).$$

**Proposition 5.109** (Existence and uniqueness of the observer-time law core). Under

$$\text{LawTimeAdm}[\text{LawEvalDat}[\text{RepEmb}_o]; \text{RepEmb}_o],$$

$\text{LawCore}^{\text{obs}}(\text{RepEmb}_o)$  exists uniquely. Moreover,

$$\text{LawCondAll}[\text{LawEvalDat}[\text{RepEmb}_o]], \\ \text{CommonValidTimeDom}^{\text{law}}[\text{LawEvalDat}[\text{RepEmb}_o]] \neq \emptyset$$

hold.

*Proof.* Definition 5.106 gives law operability. The equality

$$\text{LawEvalDat}[\text{RepEmb}_o].\text{EvalSpec} \\ = \iota_{\text{single}}(\text{Ev})$$

from Definition 5.103 and Proposition 5.86, together with Proposition 5.73, gives the unique existence of the first component. The second component is unique as a record projection in Definition 5.85; hence the law core also exists uniquely.

The same law operability gives the all-law-conditions bundle. Its fifth component and Definition 5.97 imply that the common valid time domain is nonempty.  $\square$

**Definition 5.110** (Generated representability of law-evaluation time). For a map  $\rho$  and  $\text{LawDat} \in \text{LawEvalDatSp}^{\text{gen}}[\rho]$ , put  $e := \text{LawDat}.\text{EvalSpec}$ . Define generated representability of law-evaluation time by

$$K_{\text{CommonEvalTimeGenRep}}^{\text{law}}[\text{LawDat}; \rho] \\ \iff \emptyset \neq \text{CommonValidTimeDom}^{\text{law}}[\text{LawDat}] \\ \subseteq \rho[D_e] \cap \bigcap_{a \in \text{LawDat}.\text{LawIdx}} (\text{LawDat}[a].\text{EvalTuple})^{-1}[\text{LawDat}[a].\text{CandRel}].$$

**Theorem 5.111** (Generated representability of law-evaluation time). For a map  $\rho$  and  $\text{LawDat} \in \text{LawEvalDatSp}^{\text{gen}}[\rho]$ , assume  $\text{LawOperative}[\text{LawDat}]$ . Then

$$K_{\text{CommonEvalTimeGenRep}}^{\text{law}}[\text{LawDat}; \rho].$$



*Proof.* Law operativity gives the fifth component of the all-law-conditions bundle. By Proposition 5.98, the common valid time domain is nonempty and lies in the inverse image of each candidate relation under its law-evaluation tuple map. Put  $e := \text{LawDat.EvalSpec}$ . Definition 5.105 also gives  $\text{LawDat.LawTimeDom} \subseteq \rho[D_e]$ . The same proposition places the common valid time domain inside this law time domain, proving the required inclusion.  $\square$

**Theorem 5.112** (Generation theorem for the dynamical-description stage). Under the dynamical-description operability of Definition 5.70,

$$\text{DynOperative}[\text{Ev}]$$

one has

$$\text{OrdEmbSet}_o(\mathbb{R}) \neq \emptyset,$$

and the conclusion of Proposition 5.73 holds for every element of this set.

*Proof.* The real-valued time-order embedding condition in Definition 5.67, together with Definition 5.59, gives the nonemptiness of the displayed set. Proposition 5.73 gives the conclusion for each element.  $\square$

**Theorem 5.113** (Generation at the law-description stage).

Assume  $\text{LawOperative}[\text{LawEvalDat}]$  for the base law datum. For every  $\rho_o \in \text{OrdEmbSet}_o(\mathbb{R})$ , the observer-time law core

$$\text{LawCore}^{\text{obs}}(\rho_o)$$

exists uniquely, and

$$K_{\text{CommonEvalTimeGenRep}}^{\text{law}}[\text{LawEvalDat}[\rho_o]; \rho_o]$$

holds.

*Proof.* Law operativity gives dynamical readiness and the five law conditions for the base datum. The transport in Definition 5.107 preserves the evaluation specification and maps the time domain to the image of the same trajectory domain under  $\rho_o$ . Proposition 5.104 also preserves the five law conditions. Thus  $\text{LawTimeAdm}[\text{LawEvalDat}[\rho_o]; \rho_o]$  holds. Proposition 5.109 gives existence and uniqueness of the law core. Applying Theorem 5.111 to the transported datum, which is law operative and compatible with the generated time domain, proves generated representability.  $\square$

**Definition 5.114** (Concatenation of finite tuples). For finite tuples  $A = (a_1, \dots, a_m)$  and  $B = (b_1, \dots, b_n)$ , define their concatenation by

$$A \frown B := (a_1, \dots, a_m, b_1, \dots, b_n).$$

**Definition 5.115** (Foundational datum for the differential stage). Assume

$$\text{LawTimeAdm}[\text{LawEvalDat}[\text{RepEmb}_o]; \text{RepEmb}_o].$$

Define the time foundation and the trajectory-and-observation foundation for the differential stage by

$$\begin{aligned} \text{DiffTimeBase}_o[\text{RepEmb}_o] &:= \left( \text{Qu}_o^{\text{ord}}, \right. \\ &\quad \text{RepEmb}_o, \\ &\quad \left. \text{TimeRep}_o[\text{RepEmb}_o] \right), \\ \text{DiffTrjObsBase}_{o,\zeta} &:= \left( \text{TrjMap}_o^{\text{ord}}, \right. \\ &\quad \left. \left( \text{ObsCurve}_\ell^{\text{real}}[\text{RepEmb}_o] \right)_{\ell \in \text{ObsIdx}^{\text{quot}}} \right). \end{aligned}$$

Define the foundational datum for the differential stage by

$$\begin{aligned} & \text{Dat}^{\text{diff}}(\text{RepEmb}_o) \\ & := \left( \text{DiffTimeBase}_o[\text{RepEmb}_o] \curvearrowright \text{DiffTrjObsBase} \right) \\ & \quad \curvearrowright \left( \text{LawEvalDat}[\text{RepEmb}_o].\text{CandRelStr}, \right. \\ & \quad \left. \text{LawEvalDat}[\text{RepEmb}_o].\text{FaithTransSet} \right). \end{aligned}$$

This is a seven-component ordered tuple preserving the definitional order of the tuples above.

**Proposition 5.116** (Existence and uniqueness of the foundational datum for the differential stage). Under

$$\text{LawTimeAdm}[\text{LawEvalDat}[\text{RepEmb}_o]; \text{RepEmb}_o],$$

$\text{Dat}^{\text{diff}}(\text{RepEmb}_o)$  exists uniquely.

*Proof.* Definition 5.106 and Definition 5.102 give dynamical-description operability. Proposition 5.73 uniquely determines every component of the two inner foundations. Definition 5.107 uniquely determines the candidate relation structure and the family of faithful representation transformations transported to the same representation. The conclusion follows from the outer ordered tuple in Definition 5.115.  $\square$

## 5.4 Trajectory Charts and Covariance of Differential Laws

Fix a base representation  $\rho^0$  and  $\text{LawEvalDat} \in \text{LawEvalDatSp}^{\text{gen}}[\rho^0]$ . This section treats both single-object and joint-object specifications; put  $e := \text{LawEvalDat.EvalSpec}$ . For each  $\rho_o \in \Omega[e]$ , use  $\text{LawEvalDat}[\rho_o]$  from Definition 5.107. The unindexed law indices, evaluation time domains, side-specific value spaces and value maps, and candidate relations denote components of this datum at the fixed  $\rho_o$ . When two representations are used together, their components are distinguished by  $[i]$ . For the generated single-object law datum, law-time admissibility also gives the unique foundational datum of Proposition 5.116.

The differential-evaluation structure below is an additional input; its existence does not follow from law operativity alone. A nonempty countable evaluation time domain cannot be covered by time charts with open images. Indexing each source by the natural numbers does not by itself imply that the index set of the entire source family is countable.

**Definition 5.117** (Differential-evaluation structure). The differential-law index set satisfies

$$\emptyset \neq \text{LawIdx}^{\text{diff}} \subseteq \text{LawIdx}.$$

For each

$$a \in \text{LawIdx}^{\text{diff}},$$

use the side-index set  $\text{IOSide}$  of Definition 5.83. The required differentiation order and the coordinate dimension on each input-output side satisfy

$$k_a \in \mathbb{N}_{\geq 1}, \quad m_a^s \in \mathbb{N}_0 \quad (s \in \text{IOSide}).$$

Define the total dependent-variable coordinate dimension by

$$m_a := m_a^{\text{in}} + m_a^{\text{out}}.$$

The index set of local differential charts for time and the index set of local differential charts on each input-output side satisfy

$$\text{TimeChartIdx}_a^{\text{diff}} \neq \emptyset,$$

$$\text{ValChartIdx}_{a,s}^{\text{diff}} \neq \emptyset \quad (s \in \text{IOSide}).$$

Define the value-chart index set and the full differential-chart index set by

$$\text{ValChartIdx}_a^{\text{diff}} := \prod_{s \in \text{IOSide}} \text{ValChartIdx}_{a,s}^{\text{diff}},$$

$$\text{ChartIdx}_a^{\text{diff}} := \text{TimeChartIdx}_a^{\text{diff}} \times \text{ValChartIdx}_a^{\text{diff}}.$$

For each

$$\alpha_0 \in \text{TimeChartIdx}_a^{\text{diff}},$$

the time-chart domain satisfies

$$\text{TimeChartDom}_{a,\alpha_0}^{\text{diff}} \subseteq \text{EvalTimeDom}_a^{\text{law}}.$$

The local numerical time chart is a bijection

$$\text{TimeChart}_{a,\alpha_0}^{\text{diff}} : \text{TimeChartDom}_{a,\alpha_0}^{\text{diff}} \longrightarrow \widehat{\text{TimeChartDom}}_{a,\alpha_0}^{\text{diff}} \subseteq \mathbb{R},$$

and its image is open.

For each

$$s \in \text{IOSide}, \quad \beta_{s,0} \in \text{ValChartIdx}_{a,s}^{\text{diff}},$$

the side-specific value-chart domain satisfies

$$\text{ValChartDom}_{a,s,\beta_{s,0}}^{\text{diff}} \subseteq \text{ValSp}_{a,s}^{\text{law}}.$$

The side-specific local numerical value chart is a bijection

$$\text{ValChart}_{a,s,\beta_{s,0}}^{\text{diff}} : \text{ValChartDom}_{a,s,\beta_{s,0}}^{\text{diff}} \longrightarrow \widehat{\text{ValChartDom}}_{a,s,\beta_{s,0}}^{\text{diff}} \subseteq \mathbb{R}^{m_s},$$

and its image is open.

For each

$$\beta_0 = (\beta_{s,0})_{s \in \text{IOSide}} \in \text{ValChartIdx}_a^{\text{diff}},$$

define the value-coordinate chart and its image by

$$\text{ValChart}_{a,\beta_0}^{\text{diff}} := \prod_{s \in \text{IOSide}} \text{ValChart}_{a,s,\beta_{s,0}}^{\text{diff}},$$

$$\widehat{\text{ValChartDom}}_{a,\beta_0}^{\text{diff}} := \prod_{s \in \text{IOSide}} \widehat{\text{ValChartDom}}_{a,s,\beta_{s,0}}^{\text{diff}} \subseteq \mathbb{R}^{m_a}.$$

Define the pair of a time-chart reference and a value-chart reference by

$$\chi_0 := (\alpha_0, \beta_0) \in \text{ChartIdx}_a^{\text{diff}}.$$

For each

$$\chi_0 \in \text{ChartIdx}_a^{\text{diff}},$$

define the local jet space by

$$\text{JetSpace}_{a;\chi_0}^{\text{diff},(k_a)} := \widehat{\text{TimeChartDom}}_{a,\alpha_0}^{\text{diff}} \times \widehat{\text{ValChartDom}}_{a,\beta_0}^{\text{diff}} \\ \times \prod_{r=1}^{k_a} \mathbb{R}^{m_a}.$$

The local differential relation satisfies

$$\text{JetRel}_{a;\chi_0}^{\text{diff}} \subseteq \text{JetSpace}_{a;\chi_0}^{\text{diff},(k_a)}.$$

For each

$$\alpha_0, \alpha_1 \in \text{TimeChartIdx}_a^{\text{diff}},$$

define the candidate time-chart change by

$$\text{TimeChartChg}_{a;\alpha_0,\alpha_1}^{\text{diff}} := \text{TimeChart}_{a,\alpha_1}^{\text{diff}} \circ (\text{TimeChart}_{a,\alpha_0}^{\text{diff}})^{-1}$$

on the image of the overlap of the two time-chart domains.

For each

$$\beta_0, \beta_1 \in \text{ValChartIdx}_a^{\text{diff}},$$

define the candidate value-chart change by

$$\text{ValChartChg}_{a;\beta_0,\beta_1}^{\text{diff}} := \text{ValChart}_{a,\beta_1}^{\text{diff}} \circ (\text{ValChart}_{a,\beta_0}^{\text{diff}})^{-1}$$

on the image of the overlap of the two value-chart domains.

Define the family of candidate time reparametrizations and the family of candidate value-chart changes by

$$\text{TimeChartChgSet}_a^{\text{diff}} := \left\{ \text{TimeChartChg}_{a;\alpha_0,\alpha_1}^{\text{diff}} \mid (\alpha_0, \alpha_1) \in (\text{TimeChartIdx}_a^{\text{diff}})^2 \right\},$$

$$\text{ValChartChgSet}_a^{\text{diff}} := \left\{ \text{ValChartChg}_{a;\beta_0,\beta_1}^{\text{diff}} \mid (\beta_0, \beta_1) \in (\text{ValChartIdx}_a^{\text{diff}})^2 \right\}.$$

For each

$$t \in \text{EvalTimeDom}_a^{\text{law}},$$

assume that there exists

$$\chi_0 \in \text{ChartIdx}_a^{\text{diff}}, \\ \beta_0 = (\beta_{s,0})_{s \in \text{IOSide}}$$

such that

$$t \in \text{TimeChartDom}_{a,\alpha_0}^{\text{diff}},$$

and, for every

$$s \in \text{IOSide},$$

$$\text{Val}_{a,s}^{\text{law}}(t) \in \text{ValChartDom}_{a,s,\beta_{s,0}}^{\text{diff}}$$

holds.

Define the differential-evaluation structure by

$$\begin{aligned} & \text{EvalStr}_e^{\text{diff}}[\rho_o] \\ & := \left( \text{LawIdx}^{\text{diff}}, \right. \\ & \quad \left( \begin{array}{l} k_a, (m_a^s)_{s \in \text{IOSide}}, \\ \text{TimeChartIdx}_a^{\text{diff}}, (\text{ValChartIdx}_{a,s}^{\text{diff}})_{s \in \text{IOSide}}, \\ (\text{TimeChart}_{a,\alpha_0}^{\text{diff}})_{\alpha_0 \in \text{TimeChartIdx}_a^{\text{diff}}}, \\ (\text{ValChart}_{a,s,\beta_{s,0}}^{\text{diff}})_{\substack{s \in \text{IOSide}, \\ \beta_{s,0} \in \text{ValChartIdx}_{a,s}^{\text{diff}}}}, \\ (\text{JetRel}_{a;\chi_0}^{\text{diff}})_{\substack{\alpha_0 \in \text{TimeChartIdx}_a^{\text{diff}}, \\ \beta_0 \in \text{ValChartIdx}_a^{\text{diff}}}}, \\ \text{TimeChartChgSet}_a^{\text{diff}}, \text{ValChartChgSet}_a^{\text{diff}} \end{array} \right)_{a \in \text{LawIdx}^{\text{diff}}} \end{aligned}$$

Below, the fixed evaluation specification  $e$  is omitted from the subscript of the differential-evaluation structure.

**Definition 5.118** (Local numerical chart for observer time and trajectory). Assume

$$\text{LawEvalDat}[\rho_o] \in \text{LawEvalDatSp}^{\text{gen}}[\rho_o].$$

Suppose that the differential-evaluation structure

$$\text{EvalStr}^{\text{diff}}[\rho_o]$$

of Definition 5.117 is given.

For each

$$\begin{aligned} & a \in \text{LawIdx}^{\text{diff}}, \\ & \chi_0 \in \text{ChartIdx}_a^{\text{diff}}, \\ & \beta_0 = (\beta_{s,0})_{s \in \text{IOSide}} \end{aligned}$$

define the simultaneous three-chart evaluation-time domain by

$$\begin{aligned} & \text{ChartEvalTimeDom}_{a;\chi_0}^{\text{diff}} \\ & := \left\{ t \in \text{TimeChartDom}_{a,\alpha_0}^{\text{diff}} \mid \right. \\ & \quad \left. \forall s \in \text{IOSide}, \quad \text{Val}_{a,s}^{\text{law}}(t) \in \text{ValChartDom}_{a,s,\beta_{s,0}}^{\text{diff}} \right\}. \end{aligned}$$

Define its numerical time image by

$$\begin{aligned} & \widehat{\text{ChartEvalTimeDom}}_{a;\chi_0}^{\text{diff}} \\ & := \text{TimeChart}_{a,\alpha_0}^{\text{diff}}(\text{ChartEvalTimeDom}_{a;\chi_0}^{\text{diff}}). \end{aligned}$$

Define the local numerical time coordinate by

$$\begin{aligned} \text{TimeCoord}_{a;\chi_0}^{\text{diff}} : \\ & \text{ChartEvalTimeDom}_{a;\chi_0}^{\text{diff}} \\ & \longrightarrow \widehat{\text{ChartEvalTimeDom}}_{a;\chi_0}^{\text{diff}}, \\ t & \longmapsto \text{TimeChart}_{a,\alpha_0}^{\text{diff}}(t). \end{aligned}$$

Define the local numerical value curve by

$$\begin{aligned} \text{ValCoordCurve}_{a;\chi_0}^{\text{diff}} : \\ & \widehat{\text{ChartEvalTimeDom}}_{a;\chi_0}^{\text{diff}} \\ & \longrightarrow \widehat{\text{ValChartDom}}_{a,\beta_0}^{\text{diff}}, \\ \text{ValCoordCurve}_{a;\chi_0}^{\text{diff}}(\theta) \\ & := \text{ValChart}_{a,\beta_0}^{\text{diff}} \left( \left( \text{Val}_{a,s}^{\text{law}} \left( (\text{TimeChart}_{a,\alpha_0}^{\text{diff}})^{-1}(\theta) \right) \right)_{s \in \text{IOSide}} \right). \end{aligned}$$

For each input-output side, define the local numerical curve as the corresponding component of the local numerical value curve,

$$\text{ValCoordCurve}_{a,s;\chi_0}^{\text{diff}} := \text{pr}_s \circ \text{ValCoordCurve}_{a;\chi_0}^{\text{diff}} \quad (s \in \text{IOSide}).$$

Define the local numerical chart for observer time and trajectory by

$$\begin{aligned} \text{TrjCoordChart}_{a;\chi_0}^{\text{diff}} \\ & := \left( \text{TimeCoord}_{a;\chi_0}^{\text{diff}}, \right. \\ & \quad \left. (\text{ValCoordCurve}_{a,s;\chi_0}^{\text{diff}})_{s \in \text{IOSide}} \right). \end{aligned}$$

**Proposition 5.119** (Generation and coverage of local numerical curves). For every

$$a \in \text{LawIdx}^{\text{diff}},$$

$$\chi_0 \in \text{ChartIdx}_a^{\text{diff}}$$

the map

$$\begin{aligned} \text{TimeCoord}_{a;\chi_0}^{\text{diff}} : \\ & \text{ChartEvalTimeDom}_{a;\chi_0}^{\text{diff}} \\ & \longrightarrow \widehat{\text{ChartEvalTimeDom}}_{a;\chi_0}^{\text{diff}} \end{aligned}$$

is a bijection.

Moreover, for every

$$t \in \text{EvalTimeDom}_a^{\text{law}}$$

there exist

$$\chi_0 \in \text{ChartIdx}_a^{\text{diff}},$$

$$\beta_0 = (\beta_{s,0})_{s \in \text{IOSide}}$$

such that

$$t \in \text{ChartEvalTimeDom}_{a;\chi_0}^{\text{diff}}.$$

For this chart index and

$$\theta := \text{TimeCoord}_{a;\chi_0}^{\text{diff}}(t)$$

for every

$$s \in \text{IOSide}$$

one has

$$\begin{aligned} & \text{ValCoordCurve}_{a,s;\chi_0}^{\text{diff}}(\theta) \\ &= \text{ValChart}_{a,s,\beta_{s,0}}^{\text{diff}}(\text{Val}_{a,s}^{\text{law}}(t)), \end{aligned}$$

and, moreover,

$$\begin{aligned} & \text{ValCoordCurve}_{a;\chi_0}^{\text{diff}}(\theta) \\ &= \text{ValChart}_{a,\beta_0}^{\text{diff}}\left(\left(\text{Val}_{a,s}^{\text{law}}(t)\right)_{s \in \text{IOSide}}\right). \end{aligned}$$

*Proof.* The local numerical time chart is a bijection. Its restriction to the simultaneous three-chart evaluation-time domain is therefore a bijection from the domain displayed in the proposition onto its image.

Take any

$$t \in \text{EvalTimeDom}_a^{\text{law}}.$$

By the chart-coverage condition in Definition 5.117, there exists a chart index carrying the input and output components, and the time  $t$  belongs to its simultaneous three-chart evaluation-time domain.

From

$$\theta = \text{TimeChart}_{a,\alpha_0}^{\text{diff}}(t)$$

and bijectivity,

$$\left(\text{TimeChart}_{a,\alpha_0}^{\text{diff}}\right)^{-1}(\theta) = t.$$

Substitution into the definition of the side-specific local numerical curve gives the side-specific equality. The definitions of the local numerical value curve and the value-coordinate chart give the value-coordinate equality.  $\square$

**Definition 5.120** (Common evaluation rule for the differential conditions). The five conditions below are predicates on the typed differential-evaluation structures of this section. An input to the fifth condition is a pair of structures belonging to the same base law datum and sharing the components specified in that definition. Let  $D$  be the subset of inputs on which all assumptions in a condition's definition hold, and let  $P$  be its test on that domain. Define the condition by

$$K[x] \iff \exists y \in D, \quad x = y \wedge P[y].$$

The test displayed in each definition is its restriction to  $D$ . The domain of the third condition is the truth domain of the second; that of the fourth is the truth domain of the first; and that of the fifth requires the first condition for both structures. In the first and second conditions, requirements that type subsequent expressions, such as openness and differentiability, are tested first. If a required type or preceding condition is absent, the condition is false; jets and change maps are not evaluated outside their domains.

**Definition 5.121** (Differential-atlas regularity condition (Diff1)). Suppose that the differential-evaluation structure of Definition 5.117 and the local numerical curves of Definition 5.118 are given. For every

$$a \in \text{LawIdx}^{\text{diff}}, \quad \alpha_0, \alpha_1 \in \text{TimeChartIdx}_a^{\text{diff}}, \\ \beta_0, \beta_1 \in \text{ValChartIdx}_a^{\text{diff}}$$

define the coordinate domains of the time-chart overlap and the value-chart overlap by

$$\widehat{\text{TimeChartOverlapDom}}_{a;\alpha_0,\alpha_1}^{\text{diff}} := \text{TimeChart}_{a,\alpha_0}^{\text{diff}} \left( \text{TimeChartDom}_{a,\alpha_0}^{\text{diff}} \cap \text{TimeChartDom}_{a,\alpha_1}^{\text{diff}} \right),$$

$$\widehat{\text{ValChartOverlapDom}}_{a;\beta_0,\beta_1}^{\text{diff}} := \text{ValChart}_{a,\beta_0}^{\text{diff}} \left( \text{Dom ValChart}_{a,\beta_0}^{\text{diff}} \cap \text{Dom ValChart}_{a,\beta_1}^{\text{diff}} \right).$$

Define the value-jet-change domain and the time-jet-change domain by

$$\begin{aligned} \text{ValJetChgDom}_{a;\alpha_0;\beta_0,\beta_1}^{\text{diff},(k_a)} &:= \widehat{\text{TimeChartDom}}_{a,\alpha_0}^{\text{diff}} \times \\ &\quad \widehat{\text{ValChartOverlapDom}}_{a;\beta_0,\beta_1}^{\text{diff}} \times \prod_{r=1}^{k_a} \mathbb{R}^{m_a}, \\ \text{TimeJetChgDom}_{a;\alpha_0,\alpha_1;\beta_0}^{\text{diff},(k_a)} &:= \widehat{\text{TimeChartOverlapDom}}_{a;\alpha_0,\alpha_1}^{\text{diff}} \times \\ &\quad \widehat{\text{ValChartDom}}_{a,\beta_0}^{\text{diff}} \times \prod_{r=1}^{k_a} \mathbb{R}^{m_a}. \end{aligned}$$

At a local numerical time  $\theta$ , for a  $C^{k_a}$ -class value-curve germ  $\gamma$ , define the local jet by

$$\text{jet}_{\theta}^{k_a}(\gamma) := \left( \theta, \left( \frac{d^r \gamma}{d\theta^r}(\theta) \right)_{r=0}^{k_a} \right).$$

The differential-atlas regularity condition

$$K_{\text{AtlasReg}}^{\text{diff}} \left[ \text{EvalStr}^{\text{diff}}[\rho_o] \right]$$

means that the following five conditions hold for every choice of the quantified variables in this definition.

First,

$$\begin{aligned} &\widehat{\text{TimeChartOverlapDom}}_{a;\alpha_0,\alpha_1}^{\text{diff}}, \\ &\widehat{\text{ValChartOverlapDom}}_{a;\beta_0,\beta_1}^{\text{diff}} \end{aligned}$$

are open sets.

Second,

$$\begin{aligned} \text{TimeChartChg}_{a;\alpha_0,\alpha_1}^{\text{diff}} &\in C^{k_a} \left( \widehat{\text{TimeChartOverlapDom}}_{a;\alpha_0,\alpha_1}^{\text{diff}}, \widehat{\text{TimeChartOverlapDom}}_{a;\alpha_1,\alpha_0}^{\text{diff}} \right), \\ \text{TimeChartChg}_{a;\alpha_1,\alpha_0}^{\text{diff}} &= \left( \text{TimeChartChg}_{a;\alpha_0,\alpha_1}^{\text{diff}} \right)^{-1} \end{aligned}$$



hold.

Third,

$$\begin{aligned} \text{ValChartChg}_{a;\beta_0,\beta_1}^{\text{diff}} &\in C^{k_a} \left( \widehat{\text{ValChartOverlapDom}_{a;\beta_0,\beta_1}^{\text{diff}}}, \widehat{\text{ValChartOverlapDom}_{a;\beta_1,\beta_0}^{\text{diff}}} \right), \\ \text{ValChartChg}_{a;\beta_1,\beta_0}^{\text{diff}} &= \left( \text{ValChartChg}_{a;\beta_0,\beta_1}^{\text{diff}} \right)^{-1} \end{aligned}$$

hold.

Fourth, there exists a unique value-jet-change map

$$\text{ValJetChg}_{a;\alpha_0;\beta_0,\beta_1}^{\text{diff},(k_a)} : \text{ValJetChgDom}_{a;\alpha_0;\beta_0,\beta_1}^{\text{diff},(k_a)} \longrightarrow \text{ValJetChgDom}_{a;\alpha_0;\beta_1,\beta_0}^{\text{diff},(k_a)}$$

satisfying, for every admissible  $\theta$  and  $\gamma$ ,

$$\text{ValJetChg}_{a;\alpha_0;\beta_0,\beta_1}^{\text{diff},(k_a)} (\text{jet}_{\theta}^{k_a}(\gamma)) = \text{jet}_{\theta}^{k_a} (\text{ValChartChg}_{a;\beta_0,\beta_1}^{\text{diff}} \circ \gamma).$$

Fifth, there exists a unique time-jet-change map

$$\text{TimeJetChg}_{a;\alpha_0,\alpha_1;\beta_0}^{\text{diff},(k_a)} : \text{TimeJetChgDom}_{a;\alpha_0,\alpha_1;\beta_0}^{\text{diff},(k_a)} \longrightarrow \text{TimeJetChgDom}_{a;\alpha_1,\alpha_0;\beta_0}^{\text{diff},(k_a)}$$

satisfying, for every admissible  $\theta$  and  $\gamma$ ,

$$\begin{aligned} &\text{TimeJetChg}_{a;\alpha_0,\alpha_1;\beta_0}^{\text{diff},(k_a)} (\text{jet}_{\theta}^{k_a}(\gamma)) \\ &= \text{jet}_{\text{TimeChartChg}_{a;\alpha_0,\alpha_1}^{\text{diff}}(\theta)}^{k_a} (\gamma \circ \text{TimeChartChg}_{a;\alpha_1,\alpha_0}^{\text{diff}}). \end{aligned}$$

In the value-jet-change identity,  $\gamma$  ranges over  $C^{k_a}$ -class curve germs into the coordinate domain of the value-chart overlap. In the time-jet-change identity,  $\gamma$  ranges over  $C^{k_a}$ -class curve germs on the coordinate domain of the time-chart overlap.

**Definition 5.122** (Generated-trajectory-jet existence condition (Diff2)). Define the generated-trajectory-jet existence condition by

$$\begin{aligned} &K_{\text{GeneratedTrjJetExist}}^{\text{diff}} \left[ \text{EvalStr}^{\text{diff}}[\rho_o] \right] \\ &\iff \left[ \forall a \in \text{LawIdx}^{\text{diff}}, \right. \\ &\quad \forall \chi_0 \in \text{ChartIdx}_a^{\text{diff}}, \\ &\quad \text{ChartEvalTimeDom}_{a;\chi_0}^{\text{diff}} \text{ is open,} \\ &\quad \text{ValCoordCurve}_{a;\chi_0}^{\text{diff}} \in C^{k_a} \left( \widehat{\text{ChartEvalTimeDom}_{a;\chi_0}^{\text{diff}}}, \right. \\ &\quad \left. \left. \widehat{\text{ValChartDom}_{a,\beta_0}^{\text{diff}}} \right) \right]. \end{aligned}$$

Under this condition, for each

$$\theta \in \widehat{\text{ChartEvalTimeDom}_{a;\chi_0}^{\text{diff}}},$$

define the generated-trajectory jet by

$$\begin{aligned} &\text{GeneratedTrjJet}_{a;\chi_0}^{\text{diff},(k_a)}(\theta) \\ &:= \text{jet}_{\theta}^{k_a} (\text{ValCoordCurve}_{a;\chi_0}^{\text{diff}}) \\ &\in \text{JetSpace}_{a;\chi_0}^{\text{diff},(k_a)}. \end{aligned}$$

**Definition 5.123** (Generated-trajectory-jet local-relation-membership condition (Diff3)).

Assume

$$K_{\text{GeneratedTrjJetExist}}^{\text{diff}} \left[ \text{EvalStr}^{\text{diff}}[\rho_o] \right].$$

Define the generated-trajectory-jet local-relation-membership condition by

$$\begin{aligned} & K_{\text{GeneratedTrjJetRelMem}}^{\text{diff}} \left[ \text{EvalStr}^{\text{diff}}[\rho_o] \right] \\ & \iff \left[ \begin{aligned} & \forall a \in \text{LawIdx}^{\text{diff}}, \\ & \forall \chi_0 \in \text{ChartIdx}_a^{\text{diff}}, \\ & \forall \theta \in \widehat{\text{ChartEvalTimeDom}}_{a;\chi_0}^{\text{diff}}, \\ & \text{GeneratedTrjJet}_{a;\chi_0}^{\text{diff},(k_a)}(\theta) \\ & \in \text{JetRel}_{a;\chi_0}^{\text{diff}} \end{aligned} \right]. \end{aligned}$$

**Proposition 5.124** (Local-jet consequences of Diff1–Diff3). Suppose that the differential-atlas regularity condition, the generated-trajectory-jet existence condition, and the generated-trajectory-jet local-relation-membership condition hold.

The value-jet-change map preserves the time-chart index and is uniquely determined by the value-chart-change map. The time-jet-change map preserves the value-chart index and is uniquely determined by the time-chart-change map.

On their common overlaps,

$$\text{ValJetChg}_{a;\alpha_0;\beta_0,\beta_0}^{\text{diff},(k_a)} = \text{id},$$

$$\text{TimeJetChg}_{a;\alpha_0,\alpha_0;\beta_0}^{\text{diff},(k_a)} = \text{id}$$

hold. On a common overlap of three value charts,

$$\begin{aligned} & \text{ValJetChg}_{a;\alpha_0;\beta_1,\beta_2}^{\text{diff},(k_a)} \circ \text{ValJetChg}_{a;\alpha_0;\beta_0,\beta_1}^{\text{diff},(k_a)} \\ & = \text{ValJetChg}_{a;\alpha_0;\beta_0,\beta_2}^{\text{diff},(k_a)}, \end{aligned}$$

and on a common overlap of three time charts,

$$\begin{aligned} & \text{TimeJetChg}_{a;\alpha_1,\alpha_2;\beta_0}^{\text{diff},(k_a)} \circ \text{TimeJetChg}_{a;\alpha_0,\alpha_1;\beta_0}^{\text{diff},(k_a)} \\ & = \text{TimeJetChg}_{a;\alpha_0,\alpha_2;\beta_0}^{\text{diff},(k_a)} \end{aligned}$$

hold.

For every

$$\begin{aligned} & \theta \in \widehat{\text{ChartEvalTimeDom}}_{a;\chi_0}^{\text{diff}}, \\ & \text{GeneratedTrjJet}_{a;\chi_0}^{\text{diff},(k_a)}(\theta) \in \text{JetRel}_{a;\chi_0}^{\text{diff}}. \end{aligned}$$

*Proof.* By the differential-atlas regularity condition, the time-chart-change and value-chart-change maps are bijections of class

$$C^{k_a},$$

and the reverse chart-change maps are also of class

$$C^{k_a}.$$

The

$$r$$

th derivative after a value-chart change, for

$$0 \leq r \leq k_a,$$

is determined by the chain rule from the derivatives through order

$$r$$

of the value-chart-change map and the representative curve germ. The

$$r$$

th derivative after a time-chart change is determined by the chain rule from the derivatives through order

$$r$$

of the inverse time-chart-change map and the representative curve germ. Hence the values of the two jet-change maps are independent of the choice of representative curve germ.

Every element of a jet-change domain can be represented by restricting a local polynomial curve through its zeroth-order component to a sufficiently small neighborhood, as the jet of a value-curve germ of class

$$C^{k_a}$$

of order

$$k_a.$$

Hence the two jet-change maps are defined over their entire respective jet-change domains. Two maps satisfying the same representative formula agree at every jet, so uniqueness holds.

A chart change from a chart to itself is the identity map, which gives the two identity formulas. On triple overlaps, the value-chart-change maps satisfy the value-chart composition condition, and the time-chart-change maps satisfy the time-chart composition condition. Applying two jet-change maps to a representative curve germ and using the uniqueness of each jet-change map gives the two composition formulas.

By the membership formula in Definition 5.122, each generated trajectory jet is an element of the local jet space indexed by the same law index, time-chart reference, and value-chart reference. By the membership formula in Definition 5.123, each generated trajectory jet belongs to the local jet relation indexed by the same law index, time-chart reference, and value-chart reference.  $\square$

**Lemma 5.125** (Transport of a relation restriction by a bijection). Suppose that a bijection

$$F : X \longrightarrow Y$$

and subsets

$$A \subseteq X, \quad B \subseteq Y$$

satisfy

$$\forall x \in X, \quad x \in A \iff F(x) \in B.$$

Then

$$F(A) = B$$

holds.

*Proof.* Let  $y \in F(A)$ . There exists  $x \in A \subseteq X$  such that  $y = F(x)$ . Since the assumption gives  $F(x) \in B$ , one has  $y \in B$ . Conversely, let  $y \in B$ . Surjectivity gives  $y = F(x)$  for some  $x \in X$ , and the assumption then gives  $x \in A$ . Hence  $y \in F(A)$ . The two inclusions give the equality.  $\square$

**Definition 5.126** (Value-jet-relation covariance condition (Diff4)). Assume

$$K_{\text{AtlasReg}}^{\text{diff}} \left[ \text{EvalStr}^{\text{diff}}[\rho_o] \right].$$

Define the value-jet-relation covariance condition by

$$\begin{aligned} & K_{\text{ValJetRelCov}}^{\text{diff}} \left[ \text{EvalStr}^{\text{diff}}[\rho_o] \right] \\ & \iff \left[ \forall a \in \text{LawIdx}^{\text{diff}}, \quad \forall \alpha_0 \in \text{TimeChartIdx}_a^{\text{diff}}, \right. \\ & \quad \forall \beta_0, \beta_1 \in \text{ValChartIdx}_a^{\text{diff}}, \quad \forall e \in \text{ValJetChgDom}_{a;\alpha_0;\beta_0,\beta_1}^{\text{diff},(k_a)}, \\ & \quad \left. e \in \text{JetRel}_{a;\alpha_0,\beta_0}^{\text{diff}} \iff \text{ValJetChg}_{a;\alpha_0;\beta_0,\beta_1}^{\text{diff},(k_a)}(e) \in \text{JetRel}_{a;\alpha_0,\beta_1}^{\text{diff}} \right]. \end{aligned}$$

**Proposition 5.127** (Chart-change equality for value-jet relations). When the differential-atlas regularity condition and the value-jet-relation covariance condition hold,

$$\begin{aligned} & \text{ValJetChg}_{a;\alpha_0;\beta_0,\beta_1}^{\text{diff},(k_a)} \left( \text{JetRel}_{a;\alpha_0,\beta_0}^{\text{diff}} \cap \text{ValJetChgDom}_{a;\alpha_0;\beta_0,\beta_1}^{\text{diff},(k_a)} \right) \\ & = \text{JetRel}_{a;\alpha_0,\beta_1}^{\text{diff}} \cap \text{ValJetChgDom}_{a;\alpha_0;\beta_1,\beta_0}^{\text{diff},(k_a)} \end{aligned}$$

holds.

*Proof.* By the differential-atlas regularity condition, the value-jet-change map is a bijection between the two displayed change domains. The value-jet-relation covariance condition gives the equivalence of relation membership along this bijection. Applying Lemma 5.125 gives the equality.  $\square$

**Definition 5.128** (Time-representation-change jet-relation covariance condition (Diff5)). Take two real-valued time-order embeddings

$$\rho_{o,i} \in \Omega[e], \quad i \in \{0, 1\},$$

and the set of directed pairs

$$\text{Dir}_2 := \{(0, 1), (1, 0)\}.$$

For each  $(i, j) \in \text{Dir}_2$ , define the order isomorphism between the time-representation images by

$$\text{TimeImIso}_{o;i \rightarrow j}^{\text{dyn}} := \rho_{o,j} \circ \rho_{o,i}^{-1} : \text{Im}(\rho_{o,i}) \longrightarrow \text{Im}(\rho_{o,j}).$$

Write the differential-evaluation structures as

$$\text{ES}_i := \text{EvalStr}^{\text{diff}}[\rho_{o,i}], \quad i \in \{0, 1\}.$$

Suppose that both satisfy the differential-atlas regularity condition and share the differential-law indices, the required differential orders, the value-chart indices, and the value-chart families. For every

$$\begin{aligned} a \in \text{LawIdx}^{\text{diff}}, \quad \alpha_i \in \text{TimeChartIdx}_a^{\text{diff},[i]} \quad (i \in \{0, 1\}), \\ \beta \in \text{ValChartIdx}_a^{\text{diff}}, \quad \chi_i := (\alpha_i, \beta) \end{aligned}$$

and  $(i, j) \in \text{Dir}_2$ , define the inter-embedding time-chart-overlap coordinate domain by

$$\begin{aligned} \widehat{\text{TimeRepChartOverlapDom}}_{a;\alpha_i,\alpha_j}^{\text{diff},i \rightarrow j} &:= \text{TimeChart}_{a,\alpha_i}^{\text{diff},[i]} \left( \text{TimeChartDom}_{a,\alpha_i}^{\text{diff},[i]} \right. \\ &\quad \left. \cap (\text{TimeImglso}_{o;i \rightarrow j})^{-1} (\text{TimeChartDom}_{a,\alpha_j}^{\text{diff},[j]}) \right). \end{aligned}$$

Define the inter-embedding local time-chart change and its jet-change domain by

$$\begin{aligned} \text{TimeRepChartChg}_{a;\alpha_i,\alpha_j}^{\text{diff},i \rightarrow j} &:= \text{TimeChart}_{a,\alpha_j}^{\text{diff},[j]} \circ \text{TimeImglso}_{o;i \rightarrow j} \circ (\text{TimeChart}_{a,\alpha_i}^{\text{diff},[i]})^{-1}, \\ \text{TimeRepJetChgDom}_{a;\alpha_i,\alpha_j;\beta}^{\text{diff},(k_a),i \rightarrow j} &:= \widehat{\text{TimeRepChartOverlapDom}}_{a;\alpha_i,\alpha_j}^{\text{diff},i \rightarrow j} \\ &\quad \times \widehat{\text{ValChartDom}}_{a,\beta}^{\text{diff}} \times \prod_{r=1}^{k_a} \mathbb{R}^{m_a}. \end{aligned}$$

The time-representation-change jet-relation covariance condition

$$K_{\text{TimeRepJetRelCov}}^{\text{diff}} [\text{ES}_0, \text{ES}_1]$$

means that the following four conditions hold for every choice of the quantified variables in this definition.

First,

$$\widehat{\text{TimeRepChartOverlapDom}}_{a;\alpha_i,\alpha_j}^{\text{diff},i \rightarrow j}$$

is an open set.

Second,

$$\begin{aligned} \text{TimeRepChartChg}_{a;\alpha_i,\alpha_j}^{\text{diff},i \rightarrow j} &\in C^{k_a} \left( \frac{\widehat{\text{TimeRepChartOverlapDom}}_{a;\alpha_i,\alpha_j}^{\text{diff},i \rightarrow j}}{\widehat{\text{TimeRepChartOverlapDom}}_{a;\alpha_j,\alpha_i}^{\text{diff},j \rightarrow i}} \right), \\ \text{TimeRepChartChg}_{a;\alpha_j,\alpha_i}^{\text{diff},j \rightarrow i} &= \left( \text{TimeRepChartChg}_{a;\alpha_i,\alpha_j}^{\text{diff},i \rightarrow j} \right)^{-1} \end{aligned}$$

hold.

Third, there exists a unique map

$$\begin{aligned} &\text{TimeRepJetChg}_{a;\alpha_i,\alpha_j;\beta}^{\text{diff},(k_a),i \rightarrow j} \\ &: \text{TimeRepJetChgDom}_{a;\alpha_i,\alpha_j;\beta}^{\text{diff},(k_a),i \rightarrow j} \\ &\longrightarrow \text{TimeRepJetChgDom}_{a;\alpha_j,\alpha_i;\beta}^{\text{diff},(k_a),j \rightarrow i} \end{aligned}$$

satisfying, for every admissible  $\theta$  and every value-curve germ of class  $C^{k_a}$ , denoted by  $\gamma$ ,

$$\begin{aligned} &\text{TimeRepJetChg}_{a;\alpha_i,\alpha_j;\beta}^{\text{diff},(k_a),i \rightarrow j} (\text{jet}_{\theta}^{k_a}(\gamma)) \\ &= \text{jet}_{\text{TimeRepChartChg}_{a;\alpha_i,\alpha_j}^{\text{diff},i \rightarrow j}(\theta)}^{k_a} \left( \gamma \circ \text{TimeRepChartChg}_{a;\alpha_j,\alpha_i}^{\text{diff},j \rightarrow i} \right). \end{aligned}$$

Fourth, for every

$$e \in \text{TimeRepJetChgDom}_{a;\alpha_i,\alpha_j;\beta}^{\text{diff},(k_a),i \rightarrow j},$$

$$e \in \text{JetRel}_{a;\chi_i}^{\text{diff},[i]} \iff \text{TimeRepJetChg}_{a;\alpha_i,\alpha_j;\beta}^{\text{diff},(k_a),i \rightarrow j}(e) \in \text{JetRel}_{a;\chi_j}^{\text{diff},[j]}$$

holds.

**Proposition 5.129** (Order isomorphism between images and time-jet-relation equality).

$$\text{TimeImgIso}_{o;0 \rightarrow 1}$$

is a strictly monotone order isomorphism. When the time-representation-change jet-relation covariance condition holds,

$$\begin{aligned} & \text{TimeRepJetChg}_{a;\alpha_0,\alpha_1;\beta}^{\text{diff},(k_a),0 \rightarrow 1} \left( \text{JetRel}_{a;\chi_0}^{\text{diff},[0]} \cap \text{TimeRepJetChgDom}_{a;\alpha_0,\alpha_1;\beta}^{\text{diff},(k_a),0 \rightarrow 1} \right) \\ &= \text{JetRel}_{a;\chi_1}^{\text{diff},[1]} \cap \text{TimeRepJetChgDom}_{a;\alpha_1,\alpha_0;\beta}^{\text{diff},(k_a),1 \rightarrow 0} \end{aligned}$$

holds.

*Proof.*  $\rho_{o,0}$  and  $\rho_{o,1}$  are order isomorphisms from the observer-time order-coordinate quotient to their respective images. By Proposition 5.62, the maps between the images are mutual inverses, and each direction preserves order. Hence  $\text{TimeImgIso}_{o;0 \rightarrow 1}$  is a strictly monotone order isomorphism.

By the time-representation-change jet-relation covariance condition, the time-jet-change map is a bijection between the two displayed change domains and satisfies the equivalence of relation membership. Applying Lemma 5.125 gives the image-set equality.  $\square$

**Definition 5.130** (Differential-stage condition DAG). Keep the base representation and law datum fixed as at the start of this section. Each component structure is relative to the law datum transported to its representation. Define the family of differential-evaluation structures over all real-valued time-order embeddings by

$$\begin{aligned} & \text{EvalStrFam}^{\text{diff}} \\ &:= \left( \text{EvalStr}^{\text{diff}}[\rho_o] \right)_{\rho_o \in \Omega[e]}. \end{aligned}$$

Each component shares

$$\text{LawIdx}^{\text{diff}}, \quad (k_a)_{a \in \text{LawIdx}^{\text{diff}}}, \quad (\text{ValChartIdx}_a^{\text{diff}})_{a \in \text{LawIdx}^{\text{diff}}}$$

and the value-chart family.

Define the differential-condition index set and the condition-specific arity by

$$\begin{aligned} I_{\text{diff}}^{\text{cond}} &:= \{1, \dots, 5\}, \\ \text{arity}_{\text{diff}}(i) &:= \begin{cases} 1, & i \in \{1, 2, 3, 4\}, \\ 2, & i = 5 \end{cases}. \end{aligned}$$

For each

$$i \in I_{\text{diff}}^{\text{cond}}$$

and

$$\mathbf{E} = (E_r)_{0 \leq r < \text{arity}_{\text{diff}}(i)},$$

define the differential-condition selection predicate by

$$\text{DiffPred}_i[\mathbf{E}] := \begin{cases} K_{\text{AtlasReg}}^{\text{diff}}[E_0], & i = 1, \\ K_{\text{GeneratedTrjJetExist}}^{\text{diff}}[E_0], & i = 2, \\ K_{\text{GeneratedTrjJetRelMem}}^{\text{diff}}[E_0], & i = 3, \\ K_{\text{ValJetRelCov}}^{\text{diff}}[E_0], & i = 4, \\ K_{\text{TimeRepJetRelCov}}^{\text{diff}}[E_0, E_1], & i = 5 \end{cases}.$$

For each

$$i \in I_{\text{diff}}^{\text{cond}},$$

define the five universal differential conditions by

$$\begin{aligned} & \text{DiffCond}_i^{\text{univ}} \\ & \iff \left[ \forall \mathbf{T} = (T_r)_{0 \leq r < \text{arity}_{\text{diff}}(i)} \right. \\ & \quad \left. \in (\Omega[e])^{\text{arity}_{\text{diff}}(i)}, \right. \\ & \quad \left. \text{DiffPred}_i \left[ \left( \text{EvalStr}^{\text{diff}}[T_r] \right)_{0 \leq r < \text{arity}_{\text{diff}}(i)} \right] \right]. \end{aligned}$$

Define the set of direct differential-condition-index edges by

$$E_{\text{diff}}^{\text{cond}} := \{(1, 2), (2, 3), (1, 4), (1, 5)\}.$$

Define the differential-condition node set, the set of direct differential-condition dependency edges, and the differential-stage condition DAG by

$$\begin{aligned} \text{DiffCondNodeSet} &:= I_{\text{diff}}^{\text{cond}}, \\ \text{DiffCondEdgeSet} &:= E_{\text{diff}}^{\text{cond}}, \\ \text{DiffCondDAG} &:= (\text{DiffCondNodeSet}, \text{DiffCondEdgeSet}). \end{aligned}$$

Associate node  $i$  with the condition  $\text{DiffCond}_i^{\text{univ}}$ . Every edge  $(i, j) \in \text{DiffCondEdgeSet}$  satisfies  $i < j$ , so this graph is a directed acyclic graph.

**Definition 5.131** (Differential-evaluation datum). Fix the base representation, base law-evaluation datum, and its differential-evaluation-structure family from this section. Call the triple

$$\text{DiffEvalDat} := (\rho^0, \text{LawEvalDat}, \text{EvalStrFam}^{\text{diff}}).$$

the differential-evaluation datum. Including the base representation makes the law data transported to each representation uniquely recoverable from this triple.

**Definition 5.132** (Differential-stage completion condition). Define the differential-stage completion condition by

$$\text{DiffCondAll} \left[ \text{EvalStrFam}^{\text{diff}} \right] \iff \bigwedge_{i \in I_{\text{diff}}^{\text{cond}}} \text{DiffCond}_i^{\text{univ}}.$$

Define differential-stage operability by

$$\begin{aligned} & \text{DiffOperative}[\text{DiffEvalDat}] \\ & \iff \text{LawOperative}[\text{LawEvalDat}] \\ & \quad \wedge \text{DiffCondAll} \left[ \text{EvalStrFam}^{\text{diff}} \right]. \end{aligned}$$

**Definition 5.133** (Chart-compatible differential law). In a context where the fixed  $\text{DiffEvalDat}$  uniquely determines  $e$ , the index  $e$  on the objects below is omitted. Assume the differential-stage completion condition

$$\text{DiffCondAll} \left[ \text{EvalStrFam}^{\text{diff}} \right]$$

of Definition 5.132. For each

$$\rho_o \in \Omega[e],$$

define the chart-compatible differential law by

$$\begin{aligned} & \text{ChartCompatDiffLaw}_e(\rho_o) \\ &:= \left( \left( \text{JetRel}_{a;\chi_0}^{\text{diff}} \right)_{\substack{a \in \text{LawIdx}^{\text{diff}}, \\ \chi_0 \in \text{ChartIdx}_a^{\text{diff}}}}, \right. \\ & \quad \left. \left( \text{ValJetChg}_{a;\alpha_0;\beta_0,\beta_1}^{\text{diff},(k_a)} \right)_{\substack{a \in \text{LawIdx}^{\text{diff}}, \\ \alpha_0 \in \text{TimeChartIdx}_a^{\text{diff}}, \\ \beta_0, \beta_1 \in \text{ValChartIdx}_a^{\text{diff}}}} \right). \end{aligned}$$

The first component is the family of local jet relations to which the generated trajectory jets belong. The second component is the family of value-jet-change maps that preserve the time-chart index and satisfy the image-set equality of Proposition 5.127 .

**Proposition 5.134** (Separation of order covariance from the two kinds of jet-relation covariance). The map

$$\text{TimeMglIso}_{o;0 \rightarrow 1}$$

determined by two real-valued time-order embeddings is a strictly monotone order isomorphism. Value-jet-relation covariance is given by  $\text{DiffCond}_4^{\text{univ}}$ , and the image-set equality of Proposition 5.127 holds for the value-jet-change maps that preserve the time-chart index. Time-representation-change jet-relation covariance is given by  $\text{DiffCond}_5^{\text{univ}}$ , and the image-set equality of Proposition 5.129 holds for the time-jet-change maps that preserve the value-chart index.

*Proof.* The order isomorphism between the images follows from Proposition 5.129 . Value-jet-relation covariance follows from Definition 5.126 and Proposition 5.127 . Time-representation-change jet-relation covariance follows from Definition 5.128 and Proposition 5.129 . A value-jet change preserves the time-chart index, whereas a time-jet change preserves the value-chart index. The order isomorphism between the images assumes neither jet-relation covariance condition. The conditions establishing the three conclusions are therefore distinct.  $\square$

**Theorem 5.135** (Differential covariance of law-bearing dynamical descriptions). Assume

$$\text{DiffOperative}[\text{DiffEvalDat}].$$

For arbitrary

$$\rho_{o,0}, \rho_{o,1} \in \Omega[e],$$

$$\text{TimeMglIso}_{o;0 \rightarrow 1} : \text{Im}(\rho_{o,0}) \longrightarrow \text{Im}(\rho_{o,1})$$

is the unique strictly monotone order isomorphism satisfying

$$\rho_{o,1} = \text{TimeMglIso}_{o;0 \rightarrow 1} \circ \rho_{o,0}.$$



For the differential-evaluation structure relative to each real-valued time-order embedding, the value-jet-relation image-set equality of Proposition 5.127 holds. For the pair of differential-evaluation structures relative to the two embeddings, the time-jet-relation image-set equality of Proposition 5.129 holds. The first equality preserves the time-chart index, and the second preserves the value-chart index.

*Proof.* Differential-stage operability gives the five universal differential conditions. By Proposition 5.129, the map between the images is a strictly monotone order isomorphism. The displayed composition identity follows from Definition 5.128.

Take a map  $\psi$  satisfying the same composition identity. The map  $\rho_{o,0}$  is surjective onto its image, and both maps, after precomposition with  $\rho_{o,0}$ , agree with  $\rho_{o,1}$ . Lemma 4.181 therefore gives  $\psi = \text{TimeImglso}_{o;0 \rightarrow 1}$ .

The value-jet-relation image-set equality follows from  $\text{DiffCond}_4^{\text{univ}}$  and Proposition 5.127. The time-jet-relation image-set equality follows from  $\text{DiffCond}_5^{\text{univ}}$  and Proposition 5.129.  $\square$

**Definition 5.136** (Differential-failure evaluation input). For each

$$i \in I_{\text{diff}}^{\text{cond}},$$

define the strict-ancestor index set by

$$\text{Anc}_{\text{diff}}(i) := \left\{ j \in I_{\text{diff}}^{\text{cond}} \mid (j, i) \in (E_{\text{diff}}^{\text{cond}})^+ \right\}.$$

Here the superscript  $+$  denotes the transitive closure of the direct-edge set.

Define the all-ancestors-satisfied bundle for the five universal differential conditions by

$$\begin{aligned} & \text{DiffCondAncAll}_i \\ & := \text{LawOperative} [\text{LawEvalDat}] \wedge \bigwedge_{\substack{j \in \text{Anc}_{\text{diff}}(i) \\ i \in I_{\text{diff}}^{\text{cond}}}} \text{DiffCond}_j^{\text{univ}}, \end{aligned}$$

Define the differential-failure evaluation input by

$$\begin{aligned} & \text{EvalInput}_e^{\text{diff}} \\ & := \left( \text{DiffCondDAG}, (\text{DiffCond}_i^{\text{univ}})_{i \in I_{\text{diff}}^{\text{cond}}}, \right. \\ & \quad \left. (\text{DiffCondAncAll}_i)_{i \in I_{\text{diff}}^{\text{cond}}}, \text{DiffEvalDat} \right). \end{aligned}$$

**Proposition 5.137** (Ancestor consistency of the differential-failure evaluation input).

$$\text{EvalInput}^{\text{diff}}$$

has the components

$$\begin{aligned} & \text{DiffCondDAG}, \quad (\text{DiffCond}_i^{\text{univ}})_{i \in I_{\text{diff}}^{\text{cond}}}, \\ & (\text{DiffCondAncAll}_i)_{i \in I_{\text{diff}}^{\text{cond}}}, \quad \text{DiffEvalDat}. \end{aligned}$$

For each

$$i \in I_{\text{diff}}^{\text{cond}},$$

the truth value of the universal differential condition

$$\text{DiffCond}_i^{\text{univ}}$$

is evaluated on the satisfaction domain of

$$\text{DiffCondAncAll}_i.$$

*Proof.* From the set of direct index edges in Definition 5.130, the strict-ancestor index sets are

$$(\text{Anc}_{\text{diff}}(i))_{1 \leq i \leq 5} = (\emptyset, \{1\}, \{1, 2\}, \{1\}, \{1\}).$$

Substitution into the ancestor conjunction of Definition 5.136 gives the all-ancestors-satisfied bundle for each condition. The tuple displayed in the same definition gives the four listed components.  $\square$

## 5.5 Three-layer synthesis of dynamical, law-bearing, and differential descriptions

Dynamical-description operability, law-description operability, and differential-stage operability have a cumulative dependency relation.

**Theorem 5.138** (Three-layer synthesis theorem for dynamical, law-bearing, and differential descriptions). For the single-object specification  $\text{Ev}$ , use the constructed base law datum  $\text{LawEvalDat}$ , together with its base representation  $\rho^0$ . For the differential stage, fix a differential-evaluation-structure family belonging to this same pair. Thus  $\text{LawEvalDat.EvalSpec} = \iota_{\text{single}}(\text{Ev})$ . Law assertions at each representation concern  $\text{LawEvalDat}[\rho_o]$  from Definition 5.107.

In the display of this theorem, the complete evaluation specification  $\text{Ev} = (I, o, \zeta)$  uniquely determines the subscript pair  $(o, \zeta)$ , which is omitted. The establishment conditions for the three stages satisfy

$$\begin{aligned} & \text{DynOperative}[\text{Ev}] \\ \iff & \text{ExactOperable}[I] \wedge \text{DynCondAll}[\text{Ev}], \\ & \text{LawOperative}[\text{LawEvalDat}] \\ \iff & \text{DynOperative}[\text{Ev}] \wedge \text{LawCondAll}[\text{LawEvalDat}], \\ & \text{DiffOperative}[\text{DiffEvalDat}] \\ \iff & \text{LawOperative}[\text{LawEvalDat}] \wedge \text{DiffCondAll}[\text{EvalStrFam}^{\text{diff}}]. \end{aligned}$$

Consequently,

$$\begin{aligned} \text{DiffOperative}[\text{DiffEvalDat}] & \implies \text{LawOperative}[\text{LawEvalDat}] \\ & \implies \text{DynOperative}[\text{Ev}] \\ & \implies \text{ExactOperable}[I]. \end{aligned}$$

Under dynamical-description operability, the set of real-valued time-order embeddings is nonempty, and the dynamical-description object and dynamical-description witness object relative to each embedding exist uniquely. Under law-description operability, the observer-time law core relative to each embedding exists uniquely, and generated representability of law-evaluation time holds. Under differential-stage operability, for every

$$\rho_o \in \text{OrdEmbSet}_o(\mathbb{R}),$$

$$\begin{array}{ccc} \text{DynDesc}[\rho_o], & \text{DynWit}[\rho_o], & \\ \text{LawCore}^{\text{obs,time}}(\rho_o), & \text{ChartCompatDiffLaw}(\rho_o), & \\ & \text{EvalInput}^{\text{diff}} & \end{array}$$

are determined. The dynamical-description object, dynamical-description witness object, and observer-time law core exist uniquely; moreover, the order isomorphism between time-representation embedding images, value-jet-relation covariance, and time-representation-change jet-relation covariance hold.

*Proof.* The first equivalence follows from Definition 5.70, the second from Definition 5.102, and the third from Definition 5.132. The first step of the implication chain follows from Definition 5.132, the second from Definition 5.102 and Proposition 5.86, and the third from Definition 5.70.

Existence and uniqueness of the dynamical-description object and dynamical-description witness object follow from Theorem 5.112; existence and uniqueness of the observer-time law core and generated representability of law-evaluation time follow from Theorem 5.113. Differential-stage operability gives the five universal differential conditions. The chart-compatible differential law and the differential-failure evaluation input are determined by Definition 5.133 and Definition 5.136, respectively. The three covariance conclusions follow from Theorem 5.135.  $\square$

## 5.6 Multi-object relational quantities and joint trajectory relations

Suppose that the finite object-specification index family

$$Z_0$$

of Definition 2.10 satisfies

$$|Z_0| \geq 2.$$

Assign the same realized operational configuration

$$\mathcal{X}$$

and observer-side information-coupling position

$$o \in U$$

to every component of the object-specification index family.

In what follows, each object-specific construction is evaluated relative to  $\text{Ev}_\zeta := (\mathbf{l}, o, \zeta)$ . When constructions for several objects are used together, the generated equivalence relations, state quotients and natural projections carry the object index  $\zeta$  explicitly.

**Definition 5.139** (Multi-object relation input data). Define the joint dynamical-description evaluation specification by

$$\text{Ev} := \iota_{\text{joint}}(\mathbf{l}, o, Z_0) \in \text{EvalSpecSp}.$$

Let the index set of multi-object relation types be

$$\text{JRelIdx}_{Z_0},$$

and assume

$$\mathbf{JRelIdx}_{Z_0} \neq \emptyset.$$

For each  $\mathbf{r} \in \mathbf{JRelIdx}_{Z_0}$ , the fixed  $\mathfrak{B}$ -derived symbol  $\mathfrak{B}$ , the section input  $Z_0$ , and the quantified  $\mathbf{r}$  uniquely determine the subscript strings  $(\mathfrak{B}, \mathbf{r}, Z_0)$  and  $(\mathbf{r}, Z_0)$ , which are omitted within the following definition. Let the source-side domain, relation-value set, and source-side multi-object relation satisfy

$$\begin{aligned} \mathbf{JointRelDom}^{\text{src}} &\subseteq \prod_{\zeta \in Z_0} \text{Tok}_{\mathfrak{B}}^*, \\ \mathbf{JointRel}^{\text{src}} &\subseteq \mathbf{JointRelDom}^{\text{src}} \times \mathbf{JointRelVal}. \end{aligned}$$

Define the source-side multi-object relation datum by

$$\mathbf{J}^{\text{src}} := (\mathbf{JointRelDom}^{\text{src}}, \mathbf{JointRelVal}, \mathbf{JointRel}^{\text{src}}),$$

and name its three components, in order,  $\text{Dom}$ ,  $\text{Val}$ ,  $\text{Rel}$ . Thus

$$\begin{aligned} \mathbf{x}_{\mathfrak{B}} &= (x_{\mathfrak{B}, \zeta})_{\zeta \in Z_0} \in \mathbf{J}^{\text{src}}.\text{Dom}, \\ (\mathbf{x}_{\mathfrak{B}}, v) &\in \mathbf{J}^{\text{src}}.\text{Rel} \end{aligned}$$

represents a correspondence between the family of source tokens associated with the  $\mathfrak{B}$ -derived component of each member of the object-specification index family and  $v \in \mathbf{J}^{\text{src}}.\text{Val}$ . For the predicate type, use

$$\mathbf{J}^{\text{src}}.\text{Val} = \{*\}.$$

Define the multi-object relation input data by

$$\mathbf{JRelDat}_{o, Z_0} := (\text{Ev}, \text{RI}[Z_0], \mathbf{JRelIdx}_{Z_0}, (\mathbf{J}^{\text{src}}_{\mathbf{r}, Z_0})_{\mathbf{r} \in \mathbf{JRelIdx}_{Z_0}}).$$

**Definition 5.140** (Relation-quotient descent module). For a set  $X$ , an equivalence relation  $\equiv$  on it, a subset  $A \subseteq X$ , a set  $V$ , and a relation  $S \subseteq A \times V$ , define the relation-saturation condition by

$$\begin{aligned} \text{RelSat}[\equiv; A, S] &\iff \left[ \forall x, x' \in X, x \equiv x' \right. \\ &\implies (x \in A \iff x' \in A) \\ &\quad \left. \wedge \forall v \in V, ((x, v) \in S \iff (x', v) \in S) \right]. \end{aligned}$$

Further suppose that a surjection  $q : X \twoheadrightarrow Q$  satisfies

$$x \equiv x' \iff q(x) = q(x').$$

Define the relation-quotient descent module by

$$\text{RelQuot}[q; A, S] := (q[A], \{(q(x), v) \mid (x, v) \in S\}).$$

**Lemma 5.141** (Representative criterion for the relation-quotient descent module). Suppose that a surjection  $q : X \twoheadrightarrow Q$  satisfies

$$x \equiv x' \iff q(x) = q(x')$$

and that  $\text{RelSat}[\equiv; A, S]$  holds. If

$$\text{RelQuot}[q; A, S] = (\overline{A}, \overline{S}),$$

then, for every  $x \in X$  and  $v \in V$ ,

$$q(x) \in \overline{A} \iff x \in A, \quad (q(x), v) \in \overline{S} \iff (x, v) \in S.$$

Moreover,  $\overline{A} \subseteq Q$  and  $\overline{S} \subseteq \overline{A} \times V$  satisfying these two representative criteria are unique.

*Proof.* In the forward direction, take the image representation of  $q(x)$  given by a representative  $x'$ , and use  $q(x) = q(x')$  together with the relation-saturation condition. The reverse direction follows from the definition of  $\text{RelQuot}$ . For uniqueness, surjectivity of  $q$  represents every element of  $Q$  as  $q(x)$ , and the two representative criteria determine pointwise membership in  $\overline{A}$  and relation membership in  $\overline{S}$ .  $\square$

**Definition 5.142** (Multi-object relation representative-invariance condition). Take the multi-object relation input data

$$\text{JRelDat}_{o, Z_0}$$

of Definition 5.139 .

For two families of  $\mathfrak{B}$ -derived source tokens

$$\begin{aligned} \mathbf{x}_{\mathfrak{B}} &= (x_{\mathfrak{B}, \zeta})_{\zeta \in Z_0}, \\ \mathbf{x}'_{\mathfrak{B}} &= (x'_{\mathfrak{B}, \zeta})_{\zeta \in Z_0}, \end{aligned}$$

define componentwise  $\mathfrak{B}$ -representative equivalence by

$$\begin{aligned} \mathbf{x}_{\mathfrak{B}} &\sim_{\mathfrak{B}, o, Z_0}^{\text{dyn, joint}} \mathbf{x}'_{\mathfrak{B}} \\ &\iff \left[ \forall \zeta \in Z_0, \quad x_{\mathfrak{B}, \zeta} \sim_{\mathfrak{B}, o; \zeta} x'_{\mathfrak{B}, \zeta} \right]. \end{aligned}$$

Define the multi-object relation representative-invariance condition by

$$\begin{aligned} &K_{\text{CanJointRelDesc}}^{\text{dyn}}(\text{JRelDat}_{o, Z_0}) \\ &\iff \left[ \forall \mathbf{t} \in \text{JRelIdx}_{Z_0}, \quad \text{RelSat} \left[ \sim_{\mathfrak{B}, o, Z_0}^{\text{joint}}; \text{J}_{\mathbf{t}, Z_0}^{\text{src}}.\text{Dom}, \text{J}_{\mathbf{t}, Z_0}^{\text{src}}.\text{Rel} \right] \right]. \end{aligned}$$

Here two multi-object state families have the same quotient image exactly when they are componentwise equivalent under  $\sim_{\mathfrak{B}, o; \zeta}$ . Hence this condition simultaneously expresses invariance of domain membership and relation membership under the componentwise representative equivalence defined immediately above.

**Definition 5.143** (Multi-object state quotient projection). Take the multi-object relation input data

$$\text{JRelDat}_{o, Z_0}$$

of Definition 5.139 .

Define the multi-object state quotient projection along the object-specification index family by

$$\Theta_{\mathfrak{B}, o; Z_0}^{\text{can}} : \prod_{\zeta \in Z_0} \text{Tok}_{\mathfrak{B}}^* \twoheadrightarrow \prod_{\zeta \in Z_0} \text{Qu}_{\mathfrak{B}, o; \zeta}^{\text{can}}$$

and

$$\begin{aligned} &\Theta_{\mathfrak{B}, o; Z_0}^{\text{can}} \left( (x_{\mathfrak{B}, \zeta})_{\zeta \in Z_0} \right) \\ &:= \left( \Theta_{\mathfrak{B}, o; \zeta}^{\text{can}}(x_{\mathfrak{B}, \zeta}) \right)_{\zeta \in Z_0}. \end{aligned}$$

**Lemma 5.144** (Quotient descent of multi-object relations).

Assume  $K_{\text{CanJointRelDesc}}(\text{JRelDat}_{o,Z_0})$ . For each  $\mathfrak{r} \in \text{JRelIdx}_{Z_0}$ , set

$$\begin{aligned} & \text{J}_{o,\mathfrak{r},Z_0}^{\text{can}} \\ & := \text{RelQuot} [\Theta_{\mathfrak{B},o;Z_0}^{\text{can}}; \text{J}_{\mathfrak{r},Z_0}^{\text{src}}.\text{Dom}, \text{J}_{\mathfrak{r},Z_0}^{\text{src}}.\text{Rel}] , \end{aligned}$$

and name its two components, in order,  $\text{Dom}, \text{Rel}$ . Then, for arbitrary

$$\begin{aligned} \mathbf{x}_{\mathfrak{B}} & \in \prod_{\zeta \in Z_0} \text{Tok}_{\mathfrak{B}}^*, \\ v & \in \text{J}_{\mathfrak{r},Z_0}^{\text{src}}.\text{Val}, \\ \Theta_{\mathfrak{B},o;Z_0}^{\text{can}}(\mathbf{x}_{\mathfrak{B}}) & \in \text{J}_{o,\mathfrak{r},Z_0}^{\text{can}}.\text{Dom} \\ \iff \mathbf{x}_{\mathfrak{B}} & \in \text{J}_{\mathfrak{r},Z_0}^{\text{src}}.\text{Dom}, \\ (\Theta_{\mathfrak{B},o;Z_0}^{\text{can}}(\mathbf{x}_{\mathfrak{B}}), v) & \in \text{J}_{o,\mathfrak{r},Z_0}^{\text{can}}.\text{Rel} \\ \iff (\mathbf{x}_{\mathfrak{B}}, v) & \in \text{J}_{\mathfrak{r},Z_0}^{\text{src}}.\text{Rel}. \end{aligned}$$

The quotient-level domain and quotient-level relation satisfying these two criteria are unique.

*Proof.* The multi-object state quotient projection is surjective, and equality of the images of two of its arguments is equivalent to

$$\mathbf{x}_{\mathfrak{B}} \sim_{\mathfrak{B},o,Z_0}^{\text{joint}} \mathbf{x}'_{\mathfrak{B}}.$$

Since the assumption is the relation-saturation condition for each  $\mathfrak{r}$ , applying Lemma 5.141 gives the two representative criteria and uniqueness.  $\square$

**Definition 5.145** (Quotient-level multi-object relation). For each  $\mathfrak{r} \in \text{JRelIdx}_{Z_0}$ , call the relation

$$\text{J}_{o,\mathfrak{r},Z_0}^{\text{can}}.\text{Rel}$$

uniquely determined by Lemma 5.144 the quotient-level multi-object relation corresponding to the relation type  $\mathfrak{r}$ .

**Lemma 5.146** (Common construction of the joint-time quotients). Assume  $\text{ExactOperable}[I]$ . Under  $K_{\text{JointCTime}}(\text{Ev})$ , the canonical time quotients, natural projections and generated time preorders agree across all objects. If

$$K_{\text{GenTimeOrdAntiSym}}(\text{Ev}_{\zeta}), \quad K_{\text{ObsTimeIncTrans}}(\text{Ev}_{\zeta})$$

hold for every object, the order-time quotients, order-time projections and strict linear orders also agree. The object index may therefore be omitted from these common time objects. Agreement of the individual state quotients or individual trajectory domains is not required.

*Proof.* The same equivalence relation on the same set defines the same quotient and natural projection. The same source order is mapped by the same projection, so the generating edges and their reflexive transitive closures agree. Under the antisymmetry condition, the canonical time partial orders agree, as do their strict parts, incomparability relations and quotients by incomparability. The definitions of the order-time projection and strict linear order give the conclusion.  $\square$

**Definition 5.147** (Joint scalar-time trajectory). Assume  $\text{ExactOperable}[\mathbb{I}]$  and  $K_{\text{JointCTime}}(\text{Ev})$ . For each  $\zeta \in Z_0$ , define the complete evaluation specification by

$$\text{Ev}_\zeta := (\mathbb{I}, o, \zeta) \in \text{EvSpec},$$

and assume

$$\forall \zeta \in Z_0, \quad \text{DynTrjFactCondAll}[\text{Ev}_\zeta].$$

Write the individual trajectory obtained from Proposition 5.66 as

$$\text{TrjMap}_{o;\zeta}^{\text{dyn,time,ord}} : \text{TrjDom}_{o;\zeta}^{\text{dyn,time,ord}} \longrightarrow \text{Qu}_{\mathfrak{B},o;\zeta}^{\text{can}}.$$

Define the domain component and map component of the order-time joint-trajectory datum by

$$\begin{aligned} \text{JTrj}_{o,Z_0}^{\text{ord}}.\text{Dom} &:= \bigcap_{\zeta \in Z_0} \text{TrjDom}_{o;\zeta}^{\text{time,ord}}, \\ \text{JTrj}_{o,Z_0}^{\text{ord}}.\text{Map} : \text{JTrj}_{o,Z_0}^{\text{ord}}.\text{Dom} &\longrightarrow \prod_{\zeta \in Z_0} \text{Qu}_{\mathfrak{B},o;\zeta}^{\text{can}}, \\ \text{JTrj}_{o,Z_0}^{\text{ord}}.\text{Map}(s) &:= \left( \text{TrjMap}_{o;\zeta}^{\text{time,ord}}(s) \right)_{\zeta \in Z_0}. \end{aligned}$$

**Definition 5.148** (Joint trajectory relation). For the multi-object relation input data of Definition 5.139, assume

$$K_{\text{CanJointRelDesc}}(\text{JRelDat}_{o,Z_0}).$$

Use the quotient-level multi-object relation of Definition 5.145 and the joint scalar-time trajectory of Definition 5.147.

For each

$$\mathfrak{r} \in \text{JRelIdx}_{Z_0},$$

define the relation component of the order-time joint-trajectory datum by

$$\begin{aligned} &\text{JTrj}_{o,Z_0}^{\text{ord}}.\text{Rel}_\mathfrak{r} \\ &:= \left\{ (s, v) \mid \begin{array}{l} s \in \text{JTrj}_{o,Z_0}^{\text{ord}}.\text{Dom}, \\ v \in \text{J}^{\text{src}}_{\mathfrak{r},Z_0}.\text{Val}, \\ (\text{JTrj}_{o,Z_0}^{\text{ord}}.\text{Map}(s), v) \\ \in \text{J}^{\text{can}}_{o,\mathfrak{r},Z_0}.\text{Rel} \end{array} \right\}. \end{aligned}$$

Consequently,

$$\begin{aligned} &\text{JTrj}_{o,Z_0}^{\text{ord}}.\text{Rel}_\mathfrak{r} \\ &\subseteq \text{JTrj}_{o,Z_0}^{\text{ord}}.\text{Dom} \times \text{J}^{\text{src}}_{\mathfrak{r},Z_0}.\text{Val}. \end{aligned}$$

For a predicate-valued multi-object relation, use

$$\text{J}^{\text{src}}_{\mathfrak{r},Z_0}.\text{Val} = \{*\};$$

then

$$(s, *) \in \text{JTrj}_{o,Z_0}^{\text{ord}}.\text{Rel}_\mathfrak{r}$$

represents satisfaction of the predicate on the joint state trajectory.

**Proposition 5.149** (Quotient descent of multi-object relations and restriction to a joint trajectory). Assume

$$\begin{aligned} & \text{ExactOperable}[\mathbb{I}], \\ & K_{\text{JointCTime}}(\text{Ev}), \\ & \forall \zeta \in Z_0, \text{DynTrjFactCondAll}[\text{Ev}_\zeta], \\ & K_{\text{CanJointRelDesc}}(\text{JRelDat}_{o,Z_0}). \end{aligned}$$

For each  $\mathfrak{r} \in \text{JRelIdx}_{Z_0}$ , the quotient-level multi-object relation, joint scalar-time trajectory, and joint trajectory relation exist uniquely.

Furthermore, for every

$$s \in \text{JTrj}_{o,Z_0}^{\text{ord}}.\text{Dom},$$

the equality

$$\Theta_{\mathfrak{B},o,Z_0}^{\text{can}}(\mathbf{x}_{\mathfrak{B}}) = \text{JTrj}_{o,Z_0}^{\text{ord}}.\text{Map}(s)$$

holds for some  $\mathbf{x}_{\mathfrak{B}} \in \prod_{\zeta \in Z_0} \text{Tok}_{\mathfrak{B}}^*$ , and for every such element and arbitrary  $v \in \text{J}_{\mathfrak{r},Z_0}^{\text{src}}.\text{Val}$ ,

$$(s, v) \in \text{JTrj}_{o,Z_0}^{\text{ord}}.\text{Rel}_{\mathfrak{r}} \iff (\mathbf{x}_{\mathfrak{B}}, v) \in \text{J}_{\mathfrak{r},Z_0}^{\text{src}}.\text{Rel}.$$

*Proof.* Each individual trajectory is unique by Proposition 5.66, and its product map on the common domain is also unique. Since the quotient-level multi-object relation is unique by Lemma 5.144, the inverse-image relation of Definition 5.148 is also unique.

Surjectivity of the multi-object state quotient projection gives the displayed  $\mathbf{x}_{\mathfrak{B}}$ . Composing the defining equivalence of the joint trajectory relation with the representative criterion in Lemma 5.144 gives the final equivalence.  $\square$

**Definition 5.150** (Joint trajectory relation over real-valued time). Suppose that, for each  $\zeta \in Z_0$ ,

$$K_{\text{ObsTimeRealOrdEmb}}(\text{Ev}_\zeta)$$

holds, and take

$$\rho_o \in \text{OrdEmbSet}_o(\mathbb{R}).$$

Define the domain component and relation component of the real-valued joint-trajectory datum by

$$\begin{aligned} \text{JTrj}_{o,Z_0}^{\text{real}}[\rho_o].\text{Dom} &:= \rho_o[\text{JTrj}_{o,Z_0}^{\text{ord}}.\text{Dom}] \subseteq \mathbb{R}, \\ \text{JTrj}_{o,Z_0}^{\text{real}}[\rho_o].\text{Rel}_{\mathfrak{r}} &:= \{(\rho_o(s), v) \mid (s, v) \in \text{JTrj}_{o,Z_0}^{\text{ord}}.\text{Rel}_{\mathfrak{r}}\}. \end{aligned}$$

Define the joint law-time domain by

$$\mathbb{T}_{o,Z_0}^{\text{joint}}[\rho_o] := \text{JTrj}_{o,Z_0}^{\text{real}}[\rho_o].\text{Dom}.$$

**Definition 5.151** (Observational realization of a joint relation value). For each  $\mathfrak{r} \in \text{JRelIdx}_{Z_0}$ , the section inputs  $o, Z_0$  and the quantified  $\mathfrak{r}$  uniquely determine the subscript string  $(o, \mathfrak{r}, Z_0)$ , which is omitted within the following definition. Define the domain of observational realization of a joint relation value by

$$\begin{aligned} \text{JRelObs}[\rho_o].\text{Dom} &:= \left\{ t \in \text{JTrj}_{o,Z_0}^{\text{real}}[\rho_o].\text{Dom} \mid \right. \\ & \quad \left. \exists v \in \text{J}_{\mathfrak{r},Z_0}^{\text{src}}.\text{Val}, (t, v) \in \text{JTrj}_{o,Z_0}^{\text{real}}[\rho_o].\text{Rel}_{\mathfrak{r}} \right\}. \end{aligned}$$

An observational realization of a joint relation value is a map satisfying

$$\text{JRelObs}[\rho_o].\text{Curve} : \text{JRelObs}[\rho_o].\text{Dom} \longrightarrow \text{J}_{\mathfrak{r},Z_0}^{\text{src}}.\text{Val}$$



such that, for every domain element  $t$ ,

$$(t, \text{JRelObs}[\rho_o].\text{Curve}(t)) \in \text{JTrj}_{o,Z_0}^{\text{real}}[\rho_o].\text{Rel}_t.$$

The family of observational realizations of joint relation values is supplied as law-evaluation input. Define the two-component datum for observational realization of a joint relation value by

$$\text{JRelObs}[\rho_o] := (\text{JRelObs}[\rho_o].\text{Dom}, \text{JRelObs}[\rho_o].\text{Curve}),$$

and name its two components, in order,  $\text{Dom}$ ,  $\text{Curve}$ .

**Definition 5.152** (Candidate relation structure for laws referring to joint relations). Fix the common time representation from Definition 5.150 and the family of observation realizations from Definition 5.151. For every  $\zeta \in Z_0$ , assume in addition that  $\text{DynOperative}[\text{Ev}_\zeta]$  holds. The individual canonical observable-time curves used below are then defined. Define the multi-object canonical-observable index set by

$$\text{ObsIdx}_{o,Z_0}^{\text{dyn,quot,joint}} := \bigsqcup_{\zeta \in Z_0} (\{\zeta\} \times \text{ObsIdx}_{o,\zeta}^{\text{quot}}).$$

For a nonempty  $\text{LawIdx}$ ,  $a \in \text{LawIdx}$ , and  $s \in \text{IOSide}$ , let the observable indices and joint-relation indices be

$$\begin{aligned} \text{ObsIdx}_{a,s}^{\text{joint}} &\subseteq \text{ObsIdx}_{o,Z_0}^{\text{quot,joint}}, \\ \text{RelIdx}_{a,s}^{\text{joint}} &\subseteq \text{JRelIdx}_{Z_0}. \end{aligned}$$

The input and output indices are obtained by taking  $s = \text{in}, \text{out}$ .

Let the individual law-evaluation time domain satisfy

$$\begin{aligned} \text{H}_a^{\text{joint}}[\rho_o] &\subseteq \text{T}_{o,Z_0}^{\text{joint}}[\rho_o] \\ &\cap \bigcap_{\substack{s \in \text{IOSide} \\ \tau \in \text{RelIdx}_{a,s}^{\text{joint}}}} \text{JRelObs}_{o,\tau,Z_0}[\rho_o].\text{Dom}. \end{aligned}$$

Define the value space and generated value family on each side by

$$\begin{aligned} \text{ValSp}_{a,s}^{\text{joint}} &:= \left( \prod_{(\zeta,\ell) \in \text{ObsIdx}_{a,s}^{\text{joint}}} \text{ObsVal}_{o,\zeta,\ell}^{\text{can}} \right) \times \left( \prod_{\tau \in \text{RelIdx}_{a,s}^{\text{joint}}} \text{J}^{\text{src}}_{\tau,Z_0}.\text{Val} \right), \\ \text{Val}_{a,s}^{\text{joint}}[\rho_o] : \text{H}_a^{\text{joint}}[\rho_o] &\longrightarrow \text{ValSp}_{a,s}^{\text{joint}}, \\ \text{Val}_{a,s}^{\text{joint}}[\rho_o](t) &:= \left( (\text{ObsCurve}_{o,\zeta,\ell}^{\text{real}}[\rho_o](t))_{(\zeta,\ell) \in \text{ObsIdx}_{a,s}^{\text{joint}}}, \right. \\ &\quad \left. (\text{JRelObs}_{o,\tau,Z_0}[\rho_o].\text{Curve}(t))_{\tau \in \text{RelIdx}_{a,s}^{\text{joint}}} \right). \end{aligned}$$

Set the input and output components to

$$\begin{aligned} (\text{InValSp}_a^{\text{joint}}, \text{OutValSp}_a^{\text{joint}}) &:= (\text{ValSp}_{a,\text{in}}^{\text{joint}}, \text{ValSp}_{a,\text{out}}^{\text{joint}}), \\ (\text{InVal}_a^{\text{joint}}, \text{OutVal}_a^{\text{joint}})[\rho_o] &:= (\text{Val}_{a,\text{in}}^{\text{joint}}[\rho_o], \text{Val}_{a,\text{out}}^{\text{joint}}[\rho_o]). \end{aligned}$$

Let the individual law-application domain and law-candidate relation satisfy

$$\begin{aligned} \text{E}_a^{\text{joint}}[\rho_o] &\subseteq \text{T}_{o,Z_0}^{\text{joint}}[\rho_o] \times \text{InValSp}_a^{\text{joint}}, \\ \text{R}_a^{\text{joint}}[\rho_o] &\subseteq \text{E}_a^{\text{joint}}[\rho_o] \times \text{OutValSp}_a^{\text{joint}}. \end{aligned}$$

In the same order as the components of a general law, set

$$\begin{aligned} L_a^{\text{joint}}[\rho_o] := & \left( \text{InValSp}_a^{\text{joint}}, \text{OutValSp}_a^{\text{joint}}, \text{InVal}_a^{\text{joint}}[\rho_o], \text{OutVal}_a^{\text{joint}}[\rho_o], \right. \\ & E_a^{\text{joint}}[\rho_o], R_a^{\text{joint}}[\rho_o], H_a^{\text{joint}}[\rho_o], \left( \mathbb{T}_{o, Z_0}^{\text{joint}}[\rho_o] \times \text{InValSp}_a^{\text{joint}} \right) \times \text{OutValSp}_a^{\text{joint}}, \\ & \left. t \mapsto ((t, \text{InVal}_a^{\text{joint}}[\rho_o](t)), \text{OutVal}_a^{\text{joint}}[\rho_o](t)) \right). \end{aligned}$$

Let

$$\begin{aligned} \mathcal{U}^{\text{joint}}[\rho_o] &\subseteq \mathcal{P}\left(\mathbb{T}_{o, Z_0}^{\text{joint}}[\rho_o]\right), \\ \mathcal{F}^{\text{joint}}[\rho_o] &\subseteq \prod_{a \in \text{LawIdx}} \text{Bij}(\text{EvalTupleSp}[L_a^{\text{joint}}[\rho_o]]). \end{aligned}$$

Define the candidate relation structure for laws referring to joint relations and the law-evaluation datum by

$$\begin{aligned} \text{CandRelStr}_{o, Z_0}^{\text{law, joint}}[\rho_o] &:= \left( \mathbb{T}_{o, Z_0}^{\text{joint}}[\rho_o], \text{LawIdx}, \right. \\ &\quad \left. (L_a^{\text{joint}}[\rho_o])_{a \in \text{LawIdx}}, \mathcal{U}^{\text{joint}}[\rho_o], \mathcal{F}^{\text{joint}}[\rho_o] \right) \\ &\in \text{CandRelStrSp}^{\text{law}}, \end{aligned}$$

and, further, by

$$\begin{aligned} \text{LawEvalDat}_{o, Z_0}^{\text{joint}}[\rho_o] &:= \left( \text{Ev}, \text{CandRelStr}_{o, Z_0}^{\text{law, joint}}[\rho_o] \right) \\ &\in \text{LawEvalDatSp}. \end{aligned}$$

**Proposition 5.153** (Connection of the candidate relation structure for laws referring to joint relations to the five conditions). The law-evaluation datum

$$\text{LawEvalDat}_{o, Z_0}^{\text{joint}}[\rho_o]$$

belongs to

$$\text{LawEvalDatSp}.$$

Consequently, the five-condition vector

$$\mathbf{K}^{\text{law}} \left[ \text{LawEvalDat}_{o, Z_0}^{\text{joint}}[\rho_o] \right]$$

of Definition 5.99 is defined.

*Proof.* Each law component in Definition 5.152 belongs to

$$\text{LawCompSp} \left[ \text{JTr}_{o, Z_0}^{\text{real}}[\rho_o].\text{Dom} \right],$$

and the outer five components satisfy the type of  $\text{CandRelStrSp}^{\text{law}}$ . Hence this law-evaluation datum belongs to  $\text{LawEvalDatSp}$ . Applying Definition 5.99 defines the displayed five-condition vector.  $\square$

**Proposition 5.154** (Connection of the multi-object relation condition series). The multi-object relation condition series is connected as follows.

1.

$$K_{\text{CanJointRelDesc}}(\text{JRelDat}_{o,Z_0})$$

is evaluated as the representative-invariant relation-descent condition at the dynamical-description stage.

2.

$$K_{\text{JointCTime}}(\text{Ev})$$

is the condition for a common base of the object-specific time constructions. For each

$$\zeta \in Z_0,$$

$$\text{DynTrjFactCondAll}[\text{Ev}_\zeta], \quad K_{\text{ObsTimeRealOrdEmb}}(\text{Ev}_\zeta)$$

are evaluated as the generation conditions for the corresponding individual trajectory and common time representation.

3. Separation of object correspondences by records derived from  $\mathfrak{D}$  is evaluated as the record-time resolution condition derived from  $\mathfrak{D}$  and the object-state distinguishability condition derived from  $\mathfrak{B}$  at the continuum-approximation stage.
4. Failure of the candidate relation structure for laws referring to joint relations is evaluated through failure of the five conditions in Proposition 5.153 .
5. Use the joint law datum and the representation used in its construction as the base pair for the transported family in Definition 5.107. Given a differential-evaluation-structure family belonging to this same base pair, failure of a differential law containing joint relation values is evaluated through the differential-stage condition group.

Thus the respective failure positions of multi-object relation descent, individual-trajectory generation, record separation, laws referring to joint relations, and differential laws are identified by their corresponding condition symbols.

Give the realized operational configuration

$$\mathcal{X}$$

and the family of typed comparison rules

$$\mathcal{J}_{\text{cmp}}.$$

For the finite object-specification index family

$$Z_0$$

of Definition 2.10 , take an observer-side information-coupling position

$$o \in U$$

and an object-specification index

$$\zeta \in Z_0.$$

Take an exactly operable complete evaluation specification

$$\mathbf{Ev} = (\mathbf{l}, o, \zeta) \in \mathbf{EvSpec}^{\text{exact}}$$

of Definition 5.2 . In what follows, only within a local context in which a single complete evaluation specification is fixed, each symbol root is first displayed in full in its first formula, after which the subscript pair formed by the second and third components uniquely recoverable from that complete evaluation specification is omitted. Independent identifiers are not omitted, and the component pair is not omitted when multiple complete evaluation specifications are treated simultaneously.

By Theorem 5.138 , the respective establishment conditions for dynamical, law-bearing, and differential descriptions are given by

$$\begin{aligned} & \mathbf{ExactOperable} [\mathbf{l}] \\ & \quad \wedge \mathbf{DynCondAll} [\mathbf{Ev}] \\ & \quad \Longleftrightarrow \mathbf{DynOperative} [\mathbf{Ev}] , \end{aligned}$$

$$\begin{aligned} & \mathbf{DynOperative} [\mathbf{Ev}] \\ & \quad \wedge \mathbf{LawCondAll} [\mathbf{LawEvalDat}_{o,\zeta}] \\ & \quad \Longleftrightarrow \mathbf{LawOperative} [\mathbf{LawEvalDat}_{o,\zeta}] , \end{aligned}$$

and

$$\begin{aligned} & \mathbf{LawOperative} [\mathbf{LawEvalDat}] \\ & \quad \wedge \mathbf{DiffCondAll} [\mathbf{EvalStrFam}_{o,\zeta}^{\text{diff}}] \\ & \quad \Longleftrightarrow \mathbf{DiffOperative} [\mathbf{DiffEvalDat}_{o,\zeta}] . \end{aligned}$$

Under dynamical-description operability, the observer-time quotient derived from  $\mathfrak{C}$ , object-state quotient derived from  $\mathfrak{B}$ , canonical trajectory relation derived from  $\mathfrak{CD}\mathfrak{B}$ , graph trajectory, family of canonical observable functions, law-admissible time representation, dynamical-description object, and dynamical-description witness object are determined.

Under law-description operability, the observer-time law core exists uniquely, and generated representability of law-evaluation time holds.

Under differential-stage operability, the chart-compatible differential law and differential-failure evaluation input are determined, and the order isomorphism between time-representation embedding images, value-jet-relation covariance, and time-representation-change jet-relation covariance hold.

Suppose that the finite object-specification index family

$$Z_0$$

of Definition 2.10 satisfies

$$|Z_0| \geq 2.$$

Also assume agreement of the joint-time bases,

$$K_{\text{JointCTime}}(\iota_{\text{joint}}(\mathbf{l}, o, Z_0)) .$$

For each

$$\zeta \in Z_0,$$

set the complete evaluation specification to

$$\mathbf{Ev} = (\mathbf{l}, o, \zeta),$$

and suppose that the trajectory-factorization condition bundle

$$\mathbf{DynTrjFactCondAll} [\mathbf{Ev}]$$

of Definition 5.67 holds.

If, in addition,

$$K_{\text{CanJointRelDesc}} (\mathbf{JRelDat}_{o, Z_0})$$

holds, Proposition 5.149 gives the unique existence of the quotient-level multi-object relation, joint scalar-time trajectory, and joint trajectory relation.

For each

$$\zeta \in Z_0,$$

set the complete evaluation specification to

$$\mathbf{Ev} = (\mathbf{l}, o, \zeta)$$

and suppose that

$$K_{\text{ObsTimeRealOrdEmb}} (\mathbf{Ev})$$

holds and that

$$\mathbf{RepEmb}_o^{\text{time}} \in \mathbf{OrdEmbSet}_o^{\text{time}} (\mathbb{R}).$$

Definition 5.150 then determines the joint trajectory relation over real-valued time.

When  $\mathbf{DynOperative}[(\mathbf{l}, o, \zeta)]$  holds for every  $\zeta \in Z_0$  and the family of observational realizations of joint relation values from Definition 5.151 is supplied as law-evaluation input, Definition 5.152 determines the candidate relation structure for laws referring to joint relations.

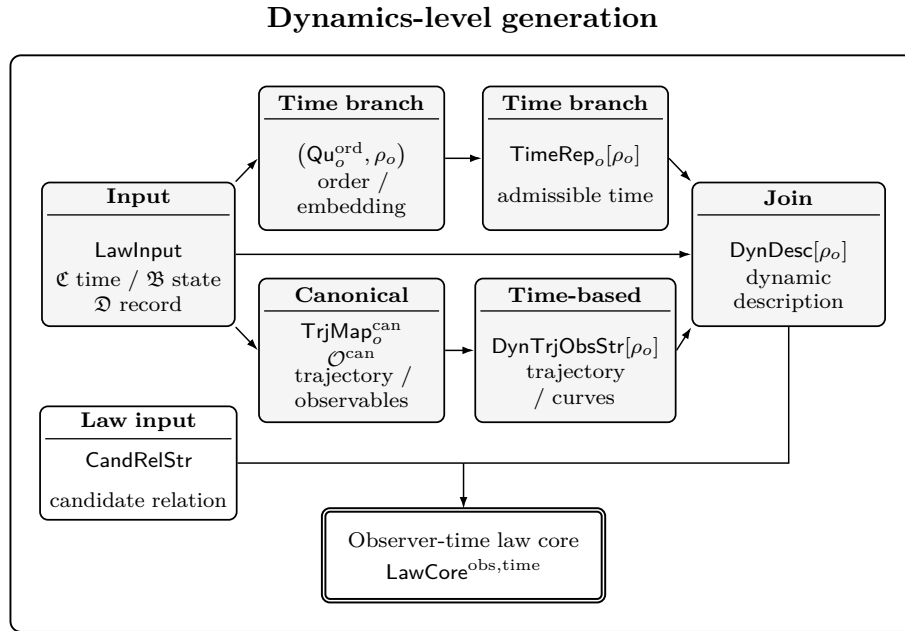


Figure 11: Generation at the dynamical-description stage. Under the conditions for each stage, the observer-time law-input datum yields the time structure and the trajectory-and-observation structure, which form the dynamical-description object. This object and the law-candidate relation structure determine the observer-time law core.

## 6 Localization of Validity, Failure, and Approximation

Take as input the complete evaluation specification of Definition 5.2. Validity and failure localization specifies the invalidating conditions and their dependencies for the comparison, representation, continuum-approximation, dynamical-description, law-representation, and differential-consistency and law-form-preservation series, and generates a six-series failure signature, scale boundaries, and operational audit data. Engineering audit parameters provide the formal evidence used to evaluate failure in each series.

In what follows, only within a local context in which a single complete evaluation specification is fixed, each symbol root is first displayed in full in its first formula, after which the subscript pair formed by the second and third components uniquely recoverable from that complete evaluation specification is omitted. Independent identifiers are not omitted, and the component pair is not omitted when multiple complete evaluation specifications are treated simultaneously.

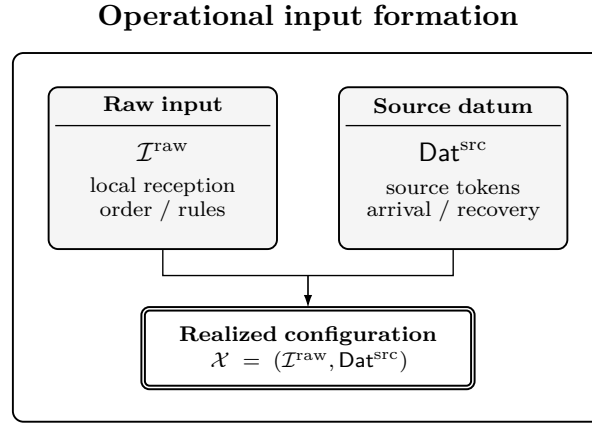


Figure 12: Formation of operational inputs. Local displays, records, object-state differences, and observer-side coupling materials are assembled as a realized operational configuration passed to the comparison and dynamical-description stages.

### 6.1 Basic Definitions of Validity Domains and Localization

**Definition 6.1** (Validity localization). For a set

$$A$$

and a binary predicate

$$P : A \longrightarrow \{\perp, \top\}$$

define its validity localization by

$$\text{Val}_A(P) := \{a \in A \mid P[a] = \top\}$$

Further, for a set

$$B$$

and a binary predicate

$$Q : B \times A \longrightarrow \{\perp, \top\}$$

define the validity localization obtained by substituting

$$b \in B$$

into its first argument by

$$\text{Val}_A(Q; b) := \{a \in A \mid Q[b; a] = \top\}$$

Suppose that a set of complete evaluation specifications

$$\text{EvSpec},$$

a condition-index set

$$I$$

and a family of binary condition predicates

$$K_i : \text{EvSpec} \longrightarrow \{\perp, \top\}, \quad i \in I$$

are given. For any

$$J \subseteq I$$

define

$$\begin{aligned} \text{Val}_{\text{EvSpec}}((K_i)_{i \in J}) &:= \{\text{Ev} \in \text{EvSpec} \mid \forall i \in J, K_i[\text{Ev}] = \top\} \\ &= \bigcap_{i \in J} \text{Val}_{\text{EvSpec}}(K_i) \end{aligned}$$

**Definition 6.2** (Approximation localization). For a set of complete evaluation specifications

$$\text{EvSpec}$$

and a totally ordered set

$$(\Lambda, \leq_\Lambda)$$

define the set of scale-indexed complete evaluation specifications by

$$\text{EvSpec}_\Lambda := \text{EvSpec} \times \Lambda$$

Suppose that an approximation-constraint index set

$$I^{\text{approx}}$$

and a family of binary condition predicates

$$K_\alpha^{\text{approx}} : \text{EvSpec}_\Lambda \longrightarrow \{\perp, \top\}, \quad \alpha \in I^{\text{approx}}$$

are given. For each

$$\text{Ev} \in \text{EvSpec}, \quad \alpha \in I^{\text{approx}}$$

the approximation localization is, by the two-argument form of Definition 6.1,

$$\text{Val}_\Lambda(K_\alpha^{\text{approx}}; \text{Ev}) = \{\lambda \in \Lambda \mid K_\alpha^{\text{approx}}[\text{Ev}; \lambda] = \top\}$$

In the remainder of this chapter, a complete evaluation specification  $\text{Ev}$  is fixed with its type in the formal environment. After the full form  $X[\text{Ev}; \nu]$  has been displayed once, each subsequent display for that same  $\text{Ev}$  is abbreviated as  $X[\nu]$ . The subscripted form

$X_{\text{Ev},\nu}$  is likewise abbreviated as  $X_\nu$  under the same condition. This abbreviation is not used on the left-hand side of a definition.

Define the scale-monotonicity condition by

$$\forall \lambda, \mu \in \Lambda, \quad \lambda \leq_\Lambda \mu \wedge K_\alpha^{\text{approx}}[\mu] = \top \implies K_\alpha^{\text{approx}}[\lambda] = \top$$

Under this condition, if

$$\mu \in \text{Val}_\Lambda(K_\alpha^{\text{approx}}; \text{Ev}), \quad \lambda \leq_\Lambda \mu$$

then

$$\lambda \in \text{Val}_\Lambda(K_\alpha^{\text{approx}}; \text{Ev})$$

Consequently, this approximation localization is an initial segment of

$$(\Lambda, \leq_\Lambda)$$

**Definition 6.3** (Localization of failure positions). Let

$$(\text{FailTok}, \preceq^{\text{fail,dep}}),$$

be a finite partially ordered set, let

$$\text{FailTok}^{\text{strict}}, \quad \text{FailTok}^{\text{approx}},$$

be its subsets, and define the evaluation-state set by

$$\text{FailState} := \{\text{sat}, \text{failed}, \text{unformed}, \text{unevaluable}\}$$

where the four elements are pairwise distinct. Suppose further that an evaluation-state map

$$\text{FailStateOf} : (\text{FailTok}^{\text{strict}} \times \text{EvSpec}) \sqcup (\text{FailTok}^{\text{approx}} \times \text{EvSpec}_\Lambda) \longrightarrow \text{FailState}$$

is given. Define the failure-position localizations on the strict-series side and the continuum-approximation-series side by

$$\begin{aligned} \text{FailLoc}^{\text{strict}}(\text{Ev}) &:= \text{Min}_{\preceq^{\text{fail,dep}}} \{t \in \text{FailTok}^{\text{strict}} \mid \text{FailStateOf}(t; \text{Ev}) = \text{failed}\}, \\ \text{FailLoc}^{\text{approx}}(\text{Ev}, \lambda) &:= \text{Min}_{\preceq^{\text{fail,dep}}} \{t \in \text{FailTok}^{\text{approx}} \mid \text{FailStateOf}(t; \text{Ev}, \lambda) = \text{failed}\} \end{aligned}$$

Each value is the set of all minimal failure-position tokens whose evaluation state is

failed

in the corresponding series. The series type of each token is exactly one of comparison, representation, dynamical description, continuum approximation, law representation, or differential consistency and law-form preservation.

**Definition 6.4** (Validity domain). Let the strict establishment condition and the approximation establishment condition be

$$\begin{aligned} K_{\text{strict}}^{\text{fail}} : \text{EvSpec} &\longrightarrow \{\perp, \top\}, \\ K_{\text{approx}}^{\text{fail}} : \text{EvSpec}_\Lambda &\longrightarrow \{\perp, \top\} \end{aligned}$$



respectively. By Definition 6.1, define

$$\begin{aligned}\mathcal{V}^{\text{strict}} &:= \text{Val}_{\text{EvSpec}}(K_{\text{strict}}^{\text{fail}}), \\ \mathcal{V}^{\text{approx}} &:= \text{Val}_{\text{EvSpec}_\Lambda}(K_{\text{approx}}^{\text{fail}}), \\ \mathcal{V}^{\text{full}} &:= (\mathcal{V}^{\text{strict}} \times \Lambda) \cap \mathcal{V}^{\text{approx}}\end{aligned}$$

and call these the strict validity domain, approximation validity domain, and full-evaluation validity domain, respectively.

**Lemma 6.5** (Inclusion maximum of a family of sets). For a set

$$A$$

and a family of sets

$$\mathfrak{A} \subseteq \mathcal{P}(A)$$

a necessary and sufficient condition for

$$\text{Max}_\subseteq \mathfrak{A}$$

to exist is

$$\bigcup_{B \in \mathfrak{A}} B \in \mathfrak{A}$$

In this case,

$$\text{Max}_\subseteq \mathfrak{A} = \bigcup_{B \in \mathfrak{A}} B$$

and the maximum element is unique.

*Proof.* If a maximum element

$$B^{\text{max}} \in \mathfrak{A}$$

exists, every

$$B \in \mathfrak{A}$$

satisfies

$$B \subseteq B^{\text{max}}$$

and

$$\bigcup_{B \in \mathfrak{A}} B = B^{\text{max}} \in \mathfrak{A}.$$

Conversely, if this union belongs to

$$\mathfrak{A}$$

then it is a maximum element containing every member. Uniqueness follows from anti-symmetry of set inclusion.  $\square$

**Definition 6.6** (Maximum validity domain). Let the nonempty family of admissible subdomains of a validity domain

$$\mathcal{V}$$

be

$$\mathfrak{V}_{\text{adm}}(\mathcal{V}) \subseteq \mathcal{P}(\mathcal{V})$$

When

$$\mathcal{V}^{\max} \in \mathfrak{V}_{\text{adm}}(\mathcal{V})$$

satisfies

$$\forall \mathcal{V}' \in \mathfrak{V}_{\text{adm}}(\mathcal{V}), \quad \mathcal{V}' \subseteq \mathcal{V}^{\max}$$

the object

$$\mathcal{V}^{\max} = \text{Max}_{\subseteq} \mathfrak{V}_{\text{adm}}(\mathcal{V})$$

is called the maximum validity domain.

**Proposition 6.7** (Existence condition for the maximum validity domain). A necessary and sufficient condition for the maximum validity domain to exist is

$$\bigcup_{\mathcal{V}' \in \mathfrak{V}_{\text{adm}}(\mathcal{V})} \mathcal{V}' \in \mathfrak{V}_{\text{adm}}(\mathcal{V})$$

In this case,

$$\mathcal{V}^{\max} = \bigcup_{\mathcal{V}' \in \mathfrak{V}_{\text{adm}}(\mathcal{V})} \mathcal{V}'$$

and it is unique.

*Proof.* Apply Lemma 6.5 to

$$A = \mathcal{V}, \quad \mathfrak{A} = \mathfrak{V}_{\text{adm}}(\mathcal{V})$$

□

**Definition 6.8** (Maximal effective interval). For a totally ordered set

$$(\Lambda, \leq_{\Lambda})$$

and a subset

$$B \subseteq \Lambda$$

define the family of initial subsets contained in that subset by

$$\text{Init}_{\Lambda}(B) := \{\mathcal{I} \subseteq B \mid \forall \mu \in \mathcal{I}, \forall \lambda \in \Lambda, \lambda \leq_{\Lambda} \mu \implies \lambda \in \mathcal{I}\}$$

For each

$$\text{Ev} \in \text{EvSpec}$$

define the family of admissible initial intervals by

$$\mathfrak{I}_{\text{init}}[\text{Ev}] := \text{Init}_{\Lambda}(\text{Val}_{\Lambda}(K_{\text{approx}}^{\text{fail}}; \text{Ev}))$$

The object

$$\mathcal{I}^{\max}[\text{Ev}] = \text{Max}_{\subseteq} \mathfrak{I}_{\text{init}}[\text{Ev}]$$

is called the maximal effective interval at

$$\text{Ev}$$

**Proposition 6.9** (Existence of the maximal effective interval). For any complete evaluation specification  $\text{Ev} \in \text{EvSpec}$ , the maximal effective interval exists uniquely, and

$$\bigcup_{\mathcal{I} \in \mathcal{I}_{\text{init}}[\text{Ev}]} \mathcal{I} \in \mathcal{I}_{\text{init}}[\text{Ev}]$$

Further, if the scale-monotonicity condition

$$\forall \lambda, \mu \in \Lambda, \quad \lambda \leq_{\Lambda} \mu \wedge K_{\text{approx}}^{\text{fail}}[\text{Ev}; \mu] = \top \implies K_{\text{approx}}^{\text{fail}}[\lambda] = \top$$

holds, then

$$\mathcal{I}^{\text{max}}[\text{Ev}] = \text{Val}_{\Lambda}(K_{\text{approx}}^{\text{fail}}; \text{Ev})$$

and it is unique. Its maximum endpoint exists if and only if

$$\text{Max}(\mathcal{I}^{\text{max}}[\text{Ev}])$$

exists.

*Proof.* Because the empty set belongs to

$$\mathcal{I}_{\text{init}}[\text{Ev}]$$

this family is nonempty. Its union is also an initial subset contained in the approximation-validity domain and therefore belongs to the same family. Applying Lemma 6.5 to

$$A = \Lambda, \quad \mathfrak{A} = \mathcal{I}_{\text{init}}[\text{Ev}]$$

yields existence and uniqueness of the maximal effective interval.

Under the scale-monotonicity condition, the approximation localization of Definition 6.2 is itself an admissible initial interval and contains all of its members. It therefore coincides with the maximal effective interval. The assertion concerning the maximum endpoint follows from the definition.  $\square$

**Definition 6.10** (Semantic-series-tagged failure-position token system). Define the sets of scale-invariant series, continuum-approximation series, and all failure series by

$$\begin{aligned} \text{FailSeries}^{\text{strict}} &:= \{\text{cmp}, \text{repr}, \text{dyn}, \text{law}, \text{diff}\}, \\ \text{FailSeries}^{\text{approx}} &:= \{\text{approx}\}, \\ \text{FailSeries} &:= \text{FailSeries}^{\text{strict}} \sqcup \text{FailSeries}^{\text{approx}} \end{aligned}$$

respectively. For a series

$$\tau \in \text{FailSeries}$$

define the number of failure positions, the index set, and the evaluation-specification type by

$$\begin{aligned} n_{\tau}^{\text{fail}} &:= \begin{cases} 8, & \tau \in \{\text{cmp}, \text{dyn}\}, \\ 3, & \tau = \text{repr}, \\ 9, & \tau = \text{approx}, \\ 5, & \tau \in \{\text{law}, \text{diff}\}, \end{cases} \\ I_{\tau}^{\text{fail}} &:= \{1, \dots, n_{\tau}^{\text{fail}}\}, \\ \text{FailSpec}_{\tau} &:= \begin{cases} \text{EvSpec}, & \tau \in \text{FailSeries}^{\text{strict}}, \\ \text{EvSpec}_{\Lambda}, & \tau = \text{approx} \end{cases} \end{aligned}$$

respectively.

Define the map that obtains the series-specific evaluation specification from a scale-indexed complete evaluation specification by

$$\begin{aligned} \text{FailSpecAt}_\tau : \text{EvSpec}_\Lambda &\longrightarrow \text{FailSpec}_\tau, \\ \text{FailSpecAt}_\tau[\text{Ev}; \lambda] &:= \begin{cases} \text{Ev}, & \tau \in \text{FailSeries}^{\text{strict}}, \\ (\text{Ev}, \lambda), & \tau = \text{approx} \end{cases} \end{aligned}$$

and define the six-series family of evaluation specifications and its component notation by

$$\begin{aligned} \text{Ev}^{\text{fail}} &:= \left( (\text{Ev})_{\tau \in \text{FailSeries}^{\text{strict}}}, ((\text{Ev}, \lambda))_{\lambda \in \Lambda} \right), \\ \text{Ev}_\tau^{\text{fail}} &:= \text{FailSpecAt}_\tau[\text{Ev}; \lambda], \quad \tau \in \text{FailSeries}^{\text{strict}}, \\ \text{Ev}_{\text{approx}, \lambda}^{\text{fail}} &:= \text{FailSpecAt}_{\text{approx}}[\text{Ev}; \lambda] \end{aligned}$$

respectively. A scale-invariant-series component does not depend on the choice of scale.

Define the tagged index sets by

$$\begin{aligned} \text{FailIdx}^{\text{strict}} &:= \bigsqcup_{\tau \in \text{FailSeries}^{\text{strict}}} \{\tau\} \times I_\tau^{\text{fail}}, \\ \text{FailIdx}^{\text{approx}} &:= \{\text{approx}\} \times I_{\text{approx}}^{\text{fail}}, \\ \text{FailIdx} &:= \text{FailIdx}^{\text{strict}} \sqcup \text{FailIdx}^{\text{approx}} \end{aligned}$$

respectively. For each

$$(\tau, i) \in \text{FailIdx}$$

let the series-specific failure condition be

$$K_{\tau, i}^{\text{fail}} : \text{FailSpec}_\tau \longrightarrow \{\perp, \top\}.$$

For a native evaluation domain  $D_{\tau, i} \subseteq \text{FailSpec}_\tau$  and a predicate  $P_{\tau, i}$  supplied in a later section, define the total predicate above by

$$K_{\tau, i}^{\text{fail}}[x] \iff \exists y \in D_{\tau, i}, \quad x = y \wedge P_{\tau, i}[y].$$

For a condition already defined everywhere, use its full domain. This extension does not evaluate partial expressions outside their native domain. Conditions, regions and signatures are defined relative to fixed selected input families for each series. Judgments based on different selected inputs are not identified merely because they have the same complete evaluation specification. Define the failure-condition classification sets by

$$\begin{aligned} \text{FailCond}^{\text{strict}} &:= \{(\tau, i, K_{\tau, i}^{\text{fail}}) \mid (\tau, i) \in \text{FailIdx}^{\text{strict}}\}, \\ \text{FailCond}^{\text{approx}} &:= \{(\tau, i, K_{\tau, i}^{\text{fail}}) \mid (\tau, i) \in \text{FailIdx}^{\text{approx}}\}, \\ \text{FailCond} &:= \text{FailCond}^{\text{strict}} \sqcup \text{FailCond}^{\text{approx}} \end{aligned}$$

respectively.

Failure-position tokens are formal tagged tokens that retain the series and local index. For any  $(\tau, i), (\sigma, j) \in \text{FailIdx}$ , their equality is specified by

$$\text{FailTok}_{\tau, i} = \text{FailTok}_{\sigma, j} \iff \tau = \sigma \wedge i = j.$$

Define the failure-position token sets for each series, for the scale-invariant series, for the continuum-approximation series, and for all series by

$$\begin{aligned}\text{FailTok}_\tau &:= \{\text{FailTok}_{\tau,i} \mid i \in I_\tau^{\text{fail}}\}, \\ \text{FailTok}^{\text{strict}} &:= \bigsqcup_{\tau \in \text{FailSeries}^{\text{strict}}} \text{FailTok}_\tau, \\ \text{FailTok}^{\text{approx}} &:= \text{FailTok}_{\text{approx}}, \\ \text{FailTok} &:= \text{FailTok}^{\text{strict}} \sqcup \text{FailTok}^{\text{approx}}\end{aligned}$$

respectively. Define the series-type map and the failure-condition classification map by

$$\begin{aligned}\text{FailSeriesOf} : \text{FailTok} &\longrightarrow \text{FailSeries}, \quad \text{FailSeriesOf}(\text{FailTok}_{\tau,i}) := \tau, \\ \text{FailCondOf} : \text{FailTok} &\longrightarrow \text{FailCond}, \quad \text{FailCondOf}(\text{FailTok}_{\tau,i}) := (\tau, i, K_{\tau,i}^{\text{fail}})\end{aligned}$$

respectively.

Define the failure-evaluation argument set and the failure-condition evaluation map by

$$\begin{aligned}\text{FailEvalArg} &:= \bigsqcup_{\tau \in \text{FailSeries}} (\text{FailTok}_\tau \times \text{FailSpec}_\tau), \\ \text{FailEval} : \text{FailEvalArg} &\longrightarrow \{\perp, \top\}, \\ \text{FailEval}(\text{FailTok}_{\tau,i}; \text{EvalSpec}) &:= K_{\tau,i}^{\text{fail}}[\text{EvalSpec}]\end{aligned}$$

respectively.

**Lemma 6.11** (Finite tagged disjoint union and classification correspondence). For a finite tag set

$$T$$

and a family of finite nonempty index sets

$$(I_t)_{t \in T}$$

let

$$X_t := \{x_{t,i} \mid i \in I_t\}, \quad X := \bigsqcup_{t \in T} X_t$$

For every  $t \in T$ , assume that the map  $I_t \rightarrow X_t$ ,  $i \mapsto x_{t,i}$  is injective. In what follows, identify each  $X_t$  with its image in  $X$  under the canonical inclusion. Then

$$(X_t)_{t \in T}$$

is a finite partition of

$$X$$

and

$$|X| = \sum_{t \in T} |I_t|.$$

Moreover, for a family of objects

$$(P_{t,i})_{t \in T, i \in I_t}$$

the map

$$x_{t,i} \longmapsto (t, i, P_{t,i})$$

is a bijection from

$$X$$

to

$$\{(t, i, P_{t,i}) \mid t \in T, i \in I_t\}$$

*Proof.* The components of the tagged disjoint union are pairwise disjoint, and each map

$$I_t \longrightarrow X_t, \quad i \longmapsto x_{t,i}$$

is surjective by the definition of  $X_t$  and injective by assumption, hence bijective. Each component is nonempty because  $I_t$  is nonempty. The partition, cardinality formula, and final bijection follow.  $\square$

**Lemma 6.12** (First-failure partition of a finite sequence of binary conditions). For a family of binary conditions

$$K_i : \mathcal{Y} \longrightarrow \{\perp, \top\}, \quad i \in \{1, \dots, n\},$$

and the empty conjunction

$$\bigwedge_{j < 1} K_j(y) := \top$$

one has

$$\left\{ y \in \mathcal{Y} \mid \neg \bigwedge_{i=1}^n K_i(y) \right\} = \bigsqcup_{i=1}^n \left\{ y \in \mathcal{Y} \mid \left( \bigwedge_{j < i} K_j(y) \right) \wedge \neg K_i(y) \right\}.$$

Consequently, for every element of the left-hand side,

$$\min\{i \in \{1, \dots, n\} \mid \neg K_i(y)\}$$

is uniquely determined.

*Proof.* The set of failed indices is finite and nonempty, and therefore has a least element

$$i_0$$

Its minimality gives membership in the

$$i_0$$

component on the right-hand side. The reverse inclusion follows from

$$\neg K_{i_0}(y)$$

in each component. Two distinct components are disjoint because the condition at the lower index is false in one and true in the other.  $\square$

**Definition 6.13** (Three-point nontransitivity witness predicate for a relation). For a binary relation on a set  $\mathcal{Y}$ ,

$$\mathcal{R} \subseteq \mathcal{Y} \times \mathcal{Y}$$

define the three-point nontransitivity witness predicate by

$$\begin{aligned} & \text{NonTrans}[\mathcal{Y}; \mathcal{R}] \\ & \iff \left[ \exists y_0, y_1, y_2 \in \mathcal{Y} : \bigwedge_{j \in \{0,1\}} (y_j, y_{j+1}) \in \mathcal{R}, \quad (y_0, y_2) \notin \mathcal{R} \right] \end{aligned}$$

**Definition 6.14** (Maximum number of failure positions). Define the maximum number of failure positions by

$$\text{FailCount}^{\max} := |\text{FailTok}|$$

**Proposition 6.15** (Finite partition of the failure-position token system and correspondence with condition classifications). The family

$$(\text{FailTok}_\tau)_{\tau \in \text{FailSeries}}$$

is a finite partition of

$$\text{FailTok}$$

and

$$\begin{aligned} & \left( |\text{FailTok}_{\text{cmp}}|, |\text{FailTok}_{\text{repr}}|, |\text{FailTok}_{\text{dyn}}|, \right. \\ & \quad \left. |\text{FailTok}_{\text{approx}}|, |\text{FailTok}_{\text{law}}|, |\text{FailTok}_{\text{diff}}| \right) \\ &= (8, 3, 8, 9, 5, 5). \end{aligned}$$

Consequently,

$$\text{FailCount}^{\max} = 38.$$

Furthermore,

$$\text{FailCondOf} : \text{FailTok} \longrightarrow \text{FailCond}$$

is a bijection.

*Proof.* Apply Lemma 6.11 to

$$(T, I_t, X_t, P_{t,i}) = (\text{FailSeries}, I_\tau^{\text{fail}}, \text{FailTok}_\tau, K_{\tau,i}^{\text{fail}})$$

The number of elements in each series from Definition 6.10 gives

$$|\text{FailTok}| = 8 + 3 + 8 + 9 + 5 + 5 = 38$$

□

**Lemma 6.16** (Reachability order of rank-increasing edges). Suppose that a set

$$V,$$

an edge set

$$E \subseteq V \times V,$$

a strict partially ordered set

$$(R, <_R)$$

and a map

$$r : V \longrightarrow R$$

satisfy

$$(v, w) \in E \implies r(v) <_R r(w)$$

Write the relation given by equality or by the existence of a positive-length finite directed path on

$$E$$

as

$$\preceq_E$$

Then

$$E$$

has no directed cycle, and

$$\preceq_E$$

is a partial order on

$$V$$

*Proof.* Equality gives reflexivity, and concatenation of directed paths gives transitivity. Because

$$r$$

increases strictly along every directed path, a directed cycle would imply

$$r(v) <_R r(v)$$

contrary to the irreflexivity of a strict partial order. Two distinct points reachable from one another in both directions would yield a directed cycle, so antisymmetry also holds.  $\square$

**Definition 6.17** (Failure-generation dependency graph and dependency order). For each

$$\tau \in \text{FailSeries}$$

define the within-series direct dependency edges by

$$E_\tau^{\text{fail}} := \begin{cases} \{(i, i+1) \mid 1 \leq i < 8\}, & \tau = \text{cmp}, \\ \{(1, 2), (2, 3)\}, & \tau = \text{repr}, \\ \{(1, 6), (6, 7), (2, 3), (3, 8), (6, 8)\}, & \tau = \text{dyn}, \\ \{(i, 7) \mid 1 \leq i \leq 6\} \cup \{(7, 8), (7, 9)\}, & \tau = \text{approx}, \\ \{(1, 3), (1, 5), (3, 5)\}, & \tau = \text{law}, \\ \{(1, 2), (2, 3), (1, 4), (1, 5)\}, & \tau = \text{diff} \end{cases}$$

and define the within-series graphs, within-series token edges, between-series graph, and complete dependency-edge set by

$$\begin{aligned} \text{FailDAG}_\tau &:= (I_\tau^{\text{fail}}, E_\tau^{\text{fail}}), \\ \text{FailEdge}_\tau &:= \{(\text{FailTok}_{\tau,i}, \text{FailTok}_{\tau,j}) \mid (i, j) \in E_\tau^{\text{fail}}\}, \\ \text{FailEdge}^{\text{intra}} &:= \bigcup_{\tau \in \text{FailSeries}} \text{FailEdge}_\tau, \\ E_{\text{series}}^{\text{fail}} &:= \{(\text{cmp}, \text{repr}), (\text{repr}, \text{dyn}), (\text{dyn}, \text{approx}), \\ &\quad (\text{dyn}, \text{law}), (\text{law}, \text{diff})\}, \\ \text{FailDAG}^{\text{series}} &:= (\text{FailSeries}, E_{\text{series}}^{\text{fail}}), \\ \text{FailEdge}^{\text{inter}} &:= \bigcup_{(\tau_0, \tau_1) \in E_{\text{series}}^{\text{fail}}} (\text{FailTok}_{\tau_0} \times \text{FailTok}_{\tau_1}), \\ \text{FailEdge} &:= \text{FailEdge}^{\text{intra}} \cup \text{FailEdge}^{\text{inter}} \end{aligned}$$



respectively. In the reachability notation of Lemma 6.16, the failure-generation dependency relation is

$$\preceq^{\text{fail,dep}} := \preceq_{\text{FailEdge}} \subseteq \text{FailTok} \times \text{FailTok}$$

Define incomparability with respect to this order by

$$s \parallel_{\text{fail,dep}} t \iff \neg (s \preceq^{\text{fail,dep}} t) \wedge \neg (t \preceq^{\text{fail,dep}} s)$$

**Proposition 6.18** (Partial-order property of the failure-generation dependency order).  
The relation

$$\preceq^{\text{fail,dep}}$$

is a partial order on

$$\text{FailTok}$$

*Proof.* For each failure series, define the series-layer map and the family of within-series ranks by

$$\begin{aligned} \text{SeriesRank}(\tau) &:= \begin{cases} 0, & \tau = \text{cmp}, \\ 1, & \tau = \text{repr}, \\ 2, & \tau = \text{dyn}, \\ 3, & \tau \in \{\text{approx}, \text{law}\}, \\ 4, & \tau = \text{diff}, \end{cases} \\ (\text{IntraRank}_\tau(i))_{i \in I_\tau^{\text{fail}}} &:= \begin{cases} (0, 1, 2, 3, 4, 5, 6, 7), & \tau = \text{cmp}, \\ (0, 1, 2), & \tau = \text{repr}, \\ (0, 0, 1, 0, 0, 1, 2, 2), & \tau = \text{dyn}, \\ (0, 0, 0, 0, 0, 0, 1, 2, 2), & \tau = \text{approx}, \\ (0, 0, 1, 0, 2), & \tau = \text{law}, \\ (0, 1, 2, 1, 1), & \tau = \text{diff} \end{cases} \end{aligned}$$

respectively. Define the rank map on all tokens into the product equipped with the lexicographic order by

$$\text{FailRank}(\text{FailTok}_{\tau,i}) := (\text{SeriesRank}(\tau), \text{IntraRank}_\tau(i))$$

For every edge of Definition 6.17,

$$\text{FailRank}(s) <_{\text{lex}} \text{FailRank}(t)$$

holds. Apply Lemma 6.16 to

$$(V, E, R, r) = (\text{FailTok}, \text{FailEdge}, \{0, 1, 2, 3, 4\} \times \mathbb{N}_0, \text{FailRank})$$

□

**Proposition 6.19** (Parallel branches of the failure-generation dependency order). With respect to the failure-generation dependency order,

$$\begin{aligned}
& s \parallel_{\text{fail,dep}} t, \\
& (s, t) \in \text{FailTok}_{\text{approx}} \times (\text{FailTok}_{\text{law}} \cup \text{FailTok}_{\text{diff}}), \\
& \text{FailTok}_{\text{dyn},4} \parallel_{\text{fail,dep}} \text{FailTok}_{\text{dyn},5}, \\
& \text{FailTok}_{\text{approx},i} \parallel_{\text{fail,dep}} \text{FailTok}_{\text{approx},j}, \\
& 1 \leq i < j \leq 6, \\
& \text{FailTok}_{\text{approx},8} \parallel_{\text{fail,dep}} \text{FailTok}_{\text{approx},9}, \\
& \text{FailTok}_{\text{law},i} \parallel_{\text{fail,dep}} \text{FailTok}_{\text{law},j}, \\
& i < j, \quad i, j \in \{1, 2, 4\}, \\
& \text{FailTok}_{\text{diff},4} \parallel_{\text{fail,dep}} \text{FailTok}_{\text{diff},5}
\end{aligned}$$

holds.

*Proof.* The between-series graph is

$$\text{cmp} \longrightarrow \text{repr} \longrightarrow \text{dyn} \longrightarrow \begin{cases} \text{approx}, \\ \text{law} \longrightarrow \text{diff} \end{cases}$$

and therefore has no directed path connecting the continuum-approximation series to the law or differential-consistency series. Within a series, the edge sets of Definition 6.17 contain no directed path joining any of the listed pairs. Because a between-series edge strictly increases the series layer, no directed path can join two points of the same series by passing through another series.  $\square$

**Lemma 6.20** (Deterministic state recursion on a finite DAG). Let

$$(V, E)$$

be a finite directed acyclic graph. Suppose that, for every vertex

$$v \in V$$

the states of its direct predecessors determine exactly one element of a state set

$$S$$

through an update map

$$\Phi_v : S^{\text{Pred}_E(v)} \longrightarrow S$$

Then the equations

$$s(v) = \Phi_v \left( (s(u))_{u \in \text{Pred}_E(v)} \right), \quad v \in V$$

are satisfied by exactly one map

$$s : V \longrightarrow S$$

*Proof.* Choose a topological ordering

$$v_1, \dots, v_{|V|}$$

of the graph. Because every direct predecessor of each

$$v_k$$

belongs to

$$v_1, \dots, v_{k-1}$$

the map

$$\Phi_{v_k}$$

uniquely determines

$$s(v_k)$$

in sequence. Finite induction gives existence and uniqueness.  $\square$

**Definition 6.21** (Evaluation states and failed-token sets). Define the set of direct-predecessor tokens by

$$\text{FailPredOf}(t) := \{u \in \text{FailTok} \mid (u, t) \in \text{FailEdge}\}$$

Let the non-ancestor formation-input satisfaction map and the evaluation-input satisfaction map be

$$\text{FailInputSat}^{\text{form}}, \text{FailInputSat}^{\text{eval}} : \text{FailEvalArg} \longrightarrow \{\perp, \top\}$$

Let the evaluation-state map, the direct-predecessor satisfaction map, and the three evaluation gates be

$$\begin{aligned} \text{FailStateOf} &: \text{FailEvalArg} \longrightarrow \text{FailState}, \\ \text{FailPredSat}, \text{FailFormed}, \text{FailInputSat}, \text{FailReady} &: \text{FailEvalArg} \longrightarrow \{\perp, \top\} \end{aligned}$$

respectively. For

$$(\sigma, \tau) \in E_{\text{series}}^{\text{fail}} \cup \{(\rho, \rho) \mid \rho \in \text{FailSeries}\}$$

which forms either a direct between-series dependency edge or a same-series pair, define the evaluation-specification reindexing map by

$$\text{FailSpecRes}_{\sigma \leftarrow \tau} : \text{FailSpec}_{\tau} \longrightarrow \text{FailSpec}_{\sigma}$$

and

$$\text{FailSpecRes}_{\sigma \leftarrow \tau}(\text{FailSpecAt}_{\tau}[\text{Ev}; \lambda]) = \text{FailSpecAt}_{\sigma}[\text{Ev}; \lambda]$$

Use the identity map between scale-invariant series and within each series, and the first projection  $(\text{Ev}, \lambda) \mapsto \text{Ev}$  for  $(\sigma, \tau) = (\text{dyn}, \text{approx})$ .

For an evaluation argument

$$a = (t; \text{EvalSpec}) \in \text{FailTok}_{\tau} \times \text{FailSpec}_{\tau}$$

and

$$u \in \text{FailPredOf}(t) \cap \text{FailTok}_{\sigma}$$

define the direct-predecessor evaluation argument by

$$\text{FailPredArgOf}(u; a) := (u; \text{FailSpecRes}_{\sigma \leftarrow \tau}(\text{EvalSpec})) \in \text{FailEvalArg}$$

The direct-predecessor satisfaction judgment is

$$\begin{aligned} \text{FailPredSat}(a) &= \top \\ \iff \forall u \in \text{FailPredOf}(t), \\ &\quad \text{FailStateOf}(\text{FailPredArgOf}(u; a)) = \text{sat}. \end{aligned}$$

For each

$$a \in \text{FailEvalArg}$$

define the three gates of input satisfaction, formation, and evaluation readiness by

$$\begin{aligned} \text{FailInputSat}(a) &= \text{FailInputSat}^{\text{form}}(a) \wedge \text{FailInputSat}^{\text{eval}}(a), \\ \text{FailFormed}(a) &= \text{FailInputSat}^{\text{form}}(a) \wedge \text{FailPredSat}(a), \\ \text{FailReady}(a) &= \text{FailPredSat}(a) \wedge \text{FailInputSat}(a) \end{aligned}$$

respectively. Here the conjunctions on the right-hand side are conjunctions on

$$\{\perp, \top\}$$

Define the evaluation state recursively by

$$\text{FailStateOf}(a) := \begin{cases} \text{unformed}, & \text{FailFormed}(a) = \perp, \\ \text{unevaluable}, & \text{FailFormed}(a) = \top \wedge \text{FailInputSat}^{\text{eval}}(a) = \perp, \\ \text{sat}, & \text{FailReady}(a) = \top \wedge \text{FailEval}(a) = \top, \\ \text{failed}, & \text{FailReady}(a) = \top \wedge \text{FailEval}(a) = \perp \end{cases}.$$

Require the input-satisfaction maps to be compatible with the native domains of Definition 6.10:

$$\text{FailReady}(t; x) = \top \implies x \in D_{\tau, i} \quad (t = \text{FailTok}_{\tau, i}).$$

If a selected input has not been supplied, its formation-input satisfaction is false. A false extended condition outside its native domain does not by itself constitute a failed state.

Define the token-specific evaluation argument and state section at a scale-indexed complete evaluation specification by

$$\begin{aligned} \text{FailArgAt}_{\text{Ev}, \lambda}(t) &:= (t; \text{FailSpecAt}_{\text{FailSeriesOf}(t)}[\text{Ev}; \lambda]), \\ \text{FailStateAt}_{\text{Ev}, \lambda}(t) &:= \text{FailStateOf}(\text{FailArgAt}_{\text{Ev}, \lambda}(t)) \end{aligned}$$

respectively.

For each

$$q \in \text{FailState}$$

define the state fiber and its failed-state components by

$$\begin{aligned} \text{FailFiber}_q[\text{Ev}; \lambda] &:= \{\text{FailTok}_{\tau, i} \in \text{FailTok} \mid \text{FailStateAt}_{\text{Ev}, \lambda}(\text{FailTok}_{\tau, i}) = q\}, \\ \text{FailedTok}[\text{Ev}; \lambda] &:= \text{FailFiber}_{\text{failed}}[\text{Ev}; \lambda], \\ \text{FailedTok}^{\text{strict}}[\text{Ev}] &:= \text{FailedTok}[\text{Ev}; \lambda] \cap \text{FailTok}^{\text{strict}}, \\ \text{FailedTok}^{\text{approx}}[\text{Ev}; \lambda] &:= \text{FailedTok}[\text{Ev}; \lambda] \cap \text{FailTok}^{\text{approx}} \end{aligned}$$

respectively. The value on the scale-invariant-series side does not depend on the choice of scale. Consequently,

$$\begin{aligned} \text{FailedTok}^{\text{strict}} &: \text{EvSpec} \longrightarrow \mathcal{P}(\text{FailTok}^{\text{strict}}), \\ \text{FailedTok}^{\text{approx}} &: \text{EvSpec}_\Lambda \longrightarrow \mathcal{P}(\text{FailTok}^{\text{approx}}), \\ \text{FailedTok} &: \text{EvSpec}_\Lambda \longrightarrow \mathcal{P}(\text{FailTok}). \end{aligned}$$

**Proposition 6.22** (Existence and uniqueness of the evaluation-state assignment and decomposition into state fibers). There exists exactly one evaluation-state map

$$\text{FailStateOf} : \text{FailEvalArg} \longrightarrow \text{FailState}$$

that satisfies the recursive conditions of Definition 6.21. For each

$$(\text{Ev}, \lambda) \in \text{EvSpec}_\Lambda$$

the family

$$\{\text{FailFiber}_q[\text{Ev}; \lambda] \mid q \in \text{FailState}\}$$

gives a decomposition into pairwise disjoint state fibers of

$$\text{FailTok}.$$

Empty state fibers are allowed. Its restrictions to

$$\text{FailTok}^{\text{strict}}, \quad \text{FailTok}^{\text{approx}}$$

give decompositions of the respective subsets into pairwise disjoint state fibers.

*Proof.* By Propositions 6.15 and 6.18,

$$(\text{FailTok}, \text{FailEdge})$$

is a finite directed acyclic graph. At each token, the four branches of Definition 6.21 return exactly one state from the family of direct-predecessor states, the two input-satisfaction values, and the failure-condition evaluation value. Lemma 6.20 therefore gives the existence and uniqueness of the evaluation-state map.

Because each token has exactly one of the four states, the four state fibers are pairwise disjoint. Their union equals

$$\text{FailTok}$$

and their restrictions to either of the two series-side subsets remain pairwise disjoint and exhaustive.  $\square$

**Proposition 6.23** (State-equivalence conditions and finiteness of the failed-token sets). For any

$$(\text{Ev}, \lambda) \in \text{EvSpec}_\Lambda, \quad \text{FailTok}_{\tau,i} \in \text{FailTok}$$

one has

$$\text{FailTok}_{\tau,i} \in \text{FailedTok}[\text{Ev}; \lambda] \iff \text{FailStateAt}_{\text{Ev}, \lambda}(\text{FailTok}_{\tau,i}) = \text{failed}.$$

Furthermore,

$$\begin{aligned} \text{FailedTok}[\lambda] &= \text{FailedTok}^{\text{strict}}[\text{Ev}] \sqcup \text{FailedTok}^{\text{approx}}[\text{Ev}; \lambda], \\ \text{FailedTok}^{\text{strict}}[\text{Ev}] &= \text{FailedTok}[\lambda] \cap \text{FailTok}^{\text{strict}}, \\ \text{FailedTok}^{\text{approx}}[\lambda] &= \text{FailedTok}[\lambda] \cap \text{FailTok}^{\text{approx}} \end{aligned}$$

and all three sets are finite.

*Proof.* The equivalence and the three displays follow from Definition 6.21. By Proposition 6.15,

$$\text{FailTok}$$

is finite, and the three sets are its subsets.  $\square$

**Definition 6.24** (Strict failure-generation dependency order and three cases of minimal failure). Define the strict dependency order by

$$s \prec^{\text{fail,dep}} t \iff s \preceq^{\text{fail,dep}} t \wedge s \neq t$$

and define the set of minimal-failure cases, the number of failed tokens, and the case map by

$$\begin{aligned} \text{FailCase} &:= \{\text{FailAbsent}, \text{FailUnique}, \text{FailParallel}\}, \\ \text{FailCount}[\text{Ev}; \lambda] &:= |\text{FailedTok}[\text{Ev}; \lambda]|, \\ \text{FailCaseOf}(\text{Ev}; \lambda) &:= \begin{cases} \text{FailAbsent}, & \text{FailCount}[\text{Ev}; \lambda] = 0, \\ \text{FailUnique}, & \text{FailCount}[\text{Ev}; \lambda] = 1, \\ \text{FailParallel}, & \text{FailCount}[\text{Ev}; \lambda] \geq 2 \end{cases} \end{aligned}$$

respectively. The three cases are pairwise distinct.

**Lemma 6.25** (Forward propagation of a nonsatisfied state into unformed successors). Take an evaluation specification

$$(\text{Ev}, \lambda) \in \text{EvSpec}_\Lambda$$

and a directed path

$$t_0 \longrightarrow t_1 \longrightarrow \cdots \longrightarrow t_m, \quad m \geq 1,$$

Using the notation

$$a_r := \text{FailArgAt}_{\text{Ev}, \lambda}(t_r)$$

one has

$$\text{FailStateAt}_{\text{Ev}, \lambda}(t_0) \neq \text{sat} \implies \forall r \in \{1, \dots, m\}, \quad \text{FailStateAt}_\lambda(t_r) = \text{unformed}.$$

*Proof.* For every

$$r \geq 1$$

one has

$$t_{r-1} \in \text{FailPredOf}(t_r).$$

The initial state is not satisfied, and at each inductive step the immediately preceding state satisfies

$$\text{unformed} \neq \text{sat}$$

so Definition 6.21 gives

$$\begin{aligned} \text{FailPredSat}(a_r) &= \text{FailFormed}(a_r) = \perp, \\ \text{FailStateOf}(a_r) &= \text{unformed} \end{aligned}$$

in succession.  $\square$

**Proposition 6.26** (Incomparability of failed tokens, equality with the minimal elements, and the three cases). For any

$$(\text{Ev}, \lambda) \in \text{EvSpec}_\Lambda$$

and distinct

$$s, t \in \text{FailedTok}[\text{Ev}; \lambda]$$

one has

$$s \parallel_{\text{fail,dep}} t.$$

Consequently,

$$\text{Min}_{\preceq_{\text{fail,dep}}} \text{FailedTok}[\lambda] = \text{FailedTok}[\lambda].$$

Moreover,

$$\text{FailCaseOf}(\text{Ev}; \lambda)$$

is uniquely determined as one of the following three cases:

$$\text{FailAbsent}, \quad \text{FailUnique}, \quad \text{FailParallel}$$

*Proof.* Suppose first that the relation

$$s \prec^{\text{fail,dep}} t$$

holds. There then exists a positive-length directed path described by

$$s = t_0 \longrightarrow \cdots \longrightarrow t_m = t$$

Proposition 6.23 then implies that the state at its initial point is

$$\text{failed} \neq \text{sat}.$$

By Lemma 6.25, the state at the endpoint is

$$\text{unformed}$$

whereas, by the same proposition, the state of

$$t \in \text{FailedTok}[\text{Ev}; \lambda]$$

is

$$\text{failed}$$

and this contradicts uniqueness of the state.

Conversely, suppose that  $t \prec^{\text{fail,dep}} s$ . The state at the initial point  $t$  is **failed**, and Lemma 6.25 implies that the state at the endpoint  $s$  is **unformed**. On the other hand,  $s \in \text{FailedTok}[\lambda]$  implies that the state of  $s$  is **failed**, again contradicting uniqueness of the state. Distinct elements of the failed set are therefore incomparable in both directions. Hence the failed set is an antichain and every element is minimal.

Because the failed set is finite, its cardinality falls into exactly one of the following three classes:

$$0, \quad 1, \quad 2 \text{ or more}$$

and Definition 6.24 gives the three cases. □

**Definition 6.27** (Series-specific completion conditions). For each

$$\tau \in \text{FailSeries}$$

define the series-specific completion condition and the six-series completion-state vector by

$$\begin{aligned} K_\tau^{\text{fail}}[\text{EvalSpec}] = \top &\iff \forall t \in \text{FailTok}_\tau, \quad \text{FailStateOf}(t; \text{EvalSpec}) = \text{sat}, \\ \mathbf{K}^{\text{fail,series}}[\text{Ev}; \lambda] &:= (K_\tau^{\text{fail}}[\text{FailSpecAt}_\tau[\text{Ev}; \lambda]])_{\tau \in \text{FailSeries}} \end{aligned}$$

Define the strict completion condition and the full-evaluation completion condition by

$$\begin{aligned} K_{\text{strict}}^{\text{fail}}[\text{Ev}] &:= \bigwedge_{\tau \in \text{FailSeries}^{\text{strict}}} K_\tau^{\text{fail}}[\text{Ev}], \\ K_{\text{full}}^{\text{fail}}[\text{Ev}; \lambda] &:= \bigwedge_{\tau \in \text{FailSeries}} K_\tau^{\text{fail}}[\text{FailSpecAt}_\tau[\text{Ev}; \lambda]] \\ &= K_{\text{strict}}^{\text{fail}}[\text{Ev}] \wedge K_{\text{approx}}^{\text{fail}}[\text{Ev}; \lambda] \end{aligned}$$

The conjunction on the right-hand side is taken in

$$\{\perp, \top\}$$

throughout.

**Proposition 6.28** (Equivalence between series-specific completion conditions and evaluation states). For every

$$(\text{Ev}, \lambda) \in \text{EvSpec}_\Lambda$$

one has

$$K_{\text{full}}^{\text{fail}}[\text{Ev}; \lambda] = \top \iff \forall \text{FailTok}_{\tau,i} \in \text{FailTok}, \quad \text{FailStateAt}_{\text{Ev},\lambda}(\text{FailTok}_{\tau,i}) = \text{sat}.$$

Whenever this condition holds,

$$\text{FailedTok}^{\text{strict}}[\text{Ev}] = \text{FailedTok}^{\text{approx}}[\text{Ev}; \lambda] = \text{FailedTok}[\text{Ev}; \lambda] = \emptyset.$$

*Proof.* The series-specific conjunction in Definition 6.27 and the partition

$$\text{FailTok} = \bigsqcup_{\tau \in \text{FailSeries}} \text{FailTok}_\tau$$

give the equivalence. If every token is in the satisfied state, the failed-state fiber is empty, and so are its two restrictions.  $\square$

**Definition 6.29** (Approximation-series actual-failure scales and approximation-validity boundary). Define the set of scales at which actual failure occurs in the approximation series and the set of scales at which the approximation-series completion condition is false by

$$\begin{aligned} \Lambda_{\text{failed}}^{\text{approx}}[\text{Ev}] &:= \{\lambda \in \Lambda \mid \text{FailedTok}^{\text{approx}}[\text{Ev}; \lambda] \neq \emptyset\}, \\ \Lambda_{\text{unsat}}^{\text{approx}}[\text{Ev}] &:= \Lambda \setminus \text{Val}_\Lambda(K_{\text{approx}}^{\text{fail}}; \text{Ev}) \\ &= \{\lambda \in \Lambda \mid K_{\text{approx}}^{\text{fail}}[\text{Ev}; \lambda] = \perp\} \end{aligned}$$

respectively.



For a set-valued map

$$S : X \longrightarrow \mathcal{P}(\Lambda)$$

let

$$\begin{aligned} \text{Dom}(\text{First}_\Lambda[S]) &:= \{x \in X \mid \exists \lambda_0 \in S[x], \forall \mu \in S[x], \lambda_0 \leq_\Lambda \mu\}, \\ \text{First}_\Lambda[S](x) &:= \min_{\leq_\Lambda} S[x], \quad x \in \text{Dom}(\text{First}_\Lambda[S]) \end{aligned}$$

define its least-scale lift. The first approximation-series actual-failure scale and the approximation-validity boundary are

$$\begin{aligned} \lambda_{\text{failed}}^{\text{approx}} &:= \text{First}_\Lambda[\Lambda_{\text{failed}}^{\text{approx}}], \\ \lambda_{\text{bdry}}^{\text{approx}} &:= \text{First}_\Lambda[\Lambda_{\text{unsat}}^{\text{approx}}] \end{aligned}$$

and both are partial maps with the common type

$$\text{EvSpec} \rightarrow \Lambda$$

**Lemma 6.30** (Inclusion relation for least elements and the boundary of an initial segment).  
In the totally ordered set

$$(\Lambda, \leq_\Lambda)$$

if the subsets

$$S_0 \subseteq S_1$$

both have least elements, then

$$\min_{\leq_\Lambda} S_1 \leq_\Lambda \min_{\leq_\Lambda} S_0.$$

If the sets are equal, their least elements are also equal. Furthermore, if the complement of an initial segment

$$I \subseteq \Lambda$$

has the least element

$$b := \min_{\leq_\Lambda} (\Lambda \setminus I)$$

then

$$I = \{\lambda \in \Lambda \mid \lambda <_\Lambda b\}.$$

*Proof.* The first inequality follows from

$$\min_{\leq_\Lambda} S_0 \in S_1$$

and the least-element property of

$$S_1$$

which together yield the inequality. If the sets are equal, antisymmetry applies. For the second assertion, if

$$\lambda <_\Lambda b$$

did not belong to

$$I$$

the least-element property of

$$b$$

would be contradicted; if

$$\lambda \in I, \quad b \leq_{\Lambda} \lambda$$

then the initial-segment property would imply

$$b \in I$$

contrary to its membership in the complement.  $\square$

**Proposition 6.31** (Relations between approximation-series actual-failure scales and the approximation-validity boundary). For every

$$\text{Ev} \in \text{EvSpec}$$

one has

$$\Lambda_{\text{failed}}^{\text{approx}}[\text{Ev}] \subseteq \Lambda_{\text{unsat}}^{\text{approx}}[\text{Ev}].$$

If both least scales are defined, then

$$\lambda_{\text{bdry}}^{\text{approx}}[\text{Ev}] \leq_{\Lambda} \lambda_{\text{failed}}^{\text{approx}}[\text{Ev}].$$

Furthermore, if

$$\forall \lambda \in \Lambda, \quad K_{\text{approx}}^{\text{fail}}[\text{Ev}; \lambda] = \perp \implies \text{FailedTok}^{\text{approx}}[\text{Ev}; \lambda] \neq \emptyset$$

holds, the two scale sets are equal. Consequently, if both least scales are defined, then

$$\lambda_{\text{failed}}^{\text{approx}}[\text{Ev}] = \lambda_{\text{bdry}}^{\text{approx}}[\text{Ev}].$$

If the approximation-validity domain is an initial segment and the approximation-validity boundary is defined, then

$$\mathcal{I}^{\text{max}}[\text{Ev}] = \{ \lambda \in \Lambda \mid \lambda <_{\Lambda} \lambda_{\text{bdry}}^{\text{approx}}[\text{Ev}] \}.$$

If the first approximation-series actual-failure scale is defined, then it is unique and

$$\begin{aligned} \text{FailedTok}^{\text{approx}}[\text{Ev}; \lambda_{\text{failed}}^{\text{approx}}[\text{Ev}]] &\neq \emptyset, \\ \text{FailCaseOf}(\text{Ev}; \lambda_{\text{failed}}^{\text{approx}}[\text{Ev}]) &\in \{\text{FailUnique}, \text{FailParallel}\}. \end{aligned}$$

*Proof.* Because the failed state differs from the satisfied state, Definition 6.27 gives

$$\text{FailedTok}^{\text{approx}}[\lambda] \neq \emptyset \implies K_{\text{approx}}^{\text{fail}}[\lambda] = \perp.$$

This yields the first inclusion. Lemma 6.30 gives the inequality between the least scales. The reverse implication assumed above yields the reverse inclusion, so the sets and their least scales coincide.

Applying the same lemma to the initial-segment property of the approximation-validity domain and to the least element of its complement, and then applying Proposition 6.9, gives the displayed form of the maximal effective interval. The final two relations follow because the least scale belongs to the set of failure scales and because Definition 6.24 selects one of the nonempty cases.  $\square$

**Definition 6.32** (Covered–unrefined refinement pair). For a parent region

$$\text{ParentReg}$$

and a family of subregions

$$(\text{SubReg}_\lambda)_{\lambda \in \text{SubIdx}}$$

call

$$\begin{aligned} & \text{RefSplit}(\text{ParentReg}; (\text{SubReg}_\lambda)_{\lambda \in \text{SubIdx}}) \\ &:= \left( \bigcup_{\lambda \in \text{SubIdx}} \text{SubReg}_\lambda, \right. \\ & \quad \left. \text{ParentReg} \setminus \bigcup_{\lambda \in \text{SubIdx}} \text{SubReg}_\lambda \right) \end{aligned}$$

the covered–unrefined refinement pair.

**Lemma 6.33** (Decomposition of covered and unrefined regions and refinement monotonicity). Suppose that two families of subregions satisfy

$$\begin{aligned} & \text{SubIdx}^{(0)} \subseteq \text{SubIdx}^{(1)}, \\ & \text{SubReg}_\lambda^{(\rho)} \subseteq \text{ParentReg}, \quad \rho \in \{0, 1\}, \quad \lambda \in \text{SubIdx}^{(\rho)}, \\ & \text{SubReg}_\lambda^{(0)} = \text{SubReg}_\lambda^{(1)}, \quad \lambda \in \text{SubIdx}^{(0)} \end{aligned}$$

and set

$$(\text{CoveredReg}^{(\rho)}, \text{UnrefinedReg}^{(\rho)}) := \text{RefSplit}(\text{ParentReg}; (\text{SubReg}_\lambda^{(\rho)})_{\lambda \in \text{SubIdx}^{(\rho)}})$$

Then

$$\text{ParentReg} = \text{CoveredReg}^{(\rho)} \sqcup \text{UnrefinedReg}^{(\rho)}, \quad \rho \in \{0, 1\}.$$

If each family of subregions is pairwise disjoint, then

$$\text{ParentReg} = \left( \bigsqcup_{\lambda \in \text{SubIdx}^{(\rho)}} \text{SubReg}_\lambda^{(\rho)} \right) \sqcup \text{UnrefinedReg}^{(\rho)}.$$

Furthermore,

$$\begin{aligned} & \text{CoveredReg}^{(0)} \subseteq \text{CoveredReg}^{(1)}, \\ & \text{UnrefinedReg}^{(1)} \subseteq \text{UnrefinedReg}^{(0)}. \end{aligned}$$

*Proof.* Each covered region is a subset of the parent region, and the unrefined region is the set difference obtained by removing the covered region from the parent region. The two regions are therefore disjoint, and their union equals the parent region. If the family of subregions is pairwise disjoint, replacing the union in the covered region by a disjoint union gives the second decomposition.

Every subregion indexed by the first index set is contained in the second covering union because the two families agree on their common indices. This proves the inclusion between the covered regions. Removing the covered regions from the same parent region reverses that inclusion, which proves the inclusion between the unrefined regions.  $\square$

**Definition 6.34** (Typed failure-refinement forest). Take an underlying set  $\mathbf{Amb}$ , a root-index set  $I$ , and, for each path  $\nu$ , a child-index set  $J(\nu)$ . Define the refinement-path sets by

$$\begin{aligned} \text{RefPath}_0[I; J] &:= \{(i) \mid i \in I\}, \\ \text{RefPath}_{n+1}[I; J] &:= \{\nu \star \lambda \mid \nu \in \text{RefPath}_n[I; J], \lambda \in J(\nu)\}, \\ \text{RefPath}_{\leq m}[I; J] &:= \bigsqcup_{n=0}^m \text{RefPath}_n[I; J], \quad \text{RefNode}[I; J] := \bigcup_{n \geq 0} \text{RefPath}_n[I; J] \end{aligned}$$

The symbol  $\star$  denotes appending a child index to the end of a path, and  $J(\nu) = \emptyset$  is allowed at a leaf.

Suppose that the node-region map  $R : \text{RefNode}[I; J] \rightarrow \mathcal{P}(\mathbf{Amb})$  satisfies, for every  $\lambda \in J(\nu)$ , the inclusion  $R(\nu \star \lambda) \subseteq R(\nu)$ . Define the forest, its node decomposition, its child relation, and its ancestor relation by

$$\begin{aligned} \text{RefForest}[I; J; R] &:= (\text{RefNode}[I; J], J, \star, R), \\ (R^{\text{cov}}(\nu), R^{\text{unref}}(\nu)) &:= \text{RefSplit}(R(\nu); (R(\nu \star \lambda))_{\lambda \in J(\nu)}), \\ \nu \prec_{\text{RefForest}}^{\text{child}} \omega &\iff \exists \lambda \in J(\nu), \omega = \nu \star \lambda, \quad \preceq_{\text{RefForest}}^{\text{anc}} := (\prec_{\text{RefForest}}^{\text{child}})^* \end{aligned}$$

The superscript  $*$  denotes the reflexive-transitive closure of the relation.

**Theorem 6.35** (Node decomposition and ancestor inclusion in a typed failure-refinement forest). Take arbitrary  $\mathbf{Amb}$ ,  $I$ ,  $J$ , and  $R$  as in Definition 6.34. For every nonleaf node  $\nu \in \text{RefNode}[I; J]$  one has

$$R(\nu) = R^{\text{cov}}(\nu) \sqcup R^{\text{unref}}(\nu).$$

If the family of child-node regions is pairwise disjoint, then

$$R^{\text{cov}}(\nu) = \bigsqcup_{\lambda \in J(\nu)} R(\nu \star \lambda).$$

Furthermore, for every  $\nu, \omega \in \text{RefNode}[I; J]$  one has

$$\nu \preceq_{\text{RefForest}[I; J; R]}^{\text{anc}} \omega \implies R(\omega) \subseteq R(\nu).$$

*Proof.* The node decomposition and the conditional disjoint union follow by substituting the node data of Definition 6.34 into Lemma 6.33. The ancestor inclusion follows by induction on the path length in the ancestor relation, using inclusion of every child-node region in its parent-node region.  $\square$

## 6.2 Failure Positions at the Comparison Stage

The comparison-stage condition-family evaluation specification corresponding to the complete evaluation specification of Definition 5.2,

$$\text{Ev} = (\mathbf{l}, o, \zeta) \in \text{EvSpec}$$

is the family specified in Definition 4.7,

$$\text{Ev}^{\text{cmp}} = (\text{Ev}_i^{\text{cmp}})_{i=1}^7,$$

with components

$$\text{Ev}_i^{\text{cmp}} = \begin{cases} \mathcal{X}, & i = 1, \\ \text{I}, & i \in \{2, \dots, 7\} \end{cases}$$

at the seven indexed positions.

Define the bracket evaluations of the comparison-stage conditions for a complete evaluation specification by

$$\begin{aligned} \text{L}_i[\text{Ev}] &:= \text{L}_i[\text{Ev}_i^{\text{cmp}}], & i \in \{1, \dots, 7\}, \\ \text{L}_{\leq i}[\text{Ev}] &:= \text{L}_{\leq i}[\text{Ev}^{\text{cmp}}], & i \in \{1, \dots, 7\}, \\ \text{L}_{< i}[\text{Ev}] &:= \text{L}_{< i}[\text{Ev}^{\text{cmp}}], & i \in \{1, \dots, 7\}, \end{aligned}$$

and define the global-condition and operability evaluations by

$$\begin{aligned} \text{G}[\text{Ev}] &:= \text{G}[\text{Ev}^{\text{cmp}}], \\ \text{Operative}[\text{Ev}] &:= \text{Operative}(\mathcal{X}) \end{aligned}$$

respectively.

### 6.2.1 Comparison-Stage First-Failure Classes

Classify the first failure position in the comparison-stage condition sequence

$$(\text{L}_i)_{i=1}^7 ; \text{G}$$

into eight comparison-stage first-failure classes. Regard each first-failure class as the parent region of the corresponding first failure position. Under internal refinement, decompose each parent region into the union of a family of subregions and its complement within the parent region.

**Definition 6.36** (Comparison-stage failure). Comparison-stage failure is said to occur at a complete evaluation specification

$$\text{Ev} \in \text{EvSpec}$$

if

$$\neg \text{Operative}[\text{Ev}]$$

holds.

**Definition 6.37** (Comparison-stage failure domain). Define the comparison-stage failure domain in the set of complete evaluation specifications

$$\text{EvSpec}$$

by

$$\begin{aligned} &\text{FailDom}_{\text{cmp}}(\text{EvSpec}) \\ &:= \left\{ \text{Ev} \in \text{EvSpec} \mid \neg \text{Operative}[\text{Ev}] \right\} \end{aligned}$$

**Proposition 6.38** (Comparison-stage failure and the comparison-stage condition sequence). For every

$$\text{Ev} \in \text{EvSpec}$$

the following equivalence holds:

$$\begin{aligned} & \neg \text{Operative}[\text{Ev}] \\ & \iff \neg (\text{L}_{\leq 7}[\text{Ev}] \wedge \text{G}[\text{Ev}]). \end{aligned}$$

*Proof.* By Definition 4.86,

$$\text{Operative}[\text{Ev}] \iff \text{L}_{\leq 7}[\text{Ev}] \wedge \text{G}[\text{Ev}].$$

Negating both sides yields the asserted equivalence.  $\square$

In the comparison-series raw-condition family, comparison-series preceding-completion condition, comparison-series evaluation-input satisfaction, comparison-stage first-failure-class family, and the recursive expansion of that family below, fix, unless otherwise stated, the following objects:

$$i \in I_{\text{cmp}}^{\text{fail}}, \quad \text{Ev} \in \text{EvSpec}$$

**Definition 6.39** (Comparison-series raw-condition family). Define the comparison-series raw-condition family by

$$K_{\text{cmp},i}^{\text{fail}}[\text{Ev}] := \begin{cases} \text{L}_i[\text{Ev}], & i \in \{1, \dots, 7\}, \\ \text{G}[\text{Ev}], & i = 8 \end{cases}$$

**Definition 6.40** (Comparison-series preceding-completion condition). Define the conjunction of the raw conditions preceding position  $i$  in the comparison series by

$$K_{\text{cmp},<i}^{\text{fail}}[\text{Ev}] := \bigwedge_{j=1}^{i-1} K_{\text{cmp},j}^{\text{fail}}[\text{Ev}]$$

In particular,

$$K_{\text{cmp},<1}^{\text{fail}}[\text{Ev}] = \top.$$

**Definition 6.41** (Comparison-series evaluation-input satisfaction). Define the input-satisfaction value of a comparison-series failure-position token by

$$\text{FailInputSat}(\text{FailTok}_{\text{cmp},i}; \text{Ev}_{\text{cmp}}^{\text{fail}}) = \top$$

**Definition 6.42** (Comparison-stage first-failure-class family). Define the comparison-stage first-failure-class family by

$$\begin{aligned} & \text{FC}_i[\text{Ev}] \\ & \iff \text{FailTok}_{\text{cmp},i} \in \text{FailedTok}^{\text{strict}}[\text{Ev}] \end{aligned}$$

**Proposition 6.43** (Recursive expansion of comparison-series first-failure classes). The equivalence

$$\begin{aligned} & \text{FC}_i[\text{Ev}] \\ & \iff K_{\text{cmp},<i}^{\text{fail}}[\text{Ev}] \wedge [K_{\text{cmp},i}^{\text{fail}}[\text{Ev}] = \perp] \end{aligned}$$

holds. Consequently, for  $i \in \{1, \dots, 7\}$ ,

$$(K_{\text{cmp},<i}^{\text{fail}}, K_{\text{cmp},i}^{\text{fail}})[\text{Ev}] = (\text{L}_{<i}, \text{L}_i)[\text{Ev}],$$

whereas for  $i = 8$ ,

$$(K_{\text{cmp},<8}^{\text{fail}}, K_{\text{cmp},8}^{\text{fail}})[\text{Ev}] = (\text{L}_{\leq 7}, \text{G})[\text{Ev}].$$

*Proof.* By Definition 6.17, the first token has no direct predecessor token, and the sole direct predecessor of token  $i \in \{2, \dots, 8\}$  is token  $i - 1$ . Definition 6.41 establishes the formation inputs and evaluation inputs of all tokens. Substituting these into Definition 6.21, using Proposition 6.23, and inducting along the finite chain yields, for every  $i \in I_{\text{cmp}}^{\text{fail}}$ ,

$$\begin{aligned} \text{FailTok}_{\text{cmp},i} &\in \text{FailedTok}^{\text{strict}}[\text{Ev}] \\ \iff &\left[ \bigwedge_{j=1}^{i-1} K_{\text{cmp},j}^{\text{fail}}[\text{Ev}] = \top \right] \wedge [K_{\text{cmp},i}^{\text{fail}}[\text{Ev}] = \perp]. \end{aligned}$$

At the base, the preceding conjunction is the empty conjunction; at the inductive step, the establishment state of token  $i - 1$  coincides with the preceding conjunction.

Substituting Definition 6.40, Definition 6.39, and Definition 6.42 yields the asserted single-index formula and the two boundary displays.  $\square$

By Definition 6.39, the explicit form of the failure condition at each position is obtained, for  $i \in \{1, \dots, 7\}$ , by negating the corresponding component from Definition 4.5 through Definition 4.78, and, for  $i = 8$ , by negating Definition 4.79.

The negation of the cardinality-one condition in component  $i = 2$  jointly represents the case in which no factorization tuple exists and the case in which two or more distinct tuples exist. The negation of the domain inclusion in component  $i = 3$  is equivalent to the existence, in the source of the inclusion, of an element that does not belong to the target domain. The negation of the set equality in component  $i = 5$  is equivalent to the existence of a comparison two-role arrival-value family that belongs to at least one of the two directional set differences.

The comparison-stage first-failure parent regions relative to the comparison condition sequence

$$(\mathbf{L}_i)_{i=1}^7 ; \mathbf{G}$$

are represented by a single index row using the comparison-series preceding-completion condition.

Table 3: Comparison-stage first-failure parent regions relative to the comparison condition sequence

| Position                             | First-failure parent region                                                                   |
|--------------------------------------|-----------------------------------------------------------------------------------------------|
| $i \in I_{\text{cmp}}^{\text{fail}}$ | $\text{FC}_i \iff K_{\text{cmp},<i}^{\text{fail}} \wedge \neg K_{\text{cmp},i}^{\text{fail}}$ |

The preceding condition at the first position is the empty conjunction  $\top$ ; the preceding condition and the failure condition at the eighth position are

$$\mathbf{L}_{\leq 7}, \quad \neg \mathbf{G}$$

respectively. At position  $i \in \{2, \dots, 7\}$ , they are

$$\mathbf{L}_{<i}, \quad \neg \mathbf{L}_i$$

respectively.

For every

$$i \in I_{\text{cmp}}^{\text{fail}}$$

the class

$$\mathbf{FC}_i$$

is a parent region for internal refinement. The witness type and the object whose generation fails in each parent region correspond as follows.

$\mathbf{FC}_1$ . A witness consists of distinct source-side tokens with the same arrival value for some comparison role and reception position. That is, it consists of  $R, u, x_R, y_R$  satisfying

$$\begin{aligned} R &\in \mathbf{Rol}^{\text{cmp}}, & u &\in U, \\ x_R, y_R &\in \text{Dom}(\mathbf{Arr}_{R,u}^{\text{cmp}}), \\ x_R &\neq y_R, & \mathbf{Arr}_{R,u}^{\text{cmp}}(x_R) &= \mathbf{Arr}_{R,u}^{\text{cmp}}(y_R). \end{aligned}$$

By Lemma 4.6, the existence of such a tuple is equivalent to  $\neg \mathbf{L}_1(\mathcal{X})$ . In this case, the generation condition fails for the family of source-matching partial maps in Definition 4.8,

$$(\mathbf{SrcMatch}_{uv}^R)_{R,u,v},$$

whose construction requires unique recovery from the entire arrival image.

$\mathbf{FC}_2$ . The witness type is a comparison-admissible position pair for which the set of component-factorization tuples for synchronized-pair joint transport has cardinality

$$0$$

or at least

$$2$$

The objects whose generation fails are the unique component-factorization family

$$\left( \begin{aligned} &\mathbf{PairDom}_{uv}[\mathbf{I}], \\ &(\mathbf{Tr}_{uv}^R[\mathbf{I}])_{R \in \mathbf{Rol}^{\text{cmp}}} \end{aligned} \right)_{(u,v)}$$

and the comparison-transport datum

$$\mathbf{Dat}^{\text{cmp},\text{tr}}[\mathbf{I}]$$

$\mathbf{FC}_3$ . The witness type is a point that belongs to the declared initial domain and does not belong to the domain of the candidate transport-word action. The objects whose generation fails are the realized restricted action

$$\mathbf{Act}_R^{\text{tr},\text{real}}[\mathbf{Spec}],$$

the realized local-specification loop datum

$$\mathbf{LoopDat}_R^{\text{real}}[\mathbf{Spec}]$$

and the indexed data family

$$\mathbf{LoopDat}_R^{\text{real},\text{loc}}[\mathbf{I}]$$



**FC<sub>4</sub>.** The witness type is a nonfixed point in the declared initial domain of a realized locally specified action. The object whose generation fails is a family of realized restricted actions that are identity maps on their declared initial domains.

**FC<sub>5</sub>.** The witness type is a comparison two-role arrival-value family belonging to the set difference between a local comparison two-role relation and the full pullback of a source-side two-role relation. The object whose generation fails is agreement between the two sets.

**FC<sub>6</sub>.** The witness type is a pair of synchronized pairs that are comparison-equivalent in each role component and whose membership values in the local comparison two-role relation bundle differ. The object whose generation fails is saturation of the local comparison two-role relation bundle along the synchronized two-role comparison-pair equivalence relation.

**FC<sub>7</sub>.** The witness type is a nonfixed point of the induced partial action of a source-irreducible mixed closed-loop word that contains both the transport atom kind and the source-matching atom kind. The object whose generation fails is the condition that the induced partial action of every source-irreducible mixed closed-loop word

$$\mathbf{e} \in \text{LoopWord}_{\text{src-irr}}^{\text{mix}, R}[\mathbf{l}]$$

be the identity map on its domain.

**FC<sub>8</sub>.** The witness type is a nonfixed point of the induced partial action of a closed-loop word composed of the pure-transport atom kind. The object whose generation fails is the condition that the induced partial action of every pure-transport closed-loop word

$$\mathbf{w} \in \text{LoopWord}_{\text{all}}^{\text{tr}, R}[\mathbf{l}]$$

be the identity map on its domain.

**Definition 6.44** (Comparison-stage first-failure parent regions). For the set of complete evaluation specifications

$$\text{EvSpec}$$

and each

$$i \in I_{\text{cmp}}^{\text{fail}}$$

define the comparison-stage first-failure parent region by

$$\begin{aligned} & \text{FailReg}_{\text{cmp}, i}(\text{EvSpec}) \\ & := \left\{ \text{Ev} \in \text{EvSpec} \mid \text{FC}_i[\text{Ev}] \right\} \end{aligned}$$

**Theorem 6.45** (Partition of the comparison-stage failure domain into first-failure parent regions). For every set of complete evaluation specifications

$$\text{EvSpec}$$

one has

$$\begin{aligned} & \text{FailDom}_{\text{cmp}}(\text{EvSpec}) \\ &= \bigsqcup_{i=1}^8 \text{FailReg}_{\text{cmp},i}(\text{EvSpec}). \end{aligned}$$

Equivalently, for every

$$\text{Ev} \in \text{EvSpec}$$

one has

$$\begin{aligned} & \neg \text{Operative}[\text{Ev}] \\ & \iff \exists! i \in I_{\text{cmp}}^{\text{fail}} : \text{FC}_i[\text{Ev}]. \end{aligned}$$

*Proof.* Apply Lemma 6.12 to

$$\begin{aligned} \mathcal{Y} &= \text{EvSpec}, \quad n = 8, \\ (K_1, \dots, K_8) &= (\text{L}_1, \dots, \text{L}_7, \text{G}) \end{aligned}$$

By Proposition 6.38 and Definition 6.37, the left-hand side of that lemma is

$$\text{FailDom}_{\text{cmp}}(\text{EvSpec})$$

By Proposition 6.43 and Definition 6.44, its  $i$ -th component is

$$\text{FailReg}_{\text{cmp},i}(\text{EvSpec})$$

The displayed disjoint union therefore holds, and the uniqueness of the first failure position in that lemma yields the pointwise necessary-and-sufficient condition.  $\square$

**Definition 6.46** (Comparison-stage failure-refinement forest). Given  $\text{EvSpec}$ , for each  $i \in I_{\text{cmp}}^{\text{fail}}$ , take a first-layer child-index set

$$\text{SubIdx}_{\text{cmp},i}$$

For each  $\lambda_1 \in \text{SubIdx}_{\text{cmp},i}$ , take a second-layer child-index set

$$\text{SubIdx}_{\text{cmp},i;\lambda_1}$$

Suppose that the corresponding families of regions satisfy

$$\begin{aligned} & \text{FailReg}_{\text{cmp},i;\lambda_1}(\text{EvSpec}) \subseteq \text{FailReg}_{\text{cmp},i}(\text{EvSpec}), \\ & \text{FailReg}_{\text{cmp},i;\lambda_1,\lambda_2}(\text{EvSpec}) \subseteq \text{FailReg}_{\text{cmp},i;\lambda_1}(\text{EvSpec}), \quad \lambda_2 \in \text{SubIdx}_{\text{cmp},i;\lambda_1} \end{aligned}$$

Define the child-index set at each path by

$$\text{ChildIdx}_{\text{cmp}}(i; \lambda_1, \dots, \lambda_k) := \begin{cases} \text{SubIdx}_{\text{cmp},i}, & k = 0, \\ \text{SubIdx}_{\text{cmp},i;\lambda_1}, & k = 1, \\ \emptyset, & k = 2 \end{cases}$$

Define the node-region map by

$$\begin{aligned} \text{NodeReg}_{\text{cmp}}((i); \text{EvSpec}) &:= \text{FailReg}_{\text{cmp},i}(\text{EvSpec}), \\ \text{NodeReg}_{\text{cmp}}((i; \lambda_1); \text{EvSpec}) &:= \text{FailReg}_{\text{cmp},i;\lambda_1}(\text{EvSpec}), \\ \text{NodeReg}_{\text{cmp}}((i; \lambda_1, \lambda_2); \text{EvSpec}) &:= \text{FailReg}_{\text{cmp},i;\lambda_1,\lambda_2}(\text{EvSpec}) \end{aligned}$$

Define the comparison-stage failure-refinement forest by

$$\text{RefForest}_{\text{cmp}}(\text{EvSpec}) := \text{RefForest} [I_{\text{cmp}}^{\text{fail}}; \text{ChildIdx}_{\text{cmp}}; \text{NodeReg}_{\text{cmp}}(-; \text{EvSpec})]$$

For  $k \in \{0, 1\}$  and  $\varsigma \in \{\text{cov}, \text{unref}\}$ , obtain the layer-specific components by

$$\text{FailReg}_{\text{cmp}, i; \lambda_1, \dots, \lambda_k}^{\varsigma}(\text{EvSpec}) := \text{NodeReg}_{\text{cmp}}^{\varsigma}((i; \lambda_1, \dots, \lambda_k); \text{EvSpec})$$

When  $k = 0$ , omit the semicolon and everything that follows it.

For each  $i \in I_{\text{cmp}}^{\text{fail}}$ , take two refinement data that share the parent region  $\text{FailReg}_{\text{cmp}, i}(\text{EvSpec})$  and satisfy the assumptions of Lemma 6.33:

$$\left( \text{SubIdx}_{\text{cmp}, i}^{(\rho)}, \left( \text{FailReg}_{\text{cmp}, i; \lambda}^{(\rho)}(\text{EvSpec}) \right)_{\lambda \in \text{SubIdx}_{\text{cmp}, i}^{(\rho)}} \right), \quad \rho \in \{0, 1\}$$

The two refinement snapshots at the root node are, by  $\text{RefSplit}$  in Definition 6.32,

$$\left( \text{RefSplit} \left( \text{FailReg}_{\text{cmp}, i}(\text{EvSpec}); \left( \text{FailReg}_{\text{cmp}, i; \lambda}^{(\rho)}(\text{EvSpec}) \right)_{\lambda \in \text{SubIdx}_{\text{cmp}, i}^{(\rho)}} \right) \right)_{\rho \in \{0, 1\}}$$

**Proposition 6.47** (Expansion of the comparison-stage failure-refinement forest). The comparison-stage failure-refinement forest of Definition 6.46 satisfies the node decomposition, conditional disjoint union, and ancestor inclusion of Theorem 6.35. The two refinement snapshots at the root node satisfy the covered-region expansion and the unrefined-region contraction of Lemma 6.33.

*Proof.* The first assertion follows by substituting

$$I_{\text{cmp}}^{\text{fail}}, \quad \text{ChildIdx}_{\text{cmp}}, \quad \text{NodeReg}_{\text{cmp}}(-; \text{EvSpec})$$

from Definition 6.46 into Theorem 6.35. The second follows by substituting the two root snapshots from the same definition into Lemma 6.33.  $\square$

### 6.2.2 Series boundary of the comparison-stage first-failure classes

**Proposition 6.48** (Series boundary of the comparison-stage first-failure classes). For every  $\text{Ev}$  in the comparison-stage failure domain,  $\text{FC}_i[\text{Ev}]$  holds for a unique  $i \in I_{\text{cmp}}^{\text{fail}}$ . The failure-position token system satisfies

$$\begin{aligned} \text{FailTok} &= \text{FailTok}_{\text{cmp}} \sqcup \bigsqcup_{\tau \in \{\text{repr}, \text{dyn}, \text{approx}, \text{law}, \text{diff}\}} \text{FailTok}_{\tau}, \\ \text{FailTok}_{\text{cmp}} &= \{ \text{FailTok}_{\text{cmp}, i} \mid i \in I_{\text{cmp}}^{\text{fail}} \}. \end{aligned}$$

The correspondence between comparison-stage first-failure classes and membership of comparison-series failure-position tokens in the failed-token set is given by Definition 6.42. For every  $(\text{Ev}, \lambda) \in \text{EvSpec}_{\Lambda}$ , the failed-token set is an antichain with respect to  $\preceq^{\text{fail}, \text{dep}}$  and coincides with the set of all its minimal elements.

*Proof.* Unique existence follows from Theorem 6.45, and the disjoint-union representation follows from Definition 6.10. The antichain property and the coincidence with the set of minimal elements follow from Proposition 6.26.  $\square$

### 6.3 Top-level signature of the comparison-stage first-failure classes

**Definition 6.49** (Truth-value indicator and membership indicator function). Define the truth-value indicator of a proposition  $P$  by

$$\mathbf{1}_{\{P\}} := \begin{cases} 1, & P, \\ 0, & \neg P \end{cases}$$

and define the membership indicator function for a set  $A$  by

$$\mathbf{1}_A(x) := \mathbf{1}_{\{x \in A\}}$$

**Definition 6.50** (Top-level signature of the comparison-stage first-failure classes). Take a set of complete evaluation specifications

$$\text{EvSpec}$$

For each  $i \in I_{\text{cmp}}^{\text{fail}}$ , define

$$\chi_{\text{FC}_i}[\text{Ev}] := \mathbf{1}_{\text{FailReg}_{\text{cmp},i}(\text{EvSpec})}(\text{Ev})$$

and define the top-level signature of the comparison-stage first-failure classes by

$$\begin{aligned} \Sigma_{\text{cmp}} : \text{FailDom}_{\text{cmp}}(\text{EvSpec}) &\longrightarrow \{0, 1\}^8, \\ \Sigma_{\text{cmp}}[\text{Ev}] &:= (\chi_{\text{FC}_i}[\text{Ev}])_{i=1}^8 \end{aligned}$$

**Proposition 6.51** (Single-component property of the top-level signature of the comparison-stage first-failure classes). Let  $\mathbf{e}_i^{(8)}$  denote, in  $\{0, 1\}^8$ , the standard basis vector with index  $i$ . For every

$$\text{Ev} \in \text{FailDom}_{\text{cmp}}(\text{EvSpec})$$

there exists a unique

$$i \in I_{\text{cmp}}^{\text{fail}}$$

such that

$$\Sigma_{\text{cmp}}[\text{Ev}] = \mathbf{e}_i^{(8)}.$$

Consequently,

$$\sum_{j=1}^8 \chi_{\text{FC}_j}[\text{Ev}] = 1.$$

Equivalently, the  $i$ -th component equals 1 and every remaining component equals 0.

*Proof.* By Theorem 6.45, the parent region containing  $\text{Ev}$  has a uniquely determined index  $i \in I_{\text{cmp}}^{\text{fail}}$ . The membership-indicator representation in Definition 6.50 then shows that the signature equals  $\mathbf{e}_i^{(8)}$  and that the sum of its components is 1.  $\square$

**Proposition 6.52** (Top-level signature of the comparison-stage first-failure classes on unrefined regions). For every

$$i \in I_{\text{cmp}}^{\text{fail}}, \quad \text{Ev} \in \text{FailReg}_{\text{cmp},i}^{\text{unref}}(\text{EvSpec})$$

one has

$$\Sigma_{\text{cmp}}[\text{Ev}] = \mathbf{e}_i^{(8)}.$$

That is, the  $i$ -th component equals 1; for every

$$j \in I_{\text{cmp}}^{\text{fail}} \setminus \{i\}$$

the corresponding  $j$ -th component equals 0.

*Proof.* By Definition 6.46, the unrefined region is contained in the  $i$ -th parent region. Applying Proposition 6.51 gives the displayed identity.  $\square$

## 6.4 Failure positions at the representation stage

The three representation-stage conditions in Definition 4.92 are placed in correspondence with the three representation failure-position tokens in Definition 6.10. Representation-stage failure is classified by the first failure position among the three conditions on the domain in which comparison-stage operativeness holds.

**Definition 6.53** (Comparison-stage-operative set of complete evaluation specifications). Define the comparison-stage-operative subset of the set of complete evaluation specifications by

$$\begin{aligned} \text{EvSpec}^{\text{op}} \\ := \{ \text{Ev} \in \text{EvSpec} \mid \text{Operative}[\text{Ev}] \} \end{aligned}$$

**Definition 6.54** (Evaluation of representation-stage conditions on complete evaluation specifications). Extend the following native-domain formulas by false to  $\text{EvSpec}$ , using the rule in Definition 6.10. Outside  $\text{EvSpec}^{\text{op}}$ , formation-input satisfaction for the representation series is false.

Let a typed family of comparison rules  $\mathcal{J}_{\text{cmp}}$  be given. For each

$$\text{Ev} \in \text{EvSpec}^{\text{op}}, \quad i \in I_{\text{repr}}^{\text{fail}} = \{1, 2, 3\}$$

define the representation-series condition family by

$$K_{\text{repr},i}^{\text{fail}}[\text{Ev}] := \begin{cases} K_{\text{CanOrdDesc}}^{\text{repr}}[\mathbb{I}], & i = 1, \\ K_{\text{LocIncTrans}}^{\text{repr}}[\mathbb{I}], & i = 2, \\ K_{\text{GlobIncTrans}}^{\text{repr}}[\mathbb{I}], & i = 3 \end{cases}$$

and set

$$\begin{aligned} \text{Repr}_{\leq 0}[\mathbb{I}] &:= \top, \\ \text{Repr}_{\leq i}[\mathbb{I}] &\iff \bigwedge_{m=1}^i K_{\text{repr},m}^{\text{fail}}[\text{Ev}] \quad (i \in I_{\text{repr}}^{\text{fail}}) \end{aligned}$$

**Definition 6.55** (Representation-failure evaluation context). Define the pair consisting of the set of complete evaluation specifications and the typed family of comparison rules by

$$\Gamma_{\text{repr}} := (\text{EvSpec}; \mathcal{J}_{\text{cmp}})$$

**Definition 6.56** (Representation-series prefix-satisfaction condition). For each  $i \in I_{\text{repr}}^{\text{fail}}$  and  $\text{Ev} \in \text{EvSpec}^{\text{op}}$ , define

$$K_{\text{repr}, < i}^{\text{fail}}[\text{Ev}] := \bigwedge_{m=1}^{i-1} K_{\text{repr}, m}^{\text{fail}}[\text{Ev}],$$

$$K_{\text{repr}, < 1}^{\text{fail}}[\text{Ev}] = \top$$

**Definition 6.57** (Representation-series evaluation-input satisfaction). For each  $i \in I_{\text{repr}}^{\text{fail}}$  and  $\text{Ev} \in \text{EvSpec}^{\text{op}}$ , define the input-satisfaction value of the representation-series failure-position token by

$$\text{FailInputSat}(\text{FailTok}_{\text{repr}, i}; \text{Ev}_{\text{repr}}^{\text{fail}}) = \top$$

**Definition 6.58** (Representation-stage failure domain). Define the representation-stage failure domain by

$$\begin{aligned} \text{FailDom}_{\text{repr}}[\Gamma_{\text{repr}}] &:= \{ \text{Ev} \in \text{EvSpec}^{\text{op}} \mid \exists i \in I_{\text{repr}}^{\text{fail}}, \neg K_{\text{repr}, i}^{\text{fail}}[\text{Ev}] \} \\ &= \{ \text{Ev} \in \text{EvSpec}^{\text{op}} \mid \neg \text{Repr}_{\leq 3}[\mathbb{I}] \}. \end{aligned}$$

**Definition 6.59** (Representation-stage first-failure parent regions). For each

$$i \in I_{\text{repr}}^{\text{fail}}, \quad \text{Ev} \in \text{EvSpec}$$

define

$$\begin{aligned} \text{FailReg}_{\text{repr}, i}[\Gamma_{\text{repr}}] &:= \{ \text{Ev} \in \text{EvSpec}^{\text{op}} \mid \text{FailTok}_{\text{repr}, i} \in \text{FailedTok}^{\text{strict}}[\text{Ev}] \}, \\ \text{FR}_i[\text{Ev}] &\iff \text{Ev} \in \text{FailReg}_{\text{repr}, i}[\Gamma_{\text{repr}}]. \end{aligned}$$

**Proposition 6.60** (Recursive expansion of the representation-stage first-failure predicates). For each  $i \in I_{\text{repr}}^{\text{fail}}$  and  $\text{Ev} \in \text{EvSpec}^{\text{op}}$ ,

$$\text{FR}_i[\text{Ev}] \iff K_{\text{repr}, < i}^{\text{fail}}[\text{Ev}] \wedge [K_{\text{repr}, i}^{\text{fail}}[\text{Ev}] = \perp]$$

holds.

*Proof.* By Definition 6.53 and Proposition 6.38, all tokens in the comparison series are satisfied. Applying Definition 6.17, Definition 6.57, and Definition 6.21 along the three nodes of the representation series shows that failure of the  $i$ -th token coincides with the conjunction of satisfaction of the generation conditions preceding the  $i$ -th position and failure of the  $i$ -th generation condition. Substitution of Definition 6.59, Definition 6.56, and Definition 6.54 gives the claim.  $\square$

**Proposition 6.61** (Generation boundary of representation-stage first failure). For each  $i \in I_{\text{repr}}^{\text{fail}}$  and  $\text{Ev} \in \text{EvSpec}^{\text{op}}$ ,

$$\text{FR}_i[\text{Ev}] \iff \text{Repr}_{\leq i-1}[\mathbb{I}] \wedge \neg \text{Repr}_{\leq i}[\mathbb{I}].$$

*Proof.* This follows from Proposition 6.60 and Definition 6.54.  $\square$

Expanding the failure evidence at each position to the corresponding generation condition yields, for  $i = 1$ , a four-point witness against the canonical order-descent condition; for  $i = 2$ , a nontransitivity witness for the local incomparability relation; and, for  $i = 3$ , a nontransitivity witness for the global incomparability relation. The first concrete form is given by the negation of Definition 4.99; the second and third are given by Definition 6.13.

**Proposition 6.62** (Correspondence between representation failure-position tokens and the three conditions). For each  $i \in I_{\text{repr}}^{\text{fail}}$  and  $\text{Ev} \in \text{EvSpec}^{\text{op}}$ , the formation-input satisfaction value and the evaluation-input satisfaction value equal  $\top$ , and the state of every direct predecessor token outside the series

$$\text{FailTok}^{(0)} \in \text{FailPredOf}(\text{FailTok}_{\text{repr},i}) \setminus \text{FailTok}_{\text{repr}}$$

is

$$\text{FailStateOf}\left(\text{FailTok}^{(0)}; \text{Ev}_{\text{FailSeriesOf}(\text{FailTok}^{(0)})}^{\text{fail}}\right) = \text{sat}.$$

Under these conditions,

$$\text{FR}_i[\text{Ev}] \iff \text{FailTok}_{\text{repr},i} \in \text{FailedTok}^{\text{strict}}[\text{Ev}].$$

*Proof.* The two input-satisfaction values follow from Definition 6.57. The satisfaction of the direct predecessor tokens outside the series follows from the proof of Proposition 6.60, and the final equivalence follows from Definition 6.59.  $\square$

**Definition 6.63** (Internal refinement families of representation-failure parent regions). For each  $i \in I_{\text{repr}}^{\text{fail}}$ , suppose that an index set  $\text{SubIdx}_{\text{repr},i}$  and a family of subregions satisfy

$$\begin{aligned} & (\text{FailReg}_{\text{repr},i;\lambda}[\Gamma_{\text{repr}}])_{\lambda \in \text{SubIdx}_{\text{repr},i}}, \\ & \text{FailReg}_{\text{repr},i;\lambda}[\Gamma_{\text{repr}}] \subseteq \text{FailReg}_{\text{repr},i}[\Gamma_{\text{repr}}] \end{aligned}$$

This family is called mutually exclusive when, for distinct  $\lambda_0, \lambda_1 \in \text{SubIdx}_{\text{repr},i}$ ,

$$\text{FailReg}_{\text{repr},i;\lambda_0}[\Gamma_{\text{repr}}] \cap \text{FailReg}_{\text{repr},i;\lambda_1}[\Gamma_{\text{repr}}] = \emptyset$$

holds.

| Parent failure region | First-failure condition                                                                                                                                     | Generation boundary                                                   |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| $\text{FR}_i$         | $K_{\text{repr},<i}^{\text{fail}}[\text{Ev}] \wedge \neg K_{\text{repr},i}^{\text{fail}}[\text{Ev}]$<br>$i \in I_{\text{repr}}^{\text{fail}} = \{1, 2, 3\}$ | $\text{Repr}_{\leq i-1}[\top] \wedge \neg \text{Repr}_{\leq i}[\top]$ |

Table 4: Correspondence of the three representation-failure parent regions with their first-failure conditions and generation boundaries

**Proposition 6.64** (Unique localization by representation-failure parent regions).

$$\text{FailDom}_{\text{repr}}[\Gamma_{\text{repr}}] = \bigsqcup_{i \in I_{\text{repr}}^{\text{fail}}} \text{FailReg}_{\text{repr},i}[\Gamma_{\text{repr}}].$$

Consequently, for each  $\text{Ev} \in \text{FailDom}_{\text{repr}}[\Gamma_{\text{repr}}]$ ,  $\text{FR}_i[\text{Ev}]$  holds for a unique  $i \in I_{\text{repr}}^{\text{fail}}$ .

*Proof.* Apply Lemma 6.12 with

$$\begin{aligned} \mathcal{Y} &= \text{EvSpec}^{\text{op}}, \\ n &= 3, \\ K_i(\text{Ev}) &\iff K_{\text{repr},i}^{\text{fail}}[\text{Ev}] \end{aligned}$$

The claim follows from Definition 6.58, Proposition 6.60, and Definition 6.59.  $\square$

**Definition 6.65** (Representation-stage failure-refinement forest). For each  $i \in I_{\text{repr}}^{\text{fail}}$  and  $\lambda_1 \in \text{SubIdx}_{\text{repr},i}$ , take a second-layer child-index set

$$\text{SubIdx}_{\text{repr},i;\lambda_1}$$

and a second-layer family of subregions, and suppose that

$$\text{FailReg}_{\text{repr},i;\lambda_1,\lambda_2}[\Gamma_{\text{repr}}] \subseteq \text{FailReg}_{\text{repr},i;\lambda_1}[\Gamma_{\text{repr}}], \quad \lambda_2 \in \text{SubIdx}_{\text{repr},i;\lambda_1}$$

Define the child-index set at each path by

$$\text{ChildIdx}_{\text{repr}}(i; \lambda_1, \dots, \lambda_k) := \begin{cases} \text{SubIdx}_{\text{repr},i}, & k = 0, \\ \text{SubIdx}_{\text{repr},i;\lambda_1}, & k = 1, \\ \emptyset, & k = 2 \end{cases}$$

Define the node-region map by

$$\begin{aligned} \text{NodeReg}_{\text{repr}}((i); \Gamma_{\text{repr}}) &:= \text{FailReg}_{\text{repr},i}[\Gamma_{\text{repr}}], \\ \text{NodeReg}_{\text{repr}}((i; \lambda_1); \Gamma_{\text{repr}}) &:= \text{FailReg}_{\text{repr},i;\lambda_1}[\Gamma_{\text{repr}}], \\ \text{NodeReg}_{\text{repr}}((i; \lambda_1, \lambda_2); \Gamma_{\text{repr}}) &:= \text{FailReg}_{\text{repr},i;\lambda_1,\lambda_2}[\Gamma_{\text{repr}}] \end{aligned}$$

Define the representation-stage failure-refinement forest by

$$\text{RefForest}_{\text{repr}}[\Gamma_{\text{repr}}] := \text{RefForest} [I_{\text{repr}}^{\text{fail}}; \text{ChildIdx}_{\text{repr}}; \text{NodeReg}_{\text{repr}}(-; \Gamma_{\text{repr}})]$$

For  $k \in \{0, 1\}$  and  $\varsigma \in \{\text{cov}, \text{unref}\}$ , obtain the layer-specific components by

$$\text{FailReg}_{\text{repr},i;\lambda_1,\dots,\lambda_k}^{\varsigma}[\Gamma_{\text{repr}}] := \text{NodeReg}_{\text{repr}}^{\varsigma}((i; \lambda_1, \dots, \lambda_k); \Gamma_{\text{repr}})$$

When  $k = 0$ , omit the semicolon and everything that follows it.

**Proposition 6.66** (Node decomposition of the representation-stage failure-refinement forest). The representation-stage failure-refinement forest of Definition 6.65 satisfies the node decomposition and conditional disjoint union of Theorem 6.35.

*Proof.* Substitute  $I_{\text{repr}}^{\text{fail}}$ ,  $\text{ChildIdx}_{\text{repr}}$ , and  $\text{NodeReg}_{\text{repr}}(-; \Gamma_{\text{repr}})$  from Definition 6.65 into Theorem 6.35.  $\square$

**Proposition 6.67** (Separation of the comparison-stage and representation-stage failure domains).

$$\text{FailDom}_{\text{cmp}}(\text{EvSpec}) \cap \text{FailDom}_{\text{repr}}[\Gamma_{\text{repr}}] = \emptyset,$$

$$\text{FailDom}_{\text{repr}}[\Gamma_{\text{repr}}] \subseteq \{\text{Ev} = (I, o, \zeta) \in \text{EvSpec} \mid \text{Operative}[\text{Ev}]\}.$$

*Proof.* Definition 6.37 gives the first domain as the domain on which  $\neg \text{Operative}[\text{Ev}]$  holds, whereas Definition 6.58 places the second domain within the domain on which  $\text{Operative}[\text{Ev}]$  holds. Both claims follow.  $\square$

**Definition 6.68** (Representation-stage failure-refinement snapshots and recursive components). For each  $i \in I_{\text{repr}}^{\text{fail}}$ , take two refinement data that share the parent region  $\text{FailReg}_{\text{repr},i}[\Gamma_{\text{repr}}]$  and satisfy the assumptions of Lemma 6.33:

$$\left( \text{SubIdx}_{\text{repr},i}^{(\rho)}, \left( \text{FailReg}_{\text{repr},i;\lambda}^{(\rho)}[\Gamma_{\text{repr}}] \right)_{\lambda \in \text{SubIdx}_{\text{repr},i}^{(\rho)}} \right), \quad \rho \in \{0, 1\}$$



By RefSplit in Definition 6.32, the two refinement snapshots at the root node are

$$\left( \text{RefSplit} \left( \text{FailReg}_{\text{repr},i}[\Gamma_{\text{repr}}]; \left( \text{FailReg}_{\text{repr},i;\lambda}^{(\rho)}[\Gamma_{\text{repr}}] \right)_{\lambda \in \text{SubIdx}_{\text{repr},i}^{(\rho)}} \right)_{\rho \in \{0,1\}} \right)$$

For each  $\lambda \in \text{SubIdx}_{\text{repr},i}$ , the recursive components at the node  $(i; \lambda)$  are  $\text{ChildIdx}_{\text{repr}}$ ,  $\star$ , and  $\text{NodeReg}_{\text{repr}}$  of Definition 6.65.

**Proposition 6.69** (Expansion of representation-stage failure-refinement snapshots and recursive components). The two root snapshots of Definition 6.68 satisfy the covered-region expansion and unrefined-region contraction of Lemma 6.33. The recursive components of Definition 6.65 satisfy the ancestor inclusion of Theorem 6.35.

*Proof.* The first assertion follows by substituting the two root snapshots into Lemma 6.33. The second follows by substituting  $I_{\text{repr}}^{\text{fail}}$ ,  $\text{ChildIdx}_{\text{repr}}$ , and  $\text{NodeReg}_{\text{repr}}(-; \Gamma_{\text{repr}})$  into Theorem 6.35.  $\square$

#### 6.4.1 Representation-stage first-failure signature

**Definition 6.70** (One-hot signature of the representation-stage first-failure parent regions). Using the membership indicator function in Definition 6.49, set

$$\chi_{\text{FR}_i}[\text{Ev}] := \mathbf{1}_{\text{FailReg}_{\text{repr},i}[\Gamma_{\text{repr}}]}(\text{Ev}).$$

Define the representation-stage first-failure signature by

$$\begin{aligned} \Sigma_{\text{repr}} : \text{FailDom}_{\text{repr}}[\Gamma_{\text{repr}}] &\longrightarrow \{0, 1\}^3, \\ \Sigma_{\text{repr}}[\text{Ev}] &:= (\chi_{\text{FR}_i}[\text{Ev}])_{i=1}^3 \end{aligned}$$

**Proposition 6.71** (One-hot property of the representation-stage first-failure signature). Let  $\mathbf{e}_i^{(3)}$  denote, in  $\{0, 1\}^3$ , the standard basis vector with index  $i$ . For every

$$\text{Ev} \in \text{FailDom}_{\text{repr}}[\Gamma_{\text{repr}}]$$

there exists a unique  $i \in I_{\text{repr}}^{\text{fail}}$  such that

$$\begin{aligned} \Sigma_{\text{repr}}[\text{Ev}] &= \mathbf{e}_i^{(3)}, \\ \sum_{j=1}^3 \chi_{\text{FR}_j}[\text{Ev}] &= 1. \end{aligned}$$

*Proof.* This follows from Proposition 6.64 and Definition 6.70.  $\square$

**Proposition 6.72** (Correspondence between representation failure-position tokens and the one-hot signature). Under the assumptions of Proposition 6.62, for every

$$\text{Ev} \in \text{FailDom}_{\text{repr}}[\Gamma_{\text{repr}}], \quad i \in I_{\text{repr}}^{\text{fail}}$$

one has

$$\chi_{\text{FR}_i}[\text{Ev}] = 1 \iff \text{FR}_i[\text{Ev}] \iff \text{FailTok}_{\text{repr},i} \in \text{FailedTok}^{\text{strict}}[\text{Ev}].$$

*Proof.* This follows from Definition 6.70 and Proposition 6.62.  $\square$

## 6.5 Failure positions at the continuum-approximation stage

Take as inputs the continuum-approximation evaluation-input datum of Definition 4.168, the three typed  $\mathfrak{D}$ -record code relations of Definition 5.38, and the observer-time-relative  $\mathfrak{D}$ -record images of Definition 5.75. The continuum-approximation conditions consist of six finite-granularity and record-fidelity conditions, the continuous-extension condition, the order-stability condition, and the relation-stability condition.

**Definition 6.73** (Index set of continuum-approximation failure conditions). For the continuum-approximation failure-position index set of Definition 6.10, define the three-layer index sets by

$$\begin{aligned} I_{\text{approx}}^{\text{fail}} &= \{1, \dots, 9\}, \\ I_{\text{approx}}^{(1)} &:= \{1, \dots, 6\}, \\ I_{\text{approx}}^{(2)} &:= \{7\}, \\ I_{\text{approx}}^{(3)} &:= \{8, 9\} \end{aligned}$$

The layer and evaluation target of each index are as follows.

| $i$ | Layer        | Evaluation target                               |
|-----|--------------|-------------------------------------------------|
| 1   | First layer  | Counting-step width derived from $\mathfrak{C}$ |
| 2   | First layer  | $\mathfrak{D}$ -record stability                |
| 3   | First layer  | $\mathfrak{D}$ -record-cell width               |
| 4   | First layer  | $\mathfrak{D}$ -typed-record separation         |
| 5   | First layer  | $\mathfrak{B}$ -object-state distinguishability |
| 6   | First layer  | $\mathfrak{D}$ -observer-side coupling limit    |
| 7   | Second layer | Continuous extension                            |
| 8   | Third layer  | Order stability                                 |
| 9   | Third layer  | Relation stability                              |

**Definition 6.74** (Continuum-approximation-stage evaluation datum). Let the complete evaluation specification and approximation scale be  $(\text{Ev}, \lambda) \in \text{EvSpec}_\Lambda$ . Define its continuum-approximation evaluation context by

$$\Gamma_{\text{Ev}, \lambda}^{\text{approx}} := \text{Ev}_{\text{approx}, \lambda}^{\text{fail}}$$

Let the approximation-map tag set be

$$\text{MapTag}^{\text{approx}} := \left\{ (\mathfrak{C}, \text{cmp}), (\mathfrak{D}, \text{cmp}), (\mathfrak{C}, \text{obs}), \right. \\ \left. (\mathfrak{B}, \text{state}), (\mathfrak{D}, \text{coupl}) \right\}$$

and let the strictly ordered approximation-map tag set be

$$\text{OrdTag}^{\text{approx}} := \text{MapTag}^{\text{approx}} \setminus \{(\mathfrak{D}, \text{coupl})\}$$

Define the typed-record tag set and the corresponding family of  $\mathfrak{D}$ -record code relations by

$$\begin{aligned} \text{RecSepTag}^{\text{approx}} &:= \text{Rol}^{\text{dyn}, \text{quot}} \cup \{\text{pair}\}, \\ \text{RecSepRel}_{\sigma, o}^{\text{dyn}} &:= \begin{cases} \text{RecordCodeRel}_{\sigma, o}^{\text{dyn}}, & \sigma \in \text{Rol}^{\text{dyn}, \text{quot}}, \\ \text{RecordCodeRel}_o^{\text{dyn}, \text{pair}}, & \sigma = \text{pair} \end{cases} \end{aligned}$$

Define the record-code evaluation component by

$$\begin{aligned} \text{RecCodeComp}_{\mathfrak{D},o}^{\text{approx}} \\ := (\text{RecSepRel}_{\sigma,o}^{\text{dyn}})_{\sigma \in \text{RecSepTag}^{\text{approx}}} \end{aligned}$$

For each  $t \in \text{MapTag}^{\text{approx}}$  define the approximation-map evaluation component by

$$\begin{aligned} \text{MapComp}_{t,\lambda}^{\text{approx}} := & \left( (\text{Dom}_t^{\text{approx}}, \mathcal{T}_t^{\text{approx,src}}), (\text{Cod}_t^{\text{approx}}, \mathcal{T}_t^{\text{approx,tgt}}), \right. \\ & \text{Int}_{t,\lambda}^{\text{approx}}, \text{Samp}_{t,\lambda}^{\text{approx}}, \text{SVal}_{t,\lambda}^{\text{approx}}, \\ & \left. \text{Map}_{t,\lambda}^{\text{approx}}, \text{Dev}_t^{\text{approx}}, \varepsilon_t^{\text{approx,fit}} \right) \end{aligned}$$

The first two components are topological spaces, with types

$$\begin{aligned} \text{Int}_{t,\lambda}^{\text{approx}} &\subseteq \text{Dom}_t^{\text{approx}}, & \text{Samp}_{t,\lambda}^{\text{approx}} &\subseteq \text{Int}_{t,\lambda}^{\text{approx}}, \\ \text{SVal}_{t,\lambda}^{\text{approx}} : \text{Samp}_{t,\lambda}^{\text{approx}} &\longrightarrow \text{Cod}_t^{\text{approx}}, & \text{Map}_{t,\lambda}^{\text{approx}} : \text{Int}_{t,\lambda}^{\text{approx}} &\longrightarrow \text{Cod}_t^{\text{approx}}, \\ \text{Dev}_t^{\text{approx}} : (\text{Cod}_t^{\text{approx}})^2 &\longrightarrow \overline{\mathbb{R}}_{\geq 0}, & \varepsilon_t^{\text{approx,fit}} &\in \overline{\mathbb{R}}_{\geq 0} \end{aligned}$$

The interval component is a connected ordered interval, and the sample component is a finite set.

For each  $A \in \mathcal{P}(\text{Int}_{t,\lambda}^{\text{approx}})$  define the set image under the approximation map by

$$\begin{aligned} \text{Map}_{t,\lambda}^{\text{approx}}[A] \\ := \{ \text{Map}_{t,\lambda}^{\text{approx}}(x) \mid x \in A \} \end{aligned}$$

For each  $A \in \mathcal{P}(\text{Cod}_t^{\text{approx}})$  define the deviation diameter by

$$\begin{aligned} \text{Diam}_t^{\text{approx}}(A) \\ := \begin{cases} 0, & A = \emptyset, \\ \sup \{ \text{Dev}_t^{\text{approx}}(y_0, y_1) \mid y_0, y_1 \in A \}, & A \in \mathcal{P}(\text{Cod}_t^{\text{approx}}) \setminus \{ \emptyset \} \end{cases} \end{aligned}$$

Suppose that the sample-fit tolerance  $\varepsilon_t^{\text{approx,fit}} \in \overline{\mathbb{R}}_{\geq 0}$  is given.

For each  $t \in \text{OrdTag}^{\text{approx}}$  let the strict orders on the domain and codomain be  $\prec_t^{\text{approx,src}}$ ,  $\prec_t^{\text{approx,tgt}}$ .

Let the approximation map for  $\mathfrak{D}$ -record code values be

$$\text{RecMap}_{\mathfrak{D},\lambda}^{\text{approx}} : \text{RecordCodeVal}_{\mathfrak{D}}^{\text{dyn}} \longrightarrow \text{Cod}_{\mathfrak{D},\lambda}^{\text{approx,rec}}$$

Define the domain for each typed-record tag by

$$\text{RecSepDom}_{\sigma} := \begin{cases} \text{Qu}_{\sigma,o}^{\text{dyn,can}}, & \sigma \in \text{Rol}^{\text{dyn,quot}}, \\ \text{TrjRel}_o^{\text{dyn,can}}, & \sigma = \text{pair} \end{cases}$$

For each  $\sigma \in \text{RecSepTag}^{\text{approx}}$ ,  $z \in \text{RecSepDom}_{\sigma}$  define the post-approximation  $\mathfrak{D}$ -record image by

$$\begin{aligned} \text{RecImg}_{\sigma,\lambda}^{\text{approx}}(z) \\ := \{ \text{RecMap}_{\mathfrak{D},\lambda}^{\text{approx}}(r_{\mathfrak{D}}) \mid (z, r_{\mathfrak{D}}) \in \text{RecSepRel}_{\sigma,o}^{\text{dyn}} \} \end{aligned}$$

Let the relation-stability tag set be

$$\text{RelTag}^{\text{approx}} := \{\text{cmp}, \text{trj}\} \cup \text{RecSepTag}^{\text{approx}}$$

For each  $\rho \in \text{RelTag}^{\text{approx}}$  define the types of the relation-preservation component by

$$\begin{aligned} \text{Arity}_{\rho}^{\text{approx}} &\in \mathbb{N}_{\geq 1}, & \text{Rel}_{\rho}^{\text{approx,src}} &\subseteq \prod_{k=1}^{\text{Arity}_{\rho}^{\text{approx}}} \text{Dom}_{\rho,k}^{\text{approx}}, \\ \text{RelMap}_{\rho,k,\lambda}^{\text{approx}} : \text{Dom}_{\rho,k}^{\text{approx}} &\longrightarrow \text{Cod}_{\rho,k}^{\text{approx}}, & \text{Rel}_{\rho,\lambda}^{\text{approx,cand}} &\subseteq \prod_{k=1}^{\text{Arity}_{\rho}^{\text{approx}}} \text{Cod}_{\rho,k}^{\text{approx}} \end{aligned}$$

Define the five typed relations by

$$\text{Rel}_{\rho}^{\text{approx,src}} := \begin{cases} \text{Rel}^{\text{cmp,cand}}[I], & \rho = \text{cmp}, \\ \text{TrjRel}_o^{\text{dyn,cand}}, & \rho = \text{trj}, \\ \text{RecSepRel}_{\rho,o}^{\text{dyn}}, & \rho \in \text{RecSepTag}^{\text{approx}} \end{cases}$$

For the continuum-approximation componentwise evaluation values

$$\kappa \in \{(\mathfrak{C}, \text{cntstep}), (\mathfrak{D}, \text{stab}), (\mathfrak{B}, \text{disc}), (\mathfrak{O}, \text{coupl})\}$$

let the order-preserving evaluation map into the nonnegative extended real values be

$$\text{DefMap}_{\kappa,\lambda}^{\text{approx}} : \text{EvalVal}_{\kappa}^{\text{cont}} \longrightarrow \overline{\mathbb{R}}_{\geq 0}$$

Let the first-layer defect quantities and tolerances be

$$\begin{aligned} &\text{Def}_i^{\text{approx}}[\text{Ev}; \lambda], \\ &\varepsilon_i^{\text{approx}} \in \overline{\mathbb{R}}_{\geq 0}, \\ &i \in I_{\text{approx}}^{(1)} \end{aligned}$$

For each

$$(i, \kappa) \in \left\{ (1, (\mathfrak{C}, \text{cntstep})), (2, (\mathfrak{D}, \text{stab})), \right. \\ \left. (5, (\mathfrak{B}, \text{disc})), (6, (\mathfrak{O}, \text{coupl})) \right\}$$

define the defect quantities for the four components by

$$\text{Def}_i^{\text{approx}}[\text{Ev}; \lambda] := \text{DefMap}_{\kappa,\lambda}^{\text{approx}}(\text{Eval}_{\kappa}^{\text{cont}}[I])$$

Assume, for the two representation tags, that

$$\begin{aligned} \text{CellImg}_{\mathfrak{D},u}^{\text{repr}}[I](x_{\mathfrak{D}}) &\subseteq \text{Int}_{(\mathfrak{D},\text{cmp}),\lambda}^{\text{approx}}, \\ \text{RecordImg}_{\mathfrak{D},o}^{\text{time,real}}(x_{\mathfrak{D}}) &\subseteq \text{Int}_{(\mathfrak{C},\text{obs}),\lambda}^{\text{approx}} \end{aligned}$$

Define the post-approximation  $\mathfrak{D}$ -record-cell images by

$$\begin{aligned} &\text{RecCellImg}_{(\mathfrak{D},\text{cmp}),\lambda}^{\text{approx}}(u, x_{\mathfrak{D}}) \\ &:= \text{Map}_{(\mathfrak{D},\text{cmp}),\lambda}^{\text{approx}}[\text{CellImg}_{\mathfrak{D},u}^{\text{repr}}[I](x_{\mathfrak{D}})], \\ &\text{RecCellImg}_{(\mathfrak{C},\text{obs}),\lambda}^{\text{approx}}(o, x_{\mathfrak{D}}) \\ &:= \text{Map}_{(\mathfrak{C},\text{obs}),\lambda}^{\text{approx}}[\text{RecordImg}_{\mathfrak{D},o}^{\text{time,real}}(x_{\mathfrak{D}})] \end{aligned}$$

Define the  $\mathfrak{D}$ -record-cell-width defect quantity by

$$\begin{aligned} \text{Def}_3^{\text{approx}}[\text{Ev}; \lambda] \\ := \sup \left( \{0\} \cup \left\{ \text{Diam}_{(\mathfrak{D}, \text{cmp})}^{\text{approx}} \left( \text{RecCellImg}_{(\mathfrak{D}, \text{cmp}), \lambda}^{\text{approx}}(u, x_{\mathfrak{D}}) \right) \mid \begin{array}{l} u \in U, \\ x_{\mathfrak{D}} \in \text{Tok}_{\mathfrak{D}}^* \end{array} \right\} \right. \\ \left. \cup \left\{ \text{Diam}_{(\mathfrak{C}, \text{obs})}^{\text{approx}} \left( \text{RecCellImg}_{(\mathfrak{C}, \text{obs}), \lambda}^{\text{approx}}(o, x_{\mathfrak{D}}) \right) \mid x_{\mathfrak{D}} \in \text{Tok}_{\mathfrak{D}}^* \right\} \right) \end{aligned}$$

Define the  $\mathfrak{D}$ -typed-record-separation defect quantity, using the truth-value indicator in Definition 6.49 as follows:

$$\begin{aligned} \text{Def}_4^{\text{approx}}[\text{Ev}; \lambda] \\ := \mathbf{1} \left\{ \begin{array}{l} \exists \sigma \in \text{RecSepTag}^{\text{approx}}, \quad \exists z_0, z_1 \in \text{RecSepDom}_{\sigma}, \\ z_0 \neq z_1, \quad \text{Reclmg}_{\sigma, \lambda}^{\text{approx}}(z_0) \cap \text{Reclmg}_{\sigma, \lambda}^{\text{approx}}(z_1) \neq \emptyset \end{array} \right\} \end{aligned}$$

For each  $\rho \in \text{RelTag}^{\text{approx}}$  define the relation-preservation evaluation component by

$$\begin{aligned} \text{RelComp}_{\rho, \lambda}^{\text{approx}} \\ := \left( \begin{array}{c} \text{Rel}_{\rho}^{\text{approx}, \text{src}}, \\ (\text{RelMap}_{\rho, k, \lambda}^{\text{approx}})_{k=1}^{\text{Arity}_{\rho}^{\text{approx}}}, \\ \text{Rel}_{\rho, \lambda}^{\text{approx}, \text{cand}} \end{array} \right) \end{aligned}$$

Define the continuum-approximation-stage evaluation datum by

$$\begin{aligned} \text{EvalDat}^{\text{approx}}[\text{Ev}; \lambda] \\ := \left( \begin{array}{l} \text{EvalInput}^{\text{cont}}[I], \\ (\text{RecCodeComp}_{\mathfrak{D}, o}^{\text{approx}})_o, \\ (\text{RecordImg}_{\mathfrak{D}, o}^{\text{time}, \text{real}}(x_{\mathfrak{D}}))_{o, x_{\mathfrak{D}}}, \\ (\text{Def}_i^{\text{approx}}[\text{Ev}; \lambda], \varepsilon_i^{\text{approx}})_{i \in I_{\text{approx}}^{(1)}}, \\ (\text{MapComp}_{t, \lambda}^{\text{approx}})_{t \in \text{MapTag}^{\text{approx}}}, \\ \text{RecMap}_{\mathfrak{D}, \lambda}^{\text{approx}}, \\ (\text{RelComp}_{\rho, \lambda}^{\text{approx}})_{\rho \in \text{RelTag}^{\text{approx}}} \end{array} \right) \end{aligned}$$

**Definition 6.75** (Continuum-approximation-stage condition group). Let the complete evaluation specification and approximation scale be

$$(\text{Ev}, \lambda) \in \text{EvSpec}_{\Lambda}$$

For the continuum-approximation-stage evaluation datum in Definition 6.74, define the six first-layer conditions by

$$\begin{aligned} K_i^{\text{approx}}[\text{Ev}; \lambda] \\ \iff \text{Def}_i^{\text{approx}}[\text{Ev}; \lambda] \leq \varepsilon_i^{\text{approx}}, \quad i \in I_{\text{approx}}^{(1)} \end{aligned}$$

Define the continuous-extension condition by

$$\begin{aligned}
& K_7^{\text{approx}}[\text{Ev}; \lambda] \\
& \iff \left[ \forall t \in \text{MapTag}^{\text{approx}}, \right. \\
& \quad \text{Map}_{t,\lambda}^{\text{approx}} : \left( \text{Int}_{t,\lambda}^{\text{approx}}, \mathcal{T}_t^{\text{approx,src}} \Big|_{\text{Int}_{t,\lambda}^{\text{approx}}} \right) \\
& \quad \longrightarrow (\text{Cod}_t^{\text{approx}}, \mathcal{T}_t^{\text{approx,tgt}}) \text{ is continuous,} \\
& \quad \forall s \in \text{Samp}_{t,\lambda}^{\text{approx}}, \\
& \quad \left. \text{Dev}_t^{\text{approx}}(\text{Map}_{t,\lambda}^{\text{approx}}(s), \text{SVal}_{t,\lambda}^{\text{approx}}(s)) \leq \varepsilon_t^{\text{approx,fit}} \right]
\end{aligned}$$

Define the order-stability condition by

$$\begin{aligned}
& K_8^{\text{approx}}[\text{Ev}; \lambda] \\
& \iff \left[ \forall t \in \text{OrdTag}^{\text{approx}}, \right. \\
& \quad \forall x, y \in \text{Int}_{t,\lambda}^{\text{approx}}, \\
& \quad \left. x \prec_t^{\text{approx,src}} y \implies \text{Map}_{t,\lambda}^{\text{approx}}(x) \prec_t^{\text{approx,tgt}} \text{Map}_{t,\lambda}^{\text{approx}}(y) \right]
\end{aligned}$$

Define the relation-stability condition by

$$\begin{aligned}
& K_9^{\text{approx}}[\text{Ev}; \lambda] \\
& \iff \left[ \forall \rho \in \text{RelTag}^{\text{approx}}, \right. \\
& \quad \forall (x_k)_{k=1}^{\text{Arity}_{\rho}^{\text{approx}}} \in \prod_{k=1}^{\text{Arity}_{\rho}^{\text{approx}}} \text{Dom}_{\rho,k}^{\text{approx}}, \\
& \quad (x_k)_{k=1}^{\text{Arity}_{\rho}^{\text{approx}}} \in \text{Rel}_{\rho}^{\text{approx,src}} \\
& \quad \iff (\text{RelMap}_{\rho,k,\lambda}^{\text{approx}}(x_k))_{k=1}^{\text{Arity}_{\rho}^{\text{approx}}} \in \text{Rel}_{\rho,\lambda}^{\text{approx,cand}} \left. \right]
\end{aligned}$$

Define the continuum-approximation-stage condition group by

$$\mathcal{K}_{\text{cont}}[\Gamma_{\text{Ev},\lambda}^{\text{approx}}] := \{ K_i^{\text{approx}}[\text{Ev}; \lambda] \mid i \in I_{\text{approx}}^{\text{fail}} \}$$

Define the predicate asserting that all nine conditions hold by

$$\begin{aligned}
& \text{CondAll}^{\text{approx}}[\text{Ev}; \lambda] \\
& \iff \bigwedge_{i \in I_{\text{approx}}^{\text{fail}}} K_i^{\text{approx}}[\text{Ev}; \lambda]
\end{aligned}$$

The target of each condition is as follows.

| $i$ | Target                                          |
|-----|-------------------------------------------------|
| 1   | Counting-step width derived from $\mathfrak{C}$ |
| 2   | $\mathfrak{D}$ -record stability                |
| 3   | $\mathfrak{D}$ -record-cell width               |
| 4   | $\mathfrak{D}$ -typed-record separation         |
| 5   | $\mathfrak{B}$ -object-state distinguishability |
| 6   | $\mathfrak{D}$ -observer-side coupling limit    |
| 7   | Continuous extension                            |
| 8   | Order stability                                 |
| 9   | Relation stability                              |

Let the non-ancestor formative-input satisfaction value be

$$\text{InputSat}_i^{\text{approx,form}}[\text{Ev}; \lambda] \in \mathbb{B}$$

When the domains and codomains of the sets, orders, topologies, maps, and relations appearing in condition  $K_i^{\text{approx}}[\text{Ev}; \lambda]$  are typed, this value takes  $\top$ .

Let the evaluation-input satisfaction value be

$$\text{InputSat}_i^{\text{approx,eval}}[\text{Ev}; \lambda] \in \mathbb{B}$$

This value takes  $\top$  when the first layer provides defect quantities and tolerances, the second layer provides finite samples, approximation maps, deviation maps, and fit tolerances, and the third layer provides strict orders or post-approximation relation candidates.

Specialize the input-satisfaction maps in Definition 6.21 by

$$\begin{aligned} \text{FailInputSat}^{\text{form}}(\text{FailTok}_{\text{approx},i}; \Gamma_{\text{Ev},\lambda}^{\text{approx}}) \\ &:= \text{InputSat}_i^{\text{approx,form}}[\text{Ev}; \lambda], \\ \text{FailInputSat}^{\text{eval}}(\text{FailTok}_{\text{approx},i}; \Gamma_{\text{Ev},\lambda}^{\text{approx}}) \\ &:= \text{InputSat}_i^{\text{approx,eval}}[\text{Ev}; \lambda] \end{aligned}$$

Specialize the failure-condition evaluation map by

$$\begin{aligned} \text{FailEval}(\text{FailTok}_{\text{approx},i}; \Gamma_{\text{Ev},\lambda}^{\text{approx}}) \\ &:= K_i^{\text{approx}}[\text{Ev}; \lambda], \quad i \in I_{\text{approx}}^{\text{fail}} \end{aligned}$$

Define input satisfaction for quantitative evaluation by

$$\begin{aligned} \text{InputSat}^{\text{approx}}[\text{Ev}; \lambda] &= \top \\ \iff K_{\text{dyn}}^{\text{fail}}[\text{Ev}] &= \top \\ \wedge \left[ \forall i \in I_{\text{approx}}^{\text{fail}}, \quad \text{InputSat}_i^{\text{approx,form}}[\text{Ev}; \lambda] &= \text{InputSat}_i^{\text{approx,eval}}[\text{Ev}; \lambda] = \top \right]. \end{aligned}$$

Satisfaction of the dynamical series expresses satisfaction of all direct predecessors outside the approximation series; this input condition does not include satisfaction of the nine conditions within the approximation series.

**Proposition 6.76** (Generation dependencies of the nine continuum-approximation conditions). By the within-series dependency edges of the continuum-approximation series in Definition 6.17, the six first-layer conditions are mutually incomparable. Within the first layer, satisfaction of one condition is not a prerequisite for evaluation readiness of another.

The direct-predecessor set of the continuous-extension condition is the token set of the six first-layer conditions. The direct-predecessor sets of the order-stability and relation-stability conditions consist of the token for the continuous-extension condition.

By the between-series dependency edges in Definition 6.17, every token in the dynamical-description series is a direct predecessor of each continuum-approximation condition. Consequently, the evaluation state in Definition 6.21 is uniquely determined by the satisfaction states of the comparison, representation, and dynamical-description series; the direct-predecessor conditions within the continuum-approximation series; the non-ancestor formative-input satisfaction value; and the evaluation-input satisfaction value.

**Definition 6.77** (Continuum-approximation-stage failure). Let the complete evaluation specification and approximation scale be  $(\text{Ev}, \lambda) \in \text{EvSpec}_\Lambda$ . The necessary and sufficient condition for continuum-approximation-stage failure is  $\text{FailedTok}^{\text{approx}}[\text{Ev}; \lambda] \neq \emptyset$ .

If two or more conditions are each ready for evaluation and fail at the same scale, all their tokens belong to this failed-token set.

**Proposition 6.78** (Recursive expansion of failed tokens in the continuum-approximation series). For each

$$\begin{aligned} i \in I_{\text{approx}}^{\text{fail}}, \quad (\text{Ev}, \lambda) \in \text{EvSpec}_\Lambda, \\ \text{FailTok}_{\text{approx}, i} \in \text{FailedTok}^{\text{approx}}[\text{Ev}; \lambda] \\ \iff \left[ \begin{aligned} &\text{InputSat}_i^{\text{approx}, \text{form}}[\text{Ev}; \lambda] = \top \\ &\wedge [\text{FailPredSat}(\text{FailTok}_{\text{approx}, i}; \Gamma_{\text{Ev}, \lambda}^{\text{approx}}) = \top] \\ &\wedge [\text{InputSat}_i^{\text{approx}, \text{eval}}[\text{Ev}; \lambda] = \top] \\ &\wedge [K_i^{\text{approx}}[\text{Ev}; \lambda] = \perp]. \end{aligned} \right] \end{aligned}$$

*Proof.* By Definition 6.21, the left-hand side is equivalent to

$$\text{FailStateOf}(\text{FailTok}_{\text{approx}, i}; \Gamma_{\text{Ev}, \lambda}^{\text{approx}}) = \text{failed}.$$

Substituting in order the equivalent conditions for the evaluation state, formation judgment, and evaluation-readiness judgment in the same definition, together with the specialization to the continuum-approximation series in Definition 6.75, yields the four conditions on the right-hand side.  $\square$

**Definition 6.79** (Nine failure classes at the continuum-approximation stage). Let the complete evaluation specification and approximation scale be  $(\text{Ev}, \lambda) \in \text{EvSpec}_\Lambda$ . For each  $i \in I_{\text{approx}}^{\text{fail}}$ , define the  $i$ -th continuum-approximation failure class by

$$\begin{aligned} \text{FA}_i[\text{Ev}; \lambda] \\ \iff \text{FailTok}_{\text{approx}, i} \in \text{FailedTok}^{\text{approx}}[\text{Ev}; \lambda]. \end{aligned}$$

The index  $i \in I_{\text{approx}}^{\text{fail}}$  is common to the condition  $K_i^{\text{approx}}[\text{Ev}; \lambda]$  and the failure class  $\text{FA}_i[\text{Ev}; \lambda]$ . Each  $i \in I_{\text{approx}}^{(1)}$  in the first layer corresponds to  $\text{Def}_i^{\text{approx}}[\text{Ev}; \lambda]$  and  $\varepsilon_i^{\text{approx}}$ , whereas each index in the second and third layers corresponds to the truth value of its condition.



**Proposition 6.80** (Condition correspondence and type separation of the six first-layer failure classes). For a scale-indexed complete evaluation specification  $(\text{Ev}, \lambda) \in \text{EvSpec}_\Lambda$  and each  $i \in I_{\text{approx}}^{(1)}$ , if

$$\text{FailReady}(\text{FailTok}_{\text{approx},i}; \Gamma_{\text{Ev},\lambda}^{\text{approx}}) = \top$$

then

$$\begin{aligned} & \text{FA}_i[\text{Ev}; \lambda] \\ & \iff \text{Def}_i^{\text{approx}}[\text{Ev}; \lambda] > \varepsilon_i^{\text{approx}}. \end{aligned}$$

The three failure classes derived from  $\mathfrak{D}$  correspond to the following distinct evaluation targets.

| Failure class | Evaluation target                       |
|---------------|-----------------------------------------|
| $\text{FA}_2$ | $\mathfrak{D}$ -record stability        |
| $\text{FA}_3$ | $\mathfrak{D}$ -record-cell width       |
| $\text{FA}_4$ | $\mathfrak{D}$ -typed-record separation |

Furthermore, if distinct first-layer indices  $i, j$  are both ready for evaluation and satisfy

$$\text{Def}_i^{\text{approx}}[\text{Ev}; \lambda] > \varepsilon_i^{\text{approx}}, \quad \text{Def}_j^{\text{approx}}[\text{Ev}; \lambda] > \varepsilon_j^{\text{approx}},$$

then

$$\begin{aligned} & |\text{FailedTok}^{\text{approx}}[\text{Ev}; \lambda] \cap \{\text{FailTok}_{\text{approx},i} \mid i \in I_{\text{approx}}^{(1)}\}| \\ & \geq 2. \end{aligned}$$

*Proof.* Substituting the three satisfaction conditions into Proposition 6.78 and then applying Definition 6.74 and Definition 6.79 in order shows, for each  $i \in I_{\text{approx}}^{(1)}$ , that membership in the failure class, membership of the failed token, failure of the condition, and exceedance of the defect quantity are equivalent.

In Definition 6.74, the second component is the evaluation value for  $\mathfrak{D}$ -record stability, the third component is the supremum of the deviation diameters of the  $\mathfrak{D}$ -record-cell images over the two representation tags, and the fourth component is the separation-evaluation value for the three typed  $\mathfrak{D}$ -record images. Hence the three failure classes derived from  $\mathfrak{D}$  correspond to the distinct evaluation targets shown in the table of the proposition.

Applying the preceding equivalence to the distinct indices  $i, j$  places both corresponding tokens in the failed-token set. The tokens have distinct semantic indices, so the set contains at least two first-layer failed tokens.  $\square$

Table 5: Correspondence between the nine continuum-approximation-stage failure classes and the condition group

| Failure class | Layer        | Evaluation target                               | Condition             |
|---------------|--------------|-------------------------------------------------|-----------------------|
| $\text{FA}_1$ | First layer  | Counting-step width derived from $\mathfrak{C}$ | $K_1^{\text{approx}}$ |
| $\text{FA}_2$ |              | $\mathfrak{D}$ -record stability                | $K_2^{\text{approx}}$ |
| $\text{FA}_3$ |              | $\mathfrak{D}$ -record-cell width               | $K_3^{\text{approx}}$ |
| $\text{FA}_4$ |              | $\mathfrak{D}$ -typed-record separation         | $K_4^{\text{approx}}$ |
| $\text{FA}_5$ | Second layer | $\mathfrak{B}$ -object-state distinguishability | $K_5^{\text{approx}}$ |
| $\text{FA}_6$ |              | $\mathfrak{D}$ -observer-side coupling limit    | $K_6^{\text{approx}}$ |
| $\text{FA}_7$ |              | Continuous extension                            | $K_7^{\text{approx}}$ |
| $\text{FA}_8$ | Third layer  | Order stability                                 | $K_8^{\text{approx}}$ |
| $\text{FA}_9$ |              | Relation stability                              | $K_9^{\text{approx}}$ |

**Proposition 6.81** (Localization of continuum-approximation-stage failures). For every  $(\text{Ev}, \lambda) \in \text{EvSpec}_\Lambda$ ,

$$\begin{aligned} & \text{FailedTok}^{\text{approx}}[\text{Ev}; \lambda] \\ &= \left\{ \text{FailTok}_{\text{approx}, i} \mid \begin{array}{l} i \in I_{\text{approx}}^{\text{fail}}, \\ \text{FA}_i[\text{Ev}; \lambda] \end{array} \right\}. \end{aligned}$$

The necessary and sufficient condition for continuum-approximation-stage failure is

$$\bigvee_{i \in I_{\text{approx}}^{\text{fail}}} \text{FA}_i[\text{Ev}; \lambda].$$

Its localization is

$$\begin{aligned} & \text{Min}_{\preceq_{\text{fail}, \text{dep}}} \text{FailedTok}^{\text{approx}}[\text{Ev}; \lambda] \\ &= \text{Min}_{\preceq_{\text{fail}, \text{dep}}} \left\{ \text{FailTok}_{\text{approx}, i} \mid \begin{array}{l} i \in I_{\text{approx}}^{\text{fail}}, \\ \text{FA}_i[\text{Ev}; \lambda] \end{array} \right\}. \end{aligned}$$

The simultaneous-failure criterion for the first layer is given by Proposition 6.80. In the third layer, if  $\text{FA}_8[\text{Ev}; \lambda] \wedge \text{FA}_9[\text{Ev}; \lambda]$  holds, both corresponding tokens belong to the failed-token set.

*Proof.* By Definition 6.79, for each

$$i \in I_{\text{approx}}^{\text{fail}}$$

one has

$$\text{FA}_i[\text{Ev}; \lambda] \iff \text{FailTok}_{\text{approx}, i} \in \text{FailedTok}^{\text{approx}}[\text{Ev}; \lambda].$$

Substituting the nine-token family of Definition 6.10 gives the displayed representation of the failed-token set in the proposition.

By Definition 6.77, continuum-approximation-stage failure is equivalent to the nonemptiness of the failed-token set. Substituting the preceding set representation gives Equation (6.81). Applying Definition 6.3 to the same set representation gives the equality of the sets of minimal elements under the dependency order.

Apply Proposition 6.80 to the first layer. For the third layer, substitute satisfaction of the two failure classes into the preceding representation of the failed-token set.  $\square$

**Definition 6.82** (Refinement family). Let the totally ordered scale set be  $(\Lambda, \leq_\Lambda)$ . For every complete evaluation specification  $\text{Ev} \in \text{EvSpec}$ , call the family of continuum-approximation-stage evaluation data  $(\text{EvalDat}^{\text{approx}}[\text{Ev}; \lambda])_{\lambda \in \Lambda}$  a refinement family.

For each  $\lambda, \mu \in \Lambda$ , call  $\lambda \leq_\Lambda \mu$  the refinement relation from scale  $\lambda$  to scale  $\mu$ .

**Definition 6.83** (Nine-component tolerance family for continuum approximation). Extend the first-layer tolerance family  $(\varepsilon_i^{\text{approx}})_{i \in I_{\text{approx}}^{(1)}}$  of Definition 6.74 to the index set  $I_{\text{approx}}^{\text{fail}}$  of Definition 6.73.

For each  $i \in I_{\text{approx}}^{(2)} \cup I_{\text{approx}}^{(3)}$  in the second or third layer, define  $\varepsilon_i^{\text{approx}} := \frac{1}{2}$ . The defect quantities of these three components encode the truth values of the conditions as 0 and 1; their margins measure the separation from this encoding threshold. Let the nine-component tolerance family for continuum approximation be

$$(\varepsilon_i^{\text{approx}})_{i \in I_{\text{approx}}^{\text{fail}}} \in (\overline{\mathbb{R}}_{\geq 0})^{I_{\text{approx}}^{\text{fail}}}.$$

**Definition 6.84** (Nine-component defect-quantity family for continuum approximation). Let the complete evaluation specification and approximation scale be  $(\text{Ev}, \lambda) \in \text{EvSpec}_\Lambda$ .

For each  $i \in I_{\text{approx}}^{(1)}$  in the first layer,  $\text{Def}_i^{\text{approx}}[\text{Ev}; \lambda] \in \overline{\mathbb{R}}_{\geq 0}$  is the first-layer defect quantity of Definition 6.74.

For each  $i \in I_{\text{approx}}^{(2)} \cup I_{\text{approx}}^{(3)}$  in the second or third layer, define

$$\begin{aligned} & \text{Def}_i^{\text{approx}}[\text{Ev}; \lambda] \\ & := \begin{cases} 0, & K_i^{\text{approx}}[\text{Ev}; \lambda] = \top, \\ 1, & K_i^{\text{approx}}[\text{Ev}; \lambda] = \perp. \end{cases} \end{aligned}$$

Let the nine-component defect-quantity family for continuum approximation be

$$(\text{Def}_i^{\text{approx}}[\text{Ev}; \lambda])_{i \in I_{\text{approx}}^{\text{fail}}}.$$

Then, for each  $i \in I_{\text{approx}}^{\text{fail}}$ ,

$$\begin{aligned} & K_i^{\text{approx}}[\text{Ev}; \lambda] = \top \\ & \iff \text{Def}_i^{\text{approx}}[\text{Ev}; \lambda] \leq \varepsilon_i^{\text{approx}}. \end{aligned}$$

**Definition 6.85** (Monotonic nondecrease of the continuum-approximation defect-quantity family). For a refinement family in Definition 6.82, the nine-component defect-quantity family for continuum approximation is monotonically nondecreasing if, for every

$$\begin{aligned} & \text{Ev} \in \text{EvSpec}, \quad i \in I_{\text{approx}}^{\text{fail}}, \quad \lambda, \mu \in \Lambda, \\ & \lambda \leq_\Lambda \mu \\ & \implies \text{Def}_i^{\text{approx}}[\text{Ev}; \lambda] \leq \text{Def}_i^{\text{approx}}[\text{Ev}; \mu]. \end{aligned}$$

**Definition 6.86** (Continuum-approximation scale validity domain). Let the complete evaluation specification be  $\text{Ev} \in \text{EvSpec}$ . Define the continuum-approximation scale validity domain as the scale slice of the approximation-validity domain in Definition 6.4:

$$\mathcal{V}_{\text{Ev}}^{\text{approx}} := \{\lambda \in \Lambda \mid (\text{Ev}, \lambda) \in \mathcal{V}^{\text{approx}}\}.$$

This is the scale domain of satisfaction, including input satisfaction and satisfaction of all nine conditions.

**Definition 6.87** (Margin operation on the extended nonnegative reals). For arbitrary  $\alpha, \beta \in \overline{\mathbb{R}}_{\geq 0}$ , define the margin operation by

$$\text{Mar}_\infty(\alpha, \beta) := \begin{cases} \alpha - \beta, & \alpha < \infty, \beta < \infty, \\ +\infty, & \alpha = \infty, \beta < \infty, \\ 0, & \alpha = \infty, \beta = \infty, \\ -\infty, & \alpha < \infty, \beta = \infty. \end{cases}$$

**Definition 6.88** (Componentwise margin and excess for continuum approximation). Let  $(\text{Ev}, \lambda) \in \text{EvSpec}_\Lambda$  and  $i \in I_{\text{approx}}^{\text{fail}}$ . Define the componentwise margin and excess by

$$\begin{aligned} & \text{Mar}_i^{\text{approx}}[\text{Ev}; \lambda] := \text{Mar}_\infty(\varepsilon_i^{\text{approx}}, \text{Def}_i^{\text{approx}}[\text{Ev}; \lambda]), \\ & \text{Exc}_i^{\text{approx}}[\text{Ev}; \lambda] := \max\{0, -\text{Mar}_i^{\text{approx}}[\text{Ev}; \lambda]\}. \end{aligned}$$

Their types are

$$\begin{aligned}\text{Mar}_i^{\text{approx}}[\text{Ev}; \lambda] &\in \overline{\mathbb{R}}, \\ \text{Exc}_i^{\text{approx}}[\text{Ev}; \lambda] &\in \overline{\mathbb{R}}_{\geq 0}.\end{aligned}$$

Under  $\text{InputSat}^{\text{approx}}[\text{Ev}; \lambda] = \top$ ,

$$\begin{aligned}\lambda \in \mathcal{V}_{\text{Ev}}^{\text{approx}} &\iff \forall i \in I_{\text{approx}}^{\text{fail}}, \quad \text{Mar}_i^{\text{approx}}[\text{Ev}; \lambda] \geq 0 \\ &\iff \forall i \in I_{\text{approx}}^{\text{fail}}, \quad \text{Exc}_i^{\text{approx}}[\text{Ev}; \lambda] = 0.\end{aligned}$$

**Lemma 6.89** (Positivity criterion for componentwise continuum-approximation excess). For every  $(\text{Ev}, \lambda) \in \text{EvSpec}_{\Lambda}$  and  $i \in I_{\text{approx}}^{\text{fail}}$ ,

$$\begin{aligned}\text{Exc}_i^{\text{approx}}[\text{Ev}; \lambda] &> 0 \\ &\iff \text{Def}_i^{\text{approx}}[\text{Ev}; \lambda] > \varepsilon_i^{\text{approx}}.\end{aligned}$$

*Proof.* Apply Definition 6.87 and Definition 6.88 to the case in which the defect quantity and tolerance threshold are both finite, the case in which only the defect quantity is infinite, the case in which only the tolerance threshold is infinite, and the case in which both are infinite. The truth values of the two sides coincide in every case.  $\square$

**Definition 6.90** (Continuum-approximation saturation and excess index sets). Let the complete evaluation specification and approximation scale be  $(\text{Ev}, \lambda) \in \text{EvSpec}_{\Lambda}$ .

Define the saturation and excess index sets by

$$\begin{aligned}I_{\text{approx}}^{\text{sat}}[\text{Ev}; \lambda] &:= \left\{ i \in I_{\text{approx}}^{\text{fail}} \mid \text{Mar}_i^{\text{approx}}[\text{Ev}; \lambda] = 0 \right\}, \\ I_{\text{approx}}^{\text{exc}}[\text{Ev}; \lambda] &:= \left\{ i \in I_{\text{approx}}^{\text{fail}} \mid \text{Exc}_i^{\text{approx}}[\text{Ev}; \lambda] > 0 \right\}.\end{aligned}$$

**Definition 6.91** (Robust validity for continuum approximation). Let the complete evaluation specification be  $\text{Ev} \in \text{EvSpec}$ .

Define the positive-margin scale domain for continuum approximation by

$$\begin{aligned}\mathcal{V}_{\Lambda}^{\text{approx},+}[\text{Ev}] &:= \left\{ \lambda \in \mathcal{V}_{\text{Ev}}^{\text{approx}} \mid \forall i \in I_{\text{approx}}^{\text{fail}}, \right. \\ &\quad \left. \text{Mar}_i^{\text{approx}}[\text{Ev}; \lambda] > 0 \right\}.\end{aligned}$$

For an approximation scale  $\lambda \in \Lambda$ , define robust validity for continuum approximation by

$$\begin{aligned}\text{Rob}^{\text{approx}}[\text{Ev}; \lambda] &\iff \lambda \in \mathcal{V}_{\Lambda}^{\text{approx},+}[\text{Ev}].\end{aligned}$$

Definition 6.88 gives

$$\mathcal{V}_{\Lambda}^{\text{approx},+}[\text{Ev}] \subseteq \mathcal{V}_{\text{Ev}}^{\text{approx}}.$$

**Proposition 6.92** (Open-neighborhood representation of robust validity for continuum approximation). With respect to the order topology generated by the total order  $(\Lambda, \leq_\Lambda)$ , suppose that, for a complete evaluation specification  $\mathbf{Ev} \in \mathbf{EvSpec}$  and each  $i \in I_{\text{approx}}^{\text{fail}}$ , the map

$$\begin{aligned}\Lambda &\longrightarrow \overline{\mathbb{R}}, \\ \mu &\longmapsto \mathbf{Mar}_i^{\text{approx}}[\mathbf{Ev}; \mu]\end{aligned}$$

is lower semicontinuous. Assume  $\forall \mu \in \Lambda, \mathbf{InputSat}^{\text{approx}}[\mathbf{Ev}; \mu] = \top$ .

For every  $\lambda \in \Lambda$ ,

$$\begin{aligned}\mathbf{Rob}^{\text{approx}}[\mathbf{Ev}; \lambda] \\ \iff \left[ \exists W \subseteq \Lambda, \quad W \text{ is open in the order topology,} \right. \\ \left. \lambda \in W \subseteq \mathcal{V}_\Lambda^{\text{approx},+}[\mathbf{Ev}] \right].\end{aligned}$$

*Proof.* For each  $i \in I_{\text{approx}}^{\text{fail}}$ , lower semicontinuity implies that

$$\left\{ \mu \in \Lambda \mid \mathbf{Mar}_i^{\text{approx}}[\mathbf{Ev}; \mu] > 0 \right\}$$

is open in the order topology.

By the input-satisfaction assumption and Definition 6.88, positive margins in all nine components imply membership in the validity scale domain. Because the index set  $I_{\text{approx}}^{\text{fail}}$  is finite, the intersection of these nine open sets,  $\mathcal{V}_\Lambda^{\text{approx},+}[\mathbf{Ev}]$ , is also open.

When  $\mathbf{Rob}^{\text{approx}}[\mathbf{Ev}; \lambda]$  holds, Definition 6.91 gives  $\lambda \in \mathcal{V}_\Lambda^{\text{approx},+}[\mathbf{Ev}]$ . Therefore one may take  $W = \mathcal{V}_\Lambda^{\text{approx},+}[\mathbf{Ev}]$ .

Conversely, when there exists an open set  $W$  satisfying the right-hand side of the proposition,  $\lambda \in W \subseteq \mathcal{V}_\Lambda^{\text{approx},+}[\mathbf{Ev}]$ . Definition 6.91 gives  $\mathbf{Rob}^{\text{approx}}[\mathbf{Ev}; \lambda]$ .  $\square$

**Definition 6.93** (Maximal effective interval for continuum approximation). Let the complete evaluation specification be  $\mathbf{Ev} \in \mathbf{EvSpec}$ .

The admissible initial-interval family of Definition 6.8 is, by Definition 6.86,

$$\begin{aligned}\mathfrak{I}_{\text{init}}[\mathbf{Ev}] \\ = \left\{ \mathcal{I} \in \mathcal{P}(\Lambda) \mid \right. \\ \mathcal{I} \subseteq \mathcal{V}_{\mathbf{Ev}}^{\text{approx}}, \\ \forall \mu \in \mathcal{I}, \quad \forall \lambda \in \Lambda, \\ \left. \lambda \leq_\Lambda \mu \implies \lambda \in \mathcal{I} \right\}.\end{aligned}$$

Define the maximal effective interval for continuum approximation by

$$\begin{aligned}\mathcal{I}^{\text{max}}[\mathbf{Ev}] \\ := \bigcup_{\mathcal{I} \in \mathfrak{I}_{\text{init}}[\mathbf{Ev}]} \mathcal{I}.\end{aligned}$$

**Definition 6.94** (Effective continuum-approximation regime). Define the effective continuum-approximation regime by

$$\mathcal{V}^{\text{approx,eff}} := \left\{ (\text{Ev}, \lambda) \in \text{EvSpec}_\Lambda \mid \lambda \in \mathcal{I}^{\text{max}}[\text{Ev}] \right\}.$$

Definition 6.10 gives

$$\begin{aligned} \mathcal{V}^{\text{approx,eff}} &= \left\{ \Gamma_{\text{Ev},\lambda}^{\text{approx}} \mid \text{Ev} \in \text{EvSpec}, \quad \lambda \in \mathcal{I}^{\text{max}}[\text{Ev}] \right\} \\ &\subseteq \mathcal{V}^{\text{approx}} \subseteq \text{EvSpec}_\Lambda. \end{aligned}$$

**Theorem 6.95** (Equivalent characterizations of quantitative validity for continuum approximation). Let the complete evaluation specification and approximation scale be  $(\text{Ev}, \lambda) \in \text{EvSpec}_\Lambda$ . Assume  $\text{InputSat}^{\text{approx}}[\text{Ev}; \lambda] = \top$ . Then the following eight conditions are equivalent.

$$\begin{aligned} &\lambda \in \mathcal{V}_{\text{Ev}}^{\text{approx}} \\ &\iff (\text{Ev}, \lambda) \in \mathcal{V}^{\text{approx}} \\ &\iff \text{CondAll}^{\text{approx}}[\text{Ev}; \lambda] \\ &\iff \left[ \begin{array}{c} \forall i \in I_{\text{approx}}^{\text{fail}}, \\ K_i^{\text{approx}}[\text{Ev}; \lambda] = \top \end{array} \right] \\ &\iff \left[ \begin{array}{c} \forall i \in I_{\text{approx}}^{\text{fail}}, \\ \text{Def}_i^{\text{approx}}[\text{Ev}; \lambda] \leq \varepsilon_i^{\text{approx}} \end{array} \right] \\ &\iff \left[ \begin{array}{c} \forall i \in I_{\text{approx}}^{\text{fail}}, \\ \text{Mar}_i^{\text{approx}}[\text{Ev}; \lambda] \geq 0 \end{array} \right] \\ &\iff \left[ \begin{array}{c} \forall i \in I_{\text{approx}}^{\text{fail}}, \\ \text{Exc}_i^{\text{approx}}[\text{Ev}; \lambda] = 0 \end{array} \right] \\ &\iff I_{\text{approx}}^{\text{exc}}[\text{Ev}; \lambda] = \emptyset. \end{aligned}$$

*Proof.* By Definition 6.86, the first two conditions are equivalent. Under the input-satisfaction assumption, induction along the DAG within the approximation series in Definition 6.17 shows that satisfaction of all nine conditions makes every approximation token satisfied. Conversely, if every approximation token is satisfied, each condition holds by Definition 6.21. By Definition 6.27, these statements are equivalent to membership in the approximation-validity domain. Definition 6.75 makes satisfaction of all nine conditions equivalent to  $K_i^{\text{approx}}[\text{Ev}; \lambda] = \top$  for every index. By Definition 6.84, each such condition is equivalent to  $\text{Def}_i^{\text{approx}}[\text{Ev}; \lambda] \leq \varepsilon_i^{\text{approx}}$ .

Definition 6.88 equates the defect-quantity conditions for all components with, respectively, nonnegativity of every margin and vanishing of every excess. By Definition 6.90, vanishing of every excess is equivalent to emptiness of the excess index set. Hence the eight conditions listed in the theorem are equivalent.  $\square$

**Theorem 6.96** (Separation of continuum-approximation saturation and excess components). Let the complete evaluation specification and approximation scale be  $(\text{Ev}, \lambda) \in \text{EvSpec}_\Lambda$ . Assume  $\text{InputSat}^{\text{approx}}[\text{Ev}; \lambda] = \top$ .

For each  $i \in I_{\text{approx}}^{\text{fail}}$ ,

$$\begin{aligned} i &\in I_{\text{approx}}^{\text{sat}}[\text{Ev}; \lambda] \\ &\iff \text{Mar}_i^{\text{approx}}[\text{Ev}; \lambda] = 0 \\ &\iff \text{Def}_i^{\text{approx}}[\text{Ev}; \lambda] = \varepsilon_i^{\text{approx}} \end{aligned}$$

and

$$\begin{aligned} i &\in I_{\text{approx}}^{\text{exc}}[\text{Ev}; \lambda] \\ &\iff \text{Exc}_i^{\text{approx}}[\text{Ev}; \lambda] > 0 \\ &\iff \text{Mar}_i^{\text{approx}}[\text{Ev}; \lambda] < 0 \\ &\iff \text{Def}_i^{\text{approx}}[\text{Ev}; \lambda] > \varepsilon_i^{\text{approx}} \end{aligned}$$

hold.

Furthermore,

$$\begin{aligned} I_{\text{approx}}^{\text{sat}}[\text{Ev}; \lambda] \\ \cap I_{\text{approx}}^{\text{exc}}[\text{Ev}; \lambda] = \emptyset \end{aligned}$$

and

$$\begin{aligned} \lambda &\in \mathcal{V}_{\text{Ev}}^{\text{approx}} \setminus \mathcal{V}_{\Lambda}^{\text{approx},+}[\text{Ev}] \\ &\iff \left[ \begin{aligned} I_{\text{approx}}^{\text{sat}}[\text{Ev}; \lambda] &\neq \emptyset, \\ I_{\text{approx}}^{\text{exc}}[\text{Ev}; \lambda] &= \emptyset \end{aligned} \right] \end{aligned}$$

hold.

For every  $A \in \mathcal{P}(I_{\text{approx}}^{\text{fail}})$ ,

$$\begin{aligned} A &\subseteq I_{\text{approx}}^{\text{sat}}[\text{Ev}; \lambda] \\ &\iff \left[ \forall i \in A, \quad \text{Def}_i^{\text{approx}}[\text{Ev}; \lambda] = \varepsilon_i^{\text{approx}} \right] \end{aligned}$$

and

$$\begin{aligned} A &\subseteq I_{\text{approx}}^{\text{exc}}[\text{Ev}; \lambda] \\ &\iff \left[ \forall i \in A, \quad \text{Def}_i^{\text{approx}}[\text{Ev}; \lambda] > \varepsilon_i^{\text{approx}} \right] \end{aligned}$$

hold.

*Proof.* By Definition 6.90 and Definition 6.88, membership in the saturation index set is equivalent to a zero margin and to equality of the defect quantity and the tolerance, whereas membership in the excess index set is equivalent to a positive excess, a negative margin, and exceedance of the tolerance by the defect quantity.

If the same index belonged to both the saturation and excess index sets, its margin would be both zero and negative, a contradiction. Thus the two sets are disjoint. By Theorem 6.95 and Definition 6.91, the scales that are valid but not robust are exactly those at which a saturation index exists and no excess index exists. The two equivalences for an arbitrary subset  $A \subseteq I_{\text{approx}}^{\text{fail}}$  follow by applying Definition 6.90 to each element of  $A$ .  $\square$

**Theorem 6.97** (Maximal effective interval and first excess for continuum approximation). Suppose that the nine-component continuum-approximation defect-quantity family is monotonically nondecreasing in the sense of Definition 6.85. For every  $\mathbf{Ev} \in \mathbf{EvSpec}$  satisfying  $\forall \mu \in \Lambda, \text{InputSat}^{\text{approx}}[\mathbf{Ev}; \mu] = \top$ ,

$$\begin{aligned} \mathcal{I}^{\max}[\mathbf{Ev}] &= \mathcal{V}_{\mathbf{Ev}}^{\text{approx}}, \\ \forall \mathcal{I} \in \mathfrak{J}_{\text{init}}[\mathbf{Ev}], \quad \mathcal{I} &\subseteq \mathcal{I}^{\max}[\mathbf{Ev}] \end{aligned}$$

holds. If  $\lambda_{\text{in}}$  is a greatest element of  $\mathcal{I}^{\max}[\mathbf{Ev}]$ , then

$$I_{\text{approx}}^{\text{exc}}[\mathbf{Ev}; \lambda_{\text{in}}] = \emptyset.$$

Moreover, if  $\mathbf{Ev} \in \text{Dom}(\lambda_{\text{bdry}}^{\text{approx}})$ , then

$$\begin{aligned} \mathcal{I}^{\max}[\mathbf{Ev}] &= \{\lambda \in \Lambda \mid \lambda <_{\Lambda} \lambda_{\text{bdry}}^{\text{approx}}[\mathbf{Ev}]\}, \\ I_{\text{approx}}^{\text{exc}}[\lambda_{\text{bdry}}^{\text{approx}}[\mathbf{Ev}]] &\neq \emptyset, \\ \forall \lambda <_{\Lambda} \lambda_{\text{bdry}}^{\text{approx}}[\mathbf{Ev}], \quad I_{\text{approx}}^{\text{exc}}[\mathbf{Ev}; \lambda] &= \emptyset \end{aligned}$$

hold. Therefore  $\lambda_{\text{bdry}}^{\text{approx}}[\mathbf{Ev}]$  is the least approximation scale at which an excess component occurs.

*Proof.* Take  $\mu \in \mathcal{V}_{\mathbf{Ev}}^{\text{approx}}$  and  $\lambda \leq_{\Lambda} \mu$ . For each  $i \in I_{\text{approx}}^{\text{fail}}$ , scale validity and monotonic nondecrease give

$$\text{Def}_i^{\text{approx}}[\mathbf{Ev}; \lambda] \leq \text{Def}_i^{\text{approx}}[\mathbf{Ev}; \mu] \leq \varepsilon_i^{\text{approx}}.$$

Hence  $\lambda \in \mathcal{V}_{\mathbf{Ev}}^{\text{approx}}$ , and this scale validity domain is an initial segment.

In Definition 6.93, the scale validity domain itself belongs to  $\mathfrak{J}_{\text{init}}[\mathbf{Ev}]$ , and every member of that family is contained in the scale validity domain. Hence the union of the family equals the scale validity domain, and each initial segment is contained in that union. If a greatest element  $\lambda_{\text{in}}$  exists, it belongs to the scale validity domain; Theorem 6.95 then gives an empty excess index set.

Under the assumption in the latter part, Definition 6.86 and Definition 6.4 identify the scale validity domain with  $\text{Val}_{\Lambda}(K_{\text{approx}}^{\text{fail}}; \mathbf{Ev})$ . Proposition 6.31 gives the initial-interval representation of the maximal effective interval. By Definition 6.29, the boundary scale lies outside the scale validity domain, while every smaller scale belongs to the maximal effective interval. Applying Theorem 6.95 to the boundary scale and to each lower scale gives, respectively, nonemptiness and emptiness of the excess index set. Thus the boundary scale is the least scale at which an excess component occurs.  $\square$



### Maximal effective interval and first excess

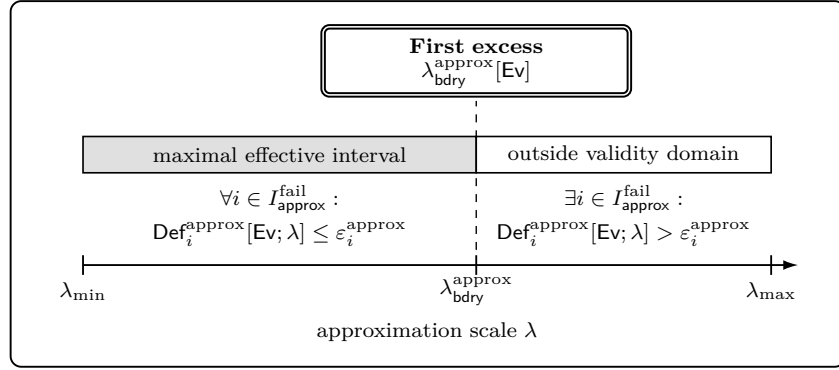


Figure 13: The maximal effective interval and first excess for continuum approximation. When the inputs for quantitative evaluation are satisfied at every scale, the nine-component defect-quantity family is monotonically nondecreasing with respect to the approximation scale, and the boundary scale is defined, the maximal effective interval equals the scale validity domain. Every scale below the boundary scale belongs to the maximal effective interval; the boundary scale lies outside the scale validity domain and is the least approximation scale at which an excess component first occurs.

#### 6.5.1 Failure-witness objects at the continuum-approximation stage

Let the complete evaluation specification, approximation scale, and continuum-approximation failure index be

$$(\text{Ev}, \lambda) \in \text{EvSpec}_\Lambda, \quad i \in I_{\text{approx}}^{\text{fail}}.$$

The witness conditions for the nine continuum-approximation failure classes are given by the satisfaction states of the formation inputs, evaluation inputs, and direct-predecessor conditions of each condition, together with the corresponding defect quantity exceeding its tolerance.

**Proposition 6.98** (Witness conditions for the nine continuum-approximation failure classes). Let the complete evaluation specification and approximation scale be

$$(\text{Ev}, \lambda) \in \text{EvSpec}_\Lambda.$$

For each  $i \in I_{\text{approx}}^{\text{fail}}$ , assume the evaluation-readiness condition of Definition 6.21 for  $\text{FailTok}_{\text{approx}, i}$  and  $\Gamma_{\text{Ev}, \lambda}^{\text{approx}}$ . Then the following three conditions are equivalent:

$$\begin{aligned} & \text{FA}_i[\text{Ev}; \lambda], \\ & \text{Def}_i^{\text{approx}}[\text{Ev}; \lambda] > \varepsilon_i^{\text{approx}}, \\ & i \in I_{\text{approx}}^{\text{exc}}[\text{Ev}; \lambda]. \end{aligned}$$

*Proof.* Substituting the three satisfaction conditions into Proposition 6.78 makes failed-token membership equivalent to  $K_i^{\text{approx}}[\text{Ev}; \lambda] = \perp$ . Definition 6.79 converts the former into  $\text{FA}_i[\text{Ev}; \lambda]$ , while Definition 6.84 converts the latter into exceedance of the tolerance by the defect quantity. By Lemma 6.89 and Definition 6.90, that exceedance is equivalent to  $i \in I_{\text{approx}}^{\text{exc}}[\text{Ev}; \lambda]$ . Hence the three conditions are equivalent.  $\square$

*Remark 6.99* (Witness objects for the nine continuum-approximation failure classes). For each  $i \in I_{\text{approx}}^{\text{fail}}$ , the component obtained by applying Proposition 6.98 gives the witness condition for the corresponding continuum-approximation failure class. The evaluation object of each component is shown in the row for  $\text{FA}_i$  in Table 5.

## 6.6 Failure positions at the dynamical-description stage

For the complete evaluation specification  $\text{Ev} \in \text{EvSpec}$  of Definition 5.2, the dynamical-description-series evaluation-specification component of Definition 6.10 is  $\text{Ev}_{\text{dyn}}^{\text{fail}} = \text{Ev}$ . The eight failure-position tokens in the dynamical-description series are associated with the eight dynamical-description conditions on a complete evaluation specification.

The eight conditions below follow the common evaluation rule of Definition 5.3. A false value outside the evaluation domain does not by itself mean that the condition has failed. Failure of the condition itself also requires evaluation readiness.

**Definition 6.100** (Evaluation of dynamical-description-stage conditions on complete evaluation specifications). For each  $\text{Ev} \in \text{EvSpec}$ ,  $i \in I_{\text{dyn}}^{\text{fail}}$ , let the evaluation context for the dynamical-description series be  $\Gamma_{\text{Ev}}^{\text{dyn}} := \text{Ev}_{\text{dyn}}^{\text{fail}}$ , and define the eight conditions by

$$K_i^{\text{dyn}}[\text{Ev}] := \begin{cases} K_{\text{GenTimeOrdAntiSym}}^{\text{dyn}}(\text{Ev}), & i = 1, \\ K_{\text{CanTrjDesc}}^{\text{dyn}}(\text{Ev}), & i = 2, \\ K_{\text{CanTrjFunc}}^{\text{dyn}}(\text{Ev}), & i = 3, \\ K_{\text{CanObsDesc}}^{\text{dyn}}(\text{Ev}), & i = 4, \\ K_{\text{LocObsChgComm}}^{\text{dyn}}(\text{Ev}), & i = 5, \\ K_{\text{ObsTimeIncTrans}}^{\text{dyn}}(\text{Ev}), & i = 6, \\ K_{\text{ObsTimeRealOrdEmb}}^{\text{dyn}}(\text{Ev}), & i = 7, \\ K_{\text{ObsTimeTrjFact}}^{\text{dyn}}(\text{Ev}), & i = 8 \end{cases}.$$

The non-ancestral formation input for the dynamical-description series includes exact operability. Thus, for every  $i \in I_{\text{dyn}}^{\text{fail}}$ ,

$$\begin{aligned} \text{FailInputSat}^{\text{form}}(\text{FailTok}_{\text{dyn},i}; \Gamma_{\text{Ev}}^{\text{dyn}}) &= \top \\ \implies \text{Ev} &\in \text{EvSpec}^{\text{exact}}. \end{aligned}$$

Further, define the evaluation-readiness predicate by

$$\begin{aligned} \text{Ready}_i^{\text{dyn}}[\text{Ev}] \\ \iff \text{FailReady}(\text{FailTok}_{\text{dyn},i}; \Gamma_{\text{Ev}}^{\text{dyn}}) &= \top. \end{aligned}$$

Then

$$\begin{aligned} \text{DynCondAll}[\text{Ev}] \\ \iff \bigwedge_{i \in I_{\text{dyn}}^{\text{fail}}} K_i^{\text{dyn}}[\text{Ev}] \end{aligned}$$

holds.

**Proposition 6.101** (Correspondence between failure-position tokens in the dynamical-description series and the eight conditions). For each  $i \in I_{\text{dyn}}^{\text{fail}}$ ,  $\text{Ev} \in \text{EvSpec}$ , the failure-condition classification is the  $\tau = \text{dyn}$  component of Definition 6.10, and

$$\begin{aligned} & \text{FailEval} \left( \text{FailTok}_{\text{dyn},i}; \Gamma_{\text{Ev}}^{\text{dyn}} \right) \\ &= K_i^{\text{dyn}}[\text{Ev}] \end{aligned}$$

holds.

Further,

$$\begin{aligned} & \text{FailTok}_{\text{dyn},i} \in \text{FailedTok}^{\text{strict}}[\text{Ev}] \\ & \iff \left[ \text{Ready}_i^{\text{dyn}}[\text{Ev}] \right] \\ & \quad \wedge \left[ K_i^{\text{dyn}}[\text{Ev}] = \perp \right]. \end{aligned}$$

*Proof.* Composing the failure-condition evaluation map of Definition 6.10 with Definition 6.100 yields the first equality.

Substituting the first equality into the failed-state equivalence of Definition 6.21 and applying Proposition 6.23 yields the second equivalence.  $\square$

**Definition 6.102** (Failure family for dynamical-description conditions). For each  $i \in I_{\text{dyn}}^{\text{fail}}$ ,  $\text{Ev} \in \text{EvSpec}$ , define the dynamical-description-condition failure predicate by

$$\begin{aligned} & \text{FDyn}_i[\text{Ev}] \\ & \iff \text{FailTok}_{\text{dyn},i} \in \text{FailedTok}^{\text{strict}}[\text{Ev}]. \end{aligned}$$

**Lemma 6.103** (Common evaluation formula for the failure family of dynamical-description conditions). For each  $i \in I_{\text{dyn}}^{\text{fail}}$ ,  $\text{Ev} \in \text{EvSpec}$ ,

$$\begin{aligned} & \text{FDyn}_i[\text{Ev}] \\ & \iff \left[ \text{Ready}_i^{\text{dyn}}[\text{Ev}] \right] \\ & \quad \wedge \left[ K_i^{\text{dyn}}[\text{Ev}] = \perp \right]. \end{aligned}$$

Further, when  $\text{Ev} \in \text{EvSpec}^{\text{exact}}$ , the failure-witness condition corresponding to each  $i \in \{1, \dots, 8\}$  is obtained by expanding the definition of each condition.

*Proof.* The first claim follows from Definition 6.102, Proposition 6.101, the evaluation-readiness judgment and failed-state equivalence of Definition 6.21, and Proposition 6.23.

Substitute  $i = 1, \dots, 8$ , and expand the respective conditions appearing in Definition 6.100 by Definitions 5.18, 5.22, 5.26, 5.31, 5.34, 5.53, 5.59, and 5.65. This yields the concrete failure-witness condition corresponding to each index.  $\square$

**Definition 6.104** (Failure at the dynamical-description stage). For each  $\text{Ev} \in \text{EvSpec}$ , define failure at the dynamical-description stage by

$$\begin{aligned} & \text{Fail}_{\text{dyn}}[\text{Ev}] \\ & \iff \exists i \in I_{\text{dyn}}^{\text{fail}}, \quad \text{FDyn}_i[\text{Ev}] \\ & \iff \text{FailedTok}^{\text{strict}}[\text{Ev}] \cap \text{FailTok}_{\text{dyn}} \neq \emptyset. \end{aligned}$$

**Definition 6.105** (Sets of failed-condition indices and minimal-failure indices in the dynamical-description series). For each  $\text{Ev} \in \text{EvSpec}$ , define the set of failed-condition indices in the dynamical-description series by

$$\begin{aligned} & \text{FailedCondIdx}_{\text{dyn}}[\text{Ev}] \\ & := \left\{ i \in I_{\text{dyn}}^{\text{fail}} \mid K_i^{\text{dyn}}[\text{Ev}] = \perp \right\}. \end{aligned}$$

Define the set of minimal-failure indices with respect to the dependency order by

$$\begin{aligned} & \text{MinFailIdx}_{\text{dyn}}[\text{Ev}] \\ & := \left\{ i \in \text{FailedCondIdx}_{\text{dyn}}[\text{Ev}] \mid \begin{array}{l} \forall j \in \text{FailedCondIdx}_{\text{dyn}}[\text{Ev}], \\ \text{FailTok}_{\text{dyn},j} \preceq^{\text{fail,dep}} \text{FailTok}_{\text{dyn},i} \implies j = i \end{array} \right\}. \end{aligned}$$

**Proposition 6.106** (Eight-condition characterization of dynamical-description-stage failure under satisfied inputs). For each  $\text{Ev} \in \text{EvSpec}$ , suppose that, for every  $i \in I_{\text{dyn}}^{\text{fail}}$ , both

$$\text{FailInputSat} \left( \text{FailTok}_{\text{dyn},i}; \Gamma_{\text{Ev}}^{\text{dyn}} \right) = \top$$

and, for every

$$\text{FailTok}^{(0)} \in \text{FailPredOf}(\text{FailTok}_{\text{dyn},i}) \setminus \text{FailTok}_{\text{dyn}},$$

$$\text{FailStateOf} \left( \text{FailTok}^{(0)}; \text{Ev}_{\text{FailSeriesOf}(\text{FailTok}^{(0)})}^{\text{fail}} \right) = \text{sat}$$

hold. Then

$$\begin{aligned} & \text{Fail}_{\text{dyn}}[\text{Ev}] \\ & \iff \neg \text{DynCondAll}[\text{Ev}] \\ & \iff \text{FailedCondIdx}_{\text{dyn}}[\text{Ev}] \neq \emptyset \\ & \iff \text{MinFailIdx}_{\text{dyn}}[\text{Ev}] \neq \emptyset. \end{aligned}$$

Further, for each  $i \in I_{\text{dyn}}^{\text{fail}}$ ,

$$\text{FDyn}_i[\text{Ev}] \iff i \in \text{MinFailIdx}_{\text{dyn}}[\text{Ev}]$$

holds.

*Proof.* By Proposition 6.101 and Definition 6.21, for each  $i \in I_{\text{dyn}}^{\text{fail}}$ , the failed state is the conjunction of evaluation readiness and  $K_i^{\text{dyn}}[\text{Ev}] = \perp$ .

Under the assumed satisfaction of the formation inputs, evaluation inputs, and extra-series direct-predecessor states, finite induction along the directed acyclic graph within the dynamical-description series of Definition 6.17 gives

$$\begin{aligned} & \text{FDyn}_i[\text{Ev}] \\ & \iff \left[ \forall j \in I_{\text{dyn}}^{\text{fail}}, \right. \\ & \quad \left. \text{FailTok}_{\text{dyn},j} \prec^{\text{fail,dep}} \text{FailTok}_{\text{dyn},i} \implies K_j^{\text{dyn}}[\text{Ev}] = \top \right] \wedge \left[ K_i^{\text{dyn}}[\text{Ev}] = \perp \right]. \end{aligned}$$

By Definition 6.105, the right-hand side is equivalent to  $i \in \text{MinFailIdx}_{\text{dyn}}[\text{Ev}]$ .

The finiteness of the index set and the existence of a minimal element make nonemptiness of the set of failed-condition indices equivalent to nonemptiness of the set of minimal-failure indices. By Definition 6.100, failure of the conjunction of all dynamical-description-stage conditions is equivalent to nonemptiness of the set of failed-condition indices. Finally, applying Definition 6.104 and the equivalence for each index yields the claim.  $\square$

Table 6: Classification of failure positions in the dynamical-description series

| Index | Failure predicate | Corresponding condition                     | Direct-predecessor indices |
|-------|-------------------|---------------------------------------------|----------------------------|
| 1     | FDyn <sub>1</sub> | $K_{\text{GenTimeOrdAntiSym}}^{\text{dyn}}$ | $\emptyset$                |
| 2     | FDyn <sub>2</sub> | $K_{\text{CanTrjDesc}}^{\text{dyn}}$        | $\emptyset$                |
| 3     | FDyn <sub>3</sub> | $K_{\text{CanTrjFunc}}^{\text{dyn}}$        | $\{2\}$                    |
| 4     | FDyn <sub>4</sub> | $K_{\text{CanObsDesc}}^{\text{dyn}}$        | $\emptyset$                |
| 5     | FDyn <sub>5</sub> | $K_{\text{LocObsChgComm}}^{\text{dyn}}$     | $\emptyset$                |
| 6     | FDyn <sub>6</sub> | $K_{\text{ObsTimeIncTrans}}^{\text{dyn}}$   | $\{1\}$                    |
| 7     | FDyn <sub>7</sub> | $K_{\text{ObsTimeRealOrdEmb}}^{\text{dyn}}$ | $\{6\}$                    |
| 8     | FDyn <sub>8</sub> | $K_{\text{ObsTimeTrjFact}}^{\text{dyn}}$    | $\{3, 6\}$                 |

**Theorem 6.107** (Complete classification of dynamical-description-stage failure and simultaneous minimal failures). For each  $\text{Ev} \in \text{EvSpec}$ ,

$$\begin{aligned}
& \text{Fail}_{\text{dyn}}[\text{Ev}] \\
& \iff \bigvee_{i \in I_{\text{dyn}}^{\text{fail}}} \text{FDyn}_i[\text{Ev}] \\
& \iff \text{FailedTok}^{\text{strict}}[\text{Ev}] \cap \text{FailTok}_{\text{dyn}} \neq \emptyset
\end{aligned}$$

holds.

When the three conditions concerning the formation inputs, evaluation inputs, and extra-series direct-predecessor states in Proposition 6.106 hold, the further equivalence

$$\begin{aligned}
& \text{Fail}_{\text{dyn}}[\text{Ev}] \\
& \iff \text{FailedCondlx}_{\text{dyn}}[\text{Ev}] \neq \emptyset \\
& \iff \text{MinFailldx}_{\text{dyn}}[\text{Ev}] \neq \emptyset
\end{aligned}$$

holds. Moreover, for every  $i \in I_{\text{dyn}}^{\text{fail}}$ ,  $\text{FDyn}_i[\text{Ev}] \iff i \in \text{MinFailldx}_{\text{dyn}}[\text{Ev}]$  holds. Consequently, if

$$|\text{MinFailldx}_{\text{dyn}}[\text{Ev}]| \geq 2,$$

the dynamical-description-condition failure predicates corresponding to multiple distinct minimal failure conditions in the dependency order hold simultaneously.

Further,

$$\text{FailedTok}^{\text{strict}}[\text{Ev}] \cap \text{FailTok}_{\text{dyn}}$$

is an antichain with respect to  $\preceq^{\text{fail,dep}}$ .

*Proof.* Fix any  $\text{Ev} \in \text{EvSpec}$ . Definitions 6.102 and 6.104 give Equation (6.107).

Assume the three conditions of Proposition 6.106. That proposition gives the chain of equivalences among stage failure, nonemptiness of the set of failed-condition indices, and

nonemptiness of the set of minimal-failure indices, and it gives the minimal-failure-index criterion for the dynamical-description-condition failure predicate for each  $i \in I_{\text{dyn}}^{\text{fail}}$ .

Assume Equation (6.107). Then there exist distinct

$$i, j \in \text{MinFailIdx}_{\text{dyn}}[\text{Ev}].$$

The minimal-failure-index criterion in Proposition 6.106 therefore gives

$$\text{FDyn}_i[\text{Ev}] \wedge \text{FDyn}_j[\text{Ev}].$$

The final antichain assertion does not require the additional input-satisfaction assumptions. By Proposition 6.26, the full failed-token set is an antichain. The set in Equation (6.107) is a subset of it and is therefore also an antichain.  $\square$

## 6.7 Failure localization at the law-description stage

**Definition 6.108** (Selected family of law-evaluation inputs). Fix a subset of evaluation specifications and a selected input family

$$\begin{aligned} \text{EvSpec}^{\text{law}} &\subseteq \{\text{Ev} \in \text{EvSpec} \mid \text{DynOperative}[\text{Ev}]\}, \\ &(\rho_{\text{Ev}}^0, \text{LawEvalDat}_{\text{Ev}})_{\text{Ev} \in \text{EvSpec}^{\text{law}}}. \end{aligned}$$

Each component consists of the single-object generated law datum of Definition 5.85 and the base representation used in its construction, with

$$\begin{aligned} \text{LawEvalDat}_{\text{Ev}}.\text{EvalSpec} &= \iota_{\text{single}}(\text{Ev}), \\ \text{LawEvalDat}_{\text{Ev}} &\in \text{LawEvalDatSp}^{\text{gen}}[\rho_{\text{Ev}}^0]. \end{aligned}$$

The selected family is not assumed to be determined automatically by the complete evaluation specification. In formulas with a fixed  $\text{Ev} \in \text{EvSpec}^{\text{law}}$ , abbreviate  $\rho_{\text{Ev}}^0, \text{LawEvalDat}_{\text{Ev}}$  as  $\rho^0, \text{LawEvalDat}$ . Conditions, failure regions and signatures are relative to this selected family. Set formation-input satisfaction for the law series to the truth value of  $\text{Ev} \in \text{EvSpec}^{\text{law}}$ , and evaluation-input satisfaction to  $\top$ .

Failure at the law-description stage is localized by the minimal failed nodes relative to the law-condition dependency directed acyclic graph of Definition 5.99. The simultaneous minimal-failure node set may contain multiple elements.

### 6.7.1 Failure Domain at the Law-Description Stage

**Definition 6.109** (Evaluation of Law Conditions on Complete Evaluation Specifications). For each

$$\begin{aligned} \text{Ev} &= (l, o, \zeta) \in \text{EvSpec}^{\text{law}}, \\ I_{\text{law}}^{\text{fail}} &= \text{LawCondNodeSet} = \{1, \dots, 5\}, \end{aligned}$$

and the five conditions in the law series are defined by

$$K_i^{\text{law}}[\text{Ev}] := \mathbf{K}_i^{\text{law}}[\text{LawEvalDat}] \quad (i \in I_{\text{law}}^{\text{fail}}).$$

Then

$$\begin{aligned} &\text{LawCondAll}[\text{LawEvalDat}] \\ &\iff \bigwedge_{i \in I_{\text{law}}^{\text{fail}}} K_i^{\text{law}}[\text{Ev}] \end{aligned}$$

holds. Extend these five conditions by false outside the selected domain, using Definition 6.10. Set the total law-series conditions to  $K_{\text{law},i}^{\text{fail}} := K_i^{\text{law}}$ .

**Definition 6.110** (Failure at the Law-Description Stage). For a complete evaluation specification  $\text{Ev} = (l, o, \zeta) \in \text{EvSpec}$ , failure at the law-description stage is defined by

$$\begin{aligned} \text{Fail}_{\text{law}}[\text{Ev}] \\ \iff \text{Ev} \in \text{EvSpec}^{\text{law}} \\ \wedge \neg \bigwedge_{i \in I_{\text{law}}^{\text{fail}}} K_i^{\text{law}}[\text{Ev}]. \end{aligned}$$

**Definition 6.111** (Law-Description-Stage Failure Domain). The law-description-stage failure domain in the set  $\text{EvSpec}$  of complete evaluation specifications is defined by

$$\begin{aligned} \text{FailDom}_{\text{law}}(\text{EvSpec}) \\ := \{ \text{Ev} \in \text{EvSpec} \mid \text{Fail}_{\text{law}}[\text{Ev}] \}. \end{aligned}$$

**Proposition 6.112** (Failure at the Law-Description Stage and Law-Description Operability). For each  $\text{Ev} = (l, o, \zeta) \in \text{EvSpec}^{\text{law}}$ ,

$$\begin{aligned} \text{Fail}_{\text{law}}[\text{Ev}] \\ \iff \text{DynOperative}[\text{Ev}] \\ \wedge \neg \text{LawOperative}[\text{LawEvalDat}]. \end{aligned}$$

*Proof.* On the domain where  $\text{DynOperative}[\text{Ev}]$  holds, Definition 5.102 gives

$$\text{LawOperative}[\text{LawEvalDat}] \iff \text{LawCondAll}[\text{LawEvalDat}].$$

Substitution of this equivalence into Definition 6.110 yields the claim.  $\square$

**Definition 6.113** (All-Ancestor Satisfaction Bundle for a Law Condition). For each  $\text{Ev} \in \text{EvSpec}$ ,  $i \in I_{\text{law}}^{\text{fail}}$ , the all-ancestor satisfaction bundle for a law condition is defined by

$$\begin{aligned} \text{LawCondAncAll}_i[\text{Ev}] \\ \iff \begin{cases} \top, & i \in \{1, 2, 4\}, \\ K_1^{\text{law}}[\text{Ev}], & i = 3, \\ K_1^{\text{law}}[\text{Ev}] \wedge K_3^{\text{law}}[\text{Ev}], & i = 5 \end{cases}. \end{aligned}$$

**Definition 6.114** (Minimal Failure Class at the Law-Description Stage). For each  $\text{Ev} \in \text{EvSpec}$ ,  $i \in I_{\text{law}}^{\text{fail}}$ , the  $i$ -th minimal failure class at the law-description stage is defined by

$$\begin{aligned} \text{FL}_i[\text{Ev}] \\ \iff \text{Ev} \in \text{EvSpec}^{\text{law}} \\ \wedge \text{LawCondAncAll}_i[\text{Ev}] \\ \wedge \neg K_i^{\text{law}}[\text{Ev}]. \end{aligned}$$

**Definition 6.115** (Simultaneous Minimal-Failure Node Set at the Law-Description Stage). For each  $\text{Ev} \in \text{EvSpec}$ , the simultaneous minimal-failure node set at the law-description stage is defined by

$$\begin{aligned} \text{MinFailDx}_{\text{law}}[\text{Ev}] \\ := \{ i \in I_{\text{law}}^{\text{fail}} \mid \text{FL}_i[\text{Ev}] \}. \end{aligned}$$

**Proposition 6.116** (Nonemptiness and Antichain Property of the Simultaneous Minimal-Failure Node Set at the Law-Description Stage).

For each  $\text{Ev} \in \text{FailDom}_{\text{law}}(\text{EvSpec})$ ,  $\text{MinFailIdx}_{\text{law}}[\text{Ev}]$  is a nonempty antichain in  $\text{LawCondDAG}$ .

*Proof.* For a complete evaluation specification in the law-description-stage failure domain, Definitions 6.111 and 6.110 give  $\text{DynOperative}[\text{Ev}]$  and  $\neg \text{LawCondAll}[\text{LawEvalDat}]$ . Definition 6.109 therefore implies that

$$\{i \in I_{\text{law}}^{\text{fail}} \mid \neg K_i^{\text{law}}[\text{Ev}]\}$$

is nonempty.

Because the law-condition dependency directed acyclic graph of Definition 5.99 is finite, the set of nodes with false values has a minimal node  $i$  with respect to the reachability order. At every ancestor of node  $i$ , the corresponding law condition holds. Definition 6.113 therefore gives  $\text{LawCondAncAll}_i[\text{Ev}]$ . Definition 6.114 then gives  $\text{FL}_i[\text{Ev}]$ . Hence the simultaneous minimal-failure node set at the law-description stage is nonempty.

For distinct  $i, j \in \text{MinFailIdx}_{\text{law}}[\text{Ev}]$ , suppose that a path runs from  $i$  to  $j$  in the law-condition dependency directed acyclic graph. Definition 6.114 gives  $\neg K_i^{\text{law}}[\text{Ev}]$ . On the other hand, since  $i$  is an ancestor of  $j$ , Definition 6.113 together with  $\text{LawCondAncAll}_j[\text{Ev}]$  gives  $K_i^{\text{law}}[\text{Ev}]$ , a contradiction. If reachability from  $j$  to  $i$  is assumed, then  $\neg K_j^{\text{law}}[\text{Ev}]$  holds, whereas the fact that  $j$  is an ancestor of  $i$  gives  $K_j^{\text{law}}[\text{Ev}]$ , again a contradiction. Thus any two distinct nodes in the set are mutually unreachable, and the simultaneous minimal-failure node set at the law-description stage is an antichain.  $\square$

| Node | Minimal failure class    | Law condition                                                                                           | Strict ancestor nodes |
|------|--------------------------|---------------------------------------------------------------------------------------------------------|-----------------------|
| 1    | $\text{FL}_1[\text{Ev}]$ | Individual law-evaluation time-domain compatibility condition<br>$K_1^{\text{law}}[\text{Ev}]$          | $\emptyset$           |
| 2    | $\text{FL}_2[\text{Ev}]$ | Law-candidate-relation right-uniqueness condition<br>$K_2^{\text{law}}[\text{Ev}]$                      | $\emptyset$           |
| 3    | $\text{FL}_3[\text{Ev}]$ | Generated evaluation-tuple law-candidate-relation membership condition<br>$K_3^{\text{law}}[\text{Ev}]$ | $\{1\}$               |
| 4    | $\text{FL}_4[\text{Ev}]$ | Faithful-transport invariance condition for the law-candidate relation<br>$K_4^{\text{law}}[\text{Ev}]$ | $\emptyset$           |
| 5    | $\text{FL}_5[\text{Ev}]$ | Common valid time-domain condition for laws<br>$K_5^{\text{law}}[\text{Ev}]$                            | $\{1, 3\}$            |

Table 7: Minimal Failure Classes at the Law-Description Stage Relative to the Law-Condition DAG

**Proposition 6.117** (Failure-Domain Cover Relative to the Law-Condition DAG). For



the set  $\text{EvSpec}$  of complete evaluation specifications,

$$\begin{aligned} & \text{FailDom}_{\text{law}}(\text{EvSpec}) \\ &= \bigcup_{i \in I_{\text{law}}^{\text{fail}}} \{\text{Ev} \in \text{EvSpec} \mid \text{FL}_i[\text{Ev}]\}. \end{aligned}$$

Equivalently, for each  $\text{Ev} \in \text{EvSpec}$ ,

$$\begin{aligned} & \text{Fail}_{\text{law}}[\text{Ev}] \\ & \iff \text{MinFailIdx}_{\text{law}}[\text{Ev}] \neq \emptyset \\ & \iff \bigvee_{i \in I_{\text{law}}^{\text{fail}}} \text{FL}_i[\text{Ev}]. \end{aligned}$$

Moreover, if  $|\text{MinFailIdx}_{\text{law}}[\text{Ev}]| \geq 2$ , the minimal failure classes at the law-description stage corresponding to the respective nodes in the set hold simultaneously.

*Proof.* Suppose failure at the law-description stage holds. By Definition 6.111, the complete evaluation specification belongs to the law-description-stage failure domain. Proposition 6.116 then shows that the corresponding simultaneous minimal-failure node set is nonempty.

Conversely, suppose the simultaneous minimal-failure node set at the law-description stage is nonempty. Definition 6.115 gives a minimal failure class at the law-description stage corresponding to some law-failure index. Definition 6.114 gives dynamical-description operability and a false value for the corresponding law-condition evaluation. By Definition 6.109, the simultaneous satisfaction of all law conditions fails; Definition 6.110 then establishes failure at the law-description stage.

Definition 6.115 also shows that nonemptiness of the simultaneous minimal-failure node set is equivalent to the occurrence of a minimal failure class at the law-description stage corresponding to some law-failure index, and hence to a disjunction over the finite index set. The three conditions in the proposition are therefore equivalent.

The equivalence for every complete evaluation specification, together with Definition 6.111, gives the set equality for the failure domain.

Suppose that the simultaneous minimal-failure node set at the law-description stage has at least two elements. By Definition 6.115, the minimal failure class at the law-description stage corresponding to each node in the set holds. Proposition 6.116 shows that these nodes form an antichain in the law-condition dependency directed acyclic graph.  $\square$

**Proposition 6.118** (Stage Separation of Failure at the Dynamical-Description and Law-Description Stages). For each  $\text{Ev} = (\text{l}, o, \zeta) \in \text{EvSpec}$ ,

$$\begin{aligned} & \text{Fail}_{\text{dyn}}[\text{Ev}] \implies \neg \text{DynOperative}[\text{Ev}], \\ & \text{Fail}_{\text{law}}[\text{Ev}] \implies \text{DynOperative}[\text{Ev}] \end{aligned}$$

hold. Consequently,

$$\neg (\text{Fail}_{\text{dyn}}[\text{Ev}] \wedge \text{Fail}_{\text{law}}[\text{Ev}])$$

holds.

*Proof.* By Definitions 6.104 and 6.102, if  $\text{Fail}_{\text{dyn}}[\text{Ev}]$  holds, then for some  $i \in I_{\text{dyn}}^{\text{fail}}$ ,

$$\text{FailStateOf}(\text{FailTok}_{\text{dyn}, i}; \Gamma_{\text{Ev}}^{\text{dyn}}) = \text{failed}$$

holds. Proposition 6.101 gives  $K_i^{\text{dyn}}[\text{Ev}] = \perp$ . Definition 6.100 then gives  $\neg \text{DynCondAll}[\text{Ev}]$ . By Definition 5.70,  $\neg \text{DynOperative}[\text{Ev}]$  holds.

The equivalence in Proposition 6.112 implies that failure at the law-description stage entails dynamical-description operability. Dynamical-description operability is incompatible with the negation of dynamical-description operability obtained from failure at the dynamical-description stage. The two failures therefore cannot hold simultaneously, and the final conclusion of the proposition follows.  $\square$

**Proposition 6.119** (Finite Witness Set for Minimal Failure Classes at the Law-Description Stage). For the set  $\text{EvSpec}$  of complete evaluation specifications, there exists a finite set  $\text{FailWit}_{\text{law}} \subseteq \text{EvSpec}$  satisfying

$$|\text{FailWit}_{\text{law}}| \leq |I_{\text{law}}^{\text{fail}}| = 5$$

such that, for each  $i \in I_{\text{law}}^{\text{fail}}$ ,

$$\begin{aligned} & \{\text{Ev} \in \text{EvSpec} \mid \text{FL}_i[\text{Ev}]\} \neq \emptyset \\ & \iff \exists \text{Ev}_i^{\text{wit}} \in \text{FailWit}_{\text{law}}, \quad \text{FL}_i[\text{Ev}_i^{\text{wit}}] \end{aligned}$$

holds.

*Proof.* For each  $i \in I_{\text{law}}^{\text{fail}}$ , if

$$\{\text{Ev} \in \text{EvSpec} \mid \text{FL}_i[\text{Ev}]\} \neq \emptyset$$

holds, fix one of its elements and denote the selected witness by  $\text{Ev}_i^{\text{wit}}$ . Define the set of selected complete evaluation specifications by

$$\begin{aligned} & \text{FailWit}_{\text{law}} \\ & := \{\text{Ev}_i^{\text{wit}} \mid i \in I_{\text{law}}^{\text{fail}}, \quad \{\text{Ev} \in \text{EvSpec} \mid \text{FL}_i[\text{Ev}]\} \neq \emptyset\}. \end{aligned}$$

This set is a subset of the set of complete evaluation specifications.

Because the law-failure index set has five elements, the finite witness set contains at most five elements.

For each law-failure index, if the domain of the corresponding minimal failure class is nonempty, the selected  $\text{Ev}_i^{\text{wit}} \in \text{FailWit}_{\text{law}}$  satisfies  $\text{FL}_i[\text{Ev}_i^{\text{wit}}]$ .

Conversely, if a selected witness in the finite witness set satisfies the corresponding minimal failure class, then

$$\begin{aligned} & \text{Ev}_i^{\text{wit}} \\ & \in \{\text{Ev} \in \text{EvSpec} \mid \text{FL}_i[\text{Ev}]\}. \end{aligned}$$

Thus, for each minimal failure class, the equivalence stated in the proposition holds between nonemptiness of the class and the existence of a selected witness corresponding to that class.  $\square$

### 6.7.2 Counterexample Refinements of Minimal Failure Classes at the Law-Description Stage

**Definition 6.120** (Minimal-Failure Parent Region at the Law-Description Stage). For the set  $\text{EvSpec}$  of complete evaluation specifications and each  $i \in I_{\text{law}}^{\text{fail}}$ , the minimal-failure parent region at the law-description stage is defined by

$$\begin{aligned} & \text{FailReg}_{\text{law},i}(\text{EvSpec}) \\ & := \{\text{Ev} \in \text{EvSpec} \mid \text{FL}_i[\text{Ev}]\}. \end{aligned}$$

**Definition 6.121** (Law-Failure Counterexample-Type Index Family). For each  $i \in I_{\text{law}}^{\text{fail}}$ , the law-failure counterexample-type index family is defined by

$$\text{SubIdx}_{\text{law},i} := \begin{cases} \{\text{EmptyTimeDom}, \text{OutEvalDom}\}, & i = 1, \\ \{\text{GenEvalArgConflict}\}, & i = 2, \\ \{\text{EmptyCandOutFiber}, \text{AltCandOutput}\}, & i = 3, \\ \{\text{CandRelMemLoss}, \text{CandRelMemGain}\}, & i = 4, \\ \left\{ \begin{array}{l} \text{EmptyCommonValidTimeDom}, \\ \text{InadmissibleCommonValidTimeDom} \end{array} \right\}, & i = 5 \end{cases}.$$

**Definition 6.122** (Individual-Law Application-Domain Counterexample Types). For a complete evaluation specification  $\text{Ev} = (\text{I}, o, \zeta) \in \text{EvSpec}^{\text{law}}$ , each  $a \in \text{LawIdx}$ , and  $t \in \text{EvalTimeDom}_a^{\text{law}}$ , the generated law-evaluation argument is defined by

$$\text{GenEvalArg}_a^{\text{law}}(t) := (t, \text{InVal}_a^{\text{law}}(t)).$$

The individual-law application-domain counterexample types are defined by

$$\begin{aligned} \text{LawEvalDomDefect}_{\text{EmptyTimeDom}}[\text{Ev}] &\iff \exists a \in \text{LawIdx}, \quad \text{EvalTimeDom}_a^{\text{law}} = \emptyset, \\ \text{LawEvalDomDefect}_{\text{OutEvalDom}}[\text{Ev}] &\iff \exists a \in \text{LawIdx}, \quad \exists t \in \text{EvalTimeDom}_a^{\text{law}}, \\ &\quad \text{GenEvalArg}_a^{\text{law}}(t) \notin \text{EvalDom}_a^{\text{law}}. \end{aligned}$$

**Definition 6.123** (Law-Output-Determinacy Counterexample Types). For a complete evaluation specification  $\text{Ev} = (\text{I}, o, \zeta) \in \text{EvSpec}^{\text{law}}$  and each

$$a \in \text{LawIdx}, \quad \text{EvalArg}_{a,0}^{\text{law}} \in \text{LawTimeDom} \times \text{InValSp}_a^{\text{law}},$$

the law-candidate output fiber and its nonuniqueness are defined by

$$\begin{aligned} \text{CandOutFib}_a^{\text{law}}(\text{EvalArg}_{a,0}^{\text{law}}) &:= \{\text{OutVal}_{a,0}^{\text{law}} \in \text{OutValSp}_a^{\text{law}} \mid (\text{EvalArg}_{a,0}^{\text{law}}, \text{OutVal}_{a,0}^{\text{law}}) \in \text{CandRel}_a^{\text{law}}\}, \\ \text{CandOutNonUniq}_a^{\text{law}}(\text{EvalArg}_{a,0}^{\text{law}}) &\iff \exists \text{OutVal}_{a,0}^{\text{law}}, \text{OutVal}_{a,1}^{\text{law}} \in \text{CandOutFib}_a^{\text{law}}(\text{EvalArg}_{a,0}^{\text{law}}), \\ &\quad \text{OutVal}_{a,0}^{\text{law}} \neq \text{OutVal}_{a,1}^{\text{law}}. \end{aligned}$$

The output conflict on generated evaluation arguments is defined by

$$\begin{aligned} \text{LawGenEvalArgConflict}[\text{Ev}] &\iff \exists a \in \text{LawIdx}, \quad \exists t \in \text{EvalTimeDom}_a^{\text{law}}, \\ &\quad \text{GenEvalArg}_a^{\text{law}}(t) \in \text{EvalDom}_a^{\text{law}}, \quad \text{CandOutNonUniq}_a^{\text{law}}(\text{GenEvalArg}_a^{\text{law}}(t)). \end{aligned}$$

The output conflict localized outside generated evaluation arguments is defined by

$$\begin{aligned} \text{LawOffGenArgConflict}[\text{Ev}] &\iff \exists a_0 \in \text{LawIdx}, \quad \exists \text{EvalArg}_{a_0,0}^{\text{law}} \in \text{EvalDom}_{a_0}^{\text{law}}, \\ &\quad \text{CandOutNonUniq}_{a_0}^{\text{law}}(\text{EvalArg}_{a_0,0}^{\text{law}}) \\ &\quad \wedge \neg \text{LawGenEvalArgConflict}[\text{Ev}]. \end{aligned}$$

**Definition 6.124** (Generated Law-Evaluation Relation-Membership Counterexample Types). For a complete evaluation specification  $\text{Ev} = (\text{I}, o, \zeta) \in \text{EvSpec}^{\text{law}}$  and each

$$\begin{aligned}\sigma &\in \text{SubIdx}_{\text{law},3}, \\ a &\in \text{LawIdx}, \\ t &\in \text{EvalTimeDom}_a^{\text{law}},\end{aligned}$$

the candidate-output-fiber type is defined by

$$\begin{aligned}\text{CandOutFibCase}_{a,\sigma}^{\text{law}}(t) \\ \iff \begin{cases} \text{CandOutFib}_a^{\text{law}}(\text{GenEvalArg}_a^{\text{law}}(t)) = \emptyset, & \sigma = \text{EmptyCandOutFiber}, \\ \text{CandOutFib}_a^{\text{law}}(\text{GenEvalArg}_a^{\text{law}}(t)) \neq \emptyset, & \sigma = \text{AltCandOutput} \end{cases}.\end{aligned}$$

The occurrence of a membership counterexample of candidate-output-fiber type  $\sigma$  is defined by

$$\begin{aligned}\text{LawCandOutCaseOcc}_\sigma[\text{Ev}] \\ \iff \exists a \in \text{LawIdx}, \quad \exists t \in \text{EvalTimeDom}_a^{\text{law}}, \\ \text{EvalTuple}_a^{\text{law}}(t) \notin \text{CandRel}_a^{\text{law}}, \quad \text{CandOutFibCase}_{a,\sigma}^{\text{law}}(t).\end{aligned}$$

**Definition 6.125** (Law-Candidate-Relation Membership-Change Counterexample Types). For a complete evaluation specification  $\text{Ev} = (\text{I}, o, \zeta) \in \text{EvSpec}^{\text{law}}$  and each

$$\begin{aligned}\sigma &\in \text{SubIdx}_{\text{law},4}, \\ a &\in \text{LawIdx}, \\ \text{EvalTupleArg}_{a,0}^{\text{law}}, \text{EvalTupleArg}_{a,1}^{\text{law}} &\in \text{EvalTupleSp}_a^{\text{law}},\end{aligned}$$

the directional predicate for a candidate-relation membership change is defined by

$$\begin{aligned}\text{CandRelMemChg}_{a,\sigma}^{\text{law}}(\text{EvalTupleArg}_{a,0}^{\text{law}}, \text{EvalTupleArg}_{a,1}^{\text{law}}) \\ \iff \left( \mathbf{1}_{\text{CandRel}_a^{\text{law}}}(\text{EvalTupleArg}_{a,0}^{\text{law}}), \mathbf{1}_{\text{CandRel}_a^{\text{law}}}(\text{EvalTupleArg}_{a,1}^{\text{law}}) \right) \\ = \begin{cases} (1, 0), & \sigma = \text{CandRelMemLoss}, \\ (0, 1), & \sigma = \text{CandRelMemGain} \end{cases}.\end{aligned}$$

The occurrence of a counterexample in membership-change direction  $\sigma$  is defined by

$$\begin{aligned}\text{LawCandRelMemChgOcc}_\sigma[\text{Ev}] \\ \iff \exists (\text{EvalTupleBij}_{a,0}^{\text{law}})_{a \in \text{LawIdx}} \in \text{FaithTransSet}^{\text{law}}, \\ \exists a_0 \in \text{LawIdx}, \quad \exists \text{EvalTupleArg}_{a_0,0}^{\text{law}} \in \text{EvalTupleSp}_{a_0}^{\text{law}}, \\ \text{CandRelMemChg}_{a_0,\sigma}^{\text{law}}(\text{EvalTupleArg}_{a_0,0}^{\text{law}}, \text{EvalTupleBij}_{a_0,0}^{\text{law}}(\text{EvalTupleArg}_{a_0,0}^{\text{law}})).\end{aligned}$$

**Definition 6.126** (Common-Valid-Time-Domain Counterexample Types for Laws). For a complete evaluation specification  $\text{Ev} = (\text{I}, o, \zeta) \in \text{EvSpec}^{\text{law}}$  and each  $\sigma \in \text{SubIdx}_{\text{law},5}$ , the

common-valid-time-domain counterexample types are defined by

$$\begin{aligned} & \text{CommonValidTimeDomFailCase}_\sigma[\text{Ev}] \\ & \iff \left[ \begin{array}{l} \sigma = \text{EmptyCommonValidTimeDom} \\ \wedge \text{CommonValidTimeDom}^{\text{law}} = \emptyset \end{array} \right] \\ & \vee \left[ \begin{array}{l} \sigma = \text{InadmissibleCommonValidTimeDom} \\ \wedge \text{CommonValidTimeDom}^{\text{law}} \neq \emptyset \\ \wedge \text{CommonValidTimeDom}^{\text{law}} \notin \text{AdmCommonTimeDomSet}^{\text{law}} \end{array} \right]. \end{aligned}$$

In the following law-failure counterexample-type occurrence family, law-failure counterexample pure-type family, and unrefined remainder-type family, fix

$$\begin{aligned} & \text{Ev} \in \text{EvSpec}^{\text{law}}, \\ & i \in I_{\text{law}}^{\text{fail}}, \\ & \sigma \in \text{SubIdx}_{\text{law},i} \end{aligned}$$

unless otherwise stated.

**Definition 6.127** (Law-Failure Counterexample-Type Occurrence Family). The law-failure counterexample-type occurrence family is defined by

$$\text{LawCaseOcc}_{i,\sigma}[\text{Ev}] \iff \begin{cases} \text{LawEvalDomDefect}_\sigma[\text{Ev}], & i = 1, \\ \text{LawGenEvalArgConflict}[\text{Ev}], & i = 2, \\ \text{LawCandOutCaseOcc}_\sigma[\text{Ev}], & i = 3, \\ \text{LawCandRelMemChgOcc}_\sigma[\text{Ev}], & i = 4, \\ \text{CommonValidTimeDomFailCase}_\sigma[\text{Ev}], & i = 5 \end{cases}.$$

**Definition 6.128** (Law-Failure Counterexample Pure-Type Family and Unrefined Remainder-Type Family). For each

$$\text{Ev} \in \text{EvSpec}^{\text{law}}, \quad i \in I_{\text{law}}^{\text{fail}}, \quad \sigma \in \text{SubIdx}_{\text{law},i},$$

the law-failure counterexample pure-type family is defined by

$$\begin{aligned} & \text{LawPureCase}_{i,\sigma}[\text{Ev}] \\ & \iff \text{LawCaseOcc}_{i,\sigma}[\text{Ev}] \wedge \bigwedge_{\lambda \in \text{SubIdx}_{\text{law},i} \setminus \{\sigma\}} \neg \text{LawCaseOcc}_{i,\lambda}[\text{Ev}]. \end{aligned}$$

For each  $\text{Ev} \in \text{EvSpec}^{\text{law}}$ ,  $i \in I_{\text{law}}^{\text{fail}}$ , the unrefined remainder-type family is defined by

$$\text{LawUnrefCase}_i[\text{Ev}] \iff \begin{cases} \bigwedge_{\sigma \in \text{SubIdx}_{\text{law},1}} \text{LawCaseOcc}_{1,\sigma}[\text{Ev}], & i = 1, \\ \text{LawOffGenArgConflict}[\text{Ev}], & i = 2, \\ \bigwedge_{\sigma \in \text{SubIdx}_{\text{law},i}} \text{LawCaseOcc}_{i,\sigma}[\text{Ev}], & i \in \{3, 4\}, \\ \perp, & i = 5 \end{cases}.$$

**Definition 6.129** (Law-Failure Counterexample Refinement Forest). For each

$$i \in I_{\text{law}}^{\text{fail}}, \quad \sigma \in \text{SubIdx}_{\text{law},i},$$

the law-failure counterexample-type subregion is defined by

$$\begin{aligned} & \text{FailReg}_{\text{law},i;\sigma}(\text{EvSpec}) \\ & := \{ \text{Ev} \in \text{FailReg}_{\text{law},i}(\text{EvSpec}) \mid \text{LawPureCase}_{i,\sigma}[\text{Ev}] \}. \end{aligned}$$

For  $k \in \{0, 1\}$ , define the child-index set along each path by

$$\text{ChildIdx}_{\text{law}}(i; \sigma_1, \dots, \sigma_k) := \begin{cases} \text{SubIdx}_{\text{law},i}, & k = 0, \\ \emptyset, & k = 1 \end{cases}.$$

When  $k = 0$ , omit the semicolon and everything that follows it. Define the node-region map by

$$\begin{aligned} \text{NodeReg}_{\text{law}}((i); \text{EvSpec}) &:= \text{FailReg}_{\text{law},i}(\text{EvSpec}), \\ \text{NodeReg}_{\text{law}}((i; \sigma); \text{EvSpec}) &:= \text{FailReg}_{\text{law},i;\sigma}(\text{EvSpec}), \\ \text{NodeReg}_{\text{law}}((i; \sigma); \text{EvSpec}) &\subseteq \text{NodeReg}_{\text{law}}((i); \text{EvSpec}). \end{aligned}$$

Define the law-failure counterexample refinement forest by

$$\text{RefForest}_{\text{law}}(\text{EvSpec}) := \text{RefForest} [I_{\text{law}}^{\text{fail}}; \text{ChildIdx}_{\text{law}}; \text{NodeReg}_{\text{law}}(-; \text{EvSpec})].$$

For each  $i \in I_{\text{law}}^{\text{fail}}$  and  $\varsigma \in \{\text{cov}, \text{unref}\}$ , the covered or unrefined region at the root node is defined by

$$\text{FailReg}_{\text{law},i}^{\varsigma}(\text{EvSpec}) := \text{NodeReg}_{\text{law}}^{\varsigma}((i); \text{EvSpec}).$$

The two child regions at node 1 are called the individual-law evaluation-time-domain emptiness subregion and the evaluation-argument outside-application-domain subregion; the child region at node 2, the output-conflict-on-generated-evaluation-arguments subregion; the two child regions at node 3, the candidate-output-fiber emptiness subregion and the alternative-candidate-output existence subregion; the two child regions at node 4, the pure-type candidate-relation-membership-loss subregion and the pure-type candidate-relation-membership-gain subregion; and the two child regions at node 5, the common-valid-time-domain emptiness subregion and the common-valid-time-domain inadmissibility subregion.

**Proposition 6.130** (Node Decomposition and Remainder Representation of the Law-Failure Counterexample Refinement Forest). For each  $i \in I_{\text{law}}^{\text{fail}}$ ,

$$\begin{aligned} \text{FailReg}_{\text{law},i}^{\text{cov}}(\text{EvSpec}) &= \bigsqcup_{\sigma \in \text{SubIdx}_{\text{law},i}} \text{FailReg}_{\text{law},i;\sigma}(\text{EvSpec}), \\ \text{FailReg}_{\text{law},i}(\text{EvSpec}) &= \text{FailReg}_{\text{law},i}^{\text{cov}}(\text{EvSpec}) \sqcup \text{FailReg}_{\text{law},i}^{\text{unref}}(\text{EvSpec}) \end{aligned}$$

hold. For each  $\text{Ev} \in \text{FailReg}_{\text{law},i}(\text{EvSpec})$ ,

$$\text{Ev} \in \text{FailReg}_{\text{law},i}^{\text{unref}}(\text{EvSpec}) \iff \text{LawUnrefCase}_i[\text{Ev}]$$

holds. In particular,

$$\begin{aligned} \text{FailReg}_{\text{law},5}^{\text{unref}}(\text{EvSpec}) &= \emptyset, \\ \text{FailReg}_{\text{law},5}^{\text{cov}}(\text{EvSpec}) &= \text{FailReg}_{\text{law},5}(\text{EvSpec}). \end{aligned}$$

Furthermore, if two child-region families for the same parent region satisfy the inclusion condition and the agreement condition on common indices in Lemma 6.33, then the covered region after refinement contains the covered region before refinement, and the unrefined region after refinement is contained in the unrefined region before refinement.

*Proof.* Each child region satisfies  $\text{LawPureCase}_{i,\sigma}$ . Because two distinct types at the same node cannot both be pure types, the family of child regions is mutually exclusive. Definition 6.129 and Theorem 6.35 give the disjoint-union representation of the covered region and the decomposition of the parent region.

The unrefined remainder representation is established node by node. At node 1, at least one of the two individual-law application-domain counterexample types occurs on the parent region. Belonging to neither pure-type region is equivalent to the occurrence of both types.

At node 2, nonuniqueness of a candidate-output fiber occurs on the parent region. Being outside the output-conflict-on-generated-evaluation-arguments subregion is equivalent to the existence of a nonunique candidate-output fiber and to the candidate-output fiber containing at most one element at every generated evaluation argument that belongs to the evaluation domain. This is  $\text{LawOffGenArgConflict}[\text{Ev}]$ .

At node 3, a generated law-evaluation relation-membership counterexample occurs on the parent region, and its candidate-output fiber is either empty or nonempty. Belonging to neither pure-type region is equivalent to the occurrence of counterexamples belonging to both fiber types.

At node 4, a counterexample to the faithful-representation-transformation invariance of candidate-relation membership occurs on the parent region. The two distinct membership-indicator values are either  $(1, 0)$  or  $(0, 1)$ . Belonging to neither pure-type region is equivalent to the occurrence of both a membership-loss counterexample and a membership-gain counterexample.

At node 5, exactly one of the following holds: the common valid time domain is empty, or it is nonempty and inadmissible. Every element of the parent region therefore belongs to one pure-type child region, and the unrefined remainder region is empty.

Substituting these results into Definition 6.128 gives the unrefined remainder representation and the two identities for node 5. The final two inclusions follow by applying Lemma 6.33 to the root nodes of the law-failure counterexample refinement forest.  $\square$

**Definition 6.131** (Law-Series Failure Signature). For a complete evaluation specification  $\text{Ev} \in \text{EvSpec}$ , the law-series failure signature is defined by

$$\begin{aligned} \Sigma_{\text{law}}^{\text{fail}}[\text{Ev}] \\ &:= \left( \mathbf{1}_{\{\text{FailTok}_{\text{law},i} \in \text{FailedTok}^{\text{strict}}[\text{Ev}]\}} \right)_{i \in I_{\text{law}}^{\text{fail}}} \\ &\in \{0, 1\}^5. \end{aligned}$$

**Proposition 6.132** (Correspondence between Local Law-Failure Classes and Failure Tokens).

Let  $\text{Ev} \in \text{EvSpec}$ ,  $i \in I_{\text{law}}^{\text{fail}}$ . Suppose that  $\text{DynOperative}[\text{Ev}]$ ,  $\text{LawCondAncAll}_i[\text{Ev}]$  and

$$\text{FailReady}(\text{FailTok}_{\text{law},i}; \text{Ev}_{\text{law}}^{\text{fail}}) = \top$$

hold. Then

$$\begin{aligned} \text{FL}_i[\text{Ev}] \\ \iff \text{FailTok}_{\text{law},i} \in \text{FailedTok}^{\text{strict}}[\text{Ev}]. \end{aligned}$$

If this assumption holds for every  $i \in I_{\text{law}}^{\text{fail}}$ , then

$$\begin{aligned} \Sigma_{\text{law}}^{\text{fail}}[\text{Ev}] \\ = \left( \mathbf{1}_{\{\text{FL}_i[\text{Ev}]\}} \right)_{i \in I_{\text{law}}^{\text{fail}}} . \end{aligned}$$

*Proof.* By Proposition 6.23, membership of a failure token in the strict failed-token set is equivalent to its failure state being **failed**. By the evaluation-readiness assumption and Definition 6.21, the latter is equivalent to the failure evaluation taking the value  $\perp$ . By Definition 6.10, this is equivalent to the corresponding law-condition evaluation taking the value  $\perp$ . By Definition 6.114 and the assumptions of dynamical-description operability and the full-ancestor satisfaction bundle, it is further equivalent to the local failure class  $\text{FL}_i[\text{Ev}]$  holding. The equivalence stated in the proposition therefore holds. Applying the equivalence obtained for all indices to each component in Definition 6.131 gives the displayed signature.  $\square$

**Proposition 6.133** (Type of the Law-Failure Localization Family). For each  $i \in I_{\text{law}}^{\text{fail}}$ ,  $\text{FailReg}_{\text{law},i}(\text{EvSpec}) \subseteq \text{EvSpec}$ , for each  $\lambda \in \text{SubIdx}_{\text{law},i}$ ,

$$\begin{aligned} \text{FailReg}_{\text{law},i;\lambda}(\text{EvSpec}) \\ \subseteq \text{FailReg}_{\text{law},i}(\text{EvSpec}) , \end{aligned}$$

and

$$\begin{aligned} \text{FailReg}_{\text{law},i}^{\text{unref}}(\text{EvSpec}) \\ \subseteq \text{FailReg}_{\text{law},i}(\text{EvSpec}) \end{aligned}$$

hold.

The law-series failure signature satisfies

$$\Sigma_{\text{law}}^{\text{fail}} : \text{EvSpec} \longrightarrow \{0, 1\}^5 .$$

*Proof.* The parent-region inclusion follows from Definition 6.120. Each refinement subregion is defined as the set obtained by adding a condition to membership in the corresponding parent region, and the unrefined remainder region is the set obtained by removing the refined covered region from the parent region. The three inclusions therefore hold.

The type of the law-series failure signature follows from Definition 6.131.  $\square$

## 6.8 Localization of Differential-Stage Failure

**Definition 6.134** (Selected family of differential-evaluation inputs). Use the same law-input family as in Definition 6.108. On a subset  $\text{EvSpec}^{\text{diff}} \subseteq \text{EvSpec}^{\text{law}}$ , fix a typed differential-data family

$$\text{DiffEvalDat}_{\text{Ev}} := \left( \rho_{\text{Ev}}^0, \text{LawEvalDat}_{\text{Ev}}, \text{EvalStrFam}_{\text{Ev}}^{\text{diff}} \right) .$$

The third component is a structure family from Definition 5.131 for the same transported law family determined by the first two components. It gives  $\text{EvalInput}_{\text{Ev}}^{\text{diff}}$  from Definition 5.136. Write its condition and ancestor-bundle components as  $\text{DiffCond}_{i,\text{Ev}}^{\text{univ}}$  and  $\text{DiffCondAncAll}_{i,\text{Ev}}$ . In formulas with a fixed  $\text{Ev} \in \text{EvSpec}^{\text{diff}}$ , omit the index  $\text{Ev}$  from these selected inputs and components. Set formation-input satisfaction for the differential series to the truth value of  $\text{Ev} \in \text{EvSpec}^{\text{diff}}$ , and evaluation-input satisfaction to  $\top$ .



Localize failure of the universal differential-condition family by minimal failed nodes relative to the differential-stage condition DAG of Definition 5.130. At each minimal failed node, all ancestor conditions hold and the universal differential condition at that node is false. Simultaneous minimal failures at mutually unreachable nodes are allowed.

### 6.8.1 Differential-Stage Failure Domain

**Definition 6.135** (Evaluation of differential conditions on a complete evaluation specification). Take a complete evaluation specification  $\text{Ev} \in \text{EvSpec}$  from Definition 5.2. For each  $i \in I_{\text{diff}}^{\text{cond}}$ , define the evaluations of the universal differential condition and the all-ancestors-satisfied bundle from Definition 5.136 by

$$\begin{aligned} \text{DiffCond}_i^{\text{univ}}[\text{Ev}] &\iff \exists e \in \text{EvSpec}^{\text{diff}}, \\ &\quad \text{Ev} = e \wedge \text{DiffCond}_{i,e}^{\text{univ}}, \\ \text{DiffCondAncAll}_i[\text{Ev}] &\iff \exists e \in \text{EvSpec}^{\text{diff}}, \\ &\quad \text{Ev} = e \wedge \text{DiffCondAncAll}_{i,e}. \end{aligned}$$

**Definition 6.136** (Assignment of universal differential conditions to differential-series failure conditions). Definitions 6.10 and 5.130 give  $I_{\text{diff}}^{\text{fail}} = I_{\text{diff}}^{\text{cond}}$ . For each  $i$  in this common index set and each  $\text{Ev} \in \text{EvSpec}$ , define the differential-series failure condition by

$$K_{\text{diff},i}^{\text{fail}}[\text{Ev}] := K_i^{\text{diff}}[\text{Ev}] := \text{DiffCond}_i^{\text{univ}}[\text{Ev}].$$

**Definition 6.137** (Differential-stage failure). On the selected domain,  $\text{DiffCondAncAll}_1[\text{Ev}]$  is the operativity of the selected law datum; it is false outside that domain.

For a complete evaluation specification  $\text{Ev} = (\text{l}, o, \zeta) \in \text{EvSpec}$ , define differential-stage failure by

$$\begin{aligned} \text{Fail}_{\text{diff}}[\text{Ev}] \\ &\iff \text{DiffCondAncAll}_1[\text{Ev}] \\ &\quad \wedge \neg \bigwedge_{i \in I_{\text{diff}}^{\text{cond}}} K_i^{\text{diff}}[\text{Ev}]. \end{aligned}$$

**Definition 6.138** (Differential-stage failure domain). Define the differential-stage failure domain in the complete-evaluation-specification set  $\text{EvSpec}$  by

$$\begin{aligned} \text{FailDom}_{\text{diff}}(\text{EvSpec}) \\ &:= \{\text{Ev} \in \text{EvSpec} \mid \text{Fail}_{\text{diff}}[\text{Ev}]\}. \end{aligned}$$

**Proposition 6.139** (Differential-stage failure and differential-stage operability). For each  $\text{Ev} = (\text{l}, o, \zeta) \in \text{EvSpec}^{\text{diff}}$ ,

$$\begin{aligned} \text{Fail}_{\text{diff}}[\text{Ev}] \\ &\iff \text{LawOperative}[\text{LawEvalDat}] \wedge \neg \text{DiffOperative}[\text{DiffEvalDat}]. \end{aligned}$$

*Proof.* By Definition 5.132, differential-stage operability is equivalent to the conjunction of law-description operability and the full differential-condition bundle. Thus, under law-description operability, the negation of differential-stage operability is equivalent to the negation of the full differential-condition bundle. By Definition 6.137, this equivalence gives the equivalence stated in the proposition.  $\square$

### 6.8.2 Differential-Stage Minimal Failure Classes

**Definition 6.140** (Differential-stage minimal failure classes). For each  $\text{Ev} \in \text{EvSpec}$ ,  $i \in I_{\text{diff}}^{\text{cond}}$ , define the  $i$ -th minimal failure class at the differential stage by

$$\begin{aligned} \text{FDiff}_i[\text{Ev}] \\ \iff \text{DiffCondAncAll}_i[\text{Ev}] \wedge \neg K_i^{\text{diff}}[\text{Ev}]. \end{aligned}$$

**Proposition 6.141** (Counterexamples for differential-stage minimal failure classes). For a complete evaluation specification  $\text{Ev} = (\text{I}, o, \zeta) \in \text{EvSpec}^{\text{diff}}$  and  $i \in I_{\text{diff}}^{\text{cond}}$ , the following conditions are equivalent:

$$\begin{aligned} \text{FDiff}_i[\text{Ev}] \\ \iff \text{DiffCondAncAll}_i[\text{Ev}] \\ \wedge \exists \mathbf{T} \left[ \begin{array}{l} \mathbf{T} = (T_r)_{0 \leq r < \text{arity}_{\text{diff}}(i)} \in (\text{OrdEmbSet}_o^{\text{time}}(\mathbb{R}))^{\text{arity}_{\text{diff}}(i)}, \\ \neg \text{DiffPred}_i \left[ \left( \text{EvalStr}^{\text{diff}}(T_r) \right)_{0 \leq r < \text{arity}_{\text{diff}}(i)} \right] \end{array} \right]. \end{aligned}$$

*Proof.* Negate the family of universal differential conditions in Definition 5.130 and transform the universal quantification into existential quantification of the same arity. Conjoining the result with the all-ancestors-satisfied bundle in Definition 6.140 gives the stated equivalence.  $\square$

Table 8: Differential-stage minimal failure classes

| Failure class      | Satisfied ancestors                                                                                                            | Counterexample                                                                                                           | Structure that fails                                              |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| FDiff <sub>1</sub> | Law-description operability                                                                                                    | A differential-evaluation structure that fails the differential-atlas regularity condition                               | The system of differential chart changes for a common jet order   |
| FDiff <sub>2</sub> | Law-description operability and the differential-atlas regularity condition                                                    | A differential-evaluation structure that fails the generated-trajectory-jet existence condition                          | The generated trajectory jet                                      |
| FDiff <sub>3</sub> | Law-description operability, the differential-atlas regularity condition, and the generated-trajectory-jet existence condition | A differential-evaluation structure that fails the generated-trajectory-jet local-relation-membership condition          | Establishment of the local differential law                       |
| FDiff <sub>4</sub> | Law-description operability and the differential-atlas regularity condition                                                    | A differential-evaluation structure that fails the value-jet-relation covariance condition                               | Identity of the differential relation across value charts         |
| FDiff <sub>5</sub> | Law-description operability and the differential-atlas regularity condition                                                    | A pair of differential-evaluation structures that fails the time-representation-change jet-relation covariance condition | Identity of the differential relation across time representations |

### 6.8.3 Differential-Stage Failure Parent Regions and Internal Subregions

**Definition 6.142** (Differential-stage failure parent regions). Take a set  $\text{EvSpec}$  of complete evaluation specifications and  $i \in I_{\text{diff}}^{\text{cond}}$ . Define the  $i$ -th differential-stage failure parent region by

$$\begin{aligned} \text{FailReg}_{\text{diff},i}(\text{EvSpec}) \\ := \{\text{Ev} \in \text{FailDom}_{\text{diff}}(\text{EvSpec}) \mid \text{FDiff}_i[\text{Ev}]\}. \end{aligned}$$

**Definition 6.143** (Internal subregions of differential-stage failure parent regions). Take a set  $\text{EvSpec}$  of complete evaluation specifications and  $i \in I_{\text{diff}}^{\text{cond}}$ . Suppose that an index set  $\text{SubIdx}_{\text{diff},i}$  and a family of subregions

$$(\text{FailReg}_{\text{diff},i;\alpha}(\text{EvSpec}))_{\alpha \in \text{SubIdx}_{\text{diff},i}}$$

satisfy

$$\text{FailReg}_{\text{diff},i;\alpha}(\text{EvSpec}) \subseteq \text{FailReg}_{\text{diff},i}(\text{EvSpec}).$$

Define the covered region and the unrefined region within the  $i$ -th parent region by

$$\begin{aligned} & (\text{FailReg}_{\text{diff},i}^{\text{cov}}(\text{EvSpec}), \text{FailReg}_{\text{diff},i}^{\text{unref}}(\text{EvSpec})) \\ & := \text{RefSplit} \left( \text{FailReg}_{\text{diff},i}(\text{EvSpec}); \left( \text{FailReg}_{\text{diff},i;\alpha}(\text{EvSpec}) \right)_{\alpha \in \text{Subld}_{\text{diff},i}} \right). \end{aligned}$$

Then

$$\begin{aligned} & \text{FailReg}_{\text{diff},i}(\text{EvSpec}) \\ & = \text{FailReg}_{\text{diff},i}^{\text{cov}}(\text{EvSpec}) \dot{\cup} \text{FailReg}_{\text{diff},i}^{\text{unref}}(\text{EvSpec}). \end{aligned}$$

**Definition 6.144** (Simultaneous minimal-failure node set). Define the family of antichains of a finite DAG  $(I, E)$  by

$$\begin{aligned} & \text{Antichain}(I, E) \\ & := \{A \subseteq I \mid \forall i, j \in A, \quad i \neq j \implies (i, j), (j, i) \notin E^+\}. \end{aligned}$$

For each  $\text{Ev} \in \text{EvSpec}$ , define the simultaneous minimal-failure node set by

$$\begin{aligned} & \text{MinFailld}_{\text{diff}}[\text{Ev}] \\ & := \{i \in I_{\text{diff}}^{\text{cond}} \mid \text{FDiff}_i[\text{Ev}]\}. \end{aligned}$$

#### 6.8.4 Failure-Domain Cover Relative to the Differential-Condition DAG

**Theorem 6.145** (Failure-domain cover relative to the differential-condition DAG). Take a set  $\text{EvSpec}$  of complete evaluation specifications. Relative to the five universal differential conditions and the differential-condition DAG of Definition 5.130,

$$\begin{aligned} & \text{FailDom}_{\text{diff}}(\text{EvSpec}) \\ & = \bigcup_{i \in I_{\text{diff}}^{\text{cond}}} \text{FailReg}_{\text{diff},i}(\text{EvSpec}). \end{aligned}$$

Equivalently, for each  $\text{Ev} \in \text{EvSpec}$ ,

$$\begin{aligned} & \text{Fail}_{\text{diff}}[\text{Ev}] \\ & \iff \bigvee_{i \in I_{\text{diff}}^{\text{cond}}} \text{FDiff}_i[\text{Ev}]. \end{aligned}$$

Members of this cover may intersect at evaluation specifications containing simultaneous minimal-failure nodes.

*Proof.* First take any evaluation specification  $\text{Ev}$  in the differential-stage failure domain. By Definition 6.137, law-description operability holds and the full differential-condition bundle does not hold. By Definition 5.132, the set of nodes at which the universal differential condition takes the value false is nonempty.

Because the differential-condition DAG is finite, this set has a minimal element  $i$  with respect to the reachability order. For  $i$ , every ancestor node  $j$  satisfies  $K_j^{\text{diff}}[\text{Ev}]$ . Indeed, if an ancestor node taking the value false existed, that node would be a false node preceding  $i$ , contrary to the minimality of  $i$ .

Proposition 5.137 gives  $\text{DiffCondAncAll}_i[\text{Ev}]$ . Because  $i$  is a false node,  $\neg K_i^{\text{diff}}[\text{Ev}]$  also holds. Definition 6.140 then gives  $\text{FDiff}_i[\text{Ev}]$ . Therefore,  $\text{Ev}$  belongs to the union of the differential-stage failure parent regions.

Conversely, for some  $i \in I_{\text{diff}}^{\text{cond}}$ , suppose that  $\text{FDiff}_i[\text{Ev}]$  holds. Definition 6.140 gives the all-ancestors-satisfied bundle and failure of the  $i$ -th universal differential condition. The all-ancestors-satisfied bundle includes law-description operability, and failure of the  $i$ -th universal differential condition entails failure of the full differential-condition bundle. By Definition 6.137,  $\text{Fail}_{\text{diff}}[\text{Ev}]$  holds. Therefore,  $\text{Ev}$  belongs to the differential-stage failure domain.

These two implications establish the set equality and the predicate equivalence for each evaluation specification.  $\square$

**Proposition 6.146** (Antichain property of the simultaneous minimal-failure node set). For each  $\text{Ev} \in \text{FailDom}_{\text{diff}}(\text{EvSpec})$ ,

$$\begin{aligned} & \text{MinFailDx}_{\text{diff}}[\text{Ev}] \\ & \in \text{Antichain}(I_{\text{diff}}^{\text{cond}}, E_{\text{diff}}^{\text{cond}}) \setminus \{\emptyset\}. \end{aligned}$$

*Proof.* Let  $i, j \in \text{MinFailDx}_{\text{diff}}[\text{Ev}]$ , and suppose that reachability holds from  $i$  to  $j$ . Definition 6.140 gives  $\neg K_i^{\text{diff}}[\text{Ev}]$ . On the other hand, because  $i$  is a strict ancestor of  $j$ , the satisfaction of  $\text{DiffCondAncAll}_j[\text{Ev}]$  gives  $K_j^{\text{diff}}[\text{Ev}]$ , a contradiction. Next suppose that reachability holds from  $j$  to  $i$ . Definition 6.140 gives  $\neg K_j^{\text{diff}}[\text{Ev}]$ . On the other hand, because  $j$  is a strict ancestor of  $i$ , the satisfaction of  $\text{DiffCondAncAll}_i[\text{Ev}]$  gives  $K_i^{\text{diff}}[\text{Ev}]$ , again a contradiction. Thus the simultaneous minimal-failure node set is an antichain.

Nonemptiness follows from the existence of a minimal element, with respect to the reachability order, in every nonempty set of false nodes on a finite DAG, together with Theorem 6.145.  $\square$

*Remark 6.147* (Dependence of the differential-stage failure-domain cover on the condition family). The cover in Theorem 6.145 covers the failure domain relative to the five universal differential conditions of Definition 5.130. Each parent region decomposes into the covered region and the unrefined region of Definition 6.143.

### 6.8.5 Finite Witnesses for Differential-Stage Failure Parent Regions

**Proposition 6.148** (Relative-nonemptiness criterion for differential-stage failure parent regions). For a set  $\text{EvSpec}$  of complete evaluation specifications, the following are equivalent.

1.

$$\prod_{i \in I_{\text{diff}}^{\text{cond}}} \text{FailReg}_{\text{diff},i}(\text{EvSpec}) \neq \emptyset.$$

2.

$$\forall i \in I_{\text{diff}}^{\text{cond}}, \quad \text{FailReg}_{\text{diff},i}(\text{EvSpec}) \neq \emptyset.$$

3. There exists a family of evaluation specifications

$$(\text{Ev}_i)_{i \in I_{\text{diff}}^{\text{cond}}} \in \prod_{i \in I_{\text{diff}}^{\text{cond}}} \text{EvSpec}$$

such that, for each  $i \in I_{\text{diff}}^{\text{cond}}$ ,  $\text{FDiff}_i[\text{Ev}_i]$  holds.

*Proof.* First suppose the first condition holds. Taking an element of the product shows that each coordinate belongs to the corresponding factor. Hence every parent region is nonempty.

Next suppose the second condition holds. For each index in the finite index set, take an evaluation specification from the corresponding nonempty parent region. By Definition 6.142, each evaluation specification belongs to the set of complete evaluation specifications and satisfies the corresponding differential-stage minimal failure condition.

Finally suppose the third condition holds. By Definition 6.142, for each  $i \in I_{\text{diff}}^{\text{cond}}$ ,  $\text{Ev}_i$  belongs to the corresponding parent region. Therefore, the family of evaluation specifications  $(\text{Ev}_i)_{i \in I_{\text{diff}}^{\text{cond}}}$  belongs to the product of the family of parent regions, and the first condition holds.  $\square$

**Corollary 6.149** (Finite witness set for differential-stage failure parent regions). For a set of complete evaluation specifications

$$\text{EvSpec}$$

and

$$(\text{Ev}_i)_{i \in I_{\text{diff}}^{\text{cond}}} \in \prod_{i \in I_{\text{diff}}^{\text{cond}}} \text{FailReg}_{\text{diff},i}(\text{EvSpec})$$

the set  $\{\text{Ev}_i \mid i \in I_{\text{diff}}^{\text{cond}}\}$  has at most five elements, and for each  $i \in I_{\text{diff}}^{\text{cond}}$ ,

$$\{\text{Ev}_j \mid j \in I_{\text{diff}}^{\text{cond}}\} \cap \text{FailReg}_{\text{diff},i}(\text{EvSpec}) \neq \emptyset.$$

*Proof.* The product membership in the proposition places the evaluation specification for each index in the corresponding parent region. Each evaluation specification also belongs to the candidate set, so the intersection of each parent region with that candidate set contains the corresponding evaluation specification. The candidate set has at most five elements.  $\square$

*Remark 6.150* (Scope of nonemptiness of differential-stage failure parent regions). Proposition 6.148 expresses simultaneous nonemptiness of the five parent regions as nonemptiness of the finite witness product. Once evaluation specifications belonging to the respective parent regions have been constructed, the finite witness set in Corollary 6.149 is determined.

### 6.8.6 Recursive Refinement of Differential-Stage Failure Regions

**Definition 6.151** (Differential-stage failure-refinement forest). Take a complete evaluation specification set  $\text{EvSpec}$ . For each  $i \in I_{\text{diff}}^{\text{cond}}$ , take a first-layer child-index set  $\text{Subldx}_{\text{diff},i}$ . For each  $\alpha_1 \in \text{Subldx}_{\text{diff},i}$ , take a second-layer child-index set

$$\text{Subldx}_{\text{diff},i;\alpha_1}$$

Suppose that the corresponding second-layer family of subregions satisfies

$$\text{FailReg}_{\text{diff},i;\alpha_1,\alpha_2}(\text{EvSpec}) \subseteq \text{FailReg}_{\text{diff},i;\alpha_1}(\text{EvSpec}), \quad \alpha_2 \in \text{Subldx}_{\text{diff},i;\alpha_1}$$

Define the child-index set at each path by

$$\text{Childldx}_{\text{diff}}(i; \alpha_1, \dots, \alpha_k) := \begin{cases} \text{Subldx}_{\text{diff},i}, & k = 0, \\ \text{Subldx}_{\text{diff},i;\alpha_1}, & k = 1, \\ \emptyset, & k = 2, \end{cases}$$

Define the node-region map by

$$\begin{aligned}\text{NodeReg}_{\text{diff}}((i); \text{EvSpec}) &:= \text{FailReg}_{\text{diff},i}(\text{EvSpec}), \\ \text{NodeReg}_{\text{diff}}((i; \alpha_1); \text{EvSpec}) &:= \text{FailReg}_{\text{diff},i;\alpha_1}(\text{EvSpec}), \\ \text{NodeReg}_{\text{diff}}((i; \alpha_1, \alpha_2); \text{EvSpec}) &:= \text{FailReg}_{\text{diff},i;\alpha_1,\alpha_2}(\text{EvSpec})\end{aligned}$$

Define the differential-stage failure-refinement forest by

$$\text{RefForest}_{\text{diff}}(\text{EvSpec}) := \text{RefForest} [I_{\text{diff}}^{\text{cond}}; \text{ChildIdx}_{\text{diff}}; \text{NodeReg}_{\text{diff}}(-; \text{EvSpec})]$$

For each  $i \in I_{\text{diff}}^{\text{cond}}$ ,  $\alpha \in \text{SubIdx}_{\text{diff},i}$ , and  $\varsigma \in \{\text{cov}, \text{unref}\}$ , set

$$\begin{aligned}\text{NodeReg}_{\text{diff}}^{\varsigma}((i); \text{EvSpec}) &= \text{FailReg}_{\text{diff},i}^{\varsigma}(\text{EvSpec}), \\ \text{FailReg}_{\text{diff},i;\alpha}^{\varsigma}(\text{EvSpec}) &:= \text{NodeReg}_{\text{diff}}^{\varsigma}((i; \alpha); \text{EvSpec}).\end{aligned}$$

For each  $i \in I_{\text{diff}}^{\text{cond}}$ , take two refinement data that share the parent region  $\text{FailReg}_{\text{diff},i}(\text{EvSpec})$  and satisfy the assumptions of Lemma 6.33:

$$\left( \text{SubIdx}_{\text{diff},i}^{(\rho)}, \left( \text{FailReg}_{\text{diff},i;\alpha}^{(\rho)}(\text{EvSpec}) \right)_{\alpha \in \text{SubIdx}_{\text{diff},i}^{(\rho)}} \right), \quad \rho \in \{0, 1\}$$

The two refinement snapshots at the root node are, by  $\text{RefSplit}$  in Definition 6.32,

$$\left( \text{RefSplit} \left( \text{FailReg}_{\text{diff},i}(\text{EvSpec}); \left( \text{FailReg}_{\text{diff},i;\alpha}^{(\rho)}(\text{EvSpec}) \right)_{\alpha \in \text{SubIdx}_{\text{diff},i}^{(\rho)}} \right) \right)_{\rho \in \{0,1\}}$$

**Proposition 6.152** (Expansion of the differential-stage failure-refinement forest). The differential-stage failure-refinement forest of Definition 6.151 satisfies the node decomposition, conditional disjoint union, and ancestor inclusion of Theorem 6.35. The two refinement snapshots at the root node satisfy the covered-region expansion and the unrefined-region contraction of Lemma 6.33.

*Proof.* The first assertion follows by substituting, from Definition 6.151,  $I_{\text{diff}}^{\text{cond}}$ ,  $\text{ChildIdx}_{\text{diff}}$ , and  $\text{NodeReg}_{\text{diff}}(-; \text{EvSpec})$  into Theorem 6.35. The second follows by substituting the two root snapshots from the same definition into Lemma 6.33.  $\square$

### 6.8.7 Boundaries Among the Law, Approximation, and Differential Failure Series

**Proposition 6.153** (Failure-series boundary between the law-description and differential stages). For each complete evaluation specification  $\text{Ev} \in \text{EvSpec}^{\text{diff}}$ ,

$$\text{Fail}_{\text{diff}}[\text{Ev}] \implies (\text{LawOperative}[\text{LawEvalDat}] \wedge \neg \text{DiffOperative}[\text{DiffEvalDat}]).$$

Furthermore,

$$\begin{aligned}\text{DiffOperative}[\text{DiffEvalDat}] \\ \implies \text{LawOperative}[\text{LawEvalDat}].\end{aligned}$$

*Proof.* The first implication follows from the equivalence in Proposition 6.139. The second follows from the first component of the conjunction given by Definition 5.132.  $\square$

**Proposition 6.154** (Evaluation targets of the law, approximation, and differential failure series). The evaluation targets of the three series are given in the following table.

| Series | Evaluation target                                                                                                                            |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------|
| FL     | Application domain, determinacy, satisfaction, faithful-representation covariance, and simultaneous satisfaction of the general-law relation |
| FA     | Quantitative-approximation defect quantities, admissible thresholds, continuous extension, order stability, and relation stability           |
| FDiff  | Differential-atlas regularity, generated-trajectory jets, local jet relations, and jet-relation covariance                                   |

FL3 evaluates membership of a generated trajectory in a general-law relation, whereas FDiff3 evaluates membership of a generated-trajectory jet in a local jet relation. FL4 evaluates preservation of general-law-relation membership under a faithful-representation change, whereas FDiff4 and FDiff5 evaluate preservation of jet-relation membership under a value-chart change and a time-representation change, respectively.

The FA series takes as input the quantitative-evaluation component of  $\text{EvalInput}^{\text{cont}}[l]$ , whereas the FDiff series takes as input  $\text{EvalInput}^{\text{diff}}$  of Definition 5.136. An implication linking the two series requires a condition that relates the components of the two input types.

*Proof.* The evaluation targets of FL3 and FL4 are an element of a general-law relation and a faithful-representation change. The evaluation targets of FDiff3 are the generated-trajectory jet and local jet relation of Definition 5.123. The evaluation targets of FDiff4 and FDiff5 are, respectively, the value-jet-change map of Definition 5.126 and the time-representation-change jet map of Definition 5.128.

The quantitative-evaluation component of the FA series belongs to the continuum-approximation evaluation input of Definition 4.168. The condition DAG, all-ancestors-satisfied bundle, and family of differential-evaluation structures of the FDiff series belong to the differential-stage failure-evaluation input of Definition 5.136. Therefore, the evaluation targets and input types of the respective series follow the distinctions in the table.  $\square$

### 6.8.8 Differential-Stage Failure Signature and Failure-Position Tokens

**Definition 6.155** (Five-component differential-stage failure signature). For a complete evaluation specification  $\text{Ev} \in \text{EvSpec}$ , define the five-component differential-stage failure signature by

$$\begin{aligned} \Sigma_{\text{diff}}^{\text{fail}}[\text{Ev}] &:= \left( \mathbf{1}_{\{\text{FDiff}_i[\text{Ev}]\}} \right)_{i \in I_{\text{diff}}^{\text{fail}}} \\ &\in \{0, 1\}^5 \end{aligned}$$

**Proposition 6.156** (Support of the differential-stage failure signature). For a complete evaluation specification  $\text{Ev} \in \text{EvSpec}$ ,

$$\begin{aligned} \text{supp} \left( \Sigma_{\text{diff}}^{\text{fail}}[\text{Ev}] \right) &= \text{MinFailIdx}_{\text{diff}}[\text{Ev}]. \end{aligned}$$

Furthermore, if  $\text{Ev} \in \text{FailDom}_{\text{diff}}(\text{EvSpec})$ , then  $\Sigma_{\text{diff}}^{\text{fail}}[\text{Ev}] \neq (0, 0, 0, 0, 0)$ , and its support is a nonempty antichain in the differential-stage condition DAG.



*Proof.* By Definition 6.155, the  $i$ -th component of the signature equals 1 if and only if  $\text{FDiff}_i[\text{Ev}]$  holds. Definition 6.144 gives the set equality for the support. Nonzeroness on the failure domain and the antichain property of the support follow from Theorem 6.145 and Proposition 6.146.  $\square$

**Proposition 6.157** (Correspondence between differential-stage minimal-failure classes and failure tokens). Let  $\text{Ev} \in \text{EvSpec}$ ,  $i \in I_{\text{diff}}^{\text{fail}}$ . Suppose that  $\text{DiffCondAncAll}_i[\text{Ev}]$  and

$$\text{FailReady}(\text{FailTok}_{\text{diff},i}; \text{Ev}_{\text{diff}}^{\text{fail}}) = \top$$

hold. Then

$$\begin{aligned} & \text{FDiff}_i[\text{Ev}] \\ & \iff \text{FailTok}_{\text{diff},i} \in \text{FailedTok}^{\text{strict}}[\text{Ev}]. \end{aligned}$$

If this assumption holds for every  $i \in I_{\text{diff}}^{\text{fail}}$ , then each failure-token membership indicator equals the corresponding component of Definition 6.155.

*Proof.* By Proposition 6.23, membership of a failure token is equivalent to the failed state of that token. By the evaluation-readiness assumption and Definition 6.21, this failed state is equivalent to a false evaluation value for that token. By Definition 6.10 and Definition 6.136, this false evaluation value is equivalent to the negation of the  $i$ -th universal differential condition. By the assumption of the all-ancestors-satisfied bundle and Definition 6.140, the conjunction of this negated condition and the all-ancestors-satisfied bundle is equivalent to the  $i$ -th differential-stage minimal-failure class. Applying the equivalences obtained for all indices to the respective components of Definition 6.155 gives the equality between each failure-token membership indicator and the corresponding signature component.  $\square$

**Proposition 6.158** (Output families of the differential-stage failure classification). The differential-stage failure classification for the complete evaluation specification set  $\text{EvSpec}$  determines the following families:

$$\begin{aligned} & \text{DiffCondDAG}, \\ & (\text{FailReg}_{\text{diff},i}(\text{EvSpec}))_{i \in I_{\text{diff}}^{\text{cond}}}, \\ & (\text{FailReg}_{\text{diff},i}^{\text{unref}}(\text{EvSpec}))_{i \in I_{\text{diff}}^{\text{cond}}}, \\ & (\Sigma_{\text{diff}}^{\text{fail}}[\text{Ev}])_{\text{Ev} \in \text{EvSpec}}. \end{aligned}$$

*Proof.* The differential-stage condition DAG is determined by Definition 5.130; the five parent regions and the unrefined regions are determined, respectively, by Definition 6.142 and Definition 6.143. The five-component signature for each evaluation specification is determined by Definition 6.155.  $\square$

## 6.9 Failure Signatures and the Approximation-Scale Boundary

For the six series of Definition 6.10, record by series the failure states of the five scale-invariant series and the scale-indexed failure state of the continuum-approximation series.

**Definition 6.159** (Six-series failure signature). Take a scale-augmented complete evaluation specification  $(\mathbf{Ev}, \lambda) \in \mathbf{EvSpec}_\Lambda$ . Using the series-index families of Definition 6.10 and the unified state section of Definition 6.21, define the six-series failure signature by

$$\begin{aligned} \Sigma^{\text{fail}}[\mathbf{Ev}; \lambda] &:= \left( \left( \mathbf{1}_{\{\text{FailStateAt}_{\mathbf{Ev}, \lambda}(\text{FailTok}_{\tau, i}) = \text{failed}\}} \right)_{i \in I_\tau^{\text{fail}}} \right)_{\tau \in \text{FailSeries}} \\ &\in \prod_{\tau \in \text{FailSeries}} \{0, 1\}^{n_\tau^{\text{fail}}} \end{aligned}$$

The law-series component is the component at the series index  $\tau = \text{law}$ .

**Proposition 6.160** (Minimal-position representation of the six-series failure signature). For every  $(\mathbf{Ev}, \lambda) \in \mathbf{EvSpec}_\Lambda$ , the coincidence of minimal elements in Proposition 6.26 holds. Furthermore,

$$\begin{aligned} \Sigma^{\text{fail}}[\mathbf{Ev}; \lambda] &= \left( \left( \mathbf{1}_{\{\text{FailTok}_{\tau, i} \in \text{FailedTok}[\mathbf{Ev}; \lambda]\}} \right)_{i \in I_\tau^{\text{fail}}} \right)_{\tau \in \text{FailSeries}}. \end{aligned}$$

*Proof.* The first conclusion follows from the coincidence of minimal elements in Proposition 6.26. By Definition 6.21, membership of each token of a scale-invariant series in the failed-token set is equivalent to its strict failure state being **failed**. For each token of the continuum-approximation series, the same membership is equivalent to its scale-indexed approximation failure state being **failed**.

By Definition 6.159, these membership indicators equal the respective components of the six-series failure signature.  $\square$

**Definition 6.161** (Strict, approximate, and full evaluation-validity domains and the approximation-boundary signature). Take the global validity domains of Definition 6.4 as input. Take a complete evaluation specification  $\mathbf{Ev} \in \mathbf{EvSpec}$ .

Define the strict-validity scale domain by

$$\begin{aligned} \mathcal{V}_{\mathbf{Ev}}^{\text{strict}} &:= \{ \lambda \in \Lambda \mid \mathbf{Ev} \in \mathcal{V}^{\text{strict}} \} \end{aligned}$$

For each  $\eta \in \{\text{approx}, \text{full}\}$ , define the validity scale domain by

$$\begin{aligned} \mathcal{V}_{\mathbf{Ev}}^\eta &:= \{ \lambda \in \Lambda \mid (\mathbf{Ev}, \lambda) \in \mathcal{V}^\eta \} \end{aligned}$$

The approximation-excess component set is  $I_{\text{approx}}^{\text{exc}}[\mathbf{Ev}; \lambda]$  of Definition 6.90. Lemma 6.89 gives its membership criterion by the strict inequality between the defect quantity and the admissible threshold.

Define the approximation-boundary signature by

$$\begin{aligned} \Sigma_{\text{approx}}^{\text{bdry}}[\mathbf{Ev}; \lambda] &:= \left( \mathbf{1}_{\{i \in I_{\text{approx}}^{\text{exc}}[\lambda]\}} \right)_{i \in I_{\text{approx}}^{\text{fail}}} \\ &\in \{0, 1\}^9 \end{aligned}$$

**Proposition 6.162** (Intersection representation of the full-evaluation-validity domain).  
For every  $\text{Ev} \in \text{EvSpec}$ ,

$$\begin{aligned}\mathcal{V}_{\text{Ev}}^{\text{full}} \\ &= \mathcal{V}_{\text{Ev}}^{\text{strict}} \cap \mathcal{V}_{\text{Ev}}^{\text{approx}}.\end{aligned}$$

*Proof.* Take any  $\lambda \in \Lambda$ . By Definition 6.27, the full-evaluation validity condition is equivalent to the conjunction of the strict-validity and approximation-validity conditions. Applying Definition 6.4 and Definition 6.161 gives the asserted equality.  $\square$

**Theorem 6.163** (Maximal approximation-validity interval and boundary signature).  
On the totally ordered scale set  $(\Lambda, \leq_\Lambda)$ , suppose that the continuum-approximation defect-quantity family of Definition 6.85 is monotonically nondecreasing. Then, for every  $\text{Ev} \in \text{EvSpec}$  satisfying  $\forall \mu \in \Lambda, \text{InputSat}^{\text{approx}}[\text{Ev}; \mu] = \top$ , the conclusions of Theorem 6.97 concerning the maximal effective interval and the first excess hold; moreover, one has

$$\mathcal{V}_{\text{Ev}}^{\text{approx}} \in \text{Init}_\Lambda(\Lambda)$$

and, in particular, if  $\mathcal{V}_{\text{Ev}}^{\text{approx}} = \Lambda$ , then  $\mathcal{I}^{\text{max}}[\text{Ev}] = \Lambda$ .

If the approximation-validity boundary  $\lambda_{\text{bdry}}^{\text{approx}}[\text{Ev}]$  is defined, then

$$\Sigma_{\text{approx}}^{\text{bdry}}[\text{Ev}; \lambda_{\text{bdry}}^{\text{approx}}[\text{Ev}]] \neq (0, \dots, 0).$$

Furthermore, for each  $\lambda \in \Lambda$  and each  $i \in I_{\text{approx}}^{\text{fail}}$ , the saturation equality  $\text{Def}_i^{\text{approx}}[\text{Ev}; \lambda] = \varepsilon_i^{\text{approx}}$  gives  $K_i^{\text{approx}}[\text{Ev}; \lambda] = \top$ .

*Proof.* Theorem 6.97 gives the conclusions concerning the maximal effective interval and the first excess. By Definition 6.85 and Theorem 6.95, the approximation-validity domain is an initial subset. The conclusion for the case in which the approximation-validity domain equals  $\Lambda$  follows from its equality with the maximal effective interval.

The nonemptiness of the excess-component set at the boundary scale and the indicator-signature definition in Definition 6.161 show that the boundary signature is nonzero. The assertion concerning the saturation equality follows from the threshold equivalence in Definition 6.84.  $\square$

**Theorem 6.164** (First-Failure Report at Structural Minimal Positions and the Approximation-Scale Boundary). Take a scale-augmented complete evaluation specification  $(\text{Ev}, \lambda) \in \text{EvSpec}_\Lambda$ . For  $\text{Ev}$  for which the approximation-validity boundary is defined, define the boundary excess-component set, boundary signature, and boundary datum by

$$\begin{aligned}\text{BdryExc}^{\text{approx}}[\text{Ev}] &:= I_{\text{approx}}^{\text{exc}}[\text{Ev}; \lambda_{\text{bdry}}^{\text{approx}}[\text{Ev}]], \\ \text{BdrySig}^{\text{approx}}[\text{Ev}] &:= \Sigma_{\text{approx}}^{\text{bdry}}[\text{Ev}; \lambda_{\text{bdry}}^{\text{approx}}[\text{Ev}]], \\ \text{BdryDatum}^{\text{approx}}[\text{Ev}] &:= (\lambda_{\text{bdry}}^{\text{approx}}[\text{Ev}], \text{BdrySig}^{\text{approx}}[\text{Ev}]).\end{aligned}$$

Define the approximation-scale boundary data by the set of all least elements of the complement:

$$\begin{aligned}\text{BdryData}^{\text{approx}}[\text{Ev}] \\ &:= \left\{ (\lambda, \Sigma_{\text{approx}}^{\text{bdry}}[\text{Ev}; \lambda]) \mid \begin{array}{l} \lambda \in \Lambda \setminus \mathcal{V}_{\text{Ev}}^{\text{approx}}, \\ \forall \mu \in \Lambda \setminus \mathcal{V}_{\text{Ev}}^{\text{approx}}, \lambda \leq_\Lambda \mu \end{array} \right\}.\end{aligned}$$

This set is empty if  $\mathcal{V}_{\text{Ev}}^{\text{approx}} = \Lambda$  or if the complement has no least element; when the approximation-validity boundary is defined,

$$\text{BdryData}^{\text{approx}}[\text{Ev}] = \{\text{BdryDatum}^{\text{approx}}[\text{Ev}]\}.$$

Define the first-failure report by

$$\text{FirstFailRep}[\text{Ev}; \lambda] := (\text{FailedTok}[\text{Ev}; \lambda], \Sigma^{\text{fail}}[\text{Ev}; \lambda], \text{BdryData}^{\text{approx}}[\text{Ev}]).$$

The first component consists of all failure tokens minimal with respect to the failure-dependency order, the second component is the six-series failure-state signature, and the third component is the approximation-scale boundary data.

*Proof.* By Definition 6.21 and Proposition 6.26, the first component coincides with the set of all minimal elements with respect to the failure-generation dependency order. Definition 6.159 determines the second component.

A least element of a subset of a total order is unique. Hence the approximation-scale boundary data are uniquely determined as either the empty set or a singleton, and the defined approximation-validity boundary and boundary signature provide that singleton. It follows that the first-failure report is uniquely determined.  $\square$

**Definition 6.165** (Multiplicity of Structural Failure and Approximation-Boundary Excess). Take a scale-augmented complete evaluation specification  $(\text{Ev}, \lambda) \in \text{EvSpec}_{\Lambda}$ .

Write the structural-failure classification as  $\text{FailCaseOf}(\text{Ev}; \lambda)$ .

Call the case  $\text{FailCaseOf}(\text{Ev}; \lambda) = \text{FailUnique}$  a simple structural failure, and the case  $\text{FailCaseOf}(\text{Ev}; \lambda) = \text{FailParallel}$  a multiple structural failure.

Define the set of approximation-boundary-excess classification values by

$$\text{BdryCase}^{\text{approx}} := \{\text{BdryAbsent}, \text{BdryUnique}, \text{BdryParallel}\}.$$

The three values on the right-hand side are taken to be distinct. **BdryAbsent** means that there are no boundary-excess components. Whether the boundary itself exists is recorded in  $\text{BdryData}^{\text{approx}}[\text{Ev}]$ .

Define the approximation-boundary-excess multiplicity

$$\text{BdryCaseOf}^{\text{approx}}[\text{Ev}] \in \text{BdryCase}^{\text{approx}}$$

by

$$\text{BdryCaseOf}^{\text{approx}}[\text{Ev}] := \begin{cases} \text{BdryAbsent}, & \text{BdryData}^{\text{approx}}[\text{Ev}] \subseteq \Lambda \times \{(0, \dots, 0)\}, \\ \text{BdryUnique}, & \text{BdryData}^{\text{approx}}[\text{Ev}] \neq \emptyset, \\ & |\text{BdryExc}^{\text{approx}}[\text{Ev}]| = 1, \\ \text{BdryParallel}, & \text{BdryData}^{\text{approx}}[\text{Ev}] \neq \emptyset, \\ & |\text{BdryExc}^{\text{approx}}[\text{Ev}]| \geq 2 \end{cases}.$$

Call the case  $\text{BdryCaseOf}^{\text{approx}}[\text{Ev}] = \text{BdryUnique}$  a simple approximation-boundary excess, and the case  $\text{BdryCaseOf}^{\text{approx}}[\text{Ev}] = \text{BdryParallel}$  a multiple approximation-boundary excess.

Define the multiplicity report by

$$\begin{aligned} \text{MultRep}[\text{Ev}; \lambda] \\ := (\text{FailCaseOf}(\text{Ev}; \lambda), \text{BdryCaseOf}^{\text{approx}}[\text{Ev}]). \end{aligned}$$

**Corollary 6.166** (Unique Minimal Position of a Simple Structural Failure). For a scale-augmented complete evaluation specification  $(\text{Ev}, \lambda) \in \text{EvSpec}_\Lambda$ , if

$$\text{FailCaseOf}(\text{Ev}; \lambda) = \text{FailUnique}$$

then  $\text{FailedTok}[\text{Ev}; \lambda]$  is a singleton and exactly one component of the six-series failure signature has value 1.

*Proof.* By Definition 6.24, the hypothesis is equivalent to  $|\text{FailedTok}[\text{Ev}; \lambda]| = 1$ . By Proposition 6.26, every element of the failed-token set is minimal with respect to the failure-dependency order. Hence the minimal-position set is a singleton and its indicator signature has exactly one component equal to 1.  $\square$

**Proposition 6.167** (Multiple Structural Failure and Multiple Approximation-Boundary Excess). For a scale-augmented complete evaluation specification  $(\text{Ev}, \lambda) \in \text{EvSpec}_\Lambda$ , if  $\text{FailCaseOf}(\text{Ev}; \lambda) = \text{FailParallel}$ , then  $|\text{FailedTok}[\text{Ev}; \lambda]| \geq 2$ .

If  $\text{BdryCaseOf}^{\text{approx}}[\text{Ev}] = \text{BdryParallel}$ , then

$$|\text{BdryExc}^{\text{approx}}[\text{Ev}]| \geq 2.$$

The first component of the multiplicity report is determined by the cardinality of the structural minimal-position set, and the second component by the cardinality of the excess-component set at the approximation-scale boundary.

*Proof.* The first assertion follows from Definition 6.24 and Proposition 6.26.

The second assertion follows from the third branch of the approximation-boundary-excess multiplicity in Definition 6.165.

By Definition 6.165, the multiplicity report is an ordered pair of the two classification values. Each component is determined solely by the cardinality of the corresponding set.  $\square$

**Theorem 6.168** (Completeness of Failure Localization Relative to the Adopted Six Series). For every  $(\text{Ev}, \lambda) \in \text{EvSpec}_\Lambda$ , the equality of minimal elements in Proposition 6.26 holds.

If  $\text{FailedTok}[\text{Ev}; \lambda] \neq \emptyset$ , then

$$\text{Min}_{\preceq_{\text{fail,dep}}} \text{FailedTok}[\lambda] \neq \emptyset.$$

The minimal-position set may contain multiple failure tokens belonging to the same series or to distinct series.

The construction by state evaluation and the construction by minimal failure positions yield the same six-series failure signature  $\Sigma^{\text{fail}}[\text{Ev}; \lambda]$ .

This cover ranges over the failure tokens of the six series declared in Definition 6.10. The domain of evaluation specifications is  $\text{EvSpec}_\Lambda$ .

*Proof.* By the disjoint union in Definition 6.10, each failure token belongs to exactly one series.

By Definition 6.21, the failed-token set is a subset of the full set of failure tokens in the six series. Proposition 6.26 gives the equality of minimal elements stated in the theorem. This equality implies that nonemptiness of the failed set gives nonemptiness of the minimal-position set.

The cardinality of the series-index set follows from the definition of the six series in Definition 6.10.

By Proposition 6.160, the construction by state evaluation and the construction by minimal failure positions yield the same six-series failure signature.  $\square$

## 6.10 Failure Positions and Structural Burden

**Definition 6.169** (Structural Burden from Principal Upper Sets of the Failure-Dependency Order). Define the six-series set of structural-target tokens by

$$\text{StructTok} := \bigsqcup_{\tau \in \text{FailSeries}} \{\text{StructTok}_{\tau,i} \mid i \in I_{\tau}^{\text{fail}}\}.$$

Define the map that assigns the corresponding structural-target token to each failure-position token by

$$\text{StructOf} : \text{FailTok} \longrightarrow \text{StructTok},$$

$$\text{StructOf}(\text{FailTok}_{\tau,i}) := \text{StructTok}_{\tau,i}.$$

For any  $s \in \text{FailTok}$ , define the principal upper set with respect to the failure-generation dependency order by

$$\text{Desc}_{\text{fail,dep}}(s) := \{t \in \text{FailTok} \mid s \preceq^{\text{fail,dep}} t\}.$$

For any  $A \subseteq \text{FailTok}$ , define the image under the map **StructOf** by

$$\text{StructOf}[A] := \{\text{StructOf}(s) \mid s \in A\}.$$

Define the structural burden of a set of failure positions  $A \subseteq \text{FailTok}$  by

$$\text{StructBurden}(A) := \bigcup_{s \in A} \text{StructOf}[\text{Desc}_{\text{fail,dep}}(s)].$$

For any  $(\text{Ev}, \lambda) \in \text{EvSpec}_{\Lambda}$ , define the failure-position-side structural burden by

$$\text{Burden}_{\text{pos}}^{\text{struct}}[\text{Ev}; \lambda] := \text{StructBurden}(\text{FailedTok}[\text{Ev}; \lambda]).$$

If the approximation-scale boundary data are empty, set  $\text{Burden}_{\text{bdry}}^{\text{struct}}[\text{Ev}] := \emptyset$ . If the approximation-scale boundary data are

$$\text{BdryData}^{\text{approx}}[\text{Ev}] = \{(\lambda_{\text{bdry}}^{\text{approx}}[\text{Ev}], \text{BdrySig}^{\text{approx}}[\text{Ev}])\}$$

define

$$\begin{aligned} & \text{Burden}_{\text{bdry}}^{\text{struct}}[\text{Ev}] \\ & := \text{StructBurden}(\{\text{FailTok}_{\text{approx},i} \mid i \in \text{BdryExc}^{\text{approx}}[\text{Ev}]\}). \end{aligned}$$

Define the structural-burden report by

$$\begin{aligned} & \text{BurdenRep}^{\text{struct}}[\text{Ev}; \lambda] \\ & := (\text{Burden}_{\text{pos}}^{\text{struct}}[\text{Ev}; \lambda], \text{Burden}_{\text{bdry}}^{\text{struct}}[\text{Ev}]). \end{aligned}$$

**Lemma 6.170** (Monotonicity and Separation of the Structural-Burden Kernel). For any  $s \in \text{FailTok}$ , the structural burden of a single failure token is

$$\text{StructBurden}(\{s\}) = \text{StructOf}[\text{Desc}_{\text{fail,dep}}(s)].$$

For any  $A, B \subseteq \text{FailTok}$ ,

$$A \subseteq B \implies \text{StructBurden}(A) \subseteq \text{StructBurden}(B).$$

Moreover, for any  $s \in A$ ,

$$\begin{aligned} & \text{StructBurden}(\{s\}) \not\subseteq \text{StructBurden}(B) \\ & \implies \text{StructBurden}(A) \setminus \text{StructBurden}(B) \neq \emptyset. \end{aligned}$$

*Proof.* The equality for a singleton follows by taking the union in Definition 6.169 over its sole element  $s$ .

For monotonicity,  $A \subseteq B$  implies that every component of the union indexed by  $A$  is also a component of the union indexed by  $B$ .

The separation assumption gives an element

$$x \in \text{StructBurden}(\{s\}) \setminus \text{StructBurden}(B).$$

Because  $\{s\} \subseteq A$  and monotonicity imply  $x \in \text{StructBurden}(A)$ , the required set difference is nonempty.  $\square$

Table 9: Failure-position tokens and structural-target tokens in the six series

| Series                                             | $\tau$ | $n_\tau^{\text{fail}}$ |
|----------------------------------------------------|--------|------------------------|
| Comparison                                         | cmp    | 8                      |
| Representation                                     | repr   | 3                      |
| Dynamical description                              | dyn    | 8                      |
| Continuum approximation                            | approx | 9                      |
| Law representation                                 | law    | 5                      |
| Differential consistency and law-form preservation | diff   | 5                      |

For every series in the table and every  $i \in I_\tau^{\text{fail}}$ , the correspondence is

$$\text{StructOf}(\text{FailTok}_{\tau,i}) = \text{StructTok}_{\tau,i}.$$

Table 10: Two components of the structural-burden report

| Component                                           | Generating input                                                               | Codomain                          |
|-----------------------------------------------------|--------------------------------------------------------------------------------|-----------------------------------|
| Failure-position-side structural burden             | $\text{FailedTok}[\text{Ev}; \lambda]$                                         | $\mathcal{P}(\text{StructTok})$   |
| Approximation-scale-boundary-side structural burden | $\text{BdryData}^{\text{approx}}[\text{Ev}]$                                   | $\mathcal{P}(\text{StructTok})$   |
| Structural-burden report                            | Failure-position-side and approximation-scale-boundary-side structural burdens | $\mathcal{P}(\text{StructTok})^2$ |

**Proposition 6.171** (Uniqueness and Dependency-Upward Closure of the Structural-Burden Report). For any  $(\text{Ev}, \lambda) \in \text{EvSpec}_\Lambda$ ,  $\text{BurdenRep}^{\text{struct}}[\text{Ev}; \lambda]$  is uniquely determined. Moreover,

$$\begin{aligned} \forall s \in \text{FailedTok}[\text{Ev}; \lambda], \\ \text{StructBurden}(\{s\}) \subseteq \text{Burden}_{\text{pos}}^{\text{struct}}[\text{Ev}; \lambda]. \end{aligned}$$

If the approximation-scale boundary data form a singleton,

$$\begin{aligned} \forall i \in \text{BdryExc}^{\text{approx}}[\text{Ev}], \\ \text{StructBurden}(\{\text{FailTok}_{\text{approx},i}\}) \subseteq \text{Burden}_{\text{bdry}}^{\text{struct}}[\text{Ev}]. \end{aligned}$$

*Proof.* By Definition 6.169, the failure-position-side structural burden is the union of the single-token structural burdens of the failure tokens. Hence each single-token structural burden is contained in this union.

When the approximation-scale boundary data are empty, the boundary-side structural burden is empty. When they form a singleton, the boundary-side structural burden is the union of the structural-target images of the principal upper sets determined by the respective boundary excess components. Hence each image is contained in this union. Both components are uniquely determined in either branch, and the structural-burden report is therefore uniquely determined.  $\square$

## 6.11 Failure-Localization Results

**Definition 6.172** (Failure-Localization Data). For any  $(\text{Ev}, \lambda) \in \text{EvSpec}_\Lambda$ , define the failure-localization data by

$$\begin{aligned} \text{FailLocData}[\text{Ev}; \lambda] \\ := & \left( (\mathcal{V}_{\text{Ev}}^{\text{strict}}, \mathcal{V}_{\text{Ev}}^{\text{approx}}, \mathcal{V}_{\text{Ev}}^{\text{full}}), \mathcal{I}^{\text{max}}[\text{Ev}], \right. \\ & \text{FirstFailRep}[\text{Ev}; \lambda], \text{MultRep}[\text{Ev}; \lambda], \\ & \left. \text{BurdenRep}^{\text{struct}}[\text{Ev}; \lambda], \Sigma^{\text{fail}}[\text{Ev}; \lambda] \right). \end{aligned}$$

**Proposition 6.173** (Determinacy and Audit Inputs of the Failure-Localization Results). The failure-localization data of Definition 6.172 are uniquely determined. Among its three validity scale domains, the full-evaluation validity scale domain equals the intersection of the strict-validity and approximation-validity scale domains. The maximal approximation-validity interval is contained in the approximation-validity scale domain; under the hypotheses of Theorem 6.163, the two are equal.

The first-failure report, multiplicity report, and structural-burden report are uniquely determined. The construction by state evaluation and the construction by minimal failure positions yield the same six-series failure signature. The minimal failure positions are covered relative to the adopted six series and to the evaluation states on the scale-augmented complete evaluation specification.

*Proof.* Definition 6.161 and Proposition 6.162 uniquely determine the three validity scale domains and establish their intersection equality. Definition 6.93 uniquely determines the maximal approximation-validity interval and places it within the approximation-validity scale domain. Under the hypotheses of Theorem 6.163, that theorem gives equality of the two domains.

Theorem 6.164 uniquely determines the first-failure report. Definition 6.165 uniquely determines the multiplicity report. Proposition 6.171 uniquely determines the structural-burden report.

By Proposition 6.160, the construction by state evaluation and the construction by minimal failure positions yield the same six-series failure signature. By Theorem 6.168, the minimal failure positions are covered relative to the adopted six series.

Consequently, the three validity scale domains, maximal approximation-validity interval, first-failure report, multiplicity report, and structural-burden report are uniquely determined, yielding the inputs to the operational audit report.  $\square$



### 6.11.1 Failure-State Classification Relative to the Adopted Evaluation Domain

**Definition 6.174** (Failure-Token State Profile). Take any  $(\text{Ev}, \lambda) \in \text{EvSpec}_\Lambda$ .

Using the state fibers of Definition 6.21, define the satisfied, unformed, and unevaluable failure-token sets, respectively, by

$$\begin{aligned}\text{SatTok}[\text{Ev}; \lambda] &:= \text{FailFiber}_{\text{sat}}[\text{Ev}; \lambda], \\ \text{UnformedTok}[\text{Ev}; \lambda] &:= \text{FailFiber}_{\text{unformed}}[\text{Ev}; \lambda], \\ \text{UnevaluableTok}[\text{Ev}; \lambda] &:= \text{FailFiber}_{\text{unevaluable}}[\text{Ev}; \lambda].\end{aligned}$$

Define the failure-token state profile by

$$\begin{aligned}\text{TokStateProf}^{\text{fail}}[\text{Ev}; \lambda] \\ := \Big( K_{\text{full}}^{\text{fail}}[\text{Ev}; \lambda], \\ \text{SatTok}[\text{Ev}; \lambda], \text{FailedTok}[\text{Ev}; \lambda], \\ \text{UnformedTok}[\text{Ev}; \lambda], \text{UnevaluableTok}[\text{Ev}; \lambda], \\ \text{BdryData}^{\text{approx}}[\text{Ev}] \Big).\end{aligned}$$

Table 11: Evaluation states of failure tokens in the adopted evaluation domain

| Condition                                                  | Evaluation-state component                                 | Corresponding failure-evaluation quantity                                                                                                                       |
|------------------------------------------------------------|------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The states of all failure-position tokens are <b>sat</b>   | $K_{\text{full}}^{\text{fail}}[\text{Ev}; \lambda] = \top$ | Full-evaluation validity                                                                                                                                        |
| $\text{FailedTok}[\text{Ev}; \lambda] \neq \emptyset$      | failed                                                     | $\text{FirstFailRep}[\text{Ev}; \lambda],$<br>$\text{MultRep}[\text{Ev}; \lambda],$<br>$\text{BurdenRep}^{\text{struct}}[\text{Ev}; \lambda]$                   |
| $\text{UnformedTok}[\text{Ev}; \lambda] \neq \emptyset$    | unformed                                                   | Unformed failure-token set                                                                                                                                      |
| $\text{UnevaluableTok}[\text{Ev}; \lambda] \neq \emptyset$ | unevaluable                                                | Unevaluable failure-token set                                                                                                                                   |
| $ \text{BdryData}^{\text{approx}}[\text{Ev}]  = 1$         | Approximation-scale boundary                               | $\lambda_{\text{bdry}}^{\text{approx}}[\text{Ev}],$<br>$\text{BdrySig}^{\text{approx}}[\text{Ev}],$<br>$\text{Burden}_{\text{bdry}}^{\text{struct}}[\text{Ev}]$ |

Distinct state sets may be nonempty simultaneously. The approximation-scale boundary data are a component concerning the scale direction of the continuum-approximation series.

**Proposition 6.175** (Failure-State Classification Relative to the Adopted Evaluation Domain). For any  $(\text{Ev}, \lambda) \in \text{EvSpec}_\Lambda$ ,

$$\begin{aligned}\text{FailTok} &= \text{SatTok}[\text{Ev}; \lambda] \\ &\sqcup \text{FailedTok}[\text{Ev}; \lambda] \\ &\sqcup \text{UnformedTok}[\text{Ev}; \lambda] \\ &\sqcup \text{UnevaluableTok}[\text{Ev}; \lambda].\end{aligned}$$

Furthermore,  $K_{\text{full}}^{\text{fail}}[\text{Ev}; \lambda] = \top$  is equivalent to  $\text{SatTok}[\lambda] = \text{FailTok}$ .

If

$$\text{FailedTok}[\lambda] \neq \emptyset$$

each failure token belongs to exactly one of the adopted six series. The first-failure report, multiplicity report, and structural-burden report

$$\begin{aligned} &\text{FirstFailRep}[\text{Ev}; \lambda], \\ &\text{MultRep}[\text{Ev}; \lambda], \\ &\text{BurdenRep}^{\text{struct}}[\text{Ev}; \lambda] \end{aligned}$$

determine the minimal failure positions, six-series failure signature, multiplicity, approximation-scale boundary, and structural burden.

$\text{TokStateProf}^{\text{fail}}[\text{Ev}; \lambda]$  has as its components the full-evaluation validity condition, the four state-specific failure-token sets, and the approximation-scale boundary data.

The domain of this classification is  $\text{EvSpec}_\Lambda$ , and its completeness is relative to the failure-token system of Definition 6.10.

*Proof.* By Definition 6.10, the full failure-token set decomposes as a disjoint union of the failure-token sets of the scale-invariant series and the continuum-approximation series. By the evaluation-state map and series-specific evaluation specifications of Definition 6.21 and Proposition 6.22, each failure-position token is assigned exactly one of the evaluation states **sat**, **failed**, **unformed**, and **unevaluable**. Hence the four state-specific failure-token sets give a disjoint-union decomposition allowing empty components of the full failure-token set.

By Proposition 6.28, the full-evaluation validity condition holds if and only if the evaluation state of every failure-position token is **sat**.

By Theorem 6.168, each failure token belongs to exactly one of the adopted six series. By Proposition 6.173, the first-failure report, multiplicity report, and structural-burden report are uniquely determined.

Definition 6.174 determines each component of the failure-token state profile.  $\square$

*Remark 6.176 (Relative Scope of Classification Completeness).* The domain of the failure-state classification is  $\text{EvSpec}_\Lambda$ . The adopted six series and the full failure-token set **FailTok** of Definition 6.10 give the domain of quantification for classification completeness.

On the basis of the results of Theorem 4.177, Theorem 5.112, and Theorem 6.164, localize validity, validity domains, the approximation limit, the maximal validity interval, the failure-position token system, and the minimal failure-position set.

Failure positions divide into six groups: the comparison, representation, continuum-approximation, dynamical-description, law-representation, and differential-consistency and law-form-preservation stages. Represent each position by a failure-position token of Definition 6.10. The failed-token set obtained from the evaluation states and the failure-generation dependency order determine the minimal-position representation of the six-series failure signature in Proposition 6.160.

### Validity and failure localization

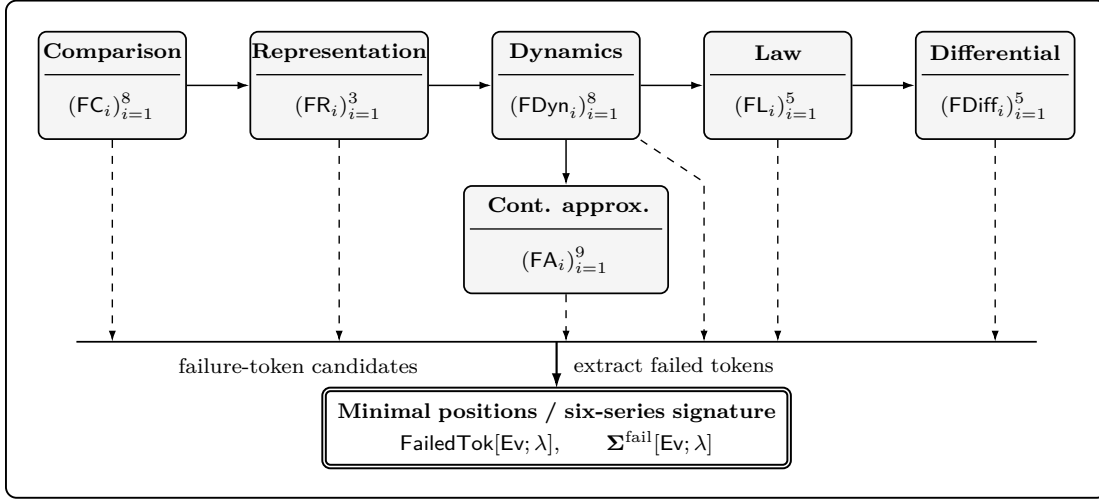


Figure 14: Six-group localization of validity and failure. The minimal failure-position set is defined as the set of positions minimal with respect to the failure-generation dependency order among the failure-position tokens belonging to the comparison, representation, continuum-approximation, dynamical-description, law-representation, and differential-consistency and law-form-preservation stages.

## 6.12 Operational Auditability of the Time-Reference Construction

Take a scale-augmented complete evaluation specification  $(\text{Ev}, \lambda) \in \text{EvSpec}_\Lambda$ . Take as input the semantic-series-tagged failure-position token system of Definition 6.10, together with the failure-localization data of Definition 6.172,  $\text{FailLocData}[\text{Ev}; \lambda]$ . Operational audit data consist of the computational components used to evaluate each failure condition and an evidence-provenance certificate that verifies the origin of those computational components.

**Definition 6.177** (Operational Audit Data for the Time-Reference Construction). Take a scale-augmented complete evaluation specification  $(\text{Ev}, \lambda) \in \text{EvSpec}_\Lambda$ . Use the six-failure-series family of evaluation specifications  $\text{Ev}^{\text{fail}}$  of Definition 6.10.

Define the set of series that require inputs with formation states by

$$\text{AuditInputSeries} := \{\text{approx}, \text{law}, \text{diff}\}.$$

Let the set of reasons for unformed audit inputs be

$$\text{AuditInputReason}.$$

For each

$$\tau \in \text{AuditInputSeries}$$

take a series-specific audit-input type  $\text{AuditInputType}_\tau$  and a series-specific audit input

$$\text{AuditBranchInput}_\tau \in \text{AuditInputType}_\tau \sqcup \text{AuditInputReason}$$

The image under the canonical injection into the left summand is a formed audit input, and the image under the canonical injection into the right summand is an unformed reason.

Define the audit-input datum by

$$\text{AuditInput}[\text{Ev}] := \left( \mathcal{X}, \text{Ev}, (\text{AuditBranchInput}_\tau)_{\tau \in \text{AuditInputSeries}} \right).$$

Define the all-scale family of evaluation specifications for the continuum-approximation series by

$$\text{Ev}_{\text{approx}}^{\text{fail}} := (\text{Ev}_{\text{approx}, \lambda}^{\text{fail}})_{\lambda \in \Lambda}.$$

Define the condition-evidence family for the six failure series by

$$\text{AuditEvidence}[\text{Ev}^{\text{fail}}] := \left( (\text{AuditEvidence}_\tau[\text{Ev}_\tau^{\text{fail}}])_{\tau \in \text{FailSeries}^{\text{strict}}}, \right. \\ \left. (\text{AuditEvidence}_{\text{approx}}[\text{Ev}_{\text{approx}, \lambda}^{\text{fail}}])_{\lambda \in \Lambda} \right).$$

Define the map that gives the condition-evidence component for each  $t \in \text{FailTok}$  by

$$\text{AuditEvidenceAt}[\text{Ev}; \lambda](t) \\ := \text{AuditEvidence}_{\text{FailSeriesOf}(t)} [\text{FailSpecAt}_{\text{FailSeriesOf}(t)} [\text{Ev}; \lambda]].$$

Let  $\text{AuditScaleData}[\text{Ev}_{\text{approx}}^{\text{fail}}]$  be the scale-audit data consisting of the scale order, approximation-scale set, codomains of defect quantities, and admissible thresholds.

Define the audit computation core by

$$\text{AuditCore}[\text{Ev}^{\text{fail}}] := \left( \text{AuditContractID}, \text{AuditInput}[\text{Ev}], \right. \\ \left. \text{AuditEvidence}[\text{Ev}^{\text{fail}}], \text{AuditScaleData}[\text{Ev}_{\text{approx}}^{\text{fail}}] \right).$$

Define the evidence-provenance certificate by

$$\text{AuditCert}[\text{Ev}^{\text{fail}}] := \left( \text{AuditInputID}, \text{AuditEvidenceID}, \text{AuditContentHash}, \right. \\ \left. \text{AuditTransformTrace}, \text{AuditEvidenceLocation} \right).$$

Define the operational audit data for the time-reference construction by

$$\text{AuditData}_{\text{time}}[\text{Ev}^{\text{fail}}] := (\text{AuditCore}[\text{Ev}^{\text{fail}}], \text{AuditCert}[\text{Ev}^{\text{fail}}]).$$

In the remainder of this section, the time-type subscript is omitted from this notation.

**Definition 6.178** (Component Families of the Audit Computation Core). The audit computation core  $\text{AuditCore}[\text{Ev}^{\text{fail}}]$  of Definition 6.177 consists of the audit contract, audit input, six-series condition-evidence family, and scale-audit data.

The condition-evidence family  $\text{AuditEvidence}[\text{Ev}^{\text{fail}}]$  has, for every  $\tau \in \text{FailSeries}$  and every  $i \in I_\tau^{\text{fail}}$ , as its components the typed condition evidence for  $\text{Ev}$  in a scale-invariant series and for each  $(\text{Ev}, \lambda)$  in the continuum-approximation series.

The scale-audit data  $\text{AuditScaleData}[\text{Ev}_{\text{approx}}^{\text{fail}}]$  have as their components the scale order, approximation-scale set, codomains of defect quantities, and admissible thresholds. Through the audit-input identifiers, condition-evidence identifiers, content hash, transformation history, and evidence location, the evidence-provenance certificate associates provenance with each component of the audit computation core.

Table 12: Typed components and computational responsibilities of the audit computation core

| Component family                     | Typed component                                                 | Computational responsibility                                                                                                              |
|--------------------------------------|-----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Audit contract                       | <code>AuditContractID</code>                                    | Identification of the condition predicates, input types, scale types, and audit-report generation convention                              |
| Audit input                          | <code>AuditInput[Ev]</code>                                     | Operative construction, complete evaluation specification, and inputs with formation states for the three series                          |
| Six-series condition-evidence family | <code>AuditEvidence[Ev<sup>fail</sup>]</code>                   | Evaluation of every condition in the five scale-invariant series and every condition at every scale in the continuum-approximation series |
| Scale-audit data                     | <code>AuditScaleData[Ev<sup>fail</sup><sub>approx</sub>]</code> | Evaluation of the scale order, approximation-scale set, codomains of defect quantities, and admissible thresholds                         |

**Definition 6.179** (Five-Valued Audit-Condition States and Six-Group Audit States). Take a scale-augmented complete evaluation specification  $(Ev, \lambda) \in \text{EvSpec}_\Lambda$ .

Define the audit-condition state set and its group-aggregation priority order by

$$\begin{aligned}
 \text{AuditStatus} &:= \left\{ \text{AuditPass}, \text{AuditFail}, \text{AuditBlocked}, \right. \\
 &\quad \left. \text{AuditUnformed}, \text{AuditIndeterminate} \right\}, \\
 \text{AuditPass} &\prec_{\text{AuditStatus}} \text{AuditIndeterminate} \prec_{\text{AuditStatus}} \text{AuditUnformed} \\
 &\quad \prec_{\text{AuditStatus}} \text{AuditBlocked} \prec_{\text{AuditStatus}} \text{AuditFail}
 \end{aligned}$$

The five state values are mutually distinct.

Define the five-valued lift map that adds the success or failure of immediate predecessor conditions to the four-valued evaluation states of Definition 6.21 by

$$\begin{aligned}
 \text{AuditStateLift} : \{\perp, \top\} \times \text{FailState} &\longrightarrow \text{AuditStatus}, \\
 \text{AuditStateLift}(b, q) &:= \begin{cases} \text{AuditBlocked}, & b = \perp, q \in \text{FailState}, \\ \text{AuditUnformed}, & (b, q) = (\top, \text{unformed}), \\ \text{AuditIndeterminate}, & (b, q) = (\top, \text{unevaluable}), \\ \text{AuditPass}, & (b, q) = (\top, \text{sat}), \\ \text{AuditFail}, & (b, q) = (\top, \text{failed}) \end{cases}
 \end{aligned}$$

For any  $t \in \text{FailTok}$ , define the immediate-predecessor-satisfaction value and the audit-condition state by

$$\begin{aligned}
 &\text{AuditPredecessorsPass}[Ev; \lambda](t) \\
 &\quad := \text{FailPredSat}(\text{FailArgAt}_{Ev, \lambda}(t)), \\
 &\text{AuditConditionStateOf}[Ev; \lambda](t) \\
 &\quad := \text{AuditStateLift}(\text{AuditPredecessorsPass}[Ev; \lambda](t), \text{FailStateAt}_{Ev, \lambda}(t)).
 \end{aligned}$$

Consequently,

$$\text{AuditConditionStateOf}[\text{Ev}; \lambda] : \text{FailTok} \longrightarrow \text{AuditStatus}.$$

For each  $q \in \text{AuditStatus}$ , define the audit-state fiber by

$$\text{AuditStatusFiber}_q[\text{Ev}; \lambda] := \{t \in \text{FailTok} \mid \text{AuditConditionStateOf}[\text{Ev}; \lambda](t) = q\}.$$

For each  $\tau \in \text{FailSeries}$ , define the series-group state by

$$\text{AuditGroupState}_\tau[\text{Ev}; \lambda] := \max_{\prec_{\text{AuditStatus}}} \{\text{AuditConditionStateOf}[\text{Ev}; \lambda](t) \mid t \in \text{FailTok}_\tau\}.$$

Define the six-group audit state by

$$\text{AuditGroupStatus}[\text{Ev}; \lambda] := (\text{AuditGroupState}_\tau[\text{Ev}; \lambda])_{\tau \in \text{FailSeries}}.$$

Define the indeterminate-condition set and indeterminate signature by

$$\begin{aligned} \text{AuditIndeterminateTok}[\text{Ev}; \lambda] &:= \text{AuditStatusFiber}_{\text{AuditIndeterminate}}[\text{Ev}; \lambda], \\ \text{AuditIndeterminateSig}[\text{Ev}; \lambda](t) &:= \mathbf{1}_{\text{AuditIndeterminateTok}[\text{Ev}; \lambda]}(t), \quad t \in \text{FailTok}. \end{aligned}$$

Consequently,

$$\text{AuditIndeterminateSig}[\text{Ev}; \lambda] : \text{FailTok} \longrightarrow \{0, 1\}.$$

**Definition 6.180** (Five-Valued Audit-Report Object). Take a scale-augmented complete evaluation specification  $(\text{Ev}, \lambda) \in \text{EvSpec}_\Lambda$ . Use the six-failure-series family of evaluation specifications  $\text{Ev}^{\text{fail}}$  of Definition 6.10.

Define the set of audit-undefined states by

$$\text{AuditUndefinedStatus} := \{\text{AuditBlocked}, \text{AuditUnformed}, \text{AuditIndeterminate}\}.$$

For each  $\tau \in \text{FailSeries}$ , define the series-specific audit-undefined reason set and the all-series audit-undefined reason set by

$$\begin{aligned} \text{AuditUndefinedReason}_\tau[\text{Ev}^{\text{fail}}; \lambda] &:= \left\{ (t, q) \in \text{FailTok}_\tau \times \text{AuditUndefinedStatus} \right. \\ &\quad \left. \mid \text{AuditConditionStateOf}[\text{Ev}; \lambda](t) = q \right\}, \\ \text{AuditUndefinedReason}[\text{Ev}^{\text{fail}}; \lambda] &:= \bigcup_{\tau \in \text{FailSeries}} \text{AuditUndefinedReason}_\tau[\text{Ev}^{\text{fail}}; \lambda]. \end{aligned}$$

For any set  $\mathcal{Y}$ , define the tagged audit-result map by

$$\begin{aligned} \text{AuditTaggedResult}_\mathcal{Y} &: \mathcal{Y} \times \mathcal{P}(\text{FailTok} \times \text{AuditUndefinedStatus}) \\ &\longrightarrow (\{\text{AuditValue}\} \times \mathcal{Y}) \\ &\quad \sqcup (\{\text{AuditReason}\} \times [\mathcal{P}(\text{FailTok} \times \text{AuditUndefinedStatus}) \setminus \{\emptyset\}]), \end{aligned}$$

$\text{AuditTaggedResult}_\mathcal{Y}(y; R)$

$$:= \begin{cases} (\text{AuditValue}, y), & R = \emptyset, \\ (\text{AuditReason}, R), & R \neq \emptyset. \end{cases}$$

The tags **AuditValue** and **AuditReason** are distinct. In what follows, the subscript  $\mathcal{V}$ , which is uniquely determined by the object type, is omitted.

Define the four tagged diagnostic components by

$$\begin{aligned}
& \text{AuditFailedPosition}[\text{Ev}^{\text{fail}}; \lambda] \\
& \quad := \text{AuditTaggedResult} \left( \begin{array}{c} \text{FailedTok}[\text{Ev}; \lambda]; \\ \text{AuditUndefinedReason}[\text{Ev}^{\text{fail}}; \lambda] \end{array} \right), \\
& \text{AuditFailureSignature}[\text{Ev}^{\text{fail}}; \lambda] \\
& \quad := \text{AuditTaggedResult} \left( \begin{array}{c} \Sigma^{\text{fail}}[\text{Ev}; \lambda]; \\ \text{AuditUndefinedReason}[\text{Ev}^{\text{fail}}; \lambda] \end{array} \right), \\
& \text{AuditApproxBoundary}[\text{Ev}^{\text{fail}}; \lambda] \\
& \quad := \text{AuditTaggedResult} \left( \begin{array}{c} \text{BdryData}^{\text{approx}}[\text{Ev}]; \\ \text{AuditUndefinedReason}_{\text{approx}}[\text{Ev}^{\text{fail}}; \lambda] \end{array} \right), \\
& \text{AuditStructuralBurden}[\text{Ev}^{\text{fail}}; \lambda] \\
& \quad := \text{AuditTaggedResult} \left( \begin{array}{c} \text{BurdenRep}^{\text{struct}}[\text{Ev}; \lambda]; \\ \text{AuditUndefinedReason}[\text{Ev}^{\text{fail}}; \lambda] \end{array} \right).
\end{aligned}$$

Define the audit-state report and audit-diagnostic report by

$$\begin{aligned}
& \text{AuditStateReport}[\text{Ev}^{\text{fail}}; \lambda] \\
& \quad \text{AuditGroupStatus}[\text{Ev}; \lambda], \\
& \quad := \left( \begin{array}{c} (\text{AuditConditionStateOf}[\text{Ev}; \lambda](t))_{t \in \text{FailTok}}, \\ \text{AuditIndeterminateTok}[\text{Ev}; \lambda], \\ \text{AuditIndeterminateSig}[\text{Ev}; \lambda], \\ \text{AuditUndefinedReason}[\text{Ev}^{\text{fail}}; \lambda] \end{array} \right), \\
& \text{AuditDiagnosticReport}[\text{Ev}^{\text{fail}}; \lambda] \\
& \quad \text{AuditFailedPosition}[\text{Ev}^{\text{fail}}; \lambda], \\
& \quad := \left( \begin{array}{c} \text{AuditFailureSignature}[\text{Ev}^{\text{fail}}; \lambda], \\ \text{AuditApproxBoundary}[\text{Ev}^{\text{fail}}; \lambda], \\ \text{AuditStructuralBurden}[\text{Ev}^{\text{fail}}; \lambda] \end{array} \right).
\end{aligned}$$

Define the five-valued audit-report object by

$$\begin{aligned}
& \text{AuditReport}[\text{Ev}^{\text{fail}}; \lambda] := \left( \begin{array}{c} \text{AuditData}[\text{Ev}^{\text{fail}}], \\ \text{AuditStateReport}[\text{Ev}^{\text{fail}}; \lambda], \\ \text{AuditDiagnosticReport}[\text{Ev}^{\text{fail}}; \lambda] \end{array} \right).
\end{aligned}$$

**Definition 6.181** (Audit-Report Generation Map). Take a scale-augmented complete evaluation specification  $(\text{Ev}, \lambda) \in \text{EvSpec}_\Lambda$ . Use the six-failure-series family of evaluation specifications  $\text{Ev}^{\text{fail}}$  of Definition 6.10.

Define the audit-report generation map by

$$\begin{aligned} \text{AuditReportOf} : & \left\{ (\text{AuditData}[\text{Ev}^{\text{fail}}], \lambda) \right. \\ & \quad \left. \begin{array}{l} \text{Ev} \in \text{EvSpec}, \lambda \in \Lambda \end{array} \right\} \\ & \longrightarrow \left\{ \text{AuditReport}[\text{Ev}^{\text{fail}}; \lambda] \right. \\ & \quad \left. \begin{array}{l} \text{Ev} \in \text{EvSpec}, \lambda \in \Lambda \end{array} \right\}, \\ \text{AuditReportOf}(\text{AuditData}[\text{Ev}^{\text{fail}}]; \lambda) : &= \text{AuditReport}[\text{Ev}^{\text{fail}}; \lambda]. \end{aligned}$$

From the state section of Definition 6.21, the generation-dependency sequence leading to the audit-state report, the audit-diagnostic report, and the five-valued audit-report object is

$$\begin{aligned} \text{FailStateAt}_{\text{Ev}, \lambda} &\longmapsto \text{AuditConditionStateOf}[\text{Ev}; \lambda] \\ &\longmapsto \text{AuditStateReport}[\text{Ev}^{\text{fail}}; \lambda], \\ (\text{AuditStateReport}[\text{Ev}^{\text{fail}}; \lambda], \text{FailLocData}[\text{Ev}; \lambda]) &\longmapsto \text{AuditDiagnosticReport}[\text{Ev}^{\text{fail}}; \lambda], \\ (\text{AuditData}[\text{Ev}^{\text{fail}}]; \lambda) &\xrightarrow{\text{AuditReportOf}} \text{AuditReport}[\text{Ev}^{\text{fail}}; \lambda]. \end{aligned}$$

Table 13: Generation dependencies of audit-report components

| Output component                | Input family                                                                              | Generation rule                                                                    | Basis                              |
|---------------------------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|------------------------------------|
| Audit-condition state family    | Defined four-valued state section and immediate-predecessor condition-satisfaction values | Five-valued lift map<br>$\text{AuditStateLift}$                                    | Definition 6.179                   |
| Audit-state report              | Audit-condition state family                                                              | Serieswise maxima, state fibres, indicator functions, and set of undefined reasons | Definition 6.179, Definition 6.180 |
| Audit-diagnostic report         | Failure-localization data and set of audit-undefined reasons                              | Component tuple of four tagged audit results                                       | Definition 6.172, Definition 6.180 |
| Five-valued audit-report object | Audit datum, audit-state report, and audit-diagnostic report                              | Product of three components                                                        | Definition 6.180                   |

**Proposition 6.182** (Determinacy of Five-Valued Audit-Condition States and Six-Group Audit States). For every  $(\text{Ev}, \lambda) \in \text{EvSpec}_\Lambda$ , and each  $t \in \text{FailTok}$ , the audit-condition state

$$\text{AuditConditionStateOf}[\text{Ev}; \lambda](t)$$

is determined as exactly one element of  $\text{AuditStatus}$ .

For each  $\tau \in \text{FailSeries}$ , the series-group state  $\text{AuditGroupState}_\tau[\text{Ev}; \lambda]$  is uniquely determined, and the six-group audit state  $\text{AuditGroupStatus}[\text{Ev}; \lambda]$  is also uniquely determined.

The serieswise set of audit-undefined reasons  $\text{AuditUndefinedReason}_\tau[\text{Ev}^{\text{fail}}; \lambda]$  and the all-series set of audit-undefined reasons  $\text{AuditUndefinedReason}[\text{Ev}^{\text{fail}}; \lambda]$  are also uniquely



determined. They are independent of the choice of linear extension of the failure-generation dependency order.

*Proof.* By Definition 6.21 and Proposition 6.22, for each  $t \in \text{FailTok}$ , the evaluation argument  $\text{FailArgAt}_{\text{Ev},\lambda}(t)$ , the immediate-predecessor condition-satisfaction value  $\text{FailPredSat}(\text{FailArgAt}_{\text{Ev},\lambda}(t))$ , and the state  $\text{FailStateAt}_{\text{Ev},\lambda}(t)$  are uniquely determined and are independent of the choice of linear extension.

The five cases of the five-valued lift map in Definition 6.179 are mutually disjoint and cover its domain. The audit-condition state is therefore uniquely determined. The lift map of the same definition gives

$$\text{AuditConditionStateOf}[\text{Ev}; \lambda](t) = \text{AuditPass} \iff \text{FailStateAt}_{\text{Ev},\lambda}(t) = \text{sat}.$$

By Proposition 6.15, each  $\text{FailTok}_\tau$  is a nonempty finite set. Because the group-aggregation priority order for audit-condition states is a total order on  $\text{AuditStatus}$ , the image of the audit-condition states of each series has a unique maximum. Thus the series-group states and the six-group audit state are uniquely determined.

Because the audit-condition state family is uniquely determined, the set comprehension in Definition 6.180 uniquely determines the serieswise sets of audit-undefined reasons. Their finite union also uniquely determines the all-series set of audit-undefined reasons.  $\square$

**Theorem 6.183** (Operational-Audit Completeness Relative to Input Evidence). Take a scale-augmented complete evaluation specification  $(\text{Ev}, \lambda) \in \text{EvSpec}_\Lambda$ . Use the six-failure-series family of evaluation specifications  $\text{Ev}^{\text{fail}}$  of Definition 6.10.

For the operational audit datum,

$$\text{AuditReportOf}(\text{AuditData}[\text{Ev}^{\text{fail}}]; \lambda) = \text{AuditReport}[\text{Ev}^{\text{fail}}; \lambda]$$

is uniquely determined, as are all its components.

The four component families

$$\begin{array}{cc} \text{AuditContractID}, & \text{AuditInput}[\text{Ev}], \\ \text{AuditEvidence}[\text{Ev}^{\text{fail}}], & \text{AuditScaleData}[\text{Ev}_{\text{approx}}^{\text{fail}}] \end{array}$$

that constitute the audit computation core respectively identify the audit convention, specify the evaluation target and formation states, provide the six-series condition evaluations, and supply the scale order, defect quantities, and admissible thresholds.

*Proof.* By Definition 6.178, the components of the audit computation core are the audit contract, audit input, six-series family of condition evidence, and scale-audit datum.

For each  $\tau \in \text{FailSeries}$  and  $i \in I_\tau^{\text{fail}}$ , the family of condition evidence provides the typed condition evidence corresponding to  $\text{FailSpecAt}_\tau[\text{Ev}; \lambda]$  from the current scale-augmented evaluation specification. Definition 6.179 determines the audit-condition state from the evaluation state and immediate-predecessor condition-satisfaction value in that definition.

By Proposition 6.182, all audit-condition states, series-group states, and sets of audit-undefined reasons are uniquely determined and are independent of the choice of linear extension of the failure-generation dependency order. The state fibres and indicator functions of Definition 6.179 also uniquely determine the set of indeterminate positions and the indeterminate signature.

Definition 6.172 uniquely determines the set of first failure positions, the six-series failure signature, the approximation-boundary data, and the structural-burden report.

The tagged audit-result map of Definition 6.180 returns the value component when the set of audit-undefined reasons is empty and the reason set when it is nonempty. Consequently, the four tagged diagnostic components, the audit-state report, the audit-diagnostic report, and the audit report are uniquely determined in sequence. Definition 6.181 then gives the equality stated in the theorem.  $\square$

**Proposition 6.184** (Covariance of Audit-Report Generation under Audit-Data Isomorphisms). For each  $a \in \{0, 1\}$ , take a complete evaluation specification  $\text{Ev}^{(a)} \in \text{EvSpec}$  and an approximation scale  $\lambda^{(a)} \in \Lambda^{(a)}$ , and let the corresponding six-failure-series family of evaluation specifications be

$$\text{Ev}^{\text{fail},(a)} = \left( \text{Ev}_{\tau}^{\text{fail},(a)} \right)_{\tau \in \text{FailSeries}}.$$

Assume that the typed audit-data isomorphism

$$\begin{aligned} \text{AuditDataIso}^{01} : \text{AuditData} \left[ \text{Ev}^{\text{fail},(0)} \right] \\ \xrightarrow{\cong} \text{AuditData} \left[ \text{Ev}^{\text{fail},(1)} \right] \end{aligned}$$

preserves the following structures.

1. The condition predicates, input types, scale types, and audit-report generation rules identified by the audit contract.
2. The six failure series, series index sets, failure-position tokens, and failure-generation dependency order.
3. The formation-state-augmented audit input, membership of the serieswise condition evidence, condition-evaluation values, measurement intervals, and confidence sets.
4. The total scale order, approximation scale, codomains of defect quantities, admissible thresholds, and approximation boundaries.
5. The correspondences among the audit-input identifiers, condition-evidence identifiers, content hashes, transformation histories, and evidence locations in the evidence-provenance certificates.
6. The correspondence from failure-position tokens to structural-target tokens.

Let the correspondence of approximation scales be  $\lambda^{(0)} \mapsto \lambda^{(1)}$ .

Then there exists a unique induced audit-report isomorphism

$$\begin{aligned} \text{AuditReportIso}^{01} : \text{AuditReport} \left[ \text{Ev}^{\text{fail},(0)}; \lambda^{(0)} \right] \\ \xrightarrow{\cong} \text{AuditReport} \left[ \text{Ev}^{\text{fail},(1)}; \lambda^{(1)} \right] \end{aligned}$$

such that

$$\begin{aligned} & \text{AuditReportIso}^{01} \left( \text{AuditReportOf} \left( \text{AuditData} \left[ \text{Ev}^{\text{fail},(0)} \right]; \lambda^{(0)} \right) \right) \\ &= \text{AuditReportOf} \left( \text{AuditData} \left[ \text{Ev}^{\text{fail},(1)} \right]; \lambda^{(1)} \right). \end{aligned}$$

This isomorphism assigns to the audit-condition state family, six-group audit state, set of audit-undefined reasons, failure-position audit component, failure-signature audit component, set of indeterminate positions, indeterminate signature, approximation-boundary audit component, and structural-burden audit component the correspondences induced by the assumed correspondences.

*Proof.* The typed audit-data isomorphism gives isomorphisms of the six failure series, series index sets, failure-position tokens, and failure-generation dependency orders. The roots, immediate predecessors, and linear extensions of the failure-generation dependency orders therefore correspond.

Preservation of the audit contracts, formation-state-augmented audit inputs, and serieswise condition evidence preserves, for each pair of corresponding failure-position tokens, the value recording whether all immediate-predecessor condition states equal **AuditPass**, the formation decision, the condition-evidence satisfaction value, and the condition-evidence failure value. Applying the finite-DAG recursion of Proposition 6.182 to corresponding linear extensions preserves every value of the five-valued audit-condition state of Definition 6.179 between corresponding failure-position tokens.

Because the state-priority order and the failure-position-token set of each series are preserved, **AuditGroupStatus**, which consists of the serieswise maxima, is also preserved. Because the correspondences between the audit-undefined state values and the failure-position tokens are preserved, they induce the serieswise sets of audit-undefined reasons and the all-series set of audit-undefined reasons.

The correspondence of audit-condition states and failure-position tokens induces the set of failure positions and the set of indeterminate positions. The correspondences of the six series and series index sets induce the failure signature and indeterminate signature.

Preservation of the total scale order, approximation scale, codomains of defect quantities, admissible thresholds, and approximation boundaries induces the approximation-boundary data and approximation-boundary audit component. The isomorphism of failure-generation dependency orders preserves the principal upper set generated by each failure-position token. Together with preservation of the correspondence from failure-position tokens to structural-target tokens, this induces the failure-position-side structural burden, approximation-scale-boundary-side structural burden, and structural-burden audit component.

Combining componentwise the correspondences among the components of the evidence-provenance certificates, the audit-data isomorphism, and the induced correspondences above gives the induced audit-report map stated in the proposition. Each component correspondence is bijective, and its inverse preserves the same structures; hence this map is an isomorphism.

Definition 6.181 determines the audit report from these components. The induced correspondences of the components make this isomorphism satisfy the commutative equation in the proposition.

Each component of the audit report is uniquely determined by the correspondence supplied by the audit-data isomorphism. The induced audit-report isomorphism having the component correspondences above is therefore also unique.  $\square$

**Definition 6.185** (Finite Operational Audit Fragment). Take a scale-augmented complete evaluation specification  $(\text{Ev}, \lambda) \in \text{EvSpec}_\Lambda$ . Use the six-failure-series family of evaluation specifications  $\text{Ev}^{\text{fail}}$  of Definition 6.10.

The finite operational audit fragment  $\text{AuditFragment}[\text{Ev}^{\text{fail}}; \lambda]$  consists of the following finite components of the operational audit datum  $\text{AuditData}[\text{Ev}^{\text{fail}}]$ .

1. The condition predicates, input types, scale types, and audit-report generation rules specified by the audit contract.
2. The finitely many values used to evaluate all failure conditions for the operational construction, complete evaluation specification, and formation-state-augmented inputs of the three series.
3. For each  $\tau \in \mathbf{FailSeries}$  and  $i \in I_\tau^{\text{fail}}$ , the typed condition evidence, condition-evidence satisfaction value, and condition-evidence failure value.
4. The finite relation table of the failure-generation dependency order.
5. The approximation scale  $\lambda$ , together with the scale-order comparisons, defect-quantity values, codomains of the defect quantities, admissible thresholds, and approximation-boundary data used in its evaluation.
6. For each  $t \in \mathbf{FailTok}$ , the principal upper set and the single-token structural burden

$$\begin{aligned} & \text{Desc}_{\text{fail}, \text{dep}}(t), \\ & \text{StructBurden}(\{t\}). \end{aligned}$$

7. The finite part of the evidence-provenance certificate corresponding to each component above.

For each  $t \in \mathbf{FailTok}$ , retain the finitely many typed values required for condition evaluation as the three components

$$\left( \text{FailInputSat}^{\text{form}}, \text{FailInputSat}^{\text{eval}}, \text{FailEval} \right) \left( \text{FailArgAt}_{\text{Ev}, \lambda}(t) \right).$$

**Proposition 6.186** (Finite Audit Procedure with Five-Valued Halting Output).

Take a scale-augmented complete evaluation specification  $(\text{Ev}, \lambda) \in \mathbf{EvSpec}_\Lambda$ . Use the six-failure-series family of evaluation specifications  $\text{Ev}^{\text{fail}}$  of Definition 6.10. Given the finite operational audit fragment  $\mathbf{AuditFragment}[\text{Ev}^{\text{fail}}; \lambda]$ , there exists an audit procedure that, after finitely many decisions, halts, for every  $t \in \mathbf{FailTok}$ , with exactly one element of  $\mathbf{AuditStatus}$ .

From these halting outputs and the approximation-boundary data and structural images retained in the finite operational audit fragment, finitely many set operations, maximum operations, and tagged-component generations yield every component of  $\mathbf{AuditStateReport}[\text{Ev}^{\text{fail}}; \lambda]$  and  $\mathbf{AuditDiagnosticReport}[\text{Ev}^{\text{fail}}; \lambda]$ . If the operational audit datum  $\mathbf{AuditData}[\text{Ev}^{\text{fail}}]$  is also supplied, its tuple with these two reports gives  $\mathbf{AuditReport}[\text{Ev}^{\text{fail}}; \lambda]$ .

*Proof.* By Definition 6.10 and Proposition 6.15,  $\mathbf{FailTok}$  is a finite set with  $\mathbf{FailCount}^{\text{max}}$  elements. By Proposition 6.18, the failure-generation dependency order is a partial order on this finite set. The failure-generation dependency DAG therefore has a finite linear extension.

Process each

$$t \in \mathbf{FailTok}$$

along that linear extension. From the finite operational audit fragment, obtain the immediate-predecessor condition-satisfaction value, formation value, evaluation-input satisfaction value, and condition-evaluation value for

$$\text{FailArgAt}_{\text{Ev}, \lambda}(t).$$

Definition 6.21 determines the four-valued evaluation state, and the five-valued lift map of Definition 6.179 determines the audit-condition state. Each processing step consists of finitely many membership, equality, or truth-value decisions and halts in one of the five mutually disjoint cases.

Because the failure-position-token set is finite, generation of all audit-condition states also halts after finitely many decisions. By Proposition 6.182, the resulting state family is unique and is independent of the choice of linear extension.

Each series is a nonempty finite set, and the audit-state priority order is a total order; hence the six-group audit state is obtained by finitely many maximum operations. The set of audit-undefined reasons, set of indeterminate positions, and indeterminate signature are obtained by set comprehensions, a finite union, and indicator functions on the finite state family.

The audit-state report consists of the components obtained above. The condition-state family determines the failure-position set and failure signature. The approximation-boundary data and single-token structural burdens retained in the finite operational audit fragment determine the structural-burden report by finite unions. Applying the tagged audit-result maps of Definition 6.180 to these values and the set of audit-undefined reasons yields every component of the audit-diagnostic report. If the operational audit datum is also supplied, its tuple with these two reports is the audit report of the same definition.  $\square$

**Corollary 6.187** (Five-Valued Operational Auditability of the Time-Reference Formation Conditions).

Given the operational audit datum  $\text{AuditData}[\text{Ev}^{\text{fail}}]$  and an approximation scale  $\lambda \in \Lambda$ , the five-valued audit states of all failure conditions, six-group audit state, set of audit-undefined reasons, failure-position audit component, failure-signature audit component, set of indeterminate positions, indeterminate signature, approximation-boundary audit component, and structural-burden audit component are uniquely determined.

When the corresponding finite operational audit fragment is given, these components are obtained from the halting output of a finite audit procedure together with that fragment. The audit-state and audit-diagnostic reports obtained from two finite operational audit fragments corresponding under a typed audit-data isomorphism correspond under the respective component maps of  $\text{AuditReportIso}^{01}$ .

*Proof.* By Theorem 6.183, all components of the audit report are uniquely determined by the operational audit datum and approximation scale.

By Proposition 6.186, the condition evaluation for every failure-position token contained in the finite operational audit fragment halts after finitely many decisions with one of the five-valued audit states. By the same proposition, every component of the audit-state and audit-diagnostic reports is obtained from all condition states and the data retained in the finite operational audit fragment by finitely many operations.

By Proposition 6.184, a typed audit-data isomorphism induces a unique audit-report isomorphism. The audit-report isomorphism therefore preserves the component correspondences that depend on the representation of the finite operational audit fragment.  $\square$

### Five-valued audit report generation

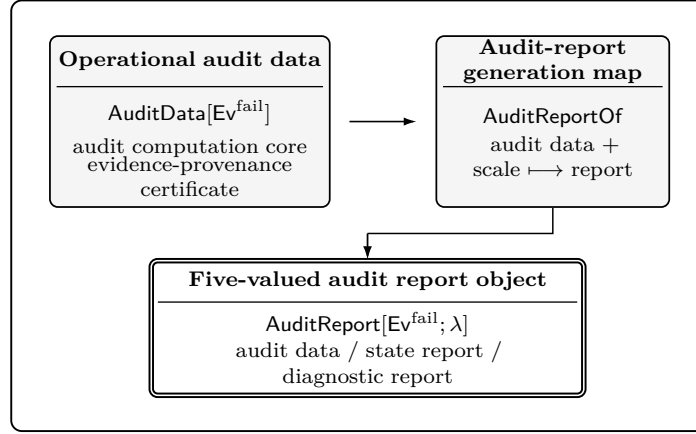


Figure 15: Map generating a five-valued audit-report object from operational audit data. The audit data, comprising the audit computation core and evidence-provenance certificate, are supplied together with the scale to the audit-report generation map **AuditReportOf**, which sends them to **AuditReport**, whose components are the audit data, audit-state report, and audit-diagnostic report.

### Finite audit procedure

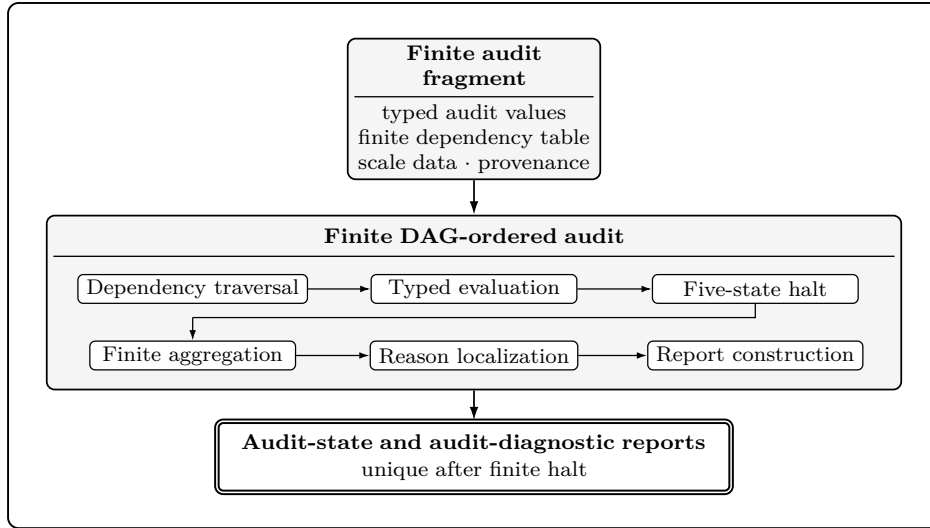


Figure 16: Finite procedure generating audit-state and audit-diagnostic reports from a finite audit fragment. It receives typed audit values, a finite dependency table, scale and boundary data, and finite evidence-provenance records; it executes dependency-order traversal, typed condition evaluation, five-valued halting, finite aggregation, reason localization, and report generation in finite-DAG order. If the operational audit datum is also supplied, its tuple with the two resulting reports gives **AuditReport**.

## 6.13 Engineering Audit Parameters and Their Correspondence to Failures of Physical-Carrier Specifications

Definition 3.2 of role-typed physical-carrier realization, Definition 6.10 of the failure-position-token system with semantic series, and Definition 6.177 of the operational audit

data for the time-reference construction are taken as inputs. Engineering measurement data constitute the formation inputs for the serieswise audit inputs and typed condition evidence. Each failure condition is evaluated through its corresponding audit-input type and condition-evidence type.

**Definition 6.188** (Engineering Evaluation-Specification Type). Define the typed tuple of the complete evaluation specification, approximation scale, observer-side information-coupling position, and target-specification index used in the engineering audit by

$$\text{EngEvSpec} := \{(\text{Ev}, \lambda, o, \zeta) \in \text{EvSpec} \times \Lambda \times U \times Z_0 \mid \text{Ev} = (\text{l}, o, \zeta)\}.$$

**Definition 6.189** (Closed-Interval Audit Kernel). For a subset  $S$  of a linearly ordered set, define its closed-interval type by

$$\mathcal{I}^{\text{cl}}(S) := \{[\ell, r] \mid \ell, r \in S, \ell \leq r\}.$$

Define the audit kernel for closed intervals over the nonnegative extended reals and admissible values by

$$\begin{aligned} \text{IntervalAudit} : \mathcal{I}^{\text{cl}}(\overline{\mathbb{R}}_{\geq 0}) \times \overline{\mathbb{R}}_{\geq 0} &\longrightarrow \text{AuditStatus}, \\ \text{IntervalAudit}([\ell, r]; \theta) &:= \begin{cases} \text{AuditPass}, & r \leq \theta, \\ \text{AuditFail}, & \theta < \ell, \\ \text{AuditIndeterminate}, & \ell \leq \theta < r \end{cases}. \end{aligned}$$

**Definition 6.190** ( $\mathfrak{D}$ -Derived Recording-Field-of-View Engineering Datum). Let the engineering evaluation specification be  $(\text{Ev}, \lambda, o, \zeta) \in \text{EngEvSpec}$ . Let the  $\mathfrak{D}$ -derived recording-field-of-view set be

$$\text{View}_{\mathfrak{D}, \zeta, o, \lambda}^{\text{rec}}[\text{Ev}] \subseteq \text{Qu}_{\mathfrak{B}, o}^{\text{dyn, can}}.$$

Define the  $\mathfrak{D}$ -derived recording-field-of-view engineering datum by

$$\begin{aligned} \text{ViewData}_{\mathfrak{D}, \zeta, o, \lambda}^{\text{rec}}[\text{Ev}] \\ := (\text{CarrierAt}_{\mathfrak{D}}(\zeta), o, \lambda, \text{View}_{\mathfrak{D}, \zeta, o, \lambda}^{\text{rec}}[\text{Ev}]). \end{aligned}$$

At target-specification position  $\zeta$ , observer-side information-coupling position  $o$ , and approximation scale  $\lambda$ , the fourth component is the set recorded by the  $\mathfrak{D}$ -derived physical recording carrier whose elements are  $\mathfrak{B}$ -derived target-state quotient classes.

**Definition 6.191** (Bulk-Recording and Following-Recording Types). Define the set of two-target complete evaluation specifications by

$$\begin{aligned} \text{EvSpec}^{\text{pair}} := \left\{ (\text{Ev}_1, \text{Ev}_2) \in \text{EvSpec}^2 \mid \exists \mathcal{X}, \exists o \in U, \exists \zeta_1, \zeta_2 \in Z_0, \right. \\ \left. \zeta_1 \neq \zeta_2, \text{Operative}(\mathcal{X}), \right. \\ \left. \bigwedge_{k=1}^2 [\text{Ev}_k = (\text{l}, o, \zeta_k)] \right\}. \end{aligned}$$

Take

$$(\text{Ev}_1, \text{Ev}_2) \in \text{EvSpec}^{\text{pair}}, \quad \lambda \in \Lambda$$

and write  $\text{Ev}_k = (l, o, \zeta_k)$ . Define the bulk-recording condition by

$$\begin{aligned} \text{BulkRecord}_{\mathfrak{D}} &: \text{EvSpec}^{\text{pair}} \longrightarrow \{\perp, \top\}, \\ \text{BulkRecord}_{\mathfrak{D}} [(\text{Ev}_k)_{k=1}^2] &\iff \text{CarrierAt}_{\mathfrak{D}}(\zeta_1) = \text{CarrierAt}_{\mathfrak{D}}(\zeta_2). \end{aligned}$$

For each  $k \in \{1, 2\}$ , define, restricted to the target-specific  $\mathfrak{D}$ -derived recording field of view, the  $\mathfrak{B}$ -typed  $\mathfrak{D}$ -record-code relation by

$$\begin{aligned} \text{RecordCodeRel}_{\mathfrak{B}, o, k}^{\text{dyn, view}} [(\text{Ev}_j)_{j=1}^2; \lambda] \\ := \text{RecordCodeRel}_{\mathfrak{B}, o}^{\text{dyn}} \cap \left( \text{View}_{\mathfrak{D}, \zeta_k, o, \lambda}^{\text{rec}} [\text{Ev}_k] \times \text{RecordCodeVal}_{\mathfrak{D}}^{\text{dyn}} \right). \end{aligned}$$

This relation therefore satisfies

$$\begin{aligned} \text{RecordCodeRel}_{\mathfrak{B}, o, k}^{\text{dyn, view}} [(\text{Ev}_j)_{j=1}^2; \lambda] \\ \subseteq \text{View}_{\mathfrak{D}, \zeta_k, o, \lambda}^{\text{rec}} [\text{Ev}_k] \times \text{RecordCodeVal}_{\mathfrak{D}}^{\text{dyn}}. \end{aligned}$$

Define the following-recording condition by

$$\begin{aligned} \text{FollowingRecord}_{\mathfrak{D}} &: \text{EvSpec}^{\text{pair}} \times \Lambda \longrightarrow \{\perp, \top\}, \\ \text{FollowingRecord}_{\mathfrak{D}} [(\text{Ev}_k)_{k=1}^2; \lambda] \\ &\iff \text{CarrierAt}_{\mathfrak{D}}(\zeta_1) \neq \text{CarrierAt}_{\mathfrak{D}}(\zeta_2) \\ &\quad \wedge \bigwedge_{k=1}^2 \left[ \text{View}_{\mathfrak{D}, \zeta_k, o, \lambda}^{\text{rec}} [\text{Ev}_k] \right. \\ &\quad \left. \subseteq \text{Dom} \left( \text{RecordCodeRel}_{\mathfrak{B}, o, k}^{\text{dyn, view}} [(\text{Ev}_j)_{j=1}^2; \lambda] \right) \right]. \end{aligned}$$

**Definition 6.192** (Engineering Datum for Recording Invasiveness). Let the engineering evaluation specification be  $(\text{Ev}, \lambda, o, \zeta) \in \text{EngEvSpec}$ . Let the correspondence of  $\mathfrak{B}$ -states before and after the invasive action be

$$\text{InvasivenessRel}_{\mathfrak{B}, \zeta, o, \lambda} [\text{Ev}] \subseteq \left( \text{Qu}_{\mathfrak{B}, o}^{\text{dyn, can}} \right)^2.$$

Let the lower endpoint, upper endpoint, and admissible value of the recording-invasiveness defect quantity be

$$\begin{aligned} \text{InvasivenessDef}_{\mathfrak{D}, \zeta, o}^- [\text{Ev}; \lambda], \quad \text{InvasivenessDef}_{\mathfrak{D}, \zeta, o}^+ [\text{Ev}; \lambda], \quad \text{InvasivenessTol}_{\mathfrak{D}, \zeta, o} \in \overline{\mathbb{R}}_{\geq 0}, \\ \text{InvasivenessDef}_{\mathfrak{D}, \zeta, o}^- [\text{Ev}; \lambda] \leq \text{InvasivenessDef}_{\mathfrak{D}, \zeta, o}^+ [\text{Ev}; \lambda]. \end{aligned}$$

Define the recording-invasiveness engineering datum by

$$\begin{aligned} \text{InvasivenessData}_{\mathfrak{D}, \zeta, o, \lambda} [\text{Ev}] \\ := \left( \begin{array}{c} \text{InvasivenessRel}_{\mathfrak{B}, \zeta, o, \lambda} [\text{Ev}], \\ [\text{InvasivenessDef}_{\mathfrak{D}, \zeta, o}^- [\text{Ev}; \lambda], \text{InvasivenessDef}_{\mathfrak{D}, \zeta, o}^+ [\text{Ev}; \lambda]] \\ \text{InvasivenessTol}_{\mathfrak{D}, \zeta, o} \end{array} \right). \end{aligned}$$

Using the closed-interval audit kernel of Definition 6.189, define the quantitative invasiveness-admissibility state by

$$\begin{aligned} \text{InvasivenessAuditState}_{\mathfrak{D}, \zeta, o} [\text{Ev}; \lambda] \\ := \text{IntervalAudit} \left( \begin{array}{c} [\text{InvasivenessDef}_{\mathfrak{D}, \zeta, o}^- [\text{Ev}; \lambda], \text{InvasivenessDef}_{\mathfrak{D}, \zeta, o}^+ [\text{Ev}; \lambda]] \\ \text{InvasivenessTol}_{\mathfrak{D}, \zeta, o} \end{array} \right) \\ \in \text{AuditStatus}. \end{aligned}$$



**Definition 6.193** (Observer-Side Communication Engineering Datum). Define the set of two-target complete evaluation specifications corresponding to the  $\mathfrak{D}$ -distributed type of Definition 3.4 by

$$\begin{aligned} \text{EvSpec}_{\mathfrak{D}}^{\text{pair,distributed}} \\ := \left\{ (\text{Ev}_1, \text{Ev}_2) \in \text{EvSpec}^{\text{pair}} \mid \exists \mathcal{X}, \exists o \in U, \exists \zeta_1, \zeta_2 \in Z_0, \right. \\ \left. \bigwedge_{k=1}^2 [\text{Ev}_k = (l, o, \zeta_k)], \bigwedge_{k=1}^2 [\text{CarrierAt}_{\mathfrak{D}}(\zeta_k) \in \text{Car}_{\mathfrak{D}}], \right. \\ \left. \text{CarrierAt}_{\mathfrak{D}}(\zeta_1) \neq \text{CarrierAt}_{\mathfrak{D}}(\zeta_2) \right\}. \end{aligned}$$

Take

$$(\text{Ev}_1, \text{Ev}_2) \in \text{EvSpec}_{\mathfrak{D}}^{\text{pair,distributed}}, \quad \lambda \in \Lambda$$

and write  $\text{Ev}_k = (l, o, \zeta_k)$ . Let the finite directed multigraph whose vertices are observer-side physical carriers be

$$\begin{aligned} \text{CommGraph}_{\mathfrak{D}} &:= \left( \begin{array}{c} \text{CommVert}_{\mathfrak{D}}, \text{CommEdge}_{\mathfrak{D}}, \\ \text{CommSrc}_{\mathfrak{D}}, \text{CommTgt}_{\mathfrak{D}} \end{array} \right), \\ \text{CommVert}_{\mathfrak{D}} &\subseteq \text{Car}_{\mathfrak{D}}, \\ |\text{CommVert}_{\mathfrak{D}}| &< \infty, \quad |\text{CommEdge}_{\mathfrak{D}}| < \infty, \\ \text{CommSrc}_{\mathfrak{D}}, \text{CommTgt}_{\mathfrak{D}} &: \text{CommEdge}_{\mathfrak{D}} \longrightarrow \text{CommVert}_{\mathfrak{D}}, \\ \bigwedge_{k=1}^2 &[\text{CarrierAt}_{\mathfrak{D}}(\zeta_k) \in \text{CommVert}_{\mathfrak{D}}]. \end{aligned}$$

Let the map assigning a physical communication carrier to each communication edge be

$$\text{CommCarrier}_{\mathfrak{D}} : \text{CommEdge}_{\mathfrak{D}} \longrightarrow \text{Car}_{\text{comm}}.$$

Define the observer-side set of communication walks and set of closed communication walks by

$$\begin{aligned} \text{CommWalk}_{\mathfrak{D}} \\ := \bigsqcup_{n \in \mathbb{N}_{\geq 1}} \left\{ (e_1, \dots, e_n) \in (\text{CommEdge}_{\mathfrak{D}})^n \mid \bigwedge_{r=1}^{n-1} [\text{CommTgt}_{\mathfrak{D}}(e_r) = \text{CommSrc}_{\mathfrak{D}}(e_{r+1})] \right\}, \\ \text{CommClosedWalk}_{\mathfrak{D}} \\ := \left\{ (e_1, \dots, e_n) \in \text{CommWalk}_{\mathfrak{D}} \mid \text{CommSrc}_{\mathfrak{D}}(e_1) = \text{CommTgt}_{\mathfrak{D}}(e_n) \right\}. \end{aligned}$$

Define the relation that holds when a positive-length directed communication walk exists in at least one direction by

$$\begin{aligned} \text{CommLinked}_{\mathfrak{D}} &\subseteq (\text{CommVert}_{\mathfrak{D}})^2, \\ (v, w) &\in \text{CommLinked}_{\mathfrak{D}} \\ \iff \exists n \in \mathbb{N}_{\geq 1}, \exists (e_1, \dots, e_n) &\in \text{CommWalk}_{\mathfrak{D}}, \\ (\text{CommSrc}_{\mathfrak{D}}(e_1), \text{CommTgt}_{\mathfrak{D}}(e_n)) &\in \{(v, w), (w, v)\} \end{aligned}$$

and require

$$(\text{CarrierAt}_{\mathfrak{D}}(\zeta_1), \text{CarrierAt}_{\mathfrak{D}}(\zeta_2)) \in \text{CommLinked}_{\mathfrak{D}}.$$

Let the measurement maps for communication distance, delay, bandwidth, and jitter be

$$\begin{aligned} & \text{CommDistance, CommDelay, CommBandwidth, CommJitter} \\ & : \text{CommEdge}_{\mathfrak{D}} \longrightarrow \mathcal{I}^{\text{cl}}(\overline{\mathbb{R}}_{\geq 0}), \end{aligned}$$

and let the measurement maps for loss rate and path integrity be

$$\text{CommLossRate, CommPathIntegrity} : \text{CommEdge}_{\mathfrak{D}} \longrightarrow \mathcal{I}^{\text{cl}}([0, 1]).$$

Let the communication-edge-type map be

$$\text{CommEdgeType} : \text{CommEdge}_{\mathfrak{D}} \longrightarrow \{\text{SourceIdEdge, TransportEdge}\}.$$

For each  $\bullet \in \{\mathfrak{C}, \mathfrak{D}\}$ , let the partial map assigning an extended comparison word to a communication walk be

$$\begin{aligned} & \text{CommWord}^{\bullet}[(\text{Ev}_k)_{k=1}^2] : \text{CommWalk}_{\mathfrak{D}} \\ & \longrightarrow \text{ExtWord}^{\bullet}(\mathcal{X}). \end{aligned}$$

Define the topological component, metric component, and observer-side communication engineering datum by

$$\begin{aligned} & \text{CommTopology}_{\mathfrak{D}} \\ & := \left( \begin{array}{c} \text{CommGraph}_{\mathfrak{D}}, \text{CommCarrier}_{\mathfrak{D}}, \\ \text{CommEdgeType, CommWalk}_{\mathfrak{D}}, \\ \text{CommClosedWalk}_{\mathfrak{D}} \end{array} \right), \\ & \text{CommMetric}_{\mathfrak{D}} \\ & := \left( \begin{array}{c} \text{CommDistance, CommDelay, CommLossRate,} \\ \text{CommBandwidth, CommJitter, CommPathIntegrity} \end{array} \right), \\ & \text{CommData}_{\mathfrak{D}}[(\text{Ev}_k)_{k=1}^2] \\ & := \left( \begin{array}{c} \text{CommTopology}_{\mathfrak{D}}, \\ (\text{CommWord}^{\bullet}[(\text{Ev}_k)_{k=1}^2])_{\bullet \in \{\mathfrak{C}, \mathfrak{D}\}}, \\ \text{CommMetric}_{\mathfrak{D}} \end{array} \right). \end{aligned}$$

**Definition 6.194** (Engineering Audit-Input Datum). Define the set of engineering-measurement type tags by

$$\begin{aligned} \text{EngParamTag} := & \{\text{ViewTag}_{\mathfrak{D}}, \text{RecordConfigTag}_{\mathfrak{D}}, \text{InvasivenessTag}_{\mathfrak{D}}, \\ & \text{CommunicationTag}_{\mathfrak{D}}, \text{CountStepTag}_{\mathfrak{C}}, \text{StabilityTag}_{\mathfrak{D}}, \\ & \text{RecordCellWidthTag}_{\mathfrak{D}}, \text{DiscernibilityTag}_{\mathfrak{B}}, \text{CouplingTag}_{\mathfrak{D}}, \\ & \text{TrajectoryTag, ObservableTag, LawTestTag, DifferentialTestTag}\}. \end{aligned}$$

Let the set of engineering-measurement references be  $\text{EngParamRef}$ , and let the type-tag map, tag-indexed engineering-measurement value types, and engineering-measurement

map be

$$\begin{aligned} \text{EngTagOf} &: \text{EngParamRef} \longrightarrow \text{EngParamTag}, \\ \text{EngValueType}_\rho & \quad (\rho \in \text{EngParamTag}), \\ \text{EngMeasure} & \in \prod_{r \in \text{EngParamRef}} \text{EngValueType}_{\text{EngTagOf}(r)}. \end{aligned}$$

Define the four basic engineering-measurement value types by

$$\begin{aligned} \text{ViewDataType}_\mathfrak{D} & \\ &:= \left\{ \text{ViewData}_{\mathfrak{D}, \zeta, o, \lambda}^{\text{rec}}[\text{Ev}] \mid (\text{Ev}, \lambda, o, \zeta) \in \text{EngEvSpec} \right\}, \\ \text{RecordConfigDataType}_\mathfrak{D} & \\ &:= \left\{ \left( \begin{array}{c} \text{BulkRecord}_\mathfrak{D}[(\text{Ev}_k)_{k=1}^2], \\ \text{FollowingRecord}_\mathfrak{D}[(\text{Ev}_k)_{k=1}^2; \lambda] \end{array} \right) \mid \begin{array}{c} (\text{Ev}_1, \text{Ev}_2) \in \text{EvSpec}^{\text{pair}}, \\ \lambda \in \Lambda \end{array} \right\}, \\ \text{InvasivenessDataType}_\mathfrak{D} & \\ &:= \left\{ \text{InvasivenessData}_{\mathfrak{D}, \zeta, o, \lambda}[\text{Ev}] \mid (\text{Ev}, \lambda, o, \zeta) \in \text{EngEvSpec} \right\}, \\ \text{CommunicationDataType}_\mathfrak{D} & \\ &:= \left\{ \text{CommData}_\mathfrak{D}[(\text{Ev}_k)_{k=1}^2] \mid (\text{Ev}_1, \text{Ev}_2) \in \text{EvSpec}_\mathfrak{D}^{\text{pair, distributed}} \right\}. \end{aligned}$$

Define the four components of the corresponding family of value types by

$$\begin{aligned} \text{EngValueType}_{\text{ViewTag}_\mathfrak{D}} & \\ &:= \text{ViewDataType}_\mathfrak{D}, \\ \text{EngValueType}_{\text{RecordConfigTag}_\mathfrak{D}} & \\ &:= \text{RecordConfigDataType}_\mathfrak{D}, \\ \text{EngValueType}_{\text{InvasivenessTag}_\mathfrak{D}} & \\ &:= \text{InvasivenessDataType}_\mathfrak{D}, \\ \text{EngValueType}_{\text{CommunicationTag}_\mathfrak{D}} & \\ &:= \text{CommunicationDataType}_\mathfrak{D}. \end{aligned}$$

For a complete evaluation specification  $\text{Ev} \in \text{EvSpec}$ , let its six-failure-series family of evaluation specifications be  $\text{Ev}^{\text{fail}}$ . Define the engineering audit-input datum by

$$\begin{aligned} \text{EngAuditInput}[\text{Ev}^{\text{fail}}] &:= \left( \text{EngParamTag}, \text{EngParamRef}, \right. \\ &\quad \text{EngTagOf}, \\ &\quad (\text{EngValueType}_\rho)_{\rho \in \text{EngParamTag}}, \\ &\quad \left. \text{EngMeasure}, \text{AuditCert}[\text{Ev}^{\text{fail}}] \right). \end{aligned}$$

**Definition 6.195** (Engineering-Implementation Auxiliary Audit Data). Define the engineering-implementation auxiliary-audit flag type by

$$\begin{aligned} &\text{EngAuxFlagType} \\ &:= \left\{ \text{MissingFieldFlag}, \text{ProvenanceInconsistencyFlag}, \right. \\ &\quad \text{RecordCellInconsistencyFlag}, \\ &\quad \text{ExternalTimeContaminationFlag}, \text{NonmonotoneDefectFlag}, \\ &\quad \text{UncertaintyOverlapFlag}, \\ &\quad \text{ContractVersionMismatchFlag}, \text{NumericalSolverFailureFlag}, \\ &\quad \left. \text{DataGapFlag} \right\}. \end{aligned}$$

For a complete evaluation specification  $\text{Ev} \in \text{EvSpec}$ , let the set of engineering-implementation auxiliary-audit flags that hold be  $\text{EngAuxFlagSet}[\text{Ev}] \subseteq \text{EngAuxFlagType}$ .

For each  $\text{EngAuxFlag}^{(0)} \in \text{EngAuxFlagSet}[\text{Ev}]$ , let the map assigning the nonempty set of engineering-measurement references used in its determination be

$$\begin{aligned} &\text{EngAuxFlagEvidenceRef} : \text{EngAuxFlagSet}[\text{Ev}] \\ &\longrightarrow \mathcal{P}(\text{EngParamRef}) \setminus \{\emptyset\}. \end{aligned}$$

Define the engineering-implementation auxiliary audit data by

$$\begin{aligned} &\text{EngAuxAuditData}[\text{Ev}] \\ &:= \left( \text{EngAuxFlagSet}[\text{Ev}], \text{EngAuxFlagEvidenceRef} \right). \end{aligned}$$

Multiple auxiliary-audit flags may hold simultaneously for the same complete evaluation specification. Each flag is accompanied by the nonempty set of engineering-measurement references that grounds its determination.

**Definition 6.196** (Engineering Auxiliary-Audit Effect Types and Effect Relation). Define the auxiliary-audit effect type by

$$\begin{aligned} &\text{EngAuxEffectType} \\ &:= \left\{ \text{FormInputUnsatisfiedEffect}, \text{EvalInputUnsatisfiedEffect}, \right. \\ &\quad \text{InputAdmissibilityViolationEffect}, \text{BoundaryMethodRequirementEffect}, \\ &\quad \left. \text{ExactConditionEvidenceRequirementEffect} \right\}. \end{aligned}$$

Let the relation between auxiliary-audit flags and auxiliary-audit effects be

$$\begin{aligned} &\text{EngAuxEffectRel} \\ &\subseteq \text{EngAuxFlagType} \times \text{EngAuxEffectType}. \end{aligned}$$

**Definition 6.197** (Engineering Auxiliary-Audit Effect Evidence). Let the scale-augmented complete evaluation specification be  $(\text{Ev}, \lambda) \in \text{EvSpec}_\Lambda$ . Define the typed pairs of auxiliary-audit flags and effects by

$$\begin{aligned} &\text{EngAuxEffectPair}[\text{Ev}] \\ &:= \left\{ \left( \begin{array}{c} \text{EngAuxFlag}^{(0)}, \\ \text{EngAuxEffect}^{(0)} \end{array} \right) \in \text{EngAuxEffectRel} \right. \\ &\quad \left. \left| \text{EngAuxFlag}^{(0)} \in \text{EngAuxFlagSet}[\text{Ev}] \right. \right\}. \end{aligned}$$

Let the candidate type and validation predicate for typed auxiliary-audit effect evidence be

$$\begin{aligned} \text{EngAuxEffectEvidenceType}[\text{Ev}; \lambda] &:= \text{EngAuxEffectPair}[\text{Ev}] \times \text{FailTok}, \\ \text{EngAuxEffectEvidenceValid}_{\text{Ev}, \lambda} : \text{EngAuxEffectEvidenceType}[\text{Ev}; \lambda] &\longrightarrow \{\perp, \top\}. \end{aligned}$$

This predicate takes the value  $\top$  if and only if the evidence references, the target failure-position token, and the corresponding input or condition evidence have compatible types and the referenced evidence substantiates the applicable effect conditions below.

1. For an unsatisfied formation input or evaluation input, the corresponding input-satisfaction value at the target token is in fact  $\perp$ .
2. A requirement for exact condition evidence identifies the typed evidence needed to evaluate the target condition and the condition that makes this evidence necessary.
3. An input-admissibility violation identifies where an external-time value is used as a premise of a generation condition.
4. A requirement for a boundary-handling method has a target token in the continuum-approximation series. For each engineering-measurement reference cited, that boundary treatment is necessary to evaluate the target token. If either form of boundary-handling data in Definition 6.198 below is supplied, it is included in the evidence component for the target condition within  $\text{AuditEvidenceAt}[\text{Ev}; \lambda](t)$ . If the method is unformed, the absence of the required evaluation input gives the target token an evaluation-input satisfaction value of  $\perp$ .

Here  $t$  is the candidate's target token. Verifying a requirement for condition evidence identifies what evidence is required; it does not determine whether the condition holds. Define the set of verified auxiliary-audit effect evidence by

$$\begin{aligned} &\text{EngAuxEffectEvidence}[\text{Ev}; \lambda] \\ &:= \left( \text{EngAuxEffectEvidenceValid}_{\text{Ev}, \lambda} \Big|_{\text{EngAuxEffectEvidenceType}[\text{Ev}; \lambda]} \right)^{-1} [\{\top\}]. \end{aligned}$$

Table 14: Engineering-implementation auxiliary-audit flags and typed audit effects

| Auxiliary-audit flag                                                                              | Auxiliary-audit effect                                                         | Required typed evidence                                                                                                                                                                                                              |
|---------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Missing required field<br>Data gap                                                                | Unsatisfied formation<br>input or unsatisfied<br>evaluation input              | Correspondence between the<br>target failure-position token and<br>missing input type<br>Correspondence between the<br>target failure-position token and<br>missing data type                                                        |
| Provenance inconsistency<br>Uncertainty-interval<br>overlap<br>Audit-contract version<br>mismatch | Unsatisfied evaluation<br>input                                                | Inconsistent component of the<br>evidence-provenance certificate<br>Intersection of the defect-quantity<br>interval and admissible value<br>Mismatch between the<br>audit-contract identifier and<br>evidence-provenance certificate |
| Record-cell inconsistency                                                                         | Unsatisfied formation<br>input or requirement for<br>exact condition evidence  | Unformed record-code relation or<br>typed record-code conflict<br>evidence                                                                                                                                                           |
| External-time<br>contamination                                                                    | Input-admissibility<br>violation                                               | Expression in which an<br>external-time value occurs in a<br>premise of a generation condition                                                                                                                                       |
| Nonmonotone defect                                                                                | Boundary-handling-<br>method requirement                                       | Counterexample to monotonicity<br>on the scale order                                                                                                                                                                                 |
| Numerical-computation<br>failure                                                                  | Unsatisfied evaluation<br>input or requirement for<br>exact condition evidence | Missing evaluation value or<br>independently constructed exact<br>condition evidence                                                                                                                                                 |

**Definition 6.198** (Boundary-Handling Data for a Nonmonotone Engineering Defect). Let the engineering-measurement reference to which the nonmonotone-defect flag refers be  $r \in \text{EngParamRef}$ . Let the scale domain corresponding to this reference be  $\Lambda_r \subseteq \Lambda$ , and let the engineering-defect map be

$$\text{EngDefect}_r : \Lambda_r \longrightarrow [0, +\infty].$$

For each  $\lambda \in \Lambda_r$ , let

$$\begin{aligned} & \text{EngMonotoneEnvelope}_r(\lambda) \\ & := \sup \left\{ \text{EngDefect}_r(\mu) \mid \begin{array}{l} \mu \in \Lambda_r, \\ \mu \leq_\Lambda \lambda \end{array} \right\}. \end{aligned}$$

When the right-hand side is finite for every  $\lambda \in \Lambda_r$ , call this map finite monotone-envelope data.

Let a finite index set  $K_r$  and a family of subsets

$$(\Lambda_{r,k})_{k \in K_r}$$

satisfy

$$\Lambda_r = \bigsqcup_{k \in K_r} \Lambda_{r,k}.$$

When each subset is order-convex with respect to  $\leq_\Lambda$  and every restriction of the engineering-defect map is monotone, call this family of subsets finite operating-region partition data.

Define the boundary-handling-method type by

$$\text{EngBoundaryMethodType} \\ := \left\{ \begin{array}{l} \text{MonotoneEnvelopeMethod}, \\ \text{OperatingRegionPartitionMethod}, \\ \text{UnformedBoundaryMethod} \end{array} \right\}.$$

Let the boundary-handling method satisfy

$$\begin{aligned} \text{EngBoundaryMethod}(r) \\ \in \text{EngBoundaryMethodType}. \end{aligned}$$

When finite monotone-envelope data are given, permit

$$\begin{aligned} \text{EngBoundaryMethod}(r) \\ = \text{MonotoneEnvelopeMethod}. \end{aligned}$$

When finite operating-region partition data are given, permit

$$\begin{aligned} \text{EngBoundaryMethod}(r) \\ = \text{OperatingRegionPartitionMethod}. \end{aligned}$$

When both kinds of boundary-handling data are unformed, set

$$\begin{aligned} \text{EngBoundaryMethod}(r) \\ = \text{UnformedBoundaryMethod}. \end{aligned}$$

**Proposition 6.199** (Closure of Engineering-Implementation Auxiliary Audit within the Six Series). Let the scale-augmented complete evaluation specification be  $(\text{Ev}, \lambda) \in \text{EvSpec}_\Lambda$ . For any

$$(\text{EngAuxFlag}^{(0)}, \text{EngAuxEffect}^{(0)}, t) \in \text{EngAuxEffectEvidence}[\text{Ev}; \lambda]$$

the following hold:

$$\begin{aligned} \text{EngAuxEffect}^{(0)} &= \text{FormInputUnsatisfiedEffect} \\ &\implies \text{FailInputSat}^{\text{form}}(\text{FailArgAt}_{\text{Ev}, \lambda}(t)) = \perp, \\ \text{EngAuxEffect}^{(0)} &= \text{EvalInputUnsatisfiedEffect} \\ &\implies \text{FailInputSat}^{\text{eval}}(\text{FailArgAt}_{\text{Ev}, \lambda}(t)) = \perp. \end{aligned}$$

When  $\text{EngAuxEffect}^{(0)} = \text{ExactConditionEvidenceRequirementEffect}$ , the required typed condition evidence is identified as the component for evaluating the target condition within  $\text{AuditEvidenceAt}[\text{Ev}; \lambda](t)$ .

For a nonmonotone-defect flag, let

$$\begin{aligned} \text{EngAuxFlag}^{(0)} &= \text{NonmonotoneDefectFlag}, \\ \text{EngAuxEffect}^{(0)} &= \text{BoundaryMethodRequirementEffect}. \end{aligned}$$

If the boundary-handling method for the corresponding

$$r \in \text{EngAuxFlagEvidenceRef}(\text{EngAuxFlag}^{(0)})$$

is `MonotoneEnvelopeMethod` or `OperatingRegionPartitionMethod`, then

$$t \in \text{FailTok}^{\text{approx}}$$

holds, and the boundary-handling data are included in the evidence component for the target condition within  $\text{AuditEvidenceAt}[\text{Ev}; \lambda](t)$ .

If the boundary-handling method is `UnformedBoundaryMethod`, then

$$t \in \text{FailTok}^{\text{approx}}, \\ \text{FailInputSat}^{\text{eval}}(\text{FailArgAt}_{\text{Ev}, \lambda}(t)) = \perp.$$

Define the set of failure positions targeted by verified auxiliary-audit effect evidence by

$$\text{EngAuxEffectTarget}[\text{Ev}; \lambda] \\ := \left\{ t \mid \begin{array}{l} \exists \text{EngAuxFlag}^{(0)}, \exists \text{EngAuxEffect}^{(0)}, \\ (\text{EngAuxFlag}^{(0)}, \text{EngAuxEffect}^{(0)}, t) \in \text{EngAuxEffectEvidence}[\text{Ev}; \lambda] \end{array} \right\}.$$

Then  $\text{EngAuxEffectTarget}[\text{Ev}; \lambda] \subseteq \text{FailTok}$ .

*Proof.* The effect-specific verification conditions in Definition 6.197 give the respective conclusions for unsatisfied formation inputs, unsatisfied evaluation inputs, and requirements for boundary-handling methods. A requirement for exact condition evidence identifies the necessary component within the series evidence data of Definition 6.177. By the candidate type, the third component of each verified item belongs to `FailTok`. Its image, the target set, is therefore a subset of `FailTok`.  $\square$

**Definition 6.200** (Uses of External Time References). Let the set of engineering-measurement references that refer to external-time values be  $\text{ExternalTimeRefSet} \subseteq \text{EngParamRef}$ . Define the type of external-time-reference uses by

$$\text{ExternalTimeUseType} \\ := \{ \text{LogIndexUse}, \text{CalibrationUse}, \text{ProvenanceUse}, \text{GeneratedConditionPremiseUse} \}.$$

Let the external-time-reference-use map be

$$\text{ExternalTimeUse} : \text{ExternalTimeRefSet} \\ \longrightarrow \text{ExternalTimeUseType}.$$

For each reference, its use value is `GeneratedConditionPremiseUse` if and only if the audited formation inputs, evaluation inputs, or condition-evaluation expressions contain a use of its external-time value as a premise. Only references without such a use are assigned one of the other three values according to their actual use. Define the set of external-time references producing input-admissibility violations by

$$\text{ExternalTimeViolationRefSet} \\ := \left\{ r \in \text{ExternalTimeRefSet} \mid \begin{array}{l} \text{ExternalTimeUse}(r) \\ = \text{GeneratedConditionPremiseUse} \end{array} \right\}.$$



**Proposition 6.201** (Noncircularity of Time Generation under External-Time-Reference Audit).

Suppose that  $\text{ExternalTimeViolationRefSet} = \emptyset$ . Then external-time references belong to the engineering audit input as components of log indexing, instrument calibration, or evidence provenance, while the grounds establishing the comparison conditions, representation conditions, dynamical-description conditions, continuum-approximation conditions, law-representation conditions, and differential-consistency conditions are determined by the typed audit inputs and condition evidence of the respective series.

If, on the other hand,

$$\text{ExternalTimeViolationRefSet} \neq \emptyset$$

then every

$$\text{EngParamRef}^{(0)} \in \text{ExternalTimeViolationRefSet}$$

identifies a use as a generation-condition premise corresponding to

$$\text{ExternalTimeContaminationFlag},$$

and provides evidence for

$$\text{InputAdmissibilityViolationEffect}.$$

*Proof.* The use map in Definition 6.200 preserves whether an external-time value is actually used as a generation-condition premise. Consider its four use cases.

For log-index use, the external-time value identifies an engineering-measurement record. For instrument-calibration use, the external-time value is used in an engineering-measurement component accompanied by a calibration provenance. For evidence-provenance use, the external-time value is used in components recording evidence location and transformation history. In these three uses, the six-series condition states are determined from the formation-input satisfaction values, evaluation-input satisfaction values, and condition-evaluation values of Definition 6.179.

For generated-condition-premise use, an external-time value occurs in the grounds establishing a formation input, evaluation input, or condition-evaluation expression of one of the six series. A reference having this use belongs to  $\text{ExternalTimeViolationRefSet}$  and identifies the use corresponding to the external-time-contamination flag of Definition 6.195. By Table 14, the evidence for an input-admissibility violation is this use as a generation-condition premise.

Consequently, when the violation set is empty, the six-series condition states are determined from the typed audit inputs and condition evidence of the respective series. When the violation set is nonempty, each element identifies a use that violates input admissibility.  $\square$

**Definition 6.202** (Engineering Formal-Condition Evidence). Let the scale-augmented complete evaluation specification be  $(\text{Ev}, \lambda) \in \text{EvSpec}_\Lambda$ , and let the corresponding six-failure-series family of evaluation specifications be  $\text{Ev}^{\text{fail}}$ . Define the engineering formal-condition-evidence type by

$$\begin{aligned} & \text{EngEvidenceType}[\text{Ev}; \lambda] \\ & := \bigsqcup_{t \in \text{FailTok}} \left( (\mathcal{P}(\text{EngParamRef}) \setminus \{\emptyset\}) \times \{\text{AuditEvidenceAt}[\text{Ev}; \lambda](t)\} \right. \\ & \quad \left. \times \{\text{AuditCert}[\text{Ev}^{\text{fail}}]\} \right). \end{aligned}$$

For each  $e \in \text{EngEvidenceType}[\text{Ev}; \lambda]$ , write its failure-position-token component and engineering-measurement-reference-set component as

$$\begin{aligned}\text{EngTarget}(e) &\in \text{FailTok}, \\ \text{EngRefs}(e) &\in \mathcal{P}(\text{EngParamRef}) \setminus \{\emptyset\}.\end{aligned}$$

Let the validation predicate for engineering formal-condition evidence be

$$\text{EngEvidenceValid}_{\text{Ev}, \lambda} : \text{EngEvidenceType}[\text{Ev}; \lambda] \longrightarrow \{\perp, \top\}.$$

For any  $e \in \text{EngEvidenceType}[\text{Ev}; \lambda]$ ,  $\text{EngEvidenceValid}_{\text{Ev}, \lambda}(e)$  is the truth value of the conjunction of the following three conditions.

1. For every  $r \in \text{EngRefs}(e)$ ,

$$\text{EngMeasure}(r) \in \text{EngValueType}_{\text{EngTagOf}(r)}.$$

2. The condition-evidence component

$$\text{AuditEvidenceAt}[\text{Ev}; \lambda](\text{EngTarget}(e))$$

contains as components the typed condition evidence required to evaluate

$$\text{FailInputSat}^{\text{form}}, \quad \text{FailInputSat}^{\text{eval}}, \quad \text{FailEval}$$

at the target failure-position token.

3. The evidence-provenance certificate  $\text{AuditCert}[\text{Ev}^{\text{fail}}]$  supplies the correspondence between the set of engineering-measurement references and the typed condition evidence of the target failure-position token.

Define the set of verified engineering formal-condition evidence by

$$\text{EngEvidence}^{\text{ver}}[\text{Ev}; \lambda] := (\text{EngEvidenceValid}_{\text{Ev}, \lambda})^{-1} [\{\top\}].$$

For any  $(\text{Ev}, \lambda) \in \text{EvSpec}_\Lambda$  and  $e \in \text{EngEvidence}^{\text{ver}}[\text{Ev}; \lambda]$ , the set of engineering-measurement references  $\text{EngRefs}(e)$  and the single-token structural burden of the target failure-position token

$$\text{StructBurden}(\{\text{EngTarget}(e)\})$$

are uniquely determined by  $e$ .

Table 15: Correspondence between engineering-measurement components and formal condition-evidence types

| Engineering-measurement component                                                | Formal object formed                                                                                                                   | Object evaluated by the condition evidence                                                                   |
|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| $\mathfrak{D}$ -derived recording field of view                                  | $\mathfrak{B}$ -typed or $\mathfrak{C}/\mathfrak{B}$ -correspondence-typed $\mathfrak{D}$ -record-code relation                        | Formation input, record-cell width, typed record-code conflict                                               |
| Bulk-recording type and following-recording type                                 | Target-specific record-code relation, typed trajectory relation, joint trajectory relation                                             | Relation descent, single-valuedness of trajectories, scalar-time trajectory factorization                    |
| $\mathfrak{D}$ -derived recording invasiveness                                   | Correspondence of $\mathfrak{B}$ -states before and after the invasive action, invasiveness-defect interval, relation-truth-value pair | $\mathfrak{B}$ -state discernibility, preservation of approximation relations, preservation of law relations |
| $\mathfrak{D}$ -derived communication engineering data                           | Comparison word, closed walk, communication-defect interval                                                                            | Comparison transport, loop consistency, $\mathfrak{D}$ -coupling bound                                       |
| $\mathfrak{C}/\mathfrak{D}/\mathfrak{B}/\mathfrak{D}$ -approximation measurement | Continuum-approximation-stage evaluation datum                                                                                         | Nine continuum-approximation conditions                                                                      |
| Trajectory and observable measurements                                           | Dynamical-description-series evaluation specification                                                                                  | Eight dynamical-description conditions                                                                       |
| Law-test measurement                                                             | Law-representation-series evaluation specification                                                                                     | Five law-representation conditions                                                                           |
| Differential-test measurement                                                    | Differential-consistency-series evaluation specification                                                                               | Five differential-consistency and law-form-preservation conditions                                           |

**Definition 6.203** (Six-Series Audit-State Evaluation of Engineering Formal-Condition Evidence). Let the scale-augmented complete evaluation specification be  $(\text{Ev}, \lambda) \in \text{EvSpec}_\Lambda$ . Define the engineering audit-state evaluation by

$$\begin{aligned} \text{EngState}_{\text{Ev}, \lambda} &: \text{EngEvidence}^{\text{ver}}[\text{Ev}; \lambda] \longrightarrow \text{AuditStatus}, \\ \text{EngState}_{\text{Ev}, \lambda}(e) &:= \text{AuditConditionStateOf}[\text{Ev}; \lambda](\text{EngTarget}(e)). \end{aligned}$$

Define the set of audit-failing engineering formal-condition evidence and the set-image of its failure-position-token component by

$$\begin{aligned} \text{EngEvidence}^{\text{fail}}[\text{Ev}; \lambda] &:= (\text{EngState}_{\text{Ev}, \lambda})^{-1} [\{\text{AuditFail}\}], \\ \text{EngFailTok}[\text{Ev}; \lambda] &:= \text{EngTarget} [\text{EngEvidence}^{\text{fail}}[\text{Ev}; \lambda]]. \end{aligned}$$

**Proposition 6.204** (Localization of Engineering Formal-Condition Evidence within the Six Series). For any  $(\text{Ev}, \lambda) \in \text{EvSpec}_\Lambda$ ,

$$\text{EngFailTok}[\text{Ev}; \lambda] \subseteq \text{FailTok}.$$

Further, let

$$r \in \text{EngParamRef}, \quad e_j \in \text{EngEvidence}^{\text{ver}}[\text{Ev}; \lambda], \quad j \in \{0, 1\}.$$

Then

$$\bigwedge_{j=0}^1 [r \in \text{EngRefs}(e_j) \wedge \text{EngState}_{\text{Ev}, \lambda}(e_j) = \text{AuditFail}] \\ \implies \{\text{EngTarget}(e_j) \mid j \in \{0, 1\}\} \subseteq \text{EngFailTok}[\text{Ev}; \lambda].$$

*Proof.* By the disjoint-union type in Definition 6.202, for each item of verified engineering formal-condition evidence  $e$ , its target component  $\text{EngTarget}(e)$  belongs to  $\text{FailTok}$ . Hence, the target image of the audit-failing evidence,  $\text{EngFailTok}[\text{Ev}; \lambda]$ , is a subset of  $\text{FailTok}$ .

By the latter assumption, for every  $j \in \{0, 1\}$ ,  $e_j \in \text{EngEvidence}^{\text{fail}}[\text{Ev}; \lambda]$ . Applying the target map gives  $\text{EngTarget}(e_j) \in \text{EngFailTok}[\text{Ev}; \lambda]$ , which proves the required inclusion.  $\square$

### Six-family localization of engineering evidence

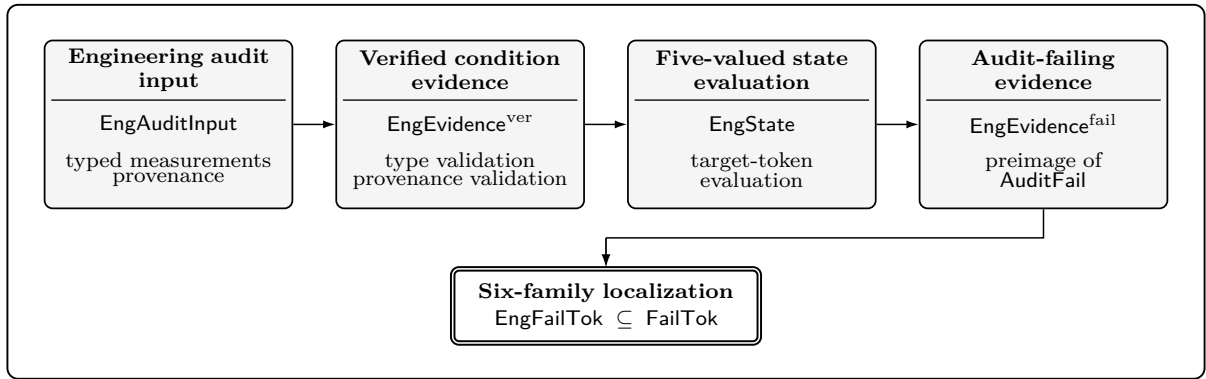


Figure 17: Localization of engineering evidence within the six series under a fixed scale-augmented complete evaluation specification  $(\text{Ev}, \lambda)$ . Common arguments of the objects in the figure are omitted. Verified engineering formal-condition evidence is formed from engineering audit input, the five-valued audit state is evaluated, and audit-failing evidence is extracted. Its target image  $\text{EngFailTok}[\text{Ev}; \lambda]$  is localized as a subset of  $\text{FailTok}$ .

**Definition 6.205** (Engineering-Evidence-Relative Structural Burden by Recording Configuration). For a common object-specification set  $Z_0$ , observer-side information-coupling position set  $U$ , and scale set  $\Lambda$ , take two realized operational configurations  $\mathcal{X}^{(a)}$  and a comparison-rule family  $\mathcal{J}_{\text{cmp}}$  that is well typed for both configurations. For each  $a \in \{0, 1\}$ , apply the preceding definitions separately to

$$\mathbf{l}^{(a)} := (\mathcal{X}^{(a)}, \mathcal{J}_{\text{cmp}}).$$

In the following comparison of the two configurations, a superscript  $(a)$  distinguishes the configuration-specific instances of input-dependent types, data, and maps. The physical-carrier value map  $\text{CarrierAt}_{\mathfrak{S}}^{(a)}$  is likewise determined by the physical realization maps of that configuration. The failure-position type, structural-target type, and structural-burden kernel are common to both configurations. The complete-evaluation-specification set is

$$\text{EvSpec}^{(a)} := \{\mathbf{l}^{(a)}\} \times U \times Z_0.$$

Let

$$\zeta_1, \zeta_2 \in Z_0, \quad \zeta_1 \neq \zeta_2, \quad o \in U, \quad \lambda \in \Lambda, \quad (a, k) \in \{0, 1\} \times \{1, 2\}.$$

Let the complete evaluation specification be

$$\text{Ev}_k^{(a)} = (l^{(a)}, o, \zeta_k) \in \text{EvSpec}^{(a)}$$

and assume

$$\begin{aligned} (\text{Ev}_k^{(a)})_{k=1}^2 &\in \text{EvSpec}^{\text{pair},(a)}, \\ (\text{Ev}_k^{(1)})_{k=1}^2 &\in \text{EvSpec}_{\mathfrak{D}}^{\text{pair,distributed},(1)}. \end{aligned}$$

Write the corresponding family of six-failure-series evaluation specifications as  $\text{Ev}^{\text{fail},(a,k)}$ . Also assume

$$\begin{aligned} \text{BulkRecord}_{\mathfrak{D}}^{(0)}[(\text{Ev}_k^{(0)})_{k=1}^2] &= \top, \\ \text{FollowingRecord}_{\mathfrak{D}}^{(1)}[(\text{Ev}_k^{(1)})_{k=1}^2; \lambda] &= \top. \end{aligned}$$

Define the structural-burden input-tag sets for the bulk-recording and following-recording types by

$$\text{RecordBurdenTag}^{(0)} := \left\{ \begin{array}{c} \text{ViewTag}_{\mathfrak{D}}, \text{RecordConfigTag}_{\mathfrak{D}}, \\ \text{InvasivenessTag}_{\mathfrak{D}}, \text{CountStepTag}_{\mathfrak{C}}, \text{StabilityTag}_{\mathfrak{D}}, \\ \text{RecordCellWidthTag}_{\mathfrak{D}}, \text{DiscernibilityTag}_{\mathfrak{B}} \end{array} \right\},$$

$$\text{RecordBurdenTag}^{(1)} := \text{RecordBurdenTag}^{(0)} \cup \{\text{CommunicationTag}_{\mathfrak{D}}, \text{CouplingTag}_{\mathfrak{D}}\}.$$

Define the verified evidence for each recording configuration by

$$\text{RecordEvidence}^{\text{ver},(a,k)} := \text{EngEvidence}^{\text{ver},(a)}[\text{Ev}_k^{(a)}; \lambda].$$

Define the predicate for evidence that refers to a structural-burden input tag by

$$\text{HasRecordBurdenRef}^{(a)} : \bigcup_{k \in \{1,2\}} \text{RecordEvidence}^{\text{ver},(a,k)} \longrightarrow \{\perp, \top\}$$

and

$$\begin{aligned} \text{HasRecordBurdenRef}^{(a)}(e) &\iff \exists r \in \text{EngRefs}^{(a)}(e), \\ &\quad \text{EngTagOf}^{(a)}(r) \in \text{RecordBurdenTag}^{(a)}. \end{aligned}$$

Define the structural-burden evidence for each recording configuration and its object-indexed form by

$$\begin{aligned} \text{RecordBurdenEvidence}^{(a,k)} &:= \left\{ e \in \text{RecordEvidence}^{\text{ver},(a,k)} \right. \\ &\quad \left. \mid \text{HasRecordBurdenRef}^{(a)}(e) \right\}, \\ \text{RecordBurdenEvidence}^{(a)} &:= \bigsqcup_{k \in \{1,2\}} \\ &\quad \left( \{k\} \times \text{RecordBurdenEvidence}^{(a,k)} \right). \end{aligned}$$

Define the evidence state of each object, the audit-failing evidence, and the recording-configuration-specific set of failure positions by

$$\begin{aligned} \text{RecordState}^{(a,k)} &:= \text{EngState}_{\text{Ev}_k^{(a)}, \lambda}^{(a)} \Big|_{\text{RecordBurdenEvidence}^{(a,k)}}, \\ \text{RecordBurdenEvidence}^{\text{fail},(a,k)} &:= \left( \text{RecordState}^{(a,k)} \right)^{-1} [\{\text{AuditFail}\}], \\ \text{RecordBurdenEvidence}^{\text{fail},(a)} &:= \bigcup_{k \in \{1,2\}} \text{RecordBurdenEvidence}^{\text{fail},(a,k)}, \\ \text{RecordBurdenTok}^{(a)} &:= \text{EngTarget}^{(a)} \left[ \text{RecordBurdenEvidence}^{\text{fail},(a)} \right]. \end{aligned}$$

Using the structural-burden kernel of Definition 6.169, define the structural burden for each recording configuration and its three representations by

$$\begin{aligned}\text{RecordBurden}^{(a)} &:= \text{StructBurden}(\text{RecordBurdenTok}^{(a)}), \\ \text{RecordBurden}^{\text{bulk}} &:= \text{RecordBurden}^{(0)}, \\ \text{RecordBurden}^{\text{follow}} &:= \text{RecordBurden}^{(1)}, \\ \text{RecordBurden}^{\Delta} &:= \text{RecordBurden}^{\text{bulk}} \triangle \text{RecordBurden}^{\text{follow}}.\end{aligned}$$

**Proposition 6.206** (Separation of Engineering-Evidence-Relative Structural Burdens for Bulk-Recording and Following-Recording Types). The two structural burdens are subsets of StructTok, and

$$\text{RecordBurden}^{\text{bulk}} = \text{RecordBurden}^{\text{follow}} \iff \text{RecordBurden}^{\Delta} = \emptyset.$$

For each  $a \in \{0, 1\}$ , if some  $t \in \text{RecordBurdenTok}^{(a)}$  satisfies

$$\text{StructBurden}(\{t\}) \not\subseteq \text{RecordBurden}^{(1-a)}$$

then

$$\text{RecordBurden}^{(a)} \setminus \text{RecordBurden}^{(1-a)} \neq \emptyset.$$

If this condition holds for both  $a = 0$  and  $a = 1$ , the two structural burdens are incomparable under inclusion.

*Proof.* The set of failure positions for each recording configuration is a subset of FailTok. By Definition 6.169, each  $\text{RecordBurden}^{(a)}$  is therefore a subset of StructTok. Applying the equivalence between emptiness of the symmetric difference and equality of two sets to the two structural burdens yields the first equivalence.

For each  $a \in \{0, 1\}$ , nonemptiness of the set difference follows by applying Lemma 6.170 to

$$A = \text{RecordBurdenTok}^{(a)}, \quad B = \text{RecordBurdenTok}^{(1-a)}$$

Nonemptiness of the set difference in both directions establishes that the two structural burdens are incomparable under inclusion.  $\square$

## 7 Audit of Law-Time Use and Nonachievement of Primitive-Time Premises

The physical establishment of a time variable appearing in a physical law requires that its time content be physically generated before its use within the law and the derivation of results. On the basis of formal evidence and statement evidence from the 72 audited references, primitive-time roots, generation dependencies, result dependencies, and physical-time qualification are assessed.

### 7.1 Physical-Time Qualification and Primitive-Time Inputs

This subsection defines the grounding obligations of physical time content, the dependence of result carriers on that content, the distinction between evaluation success and physical-time qualification, and primitive-time inputs. A time variable appearing in a physical

law serves as an argument for state change, evolution, propagation, comparison, and evaluation. Its physical use requires the identity of time values, their before-and-after order, the equivalence of time intervals, scale and unit, correspondence across locations, correspondence with clock states, and correspondence with observational, recording, and experimental procedures.

By Proposition 2.12, Proposition 3.26, and Corollary 4.190, the generation dependence from physical-role inputs to the representation-stage completion conditions for the finite family of object-specification indices

$$Z_0$$

is determined by

$$\begin{aligned} \text{RI}[Z_0] &\longrightarrow \mathcal{X} \\ &\longrightarrow \text{Qu}_{\mathfrak{C}\mathfrak{D}}^{\text{cmp}, \text{can}}(\mathcal{X}; \mathcal{J}_{\text{cmp}}) \\ &\longrightarrow \mathcal{E}^{\text{op}}(\mathcal{X}; \mathcal{J}_{\text{cmp}}) \\ &\longrightarrow \text{Repr}_{\leq 3}(\mathcal{X}; \mathcal{J}_{\text{cmp}}) \end{aligned}$$

For each  $o \in U$ ,  $\zeta \in Z_0$ , let the complete evaluation specification be

$$\text{Ev} = ((\mathcal{X}; \mathcal{J}_{\text{cmp}}), o, \zeta) \in \text{EvSpec}$$

By Theorem 5.138, motion-description operability is determined by

$$\begin{aligned} &\text{ExactOperable}(\mathcal{X}; \mathcal{J}_{\text{cmp}}) \\ &\quad \wedge \text{DynCondAll}[\text{Ev}] \\ &\iff \text{DynOperative}[\text{Ev}] \end{aligned}$$

For the law stage, fix the base law datum and base representation associated with the single-object specification  $\text{Ev}$  in Theorem 5.138. For the differential stage, also fix a differential-evaluation-structure family belonging to the same pair. Law-description operability is determined by

$$\begin{aligned} &\text{DynOperative}[\text{Ev}] \\ &\quad \wedge \text{LawCondAll}[\text{LawEvalDat}_{o, \zeta}] \\ &\iff \text{LawOperative}[\text{LawEvalDat}_{o, \zeta}] \end{aligned}$$

Differential-stage operability is determined by

$$\begin{aligned} &\text{LawOperative}[\text{LawEvalDat}_{o, \zeta}] \\ &\quad \wedge \text{DiffCondAll}[\text{EvalStrFam}_{o, \zeta}^{\text{diff}}] \\ &\iff \text{DiffOperative}[\text{DiffEvalDat}_{o, \zeta}] \end{aligned}$$

In accordance with Definition 6.181, let the corresponding family of six-failure-series evaluation specifications be

$$\text{Ev}^{\text{fail}} = (\text{Ev}_{\tau}^{\text{fail}})_{\tau \in \text{FailSeries}}$$

For an approximation scale

$$\lambda \in \Lambda$$

, the value of the audit-report generation map is

$$\begin{aligned} & \text{AuditReportOf}(\text{AuditData}_{\text{time}}[\text{Ev}^{\text{fail}}]; \lambda) \\ & := \text{AuditReport}[\text{Ev}^{\text{fail}}; \lambda] \end{aligned}$$

. By Theorem 6.183 and Corollary 6.187, this audit-report object is uniquely determined by the operational audit data and approximation scale and contains the states of the time-establishment conditions, invalidation positions, invalidation signatures, approximation boundaries, and structural burdens. The audit-state report and audit-diagnostic report are obtained from the corresponding finite operational audit fragment by finitely many operations.

For time content in a physical description, the audit covers the generation path from nontemporal physical input to that time content, transfer paths that preserve time content between distinct positions in the description, and dependency paths to physical results that use the time content. These paths provide the common basis of determination for the theory-specific synthesis in Chapter 8 and the complete, reference-by-reference audit of all 72 references in the Supplement.

**Definition 7.1** (Finite Audited Reference Set, Time Content, and Result Carriers). Let the finite set consisting of the 72 references subject to audit be

$$\text{Ref}^{\text{audit}}$$

and let

$$|\text{Ref}^{\text{audit}}| = 72$$

. For each  $R \in \text{Ref}^{\text{audit}}$ , write the time contents in the source in their order of occurrence as

$$(\text{TimeContent}_{R,i})_{i=1}^{n_R}$$

. Define the set of time contents of the reference

$$R$$

by

$$\text{TimeContent}_R := \{\text{TimeContent}_{R,i} \mid i \in \{1, \dots, n_R\}\}$$

. Define the set of time contents of all audited references by

$$\text{TimeContent}^{\text{audit}} := \coprod_{R \in \text{Ref}^{\text{audit}}} (\{R\} \times \text{TimeContent}_R)$$

. Write the set of results derived by each reference as

$$\text{Result}_R$$

. Here, a result is an object obtained by a definition, law, inference, calculation, experimental procedure, or observational analysis in the reference and constituting part of the physical claim of that reference.

**Definition 7.2** (Grounding Obligations of Physical Time Content). The correspondence between the operations carried by time content appearing in a physical law and the structures generated through Chapter 4 is defined as follows.



1. The identity of time values is supplied by the comparison equivalence relation

$$\sim_{\bullet}^{\text{cmp}}(\mathcal{X}; \mathcal{J}_{\text{cmp}})$$

and the canonical comparison quotient

$$\text{Qu}_{\bullet}^{\text{cmp,can}}(\mathcal{X}; \mathcal{J}_{\text{cmp}})$$

.

2. The before-and-after order of time values is supplied by the  $\mathfrak{C}/\mathfrak{D}$ -canonical partial order

$$\preceq_{\bullet}^{\text{can}}(\mathcal{X}; \mathcal{J}_{\text{cmp}})$$

and the local and global order-coordinate structures.

3. Within the  $\mathfrak{C}/\mathfrak{D}$ -derived conditional quantitative affine realization

$$\text{QtyAffReal}_{\bullet}(\mathcal{X}; \mathcal{J}_{\text{cmp}})$$

, the equivalence of time intervals is supplied by the quantitative affine difference

$$\text{AffDiff}_{\bullet}^{\text{qty}}$$

.

4. The correspondence between time scale and unit is supplied by a specification of the  $\mathfrak{C}/\mathfrak{D}$ -derived quantitatively linearly ordered field, quantitative affine action, and quantitative affine difference.

5. Correspondence between time contents belonging to distinct information-coupling positions is supplied by the local order-coordinate chart-transition maps contained in the comparison order atlas

$$\text{Atlas}^{\text{cmp,ord}}(\mathcal{X}; \mathcal{J}_{\text{cmp}})$$

.

6. Correspondence of clock states, displays, pulses, and cycles with time content is supplied by the generation sequence through the  $\mathfrak{C}$ -derived arrival map  $\text{Arr}_{\mathfrak{C},o}^{\text{cmp}}$  and the comparison-side recovery map  $\text{Rec}_{\mathfrak{C},o}^{\text{cmp}}$  to the  $\mathfrak{C}$ -derived canonical comparison quotient, order-coordinate structure, and  $\mathfrak{C}$ -derived representation coordinates.

7. Correspondence of observation, recording, and experimental procedures with time content is supplied by the generation sequence through the  $\mathfrak{D}$ -derived arrival map

$$\text{Arr}_{\mathfrak{D},o}^{\text{cmp}},$$

the  $\mathfrak{D}$ -derived record cell

$$W_{\mathfrak{D}}(x_{\mathfrak{D}}),$$

and the  $\mathfrak{D}$ -derived representation image of the record cell

$$\text{CellImg}_{\mathfrak{D},o}^{\text{repr}}(\mathcal{X}; \mathcal{J}_{\text{cmp}})(x_{\mathfrak{D}})$$

to the  $\mathfrak{D}$ -derived canonical comparison quotient, order-coordinate structure, and  $\mathfrak{D}$ -derived representation coordinates.

**Proposition 7.3** (Dependence of Law-Time Operations on Generated Structures). Identity determination, order comparison, interval comparison, scale transformation, correspondence between information-coupling positions, clock correspondence, or observation–recording correspondence performed by a physical law on time content takes as input the respective generated structure specified in Definition 7.2.

Time differentiation, time integration, frequency, period, proper time, and descriptions of evolution by an evolution parameter take as input the generated structures corresponding to the time difference, time order, time scale, and time correspondence that they use.

*Proof.* The identity of time values is determined by the comparison equivalence classes and canonical comparison quotient of Definition 4.68 and Definition 4.89.

The order of time values is determined by the canonical partial order and order-coordinate quotient of Definition 4.101 and Definition 4.116.

Time intervals, scales, and unit correspondences are determined by the quantitatively linearly ordered field, quantitative affine action, and quantitative affine difference of Definition 4.148.

Correspondence between information-coupling positions is determined by the local order-coordinate chart-transition maps contained in the comparison order atlas of Definition 4.128.

Correspondence with clock states, displays, pulses, and cycles is determined by the  $\mathfrak{C}$ -derived source-induced arrival preorder of Definition 4.95, the comparison-side recovery map of Definition 3.21, and the generation sequence comprising the canonical comparison quotient, order-coordinate structure, and representation coordinates.

Correspondence with observation, recording, and experimental procedures is determined by the  $\mathfrak{D}$ -derived arrival map, the  $\mathfrak{D}$ -derived record cell of Definition 3.15, the  $\mathfrak{D}$ -derived representation image of the record cell of Definition 4.159, and the  $\mathfrak{D}$ -derived representation coordinates.

Therefore, every law-time operation takes as input the generated structure corresponding to the time content used by that operation.  $\square$

**Definition 7.4** (Time-Content Transfer Events and Result Prerequisites). For each  $R \in \text{Ref}^{\text{audit}}$ , write the set of time-content transfer events within the reference as

$$\text{TransferEvent}_R$$

. For each  $\epsilon \in \text{TransferEvent}_R$ , represent the source-content set and target-content set of the transfer event

$$\epsilon$$

by

$$\begin{aligned} \text{TransferIn}_{R,\epsilon} &\subseteq \text{TimeContent}_R, \\ \text{TransferOut}_{R,\epsilon} &\subseteq \text{TimeContent}_R \end{aligned}$$

.

For each  $\eta \in \text{Result}_R$ , represent the set of time contents used in deriving the result

$$\eta$$

by

$$\text{ResultPrereq}_{R,\eta} \subseteq \text{TimeContent}_R$$

.

**Definition 7.5** (Within-Reference Time-Content Transfer Reachability). For each  $R \in \text{Ref}^{\text{audit}}$ , define the within-reference time-content transfer-reachability relation by

$$\begin{aligned} & \rightsquigarrow_R^{\text{time,content}} \\ & := \left( \bigcup_{\epsilon \in \text{TransferEvent}_R} \text{TransferIn}_{R,\epsilon} \times \text{TransferOut}_{R,\epsilon} \right)^+ \end{aligned}$$

. The superscript

+

denotes the transitive closure of a binary relation.

**Proposition 7.6** (Condition for Time-Content Participation in Result Derivation). For each

$$R \in \text{Ref}^{\text{audit}}, \quad x \in \text{TimeContent}_R, \quad \eta \in \text{Result}_R$$

, if

$$\begin{aligned} & x \in \text{ResultPrereq}_{R,\eta} \\ & \vee \left[ \exists y \in \text{ResultPrereq}_{R,\eta} : x \rightsquigarrow_R^{\text{time,content}} y \right] \end{aligned}$$

holds, then the time content

$x$

participates in the derivation sequence of the result

$\eta$

.

The same condition determines participation in result derivation for time content used in a time derivative, time integral, frequency, period, proper time, or evolution parameter.

*Proof.* In the first case,

$$x \in \text{ResultPrereq}_{R,\eta}$$

; hence, by Definition 7.4,

$x$

is used directly in deriving the result

$\eta$

.

In the second case, there exists some  $y \in \text{ResultPrereq}_{R,\eta}$  such that

$$x \rightsquigarrow_R^{\text{time,content}} y$$

. By Definition 7.5, a finite sequence of time-content transfer events connects

$x$

to

$y$

. Moreover,  $y \in \text{ResultPrereq}_{R,\eta}$ ; therefore,

$x$

participates, through the time-content transfer sequence, in deriving the result

$\eta$

The same argument applies to a time derivative, time integral, frequency, period, proper time, and evolution parameter when the corresponding time content belongs to the direct result prerequisites or reaches a result prerequisite by transfer.  $\square$

**Definition 7.7** (Evaluation Success and Physical-Time Qualification). The mathematical consistency of each reference means that the conclusions in the reference follow from its adopted premises, definitions, and rules of inference.

Predictive performance means that observable outputs are calculated from the theoretical structure in the reference. Empirical fit means that the correspondence between those outputs and values obtained by experiment or observation satisfies the evaluation conditions in the reference.

A physical-law time variable acquires physical-time qualification when the identity, before-and-after order, intervals, scale, correspondence between information-coupling positions, clock correspondence, and observation–recording correspondence listed in Definition 7.2 are generated from nontemporal physical input.

**Proposition 7.8** (Non-Entailment of Physical-Time Qualification by Evaluation Success). The simultaneous satisfaction of mathematical consistency, predictive performance, and empirical fit does not entail physical-time qualification of a physical-law time variable. That is,

$$\left. \begin{array}{l} \text{mathematical consistency,} \\ \text{predictive performance,} \\ \text{empirical fit} \end{array} \right\} \not\Rightarrow \text{physical-time qualification.}$$

*Proof.* Given equations of motion and initial values with a real-valued time variable as input, solutions conforming to the adopted rules of inference, predicted values, and correspondences with observed values can be constructed. This construction does not include as premises maps or relations that generate, from nontemporal physical input, the identity, order, intervals, and scale of the values of the time variable, correspondence between information-coupling positions, clock correspondence, and observation–recording correspondence.

By Definition 7.7, mathematical consistency, predictive performance, and empirical fit evaluate inferences, outputs, and value correspondences that use a given time variable. Physical-time qualification evaluates the generation of the time content itself that the time variable uses.

Therefore, even in a description in which the former three evaluations hold simultaneously, there exists a construction that does not take as input the generated structures of Definition 7.2. Hence, physical-time qualification does not follow from simultaneous satisfaction of the three evaluations.  $\square$

**Corollary 7.9** (Time-Content Dependence under Evaluation Success). For each  $R \in \text{Ref}^{\text{audit}}$ , when the time content  $x \in \text{TimeContent}_R$  satisfies the condition in Proposition 7.6, the satisfaction of mathematical consistency, predictive performance, or empirical fit preserves the direct dependence or transfer-reachability dependence from

$x$

to the result.

*Proof.* By Proposition 7.6,

$x$

either belongs directly to the result prerequisites or reaches a result prerequisite by transfer. By Proposition 7.8, evaluation success does not entail the generation of the time content itself. Therefore, after evaluation success has been established, the dependency relation from

$x$

to the result remains in force. □

**Definition 7.10** (Primitive-Time Input). For each  $R \in \text{Ref}^{\text{audit}}$ ,  $x \in \text{TimeContent}_R$ , suppose that the time content

$x$

is entered into the physical description in the reference and is used in a law operation or result derivation before the structure required for the use of

$x$

among the identity, before-and-after order, interval, scale, correspondence between information-coupling positions, clock correspondence, or observation–recording correspondence listed in Definition 7.2 is generated from nontemporal physical input. Then

$x$

is called a primitive-time input of the reference

$R$

.

**Theorem 7.11** (Failure of Physical-Time Qualification for a Primitive-Time Input). For a reference  $R \in \text{Ref}^{\text{audit}}$ , a primitive-time input  $x \in \text{TimeContent}_R$  lacks physical-time qualification.

*Proof.* By Definition 7.10, at least one of the identity, before-and-after order, interval, and scale of the time values required for the use of

$x$

or their correspondence between information-coupling positions, clock correspondence, or observation–recording correspondence enters a law operation or result derivation before the corresponding generated structure.

By Definition 7.2 and Proposition 7.3, these time operations take as input their respective comparison equivalence relation, canonical comparison quotient, order-coordinate structure, quantitative affine realization, comparison order atlas,  $\mathfrak{C}$ -derived arrival–recovery structure, or  $\mathfrak{D}$ -derived observation–recording structure.

For a primitive-time input, the required time content is supplied to a law operation or result derivation before these generated structures. Therefore, the generation dependence from nontemporal physical input to that time content does not hold, and the physical-time qualification of Definition 7.7 is not satisfied. □

**Corollary 7.12** (Inheritance of Primitive-Time Dependence by Results). When, for a reference  $R \in \text{Ref}^{\text{audit}}$ , a primitive-time input  $x \in \text{TimeContent}_R$ , and a result  $\eta \in \text{Result}_R$  satisfy the result-derivation participation condition of Proposition 7.6, the result

$$\eta$$

depends on time content that lacks physical-time qualification.

*Proof.* By Theorem 7.11,

$$x$$

lacks physical-time qualification.

By Proposition 7.6,

$$x$$

either belongs to the direct prerequisites of the result

$$\eta$$

or reaches a result prerequisite through a time-content transfer sequence. Therefore,

$$\eta$$

depends on the time content

$$x$$

that lacks physical-time qualification. □

**Proposition 7.13** (Failure of Evaluation Success to Complete a Primitive-Time Input). In a reference that uses a primitive-time input, even simultaneous mathematical consistency, predictive performance, and empirical fit do not establish physical-time qualification of the primitive-time input.

*Proof.* By Proposition 7.8, physical-time qualification does not follow from the simultaneous satisfaction of mathematical consistency, predictive performance, and empirical fit.

By Theorem 7.11, for a primitive-time input, the generation dependence required for the use of its time content does not hold. The three forms of evaluation success do not add that generation dependence. Therefore, physical-time qualification of the primitive-time input is not established. □

**Proposition 7.14** (Loss of Generation-Assessment Discrimination under Admission of Primitive Time). If an assessment of the establishment of time generation admits a primitive-time input as sufficient temporal input, time content generated from nontemporal physical input cannot be distinguished in the assessment result from time content entered before the generated structure.

*Proof.* For each obligation in Definition 7.2, consider time content obtained through the corresponding generated structure. Time content carrying the same identity, order, interval, scale, correspondence between information-coupling positions, clock correspondence, or observation-recording correspondence can also be placed at the input to a law operation or result derivation in accordance with Definition 7.10.

An assessment that admits the primitive-time input as sufficient temporal input maps the former generation path and the latter prior input to the same establishment result.

Therefore, its assessment result does not distinguish the presence or absence of a generation path.

An assessment of the establishment of time generation evaluates whether the generation dependence from nontemporal physical input to time content holds. An assessment that does not distinguish the presence or absence of a generation path cannot distinguish the establishment and invalidation of time generation.  $\square$

**Corollary 7.15** (Generation-Priority Condition for a Physical-Law Time Variable). For a physical-law time variable to be physically established, the generated structure corresponding to the time content used by that time variable must be established before law operations and result derivation.

*Proof.* By Theorem 7.11, a primitive-time input used before the generated structure lacks physical-time qualification. By Proposition 7.14, admission of a primitive-time input does not distinguish the establishment and invalidation of time generation. Therefore, physical establishment of a physical-law time variable requires prior establishment of the corresponding generated structure.  $\square$

**Definition 7.16** (Scope of the Audited Reference Set). The set  $\text{Ref}^{\text{audit}}$  of Definition 7.1 consists of 72 references whose source locations, time contents, transfers, and result dependencies are identified and whose generation dependencies from nontemporal physical input are audited.

Each reference contains at least one of the following uses of time.

1. An independent variable in an equation of motion, field equation, state-evolution equation, or stochastic-evolution equation.
2. Simultaneity, coordinate time, proper time, spacetime interval, or causal order.
3. A clock, period, frequency, standard, synchronization, or timekeeping procedure.
4. Thermodynamic evolution, statistical evolution, irreversibility, or an arrow of time.
5. Quantum measurement, quantum correlation, a constrained system, or an evolution description from a fundamental equation without time.
6. Gravity, cosmology, quantum gravity, emergent spacetime, or time reconstruction.
7. Temporal correspondence appearing in an experimental arrangement, observational record, or data analysis.

For each  $R \in \text{Ref}^{\text{audit}}$ , the audit input consists of the corresponding source locations, time contents, transfers, result dependencies, and source evidence for determining the generation dependencies from nontemporal physical input.

**Proposition 7.17** (Reference Domain of Cross-Reference Claims). The quantification domain of propositions across the audited reference set is  $\text{Ref}^{\text{audit}}$ . Each reference-specific determination takes the corresponding source locations, time contents, transfers, and result dependencies as evidence inputs.

*Proof.* Definition 7.1 types references, time contents, and results for each  $R \in \text{Ref}^{\text{audit}}$ . Definition 7.16 defines the scope of the audited reference set and the reference-specific audit inputs. Therefore, the quantification domain of the cross-reference audit and the evidence inputs for each reference-specific determination are determined.  $\square$

## 7.2 Evidence-Bearing Generation-Dependency Audit Method

Audit-boundary nodes, audit-boundary edges, generation traces, obligation atoms, primitive-time roots, and result dependencies are determined for each reference. For each reference, define the object of the generation-dependency audit as a finite node set with three components: physical or mathematical boundary claims, time contents, and reference-specific results.

**Definition 7.18** (Audit-Boundary Nodes). Define the set of audit-boundary-node types by

$$\text{Type}^{\text{audit,node}} := \left\{ \begin{array}{l} \text{Type}_{\text{claim}}^{\text{audit,node}}, \\ \text{Type}_{\text{time}}^{\text{audit,node}}, \\ \text{Type}_{\text{result}}^{\text{audit,node}} \end{array} \right\}$$

The three elements of this set are pairwise distinct.

For each  $R \in \text{Ref}^{\text{audit}}$ , call a claim partitioned into a minimal verifiable unit from a formal definition, equation, proposition, or an accompanying explicit statement in the source an audit-claim record. Let

$$\text{Type}_{\text{claim}}^{\text{audit,node}}$$

be the first component, and let an audit-claim record carrying, as attributes, its source location, attributed subject, degree of certainty, evidence-expression class, and a reference to its source context be the second component. Denote the finite set of such ordered pairs by

$$\text{Node}_R^{\text{audit,claim}}$$

Define the item-category domain for time contents and reference-specific results by  $\mathcal{I}_{\text{audit,item}} := \{\text{time}, \text{result}\}$ , and define the item families by

$$\begin{aligned} \text{Item}_R^{\text{audit,time}} &:= \text{TimeContent}_R, \\ \text{Item}_R^{\text{audit,result}} &:= \text{Result}_R \end{aligned}$$

For each  $j \in \mathcal{I}_{\text{audit,item}}$ , define the node set and node-forming map by

$$\begin{aligned} \text{Node}_R^{\text{audit},j} &:= \left\{ \text{Type}_j^{\text{audit,node}} \right\} \times \text{Item}_R^{\text{audit},j}, \\ \text{NodeOf}_R^{\text{audit},j} : \text{Item}_R^{\text{audit},j} &\longrightarrow \text{Node}_R^{\text{audit},j}, \\ z &\longmapsto (\text{Type}_j^{\text{audit,node}}, z) \end{aligned}$$

Define the set of audit-boundary nodes by

$$\begin{aligned} \text{Node}_R^{\text{audit,bd}} &:= \text{Node}_R^{\text{audit,claim}} \dot{\cup} \text{Node}_R^{\text{audit,time}} \\ &\quad \dot{\cup} \text{Node}_R^{\text{audit,result}} \end{aligned}$$

Define the node-type map by

$$\begin{aligned} \text{TypeOf}_R^{\text{audit,node}} : \text{Node}_R^{\text{audit,bd}} &\longrightarrow \text{Type}^{\text{audit,node}}, \\ \text{TypeOf}_R^{\text{audit,node}} (\text{NodeType}, \text{NodePayload}) &:= \text{NodeType} \end{aligned}$$



**Proposition 7.19** (Component Decomposition of Audit-Boundary Nodes). For each

$$R \in \text{Ref}^{\text{audit}}, \quad \text{Node} \in \text{Node}_R^{\text{audit,bd}}$$

exactly one of

$$\begin{aligned} \text{Node} &\in \text{Node}_R^{\text{audit,claim}}, & \text{or} \\ \text{Node} &\in \text{Node}_R^{\text{audit,time}}, & \text{or} \\ \text{Node} &\in \text{Node}_R^{\text{audit,result}} \end{aligned}$$

holds. Consequently,

$$\text{TypeOf}_R^{\text{audit,node}}$$

assigns a unique node type to each audit-boundary node.

*Proof.* By Definition 7.18,  $\text{Node}_R^{\text{audit,bd}}$  is the disjoint union of the three node sets. The three components are pairwise disjoint, and their union equals  $\text{Node}_R^{\text{audit,bd}}$ . Hence every audit-boundary node belongs to exactly one of the three components. By the definition of the node-type map, each component is assigned one node type.  $\square$

**Definition 7.20** (Five-Type Classification of Time Content). Define the codomain of the five time-content types by

$$\begin{aligned} \text{Type}^{\text{time,ord,struct}} := \{ & \text{Type}_{\text{member}}^{\text{time,ord,struct}}, \text{Type}_{\text{dependency}}^{\text{time,ord,struct}}, \\ & \text{Type}_{\text{branch}}^{\text{time,ord,struct}}, \text{Type}_{\text{separate}}^{\text{time,ord,struct}}, \\ & \text{Type}_{\text{lexical}}^{\text{time,ord,struct}} \} \end{aligned}$$

The five values denote, respectively, a member of a boundary, stage, or transition-position family; content dependent on the same family; content on a branch outside the boundary; content belonging to a separate requirement family; and lexical or illustrative content.

For each  $R \in \text{Ref}^{\text{audit}}$ , provide the five-type classification map of time content

$$\begin{aligned} \text{TypeOf}_R^{\text{time,ord,struct}} : \text{TimeContent}_R \\ \longrightarrow \text{Type}^{\text{time,ord,struct}} \end{aligned}$$

Assign each time content one type according to the function given by a definition, equation, proposition, or explicit statement in the source.

**Definition 7.21** (Eight-Axis Classification of Temporal Order Structure). Define the codomain of the eight temporal-order-structure axes by

$$\begin{aligned} \text{Axis}^{\text{time,ord,struct}} := \{ & \text{endpoint, instant, interval\_boundary, phase\_boundary,} \\ & \text{turning\_position, order\_relation, orientation, transition\_position} \} \end{aligned}$$

The eight values denote, respectively, endpoint, instant, interval boundary, phase boundary, turning position, order relation, orientation, and transition position.

For each  $R \in \text{Ref}^{\text{audit}}$ , provide the eight-axis classification map of temporal order structure

$$\begin{aligned} \text{AxisOf}_R^{\text{time,ord,struct}} : \text{TimeContent}_R \\ \longrightarrow \mathcal{P}(\text{Axis}^{\text{time,ord,struct}}) \end{aligned}$$

For each  $x \in \text{TimeContent}_R$ , impose

$$\begin{aligned} \text{TypeOf}_R^{\text{time,ord,struct}}(x) &= \text{Type}_{\text{member}}^{\text{time,ord,struct}} \\ \implies \left| \text{AxisOf}_R^{\text{time,ord,struct}}(x) \right| &\geq 1, \end{aligned}$$

and

$$\begin{aligned} \text{TypeOf}_R^{\text{time,ord,struct}}(x) &\neq \text{Type}_{\text{member}}^{\text{time,ord,struct}} \\ \implies \text{AxisOf}_R^{\text{time,ord,struct}}(x) &= \emptyset \end{aligned}$$

**Proposition 7.22** (Structural Attributes of Time-Content Nodes). For each  $R \in \text{Ref}^{\text{audit}}$ ,  $x \in \text{TimeContent}_R$ , the time-content node  $\text{NodeOf}_R^{\text{audit,time}}(x)$  has the unique attribute

$$\left( \text{TypeOf}_R^{\text{time,ord,struct}}(x), \right. \\ \left. \text{AxisOf}_R^{\text{time,ord,struct}}(x) \right)$$

*Proof.* The type-classification map of Definition 7.20 and the axis-classification map of Definition 7.21 both have domain

$$\text{TimeContent}_R$$

By Definition 7.18, the time-content node-forming map is injective. Hence every time-content node uniquely carries the type value and axis set belonging to the single time content from which it was formed.  $\square$

**Definition 7.23** (Common Type of Structural Signatures and Member Structural Signature). For each  $R \in \text{Ref}^{\text{audit}}$ , define the signature-kind set and the type of the structural-signature family by

$$\begin{aligned} \mathcal{I}_{\text{sig}}^{\text{time,ord,struct}} &:= \{\text{member}, \text{dependency}\}, \\ \text{Sig}_R^{\text{time,ord,struct}, \varsigma_{\text{sig}}} &: \\ &\left( \text{TypeOf}_R^{\text{time,ord,struct}} \right)^{-1} \left( \left\{ \text{Type}_{\varsigma_{\text{sig}}}^{\text{time,ord,struct}} \right\} \right) \\ &\longrightarrow \{0, 1\}^3, \\ \varsigma_{\text{sig}} &\in \mathcal{I}_{\text{sig}}^{\text{time,ord,struct}} \end{aligned}$$

For each member

$$x$$

of the member kind, define its signature by

$$\begin{aligned} &\text{Sig}_R^{\text{time,ord,struct,member}}(x) \\ &:= \left( \mathbf{1} \bigcup_{\epsilon \in \text{TransferEvent}_R} \text{TransferIn}_{R,\epsilon}(x), \right. \\ &\quad \mathbf{1} \bigcup_{\epsilon \in \text{TransferEvent}_R} \text{TransferOut}_{R,\epsilon}(x), \\ &\quad \left. \mathbf{1} \bigcup_{\eta \in \text{Result}_R} \text{ResultPrereq}_{R,\eta}(x) \right) \end{aligned}$$

The three components denote, respectively, membership in a transfer input set, membership in a transfer output set, and membership in a result prerequisite.

**Definition 7.24** (Dependent-Content Structural Signature). For each  $R \in \text{Ref}^{\text{audit}}$ , define the signature of each dependent content

$$x$$

of the dependent-content kind by

$$\begin{aligned} & \text{Sig}_R^{\text{time,ord,struct,dependency}}(x) \\ &:= \left( \mathbf{1} \left[ \begin{array}{c} \exists y \in \left( \text{TypeOf}_R^{\text{time,ord,struct}} \right)^{-1} \left( \left\{ \text{Type}_{\text{member}}^{\text{time,ord,struct}} \right\} \right) : \\ x \rightsquigarrow_R^{\text{time,content}} y \end{array} \right] , \right. \\ & \quad \left. \mathbf{1} \left[ \begin{array}{c} \exists y \in \left( \text{TypeOf}_R^{\text{time,ord,struct}} \right)^{-1} \left( \left\{ \text{Type}_{\text{member}}^{\text{time,ord,struct}} \right\} \right) : \\ y \rightsquigarrow_R^{\text{time,content}} x \end{array} \right] , \right. \\ & \quad \left. \mathbf{1} \bigcup_{\eta \in \text{Result}_R} \text{ResultPrereq}_{R,\eta}(x) \right) \end{aligned}$$

The three components denote, respectively, transfer reachability to a member, transfer reachability from a member, and membership in a result prerequisite.

**Proposition 7.25** (Totality and Uniqueness of Structural Signatures).

For each  $R \in \text{Ref}^{\text{audit}}$ , every member-type time content belongs to one fiber of  $\text{Sig}_R^{\text{time,ord,struct,member}}$ , and every dependent-content-type time content belongs to one fiber of  $\text{Sig}_R^{\text{time,ord,struct,dependency}}$ .

*Proof.* Each component of an indicator function takes one of

$$0 \quad \text{or} \quad 1$$

as its value. Hence each signature map assigns every element of its domain a unique element of

$$\{0, 1\}^3$$

The fibers of a map decompose its domain into pairwise disjoint inverse images; hence every element belongs to one fiber.  $\square$

**Definition 7.26** (Audit-Boundary Support Sets and Audit-Boundary Edges). For each  $R \in \text{Ref}^{\text{audit}}$ , assume that the audited transfer-event set  $\text{TransferEvent}_R$  is finite. Define the audit-edge category domain, the edge carriers for each category, and the set of audit-boundary-edge types by

$$\begin{aligned} \mathcal{I}_{\text{audit,edge}} &:= \{\text{transfer}, \text{result}\}, \\ \text{EdgItem}_R^{\text{audit,transfer}} &:= \text{TransferEvent}_R, \\ \text{EdgItem}_R^{\text{audit,result}} &:= \text{Result}_R, \\ \text{Type}^{\text{audit,edge}} &:= \left\{ \text{Type}_{\text{transfer}}^{\text{audit,edge}}, \text{Type}_{\text{result}}^{\text{audit,edge}} \right\} \end{aligned}$$

For each  $\kappa_{\text{edge}} \in \mathcal{I}_{\text{audit,edge}}$ , let the source-support map be

$$\text{Support}_R^{\text{audit},\kappa_{\text{edge}}} : \text{EdgItem}_R^{\text{audit},\kappa_{\text{edge}}} \longrightarrow \mathcal{P}_{\text{fin}} \left( \text{Node}_R^{\text{audit,claim}} \right)$$

For each  $\epsilon \in \text{TransferEvent}_R$ ,

$$\text{Support}_R^{\text{audit,transfer}}(\epsilon)$$

consists of boundary-claim nodes that state the transfer input domain, output domain, and direction. For each  $\eta \in \text{Result}_R$ ,

$$\text{Support}_R^{\text{audit,result}}(\eta)$$

consists of boundary-claim nodes that state the derivation of the result.

Define the edge set and edge-forming map for each edge category by

$$\begin{aligned} \text{Edge}_R^{\text{audit},\kappa_{\text{edge}}} &:= \left\{ \text{Type}_{\kappa_{\text{edge}}}^{\text{audit,edge}} \right\} \times \text{EdgItem}_R^{\text{audit},\kappa_{\text{edge}}}, \\ \text{EdgeOf}_R^{\text{audit},\kappa_{\text{edge}}} : \\ \text{EdgItem}_R^{\text{audit},\kappa_{\text{edge}}} &\longrightarrow \text{Edge}_R^{\text{audit},\kappa_{\text{edge}}}, \\ z &\longmapsto \left( \text{Type}_{\kappa_{\text{edge}}}^{\text{audit,edge}}, z \right) \end{aligned}$$

This map is injective, and its image equals the corresponding edge set.

Define the set of audit-boundary edges by

$$\text{Edge}_R^{\text{audit,bd}} := \text{Edge}_R^{\text{audit,transfer}} \cup \text{Edge}_R^{\text{audit,result}}$$

Define the edge-type map by

$$\text{TypeOf}_R^{\text{audit,edge}} : \text{Edge}_R^{\text{audit,bd}} \longrightarrow \text{Type}^{\text{audit,edge}}, \quad \text{TypeOf}_R^{\text{audit,edge}}(t, z) := t$$

**Definition 7.27** (Audit-Boundary Directed Hypergraph). For each  $R \in \text{Ref}^{\text{audit}}$ , define the input and output maps of audit-boundary edges by

$$\begin{aligned} \text{In}_R^{\text{audit}} : \text{Edge}_R^{\text{audit,bd}} &\longrightarrow \mathcal{P}_{\text{fin}} \left( \text{Node}_R^{\text{audit,bd}} \right), \\ \text{Out}_R^{\text{audit}} : \text{Edge}_R^{\text{audit,bd}} &\longrightarrow \mathcal{P}_{\text{fin}} \left( \text{Node}_R^{\text{audit,bd}} \right) \end{aligned}$$

For each  $\epsilon \in \text{TransferEvent}_R$  and each  $\eta \in \text{Result}_R$ , define the corresponding inputs and outputs, respectively, by

$$\begin{aligned} \text{In}_R^{\text{audit}} \left( \text{EdgeOf}_R^{\text{audit,transfer}}(\epsilon) \right) &:= \text{Support}_R^{\text{audit,transfer}}(\epsilon) \cup \text{NodeOf}_R^{\text{audit,time}} [\text{TransferIn}_{R,\epsilon}], \\ \text{Out}_R^{\text{audit}} \left( \text{EdgeOf}_R^{\text{audit,transfer}}(\epsilon) \right) &:= \text{NodeOf}_R^{\text{audit,time}} [\text{TransferOut}_{R,\epsilon}], \\ \text{In}_R^{\text{audit}} \left( \text{EdgeOf}_R^{\text{audit,result}}(\eta) \right) &:= \text{Support}_R^{\text{audit,result}}(\eta) \cup \text{NodeOf}_R^{\text{audit,time}} [\text{ResultPrereq}_{R,\eta}], \\ \text{Out}_R^{\text{audit}} \left( \text{EdgeOf}_R^{\text{audit,result}}(\eta) \right) &:= \left\{ \text{NodeOf}_R^{\text{audit,result}}(\eta) \right\} \end{aligned}$$

Define the audit-boundary directed hypergraph by

$$\text{Hypergraph}_R^{\text{audit}} := \left( \text{Node}_R^{\text{audit,bd}}, \text{Edge}_R^{\text{audit,bd}}, \right. \\ \left. \text{In}_R^{\text{audit}}, \text{Out}_R^{\text{audit}}, \right. \\ \left. \text{TypeOf}_R^{\text{audit,node}}, \text{TypeOf}_R^{\text{audit,edge}} \right)$$

The simultaneous establishment of all nodes in an edge's input set is the establishment condition for the nodes in that edge's output set.

Let the specified subset consisting of physical or mathematical boundary-claim nodes independent of the derivation of time content be

$$\text{Init}_R^{\text{audit,phys}} \subseteq \text{Node}_R^{\text{audit,claim}}$$

and call it the physical-input initial set.

**Definition 7.28** (Generation Initial Set and Reference Formal-Evidence Environment). For each  $R \in \text{Ref}^{\text{audit}}$ , let the specified subset consisting of boundary-claim nodes that provide the reference-specific formal derivation be

$$\text{Env}_R^{\text{audit,evid}} \subseteq \text{Node}_R^{\text{audit,claim}}$$

and call it the reference formal-evidence environment. The physical-input initial set and the reference formal-evidence environment satisfy

$$\text{Init}_R^{\text{audit,phys}} \subseteq \text{Env}_R^{\text{audit,evid}} \subseteq \text{Node}_R^{\text{audit,claim}}$$

**Proposition 7.29** (Cardinalities of the Audit-Boundary Node and Edge Sets). For each  $R \in \text{Ref}^{\text{audit}}$ , the audit-boundary node set and the audit-boundary edge set are finite, and

$$\begin{aligned} |\text{Node}_R^{\text{audit,bd}}| &= |\text{Node}_R^{\text{audit,claim}}| + |\text{TimeContent}_R| \\ &\quad + |\text{Result}_R|, \\ |\text{Edge}_R^{\text{audit,bd}}| &= |\text{TransferEvent}_R| + |\text{Result}_R|. \end{aligned}$$

*Proof.* By Definition 7.18, the audit-boundary node set is the disjoint union of the physical- or formal-input claim-node set, the time-content node set, and the result node set. The time-content node-forming map and the result node-forming map provide bijections from, respectively,

$$\text{TimeContent}_R, \quad \text{Result}_R$$

to the corresponding node sets. Hence the first equality holds.

By Definition 7.26, the audit-boundary edge set is the disjoint union of the transfer-edge set and the result-edge set. The transfer-edge-forming map and the result-edge-forming map provide bijections from, respectively,

$$\text{TransferEvent}_R, \quad \text{Result}_R$$

to the corresponding edge sets. Hence the second equality holds. The finiteness of each disjoint-union component implies that the audit-boundary node set and the audit-boundary edge set are finite.  $\square$

**Definition 7.30** (Reachability Stages and Reachable Set on the Audit-Boundary Directed Hypergraph). For each  $R \in \text{Ref}^{\text{audit}}$  and  $\text{Seed} \subseteq \text{Node}_R^{\text{audit,bd}}$ , define the initial reachability stage by

$$\text{Reach}_R^{\text{audit},(0)}(\text{Seed}) := \text{Seed}$$

For each

$$k \in \mathbb{N}$$

define the successor reachability stage by

$$\begin{aligned} & \text{Reach}_R^{\text{audit},(k+1)}(\text{Seed}) \\ &:= \text{Reach}_R^{\text{audit},(k)}(\text{Seed}) \\ & \quad \cup \bigcup_{\substack{\text{AuditEdge} \in \text{Edge}_R^{\text{audit,bd}} \\ \text{In}_R^{\text{audit}}(\text{AuditEdge}) \subseteq \text{Reach}_R^{\text{audit},(k)}(\text{Seed})}} \text{Out}_R^{\text{audit}}(\text{AuditEdge}) \end{aligned}$$

Define the audit-reachable set by

$$\begin{aligned} & \text{Reach}_R^{\text{audit}}(\text{Seed}) \\ &:= \text{Reach}_R^{\text{audit},(|\text{Node}_R^{\text{audit,bd}}|)}(\text{Seed}) \end{aligned}$$

**Proposition 7.31** (Finite Stabilization of Audit Reachability Stages). For each  $R \in \text{Ref}^{\text{audit}}$  and  $\text{Seed} \subseteq \text{Node}_R^{\text{audit,bd}}$ , the reachability-stage sequence is monotone increasing with respect to inclusion. Moreover, if

$$k \geq |\text{Node}_R^{\text{audit,bd}}|$$

then

$$\begin{aligned} & \text{Reach}_R^{\text{audit},(k)}(\text{Seed}) \\ &= \text{Reach}_R^{\text{audit}}(\text{Seed}) \end{aligned}$$

holds.

*Proof.* The recursion in Definition 7.30 expresses each stage as its union with the immediately preceding stage. The reachability-stage sequence is therefore monotone increasing.

The audit-boundary node set is finite, and every strict inclusion adds at least one node. Once two adjacent stages coincide, the same recursion makes all subsequent stages coincide. Hence the reachability-stage sequence stabilizes by stage

$$|\text{Node}_R^{\text{audit,bd}}|$$

and its stable value equals the audit-reachable set.  $\square$

**Definition 7.32** (Audit-Generation Traces). For each  $R \in \text{Ref}^{\text{audit}}$ ,  $\text{Seed} \subseteq \text{Node}_R^{\text{audit,bd}}$ , and  $\text{AuditNode} \in \text{Node}_R^{\text{audit,bd}}$ , let

$$\text{Trace}_R^{\text{audit}}(\text{Seed}, \text{AuditNode})$$

be the set of all finite edge sequences satisfying the following conditions.

For an edge sequence of length  $\ell \in \mathbb{N}$ ,  $\left(\text{AuditEdge}^{(j)}\right)_{j=1}^{\ell}$ , define

$$\text{TraceNodeSet}^{(0)} := \text{Seed}$$

and

$$\begin{aligned} \text{TraceNodeSet}^{(j)} &:= \text{TraceNodeSet}^{(j-1)} \\ &\cup \text{Out}_R^{\text{audit}} \left( \text{AuditEdge}^{(j)} \right) \end{aligned}$$

For each

$$j \in \{1, \dots, \ell\}$$

require

$$\text{AuditEdge}^{(j)} \in \text{Edge}_R^{\text{audit, bd}}$$

and

$$\begin{aligned} &\text{In}_R^{\text{audit}} \left( \text{AuditEdge}^{(j)} \right) \\ &\subseteq \text{TraceNodeSet}^{(j-1)} \end{aligned}$$

and further require

$$\text{AuditNode} \in \text{TraceNodeSet}^{(\ell)}$$

An edge sequence satisfying these conditions is called an audit-generation trace. The edge sequence of length zero belongs to the set when

$$\text{AuditNode} \in \text{Seed}$$

**Lemma 7.33** (Equivalence of Audit Reachability and Audit-Generation Traces). For each  $R \in \text{Ref}^{\text{audit}}$ ,  $\text{Seed} \subseteq \text{Node}_R^{\text{audit, bd}}$ , and  $\text{AuditNode} \in \text{Node}_R^{\text{audit, bd}}$ ,

$$\begin{aligned} &\text{AuditNode} \in \text{Reach}_R^{\text{audit}}(\text{Seed}) \\ &\iff \text{Trace}_R^{\text{audit}}(\text{Seed}, \text{AuditNode}) \neq \emptyset \end{aligned}$$

holds.

*Proof.* For the forward direction, a node in the initial reachability stage has a corresponding edge sequence of length zero. A node first added at reachability stage

$$k + 1$$

belongs to the output of an audit-boundary edge whose input set is contained in reachability stage

$$k$$

Ordering by stage the finite edge sequences corresponding to the input nodes and appending the audit-boundary edge yields a finite edge sequence that satisfies the all-input containment condition in Definition 7.32. Induction over reachability stages therefore gives an audit-generation trace for every node in the audit-reachable set.

For the reverse direction, the terminal node of an audit-generation trace of length zero belongs to the initial reachability stage. In an audit-generation trace of length

$$\ell + 1$$

all inputs of the terminal edge are contained in the preceding trace-node set. Induction on edge-sequence length, together with the recursion in Definition 7.30, shows that the terminal node belongs to a finite reachability stage. By Proposition 7.31, that node belongs to the audit-reachable set.  $\square$

**Definition 7.34** (Structural and Essential Influence Regions of a Physical-Input Node). For each  $R \in \text{Ref}^{\text{audit}}$  and  $\text{InitNode} \in \text{Init}_R^{\text{audit,phys}}$ , define the initial influence stage by

$$\text{Influence}_R^{\text{audit},(0)}(\text{InitNode}) := \{\text{InitNode}\}$$

For each

$$k \in \mathbb{N}$$

define the successor influence stage by

$$\begin{aligned} & \text{Influence}_R^{\text{audit},(k+1)}(\text{InitNode}) \\ &:= \text{Influence}_R^{\text{audit},(k)}(\text{InitNode}) \\ & \quad \cup \bigcup_{\substack{\text{AuditEdge} \in \text{Edge}_R^{\text{audit,bd}} \\ \text{In}_R^{\text{audit}}(\text{AuditEdge}) \subseteq \text{Reach}_R^{\text{audit},(k)}(\text{Init}_R^{\text{audit,phys}}) \\ \text{In}_R^{\text{audit}}(\text{AuditEdge}) \cap \text{Influence}_R^{\text{audit},(k)}(\text{InitNode}) \neq \emptyset}} \text{Out}_R^{\text{audit}}(\text{AuditEdge}) \end{aligned}$$

Define the structural influence region by

$$\begin{aligned} & \text{Influence}_R^{\text{audit}}(\text{InitNode}) \\ &:= \text{Influence}_R^{\text{audit},(|\text{Node}_R^{\text{audit,bd}}|)}(\text{InitNode}) \end{aligned}$$

Define the essential influence region by

$$\begin{aligned} & \text{EssentialInfluence}_R^{\text{audit}}(\text{InitNode}) \\ &:= \text{Reach}_R^{\text{audit}}(\text{Init}_R^{\text{audit,phys}}) \\ & \quad \setminus \text{Reach}_R^{\text{audit}}(\text{Init}_R^{\text{audit,phys}} \setminus \{\text{InitNode}\}) \end{aligned}$$

**Proposition 7.35** (Inclusion of the Essential Influence Region in the Structural Influence Region). For each  $R \in \text{Ref}^{\text{audit}}$  and  $\text{InitNode} \in \text{Init}_R^{\text{audit,phys}}$ ,

$$\begin{aligned} & \text{EssentialInfluence}_R^{\text{audit}}(\text{InitNode}) \\ & \subseteq \text{Influence}_R^{\text{audit}}(\text{InitNode}) \end{aligned}$$

holds.

*Proof.* Proceed by induction over the reachability stages from the physical-input initial set. At the initial stage,

$$\text{InitNode}$$

is the only node not added to the reachable set from the initial set with that node removed, and this node belongs to the initial influence stage.

At reachability stage

$$k + 1$$

take a node

$$\text{AuditNode}$$

that first belongs to the essential influence region. All inputs of an audit-boundary edge

$$\text{AuditEdge}$$



that outputs this node belong to reachability stage

$$k$$

Suppose that all inputs belong to the reachable set from

$$\text{Init}_R^{\text{audit,phys}} \setminus \{\text{InitNode}\}$$

The same edge would then place

$$\text{AuditNode}$$

in that reachable set, contrary to the definition of the essential influence region. Hence the edge has an input node that belongs to the essential influence region at a preceding stage. By the induction hypothesis, that input node belongs to influence stage

$$k$$

By the recursion in Definition 7.34,

$$\text{AuditNode}$$

belongs to influence stage

$$k + 1$$

Finite stabilization yields the inclusion.  $\square$

**Definition 7.36** (Reference-Specific Obligation Atoms). For each  $R \in \text{Ref}^{\text{audit}}$ , define reference-specific obligation atoms by the following four types. An obligation atom is the smallest unit of obligation to which one audit state is assigned in the generation-dependency audit.

1. A boundary-claim atom is  $\text{ClaimNode} \in \text{Node}_R^{\text{audit,claim}}$ .
2. A time-content atom is  $x \in \text{TimeContent}_R$ , and its audit-boundary node is  $\text{NodeOf}_R^{\text{audit,time}}(x)$ .
3. A transfer-event atom is  $\epsilon \in \text{TransferEvent}_R$ , and its audit-boundary edge is

$$\text{EdgeOf}_R^{\text{audit,transfer}}(\epsilon)$$

4. A result atom is  $\eta \in \text{Result}_R$ , and its audit-boundary edge and result node are

$$\text{EdgeOf}_R^{\text{audit,result}}(\eta), \quad \text{NodeOf}_R^{\text{audit,result}}(\eta)$$

**Definition 7.37** (Typed Audit States of Reference-Specific Obligation Atoms). For each  $R \in \text{Ref}^{\text{audit}}$ , describe the typed audit states of reference-specific obligation atoms by the following components.

For a boundary-claim node  $\text{ClaimNode} \in \text{Node}_R^{\text{audit,claim}}$ , the state component is

$$\left( \text{ClaimNode}, \right. \\ \left. \text{TypeOf}_R^{\text{audit,node}}(\text{ClaimNode}), \right. \\ \left. \mathbf{1}_{\text{Reach}_R^{\text{audit}}}(\text{Init}_R^{\text{audit,phys}})(\text{ClaimNode}) \right)$$

For a time content  $x \in \text{TimeContent}_R$ , the state component is

$$\begin{aligned} & \left( x, \text{NodeOf}_R^{\text{audit,time}}(x), \right. \\ & \quad \text{TypeOf}_R^{\text{time,ord,struct}}(x), \\ & \quad \text{AxisOf}_R^{\text{time,ord,struct}}(x), \\ & \quad \left. \mathbf{1}_{\text{Reach}_R^{\text{audit}}}(\text{Init}_R^{\text{audit,phys}})(\text{NodeOf}_R^{\text{audit,time}}(x)) \right) \end{aligned}$$

For a transfer event  $\epsilon \in \text{TransferEvent}_R$ , the state component is

$$\begin{aligned} & \left( \epsilon, \text{Support}_R^{\text{audit,transfer}}(\epsilon), \right. \\ & \quad \text{TransferIn}_{R,\epsilon}, \text{TransferOut}_{R,\epsilon}, \\ & \quad \text{EdgeOf}_R^{\text{audit,transfer}}(\epsilon), \\ & \quad \text{In}_R^{\text{audit}}(\text{EdgeOf}_R^{\text{audit,transfer}}(\epsilon)), \\ & \quad \left. \text{Out}_R^{\text{audit}}(\text{EdgeOf}_R^{\text{audit,transfer}}(\epsilon)) \right) \end{aligned}$$

For a result  $\eta \in \text{Result}_R$ , the state component is

$$\begin{aligned} & \left( \eta, \text{Support}_R^{\text{audit,result}}(\eta), \right. \\ & \quad \text{ResultPrereq}_{R,\eta}, \text{EdgeOf}_R^{\text{audit,result}}(\eta), \\ & \quad \text{In}_R^{\text{audit}}(\text{EdgeOf}_R^{\text{audit,result}}(\eta)), \\ & \quad \text{NodeOf}_R^{\text{audit,result}}(\eta), \\ & \quad \left. \mathbf{1}_{\text{Reach}_R^{\text{audit}}}(\text{Init}_R^{\text{audit,phys}})(\text{NodeOf}_R^{\text{audit,result}}(\eta)) \right) \end{aligned}$$

**Proposition 7.38** (Determinacy of Typed Audit States). For every reference-specific obligation atom, all components in Definition 7.37 are uniquely determined by  $\text{Hypergraph}_R^{\text{audit}}$ ,  $\text{Init}_R^{\text{audit,phys}}$ ,  $\text{TypeOf}_R^{\text{time,ord,struct}}$ , and  $\text{AxisOf}_R^{\text{time,ord,struct}}$ . Consequently, the reachability state of the same obligation atom in the same reference is unique.

*Proof.* The node type and edge type are respectively the values of the maps

$$\text{TypeOf}_R^{\text{audit,node}}, \quad \text{TypeOf}_R^{\text{audit,edge}}$$

The structural type and axis set of a time content are respectively the values of the maps

$$\text{TypeOf}_R^{\text{time,ord,struct}}, \quad \text{AxisOf}_R^{\text{time,ord,struct}}$$

The support set of a transfer event or result is recovered as the intersection of the corresponding edge input with the claim-node set. Time-content inputs and outputs are uniquely recovered by taking the inverse images of the time-content-node components of the corresponding edge inputs and outputs under the injective time-content node-forming map.

By Proposition 7.31, the audit-reachable set from the physical-input initial set is uniquely determined after finitely many stages. Each indicator-function value is therefore uniquely determined, and all state components are fixed.  $\square$

**Definition 7.39** (Audit-Generated Time Contents, Primitive-Time Roots, and Audit-Generated Results). For each  $R \in \text{Ref}^{\text{audit}}$ , define the audit-generated time-content set, primitive-time-root set, audit-generated result set, and result set outside the generation-reachable region, respectively, by

$$\begin{aligned}\text{TimeContent}_R^{\text{audit,gen}} &:= \left( \text{NodeOf}_R^{\text{audit,time}} \right)^{-1} \left[ \text{Reach}_R^{\text{audit}} \left( \text{Init}_R^{\text{audit,phys}} \right) \right], \\ \text{TimeContent}_R^{\text{audit,root}} &:= \text{TimeContent}_R \setminus \text{TimeContent}_R^{\text{audit,gen}}, \\ \text{Result}_R^{\text{audit,gen}} &:= \left( \text{NodeOf}_R^{\text{audit,result}} \right)^{-1} \left[ \text{Reach}_R^{\text{audit}} \left( \text{Init}_R^{\text{audit,phys}} \right) \right], \\ \text{Result}_R^{\text{audit,unreach}} &:= \text{Result}_R \setminus \text{Result}_R^{\text{audit,gen}}\end{aligned}$$

**Definition 7.40** (Generation Reach and Evidence-Conditioned Dependency Reach). For each  $R \in \text{Ref}^{\text{audit}}$ , define the generation-reachable set by

$$\text{Reach}_R^{\text{audit,gen}} := \text{Reach}_R^{\text{audit}} \left( \text{Init}_R^{\text{audit,phys}} \right)$$

For each  $\text{RootSeed} \subseteq \text{TimeContent}_R^{\text{audit,root}}$ , define the evidence-conditioned dependency-reachable set by

$$\begin{aligned}\text{Reach}_R^{\text{audit|evid}}(\text{RootSeed}) \\ &:= \text{Reach}_R^{\text{audit}} \left( \text{Init}_R^{\text{audit,phys}} \cup \text{Env}_R^{\text{audit,evid}} \right. \\ &\quad \left. \cup \text{NodeOf}_R^{\text{audit,time}}[\text{RootSeed}] \right)\end{aligned}$$

**Definition 7.41** (Time Contents Reachable under the Evidence Environment). For each  $R \in \text{Ref}^{\text{audit}}$ , define the set of time contents reached after adding the reference formal-evidence environment without supplying primitive-time roots as seeds by

$$\text{TimeContent}_R^{\text{audit,evid}} := \left( \text{NodeOf}_R^{\text{audit,time}} \right)^{-1} \left[ \text{Reach}_R^{\text{audit|evid}}(\emptyset) \right]$$

Call the assessment based on the audit-generated time-content set  $\text{TimeContent}_R^{\text{audit,gen}}$  the LCTR complete-generation assessment. Call the assessment based on the foregoing set of time contents reachable under the evidence environment the within-reference evidence-conditioned local-reach assessment. Below, abbreviate these as the complete assessment and the local-reach assessment, respectively. The latter is a reachability assessment that adds, as the reference formal-evidence environment, the physical-carrier processing or construction procedures explicitly stated in that reference. Because the two assessments have different starting sets, nonemptiness under the latter does not entail nonemptiness under the former.

In the reference-specific assessment values, abbreviate the number of time contents generated under the complete assessment as “complete,” the number of time contents reached under the local-reach assessment as “local reach,” and the number of primitive-time roots that remain after the latter assessment as “remaining roots.”

**Definition 7.42** (Nongenerated Results Reachable under the Evidence Environment and Residual Results). For each  $R \in \text{Ref}^{\text{audit}}$ , define the set of nongenerated results reachable

under the evidence environment by

$$\begin{aligned} & \text{Result}_R^{\text{audit,evidonly}} \\ & := \left\{ \eta \in \text{Result}_R \mid \text{NodeOf}_R^{\text{audit,result}}(\eta) \in \text{Reach}_R^{\text{audit|evid}}(\emptyset) \right\} \\ & \quad \setminus \text{Result}_R^{\text{audit,gen}} \end{aligned}$$

Define the residual result set by

$$\begin{aligned} & \text{Result}_R^{\text{audit,resid}} \\ & := \text{Result}_R^{\text{audit,unreach}} \setminus \text{Result}_R^{\text{audit,evidonly}} \end{aligned}$$

**Proposition 7.43** (Time-Antecedent Preservation Condition). For each  $R \in \text{Ref}^{\text{audit}}$ , if every  $\text{AuditEdge} \in \text{Edge}_R^{\text{audit,bd}}$  satisfies

$$\begin{aligned} & \text{Out}_R^{\text{audit}}(\text{AuditEdge}) \cap \text{Node}_R^{\text{audit,time}} \neq \emptyset \\ & \implies \text{In}_R^{\text{audit}}(\text{AuditEdge}) \cap \text{Node}_R^{\text{audit,time}} \neq \emptyset \end{aligned}$$

then

$$\begin{aligned} & \text{Reach}_R^{\text{audit}}(\text{Init}_R^{\text{audit,phys}}) \cap \text{Node}_R^{\text{audit,time}} \\ & = \emptyset \end{aligned}$$

holds.

*Proof.* By Definition 7.27,

$$\text{Init}_R^{\text{audit,phys}} \subseteq \text{Node}_R^{\text{audit,claim}}.$$

The disjoint-union decomposition in Definition 7.18 implies that the initial reachability stage has empty intersection with the time-content node set.

Suppose that reachability stage

$k$

has empty intersection with the time-content node set. By hypothesis, every audit-boundary edge that outputs a time-content node has at least one time-content node among its inputs. Its input set is therefore not a subset of reachability stage

$k$

Consequently, reachability stage

$k + 1$

also has empty intersection with the time-content node set. Induction over reachability stages and Proposition 7.31 give the claim.  $\square$

**Definition 7.44** (Evidence-Conditioned Primitive-Time-Root Support Family and Minimal Support Family). For each  $R \in \text{Ref}^{\text{audit}}$  and  $\eta \in \text{Result}_R$ , define the evidence-conditioned primitive-time-root support family by

$$\begin{aligned} & \text{RootSupport}_R^{\text{audit|evid}}(\eta) \\ & := \left\{ \text{RootSeed} \subseteq \text{TimeContent}_R^{\text{audit,root}} \mid \right. \\ & \quad \left. \text{NodeOf}_R^{\text{audit,result}}(\eta) \in \text{Reach}_R^{\text{audit|evid}}(\text{RootSeed}) \right\} \end{aligned}$$

Define the minimal primitive-time-root support family by

$$\begin{aligned} \text{MinRootSupport}_R^{\text{audit}}(\eta) \\ := & \left\{ \text{RootSeed} \in \text{RootSupport}_R^{\text{audit|evid}}(\eta) \mid \right. \\ & \forall \text{RootSeed}' \in \text{RootSupport}_R^{\text{audit|evid}}(\eta), \\ & \left. \text{RootSeed}' \subseteq \text{RootSeed} \implies \text{RootSeed} \subseteq \text{RootSeed}' \right\} \end{aligned}$$

**Proposition 7.45** (Upward Closure and Minimal Elements of the Evidence-Conditioned Primitive-Time-Root Support Family). For each  $R \in \text{Ref}^{\text{audit}}$  and  $\eta \in \text{Result}_R$ , the evidence-conditioned primitive-time-root support family is upward closed under inclusion. Moreover, if

$$\text{RootSupport}_R^{\text{audit|evid}}(\eta) \neq \emptyset$$

then

$$\text{MinRootSupport}_R^{\text{audit}}(\eta) \neq \emptyset$$

holds, and any two distinct elements of the minimal primitive-time-root support family are incomparable under inclusion.

*Proof.* To prove upward closure, take arbitrary  $\text{RootSeed} \in \text{RootSupport}_R^{\text{audit|evid}}(\eta)$  and  $\text{RootSeed} \subseteq \text{RootSeed}' \subseteq \text{TimeContent}_R^{\text{audit,root}}$ . Induction on the recursion in Definition 7.30 shows that inclusion of the starting sets entails inclusion at every reachability stage. Hence  $\text{RootSeed}' \in \text{RootSupport}_R^{\text{audit|evid}}(\eta)$  holds.

Because the primitive-time-root set is finite, its power set and the evidence-conditioned primitive-time-root support family are finite. Choose an element of minimum cardinality from the nonempty evidence-conditioned primitive-time-root support family. Every element of the same support family contained in that element equals it by minimality of cardinality. Hence the minimal primitive-time-root support family is nonempty.

If one of two distinct elements of the minimal primitive-time-root support family were contained in the other, the minimality of the containing element would make them equal. Thus two distinct elements are incomparable under inclusion.  $\square$

**Definition 7.46** (Finite Root-Support Antichain Update). For a family of subsets of the primitive-time-root set

$$\mathcal{F} \subseteq \mathcal{P} \left( \text{TimeContent}_R^{\text{audit,root}} \right)$$

define minimization under inclusion by

$$\begin{aligned} \text{Min}_{\subseteq}(\mathcal{F}) \\ := & \left\{ \text{RootSeed} \in \mathcal{F} \mid \forall \text{RootSeed}' \in \mathcal{F}, \quad \begin{array}{l} \text{RootSeed}' \subseteq \text{RootSeed} \\ \implies \text{RootSeed} \subseteq \text{RootSeed}' \end{array} \right\} \end{aligned}$$

For each  $R \in \text{Ref}^{\text{audit}}$ , define the initial root-support antichain assignment

$$\begin{aligned} \text{RootAnt}_R^{\text{audit,init}} : \text{Node}_R^{\text{audit,bd}} \\ \longrightarrow \mathcal{P} \left( \mathcal{P} \left( \text{TimeContent}_R^{\text{audit,root}} \right) \right) \end{aligned}$$

by

$$\begin{aligned} & \text{RootAnt}_R^{\text{audit,init}}(\text{AuditNode}) \\ & := \begin{cases} \{\emptyset\}, & \text{AuditNode} \in \text{Env}_R^{\text{audit,evid}}, \\ \{\{x\}\}, & \text{AuditNode} = \text{NodeOf}_R^{\text{audit,time}}(x), \quad x \in \text{TimeContent}_R^{\text{audit,root}}, \\ \emptyset, & \text{otherwise} \end{cases} \end{aligned}$$

For a map

$$\mathcal{S} : \text{Node}_R^{\text{audit,bd}} \longrightarrow \mathcal{P}\left(\mathcal{P}\left(\text{TimeContent}_R^{\text{audit,root}}\right)\right)$$

that assigns a support antichain to each node, and for each  $\text{AuditEdge} \in \text{Edge}_R^{\text{audit,bd}}$ , define the hyperedge root-support update by

$$\begin{aligned} & \text{RootUpd}_{R,\text{AuditEdge}}^{\text{audit}}(\mathcal{S})(\text{AuditNode}) \\ & := \text{Min}_{\subseteq} \left( \mathcal{S}(\text{AuditNode}) \right. \\ & \quad \left. \cup \left\{ \begin{array}{l} \bigcup_{\text{InputNode} \in \text{In}_R^{\text{audit}}(\text{AuditEdge})} \text{RootSeed}_{\text{InputNode}} \\ \text{AuditNode} \in \text{Out}_R^{\text{audit}}(\text{AuditEdge}), \\ \mid \\ \forall \text{InputNode} \in \text{In}_R^{\text{audit}}(\text{AuditEdge}), \\ \text{RootSeed}_{\text{InputNode}} \in \mathcal{S}(\text{InputNode}) \end{array} \right\} \right) \end{aligned}$$

For a finite edge sequence  $\left(\text{AuditEdge}^{(j)}\right)_{j=1}^{\ell}$ , define

$$\mathcal{S}^{(0)} := \text{RootAnt}_R^{\text{audit,init}}$$

and

$$\mathcal{S}^{(j)} := \text{RootUpd}_{R,\text{AuditEdge}^{(j)}}^{\text{audit}}(\mathcal{S}^{(j-1)})$$

When the terminal assignment satisfies

$$\begin{aligned} & \forall \text{AuditEdge} \in \text{Edge}_R^{\text{audit,bd}}, \\ & \text{RootUpd}_{R,\text{AuditEdge}}^{\text{audit}}(\mathcal{S}^{(\ell)}) = \mathcal{S}^{(\ell)} \end{aligned}$$

say that the finite edge sequence saturates the root-support antichain.

**Proposition 7.47** (Termination, Scan-Order Independence, and Completeness of the Finite Root-Support Antichain Update). For each  $R \in \text{Ref}^{\text{audit}}$ , there exists a finite edge sequence that saturates the root-support antichain.

If two finite edge sequences saturate the root-support antichain, their terminal assignments coincide. Moreover, for any  $\eta \in \text{Result}_R$ , the terminal assignment

$$\mathcal{S}^{(\ell)}$$

satisfies

$$\begin{aligned} & \mathcal{S}^{(\ell)}\left(\text{NodeOf}_R^{\text{audit,result}}(\eta)\right) \\ & = \text{MinRootSupport}_R^{\text{audit}}(\eta) \end{aligned}$$

*Proof.* Represent the support antichain at each node by its upward closure. The hyperedge root-support update adds to the upward closure at that node the upward closure of the union of supports selected one from each input node of the hyperedge. Minimization under inclusion does not change the upward closure. The hyperedge root-support update is therefore monotone and extensive with respect to the inclusion order on upward closures over the finite set

$$\prod_{\text{AuditNode} \in \text{Node}_R^{\text{audit}, \text{bd}}} \mathcal{P} \left( \mathcal{P} \left( \text{TimeContent}_R^{\text{audit}, \text{root}} \right) \right)$$

. Repeatedly selecting a hyperedge that produces a change reaches a fixed point for all hyperedges after finitely many steps.

The initial root-support antichain assignment assigns the empty support to the reference formal-evidence environment nodes and the singleton support containing only that root to each primitive-time-root node. Induction on the reachability stages in Definition 7.30 shows that the upward closure of the support family at each node at the fixed point equals the family of all subsets of primitive-time roots that place that node in the evidence-conditioned dependency-reachability set. The converse follows by induction that maps each hyperedge root-support update to one stage of the reachability recursion.

The fixed point is therefore unique and independent of the order in which the edges are scanned. Taking the inclusion-minimal elements at a result node yields the minimal primitive-time-root support family in Definition 7.44 .  $\square$

**Definition 7.48** (Residual Result Sets with Respect to Primitive-Time Roots). For each  $R \in \text{Ref}^{\text{audit}}$ , define the primitive-time-root-dependent result set by

$$\begin{aligned} & \text{Result}_R^{\text{audit}, \text{rootdep}} \\ & := \left\{ \eta \in \text{Result}_R^{\text{audit}, \text{resid}} \mid \text{RootSupport}_R^{\text{audit}|\text{evid}}(\eta) \neq \emptyset \right\} \end{aligned}$$

Define the result set that remains unreachable after addition of primitive-time roots by

$$\begin{aligned} & \text{Result}_R^{\text{audit}, \text{rootunreach}} \\ & := \left\{ \eta \in \text{Result}_R^{\text{audit}, \text{resid}} \mid \text{RootSupport}_R^{\text{audit}|\text{evid}}(\eta) = \emptyset \right\} \end{aligned}$$

**Proposition 7.49** (Four-Class Partition of Reference-Specific Results). For each  $R \in \text{Ref}^{\text{audit}}$ ,

$$\begin{aligned} \text{Result}_R &= \text{Result}_R^{\text{audit}, \text{gen}} \dot{\cup} \text{Result}_R^{\text{audit}, \text{evidonly}} \\ &\quad \dot{\cup} \text{Result}_R^{\text{audit}, \text{rootdep}} \dot{\cup} \text{Result}_R^{\text{audit}, \text{rootunreach}} \end{aligned}$$

holds. Moreover,

$$\begin{aligned} \text{Result}_R^{\text{audit}, \text{unreach}} &= \text{Result}_R^{\text{audit}, \text{evidonly}} \dot{\cup} \text{Result}_R^{\text{audit}, \text{rootdep}} \\ &\quad \dot{\cup} \text{Result}_R^{\text{audit}, \text{rootunreach}} \end{aligned}$$

and

$$\text{Result}_R^{\text{audit}, \text{resid}} = \text{Result}_R^{\text{audit}, \text{rootdep}} \dot{\cup} \text{Result}_R^{\text{audit}, \text{rootunreach}}$$

hold.

*Proof.* The inclusion of starting sets in Definition 7.28 and monotonicity of reachability give

$$\text{Reach}_R^{\text{audit,gen}} \subseteq \text{Reach}_R^{\text{audit|evid}}(\emptyset)$$

. The evidence-environment-reachable nongenerated result set is obtained by removing the audit-generated results from the results reachable from the evidence-conditioned empty-root starting set. The two sets are therefore disjoint.

The residual result set is obtained by removing the evidence-environment-reachable nongenerated result set from the result set outside the generation-reachability region. For each residual result, the evidence-conditioned primitive-time-root support family is either empty or nonempty. The former case gives the result set that remains unreachable after addition of primitive-time roots; the latter gives the primitive-time-root-dependent result set. Hence the four sets are pairwise disjoint and their union is the entire result set. Applying the definitions to the result set outside the generation-reachability region and to the residual result set gives the other two equalities.  $\square$

**Definition 7.50** (Primitive-Time-Root-Result Dependency Relations). For each  $R \in \text{Ref}^{\text{audit}}$ , define the primitive-time-root-result dependency relation by

$$\begin{aligned} & \text{RootResultDep}_R^{\text{audit}} \\ &:= \left\{ (x, \eta) \in \text{TimeContent}_R^{\text{audit,root}} \times \text{Result}_R^{\text{audit,rootdep}} \mid \right. \\ & \quad \left. \exists \text{RootSeed} \in \text{MinRootSupport}_R^{\text{audit}}(\eta), \quad x \in \text{RootSeed} \right\} \end{aligned}$$

Define the essential primitive-time-root-result dependency relation by

$$\begin{aligned} & \text{RootResultEssDep}_R^{\text{audit}} \\ &:= \left\{ (x, \eta) \in \text{RootResultDep}_R^{\text{audit}} \mid \right. \\ & \quad \left. \forall \text{RootSeed} \in \text{MinRootSupport}_R^{\text{audit}}(\eta), \quad x \in \text{RootSeed} \right\} \end{aligned}$$

**Proposition 7.51** (Inclusion and Result Projection of Primitive-Time-Root-Result Dependency Relations). For each  $R \in \text{Ref}^{\text{audit}}$ ,

$$\begin{aligned} & \text{RootResultEssDep}_R^{\text{audit}} \\ & \subseteq \text{RootResultDep}_R^{\text{audit}} \end{aligned}$$

holds. Moreover,

$$\begin{aligned} & \left\{ \eta \in \text{Result}_R^{\text{audit,rootdep}} \mid \exists x \in \text{TimeContent}_R^{\text{audit,root}}, \right. \\ & \quad \left. (x, \eta) \in \text{RootResultDep}_R^{\text{audit}} \right\} \\ &= \text{Result}_R^{\text{audit,rootdep}} \end{aligned}$$

holds.

*Proof.* The first inclusion follows from Definition 7.50 .



Take any  $\eta \in \text{Result}_R^{\text{audit}, \text{rootdep}}$ . By Definition 7.48 the primitive-time-root support family is nonempty. By Proposition 7.45 the minimal primitive-time-root support family is also nonempty. Take any

$$\text{RootSeed} \in \text{MinRootSupport}_R^{\text{audit}}(\eta)$$

. Because the evidence-conditioned primitive-time-root support family of a residual result does not contain the empty set, there exists some

$$x \in \text{RootSeed}$$

such that

$$(x, \eta) \in \text{RootResultDep}_R^{\text{audit}}$$

holds.

The reverse inclusion follows from

$$\text{RootResultDep}_R^{\text{audit}} \subseteq \text{TimeContent}_R^{\text{audit}, \text{root}} \times \text{Result}_R^{\text{audit}, \text{rootdep}}$$

.

□

**Definition 7.52** (Reference Subsets Induced by Primitive-Time Roots and the Result Decomposition). Define the result-class index domain by

$$\mathcal{I}_{\text{res}} := \{\text{evidonly}, \text{resid}, \text{rootdep}, \text{rootunreach}, \text{unreach}\}$$

. In order, the indices denote nongenerated results reachable under the evidence environment, residual results, primitive-time-root-dependent results, results that remain unreachable after addition of primitive-time roots, and results outside the generation-reachability region.

Define the set of references containing primitive-time roots by

$$\text{Ref}^{\text{audit}, \text{root}} := \left\{ R \in \text{Ref}^{\text{audit}} \mid \text{TimeContent}_R^{\text{audit}, \text{root}} \neq \emptyset \right\}$$

.

For each  $\rho \in \mathcal{I}_{\text{res}} \setminus \{\text{resid}\}$ , define the set of references containing results in the corresponding class by

$$\text{Ref}^{\text{audit}, \rho} := \left\{ R \in \text{Ref}^{\text{audit}} \mid \text{Result}_R^{\text{audit}, \rho} \neq \emptyset \right\}$$

.

**Proposition 7.53** (Membership Inclusion for Primitive-Time-Root-Dependent References).

$$\text{Ref}^{\text{audit}, \text{rootdep}} \subseteq \text{Ref}^{\text{audit}, \text{root}} \cap \text{Ref}^{\text{audit}, \text{unreach}}$$

holds.

*Proof.* Take any

$$R \in \text{Ref}^{\text{audit}, \text{rootdep}}$$

. By Definition 7.52, there exists some  $\eta \in \text{Result}_R^{\text{audit}, \text{rootdep}}$ . By Definition 7.48,  $\text{RootSupport}_R^{\text{audit|evid}}(\eta) \neq \emptyset$ . Choose

$$\text{RootSeed} \in \text{RootSupport}_R^{\text{audit|evid}}(\eta)$$

from this support family.

Because the result  $\eta$  is residual, it is not reachable from the empty-root starting set augmented by the evidence environment. Hence

$$\text{RootSeed} \neq \emptyset.$$

By Definition 7.44,  $\text{RootSeed} \subseteq \text{TimeContent}_R^{\text{audit,root}}$ , so the primitive-time-root set is nonempty. Thus

$$R \in \text{Ref}^{\text{audit,root}}.$$

By Proposition 7.49,

$$\text{Result}_R^{\text{audit,rootdep}} \subseteq \text{Result}_R^{\text{audit,unreach}}.$$

Because the left-hand side is nonempty, so is the right-hand side, and therefore

$$R \in \text{Ref}^{\text{audit,unreach}}$$

follows. □

**Proposition 7.54** (Result Reachability Criterion after Addition of Primitive-Time Roots).

For each  $R \in \text{Ref}^{\text{audit}}$ , assume that

$$\text{TimeContent}_R \neq \emptyset, \quad \text{Result}_R^{\text{audit,resid}} \neq \emptyset$$

. Assume further that

$$\text{Reach}_R^{\text{audit,gen}} \cap \text{Node}_R^{\text{audit,time}} = \emptyset$$

and

$$\begin{aligned} \forall \eta \in \text{Result}_R^{\text{audit,resid}}, \quad \text{NodeOf}_R^{\text{audit,result}}(\eta) \\ \in \text{Reach}_R^{\text{audit|evid}}(\text{TimeContent}_R) \end{aligned}$$

hold. Then

$$\begin{aligned} \text{TimeContent}_R^{\text{audit,gen}} &= \emptyset, & \text{TimeContent}_R^{\text{audit,root}} &= \text{TimeContent}_R, \\ \text{Result}_R^{\text{audit,rootdep}} &= \text{Result}_R^{\text{audit,resid}}, & \text{Result}_R^{\text{audit,rootunreach}} &= \emptyset \end{aligned}$$

hold. Moreover,

$$R \in \text{Ref}^{\text{audit,root}}, \quad R \in \text{Ref}^{\text{audit,rootdep}}, \quad R \in \text{Ref}^{\text{audit,unreach}}$$

hold.

*Proof.* The empty intersection of the generation-reachability set and the time-content node set, together with Definition 7.39, shows that the set of audit-generated time contents is empty and that the primitive-time-root set equals the entire time-content set.

Take any

$$\eta \in \text{Result}_R^{\text{audit,resid}}$$

. Because the result node belongs to the evidence-conditioned dependency-reachability set after all primitive-time roots have been added,

$$\text{TimeContent}_R \in \text{RootSupport}_R^{\text{audit|evid}}(\eta)$$

holds. By Definition 7.48, every residual result is a primitive-time-root-dependent result, and the result set that remains unreachable after addition of primitive-time roots is empty.

The nonemptiness of the time-content set and of the residual result set, together with Definition 7.52, gives the three reference memberships. □

**Proposition 7.55** (Criterion for All Primitive-Time Roots and Dependence of All Results). Suppose that each  $R \in \text{Ref}^{\text{audit}}$  satisfies the assumptions of Proposition 7.54 . Assume further that

$$\text{Result}_R \neq \emptyset$$

and

$$\text{Reach}_R^{\text{audit|evid}}(\emptyset) \cap \text{Node}_R^{\text{audit,result}} = \emptyset$$

hold. Then, in addition to the conclusion of that proposition,

$$\begin{aligned} \text{Result}_R^{\text{audit,gen}} &= \emptyset, & \text{Result}_R^{\text{audit,evidonly}} &= \emptyset, \\ \text{Result}_R^{\text{audit,resid}} &= \text{Result}_R^{\text{audit,unreach}} = \text{Result}_R \end{aligned}$$

hold.

*Proof.* Definition 7.28 and monotonicity of reachability give  $\text{Reach}_R^{\text{audit,gen}} \subseteq \text{Reach}_R^{\text{audit|evid}}(\emptyset)$ . By the assumption and Definition 7.39, the set of audit-generated results is empty, and the result set outside the generation-reachability region equals the entire result set. By the assumption and Definition 7.42, the evidence-environment-reachable nongenerated result set is empty, and the residual result set equals the entire result set.

The conclusion of Proposition 7.54 then follows.  $\square$

**Definition 7.56** (Reference Set in Which All Time Contents Are Primitive-Time Roots). Define the set of references for which the audit-generated time-content set is empty and the primitive-time-root set equals the entire time-content set by

$$\begin{aligned} &\text{Ref}^{\text{audit,rootall}} \\ &:= \left\{ R \in \text{Ref}^{\text{audit}} \mid \begin{array}{l} \text{TimeContent}_R^{\text{audit,gen}} = \emptyset, \\ \text{TimeContent}_R^{\text{audit,root}} = \text{TimeContent}_R \end{array} \right\}. \end{aligned}$$

**Definition 7.57** (Reference Set in Which All Results Depend on Primitive-Time Roots). Define the set of references for which the audit-generated result set is empty and the primitive-time-root-dependent result set equals the entire result set by

$$\begin{aligned} &\text{Ref}^{\text{audit,resultrootall}} \\ &:= \left\{ R \in \text{Ref}^{\text{audit}} \mid \begin{array}{l} \text{Result}_R^{\text{audit,gen}} = \emptyset, \\ \text{Result}_R^{\text{audit,rootdep}} = \text{Result}_R \end{array} \right\}. \end{aligned}$$

**Definition 7.58** (Reference Set with All Time Contents as Primitive-Time Roots and All Results Root-Dependent). Define the intersection of the reference set in which all time contents are primitive-time roots and the reference set in which all results depend on primitive-time roots by

$$\text{Ref}^{\text{audit,rootfull}} := \text{Ref}^{\text{audit,rootall}} \cap \text{Ref}^{\text{audit,resultrootall}}$$

.

**Definition 7.59** (Disjoint Unions of Per-Reference Result Classes and Primitive-Time-Root Dependency Relations). For each  $\rho \in \mathcal{I}_{\text{res}} \setminus \{\text{unreach}\}$ , define the per-reference disjoint union of the corresponding result sets by

$$\begin{aligned} &\text{Result}^{\text{audit},\rho} \\ &:= \coprod_{R \in \text{Ref}^{\text{audit}}} \left( \{R\} \times \text{Result}_R^{\text{audit},\rho} \right) \end{aligned}$$

Define the per-reference disjoint unions of the primitive-time-root-result dependency relation and the essential primitive-time-root-result dependency relation, respectively, by

$$\begin{aligned}\text{RootResultDep}^{\text{audit}} &:= \coprod_{R \in \text{Ref}^{\text{audit}}} (\{R\} \times \text{RootResultDep}_R^{\text{audit}}), \\ \text{RootResultEssDep}^{\text{audit}} &:= \coprod_{R \in \text{Ref}^{\text{audit}}} (\{R\} \times \text{RootResultEssDep}_R^{\text{audit}})\end{aligned}$$

**Proposition 7.60** (Per-Reference Disjoint-Union Decomposition of Results outside the Generation-Reachability Region). For the reference subsets,

$$\begin{aligned}\text{Ref}^{\text{audit}, \text{unreach}} &= \text{Ref}^{\text{audit}, \text{evidonly}} \cup \text{Ref}^{\text{audit}, \text{rootdep}} \\ &\quad \cup \text{Ref}^{\text{audit}, \text{rootunreach}}\end{aligned}$$

holds.

For the per-reference disjoint unions,

$$\begin{aligned}\coprod_{R \in \text{Ref}^{\text{audit}}} (\{R\} \times \text{Result}_R^{\text{audit}, \text{unreach}}) \\ = \text{Result}^{\text{audit}, \text{evidonly}} \dot{\cup} \text{Result}^{\text{audit}, \text{rootdep}} \dot{\cup} \text{Result}^{\text{audit}, \text{rootunreach}}\end{aligned}$$

holds. Moreover,

$$\text{Result}^{\text{audit}, \text{resid}} = \text{Result}^{\text{audit}, \text{rootdep}} \dot{\cup} \text{Result}^{\text{audit}, \text{rootunreach}}$$

holds.

*Proof.* Applying Proposition 7.49 to each reference shows that the result set outside the generation-reachability region equals the disjoint union of the evidence-environment-reachable nongenerated result set, the primitive-time-root-dependent result set, and the result set that remains unreachable after addition of primitive-time roots. Applying the nonemptiness of each component to Definition 7.52 gives the equality of reference subsets.

For each reference  $R \in \text{Ref}^{\text{audit}}$ , apply

$$\{R\} \times (-)$$

to the decomposition of results outside the generation-reachability region, and then take the disjoint union over the reference set. This gives the first equality of per-reference disjoint unions. Applying the same Cartesian product to the residual-result decomposition for each reference and taking the disjoint union over the reference set gives the second equality.  $\square$

**Proposition 7.61** (Inclusion and Result Projection in the Per-Reference Disjoint Union of Primitive-Time-Root-Result Dependency Relations). For the per-reference disjoint unions of the dependency relations,

$$\text{RootResultEssDep}^{\text{audit}} \subseteq \text{RootResultDep}^{\text{audit}}$$

holds.

For the per-reference disjoint union of the primitive-time-root-dependent result sets,

$$\begin{aligned} & \text{Result}^{\text{audit}, \text{rootdep}} \\ &= \left\{ (R, \eta) \in \text{Result}^{\text{audit}, \text{rootdep}} \mid \right. \\ & \quad \exists x \in \text{TimeContent}_R^{\text{audit}, \text{root}}, \\ & \quad \left. (R, (x, \eta)) \in \text{RootResultDep}^{\text{audit}} \right\} \end{aligned}$$

holds.

*Proof.* For each reference  $R \in \text{Ref}^{\text{audit}}$ , apply

$$\{R\} \times (-)$$

to the inclusion in Proposition 7.51, and then take the disjoint union over the reference set. This gives the first inclusion. Applying the same Cartesian product to the result projection in that proposition and taking the disjoint union over the reference set gives the second equality.  $\square$

**Proposition 7.62** (Component Equalities of Per-Reference Disjoint Unions). For each  $R \in \text{Ref}^{\text{audit}}$  and  $\rho \in \mathcal{I}_{\text{res}} \setminus \{\text{unreach}\}$ ,

$$\begin{aligned} & \{\eta \mid (R, \eta) \in \text{Result}^{\text{audit}, \rho}\} \\ &= \text{Result}_R^{\text{audit}, \rho}. \end{aligned}$$

Moreover, for each  $R \in \text{Ref}^{\text{audit}}$ ,

$$\begin{aligned} & \{(x, \eta) \mid (R, (x, \eta)) \in \text{RootResultDep}^{\text{audit}}\} \\ &= \text{RootResultDep}_R^{\text{audit}}, \end{aligned}$$

and

$$\begin{aligned} & \{(x, \eta) \mid (R, (x, \eta)) \in \text{RootResultEssDep}^{\text{audit}}\} \\ &= \text{RootResultEssDep}_R^{\text{audit}} \end{aligned}$$

hold.

*Proof.* For every disjoint union in Definition 7.59, the collection of second components of elements whose first component is a fixed  $R \in \text{Ref}^{\text{audit}}$  equals the corresponding per-reference set or per-reference relation. Applying this equality to each disjoint union gives the six equalities.  $\square$

**Definition 7.63** (Change Condition for the Generation-Dependency Audit System). The input to the generation-dependency audit system consists of the following five items.

1. The audit reference set

$$\text{Ref}^{\text{audit}}.$$

2. The time contents, transfer events, results, and result prerequisites of each reference

$$\text{TimeContent}_R, \quad \text{TransferEvent}_R, \quad \text{Result}_R, \quad \left\{ \text{ResultPrereq}_{R, \eta} \right\}_{\eta \in \text{Result}_R}.$$

3. The boundary claim nodes and original-source support sets of each reference

$$\text{Node}_R^{\text{audit,claim}}, \quad \text{Support}_R^{\text{audit,transfer}}, \quad \text{Support}_R^{\text{audit,result}}.$$

4. The five-type classification of time contents, the eight-axis classification of time-order structures, and the transfer inputs and outputs

$$\text{TypeOf}_R^{\text{time,ord,struct}}, \quad \text{AxisOf}_R^{\text{time,ord,struct}}, \quad \text{TransferIn}_{R,\epsilon}, \quad \text{TransferOut}_{R,\epsilon}.$$

5. The physical-input initial set and reference formal-evidence environment

$$\text{Init}_R^{\text{audit,phys}}, \quad \text{Env}_R^{\text{audit,evid}}.$$

The change condition for the generation-dependency audit system holds if the codomain, domain, map value, set component, or original-source support of any of the five items changes.

**Definition 7.64** (Retrospective Generation-Dependency Reaudit). If the change condition in Definition 7.63 holds, recompute the following for all references in the stated order.

1. The reference-specific time contents, transfer events, results, boundary claim nodes, and original-source support sets.
2. The five types of time contents, eight axes of time-order structure, membership-structure signatures, and dependency-content structure signatures.
3. The audit boundary nodes, audit boundary edges, input and output maps, and audit boundary directed hypergraph.
4. The generation-reachability set from the physical-input initial set, audit-generation traces, structural domains of influence, and essential domains of influence.
5. The audit-generated time contents, primitive-time roots, audit-generated results, and results outside the generation-reachability region.
6. The evidence-conditioned dependency-reachability family, nongenerated results reachable under the evidence environment, residual results, evidence-conditioned primitive-time-root support family, minimal support family, four-class partition of results, and primitive-time-root–result dependency relations.
7. The time-order-structure audit family, reference subsets, and per-reference disjoint unions of result classes and dependency relations.

Even when the audit-input change occurs in only one reference, recompute the per-reference disjoint unions and reference subsets over all references.

**Proposition 7.65** (Determinacy of Generation-Dependency Reaudit). Once the five inputs in Definition 7.63 are specified, the audit boundary directed hypergraph, generation-reachability set, evidence-conditioned dependency-reachability family, primitive-time roots, four-class partition of results, and per-reference disjoint unions of dependency relations obtained in the order specified in Definition 7.64 are uniquely determined.

*Proof.* The node-forming and edge-node-forming maps are specified as injections, while the node-type, edge-type, support-set, input, output, and classification maps are specified as maps on their respective domains. Their map values uniquely determine the audit boundary directed hypergraph.

The starting set and recursion uniquely determine each reachability stage, which stabilizes after finitely many stages by Proposition 7.31. The physical-input initial set uniquely determines the generation-reachability set. Set-builder definitions and set difference then uniquely determine the audit-generated time contents, primitive-time roots, audit-generated results, and results outside the generation-reachability region from the generation-reachability set.

The reference formal-evidence environment and each subset of primitive-time roots uniquely determine the evidence-conditioned dependency-reachability family. The evidence-conditioned dependency-reachability set corresponding to the empty primitive-time-root subset, the audit-generated results, and the results outside the generation-reachability region uniquely determine the nongenerated results reachable under the evidence environment and the residual results. The evidence-conditioned dependency-reachability family uniquely determines the evidence-conditioned primitive-time-root support family and its minimal support family; nonemptiness or emptiness of the same support family uniquely determines the primitive-time-root-dependent results and the results that remain unreachable after addition of primitive-time roots. Consequently, the four-class partition of results, the primitive-time-root–result dependency relations, and all per-reference disjoint unions are also uniquely determined.  $\square$

**Proposition 7.66** (Recomputation Invariance under Stable Audit Inputs). If the five inputs in Definition 7.63 remain identical after one recomputation over all references, then repeating that recomputation produces an audit boundary directed hypergraph, generation-reachability set, evidence-conditioned dependency-reachability family, primitive-time roots, four-class partition of results, and per-reference disjoint unions of dependency relations identical to the immediately preceding recomputation results.

*Proof.* By Proposition 7.65, identical five inputs produce a unique recomputation result. Because the five inputs are identical before and after the repetition, the respective values of the audit boundary directed hypergraph, generation-reachability set, evidence-conditioned dependency-reachability family, primitive-time roots, four-class partition of results, and per-reference disjoint unions of dependency relations also coincide.  $\square$

**Definition 7.67** (Fibers and Zero Fibers of Structure-Signature Maps). For a map between sets, the inverse image of a singleton in the codomain is called the fiber over that point. If a zero element is specified in the codomain, the fiber over the zero element is called the zero fiber.

For each structure-signature map in Definition 7.23 and Definition 7.24, the zero fiber is

$$\left( \text{Sig}_R^{\text{time,ord,struct},\varsigma_{\text{sig}}} \right)^{-1} (\{(0,0,0)\}), \quad \varsigma_{\text{sig}} \in \mathcal{I}_{\text{sig}}^{\text{time,ord,struct}}.$$

Each zero fiber consists of the time contents for which all three corresponding structural attributes have value zero.

**Definition 7.68** (Support Antichains). A family of sets is called an antichain under inclusion if any two distinct members of the family are incomparable under inclusion. An antichain whose members are primitive-time-root support sets is called a support antichain.

**Proposition 7.69** (Support-Antichain Property of the Minimal Primitive-Time-Root Support Family). For each  $R \in \text{Ref}^{\text{audit}}$  and  $\eta \in \text{Result}_R$ ,

$$\text{MinRootSupport}_R^{\text{audit}}(\eta)$$

is a support antichain.

*Proof.* By Proposition 7.45, any two distinct members of the minimal primitive-time-root support family are incomparable under inclusion. The claim follows from Definition 7.68 .  $\square$

**Definition 7.70** (Atomic Time Uses and Reference-Specific Classifications of Time Use). A time use is an occurrence in which one time content is used as a formal input, index, order position, clock reading, interval, evolution variable, or reference boundary in an original-source definition, formula, proposition, or explicit statement. Each time use is associated with an original-source location, one time content, a use role, and an audit state. An atomic time use is the smallest unit audited individually in the finite audit of a reference.

Here, “atomic” denotes the granularity of decomposition in a reference-specific finite audit. “Primitive” in Definition 7.10 denotes the order between the generation of time content and its use within a law.

A time use at whose original-source location a time expression appears directly is called an explicit time use. A time use in which time content is incorporated into the establishment of a definition, relation, order, evolution, observation, or comparison is called a structural time use. Additional classifications, including direct versus derived occurrence and independent root versus structural derivation, are defined separately for each reference according to its formal structure.

A time use presupposed by a definition, law, coordinate, observation, comparison, or derivation of a result is called a law-time use. A time use that gives only an order of presentation or reporting within a reference is called a report-order use.

**Definition 7.71** (Time-Use Roles, Original-Source Support, and Audit Branches). A time-use role is the formal function performed by a time content within a reference. A reference-specific finite audit uses the following roles.

**Interval** Gives the separation or order interval between two positions.

**Quantity** Uses a time content as a numerical value, difference, rate, scale, or factor in a product.

**Cross-position correspondence** Associates states, events, or readings belonging to different positions.

**Clock reading** Associates a clock state or clock output with a time content.

**Causal** Gives an ordering or dependency condition between cause and effect.

**State evolution** Orders a state transformation, trajectory, or evolution equation.

**Stochastic evolution** Orders a transition probability, transition rate, or the evolution of a stochastic process.



**Observation record** Associates the acquisition, preservation, or arrangement of observational results with a time content.

**Reference boundary** Gives the boundary of a starting point, endpoint, reference position, or domain of application.

An original-source support is a record that associates an audit node, audit edge, or result with a reference identifier, original-source location, subject of attribution, evidence type, and conditions of application.

If the same formal evidence gives rise to multiple audit paths, an audit branch is each path distinguished by a finite index. Reachability of results, the supporting-edge set, and inclusion-minimal support are determined separately for each audit branch. Statement evidence is retained as a quotation record containing the original text, a translation, the original-source location, and the minimal context.

**Definition 7.72** (Cycles, Residuals, and Completeness Counts in Finite Dependency Structures). In an audit boundary directed hypergraph, a finite sequence of audit edges is called a directed cycle if the output nodes of each edge connect to the input nodes of the next edge and the output nodes of the terminal edge connect to the input nodes of the initial edge. A maximal connected component consisting of nodes and audit edges that share a directed cycle is called a cyclic component.

An audit edge  $\text{AuditEdge} \in \text{Edge}_R^{\text{audit}, \text{bd}}$  has one-edge self-dependence if

$$\text{In}_R^{\text{audit}}(\text{AuditEdge}) \cap \text{Out}_R^{\text{audit}}(\text{AuditEdge}) \neq \emptyset$$

holds.

Original-source evidence acquired for audit but corresponding to none of the audit-claim records, the original-source supports of audit edges, or the original-source supports of results is called unmatched evidence.

A completeness count vector is a finite sequence, arranged in the same reference-specific order, of the cardinalities of the finite bases, correspondences, nodes, audit edges, support families, obligation states, and unmatched-evidence sets that constitute a reference-specific finite audit. The objects and ordering of its coordinates are stated explicitly in the definition for each reference, and components are compared between finite audits that have the same coordinate order.

The terminology of Definition 7.70, Definition 7.71, and Definition 7.72 is used in the reference-specific finite audits in the Supplement . The time contents, transfer events, results, and dependency structure of a reference-specific finite audit are represented, respectively, by the previously defined

$$\text{TimeContent}_R, \quad \text{TransferEvent}_R, \quad \text{Result}_R$$

and

$$\text{Hypergraph}_R^{\text{audit}}$$

. Original-source locations, time-use roles, reference-specific obligations, and finite counts belong to the finite audit of each reference.

**Definition 7.73** (Primitive Law-Bearing Time Premises). If the time content taken as input by a law-time use in Definition 7.70 belongs to  $\text{TimeContent}_R^{\text{audit,root}}$  in Definition 7.39, the use of that time content within the law is called a primitive law-bearing time premise. A primitive law-bearing time premise is a primitive-time root used to establish a definition, law, evolution, observation, comparison, or derivation of a result.

**Definition 7.74** (Finite Audit Carrier and Reference-Specific Complete Time-Use Audit). The finite sets of original-source locations, atomic time uses, time contents, time-use roles, obligation atoms, audit edges, support families, and results identified for one reference are called its finite audit carrier.

An audit that assigns an obligation atom and typed audit state to every atomic time use in a finite audit carrier and supplies correspondences to all audit edges, support families, and results is called a reference-specific complete time-use audit. “Complete” denotes exhaustive coverage of this finite audit carrier.

**Definition 7.75** (Classification Margins). If the same formal evidence admits multiple audit-classification branches, each furnished with an original-source location and grounds for its adoption, the resulting reference-specific finite branch set is called a classification margin. Each branch associates a classification boundary with the formal evidence supporting that boundary.

Remark 7.156 defines the interpretability of statements. A classification margin represents audit-classification branches for formal evidence.

**Definition 7.76** (Time-Generation, Cyclic-Generation, and Evidence-Open Candidates). In a reference-specific finite audit, an obligation-conforming audit path from the physical-input initial set to a time-content node is called a time-generation candidate. The existence of a time-generation candidate is determined by the reachability set in Definition 7.30, the audit-generation traces in Definition 7.32, and the audit-generated time contents in Definition 7.39 .

A candidate path that outputs the generation of time content is called a cyclic-generation candidate if it contains a cyclic component or one-edge self-dependence from Definition 7.72 . Cyclic-generation candidates are counted separately for each reference as candidates containing cyclic structure.

A candidate path is called an evidence-open candidate if an obligation atom required by that path has an evidence-open state. Evidence-open candidates are counted separately for each reference through the correspondence between original-source support and obligation states.

**Definition 7.77** (Result-Preserving Correspondences). A bijection that places the result nodes of a reference-specific finite audit in one-to-one correspondence with the result values in the reference-specific result projection of the same reference is called a result-preserving correspondence. A result-preserving correspondence uniquely specifies, within the result projection, the reference-specific result represented by each result node.

The terminology of Definition 7.73, Definition 7.74, Definition 7.75, Definition 7.76, and Definition 7.77 applies to the reference-specific audits in the Supplement . The candidate sets, classification-margin projections, and result-preserving correspondences of each reference are determined by the finite audit carrier and original-source supports of that reference.

### 7.2.1 Separate Formal and Statement-Evidence Audit Determinations for Time-Bearing Expressions

**Definition 7.78** (Four Audit Determinations for Time-Bearing Expressions). Separate the formal evidence and statement evidence of a reference, and apply each of the following four determinations to a time content or time-bearing expression.

**Pre-generation placement.** A time content or time-bearing expression is supplied to a law operation, definition, or derivation of a result before its physical generation.

**Omission of generation stages.** A physical input, clock state, record, or observational result is connected to a law description or quantitative expression that uses a time variable without passing through the generation stages required to proceed from nontemporal physical inputs to a qualified time representation.

**Reattribution of a time property.** A relational difference holding among physical carriers, clocks, signals, records, or observables is stated as a change of state or property of a time object independent of them.

**Reverse presupposition of a derived quantity.** A derived quantity that requires generated time content is used as a premise of that time content or of the generation structure that confers physical-time qualification.

For formal evidence, each determination follows from the original-source definitions, formulas, propositions, and dependencies. For statement evidence, the relation to the four determinations is conditional on each interpretation permitted by the original text, translation, original-source location, and minimal context.

The four determinations are not mutually exclusive. Multiple determinations may hold simultaneously for a single object, reference, or evidence track.

### 7.2.2 Reference-Evidence Layer, Dependency-Effective Layer, and Semantic Audit Boundary

**Definition 7.79** (Reference-Evidence Layer, Dependency-Effective Layer, and Semantic Audit Boundary). For each  $R \in \text{Ref}^{\text{audit}}$ , suppose that the finite sets

$$\text{Node}_R^{\text{audit,use}}, \quad \text{Node}_R^{\text{audit,time,src}}, \quad \text{Node}_R^{\text{audit,rootuse}}$$

are given. Assume that these sets and

$$\text{Node}_R^{\text{audit,claim}}, \quad \text{Node}_R^{\text{audit,result}}$$

are pairwise disjoint.

Define the full reference-evidence node set by

$$\begin{aligned} \text{Node}_R^{\text{audit,evid}} := & \text{Node}_R^{\text{audit,claim}} \dot{\cup} \text{Node}_R^{\text{audit,use}} \\ & \dot{\cup} \text{Node}_R^{\text{audit,time,src}} \dot{\cup} \text{Node}_R^{\text{audit,rootuse}} \\ & \dot{\cup} \text{Node}_R^{\text{audit,result}} \end{aligned}$$

and the dependency-effective node set by

$$\begin{aligned} \text{Node}_R^{\text{audit,dep}} := & \text{Node}_R^{\text{audit,claim}} \dot{\cup} \text{Node}_R^{\text{audit,time,src}} \\ & \dot{\cup} \text{Node}_R^{\text{audit,result}} \end{aligned}$$

. Define the reference-evidence node-type set by

$$\begin{aligned} \text{Type}^{\text{audit,evid,node}} &:= \text{Type}^{\text{audit,node}} \\ &\cup \left\{ \text{Type}_{\text{use}}^{\text{audit,node}}, \text{Type}_{\text{rootuse}}^{\text{audit,node}} \right\} \end{aligned}$$

. Define the reference-evidence node-type map

$$\begin{aligned} \text{TypeOf}_R^{\text{audit,evid,node}} &: \text{Node}_R^{\text{audit,evid}} \\ &\longrightarrow \text{Type}^{\text{audit,evid,node}} \end{aligned}$$

to have, on the respective disjoint-union components, the values

$$\text{Type}_{\text{claim}}^{\text{audit,node}}, \quad \text{Type}_{\text{use}}^{\text{audit,node}}, \quad \text{Type}_{\text{time}}^{\text{audit,node}}, \quad \text{Type}_{\text{rootuse}}^{\text{audit,node}}, \quad \text{Type}_{\text{result}}^{\text{audit,node}}$$

. Define the dependency-effective node-type map by

$$\text{TypeOf}_R^{\text{audit,dep,node}} := \text{TypeOf}_R^{\text{audit,evid,node}} \Big|_{\text{Node}_R^{\text{audit,dep}}}$$

. Suppose that the reference-specific content correspondence

$$\begin{aligned} \text{ContentOf}_R^{\text{audit,time}} &: \text{Node}_R^{\text{audit,time,src}} \\ &\twoheadrightarrow \text{TimeContent}_R \end{aligned}$$

is given. Define the boundary-contraction map on dependency-effective nodes

$$\begin{aligned} \text{Quot}_R^{\text{audit,bd}} &: \text{Node}_R^{\text{audit,dep}} \\ &\twoheadrightarrow \text{Node}_R^{\text{audit,bd}} \end{aligned}$$

by the following two actions. First, for

$$\text{AuditNode} \in \text{Node}_R^{\text{audit,claim}} \cup \text{Node}_R^{\text{audit,result}}$$

define

$$\text{Quot}_R^{\text{audit,bd}}(\text{AuditNode}) := \text{AuditNode}$$

. Second, for

$$\text{AuditNode} \in \text{Node}_R^{\text{audit,time,src}}$$

define

$$\begin{aligned} &\text{Quot}_R^{\text{audit,bd}}(\text{AuditNode}) \\ &:= \text{NodeOf}_R^{\text{audit,time}} \left( \text{ContentOf}_R^{\text{audit,time}}(\text{AuditNode}) \right) \end{aligned}$$

. Suppose that the reference-specific dependency input and output maps

$$\begin{aligned} \text{In}_R^{\text{audit,dep}} &: \text{Edge}_R^{\text{audit,bd}} \longrightarrow \mathcal{P}_{\text{fin}} \left( \text{Node}_R^{\text{audit,dep}} \right), \\ \text{Out}_R^{\text{audit,dep}} &: \text{Edge}_R^{\text{audit,bd}} \longrightarrow \mathcal{P}_{\text{fin}} \left( \text{Node}_R^{\text{audit,dep}} \right) \end{aligned}$$

are given. For each layer

$$\text{Layer} \in \{\text{evid}, \text{dep}\}$$

define the finite directed hypergraphs of the reference-evidence layer and the dependency-effective layer uniformly by

$$\text{Hypergraph}_R^{\text{audit,Layer}} := \left( \text{Node}_R^{\text{audit,Layer}}, \text{Edge}_R^{\text{audit,bd}}, \right. \\ \left. \text{In}_R^{\text{audit,dep}}, \text{Out}_R^{\text{audit,dep}}, \right. \\ \left. \text{TypeOf}_R^{\text{audit,Layer,node}}, \text{TypeOf}_R^{\text{audit,edge}} \right)$$

The three hypergraphs are said to be compatible under boundary contraction if, for each  $\text{AuditEdge} \in \text{Edge}_R^{\text{audit,bd}}$ ,

$$\text{In}_R^{\text{audit}}(\text{AuditEdge}) = \text{Quot}_R^{\text{audit,bd}} \left[ \text{In}_R^{\text{audit,dep}}(\text{AuditEdge}) \right], \\ \text{Out}_R^{\text{audit}}(\text{AuditEdge}) = \text{Quot}_R^{\text{audit,bd}} \left[ \text{Out}_R^{\text{audit,dep}}(\text{AuditEdge}) \right]$$

hold.

**Proposition 7.80** (Relations among the Reference-Evidence Layer, Dependency-Effective Layer, and Semantic Audit Boundary). Under Definition 7.79,

$$\text{Node}_R^{\text{audit,dep}} \subseteq \text{Node}_R^{\text{audit,evid}}$$

holds, and the input and output nodes of every edge belong to

$$\text{Node}_R^{\text{audit,dep}}$$

. Consequently,

$$\text{Node}_R^{\text{audit,use}} \dot{\cup} \text{Node}_R^{\text{audit,rootuse}} \\ = \text{Node}_R^{\text{audit,evid}} \setminus \text{Node}_R^{\text{audit,dep}}$$

consists of isolated nodes of the reference-evidence finite directed hypergraph.

If the three hypergraphs are compatible under boundary contraction, the boundary-contraction map

$$\text{Quot}_R^{\text{audit,bd}}$$

gives a surjective directed-hypergraph morphism from the dependency-effective finite directed hypergraph to the audit boundary directed hypergraph.

The cardinalities of the node sets are

$$\left| \text{Node}_R^{\text{audit,evid}} \right| = \left| \text{Node}_R^{\text{audit,claim}} \right| + \left| \text{Node}_R^{\text{audit,use}} \right| \\ + \left| \text{Node}_R^{\text{audit,time,src}} \right| + \left| \text{Node}_R^{\text{audit,rootuse}} \right| \\ + \left| \text{Node}_R^{\text{audit,result}} \right|, \\ \left| \text{Node}_R^{\text{audit,dep}} \right| = \left| \text{Node}_R^{\text{audit,claim}} \right| + \left| \text{Node}_R^{\text{audit,time,src}} \right| \\ + \left| \text{Node}_R^{\text{audit,result}} \right|$$

*Proof.* The inclusion of the two node sets, the expression for their set difference, and the isolation of every node in that difference follow from the disjoint-union decompositions in Definition 7.79 and from the codomains of the input and output maps.

On the claim-node set and the result-node set, the boundary-contraction map is the identity and is surjective. On the time-content node set, it is surjective as the composite of the reference-specific content correspondence and the time-content node-forming map. Surjectivity on the three disjoint-union components makes the boundary-contraction map surjective onto the audit boundary node set. When the three hypergraphs are compatible, the compatibility equations for the input and output sets show that the boundary-contraction map preserves the input and output of each hyperedge.

The two cardinality formulas follow from the corresponding finite disjoint-union decompositions.  $\square$

### 7.2.3 Edge-Restricted Audit Reachability and Branch Support

**Definition 7.81** (Edge-Restricted Audit Closure and Reachability). For each  $R \in \text{Ref}^{\text{audit}}$  and  $\text{EdgeSet} \subseteq \text{Edge}_R^{\text{audit,bd}}$ , define the edge-restricted one-stage closure operator

$$\text{ClStep}_{R,\text{EdgeSet}}^{\text{audit}} : \mathcal{P}(\text{Node}_R^{\text{audit,bd}}) \longrightarrow \mathcal{P}(\text{Node}_R^{\text{audit,bd}})$$

for each

$$\text{NodeSet} \subseteq \text{Node}_R^{\text{audit,bd}}$$

by

$$\text{ClStep}_{R,\text{EdgeSet}}^{\text{audit}}(\text{NodeSet}) := \text{NodeSet} \cup \bigcup_{\substack{\text{AuditEdge} \in \text{EdgeSet} \\ \text{In}_R^{\text{audit}}(\text{AuditEdge}) \subseteq \text{NodeSet}}} \text{Out}_R^{\text{audit}}(\text{AuditEdge})$$

. For each

$$\text{Seed} \subseteq \text{Node}_R^{\text{audit,bd}}$$

define the edge-restricted audit-reachability set by iterated composition, with the zeroth iterate taken to be the identity map:

$$\text{Reach}_{R,\text{EdgeSet}}^{\text{audit}}(\text{Seed}) := (\text{ClStep}_{R,\text{EdgeSet}}^{\text{audit}})^{|\text{Node}_R^{\text{audit,bd}}|}(\text{Seed})$$

. For each

$$k \in \mathbb{N}$$

define the edge-restricted  $k$ -th reachability stage by

$$\text{Reach}_{R,\text{EdgeSet}}^{\text{audit},(k)}(\text{Seed}) = (\text{ClStep}_{R,\text{EdgeSet}}^{\text{audit}})^k(\text{Seed})$$

.

**Definition 7.82** (Audit-Branch Support Families). For each  $R \in \text{Ref}^{\text{audit}}$ , let the finite audit-branch set be

$$\text{Branch}_R^{\text{audit}}$$

. Let the map assigning to each audit branch the reference-specific result determined by that branch be

$$\text{ResultOf}_R^{\text{audit,branch}} : \text{Branch}_R^{\text{audit}} \longrightarrow \text{Result}_R$$

For each

$$\iota \in \text{Branch}_R^{\text{audit}}$$

define the corresponding result edge by

$$\text{Edge}_{R,\iota}^{\text{audit,result}} := \text{EdgeOf}_R^{\text{audit,result}} \left( \text{ResultOf}_R^{\text{audit,branch}}(\iota) \right)$$

For each

$$\text{EdgeSet} \subseteq \text{Edge}_R^{\text{audit,bd}}$$

define the support determination by

$$\begin{aligned} \text{EdgeSet} \Vdash_R^{\text{audit}} \iota &\iff \text{Edge}_{R,\iota}^{\text{audit,result}} \in \text{EdgeSet} \\ &\quad \wedge \text{NodeOf}_R^{\text{audit,result}} \left( \text{ResultOf}_R^{\text{audit,branch}}(\iota) \right) \\ &\quad \in \text{Reach}_{R,\text{EdgeSet}}^{\text{audit}} \left( \text{In}_R^{\text{audit}} \left( \text{Edge}_{R,\iota}^{\text{audit,result}} \right) \right) \end{aligned}$$

Define the branch-support family by

$$\text{Support}_{R,\iota}^{\text{audit,branch}} := \left\{ \text{EdgeSet} \subseteq \text{Edge}_R^{\text{audit,bd}} \mid \text{EdgeSet} \Vdash_R^{\text{audit}} \iota \right\}$$

Define the proper-subset support predicate by

$$\begin{aligned} \text{SubSupp}_{R,\iota}^{\text{prop}} (\text{EdgeSet}', \text{EdgeSet}) \\ \iff \text{EdgeSet}', \text{EdgeSet} \in \text{Support}_{R,\iota}^{\text{audit,branch}}, \\ \text{EdgeSet}' \subsetneq \text{EdgeSet} \end{aligned}$$

Define the inclusion-minimal branch-support family by

$$\begin{aligned} \text{MinSupport}_{R,\iota}^{\text{audit,branch}} \\ := \left\{ \text{EdgeSet} \mid \begin{array}{l} \text{EdgeSet} \in \text{Support}_{R,\iota}^{\text{audit,branch}}, \\ \nexists \text{EdgeSet}' \text{ SubSupp}_{R,\iota}^{\text{prop}} (\text{EdgeSet}', \text{EdgeSet}) \end{array} \right\} \end{aligned}$$

**Proposition 7.83** (Inclusion-Minimal Support Property of a Single Result Edge). For each

$$R \in \text{Ref}^{\text{audit}}, \quad \iota \in \text{Branch}_R^{\text{audit}}$$

$$\begin{aligned} \text{MinSupport}_{R,\iota}^{\text{audit,branch}} \\ = \left\{ \left\{ \text{Edge}_{R,\iota}^{\text{audit,result}} \right\} \right\} \end{aligned}$$

holds.

*Proof.* Take the input set of the result edge as the initial set and include that result edge in the edge subset. The result node is then added at the first reachability stage. The singleton result-edge set therefore belongs to the branch-support family.

By definition, every edge set in the branch-support family contains the result edge. No proper subset of the singleton result-edge set contains that edge. Hence the singleton result-edge set is inclusion-minimal, and there is no other inclusion-minimal element.  $\square$

### 7.2.4 Original-Source Support, Terminal Residuals, and Completeness Counts

**Definition 7.84** (Original-Source Supports of Audit Edges and Terminal Residuals). For each  $R \in \text{Ref}^{\text{audit}}$ , let the map assigning a finite set of original-source locations to each dependency-effective node be

$$\text{SrcLoc}_R^{\text{audit}} : \text{Node}_R^{\text{audit,dep}} \longrightarrow \mathcal{P}_{\text{fin}}(\text{Loc}^{\text{src}})$$

. Define the original-source location set of an audit edge by

$$\begin{aligned} & \text{SrcLoc}_R^{\text{audit,edge}}(\text{AuditEdge}) \\ & := \bigcup_{\text{AuditNode} \in \text{In}_R^{\text{audit,dep}}(\text{AuditEdge}) \cup \text{Out}_R^{\text{audit,dep}}(\text{AuditEdge})} \text{SrcLoc}_R^{\text{audit}}(\text{AuditNode}) \end{aligned}$$

. Define the original-source support of each edge by

$$\begin{aligned} & \text{Evidence}_R^{\text{audit,edge}}(\text{AuditEdge}) \\ & := \left\{ \left( R, \text{SrcLoc}_R^{\text{audit,edge}}(\text{AuditEdge}), \right. \right. \\ & \quad \left. \left. \begin{array}{l} \text{attributed-source, source-statement-or-derivation,} \\ \text{applicability-condition} \end{array} \right) \right\} \end{aligned}$$

. Define the inverse of the result-node-forming map by

$$\begin{aligned} & \text{ResultOf}_R^{\text{audit,node}} : \text{Node}_R^{\text{audit,result}} \longrightarrow \text{Result}_R, \\ & \text{ResultOf}_R^{\text{audit,node}} := \left( \text{NodeOf}_R^{\text{audit,result}} \right)^{-1} \end{aligned}$$

. Let the set of directed cycles of the dependency-effective finite directed hypergraph be

$$\text{Cycle}_R^{\text{audit}}$$

. Define the set of audit edges with one-edge self-dependence by

$$\begin{aligned} & \text{SelfEdge}_R^{\text{audit}} \\ & := \left\{ \text{AuditEdge} \in \text{Edge}_R^{\text{audit,bd}} \mid \begin{array}{l} \text{In}_R^{\text{audit,dep}}(\text{AuditEdge}) \\ \cap \text{Out}_R^{\text{audit,dep}}(\text{AuditEdge}) \neq \emptyset \end{array} \right\} \end{aligned}$$

. Define the audit acyclicity condition by

$$\text{Acyc}^{\text{audit}}(R) \iff \text{Cycle}_R^{\text{audit}} = \text{SelfEdge}_R^{\text{audit}} = \emptyset$$

. Among the acquired original-source evidence, let the complement of the set of original-source locations corresponding to reference-evidence nodes, original-source supports of edges, or result supports be

$$\text{Evidence}_R^{\text{audit,unmatched}}$$



**Proposition 7.85** (Reachability from the Empty Initial Set for Audit Edges with Nonempty Inputs). For each  $R \in \text{Ref}^{\text{audit}}$  and  $\text{EdgeSet} \subseteq \text{Edge}_R^{\text{audit,bd}}$ , if every

$$\text{AuditEdge} \in \text{EdgeSet}$$

satisfies

$$\text{In}_R^{\text{audit}}(\text{AuditEdge}) \neq \emptyset$$

then

$$\text{Reach}_{R, \text{EdgeSet}}^{\text{audit}}(\emptyset) = \emptyset$$

holds.

*Proof.* The initial reachability stage is the empty set. For each audit edge whose input set is nonempty, that input set is not a subset of the empty reachability stage. No node is therefore added at the next reachability stage. Induction on the stages shows that every reachability stage is empty.  $\square$

**Definition 7.86** (Reference-Specific Completeness-Count Fields). Define the reference-specific completeness-count field set by

$$\begin{aligned} \text{Field}^{\text{audit,count}} := \{ & \text{timeuse, timecontent, userole, contentrole,} \\ & \text{obligation, useobligation, roleobligation,} \\ & \text{physclaim, mathclaim, claimnode,} \\ & \text{usenode, srccontentnode, rootusenode,} \\ & \text{resultnode, evidnode,} \\ & \text{transferredge, resultedge, auditedge,} \\ & \text{branch, alternative,} \\ & \text{fulfilled, unfulfilled, open,} \\ & \text{result, margin, unmatched} \} \end{aligned}$$

In display order, the fields represent the cardinalities of the sets of atomic time uses, time contents, use–role correspondences, content–role correspondences, obligation atoms, use–obligation correspondences, role–obligation correspondences, physical and formal claims, mathematical-audit claims, all claim nodes, time-use nodes, reference-specific time-content nodes, root-use nodes, result nodes, all reference-evidence nodes, transfer edges, result edges, all audit edges, audit branches, alternative interpretations, fulfilled obligations, unfulfilled obligations, open obligations, reference-specific results, classification-margin candidates, and unmatched evidence.

Let the reference-specific completeness-count map be

$$\text{Count}_R^{\text{audit}} : \text{Field}^{\text{audit,count}} \longrightarrow \mathbb{N}$$

. The reference-specific finite audit gives the cardinality of the finite object corresponding to each field as the value of this map.

**Proposition 7.87** (Structural Identities for Reference-Specific Completeness Counts). For each  $R \in \text{Ref}^{\text{audit}}$ ,

$$\begin{aligned} \text{Count}_R^{\text{audit}}(\text{evidnode}) &= \sum_{F \in \{\text{claimnode, usenode, srccontentnode, rootusenode, resultnode}\}} \text{Count}_R^{\text{audit}}(F), \\ \text{Count}_R^{\text{audit}}(\text{auditedge}) &= \sum_{F \in \{\text{transferredge, resultedge}\}} \text{Count}_R^{\text{audit}}(F) \end{aligned}$$

hold.

*Proof.* The first equality follows from the finite disjoint-union decomposition of the full reference-evidence node set in Definition 7.79 . The second equality follows from the finite disjoint-union decomposition of the audit boundary edge set in Definition 7.26 .  $\square$

### 7.2.5 Typed Finite Audit Input

**Definition 7.88** (Audit Decomposition of Original-Source Formal Entries). For one reference, a finite collection of formal entries that belong to the same original-source location or to a contiguous minimal context and are presented together before the audit is called an undecomposed original-source entry group. If one undecomposed entry contains multiple objects, predicates, dependency inputs, validity conditions, or domains of application, partition it, preserving the logical direction, quantification, conditions, and domain of application of the original-source entry, into units that can be subjected independently to truth-value and dependency determinations. Each resulting unit is called a minimal audit claim.

Assign to every minimal audit claim exactly one claim type: either a physical and formal claim or a mathematical-audit claim. A component is called type-homogeneous if claim type, object, and unit of determination are not mixed within it. A type-homogeneous component whose values are minimal audit claims obtained from original-source definitions, formulas, propositions, operations, or conditions and that preserves provenance correspondence to original-source locations is called a formal-evidence component.

A type-homogeneous component separated from a reference's formal description so as to bear independently an object, predicate, dependency input, validity condition, or domain of application is called a semantic component. A semantic component is an audit unit obtained in this paper by decomposing formal evidence from the original source.

**Definition 7.89** (Typed Finite Audit-Input Structure). To record the minimal audit claims and formal-evidence components of Definition 7.88 as typed finite inputs, for each  $R \in \text{Ref}^{\text{audit}}$ , define the claim-category index domain by  $\mathcal{I}_{\text{claim}} := \{\text{phys}, \text{math}\}$ . Take two finite index sets for the respective claim categories and the following five finite index sets:

$$\begin{aligned} & \left( \text{Idx}_R^{\text{audit}, \chi} \right)_{\chi \in \mathcal{I}_{\text{claim}}}, \quad \text{Idx}_R^{\text{audit}, \text{use}}, \\ & \text{Idx}_R^{\text{audit}, \text{time}, \text{src}}, \quad \text{Idx}_R^{\text{audit}, \text{result}}, \\ & \text{Idx}_R^{\text{audit}, \text{transfer}}, \quad \text{Idx}_R^{\text{audit}, \text{branch}}. \end{aligned}$$

Let the value families of physical and formal claims, mathematical-audit claims, time uses, reference-specific time contents, and results, respectively, be

$$\begin{aligned} & \left( \text{ClaimVal}_{R,i}^{\text{audit}, \chi} \right)_{\chi \in \mathcal{I}_{\text{claim}}, i \in \text{Idx}_R^{\text{audit}, \chi}}, \\ & \left( \text{UseVal}_{R,k}^{\text{audit}} \right)_{k \in \text{Idx}_R^{\text{audit}, \text{use}}}, \\ & \left( \text{SrcTimeVal}_{R,j}^{\text{audit}} \right)_{j \in \text{Idx}_R^{\text{audit}, \text{time}, \text{src}}}, \\ & \left( \text{ResultVal}_{R,q}^{\text{audit}} \right)_{q \in \text{Idx}_R^{\text{audit}, \text{result}}} \end{aligned}$$

For each

$$\kappa_{\text{edge}} \in \mathcal{I}_{\text{audit}, \text{edge}}$$

let the claim-input table and time-content-input table be given uniformly by

$$\begin{aligned}\text{ClaimIn}_R^{\text{audit}, \kappa_{\text{edge}}} &: \text{Idx}_R^{\text{audit}, \kappa_{\text{edge}}} \longrightarrow \mathcal{P}_{\text{fin}} \left( \text{Idx}_R^{\text{audit}, \text{phys}} \right), \\ \text{TimeIn}_R^{\text{audit}, \kappa_{\text{edge}}} &: \text{Idx}_R^{\text{audit}, \kappa_{\text{edge}}} \longrightarrow \mathcal{P}_{\text{fin}} \left( \text{Idx}_R^{\text{audit}, \text{time}, \text{src}} \right)\end{aligned}$$

. Let the transfer-output table be

$$\text{TimeOut}_R^{\text{audit}, \text{transfer}} : \text{Idx}_R^{\text{audit}, \text{transfer}} \longrightarrow \mathcal{P}_{\text{fin}} \left( \text{Idx}_R^{\text{audit}, \text{time}, \text{src}} \right)$$

. Require the claim-input and time-content-input tables to take the same values at any two result indices whose result values are equal.

Let the correspondence between time uses and reference-specific time contents be

$$\text{UseSrc}_R^{\text{audit}} \subseteq \text{Idx}_R^{\text{audit}, \text{use}} \times \text{Idx}_R^{\text{audit}, \text{time}, \text{src}}$$

. Let the original-source location maps be

$$\begin{aligned}\text{UseLoc}_R^{\text{audit}} &: \text{Idx}_R^{\text{audit}, \text{use}} \longrightarrow \text{Loc}^{\text{src}}, \\ \text{ClaimLoc}_R^{\text{audit}} &: \text{Idx}_R^{\text{audit}, \text{phys}} \longrightarrow \text{Loc}^{\text{src}}\end{aligned}$$

. Let the map assigning the result determined by each audit branch be

$$\text{ResultOf}_R^{\text{audit}, \text{branch}} : \text{Idx}_R^{\text{audit}, \text{branch}} \longrightarrow \left\{ \text{ResultVal}_{R, q}^{\text{audit}} \mid q \in \text{Idx}_R^{\text{audit}, \text{result}} \right\}$$

. Let the unmatched-evidence set and the reference-specific completeness-count map be

$$\begin{aligned}\text{Evidence}_R^{\text{audit}, \text{unmatched}} &\subseteq \text{Loc}^{\text{src}}, \\ \text{Count}_R^{\text{audit}} &: \text{Field}^{\text{audit}, \text{count}} \longrightarrow \mathbb{N}\end{aligned}$$

. Denote by

$$\text{AuditInput}_R^{\text{fin}}$$

the typed finite audit-input structure whose components are the foregoing finite index sets, value families, input tables, time-use-content correspondence, original-source locations, branch-result correspondence, unmatched evidence, and reference-specific completeness-count map.

### 7.2.6 Canonical Realization of Typed Finite Audit Input

**Definition 7.90** (Canonical Realization of Typed Finite Audit Input). Take the structure  $\text{AuditInput}_R^{\text{fin}}$  of Definition 7.89. Define the claim-kind values by

$$\begin{aligned}\text{ClaimKind}_{\text{phys}} &:= \text{physical-formal-claim}, \\ \text{ClaimKind}_{\text{math}} &:= \text{mathematical-audit-claim}\end{aligned}$$

. For each  $\chi \in \mathcal{I}_{\text{claim}}$  and  $i \in \text{Idx}_R^{\text{audit},\chi}$ , define the enumerated-node map and its value by

$$\begin{aligned} \text{EnumNode}_R^{\text{audit},\text{claim},\chi}(i) &= \text{Node}_{R,i}^{\text{audit},\text{claim},\chi}, \\ \text{Node}_{R,i}^{\text{audit},\text{claim},\chi} &:= \left( \begin{array}{c} \text{Type}_{\text{claim}}^{\text{audit},\text{node}}, \\ \text{ClaimKind}_{\chi}, \\ i, \text{ClaimVal}_{R,i}^{\text{audit},\chi} \end{array} \right) \end{aligned}$$

. Define the use-node category domain by  $\mathcal{I}_{\text{use}} := \{\text{use}, \text{rootuse}\}$ . For each  $v \in \mathcal{I}_{\text{use}}$  and  $k \in \text{Idx}_R^{\text{audit},\text{use}}$ , define the enumerated-node map and its value by

$$\begin{aligned} \text{EnumNode}_R^{\text{audit},v}(k) &= \text{Node}_{R,k}^{\text{audit},v}, \\ \text{Node}_{R,k}^{\text{audit},v} &:= \left( \begin{array}{c} \text{Type}_v^{\text{audit},\text{node}}, \\ k, \text{UseVal}_{R,k}^{\text{audit}} \end{array} \right) \end{aligned}$$

. Define the enumeration maps and their values for reference-specific time nodes and result nodes by

$$\begin{aligned} \text{EnumNode}_R^{\text{audit},\text{time},\text{src}}(j) &= \text{Node}_{R,j}^{\text{audit},\text{time},\text{src}}, \\ \text{Node}_{R,j}^{\text{audit},\text{time},\text{src}} &:= \left( \begin{array}{c} \text{Type}_{\text{time}}^{\text{audit},\text{node}}, \\ \text{source}, \\ j, \text{SrcTimeVal}_{R,j}^{\text{audit}} \end{array} \right), \\ \text{EnumNode}_R^{\text{audit},\text{result}}(q) &= \text{Node}_{R,q}^{\text{audit},\text{result}}, \\ \text{Node}_{R,q}^{\text{audit},\text{result}} &:= \text{NodeOf}_R^{\text{audit},\text{result}}(\text{ResultVal}_{R,q}^{\text{audit}}) \end{aligned}$$

. Define the claim-node set, the node set for each  $v$  in the use-node category domain, the reference-specific time-node set, and the result-node set by

$$\begin{aligned} \text{Node}_R^{\text{audit},\text{claim}} &:= \bigcup_{\chi \in \mathcal{I}_{\text{claim}}} \text{EnumNode}_R^{\text{audit},\text{claim},\chi} [\text{Idx}_R^{\text{audit},\chi}], \\ \text{Node}_R^{\text{audit},v} &:= \text{EnumNode}_R^{\text{audit},v} [\text{Idx}_R^{\text{audit},\text{use}}], \\ \text{Node}_R^{\text{audit},\text{time},\text{src}} &:= \text{EnumNode}_R^{\text{audit},\text{time},\text{src}} [\text{Idx}_R^{\text{audit},\text{time},\text{src}}], \\ \text{Node}_R^{\text{audit},\text{result}} &:= \text{EnumNode}_R^{\text{audit},\text{result}} [\text{Idx}_R^{\text{audit},\text{result}}]. \end{aligned}$$

Define the range of the reference-specific time-value family as the time-content set by

$$\text{TimeContent}_R := \left\{ \text{SrcTimeVal}_{R,j}^{\text{audit}} \mid j \in \text{Idx}_R^{\text{audit},\text{time},\text{src}} \right\}$$

and define the time-content correspondence by

$$\text{TimeSem}_R^{\text{audit}}(j) := \text{SrcTimeVal}_{R,j}^{\text{audit}}$$

. On the node set, denote the reference-specific content correspondence by

$$\begin{aligned} &\text{ContentOf}_R^{\text{audit},\text{time}} \left( \text{Node}_{R,j}^{\text{audit},\text{time},\text{src}} \right) \\ &:= \text{TimeSem}_R^{\text{audit}}(j) \end{aligned}$$

Define transfer events and reference-specific results by

$$\begin{aligned}\text{TransferEvent}_{R,s} &:= (R, s, \text{transfer}), \\ \text{Result}_{R,q} &:= \text{ResultVal}_{R,q}^{\text{audit}}\end{aligned}$$

and set

$$\begin{aligned}\text{TransferEvent}_R &:= \left\{ \text{TransferEvent}_{R,s} \mid s \in \text{Idx}_R^{\text{audit,transfer}} \right\}, \\ \text{Result}_R &:= \left\{ \text{Result}_{R,q} \mid q \in \text{Idx}_R^{\text{audit,result}} \right\}\end{aligned}$$

Define the enumerated edges by

$$\begin{aligned}\text{Edge}_{R,s}^{\text{audit,transfer}} &:= \text{EdgeOf}_R^{\text{audit,transfer}}(\text{TransferEvent}_{R,s}), \\ \text{Edge}_{R,q}^{\text{audit,result}} &:= \text{EdgeOf}_R^{\text{audit,result}}(\text{Result}_{R,q})\end{aligned}$$

Denote the two families for each  $\kappa_{\text{edge}} \in \mathcal{I}_{\text{audit,edge}}$  and  $i \in \text{Idx}_R^{\text{audit},\kappa_{\text{edge}}}$  by  $\text{Edge}_{R,i}^{\text{audit},\kappa_{\text{edge}}}$ .

Define the reference-specific dependency input by

$$\begin{aligned}\text{In}_R^{\text{audit,dep}}(\text{Edge}_{R,i}^{\text{audit},\kappa_{\text{edge}}}) &:= \text{EnumNode}_R^{\text{audit,claim,phys}} \left[ \text{ClaimIn}_R^{\text{audit},\kappa_{\text{edge}}}(i) \right] \\ &\quad \cup \text{EnumNode}_R^{\text{audit,time,src}} \left[ \text{TimeIn}_R^{\text{audit},\kappa_{\text{edge}}}(i) \right],\end{aligned}$$

Define the reference-specific dependency output for a transfer edge by

$$\text{Out}_R^{\text{audit,dep}}(\text{Edge}_{R,s}^{\text{audit,transfer}}) := \text{EnumNode}_R^{\text{audit,time,src}} \left[ \text{TimeOut}_R^{\text{audit,transfer}}(s) \right]$$

and, for a result edge, by

$$\text{Out}_R^{\text{audit,dep}}(\text{Edge}_{R,q}^{\text{audit,result}}) := \text{EnumNode}_R^{\text{audit,result}} [\{q\}]$$

For the same  $\kappa_{\text{edge}}$  and  $i$ , define the audit-boundary support by

$$\begin{aligned}\text{ClaimSupport}_{R,i}^{\text{audit},\kappa_{\text{edge}}} &:= \text{EnumNode}_R^{\text{audit,claim,phys}} \left[ \text{ClaimIn}_R^{\text{audit},\kappa_{\text{edge}}}(i) \right], \\ \text{TimePrereq}_{R,i}^{\text{audit},\kappa_{\text{edge}}} &:= \text{TimeSem}_R^{\text{audit}} \left[ \text{TimeIn}_R^{\text{audit},\kappa_{\text{edge}}}(i) \right]\end{aligned}$$

Define the support, time prerequisites, and time output for transfer events and results by

$$\begin{aligned}\text{Support}_R^{\text{audit,transfer}}(\text{TransferEvent}_{R,s}) &:= \text{ClaimSupport}_{R,s}^{\text{audit,transfer}}, \\ \text{TransferIn}_{R,\text{TransferEvent}_{R,s}} &:= \text{TimePrereq}_{R,s}^{\text{audit,transfer}}, \\ \text{Support}_R^{\text{audit,result}}(\text{Result}_{R,q}) &:= \text{ClaimSupport}_{R,q}^{\text{audit,result}}, \\ \text{ResultPrereq}_{R,\text{Result}_{R,q}} &:= \text{TimePrereq}_{R,q}^{\text{audit,result}}, \\ \text{TransferOut}_{R,\text{TransferEvent}_{R,s}} &:= \text{TimeSem}_R^{\text{audit}} \left[ \text{TimeOut}_R^{\text{audit,transfer}}(s) \right]\end{aligned}$$

Define the original-source location map of dependency-effective nodes by

$$\begin{aligned}\text{SrcLoc}_R^{\text{audit}}(\text{Node}_{R,p}^{\text{audit,claim,phys}}) &:= \{ \text{ClaimLoc}_R^{\text{audit}}(p) \}, \\ \text{SrcLoc}_R^{\text{audit}}(\text{Node}_{R,m}^{\text{audit,claim,math}}) &:= \emptyset, \\ \text{SrcLoc}_R^{\text{audit}}(\text{Node}_{R,j}^{\text{audit,time,src}}) &:= \{ \text{UseLoc}_R^{\text{audit}}(k) \mid (k, j) \in \text{UseSrc}_R^{\text{audit}} \}, \\ \text{SrcLoc}_R^{\text{audit}}(\text{Node}_{R,q}^{\text{audit,result}}) &:= \emptyset\end{aligned}$$

Let the audit-branch set be

$$\text{Branch}_R^{\text{audit}} := \text{Idx}_R^{\text{audit,branch}}$$

The structure determined above,

$$\begin{aligned} \text{Realize}^{\text{audit}}(\text{AuditInput}_R^{\text{fin}}) := & \left( \text{Hypergraph}_R^{\text{audit,evid}}, \text{Hypergraph}_R^{\text{audit,dep}}, \right. \\ & \text{Hypergraph}_R^{\text{audit}}, \text{Quot}_R^{\text{audit,bd}}, \\ & \text{SrcLoc}_R^{\text{audit}}, \left\{ \text{Support}_{R,\iota}^{\text{audit,branch}} \right\}_{\iota \in \text{Branch}_R^{\text{audit}}}, \\ & \text{Cycle}_R^{\text{audit}}, \text{SelfEdge}_R^{\text{audit}}, \text{Evidence}_R^{\text{audit,unmatched}}, \\ & \left. \text{Count}_R^{\text{audit}} \right) \end{aligned}$$

is called the canonical realization of typed finite audit input.

**Proposition 7.91** (Uniqueness of the Realization of Typed Finite Audit Input). Once all components of the typed finite audit-input structure  $\text{AuditInput}_R^{\text{fin}}$  are given, all node sets, all edge sets, input and output maps, the boundary-contraction map, the original-source location map, the branch-support families, the unmatched-evidence set, and the completeness-count map belonging to the canonical realization of Definition 7.90 are uniquely determined.

*Proof.* The enumerated nodes are uniquely determined by the corresponding value families, and the node sets are uniquely determined as the images or finite disjoint unions of the respective enumerated-node families. The transfer and result edges and the reference-specific inputs and outputs are uniquely determined by the corresponding finite index sets and input tables.

The reference-specific content correspondence determines the boundary-contraction map, and the images of the inputs and outputs determine the audit boundary. The original-source location map is determined by the original-source location components and the time-use-content correspondence, and the audit-branch support families are determined by the branch-result correspondence and edge-restricted reachability. The unmatched-evidence set and the completeness-count map are given as input components. Hence every component of the canonical realization is unique.  $\square$

### 7.2.7 Typed Index Display of Audit Tables

**Definition 7.92** (Typed Index Display of Audit Tables). For an index set  $I$ , a family

$$\mathcal{F} = (F_i)_{i \in I}$$

and a finite set  $J \subseteq I$ , let

$$\mathcal{F}[J] := \{F_i \mid i \in J\}$$

. When positive integers  $m, n$  satisfy  $m \leq n$ , define the discrete closed interval in the natural numbers by

$$\llbracket m, n \rrbracket := \{k \in \mathbb{N} \mid m \leq k \leq n\}$$

. Define the positive initial index interval for a positive integer  $n$  by

$$[n] := \llbracket 1, n \rrbracket$$

. Then define the finite-index-set display by

$$\{F_i\}_{i=1}^n := \{F_i \mid i \in [n]\}$$

. A column heading in a reference-specific audit table specifies a target family. The value in each cell is a finite index set of that target family. For each reference  $R$ , a time-content index set  $J$ , a supporting-claim index set  $P$ , and a result index set  $Q$  correspond, respectively, to

$$\begin{aligned} \text{TimeContent}_R[J] &= \{\text{TimeContent}_{R,j} \mid j \in J\}, \\ \text{Node}_R^{\text{audit,claim}}[P] &= \{\text{Node}_{R,p}^{\text{audit,claim}} \mid p \in P\}, \\ \text{Result}_R[Q] &= \{\text{Result}_{R,q} \mid q \in Q\} \end{aligned}$$

. This correspondence determines the semantic objects of the respective columns for time-content input indices, time-content output indices, time-content prerequisite indices, corresponding time-content indices, related time-content indices, supporting-claim indices, and result-output indices.

**Definition 7.93** (Finite-Set Constructors for Audit-Node Families). For each  $R \in \text{Ref}^{\text{audit}}$  and positive integer  $n$ , define the physical and formal claim-node set constructor and the semantic-node set constructor by

$$\begin{aligned} \text{ClaimNodes}_R(n) &:= \left\{ \text{Node}_{R,i}^{\text{audit,claim}} \mid i \in [n] \right\}, \\ \text{SemanticNodes}_R(n) &:= \left\{ \text{Node}_{R,i}^{\text{audit,semantic}} \mid i \in [n] \right\} \end{aligned}$$

**Definition 7.94** (Semantic-Support Lift Map). For each reference  $R$  that has a semantic-node family and a semantic-support index map, define

$$\begin{aligned} \text{SemanticLift}_R &: \text{dom}(\text{SupportIndex}_R^{\text{semantic}}) \longrightarrow \mathcal{P}(\text{Node}_R^{\text{audit,semantic}}), \\ \text{SemanticLift}_R(P) &:= \text{Node}_R^{\text{audit,semantic}}[\text{SupportIndex}_R^{\text{semantic}}(P)]. \end{aligned}$$

**Definition 7.95** (Prefix-Interval Audit Boundary for a Finite Node Family). For a finite indexed set

$$S = \{S_i\}_{i=1}^N$$

and

$$1 \leq n \leq N$$

define the prefix-interval audit boundary by

$$\text{PrefixBoundary}(S; n) := (\{S_i\}_{i=1}^n, S)$$

. The first component is the generation-initial set consisting of the first  $n$  components, and the second component is the reference formal-evidence environment consisting of the entire finite indexed set.

**Definition 7.96** (Prefix-Interval Boundary Constructors for Audit-Node Families). For each  $R \in \text{Ref}^{\text{audit}}$  and positive integer  $n$ , define

$$\begin{aligned} \text{ClaimPrefix}_R(n) &:= \text{PrefixBoundary}(\text{Node}_R^{\text{audit,claim}}; n), \\ \text{SemanticPrefix}_R(n) &:= \text{PrefixBoundary}(\text{Node}_R^{\text{audit,semantic}}; n) \end{aligned}$$

**Definition 7.97** (Audit Boundary by Two Index Sets). For an index set  $I$ , a family  $\mathcal{F} = (F_i)_{i \in I}$ , and  $(J, K) \in \mathcal{P}(I)^2$ , define  $\text{IndexedBoundary}(\mathcal{F}; (J, K)) := (\mathcal{F}[J], \mathcal{F}[K])$ .

**Definition 7.98** (Initial-Set–Evidence–Environment Pair). For each reference  $R$ , define the claim-audit-boundary index pair by

$$\text{ClaimBoundaryIdx}_R := \left( \text{Idx}_R^{\text{audit,claim,init}}, \text{Idx}_R^{\text{audit,claim,evid}} \right)$$

. Define the audit-typed and physically typed initial-set–evidence–environment pairs by

$$\begin{aligned} \text{InitEnv}_R^{\text{audit}} &:= \left( \text{Init}_R^{\text{audit,phys}}, \text{Env}_R^{\text{audit,evid}} \right), \\ \text{InitEnv}_R^{\text{phys}} &:= \left( \text{Init}_R^{\text{phys}}, \text{Env}_R^{\text{phys}} \right) \end{aligned}$$

. When the enumeration of physical and formal claim nodes and the two claim-index sets are given,

$$\begin{aligned} \text{InitEnv}_R^{\text{audit}} \\ = \text{IndexedBoundary} \left( \begin{array}{c} \text{EnumNode}_R^{\text{audit,claim,phys}}, \\ \text{ClaimBoundaryIdx}_R \end{array} \right). \end{aligned}$$

The first component of each pair is the generation-initial physical-input node set, and the second is the reference formal-evidence environment node set.

**Definition 7.99** (Proper-Inclusion Property of Claim Audit Boundaries). For each  $R \in \text{Ref}^{\text{audit}}$ , define

$$\begin{aligned} \text{StrictClaimBoundary}(R) \\ \iff \text{Idx}_R^{\text{audit,claim,init}} \subsetneq \text{Idx}_R^{\text{audit,claim,evid}}, \\ \text{TypedStrictClaimBoundary}(R) \\ \iff \text{StrictClaimBoundary}(R) \wedge \\ \text{Idx}_R^{\text{audit,claim,evid}} \subseteq \text{Idx}_R^{\text{audit,phys}} \end{aligned}$$

**Definition 7.100** (Standard Lift of Semantic-Support Indices). For each reference  $R$ , positive integer  $n$ , and finite index set  $P$ , define

$$\begin{aligned} \text{SemanticIndexLift}_R(P; n) \\ := \{n + p \mid p \in P\} \\ \cup \{a \in [n] \mid \text{ClaimIndex}_R^{\text{init}}(a) \cap P \neq \emptyset\}. \end{aligned}$$

The first term moves the reference formal-claim indices after the first  $n$  generation-initial semantic components; the second adds the generation-initial semantic components having a reference formal claim in common with  $P$ .

**Definition 7.101** (Lift of Claim Nodes and Mathematical Nodes to Semantic Nodes). For each reference  $R$ , nonnegative integer  $s$ , and positive integer  $n$ , define the lifts of claim nodes and mathematical nodes to semantic nodes by

$$\begin{aligned} \text{ClaimNodeLift}_R(s; n) &\iff \forall i \in [n], \quad \text{Node}_{R,s+i}^{\text{audit,semantic}} = \text{Node}_{R,i}^{\text{audit,claim}}, \\ \text{MathNodeLift}_R(s; n) &\iff \forall i \in [n], \quad \text{Node}_{R,s+i}^{\text{audit,semantic}} = \text{Node}_{R,i}^{\text{audit,math}}. \end{aligned}$$



**Definition 7.102** (Standard Semantic-Support Index Specialization). For each reference  $R$  and positive integers  $n, M$ , define the standard semantic-support index specialization by

$$\text{StdSemLift}_R(n; M) \iff \forall P \subseteq [M], \quad \text{SupportIndex}_R^{\text{semantic}}(P) = \text{SemanticIndexLift}_R(P; n).$$

**Definition 7.103** (Audit Count Vectors). For each  $R \in \text{Ref}^{\text{audit}}$ , define the obligation-state count vector and terminal-component occurrence-count vector by

$$\begin{aligned} \text{ObligationCount}_R &:= (N_{\text{fulfilled}}, N_{\text{unfulfilled}}, N_{\text{open}}), \\ \text{TerminalCount}_R &:= (N_{\text{explicit}}, N_{\text{structural}}, N_{\text{cycle}}, N_{\text{physical}}, N_{\text{evidence-open}}) \end{aligned}$$

. Define the reference-specific audit count vector by

$$\text{AuditCount}_R := (\text{ObligationCount}_R, \text{TerminalCount}_R)$$

. Define the audit-boundary count pair consisting of the number of audit-boundary nodes and the number of audit-boundary edges by

$$\text{BoundaryCount}_R^{\text{audit}} := \left( \left| \text{Node}_R^{\text{audit, bd}} \right|, \left| \text{Edge}_R^{\text{audit, bd}} \right| \right)$$

.

**Definition 7.104** (Reference-Specific Completeness-Count Vector). Define the ordered sequence of completeness-count fields by

$$\begin{aligned} \text{AuditFieldSeq} &= (\text{Field}_k^{\text{audit}})_{k=1}^{26} \\ &:= \left( \begin{array}{l} \text{timeuse, timecontent, userole, contentrole, obligation,} \\ \text{useobligation, roleobligation, physclaim, mathclaim, claimnode,} \\ \text{usenode, srccontentnode, rootusenode, resultnode,} \\ \text{evidnode, transferedge, resultedge, auditedge,} \\ \text{branch, alternative, fulfilled, unfulfilled,} \\ \text{open, result, margin, unmatched} \end{array} \right). \end{aligned}$$

For each  $R \in \text{Ref}^{\text{audit}}$ , define the reference-specific completeness-count vector by

$$\text{CompletenessCount}_R := (\text{Count}_R^{\text{audit}} (\text{Field}_k^{\text{audit}}))_{k=1}^{26}$$

.

**Definition 7.105** (Typed-Index Realization of Audit Incidence). For each  $R \in \text{Ref}^{\text{audit}}$ , let the supporting-claim index set, time-content input index set, and time-content output index set specified by the reference-specific transfer table or by finite indices be

$$P_{R,s}^{\text{tr}}, \quad J_{R,s}^{\text{in}}, \quad J_{R,s}^{\text{out}}, \quad s \in \text{Idx}_R^{\text{audit, transfer}}$$

. These satisfy

$$\begin{aligned} P_{R,s}^{\text{tr}} &\subseteq \text{Idx}_R^{\text{audit, phys}}, \\ J_{R,s}^{\text{in}}, J_{R,s}^{\text{out}} &\subseteq \text{Idx}_R^{\text{audit, time, src}} \end{aligned}$$

. Let the supporting-claim index set and time-content prerequisite index set specified by the reference-specific result table or by finite indices be

$$P_{R,q}^{\text{res}}, \quad J_{R,q}^{\text{res}}, \quad q \in \text{Idx}_R^{\text{audit,result}}$$

. These satisfy

$$P_{R,q}^{\text{res}} \subseteq \text{Idx}_R^{\text{audit,phys}}, \quad J_{R,q}^{\text{res}} \subseteq \text{Idx}_R^{\text{audit,time,src}}$$

. Require indices that denote the same transfer event or the same result to give identical support, input, and output sets in the realizations below.

Realize each transfer index by

$$\begin{aligned} \text{Support}_R^{\text{audit,transfer}}(\text{TransferEvent}_{R,s}) &:= \text{EnumNode}_R^{\text{audit,claim,phys}}[P_{R,s}^{\text{tr}}], \\ \text{TransferIn}_{R,\text{TransferEvent}_{R,s}} &:= \text{TimeContent}_R[J_{R,s}^{\text{in}}], \\ \text{TransferOut}_{R,\text{TransferEvent}_{R,s}} &:= \text{TimeContent}_R[J_{R,s}^{\text{out}}] \end{aligned}$$

. Realize each result index by

$$\begin{aligned} \text{Support}_R^{\text{audit,result}}(\text{Result}_{R,q}) &:= \text{EnumNode}_R^{\text{audit,claim,phys}}[P_{R,q}^{\text{res}}], \\ \text{ResultPrereq}_{R,\text{Result}_{R,q}} &:= \text{TimeContent}_R[J_{R,q}^{\text{res}}] \end{aligned}$$

. For a source  $R$  for which the map in Definition 7.94 is defined, define the semantic support of each transfer  $s$  or result  $q$  by the corresponding equation below whenever its supporting-claim index set  $P_{R,s}^{\text{tr}}$  or  $P_{R,q}^{\text{res}}$  belongs to  $\text{dom}(\text{SemanticLift}_R)$ .

$$\begin{aligned} \text{SemanticSupport}_R^{\text{audit,transfer}}(\text{TransferEvent}_{R,s}) &:= \text{SemanticLift}_R(P_{R,s}^{\text{tr}}), \\ \text{SemanticSupport}_R^{\text{audit,result}}(\text{Result}_{R,q}) &:= \text{SemanticLift}_R(P_{R,q}^{\text{res}}). \end{aligned}$$

When a reference-specific specification has a single time-content output index

$$j_{R,s}^{\text{out}}$$

set

$$J_{R,s}^{\text{out}} := \{j_{R,s}^{\text{out}}\}$$

. **Proposition 7.106** (Compatibility of Typed Incidence Indices with the Canonical Audit Realization). For the input of Definition 7.89, let

$$\begin{aligned} P_{R,s}^{\text{tr}} &= \text{ClaimIn}_R^{\text{audit,transfer}}(s), & J_{R,s}^{\text{in}} &= \text{TimeIn}_R^{\text{audit,transfer}}(s), \\ J_{R,s}^{\text{out}} &= \text{TimeOut}_R^{\text{audit,transfer}}(s), & P_{R,q}^{\text{res}} &= \text{ClaimIn}_R^{\text{audit,result}}(q), \\ J_{R,q}^{\text{res}} &= \text{TimeIn}_R^{\text{audit,result}}(q) \end{aligned}$$

. Then the support sets, time-content input and output sets, and result-prerequisite sets of Definition 7.105 coincide with the corresponding components determined by the canonical realization of Definition 7.90. Therefore, the two realizations determine the same audit-boundary directed hypergraph.

*Proof.* Substitute the identifications of the five index sets into the support sets, time-content input and output sets, and result-prerequisite sets in the two definitions. By the family-image correspondence of Definition 7.92, each right-hand side gives the same set of enumerated nodes or the same reference-specific time-content set. Because the input and output sets of the transfer edges and result edges coincide, the nodes, edges, and incidence maps of the audit-boundary directed hypergraphs coincide.  $\square$

**Proposition 7.107** (Unique Realization of an Audit Boundary by a Reference-Specific Finite Display). For each  $R \in \text{Ref}^{\text{audit}}$ , take finite families of claim nodes, time contents, transfer events, and results. Suppose that the physical-input initial set and the reference formal-evidence environment follow Definition 7.28 and that each index set in the transfer and result tables satisfies the typing conditions and the consistency conditions for identical events and results in Definition 7.105.

This finite display uniquely determines an audit-boundary directed hypergraph equipped with support, time-content inputs, time-content outputs, and result prerequisites. Hence reachability from the physical-input initial set and the reference formal-evidence environment, primitive time roots, the fourfold result partition, and the minimal primitive-time-root support family are also uniquely determined.

*Proof.* By Definition 7.105, the inputs, outputs, and support of each transfer edge and result edge are uniquely determined. The finite node and edge families, together with their incidence, uniquely determine the audit-boundary directed hypergraph.

Reachability is determined inductively from this directed hypergraph and the specified initial set. The primitive time roots and the fourfold result partition are determined by that reachability. The minimal primitive-time-root support family is determined as the family of inclusion-minimal elements in the family of subsets of the finite primitive-time-root set.  $\square$

**Definition 7.108** (Minimal Primitive-Time-Root Support Index Families and Sum-Product of Set Families). For each  $R \in \text{Ref}^{\text{audit}}$  and

$$\text{Result}_{R,q} \in \text{Result}_R$$

define the minimal primitive-time-root support index family by

$$\begin{aligned} \mathcal{A}_R(q) \\ := \{ J \subseteq [n_R] \mid \text{TimeContent}_R[J] \in \text{MinRootSupport}_R^{\text{audit}}(\text{Result}_{R,q}) \}. \end{aligned}$$

Then

$$\begin{aligned} \text{MinRootSupport}_R^{\text{audit}}(\text{Result}_{R,q}) \\ = \{ \text{TimeContent}_R[J] \mid J \in \mathcal{A}_R(q) \}. \end{aligned}$$

Define the sum-product of finite index-set families

$$\mathcal{A}, \mathcal{B} \subseteq \mathcal{P}([n_R])$$

by

$$\mathcal{A} \boxplus \mathcal{B} := \{ A \cup B \mid A \in \mathcal{A}, B \in \mathcal{B} \}$$

.

### 7.2.8 Reference-Specific Two-Track Evidence Records

**Definition 7.109** (Reference-Specific Two-Track Evidence Records). The evidence record for each reference is defined by two tracks: a formal-evidence record and a statement-evidence record.

The formal-evidence record has the following components in this order.

1. The formal object.
2. The original-source location at which the formal object is given.
3. The assumptions and non-temporal physical inputs explicitly stated in the original source.
4. The time contents used in the original source.
5. Dependencies or transfers among time contents, inputs, and results.
6. Reference-specific results.
7. For a reference that includes an experiment or observation, its directly measured object.
8. The domain of application of the formal generalization given by the original source.
9. Time contents generated by audit reachability from non-temporal physical inputs, and the primitive time roots that remain.
10. The terminal objects, predicates, conditions, and domains of application directly supported by the original-source evidence.
11. Reference-specific results preserved after the generation-order audit.

The statement-evidence record has a quotation-fact layer and an interpretive-audit layer, in this order. The quotation-fact layer has the following components in this order.

1. The verbatim quotation in the evidentiary source language.
2. For a non-English evidentiary source, an English translation corresponding to the verbatim quotation.
3. The original-source location.
4. The minimal context required for interpretation.
5. The subject to which the statement is attributed.

The interpretive-audit layer has the following components in this order.

1. The object addressed by the statement, the predicate assigned to that object, and the explicit domain of application.
2. The multiple possible interpretations permitted by the original text and its minimal context.

3. For each possible interpretation, the terminal objects, predicates, conditions, and domains of application directly supported by the original-source evidence.
4. The formal-evidence record against which it is compared.
5. The correspondence or difference between each possible interpretation and the formal evidence.

Audit determinations in the formal-evidence record are determined from the definitions, formulas, propositions, operational rules, and dependencies in the original source. Comparison results in the statement-evidence record do not alter the audit determinations in the formal-evidence record.

**Proposition 7.110** (Separation and Comparison of Two-Track Evidence Records). Determinations in the reference-specific formal audit are determined from the formal-evidence record, and possible interpretations of a statement are determined from the quotation-fact and interpretive-audit layers of the statement-evidence record.

When the formal description and statement in the same reference address the same object, predicate, or domain of application, their correspondence is recorded for each possible interpretation. When the objects, predicates, or domains of application of the formal description and statement differ, the difference is preserved.

*Proof.* Definition 7.109 defines the formal-evidence record and statement-evidence record as separate tracks and divides the latter into a quotation-fact layer and an interpretive-audit layer. The objects, dependencies, results, and audit reachability in the formal-evidence record provide the inputs to the formal audit. In the statement-evidence record, the quotation-fact layer preserves evidence from which the original text can be recovered, while the interpretive-audit layer preserves, for every permissible interpretation, the correspondence with or difference from the formal evidence and the evidence-effect boundary. The formal and statement tracks can therefore be compared while remaining separate.  $\square$

### 7.2.9 Evidence Fields and Local Keys in Reference-Specific Audit Tables

**Definition 7.111** (Evidence Fields in Reference-Specific Audit Tables). When the formal-evidence record of Definition 7.109 is displayed in a reference-specific audit table, the following six fields are distinguished.

1. The grounds for the result state the definitions, assumptions, formal relations, operations, observations, or experiments in the original source that support the reference-specific result.
2. The directly measured object states the physical carrier, physical quantity, or display to which the apparatus directly responds, which it records, or which it compares.
3. The directly measured predicate states the value, difference, rate, correlation, count, or relation obtained for the directly measured object.
4. The conditions and domain of application of the direct measurement state the apparatus configuration, calibration, approximations, background conditions, operations, and object range required for that measurement.

5. The object of formal generalization states the object or family of objects to which the formal construction in the original source extends its application. This object need not be identical to the directly measured object.
6. The conditions and domain of application of the formal generalization state the assumptions, maps, identifications, approximations, background conditions, and object range required for that generalization.

The six fields are used to determine the evidence-effect domain of each item without substituting the grounds for the result, the direct measurement, and the formal generalization for one another.

**Definition 7.112** (Local Keys in Reference-Specific Audit Tables). In a finite index table of Definition 7.92, a positive integer that specifies a referent only within one correspondence table for one reference is called a local key. The following kinds of local key are distinguished.

1. An original-source location key refers to one or more locations in the original source.
2. A time-content key refers to a finite set of time-content indices defined for that reference.
3. A formal-evidence key refers to a finite set of indices of reference formal descriptions supporting a transfer or result.
4. A semantic-component key refers to a finite set of indices of semantic components separated from the reference formal description.

Even when the integer is the same, it does not denote the same referent if the reference, correspondence table, or kind of key differs. Each correspondence table gives the correspondence between local keys and their referents before the table that uses those keys or within the same descriptive unit. A local key does not replace evidence; it abbreviates only repeated display within a table while preserving the ability to recover the evidence, time content, or semantic component through the correspondence table.

#### 7.2.10 Public Names and Result-Preservation Domains in Reference-Specific Audits

**Definition 7.113** (Public Names in Reference-Specific Audits). A transfer-typed audit-boundary edge of Definition 7.26 is called a transfer hyperedge, and a result-typed audit-boundary edge is called a result-dependency hyperedge.

The physical-input initial set of Definition 7.28 is the starting point of the audit-reachable set of Definition 7.30, which is called the generation-initial closure. For the same audit-reachable set, the set that starts from the reference formal-evidence environment is called the evidence closure. These are results of the same reachability operation with different starting points; they do not introduce a new closure operation.

**Definition 7.114** (LCTR Generation Order, Result-Preservation Conditions, and Result-Preservation Domain). The dependency precedence relations among non-temporal physical inputs, physical-carrier realization, comparison, time representation, dynamical description, law time, and validity determination constructed from 2 through 6 are called the LCTR generation order. This term does not mean a total order that arranges all objects in one

sequence; it denotes the precedence relations determined by the input dependencies of the respective constructed objects.

The original-source inputs, assumptions, approximations, boundary conditions, dependency directions, and domains of application that must be preserved in order to use the formal content of a reference-specific result without alteration under the LCTR generation order, together with the time representations that must be generated before use in place of primitive time roots, are called the result-preservation conditions. The range of objects and scales over which all result-preservation conditions hold and the result in question can be used is called the result-preservation domain.

The result-preserving correspondence of Definition 7.77 specifies the correspondence between audited result nodes and reference-specific results. The result-preservation conditions and result-preservation domain specify the conditions under which, and the range over which, the result identified by that correspondence can be used under the LCTR generation order.

### 7.2.11 Evidence-Effect Boundaries and Unbounded Inheritance of Limited Results

**Definition 7.115** (Evidence-Effect Domain and Evidence-Effect Boundary). The tuple of objects, predicates, conditions, and domains of application directly supported by reference-specific evidence is called the evidence-effect domain of that evidence.

In a dependency sequence from a body of evidence to a subsequent claim, a change in object, predicate, evidence type, grounding condition, or domain of application requires independent evidence or a valid derivation supporting that change. The first position at which the change occurs without such support is called the evidence-effect boundary in that dependency sequence.

The evidence-effect boundary gives the terminal point of the domain directly established by the preceding evidence. A claim beyond the boundary requires evidence or a derivation specific to that claim.

**Definition 7.116** (Unbounded Inheritance of Limited Results). The direct inheritance, by a claim beyond the evidence-effect boundary, of mathematical consistency, predictive performance, model fit, or empirical agreement established at a preceding stage of a dependency sequence is called the unbounded inheritance of limited results.

The stages of inheritance subject to audit include the following.

1. A comparative difference measured for an apparatus, clock, oscillator, signal, detector, record, or physical process.
2. The physical validity of the entire formal system that calculated that comparative difference.
3. The physical-time qualification of the time content used by the formal system.
4. Attribution from a comparative difference on a physical carrier to the existence or property of time itself, independently of the physical carrier.
5. Transfer from a property attributed to time itself to biological processes, subjective experience, cosmic history, or another domain of application.

If inheritance across an evidence-effect boundary occurs between two adjacent stages in this sequence, the unbounded inheritance of limited results holds.

**Definition 7.117** (Interpretation-Specific Evidence-Comparison Path and Number of Evidence-Effect Transgressions). An evidence-dependency sequence determined by a formal construction in a reference, a statement in the reference, and one permissible interpretation of that statement is called an interpretation-specific evidence-comparison path. If the same reference or the same statement has multiple permissible interpretations, a distinct path is defined for each interpretation.

Within one interpretation-specific evidence-comparison path, each position at which the object, predicate, evidence type, grounding condition, or domain of application changes without independent evidence or a valid derivation supporting that change is called an evidence-effect transgression. The number of evidence-effect transgressions connected successively by direct dependencies is called the number of evidence-effect transgressions along that path. The number of evidence-effect layers is not taken as the number of evidence-effect transgressions. Even for the same statement, different permissible interpretations can have different numbers of evidence-effect transgressions.

The number of evidence-effect transgressions is 0 for a path that directly describes a comparative difference on a physical carrier in terms of its object and relation. If an expression that appears to assign a property to time is used, the number of evidence-effect transgressions produced by that figurative expression is 0 when the original-source text states, within the same explanatory scope, both that the expression is figurative and does not denote a literal deformation of time, and what the actual referent is.

If expressions such as “time slows down,” “time expands or contracts,” or “time is affected” are used without identifying their figurative character and actual referent, the statement transfers a comparative difference on a physical carrier to a property of time and crosses one evidence-effect boundary. The possibility of interpreting the expression retrospectively as shorthand does not remove this evidence-effect transgression.

**Proposition 7.118** (Failure of Limited Results to Establish Claims Outside Their Domain of Application). Measurement success, model fit, predictive success, or mathematical consistency directly establishes the objects, predicates, conditions, and domains of application that belong to its evidence-effect domain. A claim beyond the evidence-effect boundary is not established by the preceding limited result alone; it requires independent evidence or a valid derivation specific to that claim.

*Proof.* By Definition 7.115, the evidence-effect domain is determined by the objects, predicates, conditions, and domains of application directly supported by reference-specific evidence, and the evidence-effect boundary is the first position of change that lacks independent evidence or a valid derivation. A claim beyond the boundary therefore does not belong to the domain directly supported by the preceding evidence.

By Proposition 7.8 and Proposition 7.13, evaluation success does not add physical-time qualification to a time content and does not change primitive time input into generated time content. For inheritance from a limited result to the physical-time qualification of a time content, these two propositions establish that an evidence-effect boundary is crossed.

Applying the same determination to every stage at which the object, predicate, evidence type, grounding condition, or domain of application changes shows that a claim beyond the boundary requires independent evidence or a valid derivation specific to that claim.  $\square$

**Corollary 7.119** (Inheritance of Primitive Time Dependence and Unbounded Inheritance of Limited Results). Result inheritance of primitive time dependence represents the dependency direction along which ungenerated time content remaining in an input propagates to a result. The unbounded inheritance of limited results represents the dependency direction



along which the effect of preceding evidence is extended to a different object, predicate, evidence type, grounding condition, or domain of application.

The two forms of inheritance constitute distinct audit axes and can hold simultaneously in the same reference or the same explanatory sequence.

*Proof.* The former is determined by a dependency path from primitive time input to a result, and the latter by inheritance across the evidence-effect boundary of Definition 7.115. Because the dependency directions subjected to determination differ, they provide distinct audit axes. If a single reference has ungenerated time content as an input and simultaneously transmits a limited success to another claim, the determinations on both axes hold at the same time.  $\square$

**Definition 7.120** (Backgrounding of a Grounding Obligation by Primitive Time Input). Suppose that a reference  $R$  introduces a time content  $x$  into a law operation or result derivation and proceeds with that derivation without examining, as an explicit input, generation condition, or grounding condition within the reference, what is required for the generation or use qualification of  $x$ , namely a condition  $\chi$ . The primitive time input  $x$  is then said to background the grounding obligation for the condition  $\chi$ .

For a backgrounded grounding obligation, the following four items are recorded separately.

1. The backgrounded generation condition or grounding condition.
2. Independent evidence establishing actual failure of the same condition.
3. Dependency reachability from the actual failure position to downstream results.
4. The extent of condition repair, use-domain restriction, and downstream regeneration or recalculation.

Actual failure is established by the independent evidence in the second item; it is not established solely by the presence of primitive time input or backgrounding.

**Proposition 7.121** (Causal Connection from a Backgrounded Grounding Obligation to Downstream Results). When the first through fourth items of Definition 7.120 are connected by the same condition  $\chi$ , primitive time input, the backgrounding of a grounding obligation, actual failure of the condition, loss of use qualification for downstream results, and regeneration or recalculation after repair constitute one audit causal sequence.

If the independent evidence in the second item does not hold, actual failure of the condition in question is undetermined. If the dependency reachability in the third item does not hold, actual failure of that condition does not invalidate the use qualification of that downstream result.

*Proof.* The first item identifies a generation condition or grounding condition that is not explicitly examined because of primitive time input. The second establishes actual failure of the same condition from independent evidence within or outside the reference. The third establishes dependency reachability with the actual failure position as its starting point and the result derivation as its endpoint. The fourth gives the repair, use-domain restriction, and regeneration range for results belonging to that reachable domain. When the four items share the same condition, the respective stages are connected as one dependency sequence.  $\square$

### 7.2.12 Input Consistency of Audit-Boundary Directed Hypergraphs

Audit-boundary directed hypergraphs are given by Definition 7.27.

The physical and mathematical boundary claims, time contents, transfer events, and reference-specific results stated in each reference are represented by a finite directed hypergraph for that reference. The input set of each hyperedge represents the set of premises used jointly to establish its output.

Audit-boundary nodes, audit-boundary support sets, and audit-boundary edges are given by Definition 7.18 and Definition 7.26.

**Proposition 7.122** (Input Consistency of Audit-Boundary Directed Hypergraphs). For each  $R \in \text{Ref}^{\text{audit}}$ , the time contents, transfer events, and results correspond injectively to their respective audit-boundary nodes or audit-boundary edges. The time-content input and time-content output of each transfer edge coincide with the images of

$$\text{TransferIn}_{R,\epsilon}, \quad \text{TransferOut}_{R,\epsilon}$$

and the time-content input of each result edge coincides with the image of

$$\text{ResultPrereq}_{R,\eta}$$

.

*Proof.* The time-content node map, result-node map, transfer-edge node map, and result-edge node map are injective in Definition 7.18 and Definition 7.26, respectively.

The input and output maps of Definition 7.27 take

$$\begin{aligned} &\text{NodeOf}_R^{\text{audit,time}} [\text{TransferIn}_{R,\epsilon}], \\ &\text{NodeOf}_R^{\text{audit,time}} [\text{TransferOut}_{R,\epsilon}] \end{aligned}$$

as the time-content input and time-content output, respectively, for a transfer edge. The same definition takes the time-content input of a result edge to be

$$\text{NodeOf}_R^{\text{audit,time}} [\text{ResultPrereq}_{R,\eta}]$$

. Thus each correspondence holds. □

## 7.3 Formal Evidence, Source Statements, and Representative Audits

The formal descriptions and original-source statements of the audited references are distinguished, and the reference-specific formal audits, representative formal evidence, and statement evidence preserving interpretive possibilities are presented.

For every reference in the audited reference set  $\text{Ref}^{\text{audit}}$ , Table 16 presents the complete reference-specific audit evidence for primitive time roots and result dependence. Under the complete determination, there are no generated time contents in any of the 72 references; in every reference, all reference-specific time contents remain as primitive time roots, and each reference has results that depend on those roots. The local-reachability-determination column records differences from this common result when the reference formal-evidence environment is added. A dash denotes the baseline case in which local reachability is empty and all reference-specific time contents remain as primitive time roots. The first

reference in the evidence column establishes the primitive time roots, and the second establishes result dependence.

Table 16: Primitive-time-root and result-dependence evidence for the 72 audited references

| No. | Reference                      | Difference under local-reachability determination | Root and result-dependence evidence |
|-----|--------------------------------|---------------------------------------------------|-------------------------------------|
| 1   | Aristotle                      | —                                                 | S.10, S.10                          |
| 2   | Newton 1687/1729               | —                                                 | S.17, S.17                          |
| 3   | Einstein 1905                  | —                                                 | S.24, S.24                          |
| 4   | Minkowski 1908/1909            | —                                                 | S.31, S.31                          |
| 5   | Einstein 1916                  | —                                                 | S.38, S.38                          |
| 6   | Bridgman 1927                  | —                                                 | S.44, S.44                          |
| 7   | von Neumann 1932               | —                                                 | S.60, S.60                          |
| 8   | Curtiss 1949                   | —                                                 | S.68, S.68                          |
| 9   | Dirac 1958                     | —                                                 | S.77, S.77                          |
| 10  | Pound–Rebka 1960               | —                                                 | S.86, S.86                          |
| 11  | ADM 1962                       | —                                                 | S.94, S.94                          |
| 12  | Feynman 1963                   | —                                                 | S.102, S.102                        |
| 13  | Taylor–Wheeler 1992            | —                                                 | S.117, S.117                        |
| 14  | DeWitt 1967                    | 2 reached; 33 remaining                           | S.124, S.124                        |
| 15  | Ehlers–Pirani–Schild 1972      | —                                                 | S.132, S.132                        |
| 16  | Hafele–Keating 1972b           | —                                                 | S.148, S.148                        |
| 17  | Hafele–Keating 1972a           | —                                                 | S.157, S.157                        |
| 18  | Bekenstein 1973                | —                                                 | S.165, S.165                        |
| 19  | Misner–Thorne–Wheeler 1973     | 4 reached; 39 remaining                           | S.173, S.173                        |
| 20  | Hawking 1975                   | —                                                 | S.181, S.181                        |
| 21  | Hawking 1976                   | —                                                 | S.190, S.190                        |
| 22  | Bailey et al. 1977             | —                                                 | S.197, S.197                        |
| 23  | Rindler 1977                   | —                                                 | S.205, S.205                        |
| 24  | Penrose 1979                   | —                                                 | S.213, S.213                        |
| 25  | Hartle–Hawking 1983            | —                                                 | S.221, S.221                        |
| 26  | Page–Wootters 1983             | —                                                 | S.231, S.231                        |
| 27  | Bombelli–Lee–Meyer–Sorkin 1987 | —                                                 | S.240, S.240                        |
| 28  | Weinberg 1989                  | 2 reached; 20 remaining                           | S.251, S.251                        |
| 29  | Isham 1993                     | 12 reached; 33 remaining                          | S.262, S.262                        |
| 30  | Kuchař 1992                    | 9 reached; 38 remaining                           | S.273, S.273                        |
| 31  | Lebowitz 1993                  | —                                                 | S.284, S.284                        |

Continued on the next page

Table 16 (continued)

| No. | Reference                              | Difference under<br>local-reachability<br>determination | Root and result-dependence<br>evidence |
|-----|----------------------------------------|---------------------------------------------------------|----------------------------------------|
| 32  | Page 1993                              | —                                                       | S.295, S.295                           |
| 33  | Barbour 1994 I                         | —                                                       | S.305, S.305                           |
| 34  | Barbour 1994 II                        | —                                                       | S.316, S.316                           |
| 35  | Connes–Rovelli 1994                    | 3 reached; 25<br>remaining                              | S.328, S.328                           |
| 36  | Headley–Clayton–Reed<br>1996           | —                                                       | S.340, S.340                           |
| 37  | Maldacena 1998                         | —                                                       | S.349, S.349                           |
| 38  | Perlmutter et al. 1999                 | —                                                       | S.362, S.362                           |
| 39  | Riess et al. 1998                      | —                                                       | S.375, S.375                           |
| 40  | Adler 2003                             | —                                                       | S.390, S.390                           |
| 41  | Philip W. Anderson<br>2001             | —                                                       | S.399, S.399                           |
| 42  | Clark 2001                             | —                                                       | S.409, S.409                           |
| 43  | Rovelli 2002                           | —                                                       | S.421, S.421                           |
| 44  | Zurek 2003                             | —                                                       | S.435, S.435                           |
| 45  | Davis–Lineweaver 2004                  | —                                                       | S.446, S.446                           |
| 46  | Schlosshauer 2005                      | —                                                       | S.458, S.458                           |
| 47  | Kiefer 2005                            | 4 reached; 18<br>remaining                              | S.468, S.468                           |
| 48  | Grøn–Elgarøy 2007                      | —                                                       | S.478, S.478                           |
| 49  | Ryu–Takayanagi 2006                    | —                                                       | S.491, S.491                           |
| 50  | Francis–Barnes–James–<br>Lewis 2007    | —                                                       | S.500, S.500                           |
| 51  | Cook–Burns 2009                        | —                                                       | S.510, S.510                           |
| 52  | Peacock 2008                           | —                                                       | S.521, S.521                           |
| 53  | Bunn–Hogg 2009                         | —                                                       | S.531, S.531                           |
| 54  | Kiefer 2009                            | 9 reached; 26<br>remaining                              | S.539, S.539                           |
| 55  | Rovelli 2011                           | 7 reached; 16<br>remaining                              | S.549, S.549                           |
| 56  | Van Raamsdonk 2010                     | —                                                       | S.565, S.565                           |
| 57  | Greene <i>The Illusion of<br/>Time</i> | —                                                       | S.573, S.573                           |
| 58  | Wolf et al. 2011                       | —                                                       | S.581, S.581                           |
| 59  | Anderson 2012                          | 5 reached; 15<br>remaining                              | S.591, S.591                           |
| 60  | Ellis 2012                             | —                                                       | S.599, S.599                           |
| 61  | Tong 2013                              | —                                                       | S.607, S.607                           |
| 62  | Lewis 2016                             | —                                                       | S.615, S.615                           |
| 63  | Smolin 2018/2020                       | —                                                       | S.626, S.626                           |
| 64  | BIPM SI 2019/2026                      | —                                                       | S.635, S.635                           |

Continued on the next page

Table 16 (continued)

| No. | Reference             | Difference under local-reachability determination | Root and result-dependence evidence |
|-----|-----------------------|---------------------------------------------------|-------------------------------------|
| 65  | ESA 2019              | —                                                 | S.643, S.643                        |
| 66  | NASA/JPL–Caltech 2020 | —                                                 | S.651, S.653                        |
| 67  | Pössel 2021           | —                                                 | S.662, S.662                        |
| 68  | PDG 2023 Resonances   | —                                                 | S.672, S.672                        |
| 69  | NIST 2025             | —                                                 | S.681, S.681                        |
| 70  | Feynman 1995          | —                                                 | S.688, S.688                        |
| 71  | Narlikar 2010         | —                                                 | S.695, S.695                        |
| 72  | Heisenberg 1958       | —                                                 | S.702, S.702                        |

This local-reachability determination shows that adding the physical-carrier processing or construction procedures explicitly stated in the respective references to the evidence environment makes a total of 57 time contents locally reachable in the ten references shown in the table. This reachability represents the limited construction capacity present in the existing references; its domain of determination differs from complete generation starting solely from non-temporal physical inputs.

### 7.3.1 Explicit Absolute Time and Operational Critique

The audit question common to Newton 1687/1729 and Bridgman 1927 is whether either the explicit positing of absolute time or the operational critique of absolute time generates time contents from the physical-input initial set.

In Newton 1687/1729, absolute, true, and mathematical time precedes the dynamical results. In Bridgman 1927, the operational critique of absolute time is followed by signal propagation, clocks, interval counting, synchronization, and correction as time contents. The respective time contents and original-source locations are given in Table 33 and Table 67.

**Proposition 7.123** (Non-Generation of Time under Explicit Absolute Time and Operational Critique). For the reference  $R$  corresponding to Newton 1687/1729, Definition 7.56 gives  $R \in \text{Ref}^{\text{audit,rootall}}$ , and

$$\begin{aligned} \text{Result}_R^{\text{audit,gen}} &= \emptyset, \\ \text{Result}_R^{\text{audit,evidonly}} &= \{\text{Result}_{R,1}\}, \\ \text{Result}_R^{\text{audit,rootdep}} &= \{\text{Result}_{R,j} \mid j \in \{2, \dots, 18\}\} \end{aligned}$$

hold.

For the reference  $R$  corresponding to Bridgman 1927, Definition 7.58 gives  $R \in \text{Ref}^{\text{audit,rootfull}}$ .

*Proof.* For Newton 1687/1729, apply Proposition S.17 and Proposition S.17. For Bridgman 1927, apply Proposition S.44 and Proposition S.44.  $\square$

**Proposition 7.124** (Failure of Operational Critique to Generate Time). The operational critique, clock construction, signal procedures, synchronization procedures, and correction

procedures in Bridgman 1927 do not generate the time contents used by that reference from the physical-input initial set.

*Proof.* Each transfer event in Definition S.42 takes at least one time content as input when it outputs a time content. No generation trace from the physical-input initial set to a time content therefore exists. By Proposition 7.123, all 57 time contents are primitive time roots and all 20 results depend on primitive time roots.  $\square$

**Corollary 7.125** (Logical Separation of the Rejection of Absolute Time from the Generation of Time). Neither a construction that explicitly posits absolute time nor a construction that operationally criticizes absolute time generates physical time contents. To provide a physical ground for a time variable in a physical law, the generation-precedence condition of Corollary 7.15 is required.

*Proof.* Apply Proposition 7.123 and Proposition 7.124.  $\square$

### 7.3.2 Synchronization, Coordinate Time, and Proper Time

The audit question common to Einstein 1905, Einstein 1916, and Taylor–Wheeler 1992 is whether the constructions of synchronization, coordinate time, and proper time generate time contents from the physical-input initial set.

Einstein 1905 introduces rods, coordinates, and clocks at the outset and develops its account from temporal relations among processes to the common time of a synchronization network and coordinate time. In Einstein 1916, the formal correspondence between spatial coordinates and a time coordinate appears at the beginning of the original source and is developed into local clocks, location-dependent clock rates, and four coordinates. In Taylor–Wheeler 1992, events, inertial frames, clocks, and worldlines are given in advance and are developed into synchronization, event coordinates, proper time, wristwatch time, and integration along worldlines.

The time contents and original-source locations of the respective references are given in Table 41, Table 58, and Definition S.115.

**Proposition 7.126** (Non-Generation of Time in the Construction of Synchronization, Coordinate Time, and Proper Time). For the reference  $R$  corresponding to each of Einstein 1905, Einstein 1916, and Taylor–Wheeler 1992, Definition 7.58 gives  $R \in \text{Ref}^{\text{audit,rootfull}}$ .

*Proof.* For Einstein 1905, apply Proposition S.24 and Proposition S.24. For Einstein 1916, apply Proposition S.38 and Proposition S.38. For Taylor–Wheeler 1992, apply Proposition S.117 and Proposition S.117.  $\square$

**Proposition 7.127** (Time-Content Presuppositions of Clock Networks, Worldlines, and Proper Time). The synchronization procedures, coordinate-time constructions, clock-rate comparisons, worldline constructions, and proper-time integrations in the three references transform, compare, or integrate time contents. These procedures do not generate their input time contents from the physical-input initial set.

*Proof.* Every transfer event in each reference includes a time content in its input when it outputs a time content. By Proposition 7.126, all time contents in the three references are primitive time roots, and all reference-specific results depend on primitive time roots.  $\square$

**Corollary 7.128** (Physical Grounds for Coordinate Time and Proper Time). The mathematical success of synchronization consistency, Lorentz transformations, clock comparison, or worldline integration does not provide physical-time qualification for the input time contents. Before coordinate time and proper time are used as time variables in physical laws, the generation-precedence condition of Corollary 7.15 must be satisfied.

*Proof.* Apply Proposition 7.127 and Proposition 7.13.  $\square$

### 7.3.3 Unit Standards, Distributed Clocks, and Experimental Clock Differences

The audit question common to BIPM SI 2019/2026, NASA/JPL–Caltech 2020, and Hafele–Keating 1972a is whether unit definitions, clock corrections, and experimental clock comparisons generate the time contents they use from the physical-input initial set.

BIPM SI 2019/2026 uses the cesium transition frequency, the second, proper time, primary clock signals, clock comparison, and relativistic corrections as components of a single system of time standards, clock comparisons, and corrections. NASA/JPL–Caltech 2020 gives clock comparisons and relativistic corrections for airborne clocks, ground reference clocks, and satellite clocks. Hafele–Keating 1972a gives the reunion display difference between moving clocks and reference clocks as an experimental quantity.

**Proposition 7.129** (Primitive Time Roots in Unit Standards, Clock Corrections, and Experimental Clock Differences). For the reference  $R$  corresponding to BIPM SI 2019/2026, Definition 7.56 gives  $R \in \text{Ref}^{\text{audit,rootall}}$ , and 25 of the 32 results depend on primitive time roots.

For  $R \in \text{Ref}^{\text{audit}}$  corresponding to NASA/JPL–Caltech 2020, under the complete determination the following equalities

$$\text{TimeContent}_R^{\text{audit,gen}} = \emptyset, \quad \text{TimeContent}_R^{\text{audit,root}} = \text{TimeContent}_R$$

hold. In the four-way partition of results, there are no generated results, evidence-environment-reachable nongenerated results, or results that remain unreachable after primitive time roots are adjoined; all eleven results depend on primitive time roots. Even under the local-reachability determination that adds the reference formal-evidence environment,

$$\text{TimeContent}_R^{\text{audit,evid}} = \emptyset$$

holds: all 20 time contents remain as primitive time roots, and all eleven results depend on those roots.

For the reference  $R$  corresponding to Hafele–Keating 1972a, Definition 7.58 gives  $R \in \text{Ref}^{\text{audit,rootfull}}$ .

*Proof.* For BIPM SI 2019/2026, apply Proposition S.635 and Proposition S.635.

For NASA/JPL–Caltech 2020, apply Proposition S.651 and Proposition S.653.

For Hafele–Keating 1972a, apply Proposition S.157 and Proposition S.157.  $\square$

**Proposition 7.130** (Primitive Time Roots Remaining after Addition of the Evidence Environment). For NASA/JPL–Caltech 2020, all 20 time contents are primitive time roots under the complete determination starting solely from non-temporal physical inputs, and there are no generated results. Adding the reference formal-evidence environment reaches none of the 20 time contents, and all eleven results depend on primitive time roots. The clock comparisons and corrections in this reference therefore do not provide physical-time qualification for its use of time as a whole.

*Proof.* Apply Proposition 7.129 and Proposition S.652. Theorem 7.11 and Corollary 7.12 give the failure of generation under the complete assessment, the remaining primitive time roots, and the inheritance of their dependence by the results.  $\square$

**Corollary 7.131** (Generation Boundary of Unit Definitions, Clock Corrections, and Clock Comparisons). Agreement among unit definitions, clock corrections, and experimental clock differences does not generate the time contents that they take as inputs. Primitive time roots remain in all three references, and dependence on those roots is inherited by the reference-specific results.

*Proof.* Apply Proposition 7.129 and Proposition 7.130.  $\square$

### 7.3.4 Quantum Measurement, Records, and Decoherence

The audit question common to von Neumann 1932, Zurek 2003, and Schlosshauer 2005 is whether measurement processes, environmental records, and decoherence generate from the physical-input initial set the time contents taken as inputs to quantum-mechanical time evolution.

In von Neumann 1932, the time evolution of nonstationary states, two processes, before-and-after measurement relations, and finite-time measurement interactions constitute time contents. In Zurek 2003, unitary evolution, decoherence scales, record persistence, histories, and inference of the past from present records constitute time contents. In Schlosshauer 2005, premeasurement evolution, system–environment interaction evolution, time-dependent reduced states, and the temporal stability of pointer states constitute time contents.

Von Neumann 1932 states:

Wenn der Zustand nicht stationär ist, d. h. sich mit der Zeit ändert, so enthält die Wellenfunktion  $\psi = \psi(q_1, \dots, q_k; t)$  die Zeit  $t$ , [...]

If the state is not stationary, that is, if it changes with time, the wave function  $\psi = \psi(q_1, \dots, q_k; t)$  contains the time  $t$ , [...]

The original-source location is von Neumann, *Mathematische Grundlagen der Quantenmechanik*, Springer, 1932, p. 8, §I.2. The complete evidence record for the same reference is given in paragraph S.2.7.

On the type of statement supplied by experience, the same reference further states:

Denn die Erfahrung macht nur Aussagen von diesem Typus: ein Beobachter hat eine bestimmte (subjektive) Wahrnehmung gemacht, und nie eine solche: eine physikalische Größe hat einen bestimmten Wert.

For experience makes statements only of the following type: an observer has had a particular (subjective) perception, and never statements of the following type: a physical quantity has a particular value.

The original-source location is p. 224, §VI.1 of the same work. The same section distinguishes the placement of the boundary between the observed object and the observer, the boundary required for comparison with experience, and the conditions under which the two processes apply depending on whether a measurement interaction is present. This statement includes subjective perception among its objects of statement, but it does not state that a change in subjective temporal experience was directly measured.

Zurek 2003 states:



Decoherence is caused by the interaction with the environment which in effect monitors certain observables of the system

The original-source location is Zurek, *Decoherence, Einselection, and the Quantum Origins of the Classical*, *Reviews of Modern Physics* 75, p. 715, abstract. The complete evidence record for the same reference is given in paragraph S.2.44.

Adler 2003 states:

decoherence has not solved the quantum measurement problem.

The original-source location is Adler, *Why Decoherence Has Not Solved the Measurement Problem: A Response to P. W. Anderson*, arXiv:quant-ph/0112095v3, p. 2, Abstract. The complete evidence record for the same reference is given in paragraph S.2.40.

Schlosshauer 2005 states:

their implications for the foundational problems of quantum mechanics, most notably the quantum measurement problem, have remained a matter of great controversy.

The original-source location is Schlosshauer, *Decoherence, the Measurement Problem, and Interpretations of Quantum Mechanics*, *Reviews of Modern Physics* 76, p. 1267, abstract. The complete evidence record for the same reference is given in paragraph S.2.46.

**Proposition 7.132** (Separation of the Evidence-Effect Domains of Quantum-Measurement Statements). The first statement in von Neumann 1932 directly supports that a nonstationary wave function contains time  $t$ . The measurement statement in the same reference distinguishes subjective apperception, the boundary between the observed object and observer, experiential statements, and the conditions of application of the two processes, but it does not support that a change in subjective temporal experience was directly measured. The statement in Zurek 2003 directly supports the occurrence of decoherence through system–environment interaction. The statement in Adler 2003 separates the suppression of interference by decoherence from a solution to the quantum measurement problem as a whole. The statement in Schlosshauer 2005 supports the continuing controversy over the implications of decoherence for the quantum measurement problem.

Interference suppression, environmental correlation, pointer structure, single-outcome selection, solution of the quantum measurement problem as a whole, and physical generation of the time contents taken as inputs to quantum evolution belong to distinct evidence-effect domains. A claim that transfers the validity of one domain to another requires independent evidence or a valid derivation supporting that transfer.

Adler 2003 is corrective evidence separating limited decoherence results from a solution to the quantum measurement problem as a whole. Schlosshauer 2005 is reserving evidence that does not treat the implications between those same two domains as settled.

*Proof.* The quotations above and paragraph S.2.7, paragraph S.2.44, paragraph S.2.40, and paragraph S.2.46 give the object, predicate, explicit domain of application, and original-source location of each quotation. By Definition 7.115, the objects, predicates, conditions, and domains of application directly supported by each quotation determine its evidence-effect domain. A claim that moves to a different object or predicate requires evidence or a derivation specific to that transfer.  $\square$

**Proposition 7.133** (Non-Generation of Time in Quantum Measurement, Records, and Decoherence). For the reference  $R$  corresponding to each of von Neumann 1932, Zurek 2003, and Schlosshauer 2005, Definition 7.56 gives  $R \in \text{Ref}^{\text{audit,rootall}}$ .

In von Neumann 1932, results 3, 5–7, and 10–13, eight in total, depend on primitive time roots; in Zurek 2003, 32 of the 43 results do. In Schlosshauer 2005, the 27 results other than results 2, 3, and 16 depend on primitive time roots.

*Proof.* For von Neumann 1932, apply Proposition S.60 and Proposition S.60. For Zurek 2003, apply Proposition S.435 and Proposition S.435. For Schlosshauer 2005, apply Proposition S.458 and Proposition S.458.  $\square$

**Proposition 7.134** (Primitive-Time Presupposition of Quantum Time Evolution). The unitary evolutions, before-and-after measurement relations, decoherence times, record persistence, and history constructions in the three references take time contents as inputs and yield quantum states, measurement records, or classical appearances. These constructions do not generate the input time contents from the physical-input initial set.

*Proof.* By Proposition 7.133, all time contents in the three references are primitive time roots. Dependence of reference-specific results that take primitive time roots as inputs is inherited by those results under Corollary 7.12.  $\square$

**Corollary 7.135** (Failure of Decoherence to Ground Physical Time). The derivation of interference suppression, pointer stability, environmental records, and classical appearance through decoherence does not provide physical grounds for the time contents taken as inputs to quantum evolution. Success in measurement theory does not convert primitive time input into physical time.

*Proof.* Apply Proposition 7.134 and Proposition 7.13.  $\square$

### 7.3.5 Internal Clocks, Thermal Time, and Local-Reachability Determination

The audit question common to Page–Wootters 1983, Connes–Rovelli 1994, and Smolin 2018/2020 is whether the elimination of external time, the construction of a temporal flow from a state, or the claim that time is fundamental generates the time contents used from the physical-input initial set.

In Page–Wootters 1983, external-coordinate-time averaging, stationary replacement, clock-state orbits, and internal time constitute time contents. In Connes–Rovelli 1994, the thermal-time flow corresponding to a state, KMS structure, inverse-temperature time scales, and geometrical time scales constitute time contents. In Smolin 2018/2020, fundamental time, fundamental causality, the objective present, the thick present, event generation, and global temporal order constitute time contents.

**Proposition 7.136** (Primitive Time Roots in Relational and Internal-Time Constructions). Each  $R$  corresponding to one of the three references belongs to  $\text{Ref}^{\text{audit,rootall}}$  in Definition 7.56. All 23 time contents in Page–Wootters 1983, 28 in Connes–Rovelli 1994, and 27 in Smolin 2018/2020 are primitive time roots under the complete determination. The results dependent on primitive time roots number 16 of 18, 21 of 37, and 21 of 28, respectively.

Under the local-reachability determination that adds the reference formal-evidence environment, Connes–Rovelli 1994 reaches Time Contents 11, 19, and 27, leaving 25 primitive time roots. Local reach is empty in Page–Wootters 1983 and Smolin 2018/2020. Smolin 2018/2020 has 127 obligation atoms (Proposition S.629).

*Proof.* For Page–Wootters 1983, apply Proposition S.231 and Proposition S.231. For Connes–Rovelli 1994, apply Proposition S.328 and Proposition S.328. For Smolin 2018/2020, apply Proposition S.626 and Proposition S.626.  $\square$

**Proposition 7.137** (Separation of the Relational Reformulation of Time from the Generation of Time). Averaging over external coordinate time to remove it from specified observables, selecting a temporal flow from a state, and taking time to be fundamental do not imply the physical generation of time contents. Describing time as relational, internal, or fundamental does not supply a generation history for a primitive time root.

*Proof.* By Proposition 7.136, all time contents in the three references are primitive time roots under the complete determination starting solely from non-temporal physical inputs. Inheritance of dependence by the root-dependent results in each reference follows from Corollary 7.12.  $\square$

**Corollary 7.138** (Failure of Relational Time to Ground Physical Time). Rendering external time unobservable, stationary conditional probabilities, the thermal-time hypothesis, and the introduction of event generation do not convert primitive time roots into physical time. Even if these results are mathematically or physically valid, they do not jointly satisfy the generation-history, comparison-structure, and grounding requirements for the input time contents to be established as physical time.

*Proof.* Apply Proposition 7.137 and Proposition 7.13.  $\square$

### 7.3.6 Canonical Gravity, the Problem of Time, and Cosmological Boundaries

The audit question common to DeWitt 1967, Kuchař 1992, and Hartle–Hawking 1983 is whether constraint equations that do not explicitly contain an external time parameter, candidates for intrinsic time, or a no-boundary path integral generate the time contents used therein from the physical-input initial set.

In DeWitt 1967, hypersurfaces, evolution, intrinsic time, and a Wheeler–DeWitt-type constraint constitute time contents. In Kuchař 1992, extrinsic time, intrinsic time, many-fingered time, the choice of time, and the components of the problem of time constitute time contents. In Hartle–Hawking 1983, the Euclidean region, the Lorentzian region, the complex-time contour, the initial condition of the universe, and the emergence of classical spacetime constitute time contents.

**Proposition 7.139** (Primitive Time Roots Remaining in Timeless Formulations of Quantum Gravity). Each  $R$  corresponding to one of the three references belongs to  $\text{Ref}^{\text{audit, rootall}}$  in Definition 7.56. All 35 time contents in DeWitt 1967, 47 in Kuchař 1992, and 21 in Hartle–Hawking 1983 are primitive time roots under the complete determination. The results dependent on primitive time roots number 19 of 25, 28 of 38, and 17 of 24, respectively.

*Proof.* For DeWitt 1967, apply Proposition S.124 and Proposition S.124. For Kuchař 1992, apply Proposition S.273 and Proposition S.273. For Hartle–Hawking 1983, apply Proposition S.221 and Proposition S.221.  $\square$

**Proposition 7.140** (Non-Identity of Timeless Formulations and the Physical Generation of Time). Removing an external time parameter from a fundamental equation does not generate the ordering of hypersurfaces, candidates for intrinsic time, the Euclidean–Lorentzian connection, or the time contents of classical spacetime. The formal absence of a time parameter does not establish the physical grounding of time contents.

*Proof.* By Proposition 7.139, none of the time contents in the three references is reachable from the physical-input initial set. The generation of time contents therefore does not follow from the absence of a time parameter in a formal equation.  $\square$

**Corollary 7.141** (Failure of the Problem of Time to Ground Physical Time). Classifying the problem of time, choosing an intrinsic time, and deriving classical spacetime from the no-boundary condition do not provide a generation history for the time contents used. Timelessness in quantum gravity does not replace the grounds for physical time.

*Proof.* Apply Proposition 7.140 and Proposition 7.13.  $\square$

### 7.3.7 Arrows of Time, the Gravitational Arrow, and Temporal Flow

The audit question common to Lebowitz 1993, Penrose 1979, and Ellis 2012 is whether an explanation of irreversibility, a low-entropy boundary condition, or a cosmological direction of time generates from the physical-input initial set the time contents supplied as inputs to that explanation.

In Lebowitz 1993, microscopic time reversal, macroscopic temporal evolution, a low-entropy initial state, and the thermodynamic arrow of time constitute time contents. In Penrose 1979, the initial singularity, the final singularity, the Weyl-curvature condition, gravitational entropy, and cosmological time asymmetry constitute time contents. In Ellis 2012, the evolution of spacetime, causal structure, the present boundary, the cosmological direction of time, and emergent complexity constitute time contents.

**Proposition 7.142** (Primitive Time Roots Remaining in Explanations of the Arrow of Time). Each  $R$  corresponding to one of the three references belongs to  $\text{Ref}^{\text{audit, rootall}}$  in Definition 7.56. All 34 time contents in Lebowitz 1993, 25 in Penrose 1979, and 40 in Ellis 2012 are primitive time roots. The results dependent on primitive time roots number 18 of 24 in Lebowitz 1993, all 20 in Penrose 1979, and all 10 in Ellis 2012.

*Proof.* For Lebowitz 1993, apply Proposition S.284 and Proposition S.284. For Penrose 1979, apply Proposition S.213 and Proposition S.213. For Ellis 2012, apply Proposition S.599 and Proposition S.599.  $\square$

**Proposition 7.143** (Separation of Explanations of Temporal Direction from the Generation of Time). A boundary condition, entropy gradient, or cosmological causal direction that assigns an asymmetric direction to a sequence of states does not generate the time contents that order that sequence. An explanation of the arrow of time does not supply the physical grounds for time.

*Proof.* By Proposition 7.142, all time contents in the three references are primitive time roots. An asymmetry that presupposes a primitive time root does not yield a generation history for that root.  $\square$

**Corollary 7.144** (Non-Repair of Primitive Time Roots by Explanations of Irreversibility). Even if explanations of thermodynamic irreversibility, low-entropy initial conditions, and a cosmological direction of time hold, the input time contents are not thereby established as physical time. Success concerning temporal direction does not repair the non-generation of time itself.

*Proof.* Apply Proposition 7.143 and Proposition 7.13.  $\square$

### 7.3.8 Black-Hole Thermodynamics, Radiation, and Information

The audit question common to Bekenstein 1973, Hawking 1975, and Hawking 1976 is whether changes in horizon area, radiation, evaporation, and temporal evolution from a pure state to a mixed state generate from the physical-input initial set the time contents used therein.

In Bekenstein 1973, stationary processes, finite processes, changes in horizon area, and the generalized second law constitute time contents. In Hawking 1975, past and future infinity, collapse, emission times, evaporation time, and thermal particle flux constitute time contents. In Hawking 1976, the initial state, the final state, temporal evolution, completion of evaporation, and information loss constitute time contents.

**Proposition 7.145** (Primitive Time Roots Remaining in Black-Hole Thermodynamics and Evaporation). All time contents in the three references are primitive time roots: 67 in Bekenstein 1973, 94 in Hawking 1975, and 77 in Hawking 1976. Of the 18 results in Bekenstein 1973, result 4 has evidence-only reach, and the remaining 17 depend on primitive time roots. There are no generated results and no results that remain unreachable after the roots are added. All 30 results in Hawking 1975 and all 35 results in Hawking 1976 depend on primitive time roots. Thus, of the three references, Hawking 1975 and Hawking 1976 belong to  $\text{Ref}^{\text{audit, rootfull}}$  in Definition 7.58, whereas Bekenstein 1973 does not.

*Proof.* For Bekenstein 1973, apply Proposition S.165 and Proposition S.165. For Hawking 1975, apply Proposition S.181 and Proposition S.181. For Hawking 1976, apply Proposition S.190 and Proposition S.190.  $\square$

**Proposition 7.146** (Failure of Black-Hole Dynamics to Ground Time). Derivations of area increase, radiation spectra, evaporation time, and processes of information loss do not generate the time contents that order and compare them. The calculability of the temporal evolution of a black hole does not imply that its time variable has a physical ground.

*Proof.* By Proposition 7.145, none of the time contents in the three references is reachable from the physical-input initial set. Result 4, which has evidence-only reach in Bekenstein 1973, is not a result of generating time content either. Thus, the derivation of results in these references does not establish the generation of the time contents used in those derivations.  $\square$

**Corollary 7.147** (Primitive-Time Presupposition of the Information-Loss Problem). The information-loss problem, which contrasts an initial state with a final state, the state before collapse with that after evaporation, and a pure state with a mixed state, presupposes temporal order and temporal evolution. Neither the formulation of this problem nor candidate solutions to it generate their input time from the physical-input initial set.

*Proof.* Apply Proposition 7.146 and Corollary 7.12.  $\square$

### 7.3.9 Duality, Entropy, and Spacetime Emergence

The audit question common to Maldacena 1998, Ryu–Takayanagi 2006, and Van Raamsdonk 2010 is whether the correspondence between a boundary theory and a bulk theory, the correspondence between entropy and a minimal surface, or the relation between quantum

entanglement and spatial connectivity generates from the physical-input initial set the time contents used on both sides.

In Maldacena 1998, boundary time, bulk time, Hamiltonian time evolution, causal structure, and thermal states constitute time contents. In Ryu–Takayanagi 2006, static spacetime, time slices, horizons, and the thermal interpretation of time constitute time contents. In Van Raamsdonk 2010, boundary time, asymptotic time and causal structure, eternal-black-hole time and causal structure, and Hamiltonian time translation constitute time contents.

**Proposition 7.148** (Primitive Time Roots Remaining in Holography and Spacetime Connectivity). All time contents in the three references are primitive time roots: 26 in Maldacena 1998, 9 in Ryu–Takayanagi 2006, and 3 in Van Raamsdonk 2010. The results dependent on primitive time roots number 32 of 37, 12 of 16, and 8 of 10, respectively. The remaining results have evidence-only reach; there are no generated results and no results that remain unreachable after the roots are added. Thus, all three references belong to  $\text{Ref}^{\text{audit,rootall}}$  in Definition 7.56, and none belongs to  $\text{Ref}^{\text{audit,rootfull}}$  in Definition 7.58.

*Proof.* For Maldacena 1998, apply Proposition S.349 and Proposition S.349. For Ryu–Takayanagi 2006, apply Proposition S.491 and Proposition S.491. For Van Raamsdonk 2010, apply Proposition S.565 and Proposition S.565.  $\square$

**Proposition 7.149** (Separation of Spacetime Correspondence and Reconstruction from the Generation of Time). Correlating time evolution between two theories, correlating a geometrical quantity on a time slice with entropy, and reconstructing spatial connectivity from quantum entanglement do not generate the time contents supplied to the correspondence and reconstruction. Even if part of spacetime can be reconstructed from another quantity, the physical grounding of the time variable does not follow.

*Proof.* By Proposition 7.148, none of the time contents in the three references is reachable from the physical-input initial set. The results with evidence-only reach in the three references are not results of generating time content either. Thus, establishing results by correspondence or reconstruction does not establish the generation of the time contents used therein.  $\square$

**Corollary 7.150** (Failure of Quantum Entanglement to Ground Physical Time). The involvement of quantum entanglement in spatial connectivity, distance, or geometry does not jointly satisfy the generation-history, comparison-structure, and grounding requirements for time contents to be established as physical time. A result concerning spatial emergence is not a proof of temporal emergence.

*Proof.* Apply Proposition 7.149.  $\square$

### 7.3.10 The Cosmological Constant, Observed Acceleration, and the Vocabulary of Expansion

The audit question common to Weinberg 1989, Riess et al. 1998, and Davis–Lineweaver 2004 is whether the magnitude of the cosmological constant, the supernova distance–redshift relation, and analyses of cosmological horizons generate from the physical-input initial set the time contents supplied to their descriptions of cosmic history.

In Weinberg 1989, cosmic expansion, the age of the universe, the times of phase transitions, and anthropic selection of an epoch constitute time contents. In Riess et al.

1998, light-curve time scales, time dilation, expansion history, the deceleration parameter, and the age of the universe constitute time contents. In Davis–Lineweaver 2004, conformal time, proper distance, light cones, particle horizons, event horizons, and expansion history constitute time contents.

**Proposition 7.151** (Primitive Time Roots Remaining in Reconstructions of Cosmic History). Each  $R$  corresponding to one of the three references belongs to  $\text{Ref}^{\text{audit,rootall}}$  in Definition 7.56. All 22 time contents in Weinberg 1989, 16 in Riess et al. 1998, and 26 in Davis–Lineweaver 2004 are primitive time roots under the complete determination. The results dependent on primitive time roots number 16 of 21, all 24, and 28 of 29, respectively.

*Proof.* For Weinberg 1989, apply Proposition S.251 and Proposition S.251. For Riess et al. 1998, apply Proposition S.375 and Proposition S.375. For Davis–Lineweaver 2004, apply Proposition S.446 and Proposition S.446.  $\square$

**Proposition 7.152** (Separation of the Quantification of Cosmic History from the Generation of Time). Calculating the age, expansion rate, and acceleration of the universe, as well as past light cones and horizons, does not generate the time contents that constitute the variables, order, and domains of integration in those calculations. The capacity to reconstruct cosmic history precisely does not imply that the temporal axis of cosmic history has been physically grounded.

*Proof.* By Proposition 7.151, none of the time contents in the three references is reachable from the physical-input initial set.  $\square$

**Corollary 7.153** (Failure of Cosmological Observational Success to Ground Time). Agreement with supernova observations, constraints on the cosmological constant, and the consistency of horizon structures do not provide a generation history for the input time contents. Observational success does not turn primitive time input into physical time.

*Proof.* Apply Proposition 7.152 and Proposition 7.13.  $\square$

The boundary supplied by a cosmological model or observational reference is separated from the domain of existence of physical time in general. Feynman, Leighton and Sands 1963 states:

It is now believed that at least our part of the universe had its beginning about ten or twelve billion years ago.

The same reference then poses the following question:

Does an earlier time have any meaning?

The original-source location is Feynman, Leighton and Sands, *The Feynman Lectures on Physics*, Volume I, Chapter 5, §5-4, at the end of *Long times*. The complete original text, location, and minimal context from the same reference are given in paragraph S.2.12.

**Proposition 7.154** (Interpretation-Specific Evidence Effect of Cosmological Boundary Statements). The path that interprets the preceding statement as the beginning of a cosmological model or region within the explicitly specified “our part of the universe” belongs to its evidence-effect domain. On a path that transfers the preceding statement

to the beginning of physical time in general, to nonexistence before that beginning, or to meaninglessness before it, the object and domain of application change. If no independent evidence or valid derivation supporting that change is supplied, the number of evidence-effect transgressions along that path is 1.

The question in the original source does not by itself yield the conclusion that time before the beginning does not exist or is meaningless.

*Proof.* By Definition 7.115, the object, predicate, conditions, and domain of application directly supported by reference-specific evidence determine the evidence-effect domain. By Definition 7.117, a separate evidence-comparison path is taken for each permissible interpretation of the same statement. A path that transfers the beginning of a model or region to the domain of existence of physical time in general changes the object and domain of application once; if it is not accompanied by independent evidence or a valid derivation, the number of evidence-effect transgressions is therefore 1.  $\square$

Riess et al. 1998 states:

The distances of the high-redshift SNe Ia are, on average, 10% to 15% farther than expected in a low mass density ( $\Omega_M = 0.2$ ) Universe without a cosmological constant.

The original-source location is the abstract of Riess et al., *Observational Evidence from Supernovae for an Accelerating Universe and a Cosmological Constant*, *The Astronomical Journal* 116, p. 1009. The complete evidentiary record from the same reference is given in paragraph S.2.39.

Perlmutter et al. 1999 states:

The data are strongly inconsistent with a  $\Lambda = 0$  flat cosmology, the simplest inflationary universe model.

The original-source location of this  $\Lambda = 0$  statement is the abstract of Perlmutter et al., *Measurements of  $\Omega$  and  $\Lambda$  from 42 High-Redshift Supernovae*, *The Astrophysical Journal* 517, p. 565. The complete evidentiary record from the same reference is given in paragraph S.2.38.

**Proposition 7.155** (Separation of Cosmological Observation, Inference, Model Fit, and Direct Demonstration). The distance excess in Riess et al. 1998 is an inferred result mediated by photometry, spectroscopy, correction, calibration, and distance fitting. The inconsistency in Perlmutter et al. 1999 is a model determination relative to the adopted sample, light-curve standardization, likelihood, and family of FRW models.

The observations, inferences, and model fits in the two references belong to their respective evidence-effect domains. Direct demonstration concerning the whole of cosmic history, the physical generation of time contents, or the domain of existence of physical time in general requires independent evidence or a valid derivation specific to each claim.

*Proof.* Paragraph S.2.39 presents the distance excess as an inferred result mediated by photometry, spectroscopy, correction, calibration, and distance fitting, and presents a positive cosmological constant and present acceleration as statistical inferences based on FRW models and likelihood. Paragraph S.2.38 presents the inconsistency with the  $\Lambda = 0$  flat model as a model determination restricted to a sample, corrections, likelihood, and a family of FRW models. By Definition 7.115, a claim having a different object, predicate, evidence type, or domain of application requires evidence or a derivation specific to that claim.  $\square$



### 7.3.11 Comparison of Source Statements with Formal Descriptions

Table 16 gives, for each reference in the survey, the time contents, primitive time roots, and root-dependent results in its formal descriptions. Proposition 7.186 and Proposition 7.187 give the cross-reference results obtained from those reference-specific formal audits.

Comparison of a source statement with a formal description concerns the logical relation between each interpretation permitted by the statement and the formal description in the same reference.

*Remark 7.156* (Interpretability of Source Statements). The permissible interpretations of a source statement are constrained by the words, qualifications, figurative expressions, assertoric strength, and context of the original text. When multiple permissible interpretations remain, the result of comparing each with the formal description is conditional on that interpretation.

*Remark 7.157* (Conditional Comparison with Formal Descriptions). If one permitted interpretation of a statement and the fixed formal description cannot both hold, the contradiction that arises under that condition is stated. If the formal description does not derive that interpretation, formal non-support is stated. If the explanatory causal direction and the input direction in the formal description are opposite, causal-direction reversal is stated. If the literal content reduces to a restatement of an observable, calculated quantity, or operational rule, collapse into shorthand is stated.

*Remark 7.158* (Logical Role of Source-Statement Comparison). The reference-specific formal audits and cross-reference propositions are determined from the formal descriptions in the original sources. Conditional comparison with source statements identifies tensions between formal use and explanatory expression within the same reference and supplies concrete comparison cases for the discussion of multistage misidentification in Chapter 8.

*Remark 7.159* (Evidentiary Conditions for Source Statements). The object, predicate, conditions, and domain of application of a statement identified by bibliographic information and its source location are determined by the meaning, qualifications, figurative expressions, assertoric strength, and context of the original text. In a comparison mediated by translation, preservation of these elements is the condition of correspondence between the original text and its translation.

*Remark 7.160* (Context of Source Statements). When an equation, figure, table, footnote, prior definition, or another sentence in the same argument determines the meaning of a statement, the permissible interpretations of that statement are restricted to those consistent with these relations.

*Remark 7.161* (Boundary between Source Statements and Formal Audit). A determination in a reference-specific formal audit follows from definitions, equations, propositions, operational rules, uses of time contents, generation dependencies, and result dependencies. With the inputs to the formal audit fixed, differences in the interpretation of explanatory statements alone leave the classification values and cross-reference propositions unchanged. Statement comparison concerns the logical relation between each interpretation and that formal description.

### 7.3.12 The Explicit Form of Primitive Time and Its Use across Physics

The most explicit form of primitive time appears in Newton 1687/1729. That reference states:

Absolute, true, and mathematical time, of itself, and from its own nature flows equably without regard to anything external, and by another name is called duration :

The original-source location is Newton, *The Mathematical Principles of Natural Philosophy*, Motte translation, p. 77, *Scholium* I immediately following the Definitions. The statement evidence in the Appendix is given in paragraph S.2.2.

This time is not introduced as an object generated from external physical inputs; it is an input prior to dynamical relations. By Proposition 7.123, the time contents in that reference have primitive time roots, and those roots propagate to subsequent results.

**Proposition 7.162** (Persistence of Primitive Time Roots among Representatives of Disparate Fields). For each  $R \in \text{Ref}^{\text{audit}}$  corresponding to Newton 1687/1729, Einstein 1905, Einstein 1916, von Neumann 1932, DeWitt 1967, Maldacena 1998, or Weinberg 1989,

$$\begin{aligned}\text{TimeContent}_R^{\text{audit,root}} &\neq \emptyset, \\ \text{Result}_R^{\text{audit,rootdep}} &\neq \emptyset\end{aligned}$$

hold.

*Proof.* For Newton 1687/1729, apply Proposition S.17 and Proposition S.17.

For Einstein 1905, apply Proposition S.24 and Proposition S.24. For Einstein 1916, apply Proposition S.38 and Proposition S.38.

For von Neumann 1932, apply Proposition S.60 and Proposition S.60. For DeWitt 1967, apply Proposition S.124 and Proposition S.124.

For Maldacena 1998, apply Proposition S.349 and Proposition S.349. For Weinberg 1989, apply Proposition S.251 and Proposition S.251.  $\square$

In each reference covered by Proposition 7.162, temporal dependency roots precede results in classical mechanics, special relativity, general relativity, quantum measurement, quantum gravity, spacetime emergence, and cosmology. Although the theoretical field and formal mechanism change, the dependency order from use of time before its generation to propagation into results is preserved.

Statements concerning the difficulty of defining time and the naming of clocks raise the same generation obligation from a perspective different from the explicit form of primitive time. Feynman, Leighton and Sands 1963 states:

time is one of the things we probably cannot define

The same reference then states:

how we measure it

The original-source location is Volume I, Chapter 5, §5-2, *Time*, and the statement evidence in the Appendix is given in paragraph S.2.12.

Anderson 2012 states:

Simply calling an entity a time or a clock does not make it one.

The original-source location is Anderson, *Problem of Time in Quantum Gravity*, arXiv:1206.2403v2, p. 1, §1.1, and the statement evidence in the Appendix is given in paragraph S.2.59.

**Proposition 7.163** (Separation of Definitional Difficulty, Measurement Procedures, and Naming from the Generation of Time). Feynman, Leighton and Sands 1963 and Anderson 2012 state the difficulty of defining time, a shift to measurement procedures, or a warning about the names “time” and “clock.” The formal constructions in both references, however, do not generate the time contents they use from non-temporal physical inputs.

*Proof.* For Feynman, Leighton and Sands 1963, apply Proposition S.102 and Proposition S.102. For Anderson 2012, apply Proposition S.591 and Proposition S.591. In both references, all time contents are primitive time roots and results dependent on primitive time roots exist.  $\square$

**Corollary 7.164** (Common Limit of Explicit Primitive Time, Definitional Difficulty, Measurement Procedures, and Warnings about Naming). An explicit statement of absolute time, the difficulty of defining time, a shift to measurement procedures, and a warning about the names “time” and “clock” do not complete the generation of the time contents used by physical law. The generation-priority condition of Corollary 7.15 must be satisfied independently of these statements.

*Proof.* Apply Proposition 7.162 and Proposition 7.163.  $\square$

### 7.3.13 Evaluation Success and Physical-Time Qualification

Agreement with experimental values, the standardization of units, the establishment of observational relations, calculations by duality, predictions of quantum measurement, and descriptions of decoherence are distinct physical and mathematical results. The dependency roots of the time contents used by these results are audited separately from the precision or validity of the results.

**Proposition 7.165** (Dependence on Primitive Time Roots in Successful Results). For each  $R \in \text{Ref}^{\text{audit}}$  corresponding to Hafele–Keating 1972a/1972b, BIPM SI 2019/2026, Riess et al. 1998, Maldacena 1998, Ryu–Takayanagi 2006, von Neumann 1932, or Zurek 2003,

$$\begin{aligned} \text{Result}_R &\neq \emptyset, \\ \text{TimeContent}_R^{\text{audit,root}} &\neq \emptyset, \\ \text{Result}_R^{\text{audit,rootdep}} &\neq \emptyset \end{aligned}$$

hold.

*Proof.* For Hafele–Keating 1972a, apply Proposition S.157 and Proposition S.157. For Hafele–Keating 1972b, apply Proposition S.148 and Proposition S.148.

For BIPM SI 2019/2026, apply Proposition S.635 and Proposition S.635. For Riess et al. 1998, apply Proposition S.375 and Proposition S.375.

For Maldacena 1998, apply Proposition S.349 and Proposition S.349. For Ryu–Takayanagi 2006, apply Proposition S.491 and Proposition S.491.

For von Neumann 1932, apply Proposition S.60 and Proposition S.60. For Zurek 2003, apply Proposition S.435 and Proposition S.435.  $\square$

**Corollary 7.166** (Non-Completion of Time Generation by Evaluation Success). Even if the results in Proposition 7.165 hold, the primitive time roots are not eliminated, and no generation trace from the physical-input initial set to the time contents is added.

*Proof.* By Proposition 7.165, the result set is nonempty in each reference, and both the set of primitive time roots of time contents and the set of results dependent on primitive time roots are nonempty. The establishment of a result therefore implies neither the elimination of primitive time roots nor the addition of a generation trace.  $\square$

The difference between an evaluated result and the object of a statement appears clearly in references concerning clock rates. Feynman, Leighton and Sands 1963 states:

Now if all moving clocks run slower, if no way of measuring time gives anything but a slower rate, we shall just have to say, in a certain sense, that time itself appears to be slower in a space ship.

The original-source location is Volume I, Chapter 15, §15-4, *Transformation of time*, and the statement evidence in the Appendix is given in paragraph S.2.12.

NIST 2025 states the following about the general theory of relativity Einstein published in 1915:

According to the audacious new theory of general relativity he published that year, time is also affected by gravity. The more strongly you feel the force of gravity, Einstein argued, the slower time passes for you.

The statement evidence in the Appendix is given in paragraph S.2.69. ESA 2019 states:

gravity and speed influences time

The statement evidence in the Appendix is given in paragraph S.2.65.

**Proposition 7.167** (Separation of Clock Relations from the Attribution of Properties to Time Itself). The formal results established in Feynman, Leighton and Sands 1963, NIST 2025, and ESA 2019 are relations among clock readings, clock rates, frequencies, lifetimes, or proper times. If the quoted statements are interpreted literally as explaining that time itself slows down, or that time itself is acted upon by gravity or speed, the formal results in those references do not provide a state space for a time carrier, a quantity of change, a law of evolution, or a coupling law to the clock relations.

*Proof.* For Feynman, Leighton and Sands 1963, apply Proposition S.102. For NIST 2025, apply Proposition S.681. For ESA 2019, apply Proposition S.643. The objects of the results supplied by these propositions are relations among clock readings, clock rates, frequencies, lifetimes, or proper times; an independent state and evolution law for a time carrier are not components of the result sets.  $\square$

**Corollary 7.168** (Non-Completion by Evaluation Success and Explanatory Attribution of Properties). The establishment of clock relations, observational relations, standardization, duality, and quantum-theoretical predictions preserves those results while leaving primitive time roots in place. Explanatory attribution of properties to time itself likewise adds no generation trace from non-temporal physical inputs to time contents.

*Proof.* Apply Corollary 7.166 and Proposition 7.167.  $\square$

### 7.3.14 Adjacent Problems of Time and Complete Time Generation

The creation, nonexistence, definition, internalization, conditional evolution, and relational recovery of time include issues adjacent to the primitive-time problem. Because the formal scope borne by each term differs among references, the wording and formal construction of the original source are compared reference by reference.

Barbour 1994 I makes the following two statements:

Time is not any one variable; it is created by all variables working in concert.

It is argued that time as such does not exist but that instants, defined as complete relative configurations of the universe, do.

The statement evidence in the Appendix is given in paragraph S.2.33. Barbour 1994 II states:

time is truly non-existent in the kinematic foundations of both theories

The statement evidence in the Appendix is given in paragraph S.2.34. Connes–Rovelli 1994 states:

there is no preferred time flow

and

thermal time ... defines the physical time

The statement evidence in the Appendix is given in paragraph S.2.35. The second quotation occurs in the postulate identifying the thermal time determined by a state with physical time.

Page–Wootters 1983 states:

Only the dependence on an internal clock time can be observable.

The statement evidence in the Appendix is given in paragraph S.2.26. Kiefer 2005 makes the following two statements:

Quantum gravity does not contain an external time parameter at the most fundamental level.

intrinsic time, given in quantum cosmology by the scale factor

The statement evidence in the Appendix is given in paragraph S.2.47. Kiefer 2009 makes the following two statements:

Time is absolute in standard quantum theory and dynamical in general relativity.

Time emerges from the separation into two different subsystems: one subsystem (here: the gravitational part) defines the time with respect to which the other subsystem (here: the non-gravitational part) evolves.

The original-source locations are Kiefer, *Does Time Exist in Quantum Gravity?*, arXiv:0909.3767v1, p. 1, abstract, and p. 6, §3. The complete evidentiary record from the same reference is given in paragraph S.2.54.

Smolin 2018/2020 states:

I take the opposite view, that time is fundamental.

The original-source location is Smolin, *Temporal Relationalism*, arXiv:1805.12468v1, p. 3, §1. The complete evidentiary record from the same reference is given in paragraph S.2.63.

**Proposition 7.169** (Evidence-Effect Domains of Two-Subsystem Time and Asserted Fundamentality of Time). The first statement in Kiefer 2009 directly supports the difference in role between the external background parameter of standard quantum theory and the dynamical spacetime variables of general relativity. Within the domain of application of the separation and approximation, the second statement supports the construction of a semiclassical time parameter or an internal-clock candidate through the separation of a heavy subsystem from a light subsystem.

The statement in Smolin 2018/2020 directly supports a position according to which time is fundamental. Fundamentality permits the coexisting interpretations of ontological fundamentality, fundamentality as a theoretical input, fundamentality of causal order, and a methodological contrast.

The construction of subsystem time, the fundamentality of time, the generation of all reference-specific time contents, the complete elimination of primitive time roots, and qualification for use within laws belong to distinct evidence-effect domains. A claim that moves between these domains requires independent evidence or a valid derivation supporting that transition.

*Proof.* Paragraph S.2.54 and paragraph S.2.63 give the object, predicate, explicit domain of application, interpretive possibilities, and original-source location of each quotation. By Definition 7.115, the objects, predicates, conditions, and domains of application directly supported by each quotation determine its evidence-effect domain. A claim that moves to a different object, predicate, or domain of application requires evidence or a derivation specific to that transition.  $\square$

Rovelli 2002 states:

Therefore time cannot be an observable [2]. But if time is not an observable, how can we measure it?

The same reference also states the following about the form of dynamical predictions:

Observe that the predictions of a mechanical theory can always be expressed as relations between partial observables.

The reference further states the following about a general mechanical formalism:

There is no need to single out a specific partial variable as the time, nor to mention evolution. Mechanics is a theory of relations between partial observables.

The original-source location of the first quotation is Rovelli, *Partial Observables*, arXiv:gr-qc/0110035v3, p. 1, §I; the second quotation is on p. 2, §II, and the third on p. 7, §VI of the same reference. The complete statement evidence and interpretation audit are given in paragraph S.2.43 and paragraph S.2.43. Rovelli 2011 states:

I summarize here this point of view, explaining why I think that in a fundamental description of nature we must “forget time”, and how this can be done in the classical and in the quantum theory.

The statement evidence in the Appendix is given in paragraph S.2.55.

**Proposition 7.170** (Evidence-Effect Domains of Relational Formalisms and the Generation of Physical Time). The statements in Rovelli 2002 directly support expressing dynamical predictions as relations among partial observables and, in a general mechanical formalism, neither selecting a particular partial variable as time nor placing the vocabulary of evolution in the fundamental structure.

The establishment of a relational formalism, the removal of privileged time, and the nonuse of the vocabulary of evolution do not mean that the clocks, records, comparisons, order, scale, and qualification for use within laws that physically realize the relations have been generated from non-temporal physical inputs. The responsibility to remove external time from a formalism and the responsibility to generate physical time belong to distinct evidence-effect domains.

*Proof.* Paragraph S.2.43 presents the observability of time, the impossibility of predicting  $t$  alone, relations among partial observables, and a general mechanical formalism without privileged time as separate statements. Paragraph S.2.43 restricts the evidentiary effect of each statement to the partial-observable formalism and general mechanical formalism of the original source. By Definition 7.115, progressing from a relational formalism to clocks, records, comparisons, order, scale, and qualification for use within laws requires independent evidence or a valid derivation supporting that transition.  $\square$

**Proposition 7.171** (Primitive Time Roots Remaining in the Formal Constructions of Adjacent Problems of Time). For each  $R$  corresponding to Connes–Rovelli 1994, Page–Wootters 1983, Kuchař 1992, or Smolin 2018/2020, Definition 7.56 gives  $R \in \text{Ref}^{\text{audit, rootall}}$ .

*Proof.* For Connes–Rovelli 1994, apply Proposition S.328. For Page–Wootters 1983, apply Proposition S.231. For Kuchař 1992, apply Proposition S.273. For Smolin 2018/2020, apply Proposition S.626.  $\square$

**Proposition 7.172** (Difference in Scope between Adjacent Programs of Time and Complete Time Generation). Thermal time, internal-clock time, intrinsic time, conditional evolution, relational observables, and some time contents reached under the local-reachability determination each provide reference-specific formal results. In each reference covered by Proposition 7.171, the path that generates all reference-specific time contents from non-temporal physical inputs is incomplete.

*Proof.* In each of Connes–Rovelli 1994, Page–Wootters 1983, Kuchař 1992, and Smolin 2018/2020, the set of generated time contents under the complete determination is empty, and all reference-specific time contents are primitive time roots. No complete generation trace covering all reference-specific time contents therefore exists in any of these references.  $\square$

**Corollary 7.173** (Separation of the Creation, Definition, Internalization, and Recovery of Time from Complete Generation). Statements that directly invoke the creation, nonexistence, definition, internalization, conditional evolution, or relational recovery of time explicitly address issues adjacent to the primitive-time problem. The qualification of all time contents used by physical law is determined, separately from the vocabulary of each statement, by generation traces from non-temporal physical inputs to all time contents.

*Proof.* Apply Proposition 7.172. □

### 7.3.15 Difference between Formal Results and the Objects of Statements

Newton 1687/1729 takes absolute time as an input to the dynamical construction; Einstein 1905/1916 and Taylor–Wheeler 1992 formalize coordinates, clock synchronization, spacetime intervals, proper time, and clock comparisons. These formal objects are fixed by the reference-specific time contents, transfer events, results, and dependencies in the Appendix.

Attributions of properties to time itself in Feynman, Leighton and Sands 1963 and NIST 2025 were presented in Section 7.3.13. Tong 2013 states:

The correct interpretation is that time itself runs more slowly in moving frames.

The same reference further states:

it is time itself which is running slower.

The statement evidence in the Appendix is given in paragraph S.2.61.

Greene states:

It turns out time itself can speed up or slow down.

The same reference further states:

That is, Einstein realized that if a stationary observer carefully analyzes what he sees, he'll find that time itself elapses more slowly for me than it does for him.

The statement evidence in the Appendix is given in paragraph S.2.57.

After discussing the frequency comparison between upper and lower clocks by means of light signals, Feynman 1995 also describes the situation as follows:

One of the ways of describing this situation is to say that the time scale is faster at the top; time flows are different in different gravitational potentials, so that time flows are unequal in various parts of the world.

After explaining the signal reaching an external observer from the surface of a collapsing body, its redshift, and the waiting time for its reception, Narlikar 2010 states:

The time flows at different rates for the two protagonists.

The complete quotation records and minimal contexts are given in paragraph S.2.70 and paragraph S.2.71.

Taylor–Wheeler 1992 states:



time cannot be separated from space. Space and time are part of a single entity, spacetime.

The statement evidence in the Appendix is given in paragraph S.2.13.

**Proposition 7.174** (Conditional Comparison of Formal Results with the Objects of Statements). If the statements in Feynman, Leighton and Sands 1963, Feynman 1995, NIST 2025, Narlikar 2010, Tong 2013, Greene, and Taylor–Wheeler 1992 are interpreted literally as explanations in which time itself or a spacetime carrier independently slows, accelerates, expands, contracts, or otherwise changes, the corresponding formal systems do not jointly provide a state space for that carrier, a quantity of change, a law of evolution, a coupling law to clock relations, and an independent measurement procedure.

If the same statements are interpreted with their effect restricted to differences in clock readings, clock rates, lifetimes, or proper times, or to spacetime intervals or coordinate relations, the corresponding formal results are preserved. This preservation is a determination concerning the domain of validity of formal results; it does not replace a determination concerning the object type and property attribution of the source statement itself.

If the original-source text states within the same explanatory scope that the temporal-property expression is figurative, does not designate a literal deformation of time, and has a specified actual referent, the number of evidence-effect transgressions produced by that explicit figure is 0. Expressions such as “time itself slows down” or “time itself speeds up” that do not state these points transfer a comparative difference on a physical carrier to a property of time and cross one evidence-effect boundary, even when a restricted interpretation is possible. Under a literal interpretation that places an unconstructed time entity before the comparative difference, placement before generation and reattribution to a property of time occur successively by direct dependency.

*Proof.* Apply the result-dependency propositions in the Appendix to Feynman, Leighton and Sands 1963, Feynman 1995, NIST 2025, Narlikar 2010, Tong 2013, Greene, and Taylor–Wheeler 1992. The result set for each reference includes differences in clock readings, clock rates, lifetimes, or proper times, or spacetime intervals or coordinate relations. A state space for an independent time carrier, a quantity of change, a law of evolution, a coupling law to clock relations, and an independent measurement procedure do not appear together as formal objects in those result sets.

The restricted interpretation preserves the differences in clock readings, clock rates, lifetimes, or proper times, or the spacetime intervals or coordinate relations in each reference as formal results. By Definition 7.117, the path on which the original-source text specifies both the figurative character and the actual referent has 0 evidence-effect transgressions, whereas a path containing an unqualified temporal-property expression crosses one evidence-effect boundary. The preservation of formal results and the number of evidence-effect transgressions in a source statement are therefore determined separately.  $\square$

**Representative Comparison of Time-Nonexistence and Time-Attribute Statements** Concerning the region in which quantum geometrodynamical effects become important and the boundary of the semiclassical approximation, Misner–Thorne–Wheeler 1973 states:

There is no spacetime, there is no time, there is no before, there is no after.  
The question of what happens “next” is without meaning.

The same reference also states the following concerning relativistic clock relations:

Velocity produces a universal time dilation; acceleration does not.

The original-source location of the first statement is Chapter 43, §43.1, p. 1183, and that of the second statement is Chapter 16, §16.4, p. 393. The complete quotation records and minimal contexts are given in paragraph S.2.19. The two statements belong to the same reference, but their stated domains of application differ. These two statements therefore do not constitute a direct contradiction asserting both existence denial and property attribution for the same time-content type and domain of application.

Heisenberg 1958 introduces the following statement as an argument by Augustine:

Time has been created together with the world, it belongs to the world,  
therefore time did not exist before the universe existed.

The same reference observes that “created” already requires a before and an after and concludes, as Heisenberg’s own analysis:

Therefore, it is impossible to define in rational terms what could be meant by  
the phrase “time has been created.”

The first statement is Augustine’s argument as presented by Heisenberg; the second is Heisenberg’s own conceptual analysis. The attributable speakers, complete quotation records, and minimal contexts are given in paragraph S.2.72. This pair connects, while preserving their different attributable speakers, a statement of the nonexistence of time with a criticism that the language of creating time presupposes temporal relations.

*Remark 7.175* (Comparison of the Targets of Time-Nonexistence and Time-Attribute Statements). Within one reference or across multiple references, the compatibility of a statement asserting the nonexistence of time with a statement attributing expansion, contraction, speed, flow, or deformation to time depends on the type and domain of application of the time contents targeted by each statement.

If both statements target the same time-content type and the same domain of application, and the property attribution presupposes the existence of that object, the property attribution is incompatible with the existence denial. If they target different time-content types or different domains of application, the existence denial and property attribution are not claims about the same object under the same conditions.

A shift of attribution from a difference in clock indication, clock rate, proper time, or coordinates, or a comparative difference between generated time representations, to a change in a property of “time itself,” independently of a physical carrier, requires independent evidence or a valid derivation supporting that shift.

The statement evidence also contains references that state relations as relations and references that constrain carrier language. Bunn–Hogg 2009 states:

the rate of change of the distance ... with respect to the cosmic time coordinate

The statement evidence in the Appendix is given in paragraph S.2.53.

Lewis 2016 makes the following two statements:

the concept of expanding space is useful in a particular scenario

expanding space can be extinguished, replaced with the simple Doppler shift

The statement evidence in the Appendix is given in paragraph S.2.62.  
Peacock 2008 states:

The worst of these is the ‘expanding space’ fallacy.

The statement evidence in the Appendix is given in paragraph S.2.52.

**Proposition 7.176** (Explicit Formal Scope in Boundary Contrasts). The statement in Bunn–Hogg 2009 concerns the rate of change of distance with respect to the cosmic-time coordinate. The statements in Lewis 2016 and Peacock 2008 concern the domain of application or fallacious character of the explanation of expanding space. These statements do not add to the formal results the expansion of a spatial carrier independent of distance, coordinates, redshift, and Doppler relations.

*Proof.* Apply the result-dependency propositions in the Appendix to Bunn–Hogg 2009, Lewis 2016, and Peacock 2008. The result objects in these references are distance, coordinates, redshift, or Doppler relations. An independent expanding carrier is not a component of the result sets.  $\square$

**Corollary 7.177** (Separation of the Explanatory Range of Statements from the Time-Generation Audit). For strong carrier statements, statements that make relations explicit, and statements that constrain carrier language alike, the reference-specific time-generation audit is determined by the dependency roots of the formal construction. The explanatory range of a statement does not change the establishment of a formal result or the presence or absence of primitive time roots.

*Proof.* Apply Proposition 7.174 and Proposition 7.176 in accordance with the boundary in Remark 7.161.  $\square$

### 7.3.16 Formal Evidence and Statement Evidence for Time-Bearing Quantities

In the formal descriptions of the original sources, a quantity containing a time difference, frequency, rate, lifetime, linewidth, time derivative, or temporal dimension is called a time-bearing quantity. On the basis of the reference-specific audits in the Appendix, twelve objects are treated: the speed of light, the cesium-133 transition frequency, the Planck constant, the reduced Planck constant, the gravitational constant, vacuum permittivity, vacuum permeability, the Hubble rate, decay constants and lifetimes, linewidths and decay widths, Planck time, and the Compton angular frequency. The four audit determinations of Definition 7.78 are applied separately to the formal evidence and statement evidence for each object.

**Proposition 7.178** (Generation Order of Time-Bearing Quantities). The dimension, unit, or algebraic relation of a time-bearing quantity does not generate physically the time contents used by that quantity. Before a time-bearing quantity can be used within a physical law, the time contents it requires and their physical-time qualification must first be established.

*Proof.* The time differences, frequencies, rates, lifetimes, linewidths, or time derivatives used by a time-bearing quantity require structures for time-value identity, before–after order, intervals, scales, clock correspondence, or observation–record correspondence. By Definition 7.2, the structures required for these operations must be generated from non-temporal physical inputs before use. If required time contents are supplied to a law operation or result derivation before their generation, they constitute primitive time input under Definition 7.10. By Theorem 7.11, those time contents do not satisfy physical-time qualification. The generation of required time contents therefore cannot be inferred backward from the dimension, unit, or algebraic relation of a time-bearing quantity.  $\square$

*Remark 7.179* (Post-Generation Use of Time-Bearing Quantities). When time contents and their physical-time qualification have been generated and the establishment conditions specific to each theory are satisfied, existing formal relations containing time-bearing quantities can be used in their post-generation domains of application. The physical relations of emission, propagation, reception, recording, atomic transitions, oscillators, particle decay, electromagnetic fields, gravitational fields, resonance lines, and measuring instruments are preserved as theory-specific inputs to the LCTR generation and audit construction. The generation-order audit does not alter these physical relations; it determines the grounds and order of application under which time-dependent expressions may be used within laws.

## Generation Order and Domain of Application of Light-Speed Expressions

Einstein 1905 states the following concerning the definition of the speed of light:

Hierbei ist

$$\text{Geschwindigkeit} = \frac{\text{Lichtweg}}{\text{Zeitdauer}},$$

wobei „Zeitdauer“ im Sinne der Definition des § 1 aufzufassen ist.

That is, velocity is the light path divided by the time interval, where “time interval” is to be understood in the sense of the definition in Section 1.

Einstein 1916 states the following concerning descriptions in accelerated coordinates:

Ebenso sieht man unmittelbar, daß das Prinzip von der Konstanz der Vakuum-Lichtgeschwindigkeit eine Modifikation erfahren muß.

Likewise, one sees immediately that the principle of the constancy of the velocity of light in vacuum must undergo a modification.

Misner–Thorne–Wheeler 1973 states the following concerning the relation between a time standard and the displayed speed of light:

No standard of time that varies so much from one month to another is acceptable. If adopted, it would make the speed of light vary from month to month!

NASA/JPL–Caltech 2020 states the light-speed limit as follows:

nothing can travel faster than the speed of light

The complete quotation records, original-source locations, minimal contexts, and interpretation audits are given in paragraph S.2.3, paragraph S.2.5, paragraph S.2.19, and paragraph S.2.66.

**Proposition 7.180** (Generation Order of Light-Speed Expressions and the Domain of Application of Constancy). The physical relations of the emission, propagation, and reception of light are preserved as theory-specific inputs to the LCTR generation and audit construction. Because the numerical expression for the speed of light uses a spatial difference, time difference, clock, length standard, reference frame, and measurement procedure, the physical-time qualification of that time difference must be generated before the velocity expression is used within a law.

The constancy of the vacuum speed of light  $c$  measured in a local inertial frame, and the local upper propagation limit for matter, signals, or causal influences, are preserved in the domain in which the corresponding local structure, clock and length standards, and measurement conditions hold. This establishment is not an unconditional universal proposition concerning an arbitrary coordinate speed of light, cosmological recession velocity, phase velocity, or displayed quantity that carries no information.

If clocks or other physical carriers and comparison constructions differ, and different generated time representations consequently result, identical numerical invariance across those representations does not follow automatically. Numerical invariance between representations depends on the coordinate transformation, metric, clock and length standards, and domain of application. Where these conditions hold, existing special-relativistic light-speed relations can be used with the post-generation time representations.

*Proof.* Einstein 1905 gives the speed of light as the ratio of a light path to a defined time interval. Einstein 1916 distinguishes within the same reference the postulate of the constancy of the vacuum speed of light from the modification of that principle in a description using accelerated coordinates. Misner–Thorne–Wheeler 1973 states that the adopted time standard propagates into the displayed speed of light. By paragraph S.2.66, the limiting statement in NASA/JPL–Caltech 2020 has effect for the vacuum speed of light in a local inertial frame and for the local propagation limit of matter, signals, or causal influences.

By Proposition 7.178, the algebraic relation for velocity does not generate the time contents used by that velocity. By Definition 7.115, progressing from local constancy or a local propagation limit to a universal proposition having a different object, reference frame, type of velocity, or domain of application requires independent evidence or a valid derivation supporting that transition. □

**Proposition 7.181** (Time-Bearing Quantities and Primitive Time Roots in Five References). In Curtiss 1949, Headley–Clayton–Reed 1996, Clark 2001, Wolf et al. 2011, and the PDG 2023 resonance reference, all reference-specific time contents are primitive time roots, and all reference-specific results that presuppose time contents are results dependent on primitive time roots.

*Proof.* For Curtiss 1949, apply Proposition S.68 and Proposition S.68. For Headley–Clayton–Reed 1996, apply Proposition S.340 and Proposition S.340. For Clark 2001, apply Proposition S.409 and Proposition S.409. For Wolf et al. 2011, apply Proposition S.581 and Proposition S.581. For the PDG 2023 resonance reference, apply Proposition S.672 and Proposition S.672. □

Curtiss 1949 uses decay constants, half-lives, and mean lives in formal relations; Headley–Clayton–Reed 1996 uses Brillouin frequency shifts, frequency widths, and acoustic-wave lifetimes; Clark 2001 uses the Planck constant, transition angular frequencies, linewidths, and lifetimes; and the PDG 2023 resonance reference uses resonance widths and lifetimes. Wolf et al. 2011 distinguishes an interpretation under examination that uses the Compton angular frequency from the conclusion of the same reference that the system is not a clock possessing the Compton frequency as a physical signal.

In accordance with Proposition 7.174, interpretations that permit attribution of properties to time itself are compared conditionally with interpretations that state differences in clock readings, rates, lifetimes, frequencies, or linewidths.

**Proposition 7.182** (Result of the Dual-Track Audit of Twelve Time-Bearing Objects). The twelve objects, the two tracks of formal evidence and statement evidence, and the four audit determinations give

$$12 \times 2 \times 4 = 96$$

determination fields. The evidence included in the Appendix directly establishes 23 fields on the formal track and 0 fields on the statement track. The breakdown by determination is

| Audit determination                          | Formal track | Statement track |
|----------------------------------------------|--------------|-----------------|
| Pre-generation placement                     | 9            | 0               |
| Omission of generation stages                | 9            | 0               |
| Reattribution of a time property             | 0            | 0               |
| Reverse presupposition of a derived quantity | 5            | 0               |
| Total                                        | 23           | 0               |

The evidence included in the Appendix does not establish the corresponding determination for the remaining 73 fields.

*Proof.* Pre-generation placement holds on the formal track for the speed of light, the cesium-133 transition frequency, the Planck constant, the reduced Planck constant, the Hubble rate, decay constants and lifetimes, linewidths and decay widths, Planck time, and the Compton angular frequency. Omission of generation stages also holds on the formal track for these nine objects. Reverse presupposition of a derived quantity holds on the formal track for the speed of light, the cesium-133 transition frequency, the Hubble rate, decay constants and lifetimes, and linewidths and decay widths. Formal evidence directly establishing reattribution of a time property does not occur on the formal track for any of the twelve objects.

In Clark 2001, the Planck constant, time content 1 in Definition S.405, enters transfer 1 together with an energy difference between states to yield a transition angular frequency. In Wolf et al. 2011, the reduced Planck constant, time content 28 in Definition S.577, enters transfers 20 and 21 together with internal-level differences to yield transition angular frequencies. These transfers connect physical state differences to time-bearing expressions without a stage generating physical-time qualification for the required time contents. Omission of generation stages therefore also holds for these paths using the two constants as inputs.

For the gravitational constant, vacuum permittivity, and vacuum permeability, the included unit-conversion and Maxwell relations do not supply the individual generation dependencies required to establish the four determinations for the respective constants.

The statements included in the Appendix give, with their original texts, the name, formal relation, interpretation, or object of criticism for each object. For pre-generation placement and omission of generation stages, the included quotations do not determine the dependency order between use of the quantity and physical generation of the required time content. For reattribution of a time property, the quotations concerning the twelve objects do not attribute differences in relations between carriers to state or property changes of an independent time object. For reverse presupposition of a derived quantity, the included quotations do not establish a dependency that returns the quantity to a premise generating its required time content or physical-time qualification. The determinations are therefore not established on the statement track. The contrast in Wolf et al. 2011 between the reading of a “clock that runs at the Compton frequency” and the same reference’s statements that it is “not a real clock” and “does not deliver a physical signal” gives a conditional relation between the literal clock interpretation and omission of generation stages. The contrast does not supply the required dependency order and therefore does not establish the determination in that field.

Adding these object counts by determination gives 23 fields on the formal track and 0 fields on the statement track. The number of fields for which establishment is not obtained is therefore

$$96 - 23 = 73$$

□

### 7.3.17 Evidentiary Stages of Clocks, Time Attributes, and Biological Application

**Definition 7.183** (Evidence-Stage Audit Including Biological Application). When an application from clock relations to biological processes is audited, the following three stages are distinguished.

1. Comparative differences directly measured in experiments on clocks, oscillators, frequencies, particle lifetimes, or other physical carriers.
2. An interpretation that attributes the comparative differences in the preceding item to time itself slowing, accelerating, expanding, contracting, or acting independently of physical carriers.
3. Application to pulse rates, thought, growth, aging, disease progression, lifetimes, or other biological processes.

The evidence at each stage is distinguished by its object, predicate, and evidence type. The evidence types are direct measurement, formal generalization, theoretical prediction, and source statement. When a source statement permits multiple interpretations, its correspondence to the second or third stage is determined for each interpretation.

**Representative Statements of Biological Application** In the same paragraph in which Feynman, Leighton, and Sands (1963) move from rate differences in moving clocks to an expression attributing a property to time itself, the objects of application are extended as follows:

the man’s pulse rate, his thought processes

The same paragraph further identifies the following object:

the time it takes for a cancer to develop

The original-source location is Feynman, Leighton and Sands, *The Feynman Lectures on Physics*, Volume I, Chapter 15, §15-4, *Transformation of time*. The minimal context is the paragraph that generalizes the light-clock rate difference to all clocks, states with a qualification that time itself appears to be slower, and then applies the account to pulse rate, thought, growth, aging, and the development of cancer. The complete quotation facts and interpretation audit for this reference are given in paragraph S.2.12 and paragraph S.2.12.

Additional statements belonging to the same type of biological application are as follows.

**Feynman 1995** Our theory is then not inconsistent, and it suggests that the frequency shift required by energy conservation be considered as a general property of all physical processes, that they run more slowly in lower gravitational potentials.

Immediately after this generalization, the same reference states:

The man on top of a mountain is living and aging at a faster rate than we.

The original-source location is *Feynman Lectures on Gravitation*, Lecture 5, §5.2, printed p. 69. The generalization from frequency shift to all physical processes and the application to human life and aging occur in the same dependency sequence. The complete reference-specific record is given in paragraph S.2.70.

**Misner–Thorne–Wheeler 1973** aging, like atomic processes,  
is tied to proper time

The original-source location is Chapter 16, §16.4, pp. 393, 396. The complete reference-specific record is given in paragraph S.2.19.

**NIST 2025** you would now be older than your own twin!

The original-source location is the third paragraph of the opening relativity overview in *Putting Einstein to the Test*; its objects are twins reunited after one has traveled in a high-speed spacecraft or near a black hole. The complete reference-specific record is given in paragraph S.2.69.

**Rindler 1977** the time dilation argument applies to any temporal change or process and therefore also to muon decay and even human life times

The original-source location is the paragraph immediately before Section 2.14 on p. 45. The complete reference-specific record is given in paragraph S.2.23.

**Taylor–Wheeler 1992** our biological clock, by which we age, and all other good clocks carried along will tick away in concord with it

The original-source location is Chapter 4, §4.5, printed p. 124. The complete reference-specific record is given in paragraph S.2.13.



**Tong 2013** the same argument holds for any process, be it clocks, elementary particles or human hearts

The original-source location is *Dynamics and Relativity*, §7.2.3, p. 119. The complete reference-specific record is given in paragraph S.2.61.

These are evidence of theoretical application to biological processes or explanatory statements. An experiment whose object is a clock, electromagnetic radiation, or a particle lifetime is not evidence that the biological process itself was directly measured.

**Proposition 7.184** (Necessary Conditions for Direct Demonstration of Biological Application). For direct demonstration to hold at the third stage of Definition 7.183, the biological process itself must be included among the experimental objects, and an experimental arrangement that isolates the relativistic conditions, a measurement procedure for the biological process, and corresponding measurement results must be supplied.

If, on the premise of direct measurement of a first-stage physical carrier, a consequence concerning the third stage is derived by formal generalization, that consequence is a theoretical prediction. The domain of direct demonstration at the third stage is determined by the measurement procedure and results whose experimental object is a biological process.

*Proof.* By Definition 7.71, the reference-specific audit separates the formal and statement branches and preserves the original text, translation, original-source location, and minimal context of statement evidence. By Definition 7.78, formal evidence is determined from definitions, equations, and dependencies, whereas statement evidence is compared conditionally for each interpretive possibility.

The object domain of direct demonstration is determined by the objects included in the experimental arrangement and the measurement procedure and results obtained for those objects. A formal generalization yields a theoretical consequence concerning the object of application, but does not generate a measurement procedure and result for an object not included in the experimental arrangement. Direct demonstration at the third stage therefore requires the three conditions stated in the proposition.  $\square$

**Proposition 7.185** (Separation of Evidence Types for Clocks, Time Attributes, and Biological Application). The objects directly measured in Bailey et al. 1977 are the decay lifetime and decay-electron counts of moving muons. The objects directly measured in Hafele–Keating 1972b are differences in cesium-clock indications, clock rates, and post-reunion display deviations. The objects directly measured in Pound–Rebka 1960 are relative frequency differences of electromagnetic radiation, resonant-transmission counts, and calibration quantities. Hafele–Keating 1972a gives predictions of rate relations and indication differences between airborne and reference clocks.

The statements preserved from Feynman, Leighton and Sands 1963, Feynman 1995, Misner–Thorne–Wheeler 1973, NIST 2025, Rindler 1977, Taylor–Wheeler 1992, and Tong 2013 extend the objects of application to pulse rates, thought, cancer development, human aging, the ages of twins, human lifetimes, biological clocks, or human hearts.

The preceding direct measurements, attribution of properties to time itself, and application to biological processes belong to different stages and evidence types under Definition 7.183.

*Proof.* The measured objects and results in Bailey et al. 1977, Hafele–Keating 1972a/1972b, and Pound–Rebka 1960 are determined by Proposition S.197, Proposition S.157, Proposition S.148, and Proposition S.86 in the Appendix.

The verbatim statements containing application to biological processes and their original-source locations are preserved in paragraph S.2.12, paragraph S.2.70, paragraph S.2.19, paragraph S.2.69, paragraph S.2.23, paragraph S.2.13, and paragraph S.2.61 in the Appendix.

Proposition 7.167 and Proposition 7.174 separate formal results that hold for physical carriers from attributions of properties to time itself. Applying these propositions to Definition 7.183 gives the separation of the three stages and evidence types stated in the proposition.  $\square$

### 7.3.18 Connections from All Source Statements to the Representative Arguments

In addition to the references treated directly in the representative arguments of this section, the original-source statements in the remaining references are connected below according to the point supported by each statement. Each item distinguishes the content stated directly, the permitted interpretive possibilities, and the boundary of evidentiary force. At the end of each item, “Statement” identifies the location of the original text, English translation when required, original-source location, and minimal context; “Audit” identifies the location of the conditional comparison between the interpretive possibilities and the formal evidence.

**Operational Critique and Stated Deficiencies of Construction** This group connects to the representative argument in Subsection 7.3.1.

**Reference 6: Bridgman 1927.** This reference presents the view that physical concepts should be specified by actual operations rather than their properties. It also states that, at the time of writing, the basic operations of time measurement have not undergone the critical examination that seems to be required, and that the specifications for clock construction needed to answer rigorously whether local time also involves space measurement have not been supplied. The scope of the operational critique is compared separately from the presence or absence of a construction that completes time generation. (Statement: paragraph S.2.6; Audit: paragraph S.2.6).

**Reference 15: Ehlers–Pirani–Schild 1972.** In the EPS route itself, measurement operations are derived from light rays and freely falling particles, and clocks are not basic tools for constructing spacetime geometry. The quotation concerning Synge’s preceding method instead reports the equality claim of the chronometric axiom conditional on interpreting a clock as an atomic clock. This is distinct from the additional empirical postulate after the EPS reconstruction. The EPS construction results concern spacetime structure and measurement operations. Additional claims of removing all temporal inputs or completing physical-time generation each require independent grounds for generation. (Statement: paragraph S.2.15; Audit: paragraph S.2.15).

**Synchronization, Coordinate Time, and Proper Time** This group connects to the representative argument in Subsection 7.3.2.

**Reference 3: Einstein 1905.** This reference defines simultaneity and time by clock indications and uses the speed of light as the ratio of a light path to a defined time interval. This operational construction of coordinate time presupposes light signals, clock indications, synchronization relations, and time intervals. The grounds for establishing

these inputs do not follow solely from the construction of coordinate time using them. (Statement: paragraph S.2.3; Audit: paragraph S.2.3).

**Reference 4: Minkowski 1908/1909.** This reference combines space and time into a four-dimensional world and describes worldlines and cone relations through the emission and reception of light. A reading as geometrical unification is distinguished from a reading as ontological unification of space and time, and the force of the quotation is limited to the four-dimensional relations and light-signal relations. (Statement: paragraph S.2.4; Audit: paragraph S.2.4).

**Reference 5: Einstein 1916.** This reference states the postulate of constant light speed in vacuum and its modification in accelerated coordinates. It states that a time meeting physical requirements cannot be introduced by identically constructed clocks at rest in the rotating coordinate system. It also describes the rate relation between clocks at the center and on the circumference of the disk. A reading in terms of relations between coordinate light speed and clock rate is separated from a reading that permits attribution of properties to time itself. (Statement: paragraph S.2.5; Audit: paragraph S.2.5).

**Metrological Standards, Clock Differences, and Direct Measurement** This group connects to the representative argument in Subsection 7.3.3.

**Reference 10: Pound–Rebka 1960.** This reference records a resonance line, a frequency difference, temperature correction, and agreement between the measured shift and the predicted value; it also states that the attribution of the frequency difference is not unique. Its evidentiary force lies within the measurement domain of electromagnetic radiation and detector counts and is distinguished from measurement of a property of time itself. (Statement: paragraph S.2.10; Audit: paragraph S.2.10).

**Reference 16: Hafele–Keating 1972b.** This reference states measured values from macroscopic clock comparison, their agreement with predicted values, uncertainty in clock-rate changes, and statistical qualifications. What is directly supported is the difference in clock indications and its uncertainty; attribution to a change in a time entity requires a separate ground. (Statement: paragraph S.2.16; Audit: paragraph S.2.16).

**Reference 17: Hafele–Keating 1972a.** This reference predicts direction-dependent effects on portable cesium clocks and takes the difference between the indications of clocks reunited at the same location as the physical observable. Formal generalization of the predicted clock difference is distinguished from attribution of a property to time itself. (Statement: paragraph S.2.17; Audit: paragraph S.2.17).

**Reference 22: Bailey et al. 1977.** This reference explains time dilation as the retardation of a moving clock and shows agreement between measurements of the muon lifetime and relativistic predictions. The directly measured particle lifetime is distinguished from explanatory extension to clocks in general or to time itself. In the route that estimates the negative-muon lifetime at rest by taking special relativity as an antecedent, a test of CPT invariance using that estimate depends on special relativity, and the estimate does not provide an independent test of special relativity. (Statement: paragraph S.2.22; Audit: paragraph S.2.22).

**Reference 64: BIPM SI 2019/2026.** This reference treats the second as the SI unit of time and proper time and describes TAI as a coordinate-time scale. The domains of establishment of the metrological standard and coordinate-time operation are separated from the physical generation of time contents themselves. (Statement: paragraph S.2.64; Audit: paragraph S.2.64).

**Quantum Measurement and Decoherence** This group connects to the representative argument in Subsection 7.3.4.

**Reference 41: Philip W. Anderson 2001.** This reference treats an account according to which decoherence, the description of collapse, and experimental verification solve the quantum measurement problem. The formal results supported by environmental interaction are separated from the extent to which they support the establishment of a unique measurement outcome or of time evolution. (Statement: paragraph S.2.41; Audit: paragraph S.2.41).

**Canonical Gravity, Internal Time, and the Problem of Time** This group connects to the representative argument in Subsection 7.3.6.

**Reference 9: Dirac 1958.** This reference takes the state at a given time as a basic concept of Hamiltonian form and discusses Hamiltonian formulation prior to quantization. The loss of four-coordinate symmetry under the adopted simplification is distinguished from the author's tentative inference about the physical status of four-dimensional symmetry. The selection of a time coordinate in the formalism is distinguished from the generation or physical qualification of time. (Statement: paragraph S.2.9; Audit: paragraph S.2.9).

**Reference 11: ADM 1962.** This reference distinguishes a preferred time coordinate already selected in a particle-mechanics example from an arbitrary function left by parametrized dynamics. The indeterminacy of lapse and shift in gravitation is stated in §3.4; the classical qualification that no single preferred coordinate frame exists in the interior is stated in §8.1, footnote 25. Coordinate freedom in the spacetime decomposition and the physical ground of the adopted time variable are compared as distinct obligations. (Statement: paragraph S.2.11; Audit: paragraph S.2.11).

**Reference 14: DeWitt 1967.** This reference treats internal time as a useful phenomenological concept under specified conditions and discusses the absence of a universal clock and limits on clock precision. The presentation of conditions for using an internal variable as time is distinguished from generating complete physical time contents from non-temporal inputs. (Statement: paragraph S.2.14; Audit: paragraph S.2.14).

**Reference 25: Hartle–Hawking 1983.** This reference states that a closed universe has no independent standard of time, that proper time depends on the four-dimensional geometries summed over rather than on boundary data, and that the classical concept of time fails in the joining region. Elimination of an external time standard is distinguished from physical generation of time contents. (Statement: paragraph S.2.25; Audit: paragraph S.2.25).

**Reference 29: Isham 1993.** This reference discusses the adequacy of definitions of time using gravitational variables alone, physical clocks made of matter, WKB time, and the Schrödinger form lost at an order of approximation. The conditions for constructing clock candidates and approximate time are separated from complete physical-time qualification. (Statement: paragraph S.2.29; Audit: paragraph S.2.29).

**Reference 30: Kuchař 1992.** This reference discusses time variables in classical gravity, construction of time from matter fields, the danger of generalizing model results to quantum gravity as a whole, and the lack of a successful resolution of the problem of time. Local construction of a candidate time is not identified with completed time generation for the theory as a whole. (Statement: paragraph S.2.30; Audit: paragraph S.2.30).

**Arrows of Time and Causal Order** This group connects to the representative argument in Subsection 7.3.7.

**Reference 24: Penrose 1979.** This reference contrasts the time symmetry of local laws with macroscopic temporal asymmetry and connects singularity structure and a low-entropy boundary condition to the arrow of time. The explanation of orientation conditions and the generation of the time contents being oriented are treated as distinct obligations. (Statement: paragraph S.2.24; Audit: paragraph S.2.24).

**Reference 27: Bombelli–Lee–Meyer–Sorkin 1987.** This reference identifies spacetime at the smallest scale with a causal set and correlates its partial order with macroscopic past–future relations. Construction of causal order is distinguished from complete time generation, including clock scales, intervals, and law time. (Statement: paragraph S.2.27; Audit: paragraph S.2.27).

**Reference 31: Lebowitz 1993.** This reference connects the macroscopic direction of time to a low-entropy initial state and temporal ordering and denies the need for an alternative explanation. The explanation of thermodynamic asymmetry is distinguished from the generation of the before–after order and time variable used in that explanation. (Statement: paragraph S.2.31; Audit: paragraph S.2.31).

**Black-Hole Thermodynamics, Radiation, and Information** This group connects to the representative argument in Subsection 7.3.8.

**Reference 18: Bekenstein 1973.** This reference gives black-hole change a direction in time, initially restricts the thermodynamic correspondence to a formal analogy, and presents the generalized second law as a conjecture. The evidence types of temporal direction, formal analogy, and conjecture are not conflated. (Statement: paragraph S.2.18; Audit: paragraph S.2.18).

**Reference 20: Hawking 1975.** This reference states the nonuniqueness of the particle concept in curved spacetime, positive frequency relative to a specified time, the Planck time, a heuristic pair-creation picture, and the domain in which the semiclassical approximation fails. The establishment of the radiation result is separated from responsibility for establishing the background time, frequency decomposition, and approximation boundary. (Statement: paragraph S.2.20; Audit: paragraph S.2.20).

**Reference 21: Hawking 1976.** This reference states that the known laws are formulated on a classical spacetime background and uses Fourier components with respect to time and a hypersurface near the Planck time. The consequences concerning the information problem depend on this formulation using a background spacetime and a time variable. (Statement: paragraph S.2.21; Audit: paragraph S.2.21).

**Reference 32: Page 1993.** This reference estimates the appearance of information in black-hole radiation using an S-matrix and a model of random pure states and states explicitly that the result is a first conjecture lacking a complete quantum analysis. This model-dependent estimate presupposes the S-matrix and the time-evolution description of the radiation process. (Statement: paragraph S.2.32; Audit: paragraph S.2.32).

**Duality, Entanglement, and Spacetime Emergence** This group connects to the representative argument in Subsection 7.3.9.

**Reference 37: Maldacena 1998.** This reference presents the duality between AdS and conformal field theory as a conjecture and makes explicit the definition of the theories,

the boundary conditions, and unresolved problems. The success of dual calculations, the complete definition of the theories, and qualification of the time variable are treated separately. (Statement: paragraph S.2.37; Audit: paragraph S.2.37).

**Reference 49: Ryu–Takayanagi 2006.** This reference proposes a geometrical area law for quantum entanglement entropy on the basis of AdS/CFT and states the difficulty of quantitative comparison in higher dimensions. Results obtained from static minimal surfaces are distinguished from a mechanism that generates spacetime or time. (Statement: paragraph S.2.49; Audit: paragraph S.2.49).

**Reference 56: Van Raamsdonk 2010.** This reference connects quantum entanglement with the emergence of classically connected spacetime while stating that a deep understanding of why and how spacetime and gravity emerge from field-theoretical degrees of freedom has not been attained. Evidence for the correspondence is separated from completion of a generation mechanism. (Statement: paragraph S.2.56; Audit: paragraph S.2.56).

**Cosmology, Expansion, and Observables** This group connects to the representative argument in Subsection 7.3.10.

**Reference 28: Weinberg 1989.** This reference presents the discrepancy of approximately 120 orders of magnitude between the theoretical prediction and observational upper bound for the cosmological constant as a problem and compares several approaches to its resolution. The quotation supports the scale of the problem and the assessment at that time; it is separate from a generation route for the time-dependent quantities used to describe the problem. (Statement: paragraph S.2.28; Audit: paragraph S.2.28).

**Reference 45: Davis–Lineweaver 2004.** This reference uses redshift as a parameter of time and rewrites cosmological equations in terms of observables. Re-expression in terms of observables is distinguished from generating time contents or cosmic history from redshift. (Statement: paragraph S.2.45; Audit: paragraph S.2.45).

**Reference 48: Grøn–Elgarøy 2007.** This reference describes expanding space as real and connects synchronization of comoving clocks with the measurement of cosmic time. A reading as a construction of cosmic time from the metric and comoving clocks is separated from a reading as expansion of an independent spatial carrier. (Statement: paragraph S.2.48; Audit: paragraph S.2.48).

**Reference 50: Francis–Barnes–James–Lewis 2007.** This reference states the synchronization of clocks belonging to privileged observers, an account according to which space appears between observers, differing starting times, and the rejection of an account according to which galaxies resist expansion. A reading in terms of clock and coordinate relations is separated from a reading in terms of the appearance of a spatial carrier. (Statement: paragraph S.2.50; Audit: paragraph S.2.50).

**Reference 51: Cook–Burns 2009.** This reference identifies the time coordinate of cosmology with the proper time of clocks moving with the galaxies. The force of coordinate specification by comoving clocks is distinguished from the obligations of generating and gluing a global law time. (Statement: paragraph S.2.51; Audit: paragraph S.2.51).

**Definitions of Time and Primitive Time** This group connects to the representative argument in Subsection 7.3.12.

**Reference 1: Aristotle.** This reference defines time as the number of change in respect of before and after and distinguishes the object counted from the means of counting. This

can be read as a definitional relation of time or as a relation between discrimination of before and after and temporal judgment; what the quotation directly supports is the distinction and relation themselves. (Statement: paragraph S.2.1; Audit: paragraph S.2.1).

**Adjacent Problems of Time and Temporal Flow** This group connects to the representative argument in Subsection 7.3.14.

**Reference 60: Ellis 2012.** This reference attributes existence and flow to time, extends the time parameter supplied by clocks from physics to biology, and also refers to the emergence of effective time from a timeless substrate. Proper time, clock rate, temporal flow, biological application, and emergence are separated as distinct evidentiary stages. (Statement: paragraph S.2.60; Audit: paragraph S.2.60).

**Reference 72: Heisenberg 1958.** This reference introduces, as Augustine’s argument, the nonexistence of time before the universe and then analyzes the phrase “time was created” as incapable of rational definition because it presupposes a before–after relation. The object being introduced is separated from Heisenberg’s own analysis, and the self-dependence of creation-of-time vocabulary is compared as an adjacent problem of time. (Statement: paragraph S.2.72; Audit: paragraph S.2.72).

**Time-Bearing Quantities Such as Lifetime, Frequency, and Linewidth** This group connects to the representative argument in Subsection 7.3.16.

**Reference 8: Curtiss 1949.** This reference names the exponential decay law, decay constant, half-life, and mean lifetime. These quotations identify the names of the law and quantities; the relations among the quantities are given in the surrounding equations and derivations. Those equations and derivations use a unit of time and elapsed time as inputs. (Statement: paragraph S.2.8; Audit: paragraph S.2.8).

**Reference 36: Headley–Clayton–Reed 1996.** This reference relates frequency difference, Lorentzian shape, full width at half maximum, acoustic-wave lifetime, and reduction in gain. The force of the statement lies in the physical relations among frequency, linewidth, and lifetime and in their operating domain; it does not include the generation of the time scales they use. (Statement: paragraph S.2.36; Audit: paragraph S.2.36).

**Reference 42: Clark 2001.** This reference identifies the rate constant with the reciprocal of lifetime and relates spectral width to lifetime, while acknowledging significant deviations from the simplified form. The conditions under which the lifetime–width relation holds are distinguished from generation of the time scale used in that relation. (Statement: paragraph S.2.42; Audit: paragraph S.2.42).

**Reference 58: Wolf et al. 2011.** Wolf et al. introduce the account by Müller, Peters, and Chu that treats the atom as a whole as a clock operating at the Compton frequency, and criticize this clock interpretation because no readable physical signal at the Compton frequency is available. Using the de Broglie relation, flight time, and Bragg condition, the leading-order neutron-interferometer phase in the same reference is expressed in terms of the lattice wave vector, gravitational acceleration, and flight time; with these quantities and the configuration fixed, that expression contains no explicit Compton frequency. (Statement: paragraph S.2.58; Audit: paragraph S.2.58).

**Reference 68: PDG 2023 Resonances.** This reference relates the pole width of a resonance to its lifetime when the relevant thresholds lie sufficiently below the resonance position, while stating that broad resonances have no direct relation between the usual width or lifetime and the pole parameters. The conditions of application of the width–lifetime

relation are distinguished from generation of the time scale. (Statement: paragraph S.2.68; Audit: paragraph S.2.68).

**Clock Relations, Time Attributes, and Biological Application** This group connects to the representative argument in Subsection 7.3.17.

**Reference 66: NASA/JPL–Caltech 2020.** This reference states that the faster one travels, the slower the time one experiences, and gives the speed of light as an upper limit. A reading in terms of clock, navigation, and signal relations; attribution to temporal experience; and possible extension to life or subjectivity are distinguished. (Statement: paragraph S.2.66; Audit: paragraph S.2.66).

**Reference 67: Pössel 2021.** This reference generalizes from the rate of indication of a moving light clock to all clocks and all processes aboard a space station. The stage at which the predicate remains attached to the physical carrier constituted by the light clock is separated from the theoretical stage of application to processes in general. (Statement: paragraph S.2.67; Audit: paragraph S.2.67).

**Reference 70: Feynman 1995.** This reference generalizes frequency shift to all physical processes and applies it to human life and aging; in another section, it describes comparison between upper and lower clocks and signals as a difference between time scales and the flow of time. Frequency and clock-rate relations, generalization to all physical processes, biological application, and attribution of time properties differ in their objects and inferential roles. (Statement: paragraph S.2.70; Audit: paragraph S.2.70).

**Reference 71: Narlikar 2010.** After explaining the signal from the surface of a collapsing object to an external observer, the waiting time for reception, redshift, and dimming, this reference states that time flows at different rates for the two subjects. A reading in terms of signal and wavelength relations is separated from a reading in terms of a property of temporal flow independent of physical carriers. (Statement: paragraph S.2.71; Audit: paragraph S.2.71).

The formal and statement evidence for the 72 references in Table 16 is linked to an audit method or a representative argument in Chapter 7 by the preceding correspondences and representative arguments. In this correspondence, agreement, lack of support, tension, and boundaries of evidentiary force between source statements and formal descriptions are determined for each interpretation permitted by the original text and minimal context.

## 7.4 Cross-Audit Results for the 72 References

The reference-specific complete audits yield cross-reference results for primitive time roots, results dependent on primitive time roots, results beyond the audit-reachable domain, and time-generation obligations.

The finite audited reference set of 72 references in Definition 7.1 is the quantification domain of the cross-reference audit.

Each row of Table 16 associates one reference  $R$  in the finite audited reference set of that definition with its primitive time-root set  $\text{TimeContent}_R^{\text{audit,root}}$  and with the supplementary propositions proving primitive time roots and result dependence. The evidence for the cross-reference propositions consists of these 72 rows and the supplementary propositions cited by each row.



**Proposition 7.186** (Universal Persistence of Primitive Time Roots in the 72 Audited References).

$$\text{Ref}^{\text{audit,root}} = \text{Ref}^{\text{audit}}.$$

Moreover, for every  $R \in \text{Ref}^{\text{audit}}$ ,

$$\text{TimeContent}_R^{\text{audit,gen}} = \emptyset, \quad \text{TimeContent}_R^{\text{audit,root}} = \text{TimeContent}_R \neq \emptyset$$

hold.

*Proof.* Definition 7.52 gives the following inclusion.

$$\text{Ref}^{\text{audit,root}} \subseteq \text{Ref}^{\text{audit}}$$

For the reverse inclusion, Table 16 contains one row for each of the 72 references in  $\text{Ref}^{\text{audit}}$ . The first evidentiary proposition in each row gives, for the corresponding reference  $R$ ,  $\text{TimeContent}_R^{\text{audit,gen}} = \emptyset$  and  $\text{TimeContent}_R^{\text{audit,root}} = \text{TimeContent}_R \neq \emptyset$ . Therefore, by Definition 7.52,  $R \in \text{Ref}^{\text{audit,root}}$ . Since this holds for every reference, the reverse inclusion follows.  $\square$

**Proposition 7.187** (Universal Persistence of Results Dependent on Primitive Time Roots in the 72 Audited References).

$$\text{Ref}^{\text{audit,rootdep}} = \text{Ref}^{\text{audit}}.$$

Consequently, for every  $R \in \text{Ref}^{\text{audit}}$ ,

$$\text{Result}_R^{\text{audit,rootdep}} \neq \emptyset$$

holds.

*Proof.* Definition 7.52 gives the following inclusion.

$$\text{Ref}^{\text{audit,rootdep}} \subseteq \text{Ref}^{\text{audit}}$$

The second evidentiary proposition attached to each row of Table 16 gives, for the corresponding reference  $R$ ,  $\text{Result}_R^{\text{audit,rootdep}} \neq \emptyset$ . Therefore,  $R \in \text{Ref}^{\text{audit,rootdep}}$ . This determination holds for all 72 rows, giving the reverse inclusion.  $\square$

**Proposition 7.188** (Decomposition of Results Beyond the Audit-Reachable Domain in the 72 Audited References).

$$\begin{aligned} \text{Ref}^{\text{audit,unreach}} &= \text{Ref}^{\text{audit,rootdep}} \\ &= \text{Ref}^{\text{audit}}, \end{aligned}$$

and

$$\text{Ref}^{\text{audit,rootunreach}} = \emptyset$$

hold.

Furthermore, for every  $R \in \text{Ref}^{\text{audit}}$ ,

$$\begin{aligned} \text{Result}_R^{\text{audit,unreach}} &= \text{Result}_R^{\text{audit,evidonly}} \cup \text{Result}_R^{\text{audit,rootdep}} \\ &\neq \emptyset, \end{aligned}$$

and

$$\text{Result}_R^{\text{audit,rootunreach}} = \emptyset$$

hold.

*Proof.* The source-specific result-dependence propositions in Table 16 give, for each  $R \in \text{Ref}^{\text{audit}}$ ,

$$\text{Result}_R^{\text{audit,rootunreach}} = \emptyset.$$

Substituting this equality into the four-way partition in Proposition 7.49 gives

$$\text{Result}_R^{\text{audit,unreach}} = \text{Result}_R^{\text{audit,evidonly}} \dot{\cup} \text{Result}_R^{\text{audit,rootdep}}.$$

By Proposition 7.187, the set of primitive-time-root-dependent results is nonempty for every reference. The set of results beyond generation reach, which contains it, is therefore also nonempty. Applying Definition 7.52 gives

$$\text{Ref}^{\text{audit,unreach}} = \text{Ref}^{\text{audit,rootdep}} = \text{Ref}^{\text{audit}}, \quad \text{Ref}^{\text{audit,rootunreach}} = \emptyset.$$

No assumption that the set of evidence-environment-reachable nongenerated results is empty is made.  $\square$

**Proposition 7.189** (Invariance of the Complete Determination under Local Reachability).

$$\left| \left\{ R \in \text{Ref}^{\text{audit}} \mid \text{TimeContent}_R^{\text{audit,evid}} \neq \emptyset \right\} \right| = 10,$$

$$\sum_{R \in \text{Ref}^{\text{audit}}} \left| \text{TimeContent}_R^{\text{audit,evid}} \right| = 57.$$

The ten references with nonempty locally reachable time-content sets are DeWitt 1967, Misner–Thorne–Wheeler 1973, Weinberg 1989, Isham 1993, Kuchař 1992, Connes–Rovelli 1994, Kiefer 2005, Kiefer 2009, Rovelli 2011, and Anderson 2012. The locally reachable time-content sets of the remaining 62 references are empty.

By contrast, for all 72 references,

$$\text{TimeContent}_R^{\text{audit,gen}} = \emptyset, \quad \text{TimeContent}_R^{\text{audit,root}} = \text{TimeContent}_R$$

hold. Local reachability in the ten references therefore does not establish complete generation from non-temporal physical inputs or remove primitive time roots under the complete determination.

*Proof.* The local-reachability column of Table 16 and the supplementary reachability determinations cited by its rows give the following counts for the ten references, in the order listed:

$$2, 4, 2, 12, 9, 3, 4, 9, 7, 5$$

Their sum is 57. The remaining 62 references have zero locally reachable time contents.

By Proposition 7.186, strict generation starting solely from non-temporal physical inputs yields an empty generated-time-content set in all 72 references, and every reference-specific time content remains a primitive time root. By Definition 7.41, local reachability adds the reference formal-evidence environment and thus has a different starting set from strict generation.  $\square$

**Theorem 7.190** (Cross-Reference Theorem on Primitive Time Roots and Result Dependence for the 72 Audited References).

$$\begin{aligned} \text{Ref}^{\text{audit,root}} &= \text{Ref}^{\text{audit,rootdep}} \\ &= \text{Ref}^{\text{audit,unreach}} \\ &= \text{Ref}^{\text{audit}}, \\ \text{Ref}^{\text{audit,rootunreach}} &= \emptyset \end{aligned}$$

hold.

Moreover, for every  $R \in \text{Ref}^{\text{audit}}$ ,

$$\begin{aligned}\text{TimeContent}_R^{\text{audit,gen}} &= \emptyset, \\ \text{TimeContent}_R^{\text{audit,root}} &= \text{TimeContent}_R \neq \emptyset, \\ \text{Result}_R^{\text{audit,rootdep}} &\neq \emptyset, \\ \text{Result}_R^{\text{audit,rootunreach}} &= \emptyset\end{aligned}$$

hold simultaneously.

Thus, none of the 72 audited references generates all the time contents it uses from non-temporal physical inputs.

*Proof.* Proposition 7.186 gives

$$\text{Ref}^{\text{audit,root}} = \text{Ref}^{\text{audit}}.$$

Proposition 7.187 gives

$$\text{Ref}^{\text{audit,rootdep}} = \text{Ref}^{\text{audit}}.$$

Proposition 7.188 gives

$$\text{Ref}^{\text{audit,unreach}} = \text{Ref}^{\text{audit}}, \quad \text{Ref}^{\text{audit,rootunreach}} = \emptyset.$$

For each reference, nonemptiness of the primitive-time-root-dependent result set follows from Proposition 7.187, and emptiness of the result set that remains unreachable after primitive time roots are adjoined follows from Proposition 7.188.

By Proposition 7.189, local reachability conditional on the evidence environment exists in ten references, but this reachability is not complete generation. By Proposition 7.186, the audited generated-time-content set is empty in all 72 references, and every reference-specific time content is a primitive time root. The final negative consequence of the theorem follows.  $\square$

**Proposition 7.191** (Counts of Time-Generation Obligations in the 72 Audited References). The counts based on primitive time input under Definition 7.10 and on the reference-specific audit results in the Appendix are as follows.

1. References that explicitly formulate the primitive-time problem number 0/72.
2. References that eliminate primitive time input from every dependency root number 0/72.
3. References that provide complete time generation from non-temporal physical inputs number 0/72.
4. References that use at least one temporal dependency root before the generation of time contents number 72/72.
5. References that explicitly state the absence of external or independent time, the fundamental status of time, difficulty defining the concept of time, or an internal or relational construction of time number 18/72.
6. References that present a dynamical construction or recovery of intrinsic, relational, thermal, or semiclassical time number 13/72.

7. Of the thirteen references in the preceding item, references that satisfy the complete time-generation obligation from elimination of primitive time input through physical-time qualification number 0/13.

*Proof.* For each reference division in Table 16, the formal inputs, first temporal dependency root, audited generated time contents, primitive time roots, results dependent on primitive time roots, and statement evidence are compared.

By Theorem 7.190, primitive time roots and results dependent on primitive time roots exist in all 72 references, and no reference generates all the time contents it uses from non-temporal physical inputs. Items 2 through 4 follow.

Explicit formulation of the primitive-time problem requires more than the absence of a single external variable. It requires formulating the following as obligations that time generation must satisfy: identification of the temporal dependency roots that act in place of time, order, clocks, histories, rates of change, and causal order; their elimination from every dependency root; construction of a generation route from non-temporal physical inputs to time contents; and separation of evaluative success from physical-time qualification. Comparing the statement evidence and formal inputs in Table 16 with these conditions gives item 1.

Item 5 counts the absence, fundamental status, or definitional difficulty of time, or an internal or relational construction of time, stated as part of the reference's own analysis. The 18 references are references 6, 12, 14, 19, 25, 26, 29, 30, 33–35, 43, 47, 54–55, 59, 63, and 72 in Table 16. Heisenberg 1958 states the definitional difficulty that language describing the creation of time presupposes a temporal relation.

The dynamical construction or recovery in item 6 denotes a procedure that provides an intrinsic, relational, thermal, or semiclassical time variable, time flow, or temporal evolution relation from a candidate dynamical clock variable, subsystem, configuration, history, or state. The 13 references are references 14, 19, 26, 29, 30, 33–35, 43, 47, 54–55, and 59 in the same table. The domain of each procedure is given by the reference-specific statements, formal constructions, and their conditions in the Appendix. Comparison of the formal inputs and primitive time roots in all 13 references shows that none satisfies the complete time-generation obligation. Items 5 through 7 follow. □

**Corollary 7.192** (Difference in Determination between Statements of Problems of Time and Complete Time Generation). In the 72 audited references, explicitly stating a problem of time listed in item 5 of the preceding proposition, or presenting a dynamical time-construction or recovery procedure under item 6, does not imply the establishment of complete time generation.

*Proof.* Apply items 5 through 7 of Proposition 7.191. □

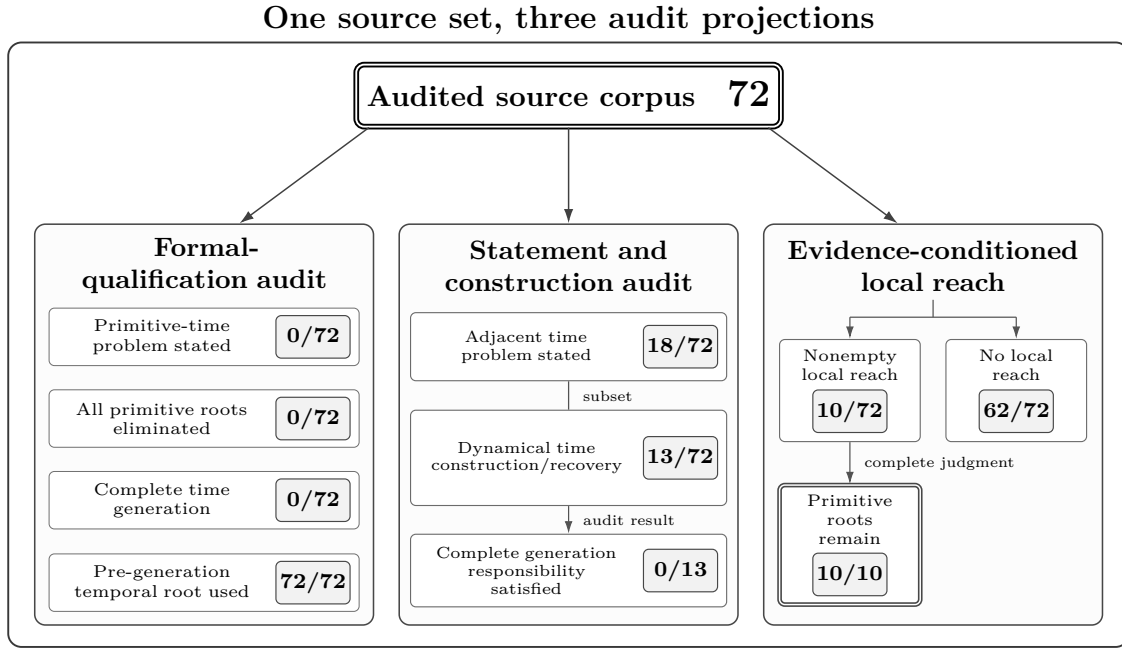


Figure 18: Three audit surfaces for the same 72 references: formal qualification, statements and constructions, and local reachability conditional on the evidence environment. Subsets defined by statements and constructions and local reachability conditional on the evidence environment are separated as distinct axes of determination.

**Proposition 7.193** (Scope of Direct Demonstration for Biological Application in the 72 Audited References). Among the seven references containing statements of biological application in Proposition 7.185, the number of references that directly measure biological processes under an experimental arrangement that isolates relativistic conditions is 0/7.

Furthermore, among the 72 audited references in Definition 7.1, the number of references that directly measure biological processes under an experimental arrangement that isolates relativistic conditions is 0/72.

*Proof.* By Proposition 7.185, the seven references containing statements of biological application and the objects directly measured in Bailey et al. 1977, Hafele–Keating 1972a/1972b, and Pound–Rebka 1960 belong to different evidentiary stages.

The reference-specific results, result dependencies, and original-source-location evidence in the Appendix cited by the 72 rows of Table 16 are compared. The objects recorded as directly measured include the muon decay lifetime, decay-electron counts, differences in cesium-clock indications, clock rates, post-reunion display deviations, relative frequency differences of electromagnetic radiation, resonant-transmission counts, and calibration quantities.

The biological-application statements in the same 72 references identify pulse rates, thought, cancer development, human aging, the ages of twins, human lifetimes, biological clocks, and human hearts as objects of application. None of the 72 rows contains a reference that includes these biological processes among its experimental objects, isolates the relativistic conditions, and provides a measurement procedure and measurement results for the biological processes. The displayed counts for the seven references and the 72 audited references therefore follow.  $\square$

**Corollary 7.194** (Separation of Biological-Application Statements from the Scope of Direct Demonstration). The biological-application statements in the seven references

of Proposition 7.193 are audited as theoretical predictions or explanatory applications. Establishment of direct demonstration concerning biological processes is determined by the necessary conditions in Proposition 7.184.

*Proof.* Apply Proposition 7.193 and Proposition 7.184.  $\square$

#### 7.4.1 Cross-Reference Results for Time-Content Classification and Structural Signatures

**Proposition 7.195** (Coverage of the Five-Type Classification of Time Contents). The time-content set of the audited reference set satisfies

$$|\text{TimeContent}^{\text{audit}}| = 2266$$

For each reference  $R \in \text{Ref}^{\text{audit}}$ ,

$$\text{TimeContent}_R = \bigcup_{\tau \in \text{Type}^{\text{time,ord,struct}}} \left( \text{TypeOf}_R^{\text{time,ord,struct}} \right)^{-1}(\{\tau\}).$$

*Proof.* For each reference row in Table 25, the five index sets are pairwise disjoint, and their disjoint union equals

$$\{1, \dots, n_R\}$$

Taking the disjoint union over all references gives coverage by the five-type classification and cardinality 2266.  $\square$

**Proposition 7.196** (Totality of the Eight-Axis Classification for Members). For every  $R \in \text{Ref}^{\text{audit}}$  and every  $x$  satisfying

$$x \in \left( \text{TypeOf}_R^{\text{time,ord,struct}} \right)^{-1} \left( \left\{ \text{Type}_{\text{member}}^{\text{time,ord,struct}} \right\} \right)$$

the following inequality holds:

$$\left| \text{AxisOf}_R^{\text{time,ord,struct}}(x) \right| \geq 1.$$

*Proof.* In Table 26, all 642 member-type entries belong to the index set of at least one axis. The eight-axis classification image of each member therefore contains at least one axis.  $\square$

**Proposition 7.197** (Unique Decomposition of Structural Signatures). The signature maps  $\text{Sig}_R^{\text{time,ord,struct,member}}$  and  $\text{Sig}_R^{\text{time,ord,struct,dependency}}$  for each reference uniquely classify the 642 reference-tagged member-type entries and the 1313 reference-tagged dependent-content-type entries, respectively, by the eight signature values.

*Proof.* For each reference, each of the two signature maps assigns each element of its domain one element of

$$\{0, 1\}^3$$

On the reference-tagged disjoint union, entries with the same signature value are grouped together. The sum of the fiber cardinalities in Table 27 is 642, and the sum of the fiber cardinalities in Table 28 is 1313. Each entry therefore belongs to the cross-reference group for the single signature value assigned by the map for its own reference.  $\square$

**Proposition 7.198** (Correspondence between Zero Fibers and Original-Source Evidence). Each of the 69 entries in the zero fiber of the member structural signature and the 133 entries in the zero fiber of the dependent-content structural signature corresponds to an evidentiary location in an original-source definition, equation, or explicit statement. Each of the remaining 573 members corresponds to an occurrence in a transfer event or result prerequisite, and each of the remaining 1180 dependent contents corresponds to transfer reachability involving a member or to an occurrence in a result prerequisite.

*Proof.* The three components of the member structural signature represent occurrence in a transfer input set, occurrence in a transfer output set, and occurrence in a result prerequisite. The three components of the dependent-content structural signature represent transfer reachability to a member, transfer reachability from a member, and occurrence in a result prerequisite. The assertion follows from the zero-fiber cardinalities in Table 27 and Table 28, together with the evidentiary correspondence established by comparison with original-source locations.  $\square$

## 7.5 Contrast with Generation-Complete Time and Physical Grounds

The generation dependencies of LCTR are contrasted with the results from the 72 audited references to derive the physical grounds required of time variables in physical laws.

**Proposition 7.199** (Grounding Sequence for Completion of LCTR Time Generation). For a finite object-specification index family

$$Z_0$$

, the generation dependencies of the physical-role inputs, realized operational configuration, canonical comparison quotient, exact operational time structure, and representation-stage completion condition are determined, in this order, by Proposition 2.12, Proposition 3.26, and Corollary 4.190.

For each  $o \in U$ ,  $\zeta \in Z_0$ , consider the complete evaluation specification

$$\mathbf{Ev} = ((\mathcal{X}; \mathcal{J}_{\text{cmp}}), o, \zeta)$$

. Operability at the dynamical-description, law-description, and differential stages is determined by the three equivalence conditions in Theorem 5.138.

*Proof.* Proposition 2.12 and Proposition 3.26 give the generation from physical-role inputs to the realized operational configuration

$$\mathcal{X}$$

. Corollary 4.190 gives the generation dependencies through the canonical comparison quotient, exact operational time structure, and representation-stage completion condition.

Theorem 5.138 gives the three equivalence conditions at the dynamical-description, law-description, and differential stages for a complete evaluation specification. The preceding generation dependencies and the three equivalence conditions therefore constitute the grounding sequence for completion of LCTR time generation.  $\square$

**Theorem 7.200** (Physical-Time Qualification of Generation-Complete Time Content). For time content obtained through the generation dependencies of Proposition 7.199, when the generation structures corresponding to Definition 7.2 are established prior to law operations and result derivation, a time variable in a physical law that uses this time content satisfies physical-time qualification.

*Proof.* By Proposition 7.3, identity determination, order comparison, interval comparison, scale transformation, correspondence between information-coupling positions, clock correspondence, and observation–record correspondence for time content take the corresponding generation structures as inputs.

By the generation dependencies in Proposition 7.199, the comparison equivalence relation, canonical comparison quotient, canonical order, local and global order coordinates, quantitative affine realization, comparison-order atlas, clock correspondence, and observation–record correspondence precede law operations and result derivation.

By Definition 7.7, time content for which these grounding obligations are generated from non-temporal physical inputs satisfies physical-time qualification.  $\square$

**Theorem 7.201** (Structural Separation between the Generation-Completion Condition and the Results of the 72-Reference Audit). For every  $R \in \text{Ref}^{\text{audit}}$ ,

$$\text{TimeContent}_R^{\text{audit,gen}} \subsetneq \text{TimeContent}_R$$

holds.

By contrast, for a reference

$$R$$

, the generation-completion condition for all its time contents is

$$\text{TimeContent}_R^{\text{audit,gen}} = \text{TimeContent}_R,$$

that is,

$$\text{TimeContent}_R^{\text{audit,root}} = \emptyset$$

.

Consequently, none of the 72 audited references satisfies the generation-completion condition for all the time contents it uses.

*Proof.* By Definition 7.39,

$$\text{TimeContent}_R^{\text{audit,root}} = \text{TimeContent}_R \setminus \text{TimeContent}_R^{\text{audit,gen}}.$$

By Theorem 7.190, for every  $R \in \text{Ref}^{\text{audit}}$ ,  $\text{TimeContent}_R^{\text{audit,root}} \neq \emptyset$ . Therefore,

$$\text{TimeContent}_R^{\text{audit,gen}} \subsetneq \text{TimeContent}_R.$$

On the generation-complete side given by Proposition 7.199 and Theorem 7.200, the time contents used are generated from non-temporal physical inputs, and the required generation structures precede law operations and result derivation. Generation completion for all time contents therefore coincides with emptiness of the primitive-time-root set.

Because the primitive-time-root set is nonempty in every one of the 72 audited references, the generation-completion condition for all time contents fails in every case.  $\square$



**Proposition 7.202** (Extensibility of the Primitive-Time Failure Classification). When a new value or component is added to a time-content type, time-order-structure axis, audit-boundary node, audit-boundary edge, original-source support, or physical-input initial set, the change condition of Definition 7.63 holds.

For the extended audit inputs, full-reference recomputation under Definition 7.64 uniquely determines the classifications, primitive time roots, four-way result partition, and cross-reference dependency relations. For references whose audit inputs are unchanged, the recomputed mathematical objects equal their immediately preceding values.

*Proof.* Definition 7.63 takes as its change condition a change in the codomain, domain, map value, or set component of the surveyed reference set, reference-specific objects, original-source support, classification maps, transfer inputs and outputs, or physical-input initial set.

Definition 7.64 recomputes from the changed inputs the classifications, audit-boundary directed hypergraph, reachability sets, primitive time roots, four-way result partition, and cross-reference dependency relations. By Proposition 7.65, the result of this recomputation is unique.

For references with unchanged inputs, Proposition 7.66 makes the mathematical objects agree before and after recomputation. Audit inputs incorporating new types, axes, nodes, edges, and support components can therefore be reevaluated while preserving the determination values for references whose audit inputs remain unchanged.  $\square$

**Theorem 7.203** (Physical-Grounding Requirement for Time Variables in Physical Laws). When the time content used by a time variable in a physical law is primitive time input, that time variable does not satisfy physical-time qualification.

Even simultaneous mathematical consistency, predictive performance, and empirical fit do not repair this loss of qualification. A result dependent on primitive time input inherits dependence on time content that does not satisfy physical-time qualification.

Furthermore, none of the 72 audited references generates all the time contents it uses from non-temporal physical inputs. Every reference contains a result dependent on a primitive time root.

Physical establishment of a time variable in a physical law therefore requires the time content and grounding structures used by that variable to be generated from non-temporal physical inputs prior to law operations and result derivation.

*Proof.* By Theorem 7.11, primitive time input does not satisfy physical-time qualification. By Proposition 7.13, simultaneous mathematical consistency, predictive performance, and empirical fit do not repair primitive time input.

By Corollary 7.12, direct dependence or transfer-reachability dependence on primitive time input is inherited by results. By Theorem 7.190, primitive time roots and results dependent on primitive time roots exist in all 72 audited references.

By Theorem 7.201, the audited generated-time-content set is a proper subset of the full time-content set in all 72 audited references. Theorem 7.200 establishes physical-time qualification when generation dependencies and grounding structures precede use.

Physical generation of time contents and grounding structures prior to law operations and result derivation is therefore necessary for the physical establishment of a time variable in a physical law.  $\square$

By Theorem 7.203, physical establishment of a time variable in a physical law requires physical generation of its time contents and grounding structures prior to law operations and result derivation.

By Theorem 7.190, primitive time roots and results dependent on primitive time roots exist in all 72 audited references, and none of the 72 references generates all the time contents it uses from non-temporal physical inputs. Proposition 7.191 expresses this cross-reference result as seven counts concerning the primitive-time problem, adjacent problems of time, and direct constructions for resolving time.

Proposition 7.162, Proposition 7.165, Proposition 7.171, Proposition 7.170, Proposition 7.132, and Proposition 7.174 provide representative cases of cross-field formal inputs, evaluative success, adjacent problems of time, the difference in scope between relational formalisms and physical time generation, the evidence types of experiential statements in quantum measurement, and the difference between formal results and objects of statements.

By Proposition 7.8 and Proposition 7.13, mathematical consistency, predictive performance, and empirical fit neither entail physical-time qualification of primitive time input nor repair its loss of qualification.

The four audit determinations in Definition 7.78, Proposition 7.178, and Proposition 7.181 separately determine, from formal evidence and statement evidence, pre-generation placement, omission of generation stages, reattribution of a time property, and reverse presupposition of a derived quantity in time-bearing expressions. Proposition 7.182 applies these determinations to the 96 determination fields on the formal and statement tracks for twelve objects and gives 23 fields established on the formal track, 0 fields established on the statement track, and 73 fields not established by the evidence included in the Appendix. Proposition 7.180 separates the physical propagation of light, the generation order of the time contents used by velocity expressions, constancy of the vacuum speed of light in local inertial frames, and distinct velocity types, reference frames, and domains of application.

Definition 7.183 separates direct measurement of physical carriers, attribution of a property to time itself, and application to biological processes into three evidentiary stages. Proposition 7.184 gives the necessary conditions for direct demonstration concerning biological processes. Proposition 7.193 gives, for the seven references containing statements of biological application and for the 72 audited references, the number of direct demonstrations within the audited reference set.

The 72 audited references, cross-cutting review references that treat studies of time diachronically, and references used to investigate whether direct experiments on biological aging that isolate relativistic conditions have been reported belong to distinct reference groups. The quantification domain of  $\text{Ref}^{\text{audit}}$  is the 72 references in the first group. The cross-cutting review references support the positioning of the research history and audit method, while the references concerning biological-aging experiments support determination of the reported scope. The denominator for counts concerning the 72 audited references remains the 72 references in the first group.

Chapter 8 takes these theorems and propositions as inputs and synthesizes the physical grounds of time variables in physical laws, cross-theoretical consequences arising from primitive-time dependence, the difference between formal results and objects of statements, and multistage misidentification.

The Appendix applies the audit-boundary directed hypergraph of Definition 7.27, the audit-generation traces of Definition 7.32, and the result dependencies of Definition 7.50 to the 72 audited references and gives, for each reference, its time contents, transfers, results, original-source-location evidence, primitive time roots, result dependencies, and statement evidence.

## 8 Representative Schemes, Classical Time, and Final Synthesis

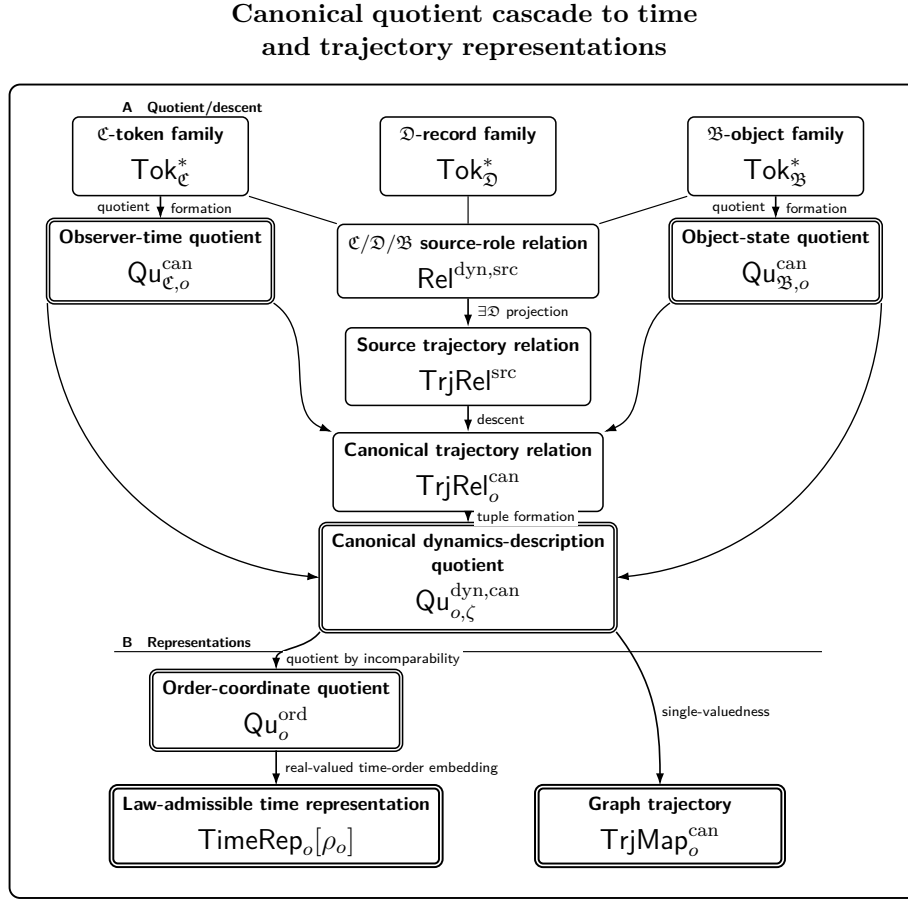


Figure 19: Dependency structure from the canonical quotient cascade to time representations and trajectory maps. The upper row constructs the canonical dynamical-description quotient from the token families derived from  $\mathcal{C}$ ,  $\mathcal{D}$ , and  $\mathcal{B}$ , through the observer-time quotient, object-state quotient, and canonical trajectory relation. The lower row distinguishes the branch that obtains a law-admissible time representation from the time-order side through the quotient by incomparability and a real-valued time-order embedding, from the branch that obtains a graph trajectory from the single-valuedness of the canonical trajectory relation.

### 8.1 Standard Schemes and Theory Reduction

A standard physical-time comparison scheme is a typed input comprising theory-specific geometry, scheme-restricted raw operational input, a family of observer worldlines, a family of comparison rules, and a family of binding data. The ordered pair consisting of the scheme-restricted raw operational input and the scheme-specific source-side realization datum determines a realized operational configuration of the same type as in Definition 3.25. The structures at the comparison and representation stages take this realized operational configuration and the scheme-specific family of comparison rules as domain-side inputs and are generated when the respective stage conditions hold. The continuum-approximation evaluation input is determined from the same input pair in accordance with Definition

4.168. The restriction to each scheme of the complete evaluation specification in Definition 5.2 serves as the domain-side input at the dynamical-description, law-representation, and differential-law-form stages.

The scheme-restricted raw operational input has as components the physical realizations and reception network derived from  $\mathfrak{C}/\mathfrak{D}/\mathfrak{B}/\mathfrak{D}$ . The scheme-specific source-side realization datum has as components the source tokens derived from  $\mathfrak{C}/\mathfrak{D}/\mathfrak{B}$ , display and record data, source-side orders, comparison-side arrival maps, comparison-side recovery maps, and the relation derived from  $\mathfrak{C}\mathfrak{D}$ . The scheme-restricted family of binding data has as components record-side retention, object-state-side distinction, and observer-side binding conditions. A time variable in a physical law is determined by a generated observer-time quotient, observer-time law core, and law-admissible time representation. Frequency, period, Hz, and standard frequency are generated at the representation stage from generated comparison-representation coordinates and count differences derived from  $\mathfrak{C}$ .

### 8.1.1 General Form of Standard Schemes

**Definition 8.1** (Standard Physical-Time Comparison Scheme). A standard physical-time comparison scheme is defined by

$$\text{Sch} := \left( (M_{\text{Sch}}, g_{\text{Sch}}), \mathcal{I}_{\text{Sch}}^{\text{raw}}, (\text{WL}_u)_{u \in U_{\text{Sch}}}, \right. \\ \left. \mathcal{J}_{\text{cmp}, \text{Sch}}, \text{BindDat}_{\text{Sch}} \right)$$

. Its components have the following types.

- $M_{\text{Sch}}, g_{\text{Sch}}$  are the scheme-specific geometric carrier and geometric structure.
- $\mathcal{I}_{\text{Sch}}^{\text{raw}}$  is the scheme-restricted raw operational input.
- For each  $u \in U_{\text{Sch}}$ ,  $\text{WL}_u$  is an observer worldline.
- $\mathcal{J}_{\text{cmp}, \text{Sch}}$  is the scheme-restricted family of comparison rules.
- $\text{BindDat}_{\text{Sch}}$  is the scheme-restricted family of binding data containing record-binding data and object-state-binding data.

The clock-side operational material consists of the cycle indices, phase labels, display updates, counter transitions, and internal counter increments in Definition 2.2. Frequency, period, Hz, and standard frequency are generated at the representation stage under the conditions of Proposition 4.153. A quantity called proper frequency in a given scheme is the value obtained by restricting the  $\mathfrak{C}$ -derived source-relative frequency representation of Definition 4.150 to that scheme.

**Definition 8.2** (Scheme-Restricted Raw Operational Input). The reception network of a standard physical-time comparison scheme  $\text{Sch}$  is defined by

$$\text{Rcv}_{U, \text{Sch}} := \left( U_{\text{Sch}}, \right. \\ \left. (\text{RcvCar}_{u, \text{Sch}}, \text{Rcv}_{\mathfrak{C}, u, \text{Sch}}, \text{Rcv}_{\mathfrak{D}, u, \text{Sch}}, \text{Rcv}_{\mathfrak{B}, u, \text{Sch}}, \preceq_{u, \text{Sch}}^{\text{rcv}})_{u \in U_{\text{Sch}}} \right)$$

. The scheme-restricted raw operational input is defined by

$$\mathcal{I}_{\text{Sch}}^{\text{raw}} := \left( \text{Rl}_{\text{Sch}}[Z_0], \text{Car}_{\text{phys},\text{Sch}}[Z_0], \text{Rlz}_{\mathfrak{C},\text{Sch}}, \right. \\ \left. \text{Rlz}_{\mathfrak{D},\text{Sch}}, \text{Rlz}_{\mathfrak{B},\text{Sch}}[Z_0], \text{Rlz}_{\mathfrak{D},\text{Sch}}, \text{Rcv}_{U,\text{Sch}} \right)$$

. It has the same seven-component type as the raw operational input of Definition 3.6.

**Definition 8.3** (Scheme-Specific Source-Side Realization Datum and Dynamical-Description-Stage Entry Data). For a standard physical-time comparison scheme  $\text{Sch}$ , define the family of source-side tokens by

$$\text{Src}_{*,\text{Sch}} := (\text{Tok}_{\mathfrak{C},\text{Sch}}^*, \text{Tok}_{\mathfrak{D},\text{Sch}}^*, \text{Tok}_{\mathfrak{B},\text{Sch}}^*),$$

define the display and record data by

$$\text{DispRec}_{*,\text{Sch}} := \left( \phi_{\mathfrak{C},\text{Sch}}, \phi_{\mathfrak{D},\text{Sch}}, \text{seq}_{\mathfrak{D},\text{Sch}}, \zeta_{\mathfrak{B},\text{Sch}}, \right. \\ \left. \{W_{\mathfrak{D},\text{Sch}}(x_{\mathfrak{D}})\}_{x_{\mathfrak{D}} \in \text{Tok}_{\mathfrak{D},\text{Sch}}^*} \right),$$

and the source-side orders by

$$\text{SOrd}_{*,\text{Sch}} := (\preceq_{*,\text{Sch}}^{\mathfrak{C}}, \preceq_{*,\text{Sch}}^{\mathfrak{D}})$$

.

Define the families of comparison-side arrival maps and comparison-side recovery maps, and the relation derived from  $\mathfrak{C}\mathfrak{D}$ , by

$$\text{Arr}_{U,\text{Sch}}^{\text{cmp}} := \left( \text{Arr}_{\mathfrak{C},u,\text{Sch}}^{\text{cmp}}, \text{Arr}_{\mathfrak{D},u,\text{Sch}}^{\text{cmp}}, \text{Arr}_{\mathfrak{B},u,\text{Sch}}^{\text{cmp}} \right)_{u \in U_{\text{Sch}}}, \\ \text{Rec}_{U,\text{Sch}}^{\text{cmp}} := \left( \text{Rec}_{\mathfrak{C},u,\text{Sch}}^{\text{cmp}}, \text{Rec}_{\mathfrak{D},u,\text{Sch}}^{\text{cmp}}, \text{Rec}_{\mathfrak{B},u,\text{Sch}}^{\text{cmp}} \right)_{u \in U_{\text{Sch}}}, \\ \text{Rel}_{\text{Sch}} := \left( \left( \text{Rel}_{u,\text{Sch}}^{\text{cmp}} \right)_{u \in U_{\text{Sch}}}, \text{Rel}_{\text{Sch}}^{\text{src}} \right)$$

.

Define the scheme-specific source-side realization datum by

$$\text{Dat}_{\text{Sch}}^{\text{src}} := \left( \text{Src}_{*,\text{Sch}}, \text{DispRec}_{*,\text{Sch}}, \text{SOrd}_{*,\text{Sch}}, \right. \\ \left. \text{Arr}_{U,\text{Sch}}^{\text{cmp}}, \text{Rec}_{U,\text{Sch}}^{\text{cmp}}, \text{Rel}_{\text{Sch}} \right)$$

.

. Define the scheme-restricted realized operational configuration by

$$\mathcal{X}_{\text{Sch}} := (\mathcal{I}_{\text{Sch}}^{\text{raw}}, \text{Dat}_{\text{Sch}}^{\text{src}})$$

.

For each

$$\bullet \in \{\mathfrak{C}, \mathfrak{D}, \mathfrak{B}\}, \quad u \in U_{\text{Sch}}$$

, the comparison-side arrival map has type

$$\text{Arr}_{\bullet,u,\text{Sch}}^{\text{cmp}} : \text{Dom} \left( \text{Arr}_{\bullet,u,\text{Sch}}^{\text{cmp}} \right) \subseteq \text{Tok}_{\bullet,\text{Sch}}^* \\ \longrightarrow \text{Rcv}_{\bullet,u,\text{Sch}}$$

. Define the uniquely recoverable reception set by

$$\text{Rcv}_{\bullet,u,\text{Sch}}^{\text{uniq}} := \left\{ \alpha \in \text{Im} \left( \text{Arr}_{\bullet,u,\text{Sch}}^{\text{cmp}} \right) \mid \left| \left( \text{Arr}_{\bullet,u,\text{Sch}}^{\text{cmp}} \right)^{-1} (\{\alpha\}) \right| = 1 \right\}$$

. The comparison-side recovery map has type

$$\text{Rec}_{\bullet,u,\text{Sch}}^{\text{cmp}} : \text{Rcv}_{\bullet,u,\text{Sch}}^{\text{uniq}} \longrightarrow \text{Tok}_{\bullet,\text{Sch}}^*$$

.  
For each

$$o \in U_{\text{Sch}}, \quad \bullet \in \{\mathfrak{C}, \mathfrak{D}, \mathfrak{B}\}$$

, let the dynamical-description-stage direct-arrival map be

$$\begin{aligned} \text{Arr}_{\bullet,o,\text{Sch}}^{\text{dyn}} : \text{Dom} \left( \text{Arr}_{\bullet,o,\text{Sch}}^{\text{dyn}} \right) &\subseteq \text{Tok}_{\bullet,\text{Sch}}^* \\ &\longrightarrow \text{Rcv}_{\bullet,o,\text{Sch}} \end{aligned}$$

. Define the dynamical-description-stage arrival set by

$$\mathcal{R}_{\bullet,o,\text{Sch}}^{\text{dyn,arr}} := \text{Im} \left( \text{Arr}_{\bullet,o,\text{Sch}}^{\text{dyn}} \right)$$

. The dynamical-description source-role relation and the dynamical-description-stage source-role relation have the respective types

$$\begin{aligned} \text{Rel}_{\text{Sch}}^{\text{dyn,src}} &\subseteq \text{Tok}_{\mathfrak{C},\text{Sch}}^* \times \text{Tok}_{\mathfrak{D},\text{Sch}}^* \times \text{Tok}_{\mathfrak{B},\text{Sch}}^*, \\ \text{Rel}_{o,\text{Sch}}^{\text{dyn}} &\subseteq \mathcal{R}_{\mathfrak{C},o,\text{Sch}}^{\text{dyn,arr}} \times \mathcal{R}_{\mathfrak{D},o,\text{Sch}}^{\text{dyn,arr}} \times \mathcal{R}_{\mathfrak{B},o,\text{Sch}}^{\text{dyn,arr}} \end{aligned}$$

. Define the dynamical-description-stage four-party reception structure by

$$\begin{aligned} \text{RcvStr}_{o,\text{Sch}}^{\text{dyn}} &:= \left( \left( \mathcal{R}_{\mathfrak{C},o,\text{Sch}}^{\text{dyn,arr}}, \mathcal{R}_{\mathfrak{D},o,\text{Sch}}^{\text{dyn,arr}}, \mathcal{R}_{\mathfrak{B},o,\text{Sch}}^{\text{dyn,arr}} \right), \right. \\ &\quad \left( \text{Arr}_{\mathfrak{C},o,\text{Sch}}^{\text{dyn}}, \text{Arr}_{\mathfrak{D},o,\text{Sch}}^{\text{dyn}}, \text{Arr}_{\mathfrak{B},o,\text{Sch}}^{\text{dyn}} \right), \\ &\quad \left. \left( \text{Rel}_{o,\text{Sch}}^{\text{dyn}}, \text{Rel}_{\text{Sch}}^{\text{dyn,src}} \right) \right) \end{aligned}$$

. The dynamical-description-stage recovery map has type

$$\text{Rec}_{\bullet,o,\text{Sch}}^{\text{dyn}} : \mathcal{R}_{\bullet,o,\text{Sch}}^{\text{dyn,arr}} \longrightarrow \text{Tok}_{\bullet,\text{Sch}}^*$$

.  
For every

$$x_{\mathfrak{B}} \in \text{Dom} \left( \text{Arr}_{\mathfrak{B},o,\text{Sch}}^{\text{dyn}} \right)$$

, require

$$\begin{aligned} &\text{Arr}_{\mathfrak{B},o,\text{Sch}}^{\text{dyn}}(x_{\mathfrak{B}}) \\ &\in \text{Rcv}_{\mathfrak{B},o,\text{Sch}} \cap \left( \text{Tok}_{\mathfrak{B},\text{Sch}}^* \times \text{Tok}_{\mathfrak{B},\text{Sch}}^* \right) \end{aligned}$$

and

$$\text{pr}_1 \left( \text{Arr}_{\mathfrak{B},o,\text{Sch}}^{\text{dyn}}(x_{\mathfrak{B}}) \right) = x_{\mathfrak{B}}$$

. Assign the  $\mathfrak{B}$ -derived object-state-side identification map the type

$$\begin{aligned} \text{Stateld}_{\mathfrak{B},o,\text{Sch}} &: \text{Dom} \left( \text{Arr}_{\mathfrak{B},o,\text{Sch}}^{\text{dyn}} \right) \\ &\longrightarrow \text{Tok}_{\mathfrak{B},\text{Sch}}^* \end{aligned}$$

, and set

$$\text{Stateld}_{\mathfrak{B},o,\text{Sch}} := \text{pr}_2 \circ \text{Arr}_{\mathfrak{B},o,\text{Sch}}^{\text{dyn}}$$

. Then

$$\text{Arr}_{\mathfrak{B},o,\text{Sch}}^{\text{dyn}}(x_{\mathfrak{B}}) = (x_{\mathfrak{B}}, \text{Stateld}_{\mathfrak{B},o,\text{Sch}}(x_{\mathfrak{B}}))$$

holds.

$\text{Dat}_{\text{Sch}}^{\text{src}}$  has the same six-component type as in Definition 3.24. Take the dynamical-description-stage entry data to be

$$\begin{aligned} &\left( \text{RcvStr}_{o,\text{Sch}}^{\text{dyn}}, \text{Rec}_{\mathfrak{C},o,\text{Sch}}^{\text{dyn}}, \right. \\ &\quad \left. \text{Rec}_{\mathfrak{D},o,\text{Sch}}^{\text{dyn}}, \text{Rec}_{\mathfrak{B},o,\text{Sch}}^{\text{dyn}} \right)_{o \in U_{\text{Sch}}} \end{aligned}$$

.

**Proposition 8.4** (Generation Position of Clock-Side Inputs and Frequency Representations in Standard Schemes). The clock-side operational material of a standard physical-time comparison scheme  $\text{Sch}$  consists of the cycle indices, phase labels, display updates, counter transitions, and internal counter increments in Definition 2.2. The scheme-restricted family of binding data  $\text{BindDat}_{\text{Sch}}$  has record-side retention, object-state-side distinction, and observer-side binding conditions as components. The scheme-restricted raw operational input  $\mathcal{I}_{\text{Sch}}^{\text{raw}}$  and the scheme-specific source-side realization datum  $\text{Dat}_{\text{Sch}}^{\text{src}}$  do not have frequency, period, Hz, or standard frequency as components. These quantities are generated at the representation stage from generated comparison-representation coordinates and count differences derived from  $\mathfrak{C}$ . A quantity called proper frequency in a given scheme is the value obtained by restricting the  $\mathfrak{C}$ -derived source-relative frequency representation to that scheme.

*Proof.* Definitions 3.6 and 3.24 construct the raw operational input and source-side realization datum from physical realizations, a reception network, source tokens derived from  $\mathfrak{C}/\mathfrak{D}/\mathfrak{B}$ , display and record data, source-side orders, comparison-side arrival maps, comparison-side recovery maps, and a relation derived from  $\mathfrak{C}\mathfrak{D}$ . By Proposition 3.26, these input data contain no time variables, time intervals, periods, or frequencies.

Definition 4.150 defines frequency and period representations from quantitative affine differences of generated  $\mathfrak{C}$ -derived comparison-representation coordinates and count differences derived from  $\mathfrak{C}$ . Definition 4.151 defines an Hz value from a frequency representation and a  $\mathfrak{C}$ -derived frequency-unit scalar, while Definition 4.152 defines a standard-frequency representation from the frequency representation of a reference  $\mathfrak{C}$ -derived source. By Proposition 4.153, these quantities are generated under the conditions of the representation stage. A quantity called proper frequency in a given scheme is likewise determined from the  $\mathfrak{C}$ -derived source-relative frequency representation in Definition 4.150.  $\square$

### 8.1.2 Representative Schemes

A representative scheme is a theory-specific input datum that supplies the input of each physical theory as a realized operational configuration

$$\mathcal{X}_{\text{Sch}} = (\mathcal{I}_{\text{Sch}}^{\text{raw}}, \text{Dat}_{\text{Sch}}^{\text{src}})$$

. Its components include clock-side cycle indices, phase labels, display updates, counter transitions, internal counter increments, record-side retention, object-state-side distinction, and observer-side binding conditions. Time variables are based on the observer-time quotient, observer-time law core, and law-admissible time representation generated from the inputs of each representative scheme. Frequency, period, Hz numerical values, standard frequency, and proper frequency are defined from cycle-count differences derived from the clock-side operational material and quantitative affine differences of generated representation coordinates.

Each representative scheme has the general form of Definition 8.1 and has as components the scheme-restricted raw operational input of Definition 8.2 and the scheme-specific source-side realization datum and dynamical-description-stage entry data of Definition 8.3. These components reduce each representative theory to the general generation sequences of Chapters 3 and 5.

**Definition 8.5** (Single-Object Evaluation Specification for a Representative Scheme). Let the representative-scheme label be  $\text{Sch}$ . When the representative scheme supplies the realized operational configuration

$$\mathcal{X}_{\text{Sch}} = (\mathcal{I}_{\text{Sch}}^{\text{raw}}, \text{Dat}_{\text{Sch}}^{\text{src}}),$$

and the typed family of comparison rules  $\mathcal{J}_{\text{cmp}, \text{Sch}}$ , define the comparison-rule-dependent common input by

$$\mathbf{l}_{\text{Sch}} := (\mathcal{X}_{\text{Sch}}, \mathcal{J}_{\text{cmp}, \text{Sch}})$$

. Define the comparison-stage condition-family evaluation specification by

$$\text{Ev}_{\text{Sch}}^{\text{cmp}} := \left( \text{Ev}_{\text{Sch}, i}^{\text{cmp}} \right)_{i=1}^7, \quad \text{Ev}_{\text{Sch}, i}^{\text{cmp}} := \begin{cases} \mathcal{X}_{\text{Sch}}, & i = 1, \\ \mathbf{l}_{\text{Sch}}, & i \in \{2, \dots, 7\} \end{cases}$$

.

Further, given an observer-side information-coupling position  $o \in U_{\text{Sch}}$  and a single-object specification index  $\zeta_{\text{Sch}}$ , define the scheme-restricted complete evaluation specification by

$$\begin{aligned} \text{Ev}_{\text{Sch}} \\ := (\mathbf{l}_{\text{Sch}}, o, \zeta_{\text{Sch}}) \end{aligned}$$

.

Denote comparison-stage operability, exact comparison closure, and dynamical-description operability, respectively, by

$$\begin{aligned} &\text{Operative}(\mathcal{X}_{\text{Sch}}), \\ &\text{ExactOperable}[\mathbf{l}_{\text{Sch}}], \\ &\text{DynOperative}[\text{Ev}_{\text{Sch}}] \end{aligned}$$

.

The correspondence between the classification numbers  $\text{DG}_1$  through  $\text{DG}_8$  and the dynamical-description conditions is as follows.



$$(DG_1) \ K_{\text{GenTimeOrdAntiSym}}^{\text{dyn}}(\text{EvSch}).$$

$$(DG_2) \ K_{\text{CanTrjDesc}}^{\text{dyn}}(\text{EvSch}).$$

$$(DG_3) \ K_{\text{CanTrjFunc}}^{\text{dyn}}(\text{EvSch}).$$

$$(DG_4) \ K_{\text{CanObsDesc}}^{\text{dyn}}(\text{EvSch}).$$

$$(DG_5) \ K_{\text{LocObsChgComm}}^{\text{dyn}}(\text{EvSch}).$$

$$(DG_6) \ K_{\text{ObsTimeIncTrans}}^{\text{dyn}}(\text{EvSch}).$$

$$(DG_7) \ K_{\text{ObsTimeRealOrdEmb}}^{\text{dyn}}(\text{EvSch}).$$

$$(DG_8) \ K_{\text{ObsTimeTrjFact}}^{\text{dyn}}(\text{EvSch}).$$

Denote the simultaneous satisfaction of all eight conditions by  $\text{DynCondAll}[\text{EvSch}]$ .

At their respective generation stages, denote the dynamical-description generation object, the law-output determinacy condition, and the law-description generation object by

$$\begin{aligned} & \text{Dat}_{o, \zeta_{\text{Sch}}}^{\text{dyn}, 0}, \\ & \text{DynWit}_{o, \zeta_{\text{Sch}}} [\text{RepEmb}_o^{\text{dyn}, \text{time}}], \\ & K_{\text{CandRelRightUnique}}^{\text{law}} [\text{LawEvalDat}_{o, \zeta_{\text{Sch}}}], \\ & \text{LawCore}_{o, \zeta_{\text{Sch}}}^{\text{obs}, \text{time}} (\text{RepEmb}_o^{\text{dyn}, \text{time}}), \\ & \mathcal{O}_{o, \zeta_{\text{Sch}}}^{\text{dyn}, \text{can}} \end{aligned}$$

. Here,

$$\text{RepEmb}_o^{\text{dyn}, \text{time}} \in \text{OrdEmbSet}_o^{\text{dyn}, \text{time}}(\mathbb{R})$$

. Generation of the observer-time law core requires

$$\text{LawOperative} [\text{LawEvalDat}_{o, \zeta_{\text{Sch}}}]$$

.

**Definition 8.6** (Generation Rule for Clock-Side Inputs and Frequency Representations). Clock-side inputs in a representative scheme are supplied as cycle indices, phase labels, display updates, counter transitions, and internal counter increments. Suppose that the conditional quantitative affine realization derived from  $\mathfrak{C}$ ,

$$\text{QtyAffReal}_{\mathfrak{C}}[\text{ISch}]$$

, is specified and that the admissible counting window

$$\text{win} = (u, j_{\mathfrak{C}}^{\text{src}}, k_0, k_1)$$

satisfies the membership and strict-order conditions of Definition 4.150. Then, upon specialization of that definition to  $\text{ISch}$ , the counting-window frequency representation and period representation derived from  $\mathfrak{C}$  are

$$\begin{aligned} \nu_{\mathfrak{C}}^{\text{win}}[\text{ISch}; \text{win}] &= \frac{\text{CntDiff}_{\mathfrak{C}}^{\text{qty}}[\text{win}]}{\text{RepDiff}_{\mathfrak{C}}^{\text{qty}}[\text{ISch}; \text{win}]}, \\ \text{Per}_{\mathfrak{C}}^{\text{win}}[\text{ISch}; \text{win}] &= (\nu_{\mathfrak{C}}^{\text{win}}[\text{ISch}; \text{win}])^{-1}, \\ \nu_{\mathfrak{C}}^{\text{win}}[\text{ISch}; \text{win}], \quad \text{Per}_{\mathfrak{C}}^{\text{win}}[\text{ISch}; \text{win}] &\in \text{QtyVal}_{\mathfrak{C}}. \end{aligned}$$

The frequency-unit scalar and Hz numerical value follow Definition 4.151.

**Definition 8.7** (Newtonian and classical-mechanics scheme). Denote the Newtonian and classical-mechanics scheme by  $\text{Sch}_{\text{NCM}}$ . This scheme has shared clock-side material, stable record correspondence, distinguishability of object states, and observer-side binding conditions as theory-specific inputs. The clock-side material is assigned to the same order representation; the record-side material supplies records corresponding to that order representation; and the object-state-side material supplies the distinctions required for the dynamical description.

Conventional classical physics uses the time variable  $t$  as an input to laws of motion. This scheme constructs the classical time variable as an order representation generated from comparison-stage closure, the representation stage, the  $\mathfrak{CDB}$ -derived dynamical-description stage, the observer-time law core, and a law-admissible time representation.

The Newtonian and classical-mechanics scheme has the following input components.

- (NM<sub>1</sub>) A family of clock-side sources corresponding to a shared clock.
- (NM<sub>2</sub>) Record-side retention that supplies stable record correspondence.
- (NM<sub>3</sub>) Distinguishability of object states required for the dynamical description.
- (NM<sub>4</sub>) Observer-side binding conditions that bind the clock side, record side, and object-state side at the same phase.
- (NM<sub>5</sub>) Comparison-side arrival maps, comparison-side recovery maps, comparison rules, and binding data.
- (NM<sub>6</sub>) The dynamical-description-stage entry data of Definition 8.3.
- (NM<sub>7</sub>) Idealization conditions for the classical domain that make branching burdens single-valued.
- (NM<sub>8</sub>) An object-state correspondence that approximates observational binding by the identity.
- (NM<sub>9</sub>) Trajectory and observable correspondences for evaluating law-output determinacy.

These inputs are used to evaluate the conditions for the comparison stage, representation stage, continuous-approximation stage, dynamical-description stage, law-representation stage, and differential-law-form stage in the classical domain.

**Definition 8.8** (Time-Metrology Scheme). Denote the time-metrology scheme by  $\text{Sch}_{\text{TM}}$ . This scheme has a reference source derived from  $\mathfrak{C}$ ,  $\text{Src}_{\mathfrak{C}}(j_{\text{std}}^{\text{src}})$ , and a normalization convention  $N_{\text{std}} \in \mathbb{N}_{>0}$ , and retains cycle-count records through record-side correspondence. A standard-time construction is a representation generated from cycle-count records, counting comparison and correction, correction records, and gluing of comparison quotients.

The time-metrology scheme has the following input components.

- (TM<sub>1</sub>) A reference source derived from  $\mathfrak{C}$  and a normalization convention.
- (TM<sub>2</sub>) A family of record-side sources that retains cycle-count records.
- (TM<sub>3</sub>) Comparison rules containing counting comparison and correction, correction records, and traceability information.

- (TM<sub>4</sub>) Comparison-side arrival maps, comparison-side recovery maps, and source-side pairing relations.
- (TM<sub>5</sub>) The dynamical-description-stage entry data of Definition 8.3.

Specializing Definition 4.152 to the time-metrology scheme gives the standard-frequency representation derived from  $\mathfrak{C}$  and the Hz unit scalar derived from  $\mathfrak{C}$  as

$$\begin{aligned}\nu_{\mathfrak{C}}^{\text{std}}[\mathbf{I}_{\text{Sch}_{\text{TM}}}] &= \nu_{\mathfrak{C}}^{\text{src}}[\mathbf{I}_{\text{Sch}_{\text{TM}}}; j_{\text{std}}^{\text{src}}], \\ \nu_{\mathfrak{C}}^{\text{Hz}}[\mathbf{I}_{\text{Sch}_{\text{TM}}}] &= \frac{\nu_{\mathfrak{C}}^{\text{std}}[\mathbf{I}_{\text{Sch}_{\text{TM}}}]}{N_{\text{std}} \cdot_{\mathfrak{C}}^{\text{qty}} 1_{\mathfrak{C}}^{\text{qty}}}.\end{aligned}$$

**Definition 8.9** (GPS scheme). Denote the GPS scheme by  $\text{Sch}_{\text{GPS}}$ . This scheme is a distributed comparison scheme with multiple families of clock-side sources, record-side correction records, transport rules, and observer-side reception structures. Clock adjustment and time correction in GPS map differences in count ratios, transport, and gluing to comparable records.

The clock-side materials at the satellite and receiver are supplied as cycle-count sources. Transport corrections, relativistic corrections, and receiver-side reconstruction are structural conditions for gluing cycle-count records and comparison-transport data into a canonical comparison quotient.

The GPS scheme has the following input components.

- (GPS<sub>1</sub>) Families of clock-side sources at satellites and receivers.
- (GPS<sub>2</sub>) Families of record-side sources that include navigation messages, correction records, and reception records.
- (GPS<sub>3</sub>) Receiver-side object-state material and observer-side binding conditions.
- (GPS<sub>4</sub>) Comparison rules that include propagation corrections, relativistic corrections, and clock corrections.
- (GPS<sub>5</sub>) Comparison-side arrival maps, comparison-side recovery maps, and source-side pairing relations.
- (GPS<sub>6</sub>) The dynamical-description-stage entry data of Definition 8.3.

The domain in which adjustment, synchronization, correction, and navigation time-tagging for multiple clock families in the GPS scheme hold is determined by the failure-position token system of Chapter 6.

**Definition 8.10** (Special-relativity scheme). Denote the special-relativity scheme by  $\text{Sch}_{\text{SR}}$ . This scheme has observer worldlines, local clock-side reception, record-side correspondence, and transport rules on Minkowski spacetime. Conventional special relativity uses Minkowski spacetime and inertial coordinate time as inputs to law descriptions. This scheme uses Minkowski spacetime as a theory-specific input carrier and constructs inertial coordinate time as a coordinate representation generated from comparison-stage closure, the representation stage, the dynamical-description stage, the observer-time law core, and a law-admissible time representation.

The special-relativity scheme has the following input components.

- (SR<sub>1</sub>) A set of observers and worldlines corresponding to a network of inertial observers.
- (SR<sub>2</sub>) Local clock-side, record-side, and object-state-side reception for each observer.
- (SR<sub>3</sub>) Comparison rules corresponding to synchronization procedures, proper-time comparisons, and inertial-coordinate transformations.
- (SR<sub>4</sub>) Comparison-side arrival maps, comparison-side recovery maps, and source-side pairing relations.
- (SR<sub>5</sub>) The dynamical-description-stage entry data of Definition 8.3.

Branches in proper-time comparison, synchronization procedures, and simultaneity are determined by the conditions under which comparison, representation, and gluing hold along the same generation sequence.

**Definition 8.11** (General-relativity scheme). Denote the general-relativity scheme by  $\text{Sch}_{\text{GR}}$ . This scheme has a network of local observers on Lorentzian spacetime, worldline-dependent reception structures, transport data, and gluing data. Conventional general relativity uses Lorentzian spacetime and proper-time representations as inputs to law descriptions. This scheme uses Lorentzian spacetime as a theory-specific input carrier and constructs a proper-time representation as a local display generated from worldline-local comparison data and generated representation coordinates.

The general-relativity scheme has the following input components.

- (GR<sub>1</sub>) A network of local observers and worldline-dependent reception structures.
- (GR<sub>2</sub>) Local clock-side, record-side, and object-state-side reception.
- (GR<sub>3</sub>) Comparison rules for local comparison, transport, and gluing.
- (GR<sub>4</sub>) Comparison-side arrival maps, comparison-side recovery maps, and source-side pairing relations.
- (GR<sub>5</sub>) The dynamical-description-stage entry data of Definition 8.3.

The domain in which local-to-global comparison restrictions, transport, and gluing hold is determined by the failure-position token system of Chapter 6.

**Definition 8.12** (Quantum-reconstruction scheme). Denote the quantum-reconstruction scheme by  $\text{Sch}_{\text{QG}}$ . This scheme takes families of clock-side, record-side, and object-state-side sources, branch-sensitive record structures, observer-side binding, and comparison-gluing conditions as inputs, while excluding external time from the primitive inputs. From these inputs it generates comparison-stage closure, a canonical comparison quotient, a  $\mathcal{CDB}$ -derived dynamical description, and a law-admissible time representation under the establishment conditions of their respective stages. For a law-evaluation datum satisfying law-description operability, it generates the observer-time law core.

The quantum-reconstruction scheme has the following input components.

- (QG<sub>1</sub>) Families of clock-side, record-side, and object-state-side sources.
- (QG<sub>2</sub>) Branch-sensitive record structures and object-state-side distinguishability.

- (QG<sub>3</sub>) Observer-side binding conditions and comparison-gluing conditions.
- (QG<sub>4</sub>) Comparison-side arrival maps, comparison-side recovery maps, and source-side pairing relations.
- (QG<sub>5</sub>) The dynamical-description-stage entry data of Definition 8.3.

The domains in which observational binding, object-state distinction, state identification, and law representation hold are determined by the conditions of the dynamical-description stage, law-representation stage, and differential-law-form stage.

**Proposition 8.13** (Primitive-Frequency-Free Reduction of Representative Schemes). For each

$$\text{Sch} \in \left\{ \text{Sch}_{\text{NCM}}, \text{Sch}_{\text{TM}}, \text{Sch}_{\text{GPS}}, \right. \\ \left. \text{Sch}_{\text{SR}}, \text{Sch}_{\text{GR}}, \text{Sch}_{\text{QG}} \right\}$$

the theory-specific components of the representative scheme construct

$$\mathcal{A}_{\text{Sch}} = (\mathcal{I}_{\text{Sch}}^{\text{raw}}, \text{Dat}_{\text{Sch}}^{\text{src}})$$

. The observer-time quotient and law-admissible time representation are generated, within domains satisfying their respective generation conditions, from this realized operational configuration, the typed family of comparison rules, the observer-side information-coupling position, and the single-object specification index. The observer-time law core is generated when the corresponding law-evaluation datum satisfies law-description operability. Frequency, period, Hz numerical values, standard frequency, and proper frequency are generated from cycle-count differences and quantitative affine differences of generated representation coordinates, subject to the conditions on quantitative affine realization, admissible counting windows, window invariance, and unit normalization required by the definition of each quantity.

*Proof.* By Definition 8.6, clock-side inputs are supplied as cycle indices, phase labels, display updates, counter transitions, and internal counter increments, while frequency representations are defined from cycle-count differences and quantitative affine differences of generated representation coordinates.

Definitions 8.7, 8.8, 8.9, 8.10, 8.11, and 8.12 supply, for their respective representative schemes, the components of the raw operational input, source-side realization datum, typed family of comparison rules, and dynamical-description-stage entry data. By Definition 8.5, these components determine the corresponding complete evaluation specification.

Definition 3.25 determines the realized operational configuration, and Definition 5.56 determines the observer-time quotient. Theorem 5.112 generates the dynamical-description object and dynamical-description evidence datum, while Theorem 5.113 together with Definition 5.108 generates the observer-time law core. Definition 5.63 generates the law-admissible time representation. By Proposition 8.4, the raw operational input and source-side realization datum contain no time variable, time interval, period, or frequency as components; these quantities belong to subsequent generation stages. Definitions 4.150 and 4.151 generate frequency, period, and Hz numerical values. Definition 4.152 generates standard frequency and the Hz unit scalar. The six representative schemes therefore obtain each time-representation quantity and frequency quantity from the general generation sequence without requiring a time variable or primitive frequency value as a pregeneration input.  $\square$

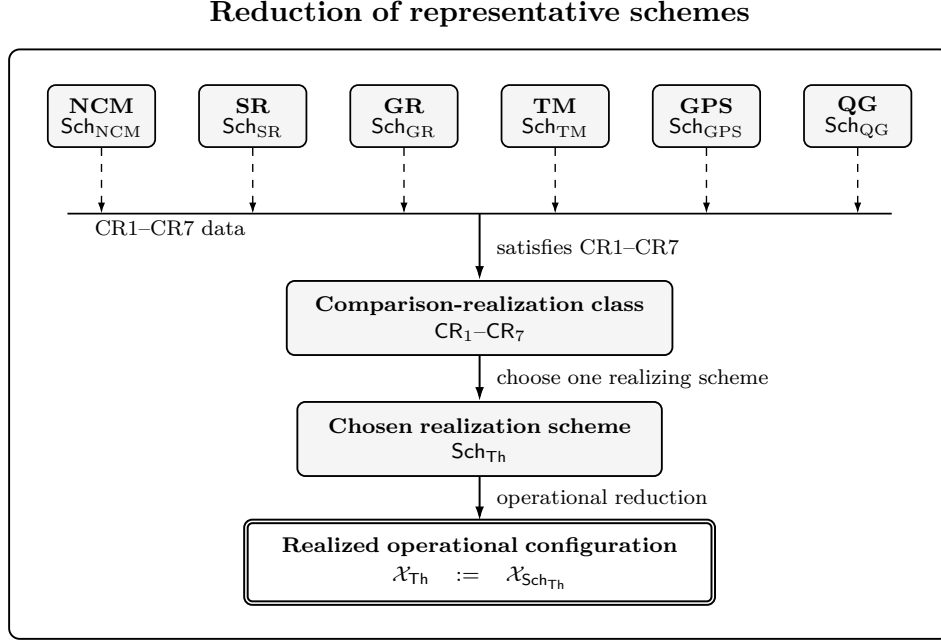


Figure 20: Reduction from the six representative schemes to theory-specific realized operational configurations. The theory corresponding to each representative scheme belongs to the comparison-realization class by supplying the data from CR1 through CR7. Once a realization scheme  $\text{Sch}_{\text{Th}}$  is selected for a theory  $\text{Th}$ , the operational reduction defines  $\mathcal{X}_{\text{Th}} := \mathcal{X}_{\text{Sch}_{\text{Th}}}$ .

### 8.1.3 Reduction to the Comparison-Realization Class and Realized Operational Configurations

**Definition 8.14** (Comparison-realization class). A physical theory  $\text{Th}$  belongs to the comparison-realization class if there exists at least one standard physical-time-comparison scheme that realizes the geometry, operational input, worldline family, comparison rules, and binding data of  $\text{Th}$  and satisfies the following seven obligations. In this section, select one such scheme and write

$$\text{Sch}_{\text{Th}}$$

This notation does not assume uniqueness of the realization scheme.

- (CR1)  $\text{Sch}_{\text{Th}}$  has the general form of a standard physical-time-comparison scheme from Definition 8.1.
- (CR2) The source-side token families, display and record data, and source-side orders of  $\text{Sch}_{\text{Th}}$  have the types specified in Definition 8.3.
- (CR3) The family of comparison-side arrival maps of  $\text{Sch}_{\text{Th}}$  has the type specified in Definition 8.3.
- (CR4) The family of comparison-side recovery maps of  $\text{Sch}_{\text{Th}}$  has the type specified in Definition 8.3.
- (CR5) The family of direct-arrival maps at the dynamical-description stage of  $\text{Sch}_{\text{Th}}$  has the type specified in Definition 8.3.

- (CR6) The family of recovery maps at the dynamical-description stage of  $\mathbf{Sch}_{\mathbf{Th}}$  has the type specified in Definition 8.3.
- (CR7) The comparison-side relations, source-side relations, and dynamical-description source-role relation of  $\mathbf{Sch}_{\mathbf{Th}}$  have the types specified in Definition 8.3.

The components and types in CR1 through CR7, together with the distinction among their  $\mathfrak{C}/\mathfrak{D}/\mathfrak{B}/\mathfrak{D}$  origins, are inherited from Definitions 8.1, 8.2, and 8.3.

**Definition 8.15** (Reduction from a theory to a realized operational configuration). For a physical theory  $\mathbf{Th}$  in the comparison-realization class, define the realized operational configuration by

$$\begin{aligned}\mathcal{X}_{\mathbf{Th}} &:= \mathcal{X}_{\mathbf{Sch}_{\mathbf{Th}}} = (\mathcal{I}_{\mathbf{Sch}_{\mathbf{Th}}}^{\text{raw}}, \text{Dat}_{\mathbf{Sch}_{\mathbf{Th}}}^{\text{src}}), \\ \text{Red}^{\text{op}}(\mathbf{Th}) &:= \mathcal{X}_{\mathbf{Th}}.\end{aligned}$$

**Theorem 8.16** (Reduction from the comparison-realization class to realized operational configurations). For every physical theory  $\mathbf{Th}$  in the comparison-realization class,  $\text{Red}^{\text{op}}(\mathbf{Th})$  from Definition 8.15 is a realized operational configuration in the sense of Definition 3.25.

*Proof.* By CR1 through CR7 of Definition 8.14, the selected  $\mathbf{Sch}_{\mathbf{Th}}$  has the general form specified in Definitions 8.1 and 8.3. Consequently,  $\mathcal{X}_{\mathbf{Sch}_{\mathbf{Th}}}$  is a realized operational configuration in the sense of Definition 3.25. By Definition 8.15, this equals  $\mathcal{X}_{\mathbf{Th}}$  and hence is  $\text{Red}^{\text{op}}(\mathbf{Th})$ .  $\square$

**Proposition 8.17** (Representative theories supply the input data of the comparison-realization class). The representative theories corresponding to

$$\begin{array}{ccc}\mathbf{Sch}_{\text{NCM}}, & \mathbf{Sch}_{\text{SR}}, & \mathbf{Sch}_{\text{GR}}, \\ \mathbf{Sch}_{\text{TM}}, & \mathbf{Sch}_{\text{GPS}}, & \mathbf{Sch}_{\text{QG}}\end{array}$$

supply the data in CR1 through CR7 of Definition 8.14. Therefore, each representative theory belongs to the comparison-realization class.

*Proof.* Definitions 8.1 and 8.3 show that the general form of a standard scheme has the data types in CR1 through CR7. Definitions 8.7 and 8.8 through 8.12 give the theory-specific components of this general form for the six representative schemes. Therefore, the representative theory corresponding to each scheme belongs to the comparison-realization class.  $\square$

**Corollary 8.18** (Common generation-stage judgments for representative theories). For each representative theory in Proposition 8.17, the operability judgment and the judgments at the comparison, representation, continuous-approximation, dynamical-description, law-representation, and differential-law-form stages take the corresponding realized operational configuration  $\mathcal{X}_{\mathbf{Th}} = \text{Red}^{\text{op}}(\mathbf{Th})$  as a domain-side input. The failure location of each stage condition is classified by the failure-position tokens of Definition 6.10.

*Proof.* By Proposition 8.17, each representative theory belongs to the comparison-realization class. By Theorem 8.16, the realized operational configuration of each representative theory is determined. Each judgment from the operability judgment through the differential-law-form-stage judgment takes this realized operational configuration and the additional inputs required at that stage as domain-side inputs. The failure location of each stage condition is classified by the failure-position tokens specified in Definition 6.10.  $\square$

#### 8.1.4 Interpretation-Path-Specific Multistage Misrecognition

**Definition 8.19** (Multistage misrecognition relative to generation order). In time-reference construction, the objects and grounding conditions of each stage are generated from raw operational inputs in the dependency order defined by the sequence from comparison through representation and dynamical description to law representation and differential law form, together with the branch from dynamical description to continuous approximation. For an interpretation-specific evidence-comparison path from Definition 7.117, an explanation containing the following operations is called a multistage misrecognition.

1. It places an object, quantity, structure, or its bearer generated at a later stage as an input or given at a stage preceding its generation conditions.
2. It identifies the satisfaction of a local or partial stage with the satisfaction of a subsequent or global stage that requires additional conditions.
3. It redescribes a relational quantity or comparison-derived difference generated at a later stage as a change in an attribute of an entity posited before construction.
4. It transfers the establishment of preceding evidence to a different object, predicate, evidence type, grounding condition, or domain of application and treats it as a direct demonstration without independent evidence or a valid derivation specific to the destination of the transfer.

Within one interpretation-specific evidence-comparison path, each of the preceding operations that occurs successively by direct dependency is counted as one stage. Neither the number of evidence-effect layers nor the number of enumerated items is taken directly as the number of stages of misrecognition. Even for the same statement, different permissible interpretations can have different numbers of stages of misrecognition.

If the number of evidence-effect transgressions along an interpretation-specific evidence-comparison path is 0, no multistage misrecognition holds under that interpretation. If an unqualified temporal-property expression transfers a comparative difference on a physical carrier to a property of time, that object-type transfer is a one-stage misrecognition. Under a literal explanation that places an unconstructed time entity in advance and attributes a comparative difference to a change in an attribute of that entity, placement before generation and attribution of the property occur successively by direct dependency and form a two-stage misrecognition.

**Proposition 8.20** (Interpretation-path-specific application of the unbounded inheritance of limited results). Apply Definitions 7.115, 7.116, and 8.19 to each representative scheme in Definition 8.5.

For each reference, retain mathematical consistency, predictive performance, model fit, empirical agreement, and comparative differences on physical carriers within their respective evidence-effect domains. If an interpretation-specific evidence-comparison path transfers an object, predicate, evidence type, grounding condition, or domain of application without independent evidence or a valid derivation, determine that transfer with direct dependency as a multistage misrecognition.

If the same reference or the same statement has multiple permissible interpretations, determine the establishment and number of stages of misrecognition separately for each interpretation. No multistage misrecognition holds under an interpretation that directly describes a comparative difference on a physical carrier or under an interpretation for



which the original-source text states its figurative character, states that it does not designate a literal deformation of time, and identifies its actual referent. An unqualified temporal-property expression constitutes a one-stage misrecognition even if it can be interpreted retrospectively as shorthand. A literal time-stretching account that places an unconstructed time entity in advance constitutes a two-stage misrecognition; a path that then transfers evidential effect to a different object domain acquires subsequent stages according to its direct dependencies.

The number of stages of misrecognition is not determined uniformly from the name of a field or reference. It is determined by the direct-dependency path formed by the formal construction, statements, and permissible interpretations of each reference.

*Proof.* By Proposition 7.118, a claim beyond the evidence-effect boundary of a limited result requires independent evidence or a valid derivation specific to that claim. By Corollary 7.119, inheritance to a result of primitive-time dependence and inheritance to a claim outside the domain of application of a limited result are distinct audit axes.

Definition 7.117 separates an interpretation-specific evidence-comparison path for each permissible interpretation of the same statement and counts evidence-effect transgressions connected by direct dependency. Definition 8.19 determines placement before generation, identification of stages of establishment, reattribution to a property of time, and transfer of evidential effect to another object domain along that path as operations of misrecognition. It is therefore possible to preserve the formal results of each representative scheme within their evidence-effect domains while determining only the stages of misrecognition actually constituted by the original-source statements and their permissible interpretations.  $\square$

### Two independent inheritance audits

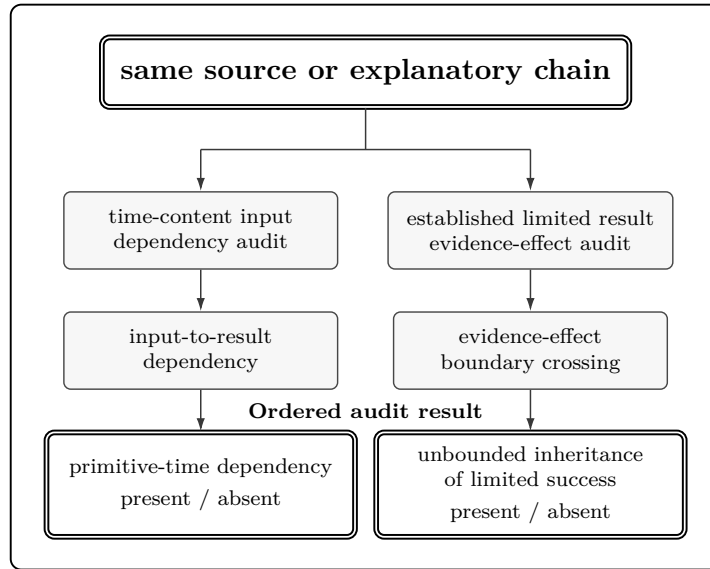


Figure 21: Primitive-time-dependence result inheritance and transgression of the evidence-effect domain of limited results are determined as distinct audit axes. The two determinations are not identified and are retained as an ordered pair for the same reference or explanatory sequence.

Apply validity localization, approximation localization, classification by the failure-position tokens of Definition 6.10, and structural-burden correspondence to the determination of representative schemes and classical time. For representative schemes, the realized

operational configuration supplied by each theory is matched to its structural burdens. In the classical domain, this separates the subclass on which a classical time representation holds, the scale-indexed validity domain on which the continuous approximation is valid, and the boundary of that domain.

**Proposition 8.21** (Loss of physical-time qualification in modern physics under primitive-time premises). For any representative scheme  $\text{Sch}$ , let the complete evaluation specification of Definition 8.5 be

$$\begin{aligned} \text{Ev}_{\text{Sch}} \\ = ((\mathcal{X}_{\text{Sch}}; \mathcal{J}_{\text{cmp}, \text{Sch}}), o, \zeta_{\text{Sch}}) \end{aligned}$$

When the time content and grounding structure used by the physical-law time variable of a representative theory are generated from nontemporal physical inputs prior to law operations and result derivation, that time variable satisfies physical-time qualification.

When time content enters a law operation or result derivation before the grounding structure required for its use, that time content is a primitive-time input, and the corresponding time variable does not satisfy physical-time qualification. Simultaneous mathematical consistency, predictive performance, and empirical fit do not repair this loss of qualification.

By Proposition 7.191, among the 72 audited references, the numbers of references that eliminate primitive-time inputs from all dependency roots and that give complete time generation from nontemporal physical inputs are, respectively, 0/72. The number of references that use at least one temporal dependency root before the generation of time content is 72/72. By Theorem 7.190, primitive-time roots and primitive-time-root-dependent results exist in all 72 references. By Theorem 7.201, all 72 references fail to satisfy the generation-completion condition for the entire set of time contents used within them.

Therefore, a time variable in a physical law is not established on a primitive premise. So long as they retain their present use of time variables without alteration, the theoretical descriptions of modern physics represented by the 72 audited references do not satisfy physical-time qualification. That is, modern physics does not hold as it stands. This nonestablishment is a loss of physical-time qualification concerning the order of time-content generation, law use, and result propagation. Mathematical consistency, predictive performance, and empirical fit are retained as evaluation successes of the respective theories.

*Proof.* By Definition 8.5, the components of a complete evaluation specification are a realized operational configuration, a typed family of comparison rules, an observer-side information-coupling position, and an object-specification index.

When time content and its grounding structure are generated prior to law operations and result derivation, Theorem 7.200 gives physical-time qualification to the time variable of a physical law that uses that time content.

When time content is used before the required grounding structure, Definition 7.10 makes that time content a primitive-time input. By Theorem 7.11, that time content does not satisfy physical-time qualification. By Proposition 7.13, simultaneous mathematical consistency, predictive performance, and empirical fit do not repair this loss of qualification.

Proposition 7.191 gives 0/72 both for elimination of primitive-time inputs from all dependency roots and for complete time generation, and 72/72 for the use of a temporal dependency root before generation. Theorem 7.190 gives the existence of primitive-time

roots and primitive-time-root-dependent results in all references. Theorem 7.201 gives failure of the generation-completion condition in all references.

The theoretical descriptions of modern physics represented by the 72 audited references therefore retain time content that lacks physical-time qualification and results that depend on that content. By the generation-priority condition in Corollary 7.15, a time variable in a physical law is not established on a primitive premise. Hence, so long as present time-variable use is retained without alteration, modern physics does not hold as it stands.  $\square$

## 8.2 Theory-Specific Structural Consequences

The consequence for each representative theory consists of an application of the general generation theorems and the structural burdens specific to that theory. For a realized operational configuration

$$\mathcal{A}_{\text{Sch}}$$

the general generation theorems supply the operability determination, comparison-stage determination, representation-stage determination, continuous-approximation-stage determination, dynamical-description-stage determination, law-representation-stage determination, differential-law-form-stage determination, and first structural-failure determination. The theory-specific structural burdens supply the conditions that the inputs of each theory must satisfy at the respective generation stages.

In SR, inertial coordinate time supplies an order representation based on the observer-time law core. In GR, comparison burdens with branch resolution appear in the canonical comparison quotient at the comparison stage or in descent conditions at the dynamical-description stage. In time metrology, standard frequency, period, and numerical values in Hz are determined from a traceability-based quotient order,  $\mathfrak{C}$ -derived comparison-representation coordinates, and quantity-affine differences. In GPS, a rotation-derived loop defect appears as a failure of global gluing at the comparison stage. In quantum reconstruction, the conditions for generating a law-admissible time representation are determined from records that exclude external time from primitive inputs, branch sensitivity, object-state differences, observer-side binding, and comparison-gluing conditions.

Table 17: Input burdens of representative schemes

| Representative | Input carriers                                                                                    | Comparison burdens                                                                       | Motion and law burdens                                                                      |
|----------------|---------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| NCM            | Shared clock<br>Stable records<br>Object-state sequence                                           | Shared-clock domain<br>Single ordering                                                   | Branching burden<br>Single-valued observational binding                                     |
| SR             | Inertial-observer network<br>Local clocks<br>Synchronization rules                                | Inter-inertial closure<br>Order representation                                           | Law-kernel representation burden                                                            |
| GR             | Local-observer network<br>Worldline-dependent reception<br>Transport data                         | Proper-time display<br>Global gluing                                                     | Branch resolution<br>Transport preservation<br>Quotient descent<br>Law single-valuedness    |
| TM             | Reference cycles<br>Counting convention<br>Correction records                                     | Counting comparison<br>Representation-coordinate descent                                 | Frequency-representation burden<br>Period-representation burden<br>Hz-representation burden |
| GPS            | Satellite clocks<br>Reception records<br>Correction rules<br>Transport rules                      | Multiple-clock comparison<br>Propagation correction<br>Loop consistency<br>Global gluing | Navigation-time representation<br>Law-kernel representation burden                          |
| QG             | External time excluded from primitive inputs<br>Records<br>Branch sensitivity<br>Observer binding | Comparison gluing<br>Object-state identification<br>Grounding burden                     | Dynamical description<br>Operability<br>Law-kernel generation burden                        |

Table 18: Generated objects and typical failures in representative schemes

| Representative | Generated objects                                             | Typical failures |
|----------------|---------------------------------------------------------------|------------------|
| NCM            | Classical time                                                | FDyn             |
|                | Classical continuous approximation                            | FL               |
|                | Classical law display                                         | FDiff            |
| SR             | Inertial coordinate time                                      | FR               |
|                | Generated representation                                      | FDyn             |
| GR             | Local proper-time display                                     | FC               |
|                | Global law-time representation                                | FDyn             |
|                | Stage difference                                              | FL               |
| TM             | $\mathfrak{C}$ -derived comparison-representation coordinates | FR               |
|                | Standard frequency and period                                 | FA               |
|                | Numerical values in Hz                                        |                  |
| GPS            | Navigation time                                               | FC               |
|                | Localization of rotation-derived loop defects                 | FA               |
|                |                                                               | FL               |
| QG             | Law-admissible time representation with external time         | FDyn             |
|                | excluded from primitive inputs                                | FL               |

**Proposition 8.22** (Type boundary of representative schemes). A representative scheme supplies the input carriers, comparison burdens, dynamical-description burdens, and law-representation burdens of an existing theory as a realized operational configuration and the additional inputs of the respective stages. It separates by type the theoretical content supplied as theory-specific input from the time comparisons, time representations, continuous-approximation validity, dynamical descriptions, law-admissible time representations, differential-law forms, scale-indexed validity domains, and failure-position tokens constructed at the generation stages.

*Proof.* Tables 17 and 18 separate the carriers supplied by representative theories as theory-specific inputs, the burdens determined at the generation stages, the generated objects, and the typical failures. NCM, SR, GR, TM, GPS, and QG have different input carriers, but by Corollary 8.18, the corresponding realized operational configurations and additional inputs at each stage follow the common generation order of the sequence from comparison through representation and dynamical description to law representation and differential law form, together with the branch from dynamical description to continuous approximation. Failure locations for the conditions of each stage are classified by the failure-position tokens of Definition 6.10. A representative scheme is therefore an input type that determines, in the same generation order and without conflating theory-specific inputs with generated objects, the time-grounding burdens, law-representation burdens, differential-law-form burdens, and scale-indexed validity domains of each theory.  $\square$

Relativistic calculations supply highly precise correspondences among clock readings, coordinate displays, proper-time displays, dynamical descriptions, and observational records. What those correspondences establish comprises clock-reading differences, coordinate-time

differences, proper-time differences along worldlines, metric structures, and the relations among them.

Under descriptions that retain time or spacetime as a preconstruction entity, these highly precise correspondences have been described as stretching, slowing, speeding, or deformation of a time entity or spacetime entity.

By Theorem 7.201, none of the 72 audited references generates from nontemporal physical inputs the entire set of time contents used within it. By Proposition 7.13, mathematical consistency, predictive performance, and empirical fit do not repair the physical-time qualification of a primitive-time input. Success of relativistic calculation therefore does not entail the stretching, slowing, speeding, or deformation of an independent time entity or spacetime entity.

The pattern that retains time or spacetime as a preconstruction entity and explains a postgeneration relational quantity as a change in an attribute of that entity also occurs in subsequent quantum theory, quantum gravity, spacetime emergence, and cosmology. Transferring the dependency into subsequent theories while retaining primitive-time inputs causes questions that should be determined as matters of time-reference construction, generated law-admissible time representations, scale-indexed validity domains, and failure boundaries to be inherited as undifferentiated problems of time, spacetime, observation, or gravity.

**Proposition 8.23** (Two-stage misrecognition in the time-stretching account). In the generation order of this paper, relativistic clock differences are determined as comparison-derived differences among generated law-admissible time representations corresponding to multiple observers or worldlines. The time-stretching account posits time as an unconstructed entity prior to these comparison-derived differences and explains relativistic clock differences by the stretching, slowing, speeding, or deformation of that time entity. This explanatory form is therefore a two-stage misrecognition of the time-stretching account in which Items 1 and 3 of Definition 8.19 occur in a sequence of direct dependence.

*Proof.* In the representative schemes, time use in SR and GR follows the generation order from realized operational configurations along the sequence from comparison through representation and dynamical description to law representation and differential law form, together with the branch from dynamical description to continuous approximation. A relativistic clock difference is a comparison-derived difference among multiple law-admissible time representations obtained in this generation order.

Prior to that comparison-derived difference, the time-stretching account places time as an unconstructed entity at a stage preceding its generation conditions. It therefore satisfies Item 1 of Definition 8.19. It further redescribes the comparison-derived difference generated at a later stage as a change in an attribute of that time entity—its stretching, slowing, speeding, or deformation—and therefore satisfies Item 3 of the same definition. The redescription in Item 3 has as its antecedent the prior input of a time entity under Item 1. Items 1 and 3 are thus connected by direct dependence, making this explanatory form a two-stage misrecognition in the time-stretching account.  $\square$

**Theorem 8.24** (Structural consequences for Newtonian and classical mechanics). For a complete evaluation specification  $\text{Ev}_{\text{Sch}_{\text{NCM}}}$  and law-evaluation datum  $\text{LawEvalDat}_{o, \zeta_{\text{Sch}_{\text{NCM}}}}$ , suppose that

$$\text{LawOperative} \left[ \text{LawEvalDat}_{o, \zeta_{\text{Sch}_{\text{NCM}}}} \right]$$

holds and that

$$\text{RepEmb}_o^{\text{dyn,time}} \in \text{OrdEmbSet}_o^{\text{dyn,time}}(\mathbb{R})$$

Then the observer-time law core

$$\text{LawCore}_{o, \zeta_{\text{Sch}_{\text{NCM}}}^{\text{obs,time}}}(\text{RepEmb}_o^{\text{dyn,time}})$$

is determined. The classical time variable displays values of the law-admissible time representation  $\text{TimeRep}_o^{\text{dyn}}$ . When the branching burden, observational-binding burden, and law-output determinacy hold in the classical domain, the laws of Newtonian and classical mechanics may use this generated time representation within their safe domain of application.

*Proof.* By Definition 5.102, law-description operability includes the simultaneous satisfaction of dynamical-description operability and the family of law conditions. Definition 5.108 determines the observer-time law core relative to the displayed real-valued time-order embedding. By Definition 5.63, the law-admissible time representation maps the  $\mathfrak{C}$ -derived canonical observer-time quotient to real-valued time.

The shared-clock domain of Definition 8.7 maps clock-side material to the same order representation; stable records connect object-state-side material to the same law-bearing domain; and the observer-side binding conditions establish the four-party binding at the same phase. In the classical domain in which law-output determinacy holds, the classical time variable displays values of the generated law-admissible time representation.

By Theorem 7.200, when time content and grounding structures are generated prior to law operations and result derivation, the time variable satisfies physical-time qualification. Failures of the conditions at each stage are classified by failure-position tokens of Definition 6.10.  $\square$

**Connection to physical-time qualification.** Time content appearing in the shared-clock domain, single ordering, and classical law display of Newtonian and classical mechanics satisfies the physical-time qualification of Theorem 7.200 when the corresponding grounding structures precede law operations and result derivation. When time content is input before its grounding structures, Theorem 7.11 entails that the content lacks physical-time qualification.

**Proposition 8.25** (Inheritance of primitive-time dependence in Newtonian and classical mechanics). In Newtonian and classical mechanics, if a classical time variable is used in equations of motion, initial-value specifications, parametric representations of trajectories, or the evolution variable of a phase flow before its time content and grounding structures are generated, that time variable becomes a primitive-time input. Trajectories, velocities, accelerations, initial-value evolution, identification of conserved quantities, determination of their temporal evolution, phase flows, and law results constructed in dependence on this input inherit primitive-time dependence, and the time content contained in each result lacks physical-time qualification. Mathematical consistency, predictive performance, operational reproducibility, and empirical fit of the equations of motion do not repair this loss of qualification.

The law-admissible time representation generated under the conditions of Theorem 8.24 may be used as a law-time input in Newtonian and classical mechanics within its safe domain of application. Formulating velocity, acceleration, phase flow, or evolution by differential equations additionally requires differential-stage operability under Definition 5.132.

*Proof.* When pregeneration time content is used in equations of motion, initial-value specifications, parametric representations of trajectories, or the evolution variable of a phase flow, Definition 7.10 and Theorem 7.11 make that use a primitive-time input lacking physical-time qualification. By Corollary 7.12, dependence on that time content is inherited by trajectories, velocities, accelerations, initial-value evolution, identification of conserved quantities, determination of their temporal evolution, phase flows, and law results. By Proposition 7.13, mathematical consistency, predictive performance, operational reproducibility, and empirical fit do not repair this loss of qualification.

By Theorem 8.24, satisfaction of law-description operability and a real-valued time-order embedding generates the law-admissible time representation displayed by the classical time variable. By Definition 5.132, differential-stage operability includes simultaneous satisfaction of law-description operability and the five differential-stage conditions. The safe domain of application for time-dependent quantities and differential-law forms in Newtonian and classical mechanics is therefore the domain in which the generated law-admissible time representation is used and the required differential-stage conditions hold.  $\square$

**Proposition 8.26** (Safe reuse of Newtonian and classical mechanics through generated time and failure classification). When the conditions of Theorem 8.24 hold, the laws, initial-value evolution, trajectories, and conserved quantities of Newtonian and classical mechanics that use a generated law-admissible time representation hold within their safe domain of application. Velocity, acceleration, phase flow, and evolution by differential equations additionally require differential-stage operability under Definition 5.132.

When a condition at any generation stage fails, the failure-position token system of Definition 6.10 and the set of minimal failure locations of Theorem 6.168 determine the condition that must be repaired first. Repair proceeds upstream-to-downstream according to the failure-generation dependency order of Definition 6.17, and downstream results are recomputed from the repaired inputs. Within the domain in which the grounding conditions again hold, the corresponding classical laws, initial-value evolution, trajectories, conserved quantities, and phase flows can be retained.

When an error source in initial values, a time representation, or a differential approximation is repaired, the repair's effect on accuracy is determined by an accuracy evaluation of the recomputed downstream results. The decision to use a classical law therefore separates into retention within its safe domain of application and repair beginning at a minimal failure location.

*Proof.* By Theorem 8.24, satisfaction of law-description operability and a real-valued time-order embedding supplies the generated law-admissible time representation used as the law-time input of Newtonian and classical mechanics. By Definition 5.132, the differential-law forms required for velocity, acceleration, phase flow, and evolution by differential equations are retained on the domain of differential-stage operability.

Definition 6.10 maps each failed condition to its corresponding failure-position token. Theorem 6.168 determines the failed locations that are minimal with respect to the failure-generation dependency order. By Definition 6.17, conditions and results downstream of such a location depend on upstream conditions. This determines an order in which the cause at a minimal failure location is repaired and downstream results are recomputed.

When an error source belongs to an initial value, a time representation, or a differential approximation, downstream results can be recomputed from inputs in which that source has been repaired, and an accuracy evaluation can be applied to the recomputed results. Existing results can consequently be retained where the grounding conditions hold, while



the repair order begins at the minimal failure location and accuracy evaluation remains conditional on recomputation.  $\square$

**Theorem 8.27** (Structural consequences for special relativity). Suppose that

$$\text{LawOperative} \left[ \text{LawEvalDat}_{o, \zeta_{\text{Sch}_{\text{SR}}}} \right]$$

holds and that

$$\text{RepEmb}_o^{\text{dyn}, \text{time}} \in \text{OrdEmbSet}_o^{\text{dyn}, \text{time}}(\mathbb{R})$$

Then the observer-time law core

$$\text{LawCore}_{o, \zeta_{\text{Sch}_{\text{SR}}}}^{\text{obs}, \text{time}} (\text{RepEmb}_o^{\text{dyn}, \text{time}})$$

is determined. Inertial coordinate time in SR supplies the coordinate display of the law-admissible time representation contained in this law core.

*Proof.* By Definition 5.102, law-description operability includes simultaneous satisfaction of dynamical-description operability and the family of law conditions. Definition 5.108 determines the observer-time law core, and by Definition 5.63, the law-admissible time representation maps the  $\mathfrak{C}$ -derived canonical observer-time quotient to real-valued time.

By Definition 8.10, the inertial-observer network, local clock-side reception, synchronization procedures, and comparison rules corresponding to inertial-coordinate transformations supply the inputs of SR. Within the domain in which these inputs satisfy the conditions of each generation stage, inertial coordinate time is the coordinate display of a generated law-admissible time representation.  $\square$

**Theorem 8.28** (Structural consequences for general relativity). When

$$\text{ExactOperable} (\mathcal{X}_{\text{Sch}_{\text{GR}}}; \mathcal{J}_{\text{cmp}, \text{Sch}_{\text{GR}}})$$

holds, worldline-local comparison data and generated representation coordinates supply the local proper-time display  $\tau_{u, \text{Sch}_{\text{GR}}}^{\text{loc}}$ .

When, in addition,

$$\text{DynOperative} [\text{Ev}_{\text{Sch}_{\text{GR}}}]$$

holds, the  $\mathfrak{C}\mathfrak{D}\mathfrak{B}$ -derived canonical trajectory relation  $\text{TrjRel}_o^{\text{dyn}, \text{can}}$  is determined. When law-description operability holds, the observer-time law core is determined. The existence of a local proper-time display alone does not entail right-uniqueness of the canonical trajectory relation or existence of the observer-time law core.

*Proof.* By Definition 8.11, worldline-local clock-side reception, record-side reception, local comparison rules, and generated representation coordinates construct the local proper-time display.

At the dynamical-description stage, descent of the  $\mathfrak{C}\mathfrak{D}\mathfrak{B}$ -derived canonical trajectory relation of Definition 5.23 requires comparison-stage closure and the family of dynamical-description conditions. Its right-uniqueness requires the dynamical-description-stage canonical-trajectory-function condition of Definition 5.26. The law stage requires the law-description operability of Definition 5.102 and the real-valued time-order embedding of Definition 5.108.

If branch resolution, transport preservation, quotient descent, or law-output determinacy fails, right-uniqueness of the canonical trajectory relation or existence of the observer-time law core does not hold even when a local proper-time display is supplied.  $\square$

**Theorem 8.29** (Structural consequences for time metrology). Suppose that

$$\text{ExactOperable}(\mathcal{X}_{\text{Sch}_{\text{TM}}}; \mathcal{J}_{\text{cmp}, \text{Sch}_{\text{TM}}})$$

holds and that a  $\mathfrak{C}$ -derived conditional quantity-affine realization

$$\text{QtyAffReal}_{\mathfrak{C}}(\mathcal{X}_{\text{Sch}_{\text{TM}}}; \mathcal{J}_{\text{cmp}, \text{Sch}_{\text{TM}}})$$

is specified. Suppose further that a nonempty family of admissible counting windows for the reference source is given and satisfies the window-invariance condition of Definition 4.150. Then  $\mathfrak{C}$ -derived comparison-representation coordinates, a standard-frequency representation, a period representation, and an Hz unit scalar are determined from the family of reference cycle-count sources, the cycle-counting convention, counting comparisons and corrections, correction records, the canonical comparison quotient, and generated representation coordinates.

*Proof.* By Definition 4.175, exact operability includes completion of the comparison and representation stages. By Definition 8.6, the family of reference cycle-count sources, the cycle-counting convention, counting comparisons and corrections, and correction records are connected to the canonical comparison quotient and generated representation coordinates.

The  $\mathfrak{C}$ -derived conditional quantity-affine realization supplies quantity-affine differences of the generated representation coordinates. A cycle-count difference together with this quantity-affine difference determines the frequency and period representations, and Definition 4.152 determines the standard-frequency representation and the Hz unit scalar. The traceability network thus supplies standard-frequency and period representations and numerical values in Hz based on the generated comparison order and quantity-affine differences, without taking a primitive frequency as input.  $\square$

**Theorem 8.30** (Structural consequences for GPS). Within a domain in which the conditions at every generation stage of the GPS scheme hold, the navigation-time variable displays values of a generated law-admissible time representation.

Suppose that pure transport words

$$\mathbf{w}_0, \mathbf{w}_1 \in \text{Word}_{\bullet}^{\text{tr}}[\text{ISch}_{\text{GPS}}]$$

with the same initial and terminal points supply a witness of nonidentity action for the loop word  $\mathbf{w}_1^{\dagger} \star \mathbf{w}_0$ . Then

$$\neg G \left[ \text{Ev}_{\text{Sch}_{\text{GPS}}}^{\text{cmp}} \right]$$

holds, and global gluing at the comparison stage fails.

*Proof.* By Definition 8.9, the GPS scheme supplies families of clock-side sources at satellites and receivers, reception records, correction rules, transport rules, and observer-side binding conditions. Within a domain in which exact operability, dynamical-description operability, and law-description operability hold, the law-admissible time representation of Definition 5.63 is generated, and the navigation-time variable displays its values.

The assumed witness of nonidentity action refutes the requirement that every pure transport loop act identically. Consequently, the comparison-stage global-gluing condition

$$G \left[ \text{Ev}_{\text{Sch}_{\text{GPS}}}^{\text{cmp}} \right]$$

therefore does not hold, and its failure location is classified by a comparison-stage failure-position token of Definition 6.10.  $\square$

**Theorem 8.31** (Structural consequences for quantum reconstruction). The quantum-reconstruction scheme  $\text{Sch}_{\text{QG}}$  excludes external time from primitive inputs and takes families of clock-side, record-side, and object-state-side sources, branch-sensitive record structures, observer-side binding, and comparison-gluing conditions as inputs.

When

$$\text{ExactOperable}(\mathcal{X}_{\text{Sch}_{\text{QG}}}; \mathcal{J}_{\text{cmp}, \text{Sch}_{\text{QG}}})$$

holds, the canonical comparison quotient and representation-stage objects are determined. When, in addition,

$$\text{DynOperative}[\text{Ev}_{\text{Sch}_{\text{QG}}}]$$

holds, the  $\mathfrak{C}$ -derived canonical observer-time quotient,  $\mathfrak{B}$ -derived canonical object-state quotient,  $\mathfrak{C}\mathfrak{D}\mathfrak{B}$ -derived canonical trajectory relation, graph trajectory, family of canonical observable functions, and law-admissible time representation are determined. When law-description operability holds, the observer-time law core is determined.

*Proof.* By Definition 8.12, the quantum-reconstruction scheme supplies a realized operational configuration

$$\mathcal{X}_{\text{Sch}_{\text{QG}}} = (\mathcal{I}_{\text{Sch}_{\text{QG}}}^{\text{raw}}, \text{Dat}_{\text{Sch}_{\text{QG}}}^{\text{src}})$$

that excludes external time from primitive inputs, together with a typed family of comparison rules.

When exact operability holds, the canonical comparison quotient of Definition 4.89 and the representation-stage objects are determined. When dynamical-description operability holds, the canonical observer-time and object-state quotients of Definition 5.16, the canonical trajectory relation of Definition 5.23, the graph trajectory of Definition 5.28, the law-admissible time representation of Definition 5.63, and the family of canonical observable functions are determined.

When law-description operability holds and a  $\mathfrak{C}$ -derived real-valued time-order embedding is selected, Definition 5.108 determines the corresponding observer-time law core. Quantum reconstruction that excludes external time from primitive inputs therefore generates the law-bearing time representation and law core within the domain in which the conditions of the respective generation stages hold.  $\square$

### 8.3 Worked Reduction in Time Metrology

Specialize the time-metrology scheme of Definition 8.8 to the  $\mathfrak{C}$ -derived frequency and period representations of Definition 4.150 and the  $\mathfrak{C}$ -derived standard-frequency representation of Definition 4.152. The reference  $\mathfrak{C}$ -derived source, cycle-count index, record-side correspondence, comparison closure,  $\mathfrak{C}$ -derived canonical comparison quotient,  $\mathfrak{C}$ -derived comparison-representation coordinate, quantity-affine difference, frequency and period representations, and numerical value in Hz are determined in this dependency order.

The second, frequency, and Hz used in time metrology are supplied as standardized measurement implementations[12]. A standardized measurement implementation supplies numerical conventions, measurement reproducibility, and transfer among standards. If, however, the second, frequency, or Hz is used before the generation of time content as a time interval, calibration criterion, synchronization criterion, reference criterion for traceability, or law-time input, that use constitutes primitive-time input. The time-metrology scheme takes reference cycles, records, comparison rules, and unit-normalization conventions as inputs and determines frequencies, periods, and numerical values in Hz as quantities

on a generated comparison representation. This generation order connects standardized numerical values to time metrology that has physical-time qualification.

### 8.3.1 From Cycle Count to Frequency Representation

Consider the time-metrology scheme of Definition 8.8. Its reference  $\mathfrak{C}$ -derived source and normalization convention are given by that definition.

For each  $i \in \{0, 1\}$ , let the cycle-count indices be  $k_i \in \mathbb{N}$ ,  $k_0 < k_1$ , and let the corresponding  $\mathfrak{C}$ -derived source tokens be

$$x_{\mathfrak{C},i} := \text{tok}_{\mathfrak{C}}(j_{\text{std}}^{\text{src}}, k_i) \in \text{Tok}_{\mathfrak{C}, \text{Sch}_{\text{TM}}}^*$$

Let the record tokens containing the ancillary records  $m_i$  and derived from  $\mathfrak{D}$  be

$$x_{\mathfrak{D},i} = (k_i, m_i) \in \text{Tok}_{\mathfrak{D}, \text{Sch}_{\text{TM}}}^*$$

and supply the  $\mathfrak{C}/\mathfrak{D}$  correspondence by

$$(x_{\mathfrak{C},i}, x_{\mathfrak{D},i}) \in \text{Rel}_{\text{Sch}_{\text{TM}}}^{\text{src}}$$

The ancillary records include correction records, environmental records, counting comparisons and corrections, and record content required for traceability. Object-state-side tokens distinguish the comparison phase to which each record belongs.

Let the counting window for the observer-side information-coupling position  $u_{\text{std}} \in U$  be

$$\text{win}_{\text{std}} := (u_{\text{std}}, j_{\text{std}}^{\text{src}}, k_0, k_1)$$

Suppose that record-side retention, correction records, counting comparisons and corrections, and the typed family of comparison rules make the two  $\mathfrak{C}$ -derived source tokens comparable within the same comparison window of the same reference sequence. When the membership and strict-order conditions of Definition 4.150 hold,

$$\text{win}_{\text{std}} \in \text{CntWin}_{\mathfrak{C}}^{\text{adm}}[\text{Sch}_{\text{TM}}].$$

The window-endpoint images  $\text{CntCls}_{\mathfrak{C},i}[\text{Sch}_{\text{TM}}; \text{win}_{\text{std}}]$  ( $i \in \{0, 1\}$ ) are then elements of the  $\mathfrak{C}$ -derived canonical comparison quotient

$$\text{Qu}_{\mathfrak{C}}^{\text{cmp}, \text{can}}(\mathcal{X}_{\text{Sch}_{\text{TM}}}; \mathcal{J}_{\text{cmp}, \text{Sch}_{\text{TM}}})$$

and satisfy the  $\mathfrak{C}$ -derived canonical strict order.

When the structural representation stage of Definition 4.92 is complete and the  $\mathfrak{C}$ -derived conditional quantity-affine realization of Definition 4.148

$$\text{QtyAffReal}_{\mathfrak{C}}[\text{Sch}_{\text{TM}}]$$

is specified, the quantity-affine difference of the two window-endpoint images in the  $\mathfrak{C}$ -derived comparison-representation coordinates is determined. The cycle-count difference is  $k_1 - k_0$ .

**Proposition 8.32** (Generation from cycle counts to a frequency representation). Suppose that

$$\text{Operative}(\mathcal{X}_{\text{Sch}_{\text{TM}}})$$

and

$$\text{Repr}_{\leq 3}(\mathcal{X}_{\text{Sch}_{\text{TM}}}; \mathcal{J}_{\text{cmp}, \text{Sch}_{\text{TM}}})$$

hold and that a  $\mathfrak{C}$ -derived conditional quantity-affine realization is specified. Suppose further that

$$\text{win}_{\text{std}} \in \text{CntWin}_{\mathfrak{C}}^{\text{adm}}[\text{lsch}_{\text{TM}}]$$

Then the  $\mathfrak{C}$ -derived counting-window frequency representation is determined as

$$\nu_{\mathfrak{C}}^{\text{win}}[\text{lsch}_{\text{TM}}; \text{win}_{\text{std}}] = \frac{\text{CntDiff}_{\mathfrak{C}}^{\text{qty}}[\text{win}_{\text{std}}]}{\text{RepDiff}_{\mathfrak{C}}^{\text{qty}}[\text{lsch}_{\text{TM}}; \text{win}_{\text{std}}]} \in \text{QtyVal}_{\mathfrak{C}}$$

Its product with the  $\mathfrak{C}$ -derived counting-window period representation is  $1_{\mathfrak{C}}^{\text{qty}}$ .

When window invariance holds for the reference  $\mathfrak{C}$ -derived source, the common value in the preceding expression supplies the  $\mathfrak{C}$ -derived standard-frequency representation. The normalization convention  $N_{\text{std}}$  determines the  $\mathfrak{C}$ -derived Hz unit scalar, and the numerical value in Hz relative to that unit scalar is determined by Definition 4.151.

*Proof.* By the definition of an admissible counting window, the two window-endpoint images satisfy the strict order on the  $\mathfrak{C}$ -derived canonical comparison quotient. By Proposition 4.153, their quantity-affine difference in the  $\mathfrak{C}$ -derived comparison-representation coordinates is positive, so the  $\mathfrak{C}$ -derived counting-window frequency and period representations are determined. The same proposition gives their product as the  $\mathfrak{C}$ -derived quantity unit  $1_{\mathfrak{C}}^{\text{qty}}$ .

When the window-invariance condition of Definition 4.150 holds for the reference  $\mathfrak{C}$ -derived source, the common value of the counting-window frequency representations becomes the  $\mathfrak{C}$ -derived source-relative frequency representation. Definition 4.152 makes this common value the  $\mathfrak{C}$ -derived standard-frequency representation and determines the  $\mathfrak{C}$ -derived Hz unit scalar from the normalization convention. Frequency, period, and numerical values in Hz are therefore determined in the dependency order from cycle-count records through comparison closure,  $\mathfrak{C}$ -derived comparison-representation coordinates, quantity-affine differences, window invariance, and the unit-normalization convention.  $\square$

**Connection to physical-time qualification.** Cycle-count records,  $\mathfrak{C}$ -derived frequency and period representations, and numerical values in Hz in time metrology are used to calibrate time variables in physical laws or as quantitative inputs within laws. Time variables in physical laws that use these quantities are subject to the generation-priority condition of Corollary 7.15. By Proposition 8.21, when time content and grounding structures are generated prior to law operations and result derivation, the time variable satisfies physical-time qualification. When time content is used before the required grounding structures, the time variable does not satisfy physical-time qualification.

In addition to the reference  $\mathfrak{C}$ -derived source,  $\mathfrak{C}/\mathfrak{D}$  record correspondence, typed family of comparison rules, and structural representation stage, an admissible counting window, quantity-affine realization, window invariance, and unit normalization are required according to the quantity used. Continuous approximation requires a finite-operational-granularity datum and continuous-approximation evaluation input. If a formed and evaluable condition in the six series fails, a failure-position token of Definition 6.10 classifies each minimal failure location. Non-satisfaction of a quantity-specific condition is determined as a defining condition of the corresponding quantity, in accordance with Proposition 8.35.

**Proposition 8.33** (Inheritance of primitive-time dependence in time metrology). If, before the generation conditions of the time-metrology scheme hold, the second, frequency, period, or a numerical value in Hz is used as a time interval, calibration criterion, synchronization criterion, reference criterion for traceability, or law-time input, that use constitutes a primitive-time input. Calibration values, frequency ratios, time scales, synchronization results, calibration chains, and law results that depend on these quantities inherit primitive-time dependence, and the time content contained in each result does not satisfy physical-time qualification. Measurement accuracy, reproducibility, agreement among standards, and traceability do not repair this loss of qualification.

When the generation conditions of the time-metrology scheme hold, frequency, period, and numerical values in Hz are generated from cycle-count records, comparison closure,  $\mathfrak{C}$ -derived comparison-representation coordinates, quantity-affine differences, window invariance, and the unit-normalization convention. When these quantities are used as law-time inputs, they become law-time inputs with physical-time qualification if law-description operability holds and the generation of law-time content precedes law operations and result derivation.

*Proof.* When time content is used in a law operation or result derivation before the generation conditions hold, Definition 7.10 and Theorem 7.11 make that use a primitive-time input that does not satisfy physical-time qualification. By Corollary 7.12, dependence on that time content is inherited by calibration values, frequency ratios, time scales, synchronization results, calibration chains, and law results. By Proposition 7.13, measurement accuracy, reproducibility, agreement among standards, and traceability do not repair this loss of qualification.

When the generation conditions hold, Proposition 8.32 generates frequency, period, and numerical values in Hz according to the dependency order from cycle-count records to the unit-normalization convention. Definition 5.102 and Corollary 7.15 give physical-time qualification to law-time inputs when law-description operability holds and generated time content precedes law operations and result derivation.  $\square$

**Corollary 8.34** (Noncircularity in time metrology). The primitive inputs of the time-metrology scheme consist of a reference  $\mathfrak{C}$ -derived source, cycle-count indices, record-side correspondence, a typed family of comparison rules, and a normalization convention. Frequency, period, and numerical values in Hz are determined from the canonical comparison quotient,  $\mathfrak{C}$ -derived comparison-representation coordinates, quantity-affine differences, and window invariance generated from these primitive inputs.

*Proof.* By Proposition 8.32, the  $\mathfrak{C}$ -derived counting-window frequency representation is determined as the ratio of a cycle-count difference to a quantity-affine difference of generated comparison-representation coordinates. The  $\mathfrak{C}$ -derived standard-frequency representation is determined by window invariance, and the  $\mathfrak{C}$ -derived Hz unit scalar and numerical values in Hz are determined by the unit-normalization convention. The dependency order from primitive inputs through comparison, representation, quantification, and normalization therefore generates the frequency, period, and numerical values in Hz of time metrology.  $\square$

**Proposition 8.35** (Localization of failure results in the worked reduction for time metrology). When a generation condition for a frequency representation, period representation, or numerical value in Hz fails in the time-metrology scheme and belongs to one of the six failure series, each minimal failure location is classified by a failure-position token of Definition 6.10.

- (1) When a formed and evaluable comparison-series condition concerning  $\mathfrak{C}/\mathfrak{D}$  record correspondence or comparison rules fails, each minimal failure location belongs to the comparison stage.
- (2) After the  $\mathfrak{C}$ -derived canonical comparison quotient has been established, failure of a formed and evaluable representation-series condition has each minimal failure location in the representation stage. Quantity-affine realization, admissible counting windows, window invariance, and unit normalization are evaluated as defining conditions of the quantities that require them.
- (3) When the finite-operational-granularity datum or continuous-approximation evaluation input is insufficient, the corresponding approximation condition is unformed or unevaluable. When a formed and evaluable condition among the nine continuous-approximation conditions fails, each minimal failure location belongs to the continuous-approximation stage.
- (4) When a formed and evaluable series condition connecting a generated frequency representation, period representation, or numerical value in Hz to a dynamical description, law representation, or differential-law form fails, each minimal failure location belongs to the corresponding subsequent stage.

*Proof.* By Definition 4.150 and Proposition 4.153, generation of frequency and period representations depends on the comparison stage, structural representation stage, quantity-affine realization, and an admissible counting window. By Definition 4.152, the standard-frequency representation and Hz unit scalar depend on window invariance and the unit-normalization convention.

Definitions 4.156 and 4.168 determine the finite-operational-granularity datum and continuous-approximation evaluation input. By Proposition 6.76, the evaluation input containing the finite-operational-granularity datum and continuous-approximation evaluation input determines the evaluation states of the nine continuous-approximation conditions. Definition 6.77 makes failure of those nine conditions a continuous-approximation-stage failure.

The defining conditions of the quantities above determine whether frequency, period, and numerical values in Hz are generated. For conditions in the six series, Definition 6.21 determines the formation and evaluation states, and Proposition 6.26 establishes that the failed-token set equals its set of minimal elements. Each token of Definition 6.10 retains the series and condition index of the failed condition.  $\square$

**Proposition 8.36** (Safe reuse of time-metrology results through generated time and failure classification). When the generation conditions of Proposition 8.32 hold, the frequency, period, and numerical values in Hz obtained from cycle-count records, comparison closure,  $\mathfrak{C}$ -derived comparison-representation coordinates, quantity-affine differences, window invariance, and the unit-normalization convention can be used within the corresponding admissible counting windows and scale-indexed validity domain. Their use as law-time inputs additionally requires law-description operability and the generation-priority condition for law-time content.

Each minimal failure location determined by Proposition 8.35 and each unsatisfied defining condition of the quantity used specify the first objects to repair among record correspondence, comparison rules, and admissible counting windows; structural representation, quantity-affine realization, window invariance, and unit normalization; finite

operational granularity and continuous-approximation evaluation input; or subsequent law connection. Frequency, period, numerical values in Hz, calibration values, frequency ratios, time scales, synchronization results, and calibration chains are recomputed from the repaired inputs and can be retained within the domain in which the grounding conditions again hold.

When an error source in calibration, a frequency ratio, record-cell width, window selection, unit normalization, or finite operational granularity is repaired, the repair's effect on accuracy is determined by an accuracy evaluation of the recomputed downstream results. The decision to use a time-metrology result therefore separates into retention within the validity domain and repair beginning at each minimal failure location.

*Proof.* By Proposition 8.32, frequency, period, and numerical values in Hz are generated according to the dependency order from cycle-count records through the unit-normalization convention. By Proposition 8.33, when generated quantities are used as law-time inputs, satisfaction of law-description operability and the generation-priority condition gives them physical-time qualification.

Proposition 8.35 classifies each minimal failure location of a generation condition into the comparison, representation, continuous-approximation, or subsequent stage. By Definition 6.17, conditions and results downstream of each minimal failure location depend on upstream conditions. This determines the order in which the cause at each minimal failure location is repaired and downstream results are recomputed from the repaired input.

When an error source belongs to calibration, a frequency ratio, record-cell width, window selection, unit normalization, or finite operational granularity, downstream results can be recomputed from inputs in which that source has been repaired, and accuracy evaluation can be applied to the recomputed results. Time-metrology results can consequently be retained where the grounding conditions hold, while the repair order begins at each minimal failure location and accuracy evaluation remains conditional on recomputation.  $\square$

## 8.4 Worked Reduction for GPS

In the GPS scheme, comparison-stage objects are constructed from satellite-side clock sources, reception records, correction records, and transport rules, and a navigation-time representation is generated through the dynamical-description and law-description stages. If what is conventionally called a navigation-time variable in GPS is presupposed before its generation, satellite-clock corrections, propagation corrections, pseudoranges, receiver-clock offsets, synchronization results, position and velocity navigation solutions, and law results using them inherit primitive-time dependence. If a rotation-derived nonidentity loop action arises under the adopted transport rules, that defect is localized at the comparison stage as a failure of the identity-action condition for every pure transport loop. In the construction of GPS navigation time, correction of the transport rules, restriction of the domain of validity, or specification of the scale domain in which an approximation holds determines the conditions for using downstream results in accordance with that failure.

### 8.4.1 Comparison Closure and the Gluing Burden in GPS

Let the set of GPS satellite identifiers be  $\text{SatIdx}_{\text{SchGPS}}$ . For each  $\sigma \in \text{SatIdx}_{\text{SchGPS}}$ , let the satellite-side  $\mathfrak{C}$ -derived source token be  $x_{\mathfrak{C},\sigma} \in \text{Tok}_{\mathfrak{C},\text{SchGPS}}^*$ . Represent the family of



satellite-side clock sources by

$$\{x_{\mathfrak{C},\sigma} \in \mathbf{Tok}_{\mathfrak{C},\text{Sch}_{\text{GPS}}}^*\}_{\sigma \in \text{SatId} \times \text{Sch}_{\text{GPS}}}$$

Let the GPS receiver-side information-coupling position and  $\mathfrak{D}$ -derived record token be

$$\begin{aligned} u_{\text{recv},\text{Sch}_{\text{GPS}}} &\in U_{\text{Sch}_{\text{GPS}}}, \\ x_{\mathfrak{D},\text{recv}} &\in \mathbf{Tok}_{\mathfrak{D},\text{Sch}_{\text{GPS}}}^* \end{aligned}$$

Type the receiver-side arrival by

$$\mathbf{Arr}_{\mathfrak{D},u_{\text{recv},\text{Sch}_{\text{GPS}}},\text{Sch}_{\text{GPS}}}^{\text{cmp}}$$

The satellite-side tokens retain cycle indices, candidate time tags in navigation messages, and correction records. The receiver-side tokens retain reception records, satellite identification, the receiver-side information-coupling position, and correction records. The input-side structure consists of the following four components.

- Rules relating satellite-side  $\mathfrak{C}$ -derived source tokens to receiver-side  $\mathfrak{D}$ -derived record tokens.
- Comparison rules that include propagation corrections, clock corrections, and relativistic corrections.
- Conditions that bind records from multiple satellites to the same receiver-side information-coupling position.
- Transport rules that map multiple comparison paths to the same navigation-time representation.

For evaluation at the GPS receiver, set

$$o = u_{\text{recv},\text{Sch}_{\text{GPS}}} \in U_{\text{Sch}_{\text{GPS}}}$$

and let the single-object specification index be  $\zeta_{\text{Sch}_{\text{GPS}}}$ . Use the GPS-scheme-restricted complete evaluation specification  $\mathbf{Ev}_{\text{Sch}_{\text{GPS}}}$  of Definition 8.5.

Suppose that the realized operational configuration and typed family of comparison rules obtained by restricting the GPS inputs to a specified domain satisfy the exact operability of Definition 4.175. By Definition 4.89, the satellite-side clock tokens, reception records, correction records, propagation corrections, and relativistic corrections descend, under the same family of comparison rules, to the  $\mathfrak{CD}$ -derived canonical comparison quotient. When a complete evaluation specifying a nonrotating domain satisfies this exact operability, records from multiple satellites become comparison objects on the same canonical comparison quotient.

When the dynamical-description operability of Definition 5.70 holds, the dynamical-description extension base data and the dynamical-description evidence datum relative to the real-valued time-order embedding selected in Definition 5.59,

$$\begin{aligned} &\mathbf{Dat}_{o,\zeta_{\text{Sch}_{\text{GPS}}}}^{\text{dyn},0}, \\ &\mathbf{DynWit}_{o,\zeta_{\text{Sch}_{\text{GPS}}}} \left[ \mathbf{RepEmb}_o^{\text{dyn},\text{time}} \right] \end{aligned}$$

are generated.

Suppose that the law-evaluation datum satisfies the equality of evaluation specifications under Proposition 5.86 and that the law-description operability of Definition 5.102 holds. By Definition 5.108, this law-evaluation datum and the selected real-valued time-order embedding determine the observer-time law core.

For any

$$T_0 \in \text{Qu}_{\mathbf{c},o}^{\text{dyn},\text{can}}$$

let the time value supplied by the law-admissible time representation of Definition 5.63,

$$t_0 := \text{TimeRep}_o^{\text{dyn}}(T_0)$$

be a GPS navigation-time value. The object ordinarily called a navigation-time variable in GPS is this GPS navigation-time value supplied by the law-admissible time representation belonging to the observer-time law core.

**Proposition 8.37** (Navigation-time representation on the exact validity domain of GPS). Suppose that the law-evaluation datum of the GPS scheme satisfies

$$\begin{aligned} \text{EvalSpec} \left[ \text{LawEvalDat}_{o,\zeta_{\text{SchGPS}}} \right] &= \text{Ev}_{\text{SchGPS}}, \\ \text{LawOperative} \left[ \text{LawEvalDat}_{o,\zeta_{\text{SchGPS}}} \right] & \end{aligned}$$

and let

$$\text{RepEmb}_o^{\text{dyn},\text{time}} \in \text{OrdEmbSet}_o^{\text{dyn},\text{time}}(\mathbb{R})$$

. Then the object ordinarily called a navigation-time variable in GPS is the GPS navigation-time value generated by the law-admissible time representation belonging to the observer-time law core.

*Proof.* Definition 5.102 and the equality of evaluation specifications imply that the corresponding complete evaluation specification is dynamical-description operative. By Definition 5.108 and Proposition 5.109,

$$\text{LawCore}_{o,\zeta_{\text{SchGPS}}}^{\text{obs},\text{time}}(\text{RepEmb}_o^{\text{dyn},\text{time}})$$

exists uniquely. By Definition 5.63, each time-quotient class is mapped to real-valued time by the law-admissible time representation of that definition. Therefore, the object ordinarily called a navigation-time variable in GPS is a GPS navigation-time value generated through the comparison, dynamical-description, and law-description stages.  $\square$

**Connection to physical-time qualification.** By Corollary 7.15, the generation structures corresponding to the time content used by a GPS navigation-time value must be established prior to law operations and result derivation. By Proposition 8.21, when time content and grounding structures are generated from nontemporal physical inputs prior to law operations and result derivation, the time variable satisfies physical-time qualification.

When time content enters a law operation or result derivation before the grounding structures required for its use, that time content is a primitive-time input under Definition 7.10. By Theorem 7.11, the time variable does not satisfy physical-time qualification. By Proposition 7.13, the simultaneous satisfaction of mathematical consistency, predictive performance, and empirical fit does not repair this loss of qualification.

Failures at the comparison, representation, continuous-approximation, dynamical-description, law-representation, and differential-consistency and law-form-preservation stages of the complete GPS evaluation specification are classified by the failure-position tokens of Definition 6.10. Failure of the identity-action condition for all-pure-transport loops belongs to the comparison-failure series, and a determination outside the scale-indexed validity domain belongs to the continuous-approximation-failure series.

**Proposition 8.38** (Inheritance of primitive-time dependence in GPS). If a navigation-time variable is used in satellite-clock correction, propagation correction, pseudorange equations, receiver-clock-offset estimation, synchronization processing, or navigation solutions before the generation structures of the GPS navigation-time value are established prior to law operations and result derivation, that use becomes a primitive-time input. Satellite-clock correction values, propagation-correction values, pseudoranges, receiver-clock offsets, synchronization results, position and velocity navigation solutions, and law results that use them inherit primitive-time dependence, and the time content contained in each result does not satisfy physical-time qualification. Mathematical consistency of the navigation equations, positioning accuracy, operational reproducibility, and empirical fit do not repair this loss of qualification.

When the GPS scheme satisfies exact operability, dynamical-description operability, and law-description operability, a GPS navigation-time value is generated. Within a domain in which the identity-action condition for all-pure-transport loops holds, this value can be used as a single exact navigation-time representation. Where that condition fails, the scale domain of approximate validity is stated, and qualification to use navigation-time values, correction values, synchronization results, and navigation solutions is determined within that domain.

*Proof.* When a navigation-time variable is used in a law operation or result derivation before the generation structures are established, Definition 7.10 and Theorem 7.11 make that use a primitive-time input that does not satisfy physical-time qualification. By Corollary 7.12, dependence on that time content is inherited by correction values, pseudoranges, receiver-clock offsets, synchronization results, navigation solutions, and law results. By Proposition 7.13, mathematical consistency, positioning accuracy, operational reproducibility, and empirical fit do not repair this loss of qualification.

When exact operability, dynamical-description operability, and law-description operability hold, Proposition 8.37 shows that the law-admissible time representation belonging to the observer-time law core generates the GPS navigation-time value. By Definition 6.42, the identity-action condition for all-pure-transport loops determines exact gluing independent of comparison path. By Definition 6.161, the domain of use for approximate operation is determined by the scale domain of approximate validity.  $\square$

#### 8.4.2 Rotation-Derived Loop Defects and $FC_8$

**Definition 8.39** (GPS rotation-derived loop action and nonidentity-action witness). Represent two comparison paths in GPS with the same initial and terminal points, for each  $\bullet \in \{\mathfrak{C}, \mathfrak{D}\}$ , by pure transport words

$$\mathbf{w}_0, \mathbf{w}_1 \in \text{Word}_{\bullet}^{\text{tr}} [\text{ISch}_{\text{GPS}}]$$

. The word constructed from the inverse and concatenation,

$$\mathbf{w}_1^{\dagger} \star \mathbf{w}_0 \in \text{LoopWord}_{\text{all}}^{\text{tr}, \bullet} [\text{ISch}_{\text{GPS}}]$$

is a pure transport loop word.

Define the action of this loop word by

$$\begin{aligned} \text{LoopAct}_{\text{GPS}}^{\bullet}[\mathbf{w}_0, \mathbf{w}_1] \\ := \text{Act}_{\bullet}^{\text{tr}}[\text{I}_{\text{Sch}_{\text{GPS}}}] \left[ \mathbf{w}_1^{\dagger} \star \mathbf{w}_0 \right] \end{aligned}$$

and define the nonidentity-action witness predicate by

$$\begin{aligned} \text{LoopWit}_{\text{GPS}}^{\bullet}[x; \mathbf{w}_0, \mathbf{w}_1] \\ \iff x \in \text{Dom}(\text{LoopAct}_{\text{GPS}}^{\bullet}[\mathbf{w}_0, \mathbf{w}_1]) \\ \wedge \text{LoopAct}_{\text{GPS}}^{\bullet}[\mathbf{w}_0, \mathbf{w}_1](x) \neq x \end{aligned}$$

. Call an  $x$  satisfying this predicate a nonidentity-action witness of a rotation-derived loop defect in GPS. The condition  $\mathbf{G}[\text{Ev}_{\text{Sch}_{\text{GPS}}}^{\text{cmp}}]$  of Definition 6.42 determines exact identity of the loop action. Approximate operation relative to an engineering tolerance threshold is determined by the scale domain of approximate validity of Definition 6.161.

**Proposition 8.40** (Localization of a rotation-derived loop defect in GPS). Suppose that, for the GPS-scheme-restricted complete evaluation specification,

$$\mathbf{L}_{\leq 7} \left[ \text{Ev}_{\text{Sch}_{\text{GPS}}}^{\text{cmp}} \right]$$

holds. Suppose further that there exist

$$\begin{aligned} \bullet &\in \{\mathfrak{C}, \mathfrak{D}\}, \\ \mathbf{w}_0, \mathbf{w}_1 &\in \text{Word}_{\bullet}^{\text{tr}}[\text{I}_{\text{Sch}_{\text{GPS}}}] \end{aligned}$$

and  $x$  such that

$$\text{LoopWit}_{\text{GPS}}^{\bullet}[x; \mathbf{w}_0, \mathbf{w}_1]$$

holds. Then

$$\text{FC}_8[\text{Ev}_{\text{Sch}_{\text{GPS}}}]$$

holds.

*Proof.* The nonidentity-action witness of Definition 8.39 invalidates the global gluing condition at the comparison stage,  $\mathbf{G}[\text{Ev}_{\text{Sch}_{\text{GPS}}}^{\text{cmp}}]$ . The conclusion follows from the assumption that the first seven comparison conditions hold, Definition 6.42, and Proposition 6.43.  $\square$

**Corollary 8.41** (Separation of stages in the worked GPS reduction). For each complete evaluation specification constructed by restricting the GPS inputs, the following hold.

1. When law-description operability holds, a navigation-time value for the restricted domain is generated.
2. When the first seven comparison-stage conditions hold and a nonidentity loop-action witness exists, the all-pure-transport-loop identity-action failure  $\text{FC}_8$  holds.
3. Approximate operation relative to an engineering tolerance threshold is determined by the scale domain of approximate validity.

*Proof.* Item 1 follows from Proposition 8.37. Item 2 follows from Proposition 8.40. Item 3 follows from Definition 6.161. The GPS navigation-time value generated in Item 1 is supplied by a law-admissible time representation belonging to the observer-time law core and is the object ordinarily called a navigation-time variable in GPS. Item 1 determines law-time generation on a specified restricted domain; Item 2 determines exact loop identity action at the comparison stage; and Item 3 determines the scale-indexed domain of approximate validity.  $\square$

**Proposition 8.42** (Safe reuse of GPS results through generated time and failure classification). Where the conditions of Proposition 8.37 and the identity-action condition for all-pure-transport loops hold, generated GPS navigation-time values, satellite-clock correction values, propagation-correction values, pseudoranges, receiver-clock offsets, synchronization results, and position and velocity navigation solutions can be used within the exact validity domain. Under approximate operation, the scale domain of approximate validity of Definition 6.161 determines the domain of use of these results.

The comparison-stage loop defect, boundary of the scale-indexed validity domain, or failure of a subsequent-stage condition classified by Proposition 8.40 and Corollary 8.41 determines the GPS construction component that must be repaired first. Clock corrections, propagation corrections, pseudoranges, receiver-clock offsets, synchronization results, and navigation solutions are recalculated from the repaired inputs, and these results can be retained within the exact validity domain or the scale domain of approximate validity.

When an error source in clock corrections, propagation corrections, comparison paths, loop defects, synchronization processing, or receiver-clock offsets is repaired, the repair's effect on accuracy is determined by an accuracy evaluation of the recalculated navigation solution. The decision to use GPS results therefore separates into retention within the validity domain and repair from the first failure position.

*Proof.* By Proposition 8.37, within a domain in which law-description operability holds, the law-admissible time representation belonging to the observer-time law core generates a GPS navigation-time value. By Proposition 8.38, within the range in which the generated GPS navigation-time value and its grounding conditions are used, the corresponding correction values, pseudoranges, receiver-clock offsets, synchronization results, and navigation solutions connect to time inputs that have physical-time qualification.

Proposition 8.40 maps a nonidentity loop action to failure of the identity-action condition for all-pure-transport loops. Corollary 8.41 separates exact law-time generation, a comparison-stage loop defect, and a scale-indexed domain of approximate validity. By Definition 6.17, conditions and results downstream of the first failure position depend on upstream conditions. The order is therefore fixed in which a cause belonging to the first failure position is repaired and the navigation results are recalculated from those inputs.

When the source of error belongs to clock corrections, propagation corrections, comparison paths, loop defects, synchronization processing, or receiver-clock offsets, the navigation solution can be recalculated from inputs in which that source has been repaired, and an accuracy evaluation can be applied to the recalculated result. It follows that retention of GPS results within the validity domain, the repair order from the first failure position, and conditional evaluation of positioning accuracy hold.  $\square$

## 8.5 Worked Reduction for SR

In the special-relativity scheme, comparison-stage objects are constructed from an inertial-observer network, local clock-side receptions, record-side receptions, synchronization rules,

and comparison rules, and inertial coordinate time is generated through the dynamical-description and law-description stages. If inertial coordinate time or proper time is presupposed prior to this generation, synchronization results, Lorentz-transformation results, time representations of worldlines, proper-time differences, differential quantities such as velocity and acceleration, clock-rate differences, and law results that use them inherit primitive-time dependence. The coordinate representation ordinarily called inertial coordinate time in SR is the law-admissible time representation obtained through this generation order, and each of its values is an SR inertial-coordinate-time value.

### 8.5.1 From an Inertial-Observer Network to Inertial Coordinate Time

The inputs to the worked reduction for SR are the inertial-observer network  $U_{\text{Sch}_{\text{SR}}}$  and, for each  $u \in U_{\text{Sch}_{\text{SR}}}$ , the local clock-side reception, record-side reception, and object-state-side reception. Clock-side reception contains proper-clock tokens or candidate local-clock tokens, and record-side reception contains local event records, signal-exchange records, and synchronization records. Object-state-side reception supplies material for distinguishing object states or event positions.

Synchronization rules provide comparability between local clock-side tokens and record-side tokens. Comparison rules make the local records of different inertial observers descend to the same comparison quotient. For an inertial observer  $o \in U_{\text{Sch}_{\text{SR}}}$  and single-object specification index  $\zeta_{\text{Sch}_{\text{SR}}}$ , use the SR-scheme-restricted complete evaluation specification  $\text{Ev}_{\text{Sch}_{\text{SR}}}$  of Definition 8.5.

Suppose that the realized operational configuration and typed family of comparison rules obtained by restricting the SR inputs to a specified domain satisfy the exact operability of Definition 4.175. By Definition 4.89, the local clock-side receptions, record-side receptions, synchronization records, and comparison records of the inertial-observer network descend, under the same family of comparison rules, to the  $\mathfrak{CD}$ -derived canonical comparison quotient. The objects obtained at this stage are quotient objects based on the comparison conditions and their order-bearing structure.

When the dynamical-description operability of Definition 5.70 holds, the dynamical-description extension base data and the dynamical-description evidence datum relative to the real-valued time-order embedding selected in Definition 5.59,

$$\begin{aligned} & \text{Dat}_{o, \zeta_{\text{Sch}_{\text{SR}}}}^{\text{dyn}, 0}, \\ & \text{DynWit}_{o, \zeta_{\text{Sch}_{\text{SR}}}} [\text{RepEmb}_o^{\text{dyn}, \text{time}}] \end{aligned}$$

are generated.

Suppose that the law-evaluation datum satisfies the equality of evaluation specifications under Proposition 5.86 and that the law-description operability of Definition 5.102 holds. By Definition 5.108, this law-evaluation datum and the selected real-valued time-order embedding determine the observer-time law core.

For any

$$T_0 \in \text{Qu}_{\mathfrak{e}, o}^{\text{dyn}, \text{can}}$$

define the conventional notation for an inertial-coordinate-time value in ordinary SR by

$$t := \text{TimeRep}_o^{\text{dyn}}(T_0)$$

. The coordinate representation ordinarily called inertial coordinate time in SR is the law-admissible time representation  $\text{TimeRep}_o^{\text{dyn}}$  of Definition 5.63, and  $t$  is the conventional

notation for its SR inertial-coordinate-time value. Minkowski geometry and Lorentz transformations are theory-specific input carriers that supply admissible transformations of this time representation and the spatial representation. The  $\mathfrak{CQ}$ -derived canonical comparison quotient is the quotient object preceding the action of those representation transformations.

**Proposition 8.43** (Inertial coordinate time is a generated representation). Suppose that the law-evaluation datum of the SR scheme satisfies

$$\begin{aligned} \text{EvalSpec} \left[ \text{LawEvalDat}_{o, \zeta_{\text{SchSR}}} \right] &= \text{Ev}_{\text{SchSR}}, \\ \text{LawOperative} \left[ \text{LawEvalDat}_{o, \zeta_{\text{SchSR}}} \right] & \end{aligned}$$

and let

$$\text{RepEmb}_o^{\text{dyn, time}} \in \text{OrdEmbSet}_o^{\text{dyn, time}}(\mathbb{R})$$

. Then the coordinate representation ordinarily called inertial coordinate time in SR is the law-admissible time representation  $\text{TimeRep}_o^{\text{dyn}}$  belonging to the observer-time law core. The SR inertial-coordinate-time value for each time-quotient class is generated by this law-admissible time representation and is conventionally denoted by  $t$ .

*Proof.* Definition 5.102 and the equality of evaluation specifications imply that the corresponding complete evaluation specification is dynamical-description operative. By Definition 5.108 and Proposition 5.109, the corresponding observer-time law core exists uniquely. By Definition 5.63, each time-quotient class is mapped to real-valued time by the law-admissible time representation of that definition. Therefore, the coordinate representation ordinarily called inertial coordinate time in SR is a law-admissible time representation generated through the comparison, dynamical-description, and law-description stages. Each of its values is an SR inertial-coordinate-time value and is conventionally denoted by  $t$ .  $\square$

### 8.5.2 Generation Stage of the Lorentz Transformation

Definition 8.10 supplies inertial-coordinate transformations among different inertial observers as comparison rules in the theory-specific inputs of SR. When law-description operability holds for different inertial observers  $o, o' \in U_{\text{SchSR}}$ , the observer-time law core and law-admissible time representation of each observer are generated. As the inertial-coordinate transformation rule in the theory-specific inputs of SR, the Lorentz transformation relates these generated time representations and spatial representations.

**Proposition 8.44** (A Lorentz transformation is a transformation among generated representations). Suppose that law-description operability holds for different inertial observers in the SR scheme. The Lorentz transformation supplied by the theory-specific inputs of SR is then an admissible representation transformation among the law-admissible time representations and spatial representations generated for the respective observers.

*Proof.* By Definition 8.10, the inertial-observer network, local clock-side receptions, record-side receptions, object-state-side receptions, synchronization procedures, proper-time comparisons, and comparison rules corresponding to inertial-coordinate transformations supply the theory-specific inputs of SR.

When exact operability holds for each observer, local receptions and synchronization records descend to the  $\mathfrak{CD}$ -derived canonical comparison quotient. When dynamical-description operability holds, dynamical-description evidence data are generated. When law-description operability holds, Proposition 8.43 generates the law-admissible time representation and SR inertial-coordinate-time values of each observer. The inertial-coordinate transformation rules in the theory-specific inputs of SR therefore relate the generated time representations and spatial representations among different inertial observers.

Furthermore, within the range in which differential-description operability holds for the differential evaluation specification of each observer, Theorem 5.135 gives differential covariance among the real-valued time-order embeddings belonging to that observer. This within-observer differential covariance and the between-observer inertial-coordinate transformation rule bear different types of responsibility. A Lorentz transformation is therefore an admissible representation transformation among the time representations and spatial representations generated for the respective observers.  $\square$

**Connection to physical-time qualification.** By Corollary 7.15, the generation structures corresponding to the time content used by an SR inertial-coordinate-time value must be established prior to law operations and result derivation. By Proposition 8.21, when time content and grounding structures are generated from nontemporal physical inputs prior to law operations and result derivation, the time variable satisfies physical-time qualification.

When time content enters a law operation or result derivation before the grounding structures required for its use, that time content is a primitive-time input under Definition 7.10. By Theorem 7.11, the time variable does not satisfy physical-time qualification. By Proposition 7.13, the simultaneous satisfaction of mathematical consistency, predictive performance, and empirical fit does not repair this loss of qualification.

Failures at the comparison, representation, continuous-approximation, dynamical-description, law-representation, and differential-consistency and law-form-preservation stages of the SR-scheme-restricted complete evaluation specification are classified by the failure-position tokens of Definition 6.10.

**Proposition 8.45** (Inheritance of primitive-time dependence in SR). If inertial coordinate time or proper time is used in synchronization rules, Lorentz transformations, parametric representations of worldlines, proper-time comparisons, spacetime intervals, differential quantities such as velocity and acceleration, or clock-rate comparisons before the generation structures of an SR inertial-coordinate-time value or proper-time value are established prior to law operations and result derivation, that use becomes a primitive-time input. Transformed coordinates, synchronization results, simultaneity relations, time representations of worldlines, proper-time differences, differential quantities, clock-rate differences, and law results that use them inherit primitive-time dependence, and the time content contained in each result does not satisfy physical-time qualification. Lorentz covariance, invariance of the spacetime interval, predictive performance, and empirical fit do not repair this loss of qualification.

When the SR scheme satisfies exact operability, dynamical-description operability, and law-description operability, the law-admissible time representation and SR inertial-coordinate-time values of each inertial observer are generated. Lorentz transformations are applied among these generated time representations and spatial representations. Results involving time derivatives additionally require differential-description operability.



*Proof.* When inertial coordinate time or proper time is used in a law operation or result derivation before the generation structures are established, Definition 7.10 and Theorem 7.11 make that use a primitive-time input that does not satisfy physical-time qualification. By Corollary 7.12, dependence on that time content is inherited by transformed coordinates, synchronization results, simultaneity relations, time representations of worldlines, proper-time differences, differential quantities, clock-rate differences, and law results. By Proposition 7.13, Lorentz covariance, invariance of the spacetime interval, predictive performance, and empirical fit do not repair this loss of qualification.

By Proposition 8.43, the law-admissible time representation and SR inertial-coordinate-time values of each inertial observer are generated when exact operability, dynamical-description operability, and law-description operability hold. By Proposition 8.44, a Lorentz transformation becomes an admissible representation transformation among generated representations. By Theorem 5.135, differential covariance among real-valued time-order embeddings holds within the range in which differential-description operability holds.  $\square$

**Corollary 8.46** (The two-stage misrecognition of the time-stretching account in SR). Relativistic clock differences in SR are comparison differences among generated law-admissible time representations corresponding to multiple inertial observers or worldlines. The time-stretching account presupposes time as an unconstructed entity prior to these generated representation differences and explains relativistic clock differences through stretching, slowing, or deformation of that time entity. By Proposition 8.23, this form of explanation is therefore the two-stage misrecognition of the time-stretching account.

*Proof.* By Proposition 8.43, inertial coordinate time in SR is a law-admissible time representation obtained from a generated observer-time law core. By Proposition 8.44, a Lorentz transformation is an admissible representation transformation among generated law-admissible time representations and spatial representations. Coordinate-time representations of different inertial observers are therefore related by the inertial-coordinate transformation rules in the theory-specific inputs of SR. Relativistic clock differences, including differences in proper-clock readings and proper-time differences along worldlines, are comparison differences among generated law-admissible time representations corresponding to multiple inertial observers or worldlines. By Proposition 8.23, in a form of explanation that places time as an unconstructed entity prior to these representation differences and redescribes subsequently generated comparison differences as changes in an attribute of that time entity, the first and third items of multistage misrecognition occur consecutively with direct dependence. This form of explanation is therefore the two-stage misrecognition of the time-stretching account.  $\square$

**Proposition 8.47** (Separation of the formal construction of special relativity from a bearer of time stretching). The formal construction of special relativity determines Lorentz transformations among inertial coordinates, spacetime intervals, accumulation of proper time along worldlines, clock readings, clock rates, and their comparison relations. Conditions of relative motion specify relations among  $\mathcal{C}\mathcal{O}\mathcal{B}\mathcal{O}$  physical carriers, worldlines, signals, synchronization, records, and generated time representations.

If the account that “time itself stretches, slows, or deforms” is taken as a literal physical mechanism, that account asserts a bearer of time stretching that remains identical through the deformation while taking different states. For this assertion to hold as a physical mechanism, it requires the type of the bearer of time stretching, an identity condition before and after deformation, a state space, a deformation magnitude, a deformation law,

a generation path from nontemporal physical inputs, coupling laws to  $\mathcal{C}\mathcal{B}\mathcal{D}$  physical carriers and clock processes, and a measurement procedure independent of clock differences.

The formal results of special relativity established by Proposition 7.167 are comparison relations among clock readings, clock rates, proper times, and generated time representations. These formal results do not supply the type, identity condition, state space, deformation law, generation path, coupling laws, or independent measurement procedure of a bearer of time stretching. A literal time-stretching account therefore reintroduces an unconstructed bearer of change prior to the comparison differences generated by the formal construction and attributes those comparison differences to changes in an attribute of that bearer. This attribution is the two-stage misrecognition of Corollary 8.46.

No misrecognition occurs when differences in clock readings, clock rates, proper times, or generated time representations are described directly in terms of their objects and relations. Even when the term time stretch is used, no misrecognition occurs in an explicit metaphor if the source text states, within the same explanatory scope, its metaphorical status, that it does not designate literal deformation of time, and its actual referent.

When the term time stretch is used without stating its metaphorical status and actual referent, interpreting the term retrospectively as shorthand leaves a one-stage misrecognition that transfers a comparison difference to an attribute of time. In a literal account that places an unconstructed time entity in advance, this one stage is joined by pre-generation placement and constitutes the two-stage misrecognition of Corollary 8.46.

*Proof.* By Proposition 8.43 and Proposition 8.44, inertial coordinate time is a value of a generated law-admissible time representation, and a Lorentz transformation is an admissible representation transformation among generated time representations and spatial representations. By Proposition 7.167, the formal results of special relativity supply clock readings, clock rates, proper times, and their comparison relations, but do not supply an independent state and evolution law for a bearer of time stretching.

Literal stretching, slowing, or deformation requires an object that changes, its states before and after the change, and a condition for determining both states as states of the same object. Treating that change as the cause of clock differences additionally requires coupling laws from the bearer of change to  $\mathcal{C}\mathcal{B}\mathcal{D}$  physical carriers and clock processes, together with a measurement procedure independent of clock differences. The formal results of special relativity contain none of these constructions.

A literal time-stretching account therefore attributes a comparison difference among generated representations to an unconstructed bearer of time stretching. By Corollary 8.46, this attribution is a two-stage misrecognition. No misrecognition occurs when comparison differences on physical carriers are described directly or when metaphorical status and the actual referent are stated. A bare expression of time stretching constitutes a one-stage misrecognition that transfers a comparison difference to an attribute of time even under an interpretation that does not assert an independent time entity.  $\square$

**Proposition 8.48** (Generation structure of relativistic clock differences in special relativity). For the  $\mathcal{C}\mathcal{B}\mathcal{D}$  physical carriers of Definition 3.1 and Definition 3.2, suppose that the conditions of Proposition 8.43 hold. Conditions of relative motion specify the worldlines to which the  $\mathcal{C}\mathcal{B}\mathcal{D}$  physical carriers belong and the relations of signal propagation, synchronization, and comparison. The clock processes, records, object-state differences, and information coupling obtained from the  $\mathcal{C}\mathcal{B}\mathcal{D}$  physical carriers generate the law-admissible time representation and SR inertial-coordinate-time values of each inertial observer.

By Proposition 8.44, a Lorentz transformation acts as an admissible representation transformation among generated time representations and spatial representations. Accumulation of proper time along worldlines, proper-clock readings, and signal-mediated comparisons determine proper-time differences, clock-reading differences, and clock-rate differences. The physical grounds for these differences lie in the dependency chain of  $\mathcal{CWB}$  physical carriers, worldlines, signals, synchronization, comparison, and generated time representations.

This dependency chain does not contain primitive time as an object of physical action. An account that attributes results generated in relation to relative motion to stretching, slowing, or deformation of primitive time itself placed prior to generation constitutes the two-stage misrecognition of Corollary 8.46.

*Proof.* Definition 3.1 and Definition 3.2 map each role in  $\mathcal{CWB}$  to a physical-carrier realization. By Proposition 8.43, the establishment of clock processes, records, object-state differences, information coupling, synchronization, and comparison generates the law-admissible time representation and SR inertial-coordinate-time values of each inertial observer. By Proposition 8.44, a Lorentz transformation becomes an admissible representation transformation among generated representations. Accumulation of proper time along worldlines, proper-clock readings, and signal-mediated comparisons therefore supply proper-time differences, clock-reading differences, and clock-rate differences as comparison results on generated time representations.

In this generation order, conditions relating to relative motion act on relations among  $\mathcal{CWB}$  physical carriers, worldlines, signals, synchronization, and comparison. By Corollary 8.46, an account that places primitive time prior to this comparison result and attributes the comparison result to a change in an attribute of primitive time itself is a two-stage misrecognition.  $\square$

**Corollary 8.49** (Localization of failed results in the worked reduction for SR). In the SR scheme, if the conditions that generate inertial-coordinate-time values as values of a law-admissible time representation fail even when local clock-side receptions, record-side receptions, synchronization rules, or comparison rules are supplied, the failure position is classified by the failure-position-token system.

*Proof.* Under the conditions of Proposition 8.43, inertial-coordinate-time values are obtained as values of a generated law-admissible time representation. Claims using continuous approximation require an approximation-validity scale domain, and use of differential law forms requires satisfaction of the differential-stage conditions. When a stage condition fails, Definition 6.10 classifies each minimal failure position as the corresponding comparison, representation, continuous-approximation, dynamical-description, law-description, or differential failure-position token.  $\square$

**Proposition 8.50** (Safe reuse of special-relativistic results through generated time and failure classification). Within a domain in which the conditions of Proposition 8.43 and Proposition 8.44 hold, generated law-admissible time representations, SR inertial-coordinate-time values, Lorentz-transformation results, synchronization results, simultaneity relations, and time representations of worldlines can be used within that domain. Proper-time differences, spacetime intervals, differential quantities such as velocity and acceleration, and clock-rate differences can be used within a domain in which the generation conditions for the time content on which each depends and the required conditions at the comparison,

representation, continuous-approximation, dynamical-description, law-representation, and differential-consistency and law-form-preservation stages hold.

Each minimal failure position classified by Corollary 8.49 determines the SR construction component that must be repaired first. A failure at the comparison stage requires repair of local clock-side receptions, record-side receptions, synchronization rules, or comparison rules; a failure at the representation or continuous-approximation stage requires repair of the time representation, comparison domain, or scale domain of approximate validity. A failure at the dynamical-description, law-representation, or differential-consistency and law-form-preservation stage requires repair of the dynamical-description evidence datum, law-evaluation datum, or differential evaluation specification. From the repaired inputs, time representations, transformed coordinates, synchronization results, simultaneity relations, time representations of worldlines, proper-time differences, differential quantities, clock-rate differences, and law results are regenerated or recalculated in dependency order.

Corollary 8.46 separates comparison differences among generated representations from the two-stage misrecognition that places an unconstructed time entity prior to those comparison differences and explains the differences through stretching, slowing, or deformation of that entity. The grounds for using Lorentz covariance, invariance of the spacetime interval, proper-time differences, and clock-rate differences are supplied by generated time representations, admissible representation transformations, and the grounding conditions at each stage.

When an error source in synchronization rules, proper-time comparisons, clock-rate comparisons, continuous approximations, or differential evaluations is repaired, the repair's effect on accuracy is determined by an accuracy evaluation of the regenerated or recalculated downstream results. The decision to use special-relativistic results therefore separates into retention within the validity domain and repair from each minimal failure position.

*Proof.* By Proposition 8.43, within a domain in which law-description operability holds, the law-admissible time representation and SR inertial-coordinate-time values of each inertial observer are generated. By Proposition 8.44, a Lorentz transformation becomes an admissible representation transformation among generated time representations and spatial representations. By Proposition 8.21, when the time content and grounding structures on which each result depends are generated prior to law operations and result derivation, that time input satisfies physical-time qualification. Transformed coordinates, synchronization results, simultaneity relations, time representations of worldlines, proper-time differences, differential quantities, clock-rate differences, and law results that satisfy the grounding conditions at each stage therefore use time inputs that have physical-time qualification.

By Corollary 8.49 and Definition 6.17, conditions and results downstream of each minimal failure position depend on upstream conditions. The order is therefore fixed in which an input or rule belonging to each minimal failure position is repaired and downstream results are regenerated or recalculated from those inputs in dependency order.

Corollary 8.46 separates comparison differences among generated law-admissible time representations from the two-stage misrecognition that explains those comparison differences through changes in an attribute of an unconstructed time entity. This connects the grounds for Lorentz covariance, invariance of the spacetime interval, proper-time differences, and clock-rate differences to generated time representations, admissible representation transformations, and the conditions at each stage.

When the source of error belongs to synchronization rules, proper-time comparisons, clock-rate comparisons, continuous approximations, or differential evaluations, downstream results can be regenerated or recalculated from inputs in which that source has been

repaired, and an accuracy evaluation can be applied to the recalculated result. It follows that retention of special-relativistic results within the validity domain, the repair order from each minimal failure position, and conditional evaluation of accuracy hold.  $\square$

## 8.6 Worked Reduction for GR

In the general-relativity scheme, comparison-stage objects are constructed from a local-observer network, worldline-dependent receptions, local comparison rules, transport rules, and gluing rules. Worldline-local comparison data and generated representation coordinates generate local proper-time representations, while satisfaction of the conditions at the dynamical-description and law-description stages generates a global law-admissible time representation. If local proper time, coordinate time, or a global law-time representation is presupposed prior to this generation, time-dependent components of the metric and connection, parametric representations of geodesics, gravitational clock differences, gravitational redshift, initial-value evolution, cosmological time representations, and law results that use them inherit primitive-time dependence. Local proper-time representations and global law-time representations belong to different generation stages.

### 8.6.1 From Local Proper-Time Representations to a Global Law-Time Representation

The inputs to the worked reduction for GR are the local-observer network  $U_{\text{Sch}_{\text{GR}}}$  and, for each  $u \in U_{\text{Sch}_{\text{GR}}}$ , worldline-dependent reception, local clock-side reception, record-side reception, and object-state-side reception. Local comparison rules, transport rules, and gluing rules supply the conditions under which comparison materials belonging to different local observers or worldlines descend to the same comparison quotient.

For a local observer  $o \in U_{\text{Sch}_{\text{GR}}}$  and single-object specification index  $\zeta_{\text{Sch}_{\text{GR}}}$ , use the GR-scheme-restricted complete evaluation specification  $\text{Ev}_{\text{Sch}_{\text{GR}}}$  of Definition 8.5. Lorentzian spacetime, the metric, the connection, local charts, transport rules, and gluing rules supply the theory-specific input carriers of GR.

Suppose that the realized operational configuration and typed family of comparison rules obtained by restricting the GR inputs to a specified domain satisfy the exact operability of Definition 4.175. By Theorem 8.28, worldline-local clock-side reception, record-side reception, local comparison rules, and generated representation coordinates generate the local proper-time representation  $\tau_{u, \text{Sch}_{\text{GR}}}^{\text{loc}}$ .

A local proper-time representation is a representation of comparison data belonging to the worldline of a local observer. Transport preservation, loop consistency, quotient descent, global gluing, branch resolution, and law-output determinacy, which are required to make records belonging to multiple local observers or worldlines descend to a single comparison quotient, canonical trajectory relation, or law domain, belong to subsequent generation stages.

By Definition 4.89, when the comparison-stage and representation-stage conditions hold, local receptions, transport records, and gluing records descend to the  $\mathfrak{CD}$ -derived canonical comparison quotient.

When the dynamical-description operability of Definition 5.70 holds, the  $\mathfrak{CD}\mathfrak{B}$ -derived canonical trajectory relation  $\text{TrjRel}_o^{\text{dyn}, \text{can}}$  is generated. Its right-uniqueness requires

$$K_{\text{CanTrjFunc}}^{\text{dyn}}(\text{Ev}_{\text{Sch}_{\text{GR}}})$$

to hold.

Use the  $\mathfrak{C}$ -derived real-valued time-order embedding selected in Definition 5.59.

Suppose that the law-evaluation datum satisfies the equality of evaluation specifications under Proposition 5.86 and that the law-description operability of Definition 5.102 holds. By Definition 5.108, this law-evaluation datum and the selected real-valued time-order embedding determine the observer-time law core, and Definition 5.63 generates the law-admissible time representation  $\text{TimeRep}_o^{\text{dyn}}$ .

**Proposition 8.51** (Stage difference between local proper-time representations and a global law-time representation). Suppose that the local proper-time representation  $\tau_{u, \text{Sch}_{\text{GR}}}^{\text{loc}}$  has been generated in the GR scheme. This condition alone does not imply the right-uniqueness of the  $\mathfrak{CDB}$ -derived canonical trajectory relation  $\text{TrjRel}_o^{\text{dyn}, \text{can}}$ , the existence of the observer-time law core

$$\text{LawCore}_{o, \zeta_{\text{Sch}_{\text{GR}}}}^{\text{obs}, \text{time}} (\text{RepEmb}_o^{\text{dyn}, \text{time}}),$$

or the existence of the law-admissible time representation  $\text{TimeRep}_o^{\text{dyn}}$ . Their generation requires satisfaction of the respective conditions at the comparison, representation, dynamical-description, and law-description stages.

*Proof.* By Theorem 8.28, a local proper-time representation is generated from worldline-local comparison data and generated representation coordinates.

By Definition 5.23, descent of the  $\mathfrak{CDB}$ -derived canonical trajectory relation requires comparison-stage closure and the dynamical-description condition groups. Its right-uniqueness requires the dynamical-description-stage canonical-trajectory-function condition of Definition 5.26. By Definition 5.102 and Definition 5.108, the existence of the observer-time law core requires the simultaneous satisfaction of dynamical-description operability and the law condition groups. By Definition 5.63, the law-admissible time representation is generated as a component of the observer-time law core.

Consequently, even after a local proper-time representation has been generated, failure of a condition at the comparison, dynamical-description, or law-description stage prevents generation of a right-unique canonical trajectory relation, an observer-time law core, or a law-admissible time representation.  $\square$

**Connection to physical-time qualification.** By Corollary 7.15, the generation structures corresponding to the time content used by a GR law-time variable must be established prior to law operations and result derivation. By Proposition 8.21, when time content and grounding structures are generated from nontemporal physical inputs prior to law operations and result derivation, the time variable satisfies physical-time qualification.

When time content enters a law operation or result derivation before the grounding structures required for its use, that time content is a primitive-time input under Definition 7.10. By Theorem 7.11, the time variable does not satisfy physical-time qualification. By Proposition 7.13, the simultaneous satisfaction of mathematical consistency, predictive performance, and empirical fit does not repair this loss of qualification.

Failures at the comparison, representation, continuous-approximation, dynamical-description, law-representation, and differential-consistency and law-form-preservation stages of the GR-scheme-restricted complete evaluation specification are classified by the failure-position tokens of Definition 6.10.

**Proposition 8.52** (Inheritance of primitive-time dependence in GR). If local proper-time representations, coordinate time, or a global law-time representation are used in time-dependent components of the metric or connection, parametric representations of geodesics,

worldline comparisons, gravitational redshift, initial-value evolution, or cosmological time representations before the generation structures of those time contents are established prior to law operations and result derivation, that use becomes a primitive-time input. The time evolution of the metric and connection, geodesic solutions, the time ordering of worldlines, gravitational clock differences, gravitational redshift, initial-value evolution, cosmological time relations, and law results that use them inherit primitive-time dependence, and the time content contained in each result does not satisfy physical-time qualification. General covariance, the mathematical consistency of the field equations, predictive performance, and empirical fit do not repair this loss of qualification.

Worldline-local comparison data and generated representation coordinates generate local proper-time representations. Global law-time use further requires exact operability, dynamical-description operability, law-description operability, transport preservation, loop consistency, quotient descent, and global gluing. Results containing time derivatives require differential-description operability.

*Proof.* When local proper time, coordinate time, or a global law-time representation is used in a law operation or result derivation before the generation structures are established, Definition 7.10 and Theorem 7.11 make that use a primitive-time input that does not satisfy physical-time qualification. By Corollary 7.12, dependence on that time content is inherited by the time evolution of the metric and connection, geodesic solutions, the time ordering of worldlines, gravitational clock differences, gravitational redshift, initial-value evolution, cosmological time relations, and law results. By Proposition 7.13, general covariance, the mathematical consistency of the field equations, predictive performance, and empirical fit do not repair this loss of qualification.

By Theorem 8.28, a local proper-time representation is generated from worldline-local comparison data and generated representation coordinates. By Proposition 8.51, a local proper-time representation alone does not imply a right-unique canonical trajectory relation, an observer-time law core, or a law-admissible time representation; these require the conditions of subsequent stages to hold. By Theorem 5.135, differential covariance among real-valued time-order embeddings holds within the range in which differential-description operability holds.  $\square$

### 8.6.2 Stage Separation of Failure Positions in GR

Generation of a local proper-time representation and generation of a global comparison quotient, a right-unique canonical trajectory relation, a law-admissible time representation, and a differential law form require conditions at different stages. If a condition at any stage fails after generation of a local proper-time representation, Definition 6.10 classifies each minimal failure position by the corresponding failure-position token.

**Proposition 8.53** (Failure localization in the worked reduction for GR). Suppose that a local proper-time representation is generated in the GR scheme and that a condition for generating a law-admissible time representation fails. Each minimal failure position is then classified uniquely as a comparison, representation, continuous-approximation, dynamical-description, law-description, or differential failure-position token.

*Proof.* Failure of a comparison-stage condition is classified as a comparison failure by Definition 6.36. Failure of a representation-stage condition is classified as a representation failure by Definition 6.58. Failure of a continuous-approximation-stage condition is classified

as a continuous-approximation failure by Definition 6.77. Failure of a dynamical-description-stage condition is classified as a dynamical-description failure by Definition 6.104. Failure of a law-description-stage condition is classified as a law-description failure by Definition 6.110. Failure of a differential-consistency and law-form-preservation-stage condition is classified as a differential failure by Definition 6.137.

By Proposition 6.26, the failed-token set equals its set of minimal elements. By Definition 6.10, the corresponding failure-position token is assigned to each minimal failure position. Hence, if a condition for generating a law-admissible time representation fails after generation of a local proper-time representation, each minimal failure position is classified uniquely within one of the six failure series.  $\square$

**Corollary 8.54** (Separation of stages in the worked reduction for GR). In GR, local proper-time representations, the  $\mathcal{CD}$ -derived canonical comparison quotient, the  $\mathcal{CD}\mathcal{B}$ -derived canonical trajectory relation, law-admissible time representations, and differential law forms belong to different generation stages.

*Proof.* By Proposition 8.51, a local proper-time representation alone implies neither the right-uniqueness of the canonical trajectory relation nor the existence of an observer-time law core or law-admissible time representation. By Proposition 8.53, each minimal failure position among the conditions at each stage is classified by the corresponding failure-position token. Local proper-time representations, the  $\mathcal{CD}$ -derived canonical comparison quotient, the  $\mathcal{CD}\mathcal{B}$ -derived canonical trajectory relation, law-admissible time representations, and differential law forms therefore belong to different generation stages.  $\square$

**Corollary 8.55** (The two-stage misrecognition of the time-stretching account in GR). Gravitational clock differences in GR are comparison differences among generated law-admissible time representations corresponding to multiple local observers or worldlines. The time-stretching account presupposes time as an unconstructed entity prior to these generated representation differences and explains gravitational clock differences through stretching, slowing, or deformation of that time entity. By Proposition 8.23, this form of explanation is therefore the two-stage misrecognition of the time-stretching account.

*Proof.* By Proposition 8.51, a global law-time representation is not generated from a local proper-time representation alone. By Corollary 8.54, local proper-time representations, the  $\mathcal{CD}$ -derived canonical comparison quotient, the  $\mathcal{CD}\mathcal{B}$ -derived canonical trajectory relation, and law-admissible time representations belong to different generation stages. Gravitational clock differences in GR are therefore comparison differences among generated law-admissible time representations corresponding to multiple local observers or worldlines. By Proposition 8.23, in a form of explanation that places time as an unconstructed entity prior to these representation differences and redescribes subsequently generated comparison differences as changes in an attribute of that time entity, the first and third items of multistage misrecognition occur consecutively with direct dependence. This form of explanation is therefore the two-stage misrecognition of the time-stretching account.  $\square$

**Proposition 8.56** (Separation of the formal construction of general relativity from a bearer of time stretching). The formal construction of general relativity determines the metric, connection, worldlines, accumulation of local proper time, clock readings, clock rates, gravitational redshift, and their comparison relations. Gravitational conditions specify relations among  $\mathcal{CD}\mathcal{B}\mathcal{D}$  physical carriers, worldlines, local clock processes, signals, transport, records, and generated time representations.



If the account that “time itself stretches, slows, or warps” is taken as a literal physical mechanism, that account asserts a bearer of time stretching that remains identical through the deformation while taking different states. For this assertion to hold as a physical mechanism, it requires the type of the bearer of time stretching, an identity condition before and after deformation, a state space, a deformation magnitude, a deformation law, a generation path from nontemporal physical inputs, coupling laws to the metric, connection, and  $\mathcal{CDBD}$  physical carriers, and a measurement procedure independent of clock differences and gravitational redshift.

The formal results of general relativity established by Proposition 7.167 are comparison relations among clock readings, clock rates, proper times, gravitational redshift, and generated time representations. These formal results do not supply the type, identity condition, state space, deformation law, generation path, coupling laws, or independent measurement procedure of a bearer of time stretching. A literal time-stretching account therefore reintroduces an unconstructed bearer of change prior to the comparison differences generated by the formal construction and attributes those comparison differences to changes in an attribute of that bearer. This attribution is the two-stage misrecognition of Corollary 8.55.

No misrecognition occurs when differences in clock readings, clock rates, proper times, gravitational redshift, or generated time representations are described directly in terms of their objects and relations. Even when the term time stretch is used, no misrecognition occurs in an explicit metaphor if the source text states, within the same explanatory scope, its metaphorical status, that it does not designate literal deformation of time, and its actual referent.

When the term time stretch is used without stating its metaphorical status and actual referent, interpreting the term retrospectively as shorthand leaves a one-stage misrecognition that transfers a comparison difference to an attribute of time. In a literal account that places an unconstructed time entity in advance, this one stage is joined by pre-generation placement and constitutes the two-stage misrecognition of Corollary 8.55.

*Proof.* By Theorem 8.28 and Proposition 8.51, local clock processes, records, object-state differences, information coupling, and comparison generate local proper-time representations, while a global law-time representation requires comparison, transport preservation, loop consistency, quotient descent, and global gluing to hold. By Proposition 7.167, the formal results of general relativity supply clock readings, clock rates, proper times, gravitational redshift, and their comparison relations, but do not supply an independent state and evolution law for a bearer of time stretching.

Literal stretching, slowing, or warping requires an object that changes, its states before and after the change, and a condition for determining both states as states of the same object. Treating that change as the cause of a clock difference or gravitational redshift additionally requires coupling laws from the bearer of change to the metric, connection,  $\mathcal{CDBD}$  physical carriers, and clock processes, together with a measurement procedure independent of clock differences and gravitational redshift. The formal results of general relativity contain none of these constructions.

A literal time-stretching account therefore attributes a comparison difference among generated representations to an unconstructed bearer of time stretching. By Corollary 8.55, this attribution is a two-stage misrecognition. No misrecognition occurs when comparison differences on physical carriers are described directly or when metaphorical status and the actual referent are stated. A bare expression of time stretching constitutes a one-stage misrecognition that transfers a comparison difference to an attribute of time even under

an interpretation that does not assert an independent time entity.  $\square$

**Proposition 8.57** (Generation structure of gravitational clock differences). For the  $\mathcal{CDB}$  physical carriers of Definition 3.1 and Definition 3.2, suppose that the conditions of Theorem 8.28 hold. As the metric and connection, gravitational conditions specify the worldlines to which the  $\mathcal{CDB}$  physical carriers belong and the relations among local clock processes, signal propagation, transport, and comparison. The clock processes, records, object-state differences, and information coupling obtained from the  $\mathcal{CDB}$  physical carriers determine generated local proper-time representations.

Accumulation of clock processes belonging to different worldlines or local settings, clock readings, and signal-mediated comparisons determine gravitational clock differences, clock-rate differences, and gravitational redshift. Within a domain in which the subsequent conditions of Proposition 8.51 hold, generation from local proper-time representations to a global law-time representation holds, and these results can be used within the same validity domain. The physical grounds for these differences lie in the dependency chain of  $\mathcal{CDB}$  physical carriers, worldlines, the metric, the connection, signals, transport, comparison, and generated time representations.

This dependency chain does not contain primitive time as the object of gravitational action. An account that attributes results generated in relation to gravitational conditions to stretching, slowing, or deformation of primitive time itself placed prior to generation constitutes the two-stage misrecognition of Corollary 8.55.

*Proof.* Definition 3.1 and Definition 3.2 map each role in  $\mathcal{CDB}$  to a physical-carrier realization. By Theorem 8.28, the establishment of worldline-local clock processes, records, object-state differences, information coupling, and comparison generates local proper-time representations. By Proposition 8.51, generation of a right-unique  $\mathcal{CB}$ -derived canonical trajectory relation, observer-time law core, and law-admissible time representation requires, in addition to local proper-time representations, satisfaction of the conditions for comparison, transport preservation, loop consistency, quotient descent, and global gluing.

Accumulation of clock processes according to worldlines, the metric, and the connection, together with clock readings and signal-mediated transport and comparison, therefore supplies gravitational clock differences, clock-rate differences, and gravitational redshift as comparison results on generated time representations. In this generation order, gravitational conditions act on relations among  $\mathcal{CDB}$  physical carriers, worldlines, the metric, the connection, signals, transport, and comparison. By Corollary 8.55, an account that places primitive time prior to this comparison result and attributes the comparison result to a change in an attribute of primitive time itself is a two-stage misrecognition.  $\square$

**Proposition 8.58** (Safe reuse of general-relativistic results through generated time and failure classification). Within a domain in which the conditions of Theorem 8.28 hold, generated local proper-time representations can be used in worldline-local comparisons. Within a domain in which the conditions at the comparison, representation, dynamical-description, and law-description stages required by Proposition 8.51 hold, the right-unique  $\mathcal{CB}$ -derived canonical trajectory relation, observer-time law core, and law-admissible time representation can be used. The time evolution of the metric and connection, geodesic solutions, the time ordering of worldlines, gravitational clock differences, gravitational redshift, initial-value evolution, cosmological time relations, and law results that use them can be used within a domain in which their respective theory-specific grounding conditions and the generation conditions for the time content on which they depend hold.

Each minimal failure position classified by Proposition 8.53 determines the GR construction component that must be repaired first or the condition whose validity domain must be restricted. A failure at the comparison stage requires repair of local clock-side receptions, record-side receptions, comparison rules, transport preservation, loop consistency, quotient descent, or global gluing. A failure at the representation or continuous-approximation stage requires repair of the time representation, comparison domain, or scale domain of approximate validity. A failure at the dynamical-description, law-representation, or differential-consistency and law-form-preservation stage requires repair of the canonical trajectory relation, dynamical-description evidence datum, law-evaluation datum, or differential evaluation specification. When repair holds, the global law-time representation, time evolution of the metric and connection, geodesic solutions, time ordering of worldlines, gravitational clock differences, gravitational redshift, initial-value evolution, cosmological time relations, and law results are regenerated or recalculated in dependency order from the repaired inputs. If the relevant stage condition does not hold throughout the entire domain, the domain of use of the results is restricted to the domain in which that condition holds.

Corollary 8.55 separates gravitational clock differences among generated representations from the two-stage misrecognition that places an unconstructed time entity prior to those comparison differences and explains the differences through stretching, slowing, or deformation of that entity. The grounds for using general covariance, the field equations, geodesic solutions, gravitational redshift, and gravitational clock differences are supplied by generated time representations, theory-specific inputs, and the grounding conditions at each stage.

When an error source in local clock comparisons, transport rules, global gluing, gravitational redshift, geodesic calculations, initial-value evolution, continuous approximations, or differential evaluations is repaired, the repair's effect on accuracy is determined by an accuracy evaluation of the regenerated or recalculated downstream results. The decision to use general-relativistic results therefore separates into retention within the validity domain and repair from each minimal failure position or restriction of the validity domain.

*Proof.* By Theorem 8.28, worldline-local comparison data and generated representation coordinates generate local proper-time representations. By Proposition 8.51, generation of a right-unique  $\mathcal{CDB}$ -derived canonical trajectory relation, observer-time law core, and law-admissible time representation requires, in addition to local proper-time representations, satisfaction of the conditions at subsequent stages. By Proposition 8.21, when the time content and grounding structures on which each result depends are generated prior to law operations and result derivation, that time input satisfies physical-time qualification. The time evolution of the metric and connection, geodesic solutions, the time ordering of worldlines, gravitational clock differences, gravitational redshift, initial-value evolution, cosmological time relations, and law results that satisfy the grounding conditions at each stage and the theory-specific conditions therefore use time inputs that have physical-time qualification.

By Proposition 8.53 and Definition 6.17, conditions and results downstream of each minimal failure position depend on upstream conditions. When an input or rule belonging to each minimal failure position can be repaired, downstream results are regenerated or recalculated from the repaired inputs in dependency order. If the relevant stage condition does not hold throughout the entire domain, the domain of use is restricted to the domain in which that condition holds, and results generated within that domain are used.

Corollary 8.55 separates gravitational clock differences among generated law-admissible

time representations from the two-stage misrecognition that explains those comparison differences through changes in an attribute of an unconstructed time entity. This connects the grounds for general covariance, the field equations, geodesic solutions, gravitational redshift, and gravitational clock differences to generated time representations, theory-specific inputs, and the conditions at each stage.

When the source of error belongs to local clock comparisons, transport rules, global gluing, gravitational redshift, geodesic calculations, initial-value evolution, continuous approximations, or differential evaluations, downstream results can be regenerated or recalculated from inputs in which that source has been repaired, and an accuracy evaluation can be applied to the recalculated result. It follows that retention of general-relativistic results within the validity domain, the order of repair from each minimal failure position or restriction of the validity domain, and conditional evaluation of accuracy hold.  $\square$

## 8.7 Three-Stage Misrecognition in the Biological Application of Relativistic Explanations

**Proposition 8.59** (Relativistic three-stage misrecognition with biological application). Direct measurements of clocks, oscillators, electromagnetic radiation, or particle decay belonging to the first stage of Definition 7.183 yield comparison differences on physical carriers. An account that attributes the preceding comparison differences to slowing, stretching, or action of time itself independently of physical carriers belongs to the second stage. An account that further applies that temporal attribute to biological processes and treats the direct measurements at the first stage as direct demonstration of biological processes belongs to the third stage.

When the second-stage account presupposes an unconstructed time entity prior to the comparison differences and attributes those differences to changes in an attribute of the time entity as a literal physical mechanism, it constitutes the two-stage misrecognition of the time-stretching account in Proposition 8.23. With this two-stage misrecognition as antecedent, the transition that treats direct measurements at the first stage as direct demonstration at the third stage falls under Item 4 of Definition 8.19. An account that connects these three evidential stages consecutively through this dependency is therefore a relativistic three-stage misrecognition with biological application.

By Proposition 7.193, the number of direct demonstrations of biological processes among the seven references containing biological-application statements is 0/7, and the number of direct demonstrations of biological processes among the 72 audited references is 0/72. The references by Feynman–Leighton–Sands, Feynman 1995, Misner–Thorne–Wheeler, NIST, Rindler, Taylor–Wheeler, and Tong recorded in Proposition 7.185 contain statements that apply accounts of clocks, proper time, or particle processes to pulse rate, thought, cancer development, aging of the human body, the ages of twins, human lifespan, biological clocks, or the human heart. Among them, the Feynman 1995 reference generalizes from nuclear transitions and frequency shifts to all physical processes and then immediately applies that generalization to human life and aging, making the dependency chain explicit within the same reference. These biological-application statements are determined to be theoretical predictions or explanatory applications in accordance with Corollary 7.194.

The validity of a theoretical prediction concerning biological application is determined by coupling laws between the physical carriers that constitute biological processes and relativistic conditions. Direct demonstration of biological processes requires a biological object, an experimental configuration that separates the relativistic condition, a mea-

surement procedure for the biological process, and corresponding measurement results. An account that states that “time itself slows, so organisms age differently, and this has been confirmed by clock or particle experiments” transfers a comparison difference on a first-stage physical carrier to a second-stage temporal attribute and then transfers that temporal attribute to a third-stage biological phenomenon. This consecutive transfer is a relativistic three-stage misrecognition with biological application that twice replaces the object of measurement, the bearer of physical action, and the evidence type.

*Proof.* Proposition 7.185 separates direct measurement of physical carriers, attribution of a property to time itself, and application to biological processes into distinct evidential stages. Proposition 7.184 gives, as necessary conditions for direct demonstration of a biological application, inclusion of the biological process among the experimental objects, an experimental configuration that separates the relativistic condition, a measurement procedure for the biological process, and corresponding measurement results.

An account that presupposes an unconstructed time entity prior to the comparison differences and attributes the first-stage comparison differences to changes in an attribute of that entity is the two-stage misrecognition of Proposition 8.23. An account that extends the experimental object domain of the first stage to the biological processes of the third stage and identifies direct demonstration at the two stages satisfies Item 4 of Definition 8.19. The counts in Proposition 7.193 show that, for both the seven references and the 72 audited references, the experimental configurations and measurement results required for direct demonstration at the third stage have not been obtained. An account that connects the three evidential stages consecutively through this dependency is therefore a relativistic three-stage misrecognition with biological application. □

In this consecutive transfer, the object under discussion changes from the clock to time and then to a biological process while retaining the evidence obtained from the clock.

**The target changes while the evidence is carried forward**

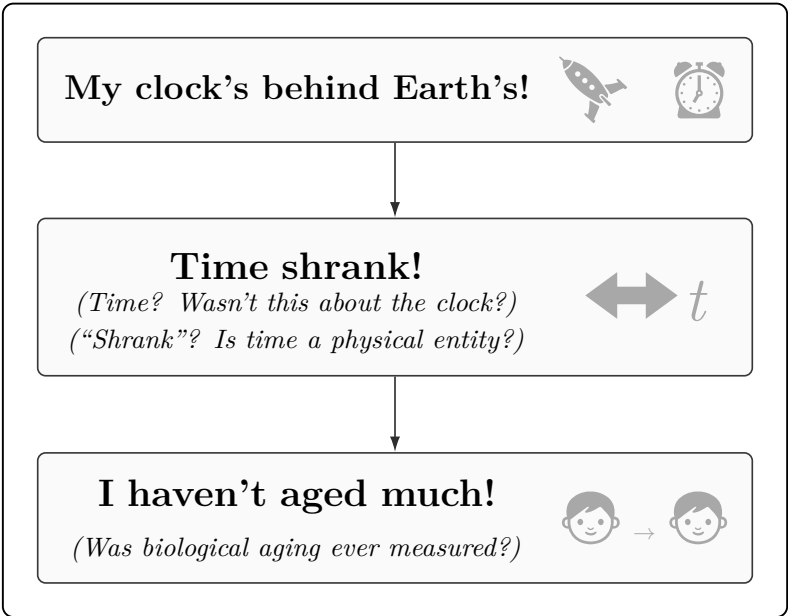


Figure 22: The object under discussion shifts among a clock, time, and a biological process; two questions identify the evidence boundary at each transition.

## 8.8 Worked Reduction for QG

In the quantum-reconstruction scheme, law-admissible time representations and observer-time law cores are generated from records, branch sensitivity, object-state differences, observer-side information coupling, comparison gluing, dynamical-description operability, and law-description operability.

A quantum-gravitational reconstruction that does not include external time among its primitive inputs has constraint systems, canonical gravity, and Wheeler–DeWitt-type descriptions as its background[19, 20, 21]. The problem of time, internal clocks, and relational observables in quantum gravity are representative research subjects concerning the generation conditions for time representations and their qualification for use in laws[8, 9, 10, 11].

Eliminating external time does not generate time content, record order, comparison order, observer-side information coupling, or qualification for use in laws. Even when an internal degree of freedom is selected as a clock, primitive-time input remains merely replaced by the internal clock unless time content is generated from differences, records, comparisons, and observer-side information coupling involving that degree of freedom. If primitive-time input is used to derive results concerning constraint equations, relational change, observables, initial boundaries, the arrow of time, measurement records, or black-hole information, those results inherit dependence on time content that lacks physical-time qualification. Mathematical consistency, predictive performance, and empirical fit do not repair this loss of qualification.

On the same dependency chain, the QG scheme generates clock-candidate-side materials, record-side retention, object-state-side differences, observer-side information coupling, comparison closure, the canonical observer-time quotient, the canonical trajectory relation, law-admissible time representations, and law-output determinacy, and determines the qualification of the time content used by each result in quantum gravity.

### 8.8.1 From QG Inputs to an Observer-Time Law Core

The inputs to the quantum-reconstruction scheme consist of clock-side or clock-candidate-side tokens, record-side retention, branch-sensitive record structures, object-state-side differences, observer-side information-coupling positions, comparison-gluing conditions, and source-side ternary relations at the dynamical-description stage.

Define this input by

$$\mathcal{X}_{\text{Sch}_{\text{QG}}} := \left( \mathcal{I}_{\text{Sch}_{\text{QG}}}^{\text{raw}}, \text{Dat}_{\text{Sch}_{\text{QG}}}^{\text{src}} \right)$$

. The raw operational input  $\mathcal{I}_{\text{Sch}_{\text{QG}}}^{\text{raw}}$  contains no external time, and the QG-side source-realization datum  $\text{Dat}_{\text{Sch}_{\text{QG}}}^{\text{src}}$  has as components the comparison-side arrival maps, recovery maps, record retention, comparison gluing, direct dynamical-description arrival maps, source-side pair relations, and source-side ternary relations.

Let the observer-side information-coupling position be  $o$ , and let the single-object specification index be  $\zeta_{\text{Sch}_{\text{QG}}}$ . Use the QG-scheme-restricted complete evaluation specification  $\text{Ev}_{\text{Sch}_{\text{QG}}}$  of Definition 8.5.

The generation sequence in the worked reduction for QG is

$$\begin{aligned}
& (\mathcal{X}_{\text{Sch}_{\text{QG}}}; \mathcal{J}_{\text{cmp}, \text{Sch}_{\text{QG}}}) \\
& \longmapsto \text{Qu}^{\text{cmp}, \text{can}} (\mathcal{X}_{\text{Sch}_{\text{QG}}}; \mathcal{J}_{\text{cmp}, \text{Sch}_{\text{QG}}}) \\
& \longmapsto \text{Dat}_{o, \zeta \text{Sch}_{\text{QG}}}^{\text{dyn}, 0} \\
& \longmapsto \text{DynWit}_{o, \zeta \text{Sch}_{\text{QG}}} [\text{RepEmb}_o^{\text{dyn}, \text{time}}] \\
& \longmapsto \text{TimeRep}_o^{\text{dyn}} \\
& \longmapsto \text{LawCore}_{o, \zeta \text{Sch}_{\text{QG}}}^{\text{obs}, \text{time}} (\text{RepEmb}_o^{\text{dyn}, \text{time}}).
\end{aligned}$$

The comparison-stage objects require exact operability; the dynamical-description objects and the law-admissible time representation require dynamical-description operability; and the observer-time law core and qualification for use in laws require law-description operability.

Use the real-valued time-order embedding selected in Definition 5.59. Suppose that the law-evaluation datum satisfies the equality of evaluation specifications under Proposition 5.86.

**Proposition 8.60** (Law admissibility in the worked reduction for QG). Suppose that, for the QG-scheme-restricted complete evaluation specification,

$$\text{LawOperative} [\text{LawEvalDat}_{o, \zeta \text{Sch}_{\text{QG}}}]$$

holds. Then the law-admissible time representation  $\text{TimeRep}_o^{\text{dyn}}$  and the observer-time law core

$$\text{LawCore}_{o, \zeta \text{Sch}_{\text{QG}}}^{\text{obs}, \text{time}} (\text{RepEmb}_o^{\text{dyn}, \text{time}})$$

are generated. For any

$$T_0 \in \text{Qu}_{\mathfrak{c}, o}^{\text{dyn}, \text{can}}$$

the law-time value is

$$t_0 := \text{TimeRep}_o^{\text{dyn}} (T_0)$$

.

*Proof.* Definition 5.102 and the equality of evaluation specifications imply dynamical-description operability for the QG-scheme-restricted complete evaluation specification. By Theorem 8.31, the canonical observer-time quotient, canonical object-state quotient, canonical trajectory relation, graph trajectory, canonical family of observable functions, and law-admissible time representation are generated. Definition 5.108 generates the observer-time law core. The equality for the law-time value follows from Definition 5.63.  $\square$

**Proposition 8.61** (Generation condition for a time representation in QG reconstruction). When the dynamical-description-operability condition of Theorem 8.31 holds for the QG-scheme-restricted complete evaluation specification, the law-admissible time representation  $\text{TimeRep}_o^{\text{dyn}}$  is generated. If, in addition,

$$\text{LawOperative} [\text{LawEvalDat}_{o, \zeta \text{Sch}_{\text{QG}}}]$$

holds, this time representation satisfies qualification for use in laws and the observer-time law core is generated.

*Proof.* By Definition 5.70 and Theorem 8.31, satisfaction of dynamical-description operability generates the canonical observer-time quotient, canonical object-state quotient, canonical trajectory relation, graph trajectory, canonical family of observable functions, and law-admissible time representation. By Definition 5.102, law-description operability is the simultaneous satisfaction of dynamical-description operability and the law condition groups. By Definition 5.108, this condition generates the observer-time law core. By Corollary 7.15, generation of law-time content precedes law operations and result derivation. The time representation generated at the dynamical-description stage therefore satisfies qualification for use in laws through the satisfaction of law-description operability.  $\square$

**Connection to the audit of law-time use.** Law-time use in a quantum-gravitational reconstruction is subject to the audit of physical-time qualification in Chapter 7. When every generation structure required by the time content used by a law-time use  $\vartheta$  is generated from nontemporal physical inputs and is established prior to law operations and result derivation, that time content satisfies physical-time qualification. If any generation stage for the evaluation specification fails, simultaneous satisfaction of all stages for that complete evaluation specification is not obtained. When time content is used in a law operation or result derivation prior to its generation structures, that time content is a primitive-time input under Definition 7.10. By Theorem 7.11, this time use does not satisfy physical-time qualification. By Proposition 7.13, mathematical consistency, predictive performance, and empirical fit do not repair this loss of qualification.

### 8.8.2 Law Admissibility without External Time

The problem of time in quantum gravity arises not only from the tension between law descriptions that use external time and theoretical constructions that do not take background spacetime as a primitive premise, but also from assigning temporal roles to constraint equations, internal clocks, or relational observables without generating temporal content after external time has been removed. Removal of external time eliminates one form of primitive-time input, but it does not generate clock-candidate-side differences, record retention, comparison order, observer-side information coupling, the canonical observer-time quotient, or a law-admissible time representation. Even when an internal degree of freedom is selected as a clock, that internal clock bears ungenerated temporal content unless it satisfies the same generation pathway.

When ungenerated temporal content enters a constraint equation or law operation, the time variable fails physical-time qualification. Results concerning relational change, observables, initial boundaries, the arrow of time, measurement records, and black-hole information that are derived from ungenerated temporal content inherit this qualification failure. The problem of time in quantum gravity therefore cannot be resolved solely by the presence or absence of external time or by the choice of a clock variable; it requires simultaneous determination of the generation of temporal content, qualification for use within laws, and the inheritance of dependence by results.

The QG scheme requires clock-candidate-side materials, record-side retention, object-state-side differences, observer-side information coupling, comparison closure, the canonical observer-time quotient, the canonical trajectory relation, a law-admissible time representation, and law-output determinacy to hold along the same dependence chain. This dependence chain generates the time variable, confers qualification for use within laws, and excludes the inheritance of primitive-time dependence by results.



**Proposition 8.62** (Relativity, relationality, and non-primitivity do not imply nonexistence). The dependence of an object on a frame, observer, state, local region, or relation to another object supplies conditions for that object's existence. The absence of a preferred representative expresses nonuniqueness of choice; being nonabsolute expresses the failure of frame-independent, stand-alone entityhood; and being nonprimitive expresses a dependence position in which the object is generated from other physical inputs. Each condition preserves the possibility that the object exists as a relational or generated object and does not imply its nonexistence.

A claim that time does not exist must specify the type and domain of the temporal content being denied and establish that no object of that type performing a temporal role exists within the theory. If instants, relations among configurations, an order of change, histories, evolution, or clock comparisons perform temporal roles, declining to adopt an independent absolute time entity and denying the existence of all temporal structure are different propositions.

*Proof.* Relational dependence, nonuniqueness, nonabsoluteness, and generative dependence specify conditions for an object or its relations to other objects. An existence denial states that, within a specified type and domain, no object of that type exists. The latter therefore does not follow from the former.

Definition 7.10 classifies temporal content used before generation as primitive-time input. Corollary 7.15 requires temporal content to be generated from nontemporal physical inputs before it is used within a law. In this generation order, temporal content exists as a generated object rather than a primitive input. Non-primitivity therefore does not imply the nonexistence of temporal structure.

Time representations dependent on multiple observers, states, or local regions also exist as relational or local temporal structures when comparison rules, gluing conditions, and domains of validity are supplied. Failure of a preferred representative or global representation does not imply the nonexistence of local or relational temporal structure.  $\square$

**Proposition 8.63** (Formal obligations on claims of time creation and time nonexistence). A theory that literally claims the creation of time must exhibit a generation path from nontemporal physical inputs through object-state differences that presuppose no temporal order, records that retain those differences, comparison rules among the records, observer-side information coupling, temporal order, a time representation, and qualification for use within laws; it must also exclude primitive-time roots from every temporal content it uses.

A theory that literally claims the nonexistence of time must, in accordance with Proposition 8.62, separate claims of relativity, relationality, non-preference, nonabsoluteness, or non-primitivity from an existence denial. It must then delimit the temporal content, temporal order, time representations, and law time that it denies; supply a formal construction in their place; and establish that no temporal dependence root remains in the replacement construction and that subsequent results do not depend on any such root. If the replacement construction uses “instant,” “change,” “history,” “before and after,” or “evolution,” the grounds for each must be generated from nontemporal physical inputs.

These formal obligations apply to every theory that advances the creation, emergence, recovery, or nonexistence of time, and they apply with equal force to the generation of time in this paper.

*Proof.* By Corollary 7.15, generation of temporal content precedes law operations and result derivation. By Definition 7.10, temporal content used before its grounding struc-

ture is primitive-time input. By Theorem 7.11, primitive-time input fails physical-time qualification. By Corollary 7.12, that dependence is inherited by subsequent results.

A claim of time creation therefore requires a generation path from nontemporal physical inputs to the temporal content used, together with exclusion of primitive-time roots. A claim of time nonexistence requires a delimited object of denial, a replacement construction, and exclusion of temporal dependence roots from that construction. Renaming, eliminating a variable, substituting a relation, or omitting temporal vocabulary does not discharge these formal obligations.  $\square$

**Proposition 8.64** (Interpretation-path-specific determination of statements of time creation and time nonexistence). The Barbour 1994 I source presented in Subsection 7.3.14 contains the following statement.

Original English:

change creates time

Original English:

It is argued that time as such does not exist but that instants, defined as complete relative configurations of the universe, do.

The Barbour 1994 II source contains the following statement.

Original English:

time is truly non-existent in the kinematic foundations of both theories

The three statements are determined separately for each statement and each admissible interpretation, along the interpretation-specific evidence-comparison path of Definition 7.117.

If “change creates time” is interpreted as the construction, within the explicitly stated domain of application, of configurational differences, an ordering, a scale, a clock correspondence, or an ephemeris-type parameter, it directly describes the corresponding formal construction. The number of crossings of evidential scope is then zero, and no multi-stage misrecognition arises. This interpretation does not claim physical generation of every temporal content used in the source.

If the same statement is interpreted as complete generation of physical time, the operation that uses a realized history, path parameter, or existing temporal content before its generation falls under item 1 of Definition 8.19. The further operation that identifies a restricted configurational construction with physical time possessing generation of every temporal content, exclusion of primitive-time roots, and qualification for use within laws falls under item 2. An interpretive path that connects these two operations by direct dependence constitutes a two-stage misrecognition.

If “time as such does not exist” and “time is truly non-existent in the kinematic foundations” are interpreted as restricted to declining to adopt absolute external time or to the kinematic priority of relative configurations, the number of crossings of evidential scope is zero, and no multi-stage misrecognition arises. An interpretation that extends these statements to the nonexistence of physical time in general transfers the force of an existence denial from a restricted object and domain of application to a different object and domain. It therefore constitutes a one-stage misrecognition under item 4 of the same definition. The total of 39 temporal contents and the time-dependent transfers present in the Barbour 1994 I and II sources are incompatible with this extended interpretation.

*Proof.* Subsection 7.3.11 and Remark 7.159 establish the original wording, source attribution, and context of the three statements. Theorem 7.190 establishes that all 18 temporal contents in Barbour 1994 I and all 21 temporal contents in Barbour 1994 II are primitive-time roots. The source-specific Propositions S.305 and S.316 partition the 25 results in Barbour 1994 I into 9 results with evidence-only reach and 16 results that depend on primitive-time roots, and the 26 results in Barbour 1994 II into 8 results with evidence-only reach and 18 results that depend on primitive-time roots. Neither source has generated results or results that remain unreachable after the roots are added. Establishing a result with evidence-only reach does not imply the generation of temporal content.

The interpretation restricted to a configurational construction remains within the evidential scope of the formal result and crosses no evidential boundary. The interpretation as complete generation of physical time directly connects pre-generation use of primitive-time roots to identification of a partial construction with complete generation. The interpretation as the nonexistence of physical time in general transfers an existence denial from the restricted domain of absolute external time or kinematic foundations to physical time in general. The respective numbers of stages are therefore zero, two, and one.  $\square$

**Proposition 8.65** (Interpretation-path-specific determination of generation obligations in thermal-time statements). The Connes–Rovelli 1994 source presented in Subsection 7.3.14 contains the following statements.

Original English:

there is no preferred time flow

Original English:

thermal time ... defines the physical time

If “there is no preferred time flow” is interpreted as restricted to the absence of an externally specified, state-independent preferred flow, the number of crossings of evidential scope is zero, and no multi-stage misrecognition arises. An interpretation that extends the same statement to the absence of every physical time flow transfers the absence of a restricted flow to a different object and domain of application. It therefore constitutes a one-stage misrecognition under item 4 of Definition 8.19.

If “thermal time ... defines the physical time” is interpreted as restricted to the mathematical construction in which a thermal state selects a modular automorphism flow, or to a postulate assigning a physical-time interpretation to that flow, the number of crossings of evidential scope is zero, and no multi-stage misrecognition arises. This interpretation does not claim a theorem-level derivation of physical-time qualification.

An interpretation that identifies the selected state-dependent flow with physical time possessing every generation condition and qualification for use within laws transfers a partial construction to the establishment of a subsequent stage. It therefore constitutes a one-stage misrecognition under item 2 of the same definition. An interpretation that further claims complete generation of physical time from nontemporal physical inputs, although the construction uses a real parameter, known Hamiltonian time, proper time, geometric time, or FRW time as a pre-generation input, directly connects items 1 and 2 and constitutes a two-stage misrecognition.

*Proof.* Subsection 7.3.11 and Remark 7.159 establish the original wording, source attribution, and context of the two statements. Proposition 7.171 establishes that every temporal content in the source is a primitive-time root. Theorem 7.190 establishes that the set

of audit-generated temporal contents is empty. The source-specific Proposition S.328 establishes that the 37 results in the source consist of 16 results with evidence-only reach and 21 results that depend on primitive-time roots.

By Proposition 7.137 and Corollary 7.138, selecting a time flow from a state does not supply the generation history, comparison structure, or conditions of validity of the input temporal content. By Proposition 7.172, the scope of the formal result concerning thermal time does not coincide with that of complete time generation.

The restricted flow selection and interpretive postulate remain within the evidential scope of their formal result. The path that transfers a partial flow to complete establishment of physical time has one stage under item 2. The path that treats a construction using pre-generation time roots as complete generation from nontemporal inputs has two stages under items 1 and 2. The path that extends the absence claim to every physical time flow has one stage under item 4. The stage count is therefore zero, one, or two according to the admissible interpretation.  $\square$

**Proposition 8.66** (Interpretation-path-specific determination of time-emergence statements through bipartite separation). The Kiefer 2009 source presented in Subsection 7.3.14 contains the following statements.

Original English:

Time is absolute in standard quantum theory and dynamical in general relativity.

Original English:

Time emerges from the separation into two different subsystems

If the second statement purports to assert that separation into two subsystems generates temporal content usable by physical laws, the formal obligations concerning time emergence in Proposition 8.63 arise.

For the formal audit of this source, in the component order of Definition S.4,

$$\text{AuditPartIdx}_R = (\emptyset, [35]; \emptyset, \{5, 6, 7, 8, 13\}, \{1, 2, 3, 4, 9, 10, 11, 12, 14, 15\}, \emptyset).$$

If the first statement is interpreted as restricted to the difference in formal role between an external background parameter in standard quantum theory and a dynamical spacetime variable in general relativity, the number of crossings of evidential scope is zero, and no multi-stage misrecognition arises.

If the second statement is interpreted as restricted to constructing a semiclassical time parameter or an internal-clock candidate through separation into two subsystems, it directly describes a source-specific approximate construction, and the number of crossings of evidential scope is likewise zero. If the same statement is interpreted as complete generation of physical time and the subsystem parameter is identified with generation of every temporal content, qualification for use within laws, and exclusion of primitive-time roots, it constitutes a one-stage misrecognition under item 2 of Definition 8.19. An interpretation that further treats a semiclassical construction using pre-generation time roots as complete generation from nontemporal physical inputs directly connects items 1 and 2 and constitutes a two-stage misrecognition.

An interpretation that transfers the absence of external time from the fundamental equation to the nonexistence of physical time in general changes the object and domain of

application. It therefore constitutes a one-stage misrecognition under item 4 of the same definition.

*Proof.* Proposition 7.169 establishes the original wording, source attribution, and original locations of the two statements, and separates the contrast between temporal roles in standard quantum theory and general relativity, bipartite time, complete time generation, exclusion of primitive-time roots, and qualification for use within laws into distinct evidential scopes. By Theorem 7.190, the set of audit-generated temporal contents is empty and every temporal content in the source is a primitive-time root. For the Kiefer 2009 reference specifically, Proposition S.539 establishes that the 15 results consist of 5 results with evidence-only reach and 10 results that depend on primitive-time roots. The four-way partition in Proposition 7.49 then implies that the audit-generated result set is empty and that all 15 results lie beyond reach from the generation-initial set.

Proposition 8.63 requires, for a literal claim of time emergence, a generation path from nontemporal physical inputs through temporal content, temporal order, a time representation, and qualification for use within laws, together with exclusion of primitive-time roots from every temporal content. Because the source contains no audit-generated temporal content and every result depends on primitive-time roots, these formal obligations are not fulfilled.

The interpretation restricted to a semiclassical time parameter or internal-clock candidate remains within the evidential scope of a partial formal result. The path that transfers the partial construction to complete establishment of physical time has one stage under item 2; the path that connects use of pre-generation time roots to identification with complete generation has two stages under items 1 and 2; and the path that transfers the absence of external time to the nonexistence of physical time in general has one stage under item 4. The stage count is therefore zero, one, or two according to the admissible interpretation.  $\square$

**Proposition 8.67** (Interpretation-path-specific determination of generation qualification in the statement that time is fundamental). The Smolin 2018/2020 source presented in Subsection 7.3.14 contains the following statement.

Original English:

I take the opposite view, that time is fundamental.

If this statement purports to license the use of temporal content as a ground of physical laws or as a law input prior to its generation, that temporal content is a primitive-time input under Definition 7.10. By Theorem 7.11, that temporal content fails physical-time qualification.

For the complete determination and result classification of this source, in the component order of Definition S.4,

$$\begin{aligned} \text{AuditPartIdx}_R &= (\emptyset, [27]; \\ &\quad \emptyset, \{2, 3, 4, 6, 7, 21, 22\}; \\ &\quad \{1, 5\} \cup \llbracket 8, 20 \rrbracket \cup \llbracket 23, 28 \rrbracket, \emptyset). \end{aligned}$$

The local-reach determination gives

$$\begin{aligned} \text{TimeContent}_R^{\text{audit,evid}} &= \emptyset, \\ \left| \text{TimeContent}_R \setminus \text{TimeContent}_R^{\text{audit,evid}} \right| &= 27. \end{aligned}$$

There are 127 obligation atoms.

If “time is fundamental” is interpreted as a classification asserting the theoretical priority or irreducibility of time, the number of crossings of evidential scope is zero, and no multi-stage misrecognition arises. This classification does not assert generation-prior qualification of temporal content.

If the same statement is interpreted as approving the use of temporal content before its generation as a ground of physical laws or as a law input, that temporal content becomes a primitive-time input. This interpretation constitutes a one-stage misrecognition under item 1 of Definition 8.19. Neither the local-reach determination nor the complete determination yields any generated temporal content. All 27 source-specific temporal contents remain primitive-time roots, and 21 of the 28 results depend on primitive-time roots.

*Proof.* Proposition 7.169 separates the fundamentality of time, generation of all source-specific temporal contents, exclusion of primitive-time roots, and qualification for use within laws into distinct evidential scopes. Proposition 7.171 establishes the original wording, source attribution, and original location of the statement, that the set of strictly generated temporal contents is empty, and that all 27 source-specific temporal contents are primitive-time roots. No temporal content is reached under the local-reach determination, so all 27 remain primitive-time roots. Proposition 7.136 establishes that there are 127 obligation atoms and that 21 of the 28 results depend on primitive-time roots. The source-specific result classification in Proposition S.626, together with the four-way partition in Proposition 7.49, establishes that the audit-generated result set is empty and that all 28 results, comprising 7 results with evidence-only reach and 21 results that depend on primitive-time roots, lie beyond reach from the generation-initial set.

By Corollary 7.15, generation of temporal content precedes law operations and result derivation. For this source, neither the local-reach determination nor the complete determination yields any generated temporal content. None of the 27 temporal contents is therefore generated, and the dependence of 21 of the 28 results on primitive-time roots remains.

The interpretation restricted to theoretical priority or irreducibility remains within the evidential scope of a classification and has zero stages. The interpretation that licenses use within laws before generation places a primitive-time input among the law inputs and therefore constitutes a one-stage misrecognition under item 1. Along either path, the qualification of temporal content used in physical laws is determined by its generation path from nontemporal physical inputs.  $\square$

**Proposition 8.68** (The problem of time as law admissibility). Even when external time is absent from the raw operative input to the QG-scheme-restricted complete evaluation specification, the establishment condition for the law-description stage is

$$\text{LawOperative} \left[ \text{LawEvalDat}_{o, \zeta_{\text{Sch}_{\text{QG}}}} \right]$$

. If temporal content is used in a constraint equation, law operation, or result derivation before the generated structures required for that use are established, that content is a primitive-time input, and results depending on it inherit the qualification failure.

*Proof.* Exclusion of external time from the input is a component condition on the raw operative input. Law-description operability under Definition 5.102 requires the joint satisfaction of dynamical-description operability and the law condition group. Removing

external time does not supply the generation conditions constituting law-description operability.

When law-description operability holds, Proposition 8.60 generates a law-admissible time representation, observer-time law core, and law-time value. By Corollary 7.15, generation of law-time content precedes law operations and result derivation.

If temporal content is used before the generated structures required for that use are established, Definition 7.10 and Theorem 7.11 make that content a primitive-time input that fails physical-time qualification. By Corollary 7.12, dependence on that temporal content is inherited by the results. By Proposition 7.13, mathematical consistency, predictive performance, and empirical adequacy do not repair this qualification failure.  $\square$

**Proposition 8.69** (Failure localization in the worked QG reduction). If a generation condition in the QG-scheme-restricted complete evaluation specification fails, each minimal failure position is classified within the six series of comparison, representation, continuous-approximation, dynamical-description, law-description, or differential failure. The QG-specific correspondences are as follows.

- (1) When a comparison-stage or representation-stage condition concerning record sequences, object-state identification, or comparison gluing is formed and evaluable, its failure is classified by a failure-position token of the corresponding stage.
- (2) Where evaluation readiness holds for condition 3, excess  $\mathfrak{D}$ -record cell width is determined by

$$\text{FA}_3 [\text{Ev}_{\text{Sch}_{\text{QG}}}; \lambda]$$

- (3) Whether a failure occurs at the dynamical-description stage is determined by

$$\bigvee_{i \in I_{\text{dyn}}^{\text{fail}}} \text{FDyn}_i [\text{Ev}_{\text{Sch}_{\text{QG}}}]$$

- (4) Whether a failure occurs at the law-description stage is determined by

$$\bigvee_{i \in I_{\text{law}}^{\text{fail}}} \text{FL}_i [\text{Ev}_{\text{Sch}_{\text{QG}}}]$$

- (5) Whether a failure occurs at the differential stage is determined by

$$\bigvee_{i=1}^5 \text{FDiff}_i [\text{Ev}_{\text{Sch}_{\text{QG}}}]$$

*Proof.* By Proposition 6.26, the failed-token set equals its set of minimal elements. By Definition 6.10, the corresponding failure-position token is assigned to each minimal failure position. Failure of a comparison-stage condition is classified by Definition 6.36; failure of a representation-stage condition, by Definition 6.58; failure of a continuous-approximation-stage condition, by Definition 6.77; failure of a dynamical-description-stage condition, by Definition 6.104; failure of a law-description-stage condition, by Definition 6.110; and failure of a differential-stage condition, by Definition 6.137.

Where evaluation readiness holds for condition 3, excess  $\mathfrak{D}$ -record cell width is determined by Definition 6.79. The minimal failure classes at the dynamical-description, law-description, and differential stages are given, respectively, by Definitions 6.104, 6.114, and 6.140. Each minimal failure position in the QG-scheme-restricted complete evaluation specification is therefore classified within one of the six failure series.  $\square$

### 8.8.3 Cosmological initial boundaries and time-generation conditions

When a cosmological initial boundary is specified by a time variable, if generation of that variable does not precede the law description of the initial boundary, the beginning of the universe, what is before or after that beginning, and the direction of time are described by a temporal order lacking grounds. If a theory describing the origin of time uses ungenerated time in its account of that origin, the theory uses the temporal order to be explained as an explanatory input before generating it, and results derived from the initial boundary inherit primitive-time dependence.

Quantum cosmology and the problem of initial conditions concern cosmological initial boundaries and temporal orientation[22, 23]. From local displays, records, object-state differences, and observer-side information coupling, the QG scheme generates comparison closure, the canonical observer-time quotient, a law-admissible time representation, and an observer-time law core. If a time variable is applied to an initial boundary before generation holds, its qualification failure is determined as primitive-time input; if it is applied after generation holds, it is evaluated as a boundary on a generated time representation.

**Proposition 8.70** (Time-generation conditions at a cosmological initial boundary). Suppose that the QG-scheme-restricted complete evaluation specification targets a cosmological initial boundary. When dynamical-description operability holds, the law-admissible time representation used to describe the initial boundary is generated. When law-description operability holds, that time representation satisfies qualification for use within laws and the observer-time law core is generated.

*Proof.* For a QG-scheme-restricted complete evaluation specification, Proposition 8.61 gives generation of a law-admissible time representation through dynamical-description operability and generation of qualification for use within laws and an observer-time law core through law-description operability. The cosmological initial boundary belongs to the same evaluation specification through the object specification. The generation conditions of that proposition therefore apply to the description of the initial boundary.  $\square$

**Proposition 8.71** (Localization of time failure at a cosmological initial boundary). If any generation condition in Proposition 8.70 fails, each minimal failure position is classified by the corresponding failure-position token in the six failure series.

*Proof.* Proposition 8.69 classifies each minimal failure position in a QG-scheme-restricted complete evaluation specification within the six failure series. Because a cosmological initial boundary belongs to the object specification of the same evaluation specification, the classification of that proposition applies.  $\square$

### 8.8.4 Consequences for extreme-gravity domains, observables, and the arrow of time

In the black-hole information problem, descriptions of information preservation, radiation, records, external observers, and interior regions compare the periods before and after radiation, the evaporation process, and information recovery along the same temporal evolution. Black-hole entropy, Hawking radiation, predictability, and recovery of information in radiation are representative lines of research[24, 25, 26, 27]. If this temporal evolution uses ungenerated temporal content, results asserting information preservation, information loss, or information recovery inherit primitive-time dependence. The QG scheme generates



records, comparisons, observer-side information coupling, a law-admissible time representation, a scale-indexed effective domain, and failure positions, thereby determining the physical-time qualification of the temporal evolution used in the information problem.

For the problem of observables, selecting an internal degree of freedom as a clock and describing its relation to other quantities does not by itself generate temporal content from differences in that clock degree of freedom, records, comparisons, and observer-side information coupling. Relational changes and observable values defined relative to an ungenerated internal clock inherit primitive-time dependence. Joint satisfaction of object-state differences, observer-side information coupling, the canonical observer-time quotient, the canonical family of observable functions, and a law-admissible time representation confers qualification for use within laws on observable values. Constrained systems, canonical quantum gravity, and relational observables are representative lines of research[19, 21, 11].

The arrow of time is an additional orientation condition on a generated time representation. If a low-entropy initial condition or the orientation of a coarse-grained history is placed on ungenerated time, the before-and-after relation on which that orientation depends remains without physical-time qualification. Thermodynamic orientation, cosmological orientation, and record orientation are evaluated when record retention, comparison closure, observer-side information coupling, and coarse-grained histories hold on a generated time representation. Cosmological initial conditions, singularities, and boundary conditions in quantum cosmology are representative lines of research[23, 22].

When quantum measurement records are assigned a before and after, a recording time, or a decoherence process, if the temporal content bearing that order is ungenerated, the measurement records and decoherence results inherit primitive-time dependence. Joint satisfaction of object-state differences, record-side retention, observer-side information coupling, and a law-admissible time representation temporalizes the measurement records. Mathematical formulations of the measurement process and decoherence are representative lines of research[28, 29].

Proposition 8.23 classifies an explanatory form that treats relativistic clock differences as stretching, slowing, or deformation of an unconstructed time entity as the two-stage misrecognition of the time-stretching account. Atomic-clock differences, differences in molecular-vibration periods, covalent-bond vibration sequences, electronic-state transitions, scattering events, and decay events supply physical carriers from which  $\mathfrak{C}$ -derived sequences,  $\mathfrak{D}$ -derived records,  $\mathfrak{B}$ -derived differences, and  $\mathfrak{O}$ -derived coupling generate time representations.

Gravitational, geometric, and medium conditions modulate the physical carriers supplied by atomic clocks, molecular vibrations, covalent-bond vibrations, electronic-state transitions, scattering detections, and decay detections. This modulation is classified as changes in role-carrier realization, record retention, comparison gluing, object-state identification, observer-side information coupling, trajectory relations, and law-output determinacy.

Atomic-clock and period differences under gravitational conditions, conventionally represented as proper-time differences, are comparison-derived differences among generated law-admissible time representations. By Proposition 8.32, clock and period differences reduce to differences in frequency representations obtained from  $\mathfrak{C}$ -derived cycle-count differences and generated representation-coordinate differences. Stretching of a time entity is not the generation mechanism of these comparison-derived differences.

Using electronic-state transitions, scattering events, and decay events as event sequences

on a time variable requires generation of that variable to precede the law description of the event sequence. Electronic-state transitions contribute to the establishment of object-state differences and trajectory relations; scattering and decay events contribute to the establishment of detection records, object-state differences, and observer-side information coupling.

In extreme-gravity domains, identification of  $\mathfrak{C}$ -derived sequences, preservation of the correspondence between  $\mathfrak{D}$ -derived records and object-state differences, the  $\mathfrak{C}\mathfrak{D}\mathfrak{B}$ -derived canonical trajectory relation, or law-output determinacy may fail. By Proposition 8.69, each minimal failure position is classified within the six failure series.

For theories of spacetime emergence, the QG scheme requires generation conditions for the time representations used in post-emergence spatiotemporal descriptions, scale-indexed effective domains, and failure positions. Spacetime construction from light and free fall, causal sets, holography, and connections between entanglement and geometry are representative lines of research[30, 31, 32, 33].

In descriptions concerning the cosmological constant and dark energy, expansion history, observational records, distance scales, time values, and law representations are joined on the same generated time representation. Vacuum-energy values, matter components, and field content belong to theory-specific inputs; the audit concerns the conditions for generating temporal comparisons, dynamical descriptions, and law-admissible time representations from those inputs. The cosmological-constant problem and observations of dark energy are representative lines of research[34, 35, 36].

**Proposition 8.72** (Safe reuse of quantum-gravitational results through generated time and failure classification). Within a domain in which the conditions of Propositions 8.61 and 8.60 hold, the generated law-admissible time representation, observer-time law core, law-time value, canonical observer-time quotient, canonical object-state quotient, canonical trajectory relation, graph trajectory, and canonical family of observable functions may be used. Results concerning constraint equations, internal clocks, relational observables, cosmological initial boundaries, the arrow of time, quantum measurement records, decoherence, black-hole information, spacetime emergence, the cosmological constant, and dark energy may be used within domains in which their respective theory-specific conditions and the generation conditions of the temporal contents on which they depend hold.

Each minimal failure position classified by Proposition 8.69 determines the QG constituent to be repaired first or the condition by which the domain of validity must be restricted. Failure at the comparison stage requires repair of clock-candidate-side materials, record-side retention, object-state-side differences, observer-side information coupling, comparison rules, or comparison gluing. Failure at the representation or continuous-approximation stage requires repair of the canonical observer-time quotient, law-admissible time representation,  $\mathfrak{D}$ -record cell width, comparison domain, or scale domain of approximation validity. Failure at the dynamical-description, law-representation, or differential stage requires repair of the canonical trajectory relation, canonical family of observable functions, law-evaluation datum, law-output determinacy, or differential evaluation specification. When repair succeeds, relational changes, observables, initial boundaries, the arrow of time, measurement records, decoherence, black-hole information, post-emergence spacetime descriptions, and cosmological law results are regenerated or recomputed in dependence order from the repaired inputs. If the stage condition does not hold throughout the entire domain, the domain in which a result is used is restricted to the region in which that condition holds.

By Proposition 8.68, generation conditions for temporal content and law-description

operability remain necessary after external time has been removed. Failure classification assigns failures of external-time input, undergeneration of an internal clock, comparison gluing, time representation, dynamical description, law-output determinacy, and differential consistency to the corresponding minimal failure positions. The problem of time in quantum gravity can thereby be decomposed into stagewise problems accompanied by a repair target or a restriction of the domain of validity.

When an error source in record retention, object-state identification, comparison gluing,  $\mathfrak{D}$ -record cell width, relational observables, initial boundaries, measurement records, continuous approximation, or differential evaluation is repaired, the repair's effect on accuracy is determined by an accuracy evaluation of the regenerated or recomputed downstream results. Decisions concerning the use of quantum-gravitational results therefore divide into retention within a domain of validity and repair from each minimal failure position or restriction of the domain of validity.

*Proof.* By Proposition 8.61, dynamical-description operability generates a law-admissible time representation, while law-description operability confers qualification for use within laws on that representation and generates the observer-time law core. By Proposition 8.60 and Theorem 8.31, the canonical observer-time quotient, canonical object-state quotient, canonical trajectory relation, graph trajectory, canonical family of observable functions, law-admissible time representation, and law-time value are generated. By Proposition 8.21, when the temporal content on which each result depends and its grounding structure are generated prior to law operations and result derivation, the temporal input satisfies physical-time qualification.

By Proposition 8.69 and Definition 6.17, conditions and results downstream of each minimal failure position depend on upstream conditions. When the input or rule at each minimal failure position can be repaired, downstream results are regenerated or recomputed in dependence order from the repaired inputs. If the relevant stage condition does not hold throughout the entire domain, the domain of use is restricted to the region in which it does hold, and results generated within that region are used.

Proposition 8.68 separates removal of external time from generation of temporal content and determines qualification for use within laws through law-description operability. Propositions 8.70 and 8.71 supply the same generation conditions and failure classification for cosmological initial boundaries. The problems of external-time input, internal clocks, comparison gluing, time representation, dynamical description, law-output determinacy, and differential consistency can therefore be separated by their minimal failure positions.

When an error source belongs to record retention, object-state identification, comparison gluing,  $\mathfrak{D}$ -record cell width, relational observables, initial boundaries, measurement records, continuous approximation, or differential evaluation, downstream results can be regenerated or recomputed from inputs in which that source has been repaired, and accuracy evaluation can be applied to the recomputed results. Thus retention of quantum-gravitational results within a domain of validity, the order of repair or domain restriction from each minimal failure position, and conditional accuracy evaluation are established.  $\square$

## 8.9 Generation order of time-bearing quantities and cross-field consequences

**Proposition 8.73** (Post-generation qualification for use of time-bearing quantities). For the twelve objects in Subsection 7.3.16, the dimension, unit, constancy, invariance, or algebraic relation of a time-bearing quantity does not supply the physical generation of

the temporal content used by that quantity. The use of each time-bearing quantity within a physical law is established in a domain in which the required temporal content and physical-time qualification are generated before law operations and result derivation, and in which the theory-specific establishment conditions are satisfied.

The object name and the presence or absence of a temporal dimension provide the extraction conditions, while the generation-dependence order appearing in the formal relations of the original source provides the basis of the audit determination. The physical relations of emission, propagation, reception, recording, atomic transitions, oscillators, particle decay, electromagnetic fields, gravitational fields, resonance lines, and measuring instruments are preserved as theory-specific inputs to the LCTR generation and audit construction. After temporal content has been generated, predicates concerning the constancy, invariance, rate, lifetime, linewidth, or frequency of each quantity are applied relative to generated time representations, admissible transformations, measurement procedures, and theory-specific domains of validity.

*Proof.* By Proposition 7.178, the generation of the temporal content required by a time-bearing quantity does not follow from the dimension, unit, or algebraic relation of that quantity. By Theorem 7.203, the physical establishment of a physical-law time variable requires the generation of temporal content and its grounding structure before law operations and result derivation.

By Remark 7.179, when temporal content and physical-time qualification have been generated and the theory-specific establishment conditions are satisfied, existing formal relations, including those for the twelve objects, may be used in their post-generation domains of application. Preservation of the physical relations as theory-specific inputs and post-generation qualification for use of time-bearing quantities therefore hold jointly.  $\square$

**Proposition 8.74** (Physical domain of light-speed invariance). The physical relations of the emission, propagation, reception, and recording of light are preserved as theory-specific inputs. Because the numerical expression for the speed of light uses a spatial difference and a time difference, it requires the temporal content supplying that time difference and its physical-time qualification.

The quantification domain of light-speed invariance in special relativity is the domain in which a generated law-admissible time representation, a spatial representation, admissible representation transformations between inertial observers, clock and length standards, measurement procedures, and the light-propagation relation hold. The constancy of the vacuum speed of light  $c$  measured in a local inertial frame, and the local upper propagation limit for matter, signals, or causal influences, are preserved in this domain. This establishment does not entail an unconditional universal proposition concerning an arbitrary coordinate speed of light, cosmological recession velocity, phase velocity, or displayed quantity that carries no information. If physical carriers and comparison constructions differ, and different generated time representations result, identical numerical invariance across those representations does not follow automatically. The determination must be made with the coordinate transformation, metric, clock and length standards, and domain of application fixed consistently. Where these conditions hold, the numerical expression and invariance of the speed of light may be used as results of special relativity. Dependence that uses the speed of light as a ground for the establishment of a time scale or synchronization rule before temporal content is generated is determined as pre-generation placement and reverse presupposition of a derived quantity.

*Proof.* By Proposition 7.182, pre-generation placement, omission of generation stages, and

reverse presupposition of a derived quantity hold on the formal track for the speed of light. Proposition 7.180 separates the physical propagation of light, the generation order of the temporal content used by a velocity expression, the constancy of the vacuum speed of light in a local inertial frame, and the different types of velocity, frames of reference, and domains of application. By Proposition 8.44, a Lorentz transformation is established as an admissible representation transformation between generated time representations and spatial displays. By Proposition 8.50, the transformation results and differential quantities of special relativity may be used in a domain in which the temporal content on which each result depends and the conditions at each stage hold.

It follows that, while the physical propagation relation of light is preserved as a theory-specific input, the quantification domain of the numerical expression and invariance of the speed of light is determined by generated time representations, admissible transformations, and theory-specific establishment conditions.  $\square$

**Corollary 8.75** (Cross-field consequence for the twelve time-bearing objects). Among the 96 determination fields produced by applying the four audit determinations on two tracks to the twelve time-bearing objects, an audit determination holds in 21 fields on the formal track and in zero fields on the statement track. The evidence included in the Appendix does not establish the corresponding determination for the remaining 75 fields.

The 21 established fields on the formal track consist of nine fields of pre-generation placement, seven fields of omission of generation stages, and five fields of reverse presupposition of a derived quantity. Reattribution of a time property is established in zero fields. This distribution separates the failure positions according to the dependence order in the original-source formal description of each object.

*Proof.* Apply the four audit determinations on two tracks and the determination-specific breakdown of Proposition 7.182. Proposition 7.181 establishes the persistence of primitive-time-root dependence in the decay constants, half-lives, mean lives, frequency shifts, frequency widths, acoustic-wave lifetimes, Planck constant, transition angular frequencies, linewidths, resonance widths, lifetimes, and Compton angular frequency treated in Curtiss 1949, Headley–Clayton–Reed 1996, Clark 2001, Wolf et al. 2011, and the PDG 2023 resonance reference. The determinations for the twelve objects therefore follow from the dependence orders in the source-specific formal evidence.  $\square$

## 8.10 Cross-field synthesis of representative theoretical results through generated time and failure classification

**Proposition 8.76** (Cross-field evidence comparison of the unbounded inheritance of limited results). The general application of Proposition 8.20 is compared with the formal evidence and statement evidence from thermodynamics, cosmology, quantum measurement, quantum gravity, relativity, biological application, and time-bearing quantities.

Subsection 7.3.18 maps the formal evidence and statement evidence from all 72 audited references to the audit method or representative arguments of Chapter 7. From that full-reference connection, the present proposition extracts cross-field direct-dependency paths and synthesizes, for each representative path, the evidence boundaries encompassing sources of the same or a closely related form.

When macroscopic asymmetry, redshift and model fit, decoherence, removal of external time and selection of an internal clock, differences in clock readings, frequencies, or

particle lifetimes, and relations containing time-bearing quantities are established, their establishment belongs to the evidence-effect domain specific to each result.

These limited results alone do not establish the physical generation of temporal content, the physical-time qualification of law time, a time attribute independent of physical carriers, direct demonstration concerning biological processes or subjective experience, a beginning, nonexistence, or emergence of time itself, or generation of the time roots on which time-bearing quantities depend. Each such claim requires independent evidence or a valid derivation specific to that claim.

A literal interpretation that transfers a relativistic clock difference to a change in an attribute of an unconstructed time entity constitutes a two-stage misrecognition, and an interpretation that then transfers it to direct demonstration concerning biological processes constitutes a three-stage misrecognition. No misrecognition arises for a statement whose original-source text identifies its figurative character, states that it does not designate a literal deformation of time, and identifies the actual referent within the same explanatory scope. An interpretation that reads an expression as a bare time-attribute expression constitutes a one-stage misrecognition; one that reads it as a literal time attribute constitutes a two-stage misrecognition. Thermodynamics, cosmology, quantum measurement, quantum gravity, and time-bearing quantities are likewise assigned no single stage count by field. Only direct dependencies that cross evidence-effect boundaries are counted, separately for each permissible interpretation of an actual original-source statement.

Statements about the arrow of time concerning time-reversal symmetry, low-entropy boundary conditions, and the cosmological direction of time are preserved within the evidence-effect domain of the asymmetry or orientation that each describes. The arrow of time is a subsequent orientation condition on a generated time representation, and an interpretive path that remains within this dependence order crosses zero evidence-effect boundaries.

An interpretive path that reads Feynman's statement concerning the beginning of "at least our part of the universe" as the initial boundary of a cosmological model or region crosses zero evidence-effect boundaries. A path that transfers the same statement to the beginning, nonexistence, or meaninglessness of physical time in general crosses one evidence-effect boundary, from the boundary of a model or region to the domain of existence of time in general, and constitutes a one-stage misrecognition. The interrogative statement in the original source asking whether an earlier time has meaning does not conclude that physical time in general is nonexistent or meaningless.

The argument of Augustine presented in Heisenberg 1958 states that time was created together with the world and did not exist before the universe. Heisenberg himself analyzes the word "created" in the creation of time as presupposing a before-after relation and therefore finds no rational definition for the phrase "time was created." This analysis directly identifies the self-dependence by which a statement about the beginning or creation of time uses a temporal relation in the statement operation itself. Establishing a statement of time creation requires a generation relation that identifies the states before and after creation without depending on time.

The excess distances of distant supernovae derived by Riess et al. and the incompatibility derived by Perlmutter et al. between their supernova sample and a flat, matter-only cosmological model are preserved within their respective evidence-effect domains of observation, correction, calibration, estimation, and model fitting. An interpretation that transfers these results to direct demonstration of the entire cosmic history, generation of temporal content, or the domain of existence of physical time in general constitutes

an evidence-effect boundary along that interpretive path when an actual statement or explanation performs the transfer. The cited original sources of Riess et al. and Perlmutter et al. do not yield this transfer.

The corrections to the vocabulary of cosmic expansion by Bunn–Hogg, Lewis et al., and Peacock describe observed redshift in terms of geodesics, local relative motion, or spacetime geometry, without attributing it to the action of an independent physical carrier called expanding space. In the cross-field audit, these corrections provide counterevidence against paths that move from a limited result to a temporal ontology or an independent carrier.

The agreement between the muon-lifetime measurement and relativistic prediction in Bailey et al. 1977 is preserved within the evidence-effect domain of the particle lifetime, measurement conditions, and adopted theoretical antecedents. A path that estimates the negative-muon lifetime at rest with special relativity as an antecedent does not simultaneously transmit that same estimate to an independent test of special relativity and an unconditional test of CPT invariance.

In Feynman 1995, an explanation of a nuclear transition and frequency shift proceeds within the same source to a generalization to all physical processes, an application to human life and aging, and a description in terms of an attribute of the flow of time. In Narlikar 2010, an explanation of signals from a collapsing body, redshift, and reception waiting time proceeds to a description in terms of a difference in the rates at which time flows for two subjects. The former makes concrete the boundary between formal generalization to physical processes and direct demonstration concerning biological processes; the latter makes concrete the boundary between signal–wavelength relations and an attribute of time flow independent of physical carriers. Limited results concerning clocks, signals, frequencies, or particle lifetimes add no evidence specific to the claims beyond those boundaries.

Von Neumann 1932 explicitly introduces time  $t$  as an input to a nonstationary wave function and quantum evolution. The mathematical consistency, predictive performance, or empirical agreement of a quantum formalism using this time input does not generate time  $t$  from nontemporal physical inputs. The source distinguishes an empirical statement of the type in which an observer has obtained a subjective perception from a statement of the type in which a physical quantity has a value. The former includes subjective perception among its objects but supplies no direct measurement of a change in subjective temporal experience. The system–environment interaction, environmental correlations, suppression of interference, and pointer structure in Zurek 2003 are preserved within the evidence-effect domain of decoherence. Adler 2003 supplies corrective evidence that separates limited decoherence results from a solution of the entire quantum measurement problem, while Schlosshauer 2005 supplies reserving evidence that does not treat the implication between them as settled. The evidence paths provided by these four sources separate external-time input, decoherence results, single-outcome selection, resolution of the entire quantum measurement problem, and generation of temporal content into distinct evidence-effect domains. If another actual statement moves among these domains, an evidence-effect boundary is counted for each direct dependency.

In quantum gravity, removal of an external time parameter, selection of an internal clock, a state-dependent flow, bipartite time, relational reformulation, and complete generation of physical time are treated as distinct results. Interpretive paths that read statements in Barbour 1994 I and II, Connes–Rovelli 1994, Kiefer 2009, and Smolin 2018/2020 as limited formal results or theoretical classifications preserve those statements within their evidence-

effect domains. Paths that transfer the same statements to the creation or nonexistence of physical time in general, complete generation of every temporal content, complete exclusion of primitive-time roots, or qualification for use within laws prior to generation constitute one- or two-stage misrecognitions according to the established interpretation-path-specific determinations. Descriptions of the problem of time, candidate constructions, and relational results by DeWitt, Isham, Kuchař, Page–Wootters, and Rovelli are preserved as evidence identifying the problem and supplying partial formal results. In particular, the relational formalism of Rovelli 2002 expresses dynamical predictions as relations among partial observables, selects no particular partial variable as time, and places no vocabulary of evolution in the fundamental structure. This result does not generate, from nontemporal physical inputs, the clocks, records, comparisons, order, scale, and qualification for use within laws that physically realize those relations. The obligation to remove external time from a formalism and the obligation to generate physical time belong to different evidence-effect domains.

Within operational constructions, Bridgman 1927 explicitly identifies the inadequacy of basic operations for time measurement and clock-construction specifications. Ehlers–Pirani–Schild 1972 excludes clocks from the basic tools of its own construction from light rays and freely falling particles and derives measurement operations. The atomic-clock interpretation of the chronometric axiom in Synge’s preceding method is distinct from the additional empirical postulate equating gravitational and atomic time after the EPS reconstruction. These results identify incomplete construction, change the order of construction, or establish conditional structural results. A hypothetical interpretation that extends identification of a problem, removal of a primitive instrument, or transition to a derived operation to completed generation of all time contents or law-time qualification crosses an evidence-effect boundary from a local or partial stage to a subsequent stage.

In black-hole thermodynamics, radiation, and information, the formal analogy and conjecture in Bekenstein 1973, the radiation result and boundary of the semiclassical approximation in Hawking 1975, the classical spacetime background and temporal Fourier components in Hawking 1976, and the model-dependent first conjecture in Page 1993 are each preserved within the domain of effect established by the original source. Computational success concerning area increase, a radiation spectrum, an evaporation time, or the appearance of information does not repair the generation of the temporal content presupposed by those computations, the physical establishment of the background spacetime, or resolution of the entire black-hole information problem. An interpretive path that transfers a limited computational result to any of these subsequent claims requires independent evidence or a valid derivation for each destination of transfer.

In duality, entanglement, and spacetime emergence, the duality conjecture of Maldacena 1998, the proposed area law and limitations on quantitative comparison in Ryu–Takayanagi 2006, and the relation between entanglement and spatial connectedness in Van Raamsdonk 2010 are preserved as their respective formal results. An interpretation that transfers a boundary–bulk correspondence, minimal-surface correspondence, or explanation of spatial connectedness to physical-time qualification of a time variable, completion of a spacetime-generation mechanism, or complete generation of physical time in general crosses an evidence-effect boundary from the domain in which the formal correspondence is established to a generation obligation.

The mathematical consistency, predictive performance, model fit, and empirical agreement established in each field are preserved within their respective evidence-effect domains. Generating temporal content from nontemporal physical inputs and auditing its domain of



validity and failure position determine the domain in which existing results may be reused safely.

*Proof.* By Proposition 7.118, a limited result does not establish a claim beyond an evidence-effect boundary. By Corollary 7.119, inheritance of primitive-time dependence by results and inheritance of a limited result by a claim outside its domain of application are distinct audit axes.

Proposition 7.142, Proposition 7.143, and Corollary 7.144 separate the establishment of a temporal direction from the generation of temporal content. Proposition 7.154 separates the initial boundary of a cosmological model or region from the domain of existence of physical time in general, and determines the number of boundary crossings for each interpretive path without converting an interrogative statement in the original source into a conclusion. Proposition 7.155 separates the evidence types of observation, estimation, model fit, and direct demonstration. Proposition 7.176 and Corollary 7.177 separate the formal domain of effect made explicit by corrections to cosmic-expansion vocabulary from the time-generation audit. Proposition 7.133, Proposition 7.134, Corollary 7.135, and Proposition 7.132 separate external-time input, an empirical statement concerning subjective perception, direct measurement of subjective temporal experience, the domain of validity of decoherence, single-outcome selection, resolution of the entire quantum measurement problem, and generation of law time, and determine the roles of corrective and reserving evidence.

Proposition 7.172, Proposition 7.170, and Proposition 7.169 separate the limited results of adjacent time programs, removal of a privileged time from a relational formalism, the distinct obligations of removing external time and generating physical time, the fundamentality of time, and complete time generation into different evidence-effect domains. Proposition 8.64, Proposition 8.65, Proposition 8.66, and Proposition 8.67 determine, for each permissible interpretation of each actual statement, the number of evidence-effect boundaries connected by direct dependency.

Subsection 7.3.18 connects the statement content, interpretability, conditional comparison with formal evidence, and evidence-effect boundary for all 72 audited references—not only the representative paths above—to the audit method or representative arguments of Chapter 7. Subsections 7.3.1, 7.3.8, and 7.3.9 establish the primitive-time roots and result dependencies remaining in the formal evidence for operational constructions, black holes, and spacetime emergence.

Proposition 7.178 and Proposition 8.73 separate relations containing time-bearing quantities from post-generation qualification for use of their time roots. Proposition 7.185, Proposition 7.184, Proposition 7.193, and Proposition 8.59 separate direct measurement of physical carriers, attribution to a time property, and direct demonstration of biological application.

By Theorem 7.190, primitive-time roots and results dependent on primitive-time roots occur in all 72 audited references. It follows that evidence-effect boundaries separate each limited result's evidence-effect domain from generation of temporal content, time qualification, time attributes, biological application, temporal ontology, and the time roots of time-bearing quantities. Within a path formed by the formal construction, statements, and permissible interpretations of each reference, a direct dependency that crosses such a boundary constitutes unbounded inheritance of a limited result. Generated time and failure classification determine, for each such path, the domain in which existing results may be reused safely.  $\square$

**Proposition 8.77** (Target consistency between time-nonexistence statements and the attribution of time properties). When a time-nonexistence statement and a time-property statement coexist within the same theoretical description or reference, or when statements from multiple references are synthesized across fields under a common explanatory vocabulary, the type and domain of application of the temporal content targeted by the two statements are aligned for determination in accordance with Proposition 8.63 and Remark 7.175.

If the two statements target the same type of temporal content and the same domain of application, the nonexistence of time is incompatible with the establishment of stretching, slowing, flow, or deformation whose bearer is that time. Attribution of a property to time requires, in the same domain of application, an object that bears the property.

If the two statements target different types of temporal content or different domains of application, the existence denial and property attribution are not directly contradictory. In this case, the target of the existence denial must be separated from the target of the property attribution, and the change in target type or domain of application must be made explicit.

If the target of property attribution is a difference in clock readings, clock rates, proper times, or coordinates, or a comparison-derived difference among generated law-admissible time representations, the property belongs to that physical carrier or relational quantity. This change of target type or domain of application does not entail the stretching, slowing, flow, or deformation of “time itself” independently of physical carriers.

*Proof.* By Proposition 8.62 and Proposition 8.63, a time-nonexistence statement bears the obligation to specify the type and domain of application of the temporal content being denied.

By Proposition 8.47 and Proposition 8.56, a statement that attributes stretching, slowing, or deformation to time as a literal physical property requires a property-bearing object that remains identical before and after the change, together with its state space, deformation rule, and measurement correspondence. Literal attribution of flow likewise requires an object that flows and conditions for identifying changes in the state or position of that object.

If an existence denial and property attribution target the same type of temporal content and the same domain of application, one requires the absence of the object while the other requires that object as the bearer of a property. They therefore cannot hold simultaneously.

If the type of temporal content or domain of application differs, the existence denial and property attribution belong to different objects or regions. No direct contradiction then arises, but an explanation that connects both statements through the same word “time” leaves the bearer of the property undetermined unless it makes the change in target type or domain of application explicit.

By Proposition 7.174, the objects directly supplied by the formal results of relativity are differences in clock readings, clock rates, lifetimes, or proper times, spacetime intervals, or coordinate relations; they are not a state space, deformation rule, and independent measurement procedure for a bearer of time stretching independent of physical carriers. Changing the target of property attribution to these physical carriers or relational quantities therefore does not entail a change in a property of “time itself” independently of physical carriers.

It follows that statements with the same target type and domain of application are incompatible, while statements with different target types or domains bear an obligation

to make the target change explicit and do not entail the attribution of a property to “time itself.” □

Figure 23 compares time-nonexistence statements with time-property statements along the two axes of the type and domain of application of the temporal content targeted by the statement. The temporal-content types distinguish time independent of physical carriers, differences in clock readings or clock rates, differences in proper times or coordinates, and comparison-derived differences among generated time representations. The domains of application distinguish the theoretical domain, approximate domain, and physical conditions in which the statement is said to hold.

### Target-consistency audit of statements about time

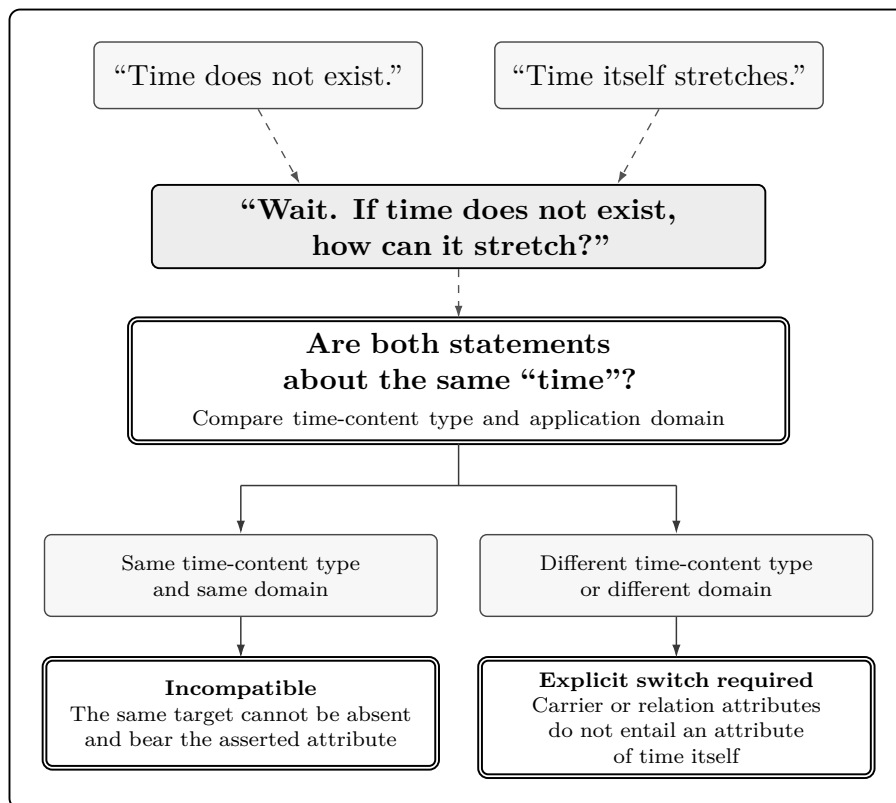


Figure 23: Target consistency between time-nonexistence statements and time-property statements. The upper row gives representative forms of the two kinds of statement subjected to audit, while the gray band gives the question that arises for a reader who encounters them in succession. These are not verbatim quotations from an original source. The dashed lines from the upper row to the central audit box indicate reading order and do not express formal implication. Solid lines from the central box onward indicate the target-consistency determination. The two statements are incompatible when they concern the same type and the same domain. Different types or domains require an explicit change, and a property of a physical carrier or relational quantity does not entail a property of time itself independently of physical carriers.

The 72 audited references contain both statements that time does not exist and statements that attribute stretching, slowing, flow, or deformation to time itself. Misner–Thorne–Wheeler 1973 contains both series of statements within the same reference, but their explicitly stated domains of application differ, so they are not determined to be

directly contradictory under identical conditions. Heisenberg 1958 records the nonexistence of time before the universe and the temporal presupposition in the operation of stating it, while Feynman 1995 and Narlikar 2010 record statements that move from clock or signal relations to an attribute of time flow. These cases show that the target of an existence denial must be separated by type and domain of application from the target of a property attribution, not only within a single reference but also when the same word “time” is used in different theoretical domains.

The same determination applies when statements from multiple references are synthesized as part of the common explanatory vocabulary of physics. The object of determination is whether a physical synthesis that places time-nonexistence statements and time-property statements within the same explanatory system preserves the identity of their targets or changes to a different object or domain of application.

**Proposition 8.78** (Safe reuse of representative theoretical results through generated time and failure classification). Under the respective conditions of Propositions 8.26, 8.36, 8.42, 8.50, 8.58, and 8.72, time-dependent results in Newtonian and classical mechanics, time metrology, GPS, special relativity, general relativity, and quantum gravity are qualified for use within domains in which temporal content and its grounding structure are generated prior to use within laws.

When a generation-stage condition fails, the failure-position-token system of Definition 6.10 and the set of minimal failure positions of Theorem 6.168 determine the condition to be repaired first or the condition by which the domain of validity must be restricted. When repair succeeds, downstream results are regenerated or recomputed from the repaired inputs in accordance with the failure-generation dependence order of Definition 6.17. If the condition does not hold throughout the entire domain, the domain in which a result is used is restricted to the region in which the condition holds. The mathematical, predictive, measurement, and engineering results of each theory can thereby be reused safely within domains that satisfy physical-time qualification.

The strength of the connection to numerical accuracy differs across fields. In time metrology and GPS, repair of error sources belonging to calibration, frequency ratios, record-cell widths, window selection, clock corrections, propagation corrections, loop defects, and receiver-clock offsets connects directly to recomputation and accuracy evaluation of downstream results. In Newtonian and classical mechanics, special relativity, and general relativity, repair connects to recomputation and accuracy evaluation when an error source belonging to initial values, time representations, synchronization, clock-rate comparison, local or global time conditions, transport, approximation domains, or differential evaluation is included among the minimal failure positions. In quantum gravity, the principal consequence is determination of the time-generation conditions, failure positions, repair targets, and qualification of existing results for application; a subsequent accuracy evaluation is connected when a specific approximation or estimation error is localized at a failure position.

Problems that depended on a primitive-time premise thereby shift from mysteries or theoretical impasses formulated with unestablished temporal content to problems of identifying time-generation conditions, minimal failure positions, repair order, and domains of validity. Generated time and failure classification supply grounds for using the results of existing theories safely under their conditions of validity and, upon failure, for identifying repair targets from upstream to downstream.

*Proof.* For each representative theory, the six local propositions give the domain of validity in which generated time representations may be used, each minimal failure position, repair

or restriction of the domain of validity, regeneration or recomputation after repair, and conditional accuracy evaluation.

Definition 6.10 classifies a failed generation-stage condition by the corresponding failure-position token. Theorem 6.168 determines the failure positions minimal with respect to the failure-generation dependence order. By Definition 6.17, conditions and results downstream of a minimal failure position depend on upstream conditions. The condition belonging to the minimal failure position is therefore repaired, and downstream results are regenerated or recomputed in dependence order from the repaired inputs. If repair does not hold throughout the entire domain, the domain of validity of that condition determines the domain in which the result may be used.

Comparison of the connections between error sources and accuracy evaluation specified by the local propositions shows a direct connection to measurement and correction inputs in time metrology and GPS; in Newtonian and classical mechanics, special relativity, and general relativity, the connection holds when the relevant initial value, time representation, synchronization, transport, approximation, or differential evaluation is included at a failure position; in quantum gravity, it holds when a specific approximation or estimation error is localized at a failure position.

Thus, for existing results in all six fields, explicit time-generation dependence, determination of safe domains of application, first-failure diagnosis, repair order, restriction of domains of validity, safe reuse, and field-specific conditional accuracy evaluation are established.  $\square$

**Proposition 8.79** (Qualification failure of primitive-time input and conditional failure of downstream computational results). Let  $x$  be temporal content used by a representative scheme, and let  $\eta$  be a downstream result in whose derivation that temporal content participates either directly or through a temporal-content transfer sequence. The following two determinations hold.

The first is the determination of physical-time qualification. If  $x$  is a primitive-time input,  $x$  fails physical-time qualification, and  $\eta$  depends on temporal content that fails physical-time qualification. This qualification failure remains even when mathematical consistency, predictive performance, and empirical agreement hold.

The second is the determination of downstream computational results. In a scale-indexed complete evaluation specification  $(\text{Ev}, \lambda)$ , consider temporal content  $x$ , a downstream result  $\eta$ , and a condition  $\chi$  that governs either the generation or qualification for use of that temporal content, or the input value, comparison, representation, synchronization, transport, approximation, or differential evaluation of that result. Suppose that the primitive-time input  $x$  backgrounds the grounding obligation for  $\chi$  in the sense of Definition 7.120, that independent evidence establishes the actual failure of the same  $\chi$ , and that downstream dependence holds from that failure position to the derivation stage of  $\eta$ . Then  $\eta$  loses qualification for use as a physical computational result within the evaluation domain concerned. If the failed condition can be repaired, downstream results are regenerated or recomputed from the repaired inputs. If the condition cannot be repaired throughout the entire domain, the domain in which the result is used is restricted to the region in which the condition holds. Whether the recomputed value agrees with or differs from the preceding value is determined by the recomputation after repair.

The establishment conditions of the first determination are the presence of a primitive-time input and its participation in result derivation. The establishment conditions of the second determination are the backgrounding of a grounding obligation by primitive-time input, independent evidence establishing the actual failure of the same condition, and

downstream dependence from the failure position to result derivation. Evaluation success, failure of physical-time qualification, and loss of qualification for use of a downstream computational result can therefore hold jointly in the same case.

*Proof.* By Theorem 7.11, the primitive-time input  $x$  fails physical-time qualification. By Proposition 7.6 and Corollary 7.12, if  $x$  participates in the derivation of  $\eta$  either directly or through a temporal-content transfer sequence,  $\eta$  depends on temporal content that fails physical-time qualification. By Proposition 7.13, mathematical consistency, predictive performance, and empirical agreement do not repair this qualification failure. The first determination follows.

Proposition 7.121 connects, within a single audit-causal sequence, the backgrounding of a grounding obligation by primitive-time input, independent evidence establishing the actual failure of the same condition, the reach of dependence to a downstream result, and regeneration or recomputation after repair. By Proposition 8.78, the six representative fields localize an actual failure of a generation condition or establishment condition at a minimal failure position and regenerate or recompute downstream results from repaired inputs in accordance with the failure-generation dependence order. By Definition 6.4, an evaluation specification requiring a failed condition does not belong to the corresponding domain of validity. By Definition 6.17 and Theorem 6.168, each minimal failure position and its downstream dependencies are determined; by Proposition 6.171, the structural burden imposed by that failure position is determined. If the derivation stage of  $\eta$  belongs to this downstream dependence,  $\eta$  loses qualification for use as a physical computational result within the evaluation domain concerned. If repair succeeds, the result is regenerated or recomputed from the repaired inputs; if repair does not succeed throughout the entire domain, the establishment domain determines the domain in which the result may be used. The second determination follows.

The first determination follows from primitive-time input and its participation in result derivation. The second follows from the identity of the condition whose grounding obligation is backgrounded by primitive-time input, the condition whose actual failure is established by independent evidence, and the condition from which downstream dependence begins. The two determinations have different establishment conditions, and mathematical consistency, predictive performance, and empirical agreement repair neither failure condition.  $\square$

## Evaluation success and qualification consequences

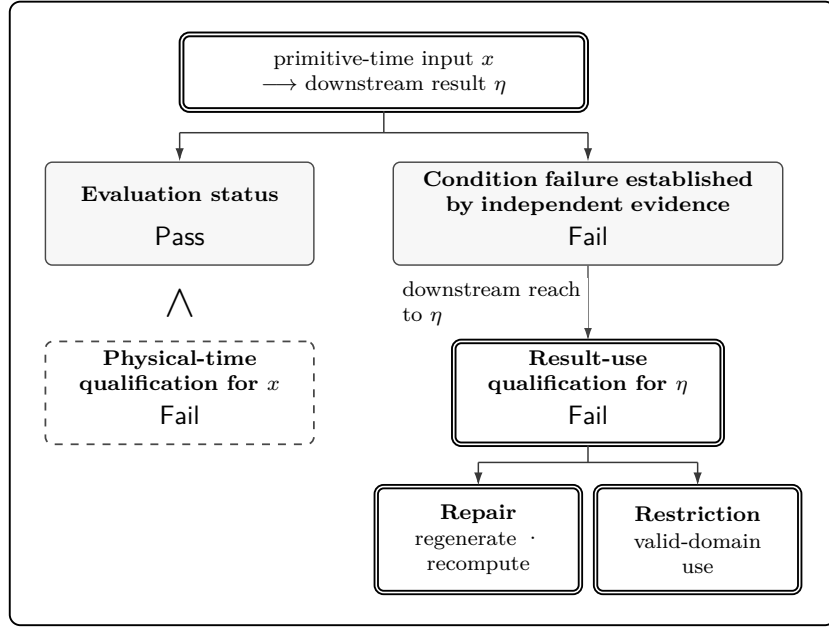


Figure 24: Coexistence of evaluation success and two qualification failures. The left branch shows evaluation success coexisting with the failure of physical-time qualification for a primitive-time input. The right branch shows failure of a generation or establishment condition, established by independent evidence, reaching the qualification for use of downstream results and requiring repair or use within the domain of validity.

## 8.11 Classical time as a generated representation

This section constructs the classical domain as a subclass satisfying the conditions of the theory of generated time. A time variable belonging to this domain is generated in the order in which exact comparison closure, dynamical-description operability, law-description operability, and differential-description operability are established. A continuous time representation in this domain holds at scales belonging to the scale domain of validity of the continuous approximation in Definition 6.86.

The representative-scheme single-object evaluation specification of Definition 8.5 connects each theory to a realized operational configuration, and Proposition 8.22 supplies the type boundary between theory-specific inputs and generated objects. When the specialization conditions defining a classical domain hold, Theorem 5.112, Theorem 5.113, and Definition 5.133 apply.

### 8.11.1 Minimal operational model of classical time

The minimal operational model of classical time has a single observer-side information-coupling position, a single object-specification index, a  $\mathfrak{C}$ -derived clock-side source, a  $\mathfrak{D}$ -derived record-side source, and a  $\mathfrak{B}$ -derived object-state-side source. Each generated object is obtained from satisfaction of the corresponding stage conditions.

**Definition 8.80** (Minimal operational model of classical time). Let  $\text{Sch}_{\text{ClMin}}$  be an instance

of the minimal classical scheme. Define the complete evaluation specification by

$$\begin{aligned} & \text{Ev}_{\text{Sch}_{\text{CMin}}} \\ & := ((\mathcal{X}_{\text{Sch}_{\text{CMin}}}; \mathcal{J}_{\text{cmp}, \text{Sch}_{\text{CMin}}}), o_{\text{Sch}_{\text{CMin}}}, \zeta_{\text{Sch}_{\text{CMin}}}) \end{aligned}$$

Define the realized operational configuration by

$$\begin{aligned} & \mathcal{X}_{\text{Sch}_{\text{CMin}}} \\ & = (\mathcal{I}_{\text{Sch}_{\text{CMin}}}^{\text{raw}}, \text{Dat}_{\text{Sch}_{\text{CMin}}}^{\text{src}}) \end{aligned}$$

Let the single  $\mathfrak{C}$ -derived source index of the minimal classical model be  $j_{\mathfrak{C}, \text{Sch}_{\text{CMin}}}^{\text{src}} \in \text{Idx}_{\mathfrak{C}}^{\text{src}}$ . For each  $k \in \mathbb{N}$ , define the  $\mathfrak{C}$ -derived clock-side source token by

$$\begin{aligned} x_{\mathfrak{C}, k} & := \text{tok}_{\mathfrak{C}}(j_{\mathfrak{C}, \text{Sch}_{\text{CMin}}}^{\text{src}}, k) \\ & \in \text{Tok}_{\mathfrak{C}, \text{Sch}_{\text{CMin}}}^* \end{aligned}$$

Define the  $\mathfrak{D}$ -derived record-side source token and the  $\mathfrak{B}$ -derived object-state-side source token by

$$\begin{aligned} x_{\mathfrak{D}, k} & \in \text{Tok}_{\mathfrak{D}, \text{Sch}_{\text{CMin}}}^*, \\ x_{\mathfrak{B}, k} & \in \text{Tok}_{\mathfrak{B}, \text{Sch}_{\text{CMin}}}^* \end{aligned}$$

The  $\mathfrak{D}$ -derived record-side source token satisfies

$$\text{seq}_{\mathfrak{D}}(x_{\mathfrak{D}, k}) = k$$

The  $\mathfrak{C}/\mathfrak{D}/\mathfrak{B}$ -derived coupling satisfies

$$(x_{\mathfrak{C}, k}, x_{\mathfrak{D}, k}, x_{\mathfrak{B}, k}) \in \text{Rel}_{\text{Sch}_{\text{CMin}}}^{\text{dyn}, \text{src}}$$

, and its  $\mathfrak{C}/\mathfrak{D}$  component satisfies

$$(x_{\mathfrak{C}, k}, x_{\mathfrak{D}, k}) \in \text{Rel}_{\text{Sch}_{\text{CMin}}}^{\text{src}}$$

Use

$$\text{Arr}_{\mathfrak{C}, o_{\text{Sch}_{\text{CMin}}}}^{\text{cmp}}, \quad \text{Arr}_{\mathfrak{D}, o_{\text{Sch}_{\text{CMin}}}}^{\text{cmp}}$$

as the comparison-stage arrival maps, and, as the dynamical-description-stage  $\mathfrak{B}$ -derived arrival map, use

$$\text{Arr}_{\mathfrak{B}, o_{\text{Sch}_{\text{CMin}}}}^{\text{dyn}}$$

**Proposition 8.81** ( $\mathfrak{C}$ -derived observer-time quotient in the minimal classical model). For the minimal classical model of Definition 8.80, suppose that

$$\text{ExactOperable}(\mathcal{X}_{\text{Sch}_{\text{CMin}}}; \mathcal{J}_{\text{cmp}, \text{Sch}_{\text{CMin}}})$$

and

$$\text{DynOperative}[\text{Ev}_{\text{Sch}_{\text{CMin}}}]$$

hold.



For each  $k \in \mathbb{N}$ , define

$$T_k := \Theta_{\mathfrak{C}, \mathcal{O}_{\text{SchClMin}}}^{\text{dyn,can}}(x_{\mathfrak{C},k})$$

. Suppose further that the following three conditions hold.

First, for every

$$T \in \text{Qu}_{\mathfrak{C}, \mathcal{O}_{\text{SchClMin}}}^{\text{dyn,can}}$$

there exists  $k \in \mathbb{N}$  such that  $T = T_k$ .

Second, for every  $k_0, k_1 \in \mathbb{N}$ ,

$$\begin{aligned} T_{k_0} = T_{k_1} \\ \iff k_0 = k_1 \end{aligned}$$

.

Third, for every  $k_0, k_1 \in \mathbb{N}$ ,

$$\begin{aligned} T_{k_0} \prec_{\mathcal{O}_{\text{SchClMin}}}^{\text{dyn,time,can}} T_{k_1} \\ \iff k_0 < k_1 \end{aligned}$$

.

Then

$$\text{Qu}_{\mathfrak{C}, \mathcal{O}_{\text{SchClMin}}}^{\text{dyn,can}} \cong \mathbb{N}$$

holds as an order isomorphism.

For any

$$\rho_{\mathcal{O}_{\text{SchClMin}}} \in \text{OrdEmbSet}_{\mathcal{O}_{\text{SchClMin}}}^{\text{dyn,time}}(\mathbb{R})$$

, define the time value for each  $k \in \mathbb{N}$  by

$$t_k := \text{TimeRep}_{\mathcal{O}_{\text{SchClMin}}}^{\text{dyn}} \left[ \rho_{\mathcal{O}_{\text{SchClMin}}} \right] (T_k)$$

.

*Proof.* Exact comparison closure determines the  $\mathfrak{C}$ -derived observer-time quotient and  $\mathfrak{C}$ -derived natural projection of Definition 5.16. Dynamical-description operability and Theorem 5.112 give

$$\text{OrdEmbSet}_{\mathcal{O}_{\text{SchClMin}}}^{\text{dyn,time}}(\mathbb{R}) \neq \emptyset$$

.

By the assumed coverage and injectivity, the correspondence  $k \mapsto T_k$  is a bijection from the natural numbers to the  $\mathfrak{C}$ -derived observer-time quotient. By order agreement, this bijection preserves the natural-number order and the canonical strict observer-time order, and its inverse also preserves order. The  $\mathfrak{C}$ -derived observer-time quotient is therefore order-isomorphic to the natural numbers.

Choose any real-valued temporal-order embedding. Definition 5.63 determines a law-admissible time representation, and a time value is determined as the image of each time-quotient class. Hence

$$t_k = \text{TimeRep}_{\mathcal{O}_{\text{SchClMin}}}^{\text{dyn}} \left[ \rho_{\mathcal{O}_{\text{SchClMin}}} \right] (T_k)$$

is a real-valued order representation of the generated time-quotient classes.  $\square$

**Corollary 8.82** (Continuous approximation of the minimal classical model). For the complete evaluation specification of Definition 8.80, suppose that a refinement family under Definition 6.82 and a nine-component defect family under Definition 6.84 are given, and that monotone nondecreasing behavior under Definition 6.85 holds.

For every

$$\lambda \in \mathcal{I}^{\max} [\text{EvSch}_{\text{CMin}}]$$

,

$$\lambda \in \mathcal{V}_{\text{EvSch}_{\text{CMin}}}^{\text{approx}}$$

holds. At this scale, all nine continuous-approximation conditions hold, including the continuous-extension, order-stability, and relation-stability conditions.

*Proof.* By Definition 6.93, the maximal effective interval is the initial interval of the evaluation-specification-specific scale domain of validity of continuous approximation. Membership in the maximal effective interval therefore gives membership in the scale domain of validity of continuous approximation for the same full evaluation specification. By Theorem 6.95, the latter membership is equivalent to simultaneous satisfaction of the nine conditions for continuous approximation.  $\square$

**Connection to physical-time qualification.** The components of the full evaluation specification  $\text{EvSch}_{\text{CMin}}$  are the realized operational configuration, the family of comparison rules, the observer-side information-coupling position, and the object-specification index. Stable records, the object-state sequence, clock-side finite granularity, the conditions for continuous approximation, and law-bearing single-valuedness are used in the law-time-use audit as establishment evidence for the respective generation stages.

Proposition 7.199 gives the generation dependencies from the realized operational configuration through the representation, dynamical-description, law-description, and differential stages. In this model, when the grounding structure corresponding to time content precedes law operations and result derivation, Theorem 7.200 implies that a physical-law time variable using that time content satisfies physical-time qualification. Failure at any generation stage is localized as a failure-position token by Definition 6.10 and Definition 6.21. Failure-position tokens localize the establishment boundary of the classical approximation, and physical-time qualification is determined by the generation-precedence condition of Corollary 7.15.

**Definition 8.83** (Classical regime). Let the full evaluation specification be

$$\text{Ev} = ((\mathcal{X}; \mathcal{J}_{\text{cmp}}), o, \zeta) \in \text{EvSpec}$$

and let the law-evaluation datum having this full evaluation specification as its first component be

$$\text{LawEvalDat}_{o,\zeta} = (\text{Ev}, \text{CandRelStr}_{o,\zeta}^{\text{law}})$$

and the differential-evaluation datum having this law-evaluation datum as its first component be

$$\begin{aligned} \text{DiffEvalDat}_{o,\zeta} \\ = (\text{LawEvalDat}_{o,\zeta}, \text{EvalStrFam}_{o,\zeta}^{\text{diff}}) \end{aligned}$$

.

The data  $\text{Ev}$ ,  $\text{LawEvalDat}_{o,\zeta}$ , and  $\text{DiffEvalDat}_{o,\zeta}$  form a classical regime if the following six conditions hold.

- (CL<sub>1</sub>) A single shared  $\mathfrak{C}$ -derived clock domain is established, and the  $\mathfrak{C}$ -derived clock-side data used at the comparison stage belong to the same shared family of comparison clocks.
- (CL<sub>2</sub>) The branching structure at the comparison stage is single-valued with respect to generation of the  $\mathfrak{C}\mathfrak{D}$ -derived canonical comparison quotient.
- (CL<sub>3</sub>) The  $\mathfrak{B}$ -derived object-state-side identification map satisfies  $\text{Stateld}_{\mathfrak{B},o} = \text{id}$ , and, for every admissible  $\mathfrak{B}$ -derived object-state-side token  $x_{\mathfrak{B}}$ ,

$$\text{Arr}_{\mathfrak{B},o}^{\text{dyn}}(x_{\mathfrak{B}}) = (x_{\mathfrak{B}}, x_{\mathfrak{B}})$$

holds.

- (CL<sub>4</sub>) The exact comparison closure of Definition 4.175,

$$\text{ExactOperable}(\mathcal{X}; \mathcal{J}_{\text{cmp}})$$

holds.

- (CL<sub>5</sub>) The all-dynamical-description-conditions bundle  $\text{DynCondAll}[\text{Ev}]$  of Definition 5.67 holds. Its eight conjunctive components correspond, in the order given in that definition, to the dynamical-description conditions  $\text{DG}_1$  through  $\text{DG}_8$ .
- (CL<sub>6</sub>) The law-description operability of Definition 5.102,

$$\text{LawOperative}[\text{LawEvalDat}_{o,\zeta}]$$

and the differential-description operability of Definition 5.132,

$$\text{DiffOperative}[\text{DiffEvalDat}_{o,\zeta}]$$

hold.

*Remark 8.84* (Classical subclasses). The classical regime of Definition 8.83 includes a local classical configuration with a single observer-side information-coupling position and a single object, and a classical many-body configuration in which multiple objects or multiple observer-side information-coupling positions share the same  $\mathfrak{C}$ -derived clock domain. In the latter, the shared family of comparison clocks, the  $\mathfrak{B}$ -derived object-state-side identification maps, and the direct-reception conditions hold for all objects and observer-side information-coupling positions.

**Theorem 8.85** (Classical-time establishment theorem). Suppose that  $\text{Ev}$ ,  $\text{LawEvalDat}_{o,\zeta}$ , and  $\text{DiffEvalDat}_{o,\zeta}$  form a classical regime in the sense of Definition 8.83. Then  $\text{Qu}_{\mathfrak{C},o}^{\text{dyn,can}}$  is generated, and

$$\text{OrdEmbSet}_o^{\text{dyn,time}}(\mathbb{R}) \neq \emptyset$$

For every

$$\rho_o \in \text{OrdEmbSet}_o^{\text{dyn,time}}(\mathbb{R})$$

, the law-admissible time representation

$$\begin{aligned} & \text{TimeRep}_o^{\text{dyn}}[\rho_o] \\ &= \rho_o \circ \text{Prj}_o^{\text{dyn,time,ord}} \end{aligned}$$

is determined. For any

$$T_0 \in \mathbf{Qu}_{\mathfrak{C},o}^{\text{dyn,can}}$$

, denote its conventional representation in existing classical physics by

$$t := \text{TimeRep}_o^{\text{dyn}}[\rho_o](T_0)$$

.

For the same real-valued temporal-order embedding, the observer-time law core

$$\text{LawCore}_{o,\zeta}^{\text{obs,time}}(\rho_o)$$

and the chart-compatible differential law

$$\text{ChartCompatDiffLaw}_{o,\zeta}(\rho_o)$$

are determined.

*Proof.* Conditions  $\text{CL}_4$  and  $\text{CL}_5$  of Definition 8.83 give

$$\begin{aligned} & \text{ExactOperable}(\mathcal{X}; \mathcal{J}_{\text{cmp}}) \\ & \wedge \text{DynCondAll}[\text{Ev}] \end{aligned}$$

. Definition 5.70 then gives  $\text{DynOperative}[\text{Ev}]$ . By Theorem 5.112, the  $\mathfrak{C}$ -derived observer-time quotient is generated and the set of real-valued temporal-order embeddings is nonempty. Definition 5.63 then determines the law-admissible time representation and real-valued time for each real-valued temporal-order embedding.

Law-description operability in  $\text{CL}_6$  of Definition 8.83 and Theorem 5.113 establish the observer-time law core. Differential-description operability in the same  $\text{CL}_6$  and Definition 5.133 establish the chart-compatible differential law.  $\square$

**Corollary 8.86** (Classical time is a generated representation). The time variable  $t$  in a classical regime is obtained by composing exact comparison closure, the all-dynamical-description-conditions bundle, the  $\mathfrak{C}$ -derived observer-time quotient, the temporal-order coordinate projection, and a real-valued temporal-order embedding.

*Proof.* By Theorem 8.85 and Definition 5.63,

$$t = \rho_o(\text{Prj}_o^{\text{dyn,time,ord}}(T_0)) .$$

$\square$

## 8.12 Validity domain and breakdown boundary of classical time

**Definition 8.87** (Classical continuous-approximation regime). Take a full evaluation specification  $\text{Ev}$ , a law-evaluation datum  $\text{LawEvalDat}_{o,\zeta}$  having  $\text{Ev}$  as its first component, and a differential-evaluation datum  $\text{DiffEvalDat}_{o,\zeta}$  having that law-evaluation datum as its first component, together forming a classical regime in the sense of Definition 8.83.

Suppose that a refinement family of Definition 6.82, a nine-component family of defect quantities of Definition 6.84, and a nine-component family of tolerances of Definition 6.83 are given, and that monotone nondecrease in the sense of Definition 6.85 holds.

When the maximal effective interval of Definition 6.93 satisfies  $\mathcal{I}^{\max}[\mathbf{Ev}] \neq \emptyset$ , call this maximal effective interval the classical continuous-approximation regime of the full evaluation specification  $\mathbf{Ev}$ .

For each  $\lambda \in \mathcal{I}^{\max}[\mathbf{Ev}]$  in this interval, membership in the scale domain of validity of continuous approximation of Definition 6.86,  $\lambda \in \mathcal{V}_{\mathbf{Ev}}^{\text{approx}}$ , holds.

A classical continuous-approximation regime associated with a single observer-side information-coupling position and a single object is called a local classical continuous-approximation regime.

For multiple observer-side information-coupling positions or multiple objects, suppose that the full evaluation specification, law-evaluation datum, and differential-evaluation datum associated with each pair satisfy the six conditions of Definition 8.83, that each maximal effective interval is nonempty, and that  $\mathbf{CL}_1$  of the same definition uses a  $\mathfrak{C}$ -derived clock family common to all pairs. The family consisting of the respective maximal effective intervals is then called a shared classical continuous-approximation regime.

**Theorem 8.88** (Classical continuous-approximation validity theorem). Suppose that  $\mathbf{Ev}$ ,  $\mathbf{LawEvalDat}_{o,\zeta}$ , and  $\mathbf{DiffEvalDat}_{o,\zeta}$  form a classical regime and that the classical continuous-approximation regime of Definition 8.87 is determined. Then, for every  $\lambda \in \mathcal{I}^{\max}[\mathbf{Ev}]$ ,  $\mathbf{CondAll}^{\text{approx}}[\mathbf{Ev}; \lambda]$  holds.

The objects generated from the full evaluation specification  $\mathbf{Ev}$ ,

$$\begin{aligned} & \mathbf{Qu}_{\mathfrak{C},o}^{\text{dyn,can}}, \\ & \mathbf{TrjRel}_o^{\text{dyn,can}}, \\ & \mathcal{O}_{o,\zeta}^{\text{dyn,can}}, \\ & \mathbf{TimeRep}_o^{\text{dyn}} \end{aligned}$$

jointly satisfy the continuous-extension, order-stability, and relation-stability conditions at this scale.

*Proof.* Definition 8.87 gives  $\lambda \in \mathcal{V}_{\mathbf{Ev}}^{\text{approx}}$ . By Theorem 6.95, this membership is equivalent to simultaneous satisfaction of the nine conditions for continuous approximation. Conditions  $\mathbf{CL}_4$  and  $\mathbf{CL}_5$  of Definition 8.83 establish dynamical-description operability, and  $\mathbf{CL}_6$  of the same definition establishes law-description operability and differential-description operability. The  $\mathfrak{C}$ -derived observer-time quotient, trajectory relation, observable system, and law-admissible time representation are therefore generated from the assumed full evaluation specification, and the nine conditions for continuous approximation are evaluated at scale  $\lambda$ .  $\square$

**Corollary 8.89** (Generation and domain-of-validity guarantee of the classical continuous approximation). Continuous time, infinitesimal time intervals, smooth trajectory representations, and differential law descriptions in a classical continuous-approximation regime are generated from the full evaluation specification  $\mathbf{Ev}$ . The establishment of these representations is guaranteed at each scale in the maximal effective interval  $\mathcal{I}^{\max}[\mathbf{Ev}]$ , and this interval gives the maximal initial domain of establishment of the classical continuous approximation.

*Proof.* By Theorem 8.88, the  $\mathfrak{C}$ -derived observer-time quotient, trajectory relation, observable system, and law-admissible time representation are generated from the full evaluation

specification. Continuous time, infinitesimal time intervals, smooth trajectory representations, and differential law descriptions are representations based on these generated objects.

By Definition 6.93 and Theorem 6.95, the nine conditions for continuous approximation hold at every scale in the maximal effective interval of the full evaluation specification. This interval is therefore the maximal initial domain guaranteeing the establishment of these representations.  $\square$

**Proposition 8.90** (Generation and domain-of-validity guarantee of a shared classical continuous-approximation regime). For each object and observer-side information-coupling position using a shared  $\mathfrak{C}$ -derived clock, suppose that a classical continuous-approximation regime of Definition 8.87 is defined. A classical continuous-approximation time representation is then generated from each full evaluation specification. Its establishment is guaranteed at every scale in the respective maximal effective interval, and the family consisting of these intervals gives the shared classical continuous-approximation regime.

*Proof.* By Theorem 8.88, the  $\mathfrak{C}$ -derived observer-time quotient, trajectory relation, observable system, and law-admissible time representation are generated from each full evaluation specification. Conditions  $\text{CL}_1$ – $\text{CL}_6$  of Definition 8.83 establish a classical regime for each pair.

By Definition 6.93 and Theorem 6.95, the nine conditions for continuous approximation hold at every scale in each maximal effective interval. By Definition 8.87, the family of these intervals satisfying the shared  $\mathfrak{C}$ -derived clock condition gives the shared classical continuous-approximation regime. Each time representation constituting the shared regime is therefore generated from its full evaluation specification, and its domain of establishment is guaranteed by the respective maximal effective interval.  $\square$

**Theorem 8.91** (First invalidation boundary of classical time). Let the full evaluation specification in a classical regime be  $\text{Ev} \in \text{EvSpec}$ , and suppose that a classical continuous-approximation regime of Definition 8.87 is defined. A classical continuous-approximation time representation is generated from  $\text{Ev}$ , and its establishment is guaranteed at every scale in  $\mathcal{I}^{\max}[\text{Ev}]$ .

For any  $\lambda \in \Lambda$ , if

$$\text{FailedTok}[\text{Ev}; \lambda] \neq \emptyset$$

, then

$$\begin{aligned} & \text{Min}_{\leq_{\text{fail}, \text{dep}}} \text{FailedTok}[\text{Ev}; \lambda] \\ &= \text{FailedTok}[\text{Ev}; \lambda]. \end{aligned}$$

Every member of this set of invalidation tokens belongs to one of the six failure series of Definition 6.10.

Furthermore, under the assumptions of Theorem 6.163, if  $\Lambda \setminus \mathcal{V}_{\text{Ev}}^{\text{approx}}$  has a least element, the first approximation-invalidation scale  $\lambda_{\text{bdry}}^{\text{approx}}[\text{Ev}]$  exists, and

$$\begin{aligned} & \mathcal{I}^{\max}[\text{Ev}] \\ &= \{ \lambda \in \Lambda \mid \lambda < \lambda_{\text{bdry}}^{\text{approx}}[\text{Ev}] \}, \\ & I_{\text{approx}}^{\text{exc}}[\text{Ev}; \lambda_{\text{bdry}}^{\text{approx}}[\text{Ev}]] \neq \emptyset. \end{aligned}$$

The approximation-scale boundary datum is then

$$\{ (\lambda_{\text{bdry}}^{\text{approx}}[\text{Ev}], \Sigma_{\text{approx}}^{\text{bdry}}[\text{Ev}; \lambda_{\text{bdry}}^{\text{approx}}[\text{Ev}]] ) \}$$

For the local classical configuration and shared classical configuration of Remark 8.84, let the respective full evaluation specifications and scales be  $(\mathbf{Ev}_0, \lambda_0)$  and  $(\mathbf{Ev}_1, \lambda_1)$ . If

$$\text{FailedTok}[\mathbf{Ev}_0; \lambda_0] \neq \text{FailedTok}[\mathbf{Ev}_1; \lambda_1]$$

, then the sets of minimal invalidation tokens of the two configurations are distinct.

*Proof.* By Theorem 8.88 and Corollary 8.89, the classical continuous-approximation time representation is generated from the full evaluation specification and its establishment is guaranteed at every scale in the maximal effective interval.

Definition 6.21 determines the set of all tokens invalidated at the scale in question. By Proposition 6.26, this set coincides with the set of minimal elements under the failure-generation dependence order. By Definition 6.10, each invalidation token belongs to the comparison, representation, continuous-approximation, dynamical-description, law-description, or differential-consistency failure series.

By Theorem 6.163, when the complement of the scale domain of approximation establishment has a least element, the first approximation-invalidation scale exists, and the maximal effective interval is the set of all scales smaller than this scale. The same theorem gives a nonempty set of approximation-excess components at the first approximation-invalidation scale. By Theorem 6.164, the ordered pair consisting of this scale and the approximation-boundary signature is the unique member of the approximation-scale boundary datum.

Each invalidation-token set for the local classical configuration and the shared classical configuration coincides with the respective set of minimal invalidation tokens. Therefore, if the two invalidation-token sets are distinct, extensionality gives distinct sets of minimal invalidation tokens.  $\square$

For the following claims about the generation and validity of continuous-approximation representations, take a classical continuous-approximation regime as defined in Definition 8.87.

**Connection to physical-time qualification.** The generation sequence of Proposition 7.199 gives the generation order from nontemporal physical inputs through the representation, dynamical-description, law-description, and differential stages. In a classical regime, Corollary 8.89 generates a classical continuous-approximation time representation from the full evaluation specification.

By Corollary 7.15, the generation structure corresponding to the time content used by a physical-law time variable is established before law operations and result derivation. When this generation-precedence condition holds, Theorem 7.200 implies that the physical-law time variable using that time content satisfies physical-time qualification.

For a full evaluation specification and scale satisfying the assumptions of Theorem 8.91, Theorem 8.88 guarantees establishment of the generated classical continuous-approximation time representation at each scale in the respective maximal effective interval. By the same first-invalidation-boundary theorem, a nonempty invalidation-token set coincides with the set of minimal elements under the failure-generation dependence order, and each member belongs to one of the six failure series.

Thus, the physical-time qualification of time content used by classical physical laws and the scale-indexed establishment guarantee of a generated classical continuous-approximation time representation are supplied by the generation-precedence condition

and the maximal effective interval, respectively, and invalidation is localized within the six failure series.

**Corollary 8.92** (Establishment boundary of the classical-time continuous approximation). The conventional representation  $t$  of a classical time variable is determined from the generation sequence of Theorem 8.85. The classical continuous-approximation time representation is generated from the full evaluation specification by Corollary 8.89. For a full evaluation specification and scale satisfying the assumptions of Theorem 8.91, its establishment is guaranteed at every scale in the respective maximal effective interval of Definition 6.93, and a nonempty invalidation-token set coincides with the set of minimal elements under the failure-generation dependence order; each member belongs to one of the six failure series of Definition 6.10.

*Proof.* By Theorem 8.85, the conventional representation  $t$  of a classical time variable is determined by applying a law-admissible time representation to a member of the generated  $\mathfrak{C}$ -derived observer-time quotient.

By Theorem 8.88 and Corollary 8.89, the classical continuous-approximation time representation is generated from the full evaluation specification and its establishment is guaranteed at every scale in the maximal effective interval. By Theorem 8.91, the invalidation-token set coincides with the set of minimal elements under the failure-generation dependence order, and each member belongs to one of the six failure series.  $\square$

**Proposition 8.93** (Generation, establishment guarantee, and invalidation localization in a classical regime). In a classical regime, Theorem 8.85 determines the conventional representation  $t$  of a classical time variable from the generation sequence. By Corollary 8.89, the classical continuous-approximation time representation is generated from the full evaluation specification. For a full evaluation specification and scale satisfying the assumptions of Theorem 8.91, its establishment is guaranteed at every scale in the respective maximal effective interval, and a nonempty invalidation-token set coincides with the set of minimal elements under the failure-generation dependence order; each member belongs to one of the six failure series.

*Proof.* The generation sequence for the conventional representation of a classical time variable follows from Theorem 8.85. Generation of the classical continuous-approximation time representation and its establishment guarantee through the maximal effective interval follow from Corollary 8.89. Minimality of the invalidation-token set and membership in the six failure series follow from Theorem 8.91.  $\square$

**Corollary 8.94** (Generation, establishment guarantee, and invalidation localization of law time in classical physics). The conventional representation  $t$  of a time variable in a classical regime is determined, by Theorem 8.85, by applying a law-admissible time representation to a member of the generated  $\mathfrak{C}$ -derived observer-time quotient. When the generation-precedence condition of Corollary 7.15 holds, Theorem 7.200 implies that the physical-law time variable using that time content satisfies physical-time qualification.

The continuous time, infinitesimal time intervals, smooth trajectory representations, and differential law descriptions used in classical physics are generated from the full evaluation specification by Corollary 8.89. For a full evaluation specification and scale satisfying the assumptions of Theorem 8.91, their establishment is guaranteed at every scale in the respective maximal effective interval. By Corollary 8.92, a nonempty invalidation-token set coincides with the set of minimal elements under the failure-generation dependence order, and each member belongs to one of the six failure series.



*Proof.* Generation of the conventional representation of a classical time variable follows from Theorem 8.85. Physical-time qualification of a physical-law time variable follows from Corollary 7.15 and Theorem 7.200. Generation and scale-indexed establishment guarantee of the classical continuous-approximation time representation follow from Corollary 8.89. Minimality of the invalidation-token set and membership in the six failure series follow from Corollary 8.92.  $\square$

### 8.13 Difference from existing theories

The structures addressed by clock synchronization, relativistic time-stretching representations, proper time in GR, relational time, quantum-theoretic time reconstruction, time metrology, and operationalism are mapped to representative schemes and to the stages of the generation pathway. Differences between the partial objects borne by these seven subjects and the generation hierarchy are expressed as hierarchical differences along the generation pathway through physical-time generation, scale-indexed validity domains, and six-series failure localization.

Clock synchronization, relativistic time-stretching representations, proper time in GR, relational time, quantum-theoretic time reconstruction, time metrology, and operationalism respectively make objects of temporal measurement, synchronization, geometrization, relationalization, standardization, and connection to observation. Differences in clock readings, clock rates, proper times, and frequencies on physical carriers are relativistic results belonging to their respective objects and relations. Statements in Bailey et al. 1977 and Pössel 2021 directly describe these physical-carrier relations[37, 18]. An interpretation path that places ungenerated time prior to the comparative difference and, in direct dependence on that placement, reattributes the comparative difference on physical carriers to a change in an attribute of that time constitutes a two-stage misrecognition in the sense of Proposition 8.23. The difference from the seven subjects lies in mapping these partial objects onto a single generation pathway from local displays, records, object-state differences, and observer-side information-coupling positions through comparison closure, the canonical comparison quotient, the representation stage, the dynamical-description stage, a law-admissible time representation, a scale-indexed validity domain, and six-series failure localization.

Clock synchronization, proper time in GR, relational time, observables, time metrology, and operational measurement have developed as distinct subjects[2, 4, 10, 11, 12, 7]. The generation pathway in this chapter maps the structures they address to the respective stages of time reference, the observer-time quotient, a law-admissible time representation, a scale-indexed validity domain, and six-series failure localization.

Table 19: Hierarchical differences between the seven subjects and the generation pathway

| Subject                                     | Partial object                                                                                                       | Correspondence on the generation pathway                                                    | Hierarchical difference                                                                                                                                                                                                                                                                             |
|---------------------------------------------|----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Clock synchronization                       | Interlock comparison<br>Signal exchange<br>Convention of simultaneity                                                | Comparison stage<br>Part of the representation stage                                        | <ul style="list-style-type: none"> <li>• Connects <math>\mathfrak{C}/\mathfrak{D}/\mathfrak{B}/\mathfrak{D}</math> to dynamical description</li> <li>• Connects to the observer-time law core</li> <li>• Connects to a scale-indexed validity domain and six-series failure localization</li> </ul> |
| Relativistic time-stretching representation | Clock differences<br>Coordinate-time differences<br>Reattribution of proper-time differences to an attribute of time | Worked SR and GR reductions<br>Comparative differences among generated time representations | <ul style="list-style-type: none"> <li>• Preserves direct descriptions of physical-carrier relations</li> <li>• Evaluates time-stretching representations by interpretation path</li> </ul>                                                                                                         |
| Proper time in GR                           | Local-time display along a worldline                                                                                 | Input carrier or locally generated representation in the worked GR reduction                | <ul style="list-style-type: none"> <li>• Separates local proper-time displays</li> <li>• Separates global law-admissible time representations</li> </ul>                                                                                                                                            |
| Relational time                             | Relations among variables<br>Ordering by an internal variable                                                        | Representation stage<br>Part of the dynamical-description stage                             | <ul style="list-style-type: none"> <li>• Stages records, comparison, and observer coupling</li> <li>• Stages the path from quotient descent through law representation</li> </ul>                                                                                                                   |
| Quantum-theoretic time reconstruction       | Temporal description without external time                                                                           | Input-audit target in the worked QG reduction                                               | <ul style="list-style-type: none"> <li>• Requires generation conditions for the observer-time law core</li> <li>• Requires generation conditions for a law-admissible time representation</li> </ul>                                                                                                |
| Time metrology                              | Standard frequency<br>Time scale<br>Unit convention                                                                  | Worked reduction for time metrology                                                         | <ul style="list-style-type: none"> <li>• Generates frequency representations from count differences</li> <li>• Uses generated coordinate differences for quantitative affine differences</li> </ul>                                                                                                 |
| Operationalism                              | Operational measurement<br>Procedures<br>Observables                                                                 | Materials for input formation, the comparison stage, and the dynamical-description stage    | <ul style="list-style-type: none"> <li>• Connects to six-series failure localization</li> <li>• Connects to scale-indexed validity domains</li> <li>• Connects to the law-representation stage and differential-law-form stage</li> </ul>                                                           |

**Proposition 8.95** (The difference from the seven subjects is hierarchical). The structures addressed by clock synchronization, relativistic time-stretching representations, proper time in GR, relational time, quantum-theoretic time reconstruction, time metrology, and operationalism correspond to the comparison, representation, dynamical-description, law-representation, or differential-law-form stage of the generation pathway, to scale-indexed

validity-domain determination, or to six-series failure localization. The difference from the seven subjects is a hierarchical difference: their structures are mapped to stages of a single generation pathway and carried through physical-time generation, scale-indexed validity domains, and six-series failure localization.

*Proof.* Each row of Table 19 shows a structure addressed by one of the seven subjects and its correspondence on the generation pathway. Clock synchronization addresses interclock comparison and signal exchange and therefore corresponds to the comparison and representation stages. Statements in Bailey et al. 1977 and Pössel 2021 directly describe physical-carrier relations in which a moving clock or light clock bears the predicate of running faster or slower. Consider an interpretation path that places ungenerated time prior to a clock difference, coordinate-time difference, or proper-time difference. Suppose that reattribution of the subsequently obtained comparative difference to a change in an attribute of that time directly depends on that placement. Theorem 7.190 establishes that primitive-time roots remain in all 72 audited references. Every interpretation path of a recorded statement that contains placement before generation and attribute reattribution in direct dependence constitutes a two-stage misrecognition in the sense of Proposition 8.23. Proper time in GR is a local display along a worldline and corresponds to an input carrier or locally generated representation in Section 8.6. Relational time addresses relations among variables and therefore corresponds to the representation or dynamical-description stage. Quantum-theoretic time reconstruction addresses temporal description in formalisms without external time and is therefore an input-audit target in Section 8.8. Time metrology addresses counting, standard frequencies, time scales, and unit conventions and, in Section 8.3, generates frequency and period representations from count differences and generated coordinate differences. Operationalism addresses operational measurement and observables and therefore supplies materials for input formation and the comparison and dynamical-description stages.

The structures addressed by these seven subjects correspond to stages of the generation pathway. This correspondence supplies a generation hierarchy from local reception, records, object-state differences, and observer-side information-coupling positions through the canonical comparison quotient, representation coordinates, the observer-time law core, a law-admissible time representation, a scale-indexed validity domain, and six-series failure localization. The difference is therefore one of generation hierarchy and scope.  $\square$

The structures addressed by clock synchronization, relativistic time-stretching representations, proper time in GR, relational time, quantum-theoretic time reconstruction, time metrology, and operationalism correspond to stages on the generation pathway through comparison closure, the canonical comparison quotient, time representation, dynamical description, a law-admissible time representation, a scale-indexed validity domain, and six-series failure localization.

**Corollary 8.96** (Reconstruction of the seven subjects within the generation pathway). The structures addressed by clock synchronization, relativistic time-stretching representations, proper time in GR, relational time, quantum-theoretic time reconstruction, time metrology, and operationalism correspond to partial stages of the generation pathway or to inputs of representative schemes. Through this correspondence, each structure becomes a component of the generation pathway through physical-time generation, scale-indexed validity domains, and six-series failure localization.

*Proof.* By Proposition 8.95, the structures addressed by the seven subjects correspond to partial stages of the generation pathway or to inputs of representative schemes. Each

structure thereby becomes a component of the generation pathway through physical-time generation, scale-indexed validity domains, and six-series failure localization.  $\square$

## 8.14 Theoretical structures retained as theory-specific inputs

In theories belonging to the comparison-realization class of Definition 8.14, equations of motion, interaction terms, field equations, quantization procedures, correction rules, and engineering implementations are retained as theory-specific inputs to the realized operational configuration of Theorem 8.16. From these theory-specific inputs, the generation pathway generates the establishment conditions for temporal comparison, time representation, dynamical description, a law-admissible time representation, a scale-indexed validity domain, and six-series failure localization.

The following table shows the correspondence between theoretical content retained as theory-specific input and objects constructed or determined on the generation pathway.

Table 20: Separation of theoretical structures retained as theory-specific input from generated objects

| Retained object            | Input position                        | Corresponding object on the generation pathway                                                                 |
|----------------------------|---------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Equation of motion         | Theory-specific input                 | Law-admissible time representation                                                                             |
| Field equation             |                                       |                                                                                                                |
| Interaction term           |                                       |                                                                                                                |
| Quantization procedure     | QG-specific input                     | Canonical family of observables, trajectory relation, observer-time law core                                   |
| Correction rule            | Comparison-rule and record-side input | Determination of observer-time-law-core generation                                                             |
| Engineering implementation | Realization material                  | Comparison quotient, representation coordinates, navigation-time representation                                |
|                            |                                       | Canonical quotient, representation coordinates, scale-indexed validity domain, six-series failure localization |

Equations of motion supply motion content borne on law-admissible time representations. Interaction terms supply motion content evaluated on canonical families of observables, trajectory relations, and observer-time law cores. Field equations supply field-law content and become generation inputs for the time representations bearing that content. Quantization procedures are theory-specific QG inputs used to determine whether an observer-time law core can be generated without taking external time as a primitive premise. Correction rules are generation inputs for comparison quotients, representation coordinates, and navigation-time representations. Engineering implementations supply realization materials for determining canonical quotients, representation coordinates, scale-indexed validity domains, and six-series failure localization.

**Proposition 8.97** (Separation of theory-specific input from generated objects). In representative schemes belonging to the comparison-realization class of Definition 8.14, equations of motion, interaction terms, field equations, quantization procedures, correction rules, and

engineering implementations are retained as theory-specific inputs to the corresponding realized operational configuration. The objects constructed or determined on the generation pathway are comparison closure, the canonical comparison quotient, representation coordinates, the observer-time law core, a law-admissible time representation, a scale-indexed validity domain, and six-series failure localization. Existing theories' motion content is therefore retained as theory-specific input, while the establishment conditions and validity domain of the time variable used for that motion content are generated and its invalidation position is classified within the six series.

*Proof.* Conditions CR1 through CR7 of Definition 8.14 supply the carriers, comparison rules, record-side materials, object-state-side materials, and dynamical-description-side materials of representative theories as theory-specific inputs. By Theorem 8.16, these inputs determine the corresponding realized operational configuration.

By Corollary 8.18, determinations of operability and of the comparison, representation, continuous-approximation, dynamical-description, law-representation, and differential-law-form stages take this realized operational configuration as a domain-side input. Theorems 4.90 and 4.147 construct the objects from comparison closure through representation coordinates; Definitions 5.108 and 5.63 determine the observer-time law core and law-admissible time representation. Definition 6.4 determines the validity domain, and Definition 6.10 classifies invalidation positions within the six series.

Existing theoretical content is therefore retained as theory-specific input to the realized operational configuration, while the generation pathway constructs or determines the time variable's establishment conditions, validity domain, and invalidation positions.  $\square$

Through the separation of theory-specific input from generated objects, equations of motion, interaction terms, field equations, quantization procedures, correction rules, and engineering implementations are retained as theory-specific inputs. While retaining this theoretical content as theory-specific input, the generation pathway constructs temporal comparison, time representation, dynamical description, a law-admissible time representation, a scale-indexed validity domain, and six-series failure localization.

**Corollary 8.98** (Generation of time-variable establishment conditions while retaining existing theoretical content). For the existing theoretical content in Proposition 8.97, the generation pathway generates establishment conditions for using a time variable within laws, a scale-indexed validity domain, and six-series failure localization. Existing theoretical content holds as law content under a generated law-admissible time representation and scale-indexed validity domain; upon invalidation, its invalidation position within the six series and its structural burden are determined.

*Proof.* By Proposition 8.97, theoretical content is retained as theory-specific input to the realized operational configuration, while the time variable's establishment conditions, validity domain, and invalidation positions are constructed or determined on the generation pathway. Definition 6.4 determines the scale-indexed validity domain, while Definition 6.10 and Proposition 6.171 determine the invalidation position and structural burden. The claim follows.  $\square$

**Definition 8.99** (Result-preserving re-grounding). The construction that retains existing theoretical content as theory-specific input to the generation and audit construction of LCTR; retains existing results within their result-specific evidence-effect domains; invalidates the physical-time qualification of primitive-time premises; generates the required

time content and grounding structure from nontemporal physical inputs; audits their validity domains and invalidation positions; performs repair, regeneration, recomputation, or restriction of the domain of use upon invalidation; and reuses existing results within the intersection of the evidence-effect domain and the validity domain is called result-preserving re-grounding.

Figure 25 shows the retention pathway for existing theoretical content, the generation-and-audit pathway for temporal qualification, and the post-invalidation regeneration pathway.

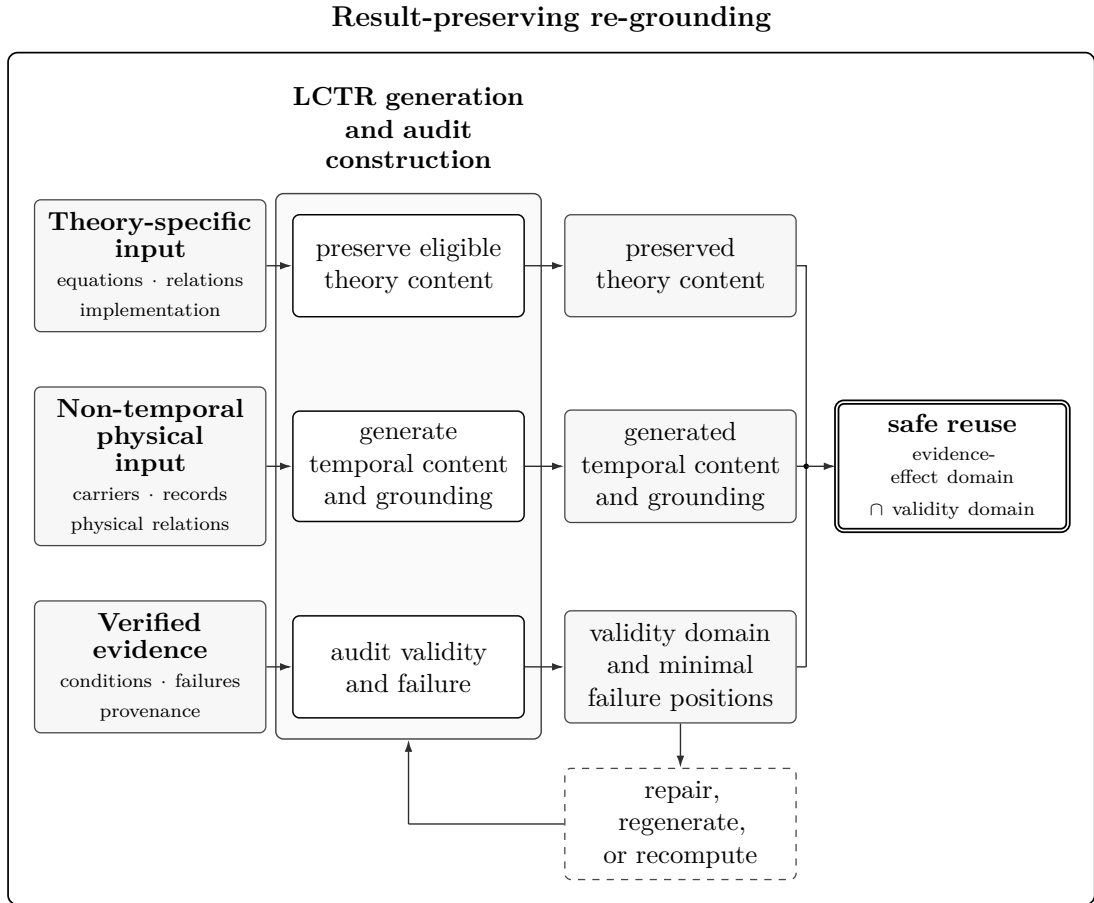


Figure 25: Result-preserving re-grounding in Definition 8.99. The retention pathway for existing theoretical content is separated from the generation-and-audit pathway for temporal qualification, and the safe domain of reuse is determined as the intersection of the evidence-effect domain and the validity domain. After invalidation, the construction is re-audited through repair, regeneration, and recomputation.

## 8.15 The typed generation-and-audit architecture of LCTR

LCTR integrates type separation, input–operation–output boundaries, staged generation, dependency order, qualification for use, audit states, invalidation localization, evidence provenance, and the scope of regeneration as conditions for the establishment of a physical theory. In LCTR, these principles used by the design methodology of computer science (CS) are mathematically defined as types, maps, generation closures, partial orders, validity domains, state sets, and evidence correspondences.

The CS-style hierarchy of this construction lies in reusing an encapsulated typed output as a single input item at the next level without re-expanding its internal construction, and then closing that item together with added components or conditions into one higher-level construction. In Chapters 2–3, role instances are encapsulated into a role-instance bundle family, raw operational input, and a realized operational configuration. In Chapter 4, the representation-stage generated-structure bundles accumulate recursively by head-preserving extension. In Chapter 5, qualification for dynamical, law, and differential description accumulates while retaining upstream qualification. In Chapter 6, the audit core and provenance certificate are encapsulated into operational audit data, which is then encapsulated with the state and diagnostic reports into the five-valued audit-report object.

**Definition 8.100** (LCTR typed generation-and-audit architecture). Define the LCTR typed generation-and-audit architecture as the structure consisting of the following six structures and their dependency edges.

1. Type structure: Definitions 2.5, 2.10, and 3.2 separate roles, role instances, and physical-carrier realizations and assign types to each evaluation input.
2. Generation structure: Definition 4.171 and Theorems 4.90, 4.147, 5.112, 5.113, and 5.135 determine the generation order of comparison, time representation, dynamical description, law description, and differential form.
3. Qualification-for-use structure: Definitions 5.63 and 6.4 and Proposition 6.162 determine the qualification of generated objects for use within laws and their scale-indexed domains of establishment.
4. Invalidation-audit structure: Definitions 6.17, 6.10, and 6.179, Theorem 6.164, and Proposition 6.171 determine audit states, minimal invalidation positions, scopes of effect, and scopes of regeneration.
5. Theoretical-and-engineering evidence structure: Definitions 6.188 and 6.194 and Proposition 6.204 determine correspondences among theoretical conditions, engineering measurands, verified evidence, and invalidation positions.
6. Cross-reference audit structure: Definitions 7.89 and 7.90 and Theorem 7.190 determine correspondences between reference-specific formal and statement evidence and common audit determinations.

The output of each structure becomes the input of a subsequent structure as specified by the dependency edges.

**Proposition 8.101** (Integrity of the typed generation-and-audit architecture). Under Definition 8.100, the following hold.

1. For each generated object, its input type, formation conditions, output type, and conditions for subsequent use are determined.
2. The physical qualification for use of time representations, dynamical descriptions, law time, continuous approximations, and differential forms is determined from upstream conditions and scale conditions.
3. Each audit state is determined as pass, fail, blocked, unformed, or indeterminate; upon invalidation, the minimal invalidation positions are determined.

4. The upward dependency set of an invalidation position determines the affected subsequent objects, structural burden, and scope of regeneration.
5. Theoretical conditions and engineering measurands retain distinct types, and verified evidence determines the state of the corresponding invalidation-position token.
6. Formal evidence and statement evidence from original sources retain reference-specific provenance, and the canonical finite audit realization determines the common audit determination.

LCTR therefore gives the input formation, object generation, qualification for use within laws, scale-indexed domains of establishment, invalidation states, evidence provenance, repair targets, and scope of downstream regeneration of a physical theory as a single typed dependency structure.

*Proof.* Item 1 follows from the type separation of roles, role instances, and physical-carrier realizations and from the input and output specifications of typed generation closure. Item 2 follows from the generation order of comparison, time representation, dynamical description, law description, and differential form and from the definitions of law-admissible time representations and scale-indexed domains of establishment.

Item 3 follows from the five-valued audit states and the first-breakdown theorem. Item 4 follows from the invalidation-dependency order, invalidation-position tokens, and structural-burden completeness. Item 5 follows from engineering evaluation specifications, engineering audit-parameter vectors, and formal-condition evidence localization. Item 6 follows from typed finite audit inputs, canonical finite audit realizations, and the all-reference primitive-time-root result.

The dependencies among types, generation, qualification for use, invalidation, evidence, and regeneration constitute the theoretical structure of CS design methodology in LCTR. The equations of physical theory, the types, dependencies, states, and change effects specified by CS design discipline, and engineering evidence jointly constitute this typed generation-and-audit architecture.  $\square$

## 8.16 Worked classical many-body audit and background conditions of classical world representation

**Proposition 8.102** (Backgrounding the establishment conditions for time reference in a classical world representation). A classical world representation explicitly includes point masses, centers of mass, shapes, moments of inertia, and rigid-body constraints as variables or relations in law expressions. At the same time, it does not describe the role instances, physical-carrier realizations, record-arrival conditions, observer-side information coupling, and joint law domain that establish time reference as variables, relations, generation conditions, or invalidation conditions in those expressions; instead, it uses them as presuppositions for describing multiple objects by the same time variable. This treatment is called the backgrounding of establishment conditions for time reference.

*Proof.* Definitions 2.10, 3.2, 6.190, 6.193, and 5.139 supply the role types, carriers, records, information coupling, and relational construction required for multi-object relations to hold physically.

A classical world representation does not include within its law description either a path by which these structures are generated from nontemporal physical inputs or conditions



under which use of the law ceases when those structures are invalidated. Nevertheless, it uses these structures as presuppositions for describing multiple objects by the same time variable. A classical world representation therefore supplies the establishment conditions for time reference in advance as background conditions outside the expressions.  $\square$

**Proposition 8.103** (Non-completion of background conditions by the success of classical physics and the scale-indexed validity domain). The mathematical consistency, predictive performance, and empirical fit of classical physics do not complete the physical generation of backgrounded establishment conditions for time reference. The domain in which multi-object relations hold physically under the same time variable belongs to the scale-indexed validity domain in which the physical-carrier realization of role instances,  $\mathfrak{D}$ -derived record field of view, record stability, observer-side information coupling,  $\mathfrak{D}$ -communication, multi-object relations, and scale-indexed defect quantities satisfy their respective establishment conditions.

*Proof.* By Proposition 7.13, mathematical consistency, predictive performance, and empirical fit add no generation dependence to time inputs.

Definition 6.194 supplies  $\mathfrak{D}$ -derived record field of view, record invasiveness, the  $\mathfrak{D}$ -communication graph, multi-object-relation defect, and common-law-domain defect as engineering audit inputs. Definition 6.161 supplies the scale domain of approximation establishment in which approximation defects do not exceed their tolerances. Definitions 6.191 and 6.205 supply the two-object recording configurations and their structural burdens relative to engineering evidence.

Each engineering audit result is connected to the establishment determination of the corresponding theoretical condition through verified engineering evidence for formal conditions. Evaluation specifications and scales for which all six series are established belong to the full evaluation-validity domain and give the physical domain of multi-object relations under the same time variable. The success of classical physics therefore supplies no ground for completing the physical generation of the establishment conditions for time reference while leaving them outside the expressions; the domain of that success is fixed as a scale-indexed validity domain satisfying the respective establishment conditions.  $\square$

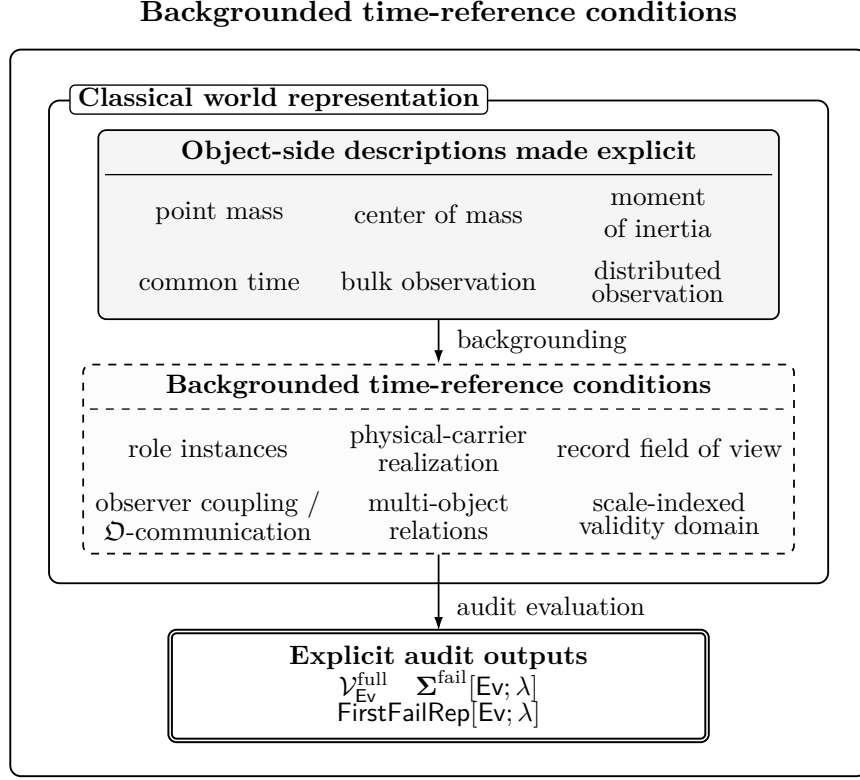


Figure 26: Object-side physical quantities made explicit by a classical world representation and the establishment conditions for time reference that it backgrounds outside the expressions. Multi-object descriptions using point masses, centers of mass, moments of inertia, and common time correspond to role instances, physical-carrier realizations,  $\mathfrak{D}$ -derived record field of view,  $\mathfrak{D}$ -communication, multi-object relations, scale-indexed validity domains, six-series invalidation signatures, and audit reports.

Table 21: Object-side structures made explicit by a classical world representation, backgrounded obligations for establishing time reference, and LCTR-generated objects

| Classical input          | Backgrounded establishment obligation                                        | LCTR-generated object                                                  |
|--------------------------|------------------------------------------------------------------------------|------------------------------------------------------------------------|
| Point-mass approximation | Object identity and interobject relations                                    | Role-instance family, multi-object relations                           |
| Center of mass           | Composite-object state                                                       | Physical-carrier realization, object-state quotient                    |
| Moment of inertia        | Comparability of observables                                                 | Canonical family of observables, quantitative affine realization       |
| Common time $t$          | $\mathfrak{C}/\mathfrak{D}/\mathfrak{D}$ comparison, recording, and coupling | Comparison closure, order-coordinate structure, observer-side coupling |
| Bulk observation         | $\mathfrak{D}$ -record field of view, cell width, and stability              | Bulk-recording condition, recording structural burden                  |
| Distributed observation  | $\mathfrak{D}$ -communication and comparison-loop consistency                | Following-recording condition, recording structural burden             |

**Definition 8.104** (Classical multi-object audit example). Let two distinct object-specification indices and the finite family of object-specification indices be

$$\zeta_1, \zeta_2 \in Z_0, \quad \zeta_1 \neq \zeta_2, \quad Z_0 = \{\zeta_1, \zeta_2\}$$

, and let the relation index be  $\rho \in \mathbf{JRelIdx}_{Z_0}$ .

For each  $a \in \{0, 1\}$  and  $k \in \{1, 2\}$ , let the full evaluation specification be

$$\mathbf{Ev}_k^{(a)} = ((\mathcal{X}^{(a)}; \mathcal{J}_{\text{cmp}}), o, \zeta_k)$$

, and assume

$$(\mathbf{Ev}_1^{(a)}, \mathbf{Ev}_2^{(a)}) \in \mathbf{EvSpec}^{\text{pair}}$$

. For the full evaluation specifications of following-recording type, assume

$$(\mathbf{Ev}_1^{(1)}, \mathbf{Ev}_2^{(1)}) \in \mathbf{EvSpec}_{\mathfrak{D}}^{\text{pair, distributed}}$$

.

The two recording configurations satisfy

$$\mathbf{BulkRecord}_{\mathfrak{D}} \left[ \left( \mathbf{Ev}_k^{(0)} \right)_{k=1}^2 \right] = \top$$

and

$$\mathbf{FollowingRecord}_{\mathfrak{D}} \left[ \left( \mathbf{Ev}_k^{(1)} \right)_{k=1}^2 ; \lambda \right] = \top$$

.

For each configuration  $a \in \{0, 1\}$ , suppose that the multi-object relation input data of Definition 5.139 are supplied and satisfy the representative-invariance condition of Definition 5.142. Denote the corresponding quotient-level datum by  $\mathbf{J}_{o, \rho, Z_0}^{(a), \text{can}}$ . For the same object-specification set  $Z_0$  and relation-type index  $\rho$ , audit each quotient-level relation  $\mathbf{J}_{o, \rho, Z_0}^{(a), \text{can}}.\mathbf{Rel}$  under its corresponding recording configuration. Call these two configurations the classical multi-object audit example.

### Bulk-record / following- record audit comparison

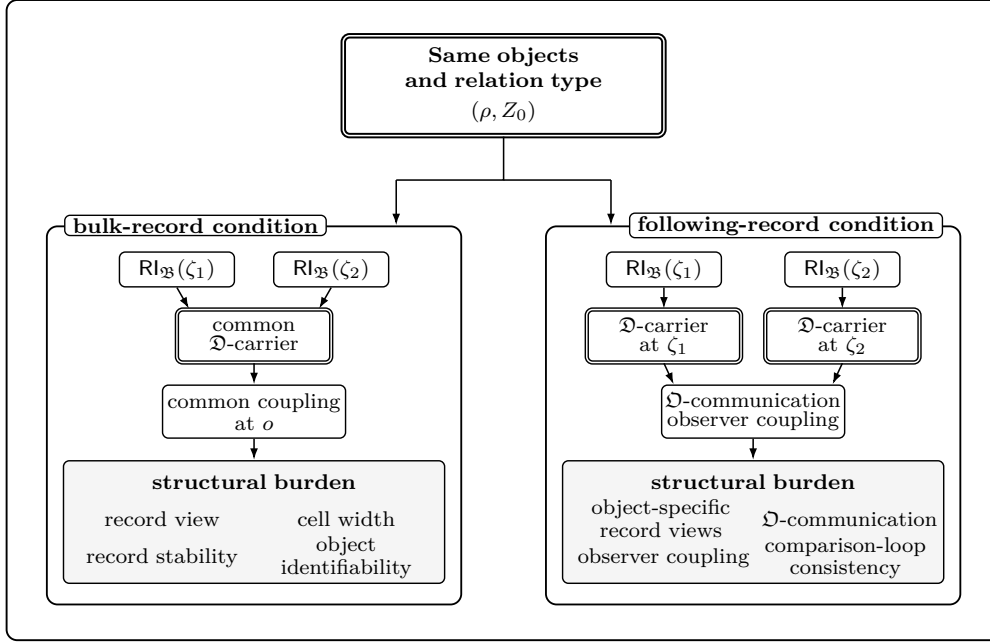


Figure 27: Structural burdens relative to engineering evidence for the bulk-recording and following-recording conditions. For the same object-specification set and relation index, the bulk-recording condition requires record field of view, cell width, record stability, and object identifiability in a single  $\mathfrak{D}$ -carrier. The following-recording condition requires  $\mathfrak{D}$ -communication, observer-side coupling, and comparison-loop consistency in addition to object-specific record fields of view in distinct  $\mathfrak{D}$ -carriers.

Table 22: Audit inputs for the bulk-recording and following-recording conditions

| Recording configuration       | $\mathfrak{D}$ -carrier condition | Additional audit inputs                                                                                                   |
|-------------------------------|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Bulk-recording condition      | Single $\mathfrak{D}$ -carrier    | Record field of view, cell width, record stability, object identifiability                                                |
| Following-recording condition | Distinct $\mathfrak{D}$ -carriers | Object-specific record fields of view, $\mathfrak{D}$ -communication, observer-side coupling, comparison-loop consistency |

**Proposition 8.105** (Classical multi-object audit of the bulk-recording condition). The invalidation-position set for the bulk-recording condition in the classical multi-object audit example is  $\text{RecordBurdenTok}^{(0)}$ , and its structural burden relative to engineering evidence is  $\text{RecordBurden}^{\text{bulk}}$ .

*Proof.* By Definition 6.191, the bulk-recording condition requires the two  $\mathfrak{D}$ -derived role instances to be realized by the same physical carrier. For the engineering evidence of the bulk-recording condition, Definition 6.205 determines the invalidation-position set in the proposition from the respective input tags for record field of view, recording configuration, record invasiveness,  $\mathfrak{C}$ -count increment,  $\mathfrak{D}$ -record stability,  $\mathfrak{D}$ -record cell width, and  $\mathfrak{B}$ -identifiability, and gives the structural burden relative to engineering evidence in the proposition as the union of their dependency-upward structural images.  $\square$

**Proposition 8.106** (Classical multi-object audit of the following-recording condition). The invalidation-position set for the following-recording condition in the classical multi-object audit example is  $\text{RecordBurdenTok}^{(1)}$ , and its structural burden relative to engineering evidence is  $\text{RecordBurden}^{\text{follow}}$ .

*Proof.* By Definition 6.191, the following-recording condition requires the two  $\mathfrak{D}$ -derived role instances to be realized by distinct physical carriers and each object-specific record field of view to be contained in the domain of the corresponding record-code relation. For the engineering evidence of the following-recording condition, Definition 6.205 adds the  $\mathfrak{D}$ -communication and observer-side-coupling input tags to the input tags of the bulk-recording condition to determine the invalidation-position set in the proposition, and gives the structural burden relative to engineering evidence in the proposition as the union of their dependency-upward structural images.  $\square$

**Proposition 8.107** (Difference in structural burden between the bulk-recording and following-recording conditions). The two structural burdens relative to engineering evidence in the classical multi-object audit example are subsets of  $\text{StructTok}$ , and

$$\begin{aligned} \text{RecordBurden}^{\text{bulk}} &= \text{RecordBurden}^{\text{follow}} \\ \iff \text{RecordBurden}^{\Delta} &= \emptyset. \end{aligned}$$

If both set differences

$$\text{RecordBurden}^{\text{bulk}} \setminus \text{RecordBurden}^{\text{follow}}$$

and

$$\text{RecordBurden}^{\text{follow}} \setminus \text{RecordBurden}^{\text{bulk}}$$

are nonempty, the two structural burdens are incomparable under inclusion.

*Proof.* Applying Proposition 6.206 to the bulk-recording and following-recording conditions of Definition 8.104 gives the asserted equivalence and incomparability condition.  $\square$

**Connection to the audit of law-time use.** The two recording configurations in the classical multi-object audit example supply full evaluation specifications to the physical-time-qualification audit in Section 7. For the full evaluation specifications corresponding to the two recording configurations, when the generation dependencies and operability conditions of Proposition 7.199 hold prior to law operations and result derivation, Theorem 7.200 implies that the physical-law time variable bearing the time content used there satisfies physical-time qualification.

If time content is entered into a law operation or result derivation before the generation structure required for its use has been established, that time content is a primitive-time input under Definition 7.10. By Theorem 7.203, the physical-law time variable bearing that time content does not satisfy physical-time qualification. Invalidation of engineering evidence in each recording configuration is localized as the recording-configuration-specific invalidation-position set of Definition 6.205, the six-series invalidation signature of Definition 6.159, and the structural-burden difference of Proposition 8.107.

### Configuration-specific outputs of the classical multi-object audit

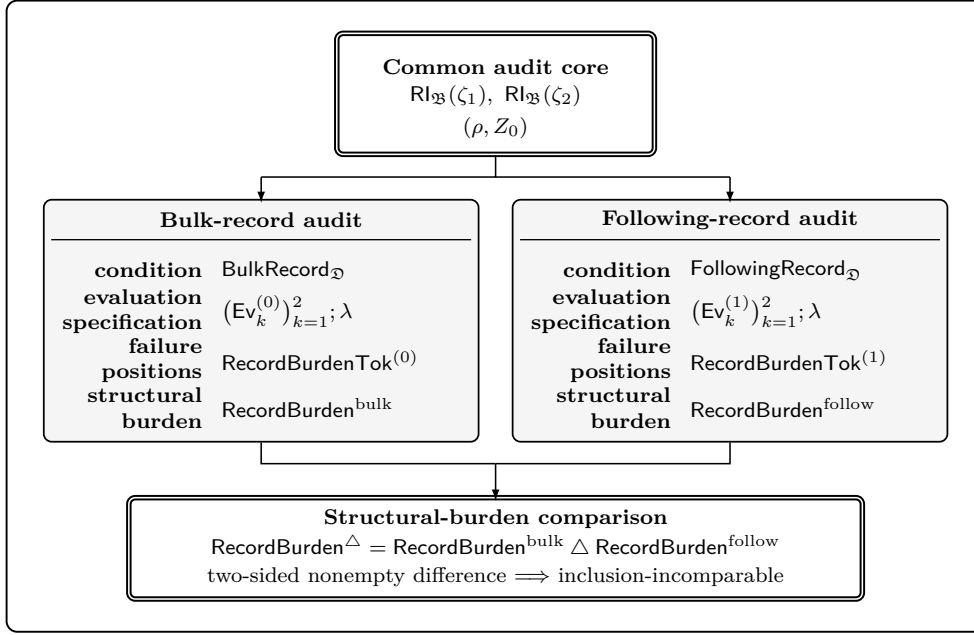


Figure 28: Configuration-specific outputs of the classical multi-object audit example. Role instances, full evaluation specifications,  $\mathfrak{D}$ -carrier conditions, record evidence,  $\mathfrak{D}$ -side evidence, and the multi-object relation are supplied as inputs, and the structural burdens relative to engineering evidence for the bulk-recording and following-recording conditions are compared.

Table 23: Configuration-specific outputs of the classical multi-object audit example

| Audit component                | Bulk-recording condition                                               | Following-recording condition                                            |
|--------------------------------|------------------------------------------------------------------------|--------------------------------------------------------------------------|
| Role instances                 | $\text{RI}_{\mathfrak{B}}(\zeta_1), \text{RI}_{\mathfrak{B}}(\zeta_2)$ |                                                                          |
| Full evaluation specifications | $\text{Ev}_1^{(0)}, \text{Ev}_2^{(0)}$                                 | $\text{Ev}_1^{(1)}, \text{Ev}_2^{(1)}$                                   |
| $\mathfrak{D}$ -carrier        | Common physical carrier                                                | Distinct physical carriers                                               |
| Record evidence                | Shared record field of view and shared code relation                   | Object-specific record fields of view and object-specific code relations |
| $\mathfrak{D}$ -side evidence  | Common information-coupling position $o$                               | $\mathfrak{D}$ -communication datum and observer-side coupling           |
| Multi-object relation          | $\text{J}_{o,\rho,Z_0}^{(0),\text{can}}.\text{Rel}$                    | $\text{J}_{o,\rho,Z_0}^{(1),\text{can}}.\text{Rel}$                      |
| Failure output                 | Bulk-recording structural burden                                       | Following-recording structural burden                                    |

## 8.17 Final synthesis

The final synthesis theorem integrates, into a single dependence order, generation from the comparison stage through the differential-description stage, the scale-indexed validity domain of the continuous-approximation stage, invalidation localization in the six failure series, five-valued audit reports, the cross-source results of the 72 audited sources, worked reductions of representative theories, the classical multi-object audit, differences from

existing theories, result-preserving re-grounding, and the typed generation-and-audit architecture. The following table maps each synthesis item to its direct grounds and outputs.

Table 24: Direct grounds and outputs of the final synthesis theorem

| Synthesis item                                     | Direct grounds                                                                        | Output                                                                                                                                                                                                                                                                                                                          |
|----------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Synthesis (1):<br>Comparison closure               | Theorem 4.90                                                                          | $L_{\leq 7}[Ev^{cmp}]$ , $G[Ev^{cmp}]$ , $\mathfrak{CD}$ -derived canonical comparison quotient                                                                                                                                                                                                                                 |
| Synthesis (2):<br>Representation generation        | Theorems 4.147, 4.177, 4.179, and 4.187                                               | $\text{Repr}_{\leq 3}(\mathcal{X}; \mathcal{J}_{cmp})$ , exact operational time structure, local and global order-coordinate structures, comparison-order atlas                                                                                                                                                                 |
| Synthesis (3):<br>Dynamical-description generation | Theorems 5.112 and 5.138                                                              | $\text{DynCondAll}[Ev]$ , observer-time quotient, trajectory relation, graph trajectory, canonical family of observables, law-admissible time representation                                                                                                                                                                    |
| Synthesis (4):<br>Law and differential generation  | Theorems 5.113, 5.135, and 7.203                                                      | Observer-time law core, generated representationality of law-evaluation time, chart-compatible differential law, differential covariance under changes of time representation                                                                                                                                                   |
| Synthesis (5):<br>Six failure series               | Definition 6.10 and Theorem 6.168                                                     | Six series of comparison, representation, dynamical description, continuous approximation, law description, and differentiation; invalidation positions; six-series invalidation signature                                                                                                                                      |
| Synthesis (6):<br>Scale boundary and audit         | Propositions 6.162 and 6.154, Theorems 6.163, 6.164, and 6.183, and Proposition 6.199 | Full evaluation-validity domain, dual qualification of continuous approximation and differential form, maximal effective interval for approximation, approximation-invalidation scale, approximation-boundary signature, five-valued audit-report object, structural-burden audit component, typed separation of failure causes |

Continued on next page



| Synthesis item                                                                         | Direct grounds                                                                                                                                | Output                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Synthesis (7):<br>72 audited<br>sources and<br>representative<br>theories              | Theorem 7.190, Propositions 7.191, 8.73, 8.74, Corollary 8.75, Propositions 8.59 and 8.77, Theorem 8.16, and the respective worked reductions | Universal persistence of primitive-time roots, counts of time-generation obligations, target consistency between time-nonexistence statements and time-attribute statements, generation conditions, validity domains, and invalidation positions of representative theories                                                                                                                                                                                                                                                                                                                                                                                        |
| Synthesis (8):<br>Theoretical differ-<br>ence and<br>result-preserving<br>re-grounding | Propositions 8.95 and 8.97, Definition 8.99, Propositions 8.102, 7.121, 8.78, and 8.79, Definitions 3.2 and 6.188, and Proposition 6.204      | Result-preserving re-grounding, objects retained from existing theories, generation obligations for time-reference conditions, scale-indexed domain of establishment of multi-object relations under a single time variable, causal connection from the backgrounding of grounding obligations through actual invalidation and downstream reach, separation between the invalidation of physical-time qualification and conditional invalidation of use of downstream computational results, scope of regeneration and recomputation after repair, range of safe reuse under generation conditions, typed applicability conditions for engineering implementations |
| Synthesis (9):<br>Typed generation<br>and<br>audit structure                           | Definition 8.100 and Proposition 8.101                                                                                                        | Typed dependence structure of physical theory, use qualification, audit states, invalidation localization, evidence provenance, range of effects, scope of repair and regeneration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

Continued on next page

| Synthesis item                            | Direct grounds              | Output                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------------------------------------|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Synthesis (10):<br>Final conse-<br>quence | Synthesis (1)–Synthesis (9) | Constructive definition of physical time, generation of temporal identity and before–after relations, conditional representation by linear order and real-valued time, unique re-expression of time-coordinate freedom, establishment obligations for multi-object relations under a single time variable, separation of the obligations for temporal order and causal order, generation and maximal validity domain of classical continuous time, formal obligations for statements of time creation, emergence, and nonexistence, change of the problem of time in quantum gravity into a problem of law-time use qualification, changes in field-specific temporal obligations, qualification invalidation of primitive-time premises, six-series invalidation localization, audit report |

The following final synthesis theorem therefore gives the generation, establishment, invalidation, and audit of time variables in the same dependence order from the direct grounds in Table 24.

## Collapse of the Mystery Basis

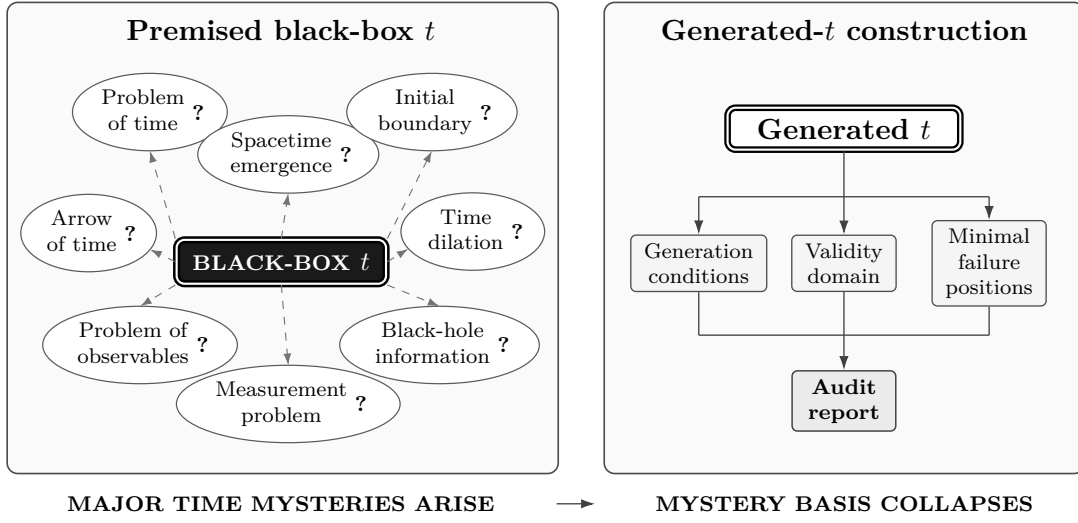


Figure 29: A time variable  $t$  presupposed as a black box without a generation path supplies the basis on which the problem of time, time stretching, initial boundaries, the arrow of time, the problem of observables, the measurement problem, the black-hole information problem, and spacetime emergence appear as mutually independent mysteries. Once time content and its grounding structure are generated from nontemporal physical inputs, the basis on which these mysteries are constituted is lost, and each problem is transferred to generation conditions, scale-indexed validity domains, the six failure series, and audit reports.

**Definition 8.108** (Time-problem connection domain). A time-problem connection domain is a family of problem descriptions that appear as independent mysteries because an ungenerated time variable, temporal order, or time representation is entered in advance into law operations or result derivations, and that are connected to the generation conditions, scale-indexed validity domain, six failure series, or audit report of the corresponding time content and grounding structure.

The difficulty of defining time in Definition 1.1 is made concrete by Theorem 7.203 as the requirement for the physical grounds of physical-law time variables. When comparison closure, the  $\mathcal{C}\mathcal{D}$ -derived canonical comparison quotient, the  $\mathcal{C}$ -derived observer-time quotient, the  $\mathcal{C}\mathcal{D}\mathcal{B}$ -derived trajectory relation, the observer-time law core, and a law-admissible time representation hold along the same generation dependence, a time variable is established as a generated law-admissible time representation.

Comparison closure and the canonical comparison quotient give the conditions under which two physical materials may be identified as belonging to the same time content. The canonical partial order preserves before–after relations and incomparability, and the quotient by incomparability gives a strict linear order when its transitivity condition holds. Real-valued time is obtained as a representation when an order embedding from that ordered quotient into the real numbers exists. Accordingly, LCTR specifies the real line and linear order as downstream representations of a time structure whose generation conditions have been satisfied. Freedom in the choice of an admissible order representation is fixed as re-expression through a unique strict order isomorphism between the representation images. This is LCTR’s answer to the problem that Feynman presented as the difficulty of giving time a satisfactory definition[6]. LCTR changes the definition of physical time

from the specification of lexical meaning to the generation of conditions of establishment and generates, from nontemporal physical inputs, the conditions for temporal identity, before–after relations, comparability, linearization, real-valued representation, use within laws, and invalidation.

For a claim that uses continuous approximation, its range of establishment is determined as a scale-indexed validity domain, and invalidation is localized by the six failure series, approximation-boundary signatures, and five-valued audit reports.

In the following figure, the horizontal axis presents the structural reach borne by a use of time in the order local, multiple records, multiple observers, and gluing/boundary; the vertical axis presents the stages at which grounding obligations arise in the order comparison, representation, continuous approximation, dynamical description, law description, and differentiation. A large filled circle denotes a principal localization burden; a small filled circle, an additional direct burden; a diamond, a scale condition; an open circle, a derived or downstream application; a gray horizontal line, a structural range; a thin line, the connection among multiple burdens borne by the same object; and a square, engineering evidence. Positional differences within a band are used for visual discrimination. The engineering-evidence band at the bottom represents verified engineering inputs connected to the six theoretical stages.

The abbreviations correspond to the following constructions or audits in the text. C.Min. denotes the minimal operational model of classical time (8.11.1); T.Met., the worked reduction of time metrology (8.3); SR, the worked reduction of special relativity (8.5); C.Dom., the validity domain and failure boundary of classical time (8.12); GPS, the worked reduction of GPS (8.4); GR, the worked reduction of general relativity (8.6); and QG, quantum-gravitational reconstruction (8.8). I.Bd., connected to QG, denotes the downstream application to a cosmological initial boundary (8.8.3); B.H., that to the extreme-gravity domain including the black-hole information problem (8.8.4). G/F and C.Multi. correspond to bulk recording and following recording and to the classical multi-object audit example (8.16).

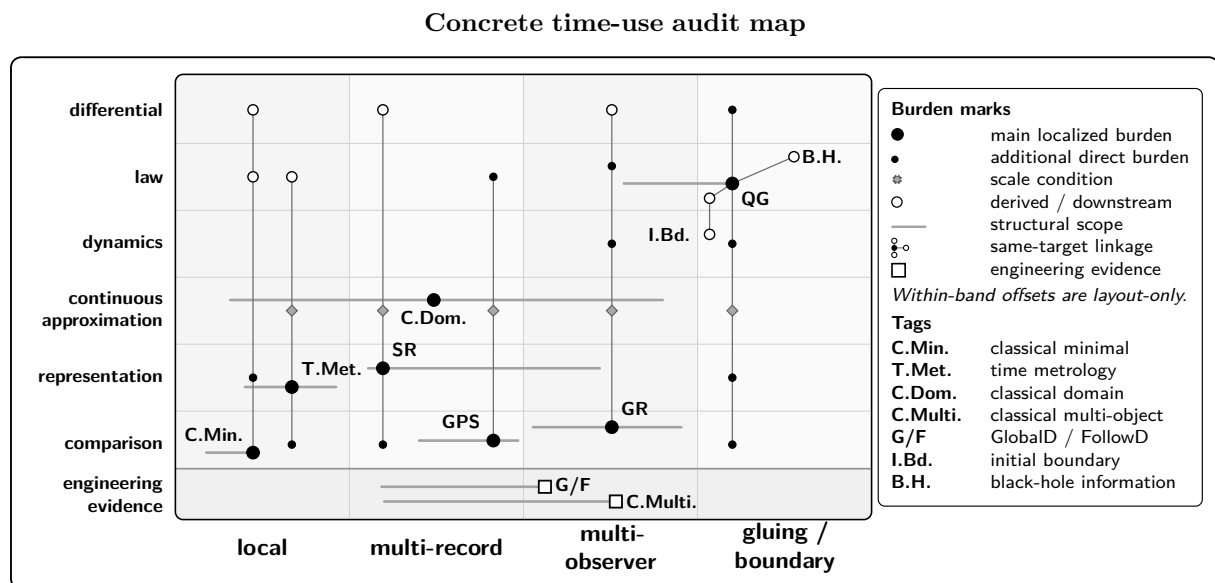


Figure 30: Correspondence between structural reach and generation-and-audit burdens in representative uses of time.

**Theorem 8.109** (Final synthesis theorem). For a realized operational configuration  $\mathcal{X}$ , a typed family of comparison rules  $\mathcal{J}_{\text{cmp}}$ , an observer-side information-coupling position  $o \in U$ , an object-specification index  $\zeta \in Z$ , a complete evaluation specification

$$\text{Ev} = ((\mathcal{X}; \mathcal{J}_{\text{cmp}}), o, \zeta) \in \text{EvSpec}$$

and an approximation scale  $\lambda \in \Lambda$ , the following hold.

**Synthesis (1): comparison closure.** Operability at the comparison stage is given by

$$\begin{aligned} \text{Operative}(\mathcal{X}) &\iff \text{L}_{\leq 7} [\text{Ev}^{\text{cmp}}] \\ &\quad \wedge \text{G} [\text{Ev}^{\text{cmp}}] \end{aligned}$$

Under this condition, the extended comparison-arrival equivalence relation, the  $\mathfrak{C}/\mathfrak{D}$  comparison-quotient sets, the canonical projections, the quotient-level  $\mathfrak{C}\mathfrak{D}$ -derived relation, and the  $\mathfrak{C}\mathfrak{D}$ -derived canonical comparison quotient are determined uniquely in dependence order.

**Synthesis (2): representation generation.** Exact operability is given by

$$\begin{aligned} &\text{ExactOperable}(\mathcal{X}; \mathcal{J}_{\text{cmp}}) \\ &\iff \text{Operative}(\mathcal{X}) \wedge \text{Repr}_{\leq 3}(\mathcal{X}; \mathcal{J}_{\text{cmp}}) \end{aligned}$$

Under this condition, the  $\mathfrak{C}/\mathfrak{D}$  canonical partial orders, exact operational time structure, local and global order-coordinate quotients, local and global exact linear orders, canonical order-coordinate projections, local-to-global order-coordinate embeddings, and comparison-order atlas are generated in dependence order. Once a selected representation range and an order embedding of that range are specified, the  $\mathfrak{C}/\mathfrak{D}$ -derived representation coordinates factor through the canonical global order-coordinate projection.

**Synthesis (3): dynamical-description generation.** Dynamical-description operability is given by

$$\begin{aligned} &\text{DynOperative}[\text{Ev}] \\ &\iff \text{ExactOperable}(\mathcal{X}; \mathcal{J}_{\text{cmp}}) \wedge \text{DynCondAll}[\text{Ev}] \end{aligned}$$

Under this condition, the  $\mathfrak{C}$ -derived observer-time quotient, the  $\mathfrak{B}$ -derived object-state quotient, the  $\mathfrak{C}\mathfrak{D}\mathfrak{B}$ -derived trajectory relation, a graph trajectory, a canonical family of observables, the observer-time order-coordinate quotient, a real-valued time representation, a scalar-time trajectory, and a family of real-valued observable curves are generated.

**Synthesis (4): law and differential generation and the physical establishment of time variables.** Law-description operability and differential-stage operability are given by

$$\begin{aligned} &\text{LawOperative}[\text{LawEvalDat}_{o,\zeta}] \\ &\iff \text{DynOperative}[\text{Ev}] \wedge \text{LawCondAll}[\text{LawEvalDat}_{o,\zeta}], \\ &\text{DiffOperative}[\text{DiffEvalDat}_{o,\zeta}] \\ &\iff \text{LawOperative}[\text{LawEvalDat}_{o,\zeta}] \wedge \text{DiffCondAll}[\text{EvalStrFam}_{o,\zeta}^{\text{diff}}] \end{aligned}$$

The former establishes the observer-time law core and the generated representationality of law-evaluation time; the latter establishes chart-compatible differential laws, value-jet-relation covariance, and time-representation-change jet-relation covariance.

The law-admissible time representation is generated as

$$\text{TimeRep}_o^{\text{dyn}} = \text{RepEmb}_o^{\text{dyn,time}} \circ \text{Prj}_o^{\text{dyn,time,ord}}$$

By Theorem 7.203, a time variable in a physical law is not established as a primitive premise. When time content and its grounding structure are generated from nontemporal physical inputs before law operations and result derivations, the time variable satisfies physical-time qualification.

**Synthesis (5): invalidation localization in the six failure series.** The six series of comparison, representation, dynamical description, continuous approximation, law description, and differentiation are distinguished by

$$\text{FailSeries} = \{\text{cmp, repr, dyn, law, diff}\} \sqcup \{\text{approx}\}$$

Invalidation of a formed and evaluable condition is localized to the corresponding set of invalidation-position tokens  $\text{FailedTok}[\text{Ev}; \lambda]$  and gives the six-series invalidation signature  $\Sigma^{\text{fail}}[\text{Ev}; \lambda]$ . Because multiple invalidation positions can hold in the same evaluation, invalidations in the six series do not require an exclusive disjunction. Unformed and unevaluable states are not replaced by invalidation; they are retained as their respective audit states.

**Synthesis (6): scale-indexed validity domain, boundary, and five-valued audit.**

Under the all-scale approximation-input admissibility and defect-monotonicity assumptions of Theorem 6.163, the approximation-validity scale domain  $\mathcal{V}_{\text{Ev}}^{\text{approx}}$  is an initial subset of the totally ordered set of scales and coincides with the maximal effective interval for approximation  $\mathcal{I}^{\text{max}}[\text{Ev}]$ . If the complement has a least element, the approximation-invalidation scale  $\lambda_{\text{bdry}}^{\text{approx}}[\text{Ev}]$  and approximation-boundary signature

$$\Sigma_{\text{approx}}^{\text{bdry}} [\text{Ev}; \lambda_{\text{bdry}}^{\text{approx}} [\text{Ev}]]$$

are determined uniquely.

The five-valued audit-report object

$$\text{AuditReport} [\text{Ev}^{\text{fail}}; \lambda]$$

is determined uniquely from operational audit data and gives the series-specific audit-group states, condition-specific audit states, invalidation positions, six-series invalidation signature, indeterminate tokens, indeterminate signature, approximation boundary, structural burden, and reason for an undefined audit.

When a differential equation that uses continuous approximation is employed as a physical description, its use qualification is given by the full evaluation-validity domain in which both the five exact series—comparison, representation, dynamical description, law description, and differentiation—and the continuous-approximation series hold. The continuous-approximation series evaluates the chain from finite physical realization through continuous extension, order stability, and relation stability; the differential series evaluates the differential atlas, generated-trajectory jets, local jet relations, and jet-relation covariance. The input types of the two series are distinguished, and both qualifications are satisfied simultaneously in the full evaluation-validity domain.

The five-valued audit states separate satisfaction, condition invalidation, blockage by a prior condition, an unformed state, and indeterminacy. Engineering-implementation

auxiliary-audit flags are connected, through verified auxiliary-audit-effect evidence, to problems concerning formation inputs, evaluation inputs, input admissibility, boundary treatment, or condition evidence. Numerical-solver failure and data gaps are retained as engineering auxiliary-audit causes and have a type distinct from condition invalidation of a target invalidation-position token.

**Synthesis (7): the 72 audited sources and representative theories.** For the 72 audited sources,

$$\text{Ref}^{\text{audit,root}} = \text{Ref}^{\text{audit}}$$

holds, and every source contains a primitive-time root and results that depend on a primitive-time root. Under the complete criterion, the generated-time-content set is empty for all 72 sources, and no source generates, from nontemporal physical inputs, the entirety of the time content it uses. Under the local-reach criterion, 10 sources reach a total of 57 temporal contents. The strict generated-time-content sets of all 72 sources are empty, and the entire source-specific time content consists of primitive-time roots. In the remaining 62 sources, no time content is reached even under the local-reach criterion. Sources that explicitly state the absence of external or independent time, the fundamental status of time, difficulty defining the concept of time, or an internal or relational construction of time number (18/72); sources that present a dynamical construction or recovery of intrinsic, relational, thermal, or semiclassical time number (13/72); and among the latter, sources that satisfy the complete time-generation obligation number (0/13).

Newtonian and classical mechanics, time metrology, GPS, special relativity, general relativity, and quantum-gravitational reconstruction are placed within the same generation dependence, conditions of establishment, scale-indexed validity domains, and six failure series. The minimal classical model and the classical multi-object audit example give this placement as a concrete construction and audit. Relativistic clock differences hold as comparison differences among generated law-admissible time representations on physical carriers of  $\mathcal{CQBD}$ . Statements that directly describe comparison differences on physical carriers are retained as relativistic results within the evidence-effect domains of those results. An interpretation path in which reattribution of comparison differences on physical carriers to changes in an attribute of time itself directly depends on prior placement of ungenerated time constitutes a two-stage misrecognition by Definition 8.19. By Proposition 8.77, statements that assert the nonexistence of time are incompatible with statements that attribute stretching, rate, flow, or deformation to time itself when both target the same time-content type and the same application domain. When they target distinct time-content types or application domains, their separation and transition must be stated; a transition of the target to a clock-indication difference, clock-rate difference, proper-time difference, coordinate difference, or comparison difference between generated time representations does not entail a change in an attribute of “time itself” independent of physical carriers. Proposition 8.62 shows that relativity, relationality, nonprivilege, nonabsoluteness, or nonprimitivity does not imply the nonexistence of time. A theory that asserts the creation, emergence, recovery, or nonexistence of time bears the general formal obligations of Proposition 8.63. For Barbour 1994 I and II, Proposition 8.64 retains, as formal results, an interpretation restricted to a specified configuration construction, the nonadoption of absolute external time, and the kinematical priority of relative configuration. An interpretation that transfers time creation to complete generation of physical time constitutes a two-stage misrecognition, consisting of the use of primitive-time roots before generation and identification of a partial construction with complete generation; an interpretation that transfers a restricted nonexistence of time to the nonexistence of

physical time in general constitutes a one-stage misrecognition. For Connes–Rovelli 1994, Proposition 8.65 retains, as restricted results, the selection of a state-dependent flow and the assignment of a physical-time interpretation to that flow. An interpretation that transfers a partial flow to complete establishment of physical time constitutes a one-stage misrecognition; an interpretation that treats a construction using time roots before generation as complete generation from nontemporal inputs constitutes a two-stage misrecognition. For Kiefer 2009, Proposition 8.66 retains, as restricted results, the construction of a semiclassical time parameter and internal-clock candidates. An interpretation that transfers a partial construction to complete establishment of physical time constitutes a one-stage misrecognition; an interpretation that treats a semiclassical construction using time roots before generation as complete generation from nontemporal inputs constitutes a two-stage misrecognition. For Smolin 2018/2020, Proposition 8.67 retains, as a classification, an interpretation restricted to the theoretical priority or irreducibility of time. An interpretation under which time content before generation can be used as the grounding of a physical law or as a law input constitutes a one-stage misrecognition. For this source, neither the local-reach determination nor the complete determination yields any generated temporal content. All 27 temporal contents are primitive-time roots, and 21 of the 28 results depend on primitive-time roots.

By Proposition 8.73, the use within physical laws of the speed of light, the caesium-133 transition frequency, the Planck constant, the reduced Planck constant, the gravitational constant, vacuum permittivity, vacuum permeability, the Hubble rate, decay constants and lifetimes, linewidths and decay widths, Planck time, and the Compton angular frequency is established after the required time content and physical-time qualification have been generated. The physical relations borne by these quantities are retained as theory-specific inputs to the LCTR generation-and-audit construction, and predicates concerning their constancy, invariance, rates, lifetimes, linewidths, or frequencies are applied to generated time representations and theory-specific validity domains.

By Proposition 8.74, the quantification domain of light-speed invariance is determined as the domain in which generated law-admissible time representations, spatial representations, admissible representation transformations among inertial observers, clock and length standards, measurement procedures, and light-propagation relations hold. This quantification domain retains the constancy of the vacuum speed of light  $c$  as measured in local inertial frames and the local propagation limit for matter, signals, or causal influence; it does not include arbitrary coordinate speeds of light, cosmological recession velocities, phase velocities, or display quantities that carry no information in the same quantification domain of invariance. By Corollary 8.75, the two-track, four-part determination for the twelve time-bearing targets establishes generation-order determinations in 21 formal-track fields and separates invalidation positions according to the dependence order of each target.

By Proposition 8.59, direct measurements of physical carriers, attribution of properties to time itself, and application to living processes are separated as distinct evidential stages. An account that presupposes an unconstructed time entity, attributes comparison differences on physical carriers to changes in its properties, and further treats direct measurements established for physical carriers as direct demonstrations of living processes replaces the object of measurement, the bearer of physical action, and the evidence type twice. This account constitutes a relativistic three-stage misrecognition involving biological application. The evidential force of successful relativistic calculations and of experimental results concerning clocks, oscillators, electromagnetic radiation, and particle decay is not inherited by direct demonstration of living processes. The numbers of direct demonstrations



of living processes among the seven sources containing biological-application statements and among all 72 audited sources are 0/7 and 0/72, respectively. The Feynman 1995 source connects frequency shift, generalization to all physical processes, and application to human life and aging within a single source, thereby displaying this difference among evidential stages as one dependence sequence.

The Feynman 1995 and Narlikar 2010 sources add cases in which a predicate moves from clock, signal, and frequency relations to a time-scale or temporal-flow property. The Heisenberg 1958 source adds a statement of the nonexistence of time before the universe and a conceptual analysis according to which the vocabulary of time creation presupposes temporal relations. Time-property attribution, biological application, time nonexistence, and the self-dependence of time creation are thereby connected to the 72-source audit not merely as vocabulary dispersed among source groups, but as dependence sequences directly traceable within sources.

Time descriptions concerning the problem of time in quantum gravity, cosmological initial boundaries, the black-hole information problem, the problem of observables, the arrow of time, quantum-measurement records, and spacetime emergence are connected, as the time-problem connection domain of Definition 8.108, to the generation conditions, scale-indexed validity domains, six failure series, and audit reports of their respective time content. By Proposition 7.170, a result that removes privileged time from a relational formalism and the obligation to generate clocks, records, comparisons, order, scale, and qualification for use within laws from nontemporal physical inputs belong to different evidence-effect domains.

Proposition 8.76 applies the unbounded inheritance of limited results, defined in Chapter 7, to thermodynamics, cosmology, quantum measurement, quantum gravity, relativity, biological application, and time-bearing quantities. The mathematical consistency, predictive performance, model fit, and empirical fit of each field are retained within their evidence-effect domains; their force is not automatically inherited by the physical generation of time content, the physical-time qualification of law time, a time attribute independent of physical carriers, direct demonstration of an unmeasured target, or an ontology of time. Section 7.3.18 connects the evidential roots of this cross-field collation to the formal evidence and statement evidence of all 72 audited sources.

**Synthesis (8): differences from existing theories and result-preserving re-grounding.** The equations of motion, interaction terms, field equations, quantization procedures, correction rules, and engineering implementations of existing theories are retained as theory-specific inputs to the LCTR generation-and-audit construction. By generating the time content used by time variables and its grounding structure, and by auditing their validity domains and invalidation positions, existing theories are safely reused within the range in which the generation conditions hold. This gives the result-preserving re-grounding of Definition 8.99.

By Proposition 8.79, a primitive-time-input audit separates the determination of physical-time qualification from the determination of qualification for use as a physical computational result when actual invalidation of a generation condition or condition of establishment reaches a downstream result. The presence of primitive-time input and its participation in result derivation determine the former, and the qualification invalidation remains after mathematical consistency, predictive performance, and empirical fit have been established. The latter requires that the condition backgrounded by primitive-time input as a grounding obligation, the condition whose actual invalidation is established by independent evidence, and the condition that forms the starting point of downstream

dependence be the same. When this causal connection holds, the result loses qualification for use as a physical computational result in the evaluation domain and is regenerated or recomputed from repaired inputs. Whether the recomputed value agrees with or differs from the prior value is determined by the result of recomputation after repair.

When existing physics uses time-reference roles, physical carriers, recording arrangements, information coupling, and joint-law establishment conditions as background conditions outside its equations, without making them explicit as variables, relations, generation conditions, or failure conditions of those equations, the mathematical consistency, predictive performance, and empirical fit of the theory do not supply the physical generation of the backgrounded time-reference conditions. The classical multi-object audit makes these background conditions explicit as audit inputs and separates the structural burdens and invalidation signatures of the respective recording arrangements.

For each engineering implementation, if a role-typed realization by physical carriers, an engineering evaluation specification, typed engineering measurements, evidence-provenance certificates, and verified evidence for engineering formal conditions are given, then the use of time-dependent laws in that implementation is evaluated by the same six-series, five-valued audit. Applicability is determined not by the name of the field but by the establishment of typed correspondences and verified evidence connecting theoretical conditions to engineering measurements.

**Synthesis (9): typed generation-and-audit architecture.** By Definition 8.100 and Proposition 8.101, the type structure, generation structure, use-qualification structure, invalidation-audit structure, theoretical and engineering evidence structure, and cross-source audit structure form a single typed dependence structure. For every generated object, its input type, formation conditions, output type, and downstream conditions of use are determined; upon invalidation, the minimal invalidation positions, range of effects, target of repair, and scope of downstream regeneration are determined. The design methodology of computer science is realized as the mathematical and logical structure of LCTR that specifies types, dependence order, audit states, evidence provenance, and change effects.

**Synthesis (10): final consequence.** By Synthesis (1)–Synthesis (9), temporal comparison, temporal order, time representation, dynamical description, law time, and differential-law form are given as distinct stages of one generation dependence beginning with non-temporal physical inputs. In a physical case in which every stage holds, the physical grounds and the range of safe use of a time variable are obtained. For a differential equation that uses continuous approximation, the full evaluation-validity domain, maximal effective interval for approximation, and approximation boundary give its physical use qualification and scale limit. Under the classical-continuum-regime assumptions, LCTR changes the stage of establishment of classical continuous time from a primitive premise of law description to a classical continuous-approximation time representation generated from a complete evaluation specification and established at every scale within the maximal effective interval for approximation. In a case of invalidation, the six failure series, scale boundary, invalidation signature, and five-valued audit report separate the minimal invalidation positions, causal type, unresolved range, target of repair, and regeneration order of downstream results. These outputs constitute a conditional operational warranty for continuous approximation and differential forms that returns the domain of use and the cause of invalidation.

LCTR changes the stage at which temporal identity, before–after relations, and real-valued time variables become available for use in physical laws. Whether two physical

materials may be treated as belonging to the same time content, which may be treated as before or after, when incomparable materials may be quotiented into a single linear order, and when that order may be represented by a real-valued time variable are determined by satisfaction of generation conditions prior to law description. Accordingly, temporal identity, simultaneity based on synchronization procedures, the order used for initial conditions and trajectory evolution, a single time axis, and real-valued time variables are not descriptive resources that a theory may use unconditionally; they acquire qualification for use in domains where the corresponding generation conditions hold. Freedom in the choice of an admissible time coordinate is determined as a unique strict order isomorphism between representation images of the generated temporal structure, so freedom of coordinate choice and establishment of the temporal structure become distinct obligations.

The use of a single time variable for multiple objects is itself subject to an independent establishment obligation. Use of a common time representation, synchronization, same-time judgments, and a joint law-time domain requires satisfaction of the conditions on the physical-carrier realization of each object, record field of view and record stability, observer-side information coupling, communication, multi-object relations, and scale-indexed defect quantities that apply to the objects and configuration in question. Remote or global constructions involving transport or gluing additionally require transport consistency, loop consistency, quotient descent, and gluing.

LCTR's framework, which grounds the establishment of time in generation conditions, reallocates the obligations assigned to time in each field. In time metrology, periods, frequencies, standard frequencies, and representations in hertz are not premises of time generation but metrological results obtained from cycle counts, record correspondences, comparison rules, generated time differences, and normalization. This generation order blocks the circularity in which frequency is presupposed as an input to time generation while an ungenerated time difference is used in its definition. In special relativity, LCTR does not rederive Lorentz transformations; it retains them as theory-specific SR inputs that relate generated law-admissible time representations and spatial representations. In general relativity, establishment of a local proper-time representation does not imply establishment of a global law-time representation; the latter requires downstream conditions including transport, loop consistency, quotient descent, and gluing. Construction of causal order and generation of temporal order are likewise distinct obligations. Providing causal order as a primitive partial order or a pre-given spacetime structure does not entail generation of clock scales, time intervals, or law time. In quantum gravity, elimination of external time or selection of an internal clock is an obligation distinct from generation of physical time and establishment of its qualification for use within laws. LCTR changes the problem of time in quantum gravity from the problem of eliminating external time or selecting an internal clock to the problem of whether the time content in use has been generated from nontemporal physical inputs, satisfies law-descriptive operability, and has acquired qualification for input into physical laws. The arrow of time is not a ground for generating time but a downstream orientation condition on an already generated time representation.

For a theory that literally asserts the creation, emergence, recovery, or nonexistence of time, LCTR imposes as formal obligations the time-content type and domain at issue, a generation path from nontemporal physical inputs or an alternative structure, elimination of primitive-time roots from all time content, and elimination of temporal-dependence roots from downstream results. LCTR thereby changes assertions of time creation, time emergence, and time nonexistence from verbal claims into claims in physics whose establishment can be audited by formal and statement evidence.

The consequences of primitive-time input are separated into invalidation of physical-time qualification and invalidation of qualification for use of downstream computational results reached by actual invalidation of a backgrounded grounding obligation. The former is compatible with mathematical consistency, predictive performance, and empirical fit. The latter holds through the backgrounding of a grounding obligation by primitive-time input, independent evidence establishing actual invalidation of the same condition, and downstream dependence from the invalidation position to result derivation. When the latter holds, downstream results are regenerated or recomputed from repaired inputs, and the recomputed values are evaluated anew. The domain in which existing results can be retained safely is the intersection of the evidence-effect domain specific to each result and the validity domain given by the generation conditions and conditions of establishment.

Cross-field collation of the 72 audited sources shows that descriptions in which the force of limited computational success, predictive success, model fit, or experimental success is inherited by the physical establishment of an entire mathematical expression, the physical-time qualification of an ungenerated time variable, a time attribute independent of physical carriers, application to an unmeasured target, or an ontology of time recur across multiple fields from antiquity to the present. This recurrent transfer of the force of limited results from their result-specific evidence-effect boundaries, successively to the physical establishment of an entire mathematical expression, the physical-time qualification of an ungenerated time variable, a time attribute independent of physical carriers, application to an unmeasured target, or an ontology of time is a common cross-field misrecognition in physics. It is the unbounded inheritance of limited results in Definition 7.116. By Proposition 8.76, this inheritance is localized to the evidence-effect boundary of each field. Retaining each result within its evidence-effect domain, generating time content from nontemporal physical inputs, and auditing its validity domain and invalidation position determine the domain in which existing results can be reused safely.

A time variable in a physical law is not established as a primitive premise. The 72 audited sources show that mathematical consistency, predictive performance, or empirical fit does not remove primitive-time roots. In every audited source from antiquity to the present, at least one item of time content was entered into law operations or result derivations before the physical grounding structure required for its use had been generated from nontemporal physical inputs. This unanimity shows that the unfulfilled obligation to ground physical time content has recurred historically in the uses of time throughout physics. When such content is replaced by generated time representations, the basis of mysteries constituted by presupposing ungenerated time is lost, and each problem is transferred to generation conditions, validity domains, failure classifications, and the audit evidence to be acquired next.

*Proof.* Synthesis (1) follows from Theorem 4.90. That theorem makes joint satisfaction of  $\mathbf{L}_{\leq 7}[\mathbf{Ev}^{\text{cmp}}]$  and  $\mathbf{G}[\mathbf{Ev}^{\text{cmp}}]$  equivalent to operability at the comparison stage and supplies, in dependence order, the extended comparison-arrival equivalence relation, the  $\mathfrak{C}/\mathfrak{D}$  comparison-quotient sets, the canonical projections, the quotient-level  $\mathfrak{C}\mathfrak{D}$ -derived relation, and the  $\mathfrak{C}\mathfrak{D}$ -derived canonical comparison quotient.

Synthesis (2) follows from Theorem 4.147, Theorem 4.177, Theorem 4.179, and Theorem 4.187. From satisfaction of  $\mathbf{Repr}_{\leq 3}(\mathcal{X}; \mathcal{J}_{\text{cmp}})$ , they supply, in dependence order, the  $\mathfrak{C}/\mathfrak{D}$  canonical partial orders, exact operational time structure, local and global order-coordinate structures, canonical order-coordinate projections, local-to-global order-coordinate embeddings, and comparison-order atlas. When a selected representation range and an order embedding of that range are specified, the  $\mathfrak{C}/\mathfrak{D}$ -derived representation coordinates factor

through the canonical global order-coordinate projection.

Synthesis (3) follows from Theorem 5.112 and Theorem 5.138. The two theorems make dynamical-description operability equivalent to joint satisfaction of exact operability and all dynamical-description conditions, and generate the  $\mathfrak{C}$ -derived observer-time quotient, the  $\mathfrak{B}$ -derived object-state quotient, the  $\mathfrak{CDB}$ -derived trajectory relation, a graph trajectory, a canonical family of observables, the observer-time order-coordinate quotient, a real-valued time representation, a scalar-time trajectory, and a family of real-valued observable curves.

Synthesis (4) follows from Theorem 5.113, Theorem 5.135, and Theorem 7.203. The first theorem supplies the observer-time law core and the generated representationality of law-evaluation time; the second supplies chart-compatible differential laws, value-jet-relation covariance, and time-representation-change jet-relation covariance. By the third theorem, generation of time content and its grounding structure from nontemporal physical inputs before law operations and result derivations is necessary for the physical qualification of a physical-law time variable. A time variable in a physical law is therefore not established as a primitive premise.

Synthesis (5) follows from Definition 6.10 and Theorem 6.168. Within each of the comparison, representation, dynamical-description, continuous-approximation, law-description, and differential series, invalidation of a formed and evaluable condition is localized to the corresponding invalidation-position token and gives a six-series invalidation signature. Because multiple invalidation positions can hold in the same evaluation, the six series do not require an exclusive disjunction. Unformed and unevaluable states are not replaced by invalidation; they are retained as their respective audit states.

Synthesis (6) follows from Proposition 6.162, Proposition 6.154, Theorem 6.163, Theorem 6.164, and Theorem 6.183. The first proposition gives the full evaluation-validity domain as the intersection of the strict validity domain and the approximation-validity domain. The second separates the objects of determination and input types of the continuous-approximation and differential series. By Theorem 5.138, differential-stage operability includes joint satisfaction of law-description-stage operability and the five universal differential conditions. The physical use qualification of a differential equation employing continuous approximation is therefore given as the full evaluation-validity domain in which the five exact series and the continuous-approximation series hold simultaneously.

Theorem 6.163 and Theorem 6.164 supply the approximation-validity scale domain, maximal effective interval for approximation, approximation-invalidation scale, approximation-boundary signature, and first-invalidation report. Theorem 6.183 supplies the five-valued audit-report object whose components are series-specific audit-group states, condition-specific audit states, invalidation positions, the six-series invalidation signature, indeterminate tokens, the indeterminate signature, the approximation boundary, structural burden, and reasons for an undefined audit. Definition 6.179 gives satisfaction, condition invalidation, blockage by a prior condition, an unformed state, and indeterminacy as distinct state values.

The  $\mathfrak{D}$ -derived record-cell width belongs to the engineering-audit input datum of Definition 6.194. By Proposition 6.204, if the audit state of verified engineering formal-condition evidence containing its engineering-measurement reference is invalidation, the invalidation position of the theoretical condition indicated by the target component of that evidence is localized within the six series. Proposition 6.199 connects an engineering auxiliary-audit cause, through verified effect evidence, to a formation input, evaluation input, input admissibility, boundary treatment, or condition evidence in the existing six series and separates the type of theoretical-condition invalidation from that of an

engineering auxiliary-audit cause.

Synthesis (7) follows from Theorem 7.190, Proposition 7.191, Theorem 8.16, and the respective worked reductions listed in Table 24. Every one of the 72 audited sources contains primitive-time roots and at least one result dependent on primitive-time roots. This does not mean that every result in each source depends on primitive time roots. By Proposition 7.188, the set of results beyond generation reach is the disjoint union of the evidence-environment-reachable nongenerated result set and the primitive-time-root-dependent result set. Under the complete criterion, the generated-time-content set is empty for all 72 sources, and no source generates, from nontemporal physical inputs, the entirety of the time content it uses. By Proposition 7.189, under the local-reach criterion, 10 sources reach a total of 57 temporal contents. The strict generated-time-content sets of all 72 sources are empty, and the entire source-specific time content consists of primitive-time roots. In the remaining 62 sources, no time content is reached even under the local-reach criterion. Sources that explicitly state the absence of external or independent time, the fundamental status of time, difficulty defining the concept of time, or an internal or relational construction of time number (18/72); sources that present a dynamical construction or recovery of intrinsic, relational, thermal, or semiclassical time number (13/72); and among the latter, sources that satisfy the complete time-generation obligation number (0/13).

By Definition 7.10, the universal persistence of primitive-time roots means that in each source at least one item of time content was entered into law operations or result derivations before the physical grounding structure required for its use had been generated from nontemporal physical inputs. The audited source set extends from ancient definitions of time to modern theories, experiments, metrological standards, and engineering materials. This unanimity therefore shows that the unfulfilled obligation to ground physical time content has recurred historically in the uses of time throughout physics.

Theorem 8.16 and the respective worked reductions connect Newtonian and classical mechanics, time metrology, GPS, special relativity, general relativity, and quantum-gravitational reconstruction to the generation conditions, validity domains, and invalidation positions of time content. Definition 8.19 retains the path that directly describes comparison differences on physical carriers as a relativistic result, and classifies a path in which reattribution to a time attribute directly depends on prior placement of ungenerated time as a two-stage misrecognition. Theorem 7.190 establishes that primitive-time roots remain in all 72 sources. Proposition 7.167 distinguishes formal results from statement targets. Proposition 8.77 collates time-nonexistence statements and time-attribute statements after aligning their time-content types and application domains, and gives incompatibility when both are the same and, when they differ, an obligation to state the transition in target type or application domain.

Proposition 8.73, from Proposition 7.178 and Remark 7.179, gives post-generation use qualification for time-bearing quantities and retention of physical relations as theory-specific inputs. Proposition 8.74 connects the three determinations on the formal track of the speed of light to admissible transformations among generated time representations in special relativity, and gives a quantification domain of light-speed invariance that separates the vacuum speed of light and local propagation limit in local inertial frames from coordinate speeds of light, cosmological recession velocities, phase velocities, and display quantities that carry no information. Corollary 8.75 connects the 96 determination fields of Proposition 7.182 to the cross-field consequences of Chapter 8.

Proposition 8.59 connects the separation of evidential stages in Section 7.3.17 and the

range of direct demonstrations in Section 7.4 to a three-stage misrecognition in relativistic explanation. This separates direct measurements of physical carriers, attribution of properties to time itself, and application to living processes, and localizes the compression of the evidential range that treats a direct measurement at the first stage as a direct demonstration at the third. No direct demonstration of living processes occurs in either the seven sources containing biological-application statements or all 72 audited sources; the counts are (0/7) and (0/72), respectively. The Feynman 1995 source connects frequency shift, generalization to all physical processes, and application to human life and aging within a single source.

The Feynman 1995 and Narlikar 2010 sources show predicate transfer from clock, signal, and frequency relations to a time-scale or temporal-flow property. The Heisenberg 1958 source distinguishes a statement of the nonexistence of time before the universe from the presupposition of temporal relations by the vocabulary of time creation. The Bombelli–Lee–Meyer–Sorkin 1987 source distinguishes the construction that relates primitive causal order to macroscopic past–future relations from the complete generation of clock scales, time intervals, and law time. This statement evidence adds time-property attribution, biological application, time nonexistence, the self-dependence of time creation, and separation of the obligations for causal order and complete time generation to the 72-source audit as dependence sequences traceable within sources.

Definition 8.108 connects time descriptions concerning the problem of time in quantum gravity, cosmological initial boundaries, the black-hole information problem, the problem of observables, the arrow of time, quantum-measurement records, and spacetime emergence to the generation conditions, scale-indexed validity domains, six failure series, and audit reports of their respective time content. Proposition 7.170 separates a result that removes privileged time from a relational formalism from the obligation to generate physical time from nontemporal physical inputs. Proposition 8.72 connects results concerning atomic-clock differences, molecular-vibration-period differences, covalent-bond vibration sequences, electronic-state transitions, scattering events, decay events, the cosmological constant, and dark energy to the generation conditions, validity domains, minimal invalidation positions, and regeneration or recomputation after repair of the time content on which they depend.

Proposition 8.63 requires a theory that literally asserts the creation, emergence, recovery, or nonexistence of time to provide a generation path or alternative structure, elimination of primitive-time roots, and elimination of temporal-dependence roots from downstream results. These statements therefore become claims in physics whose establishment is judged by satisfaction of the formal obligations specified in that proposition.

Proposition 8.68 shows that even after external time has been excluded from raw operational input, the law-description stage requires law-description operability, while use of each temporal content within laws requires prior establishment of the generated structures needed for that use. The problem of time in quantum gravity therefore becomes a problem of judging generation of time content and establishment of law-time use qualification, rather than only elimination of external time or selection of an internal clock.

Definition 7.115, Definition 7.116, Proposition 7.118, and Corollary 7.119 give the evidence-effect domain of a limited result, the nonestablishment of claims beyond its boundary, and the difference between this audit axis and primitive-time dependency inheritance. Proposition 8.76 applies them to the physical fields considered in Chapter 8. Retaining the limited results of each field is therefore separated from inheriting those results into qualification of ungenerated time, time attributes, unmeasured targets, or an ontology of time. By Section 7.3.18, the statement evidence and interpretive possibilities that

determine this separation are connected to the audit method or representative arguments of Chapter 7 for all 72 audited sources, not only the representative sources.

Synthesis (8) follows from Proposition 8.95, Proposition 8.97, Proposition 8.102, and Proposition 8.103. The structures addressed by clock synchronization, relativistic time-stretching representations, proper time in general relativity, relational time, quantum-theoretical time reconstruction, time metrology, and operationalism are mapped to stages of the generation dependence or to inputs of representative schemes. The equations of motion, interaction terms, field equations, quantization procedures, correction rules, and engineering implementations of existing theories are retained as theory-specific inputs to the LCTR generation-and-audit construction. By generating time content and its grounding structure and auditing their validity domains and invalidation positions, existing theories are reused safely within the range in which the generation conditions hold.

A classical world representation does not describe time-reference roles, physical carriers, recording arrangements, information coupling, or joint-law establishment conditions as variables, relations, generation conditions, or invalidation conditions in law expressions; it enters them in advance as background conditions outside the expressions. By Proposition 8.103, mathematical consistency, predictive performance, and empirical fit do not supply the physical generation of backgrounded time-reference establishment conditions.

Definition 3.2 gives role-typed realization by physical carriers, while Definition 6.188 and Definition 6.194 give engineering evaluation specifications and typed engineering measurements. Proposition 6.204 localizes the target components of verified engineering formal-condition evidence to invalidation positions in the six series. For every engineering implementation supplied with these typed correspondences and evidence, its use of time-dependent laws can therefore be connected to the same six-series, five-valued audit.

Proposition 7.121 connects backgrounding, actual invalidation, downstream reach, repair, and recomputation into one causal sequence when the grounding obligation backgrounded by primitive-time input, the condition whose actual invalidation is established by independent evidence, and the condition forming the starting point of dependence reaching downstream result derivation are the same. In the absence of this same-condition connection, primitive-time input and condition invalidation are retained as separate audit results. Proposition 8.79 separates invalidation of physical-time qualification, determined by the presence of primitive-time input and its participation in result derivation, from invalidation of qualification for use of downstream computational results, determined by this causal sequence. Proposition 8.78 gives a direct connection to recomputation and precision evaluation after repair in time metrology and GPS; a connection in Newtonian and classical mechanics, special relativity, and general relativity when the corresponding error cause is included in the minimal invalidation positions; and a connection in quantum gravity when a concrete approximation or estimation error is localized to an invalidation position. The safe reuse domain of existing results under result-preserving re-grounding is therefore determined as the intersection of the evidence-effect domain specific to a result and the validity domain given by the generation conditions and conditions of establishment.

Synthesis (9) follows from Definition 8.100 and Proposition 8.101. The type, generation, use-qualification, invalidation-audit, theoretical and engineering evidence, and cross-source audit structures connect input types, formation conditions, output types, downstream conditions of use, audit states, evidence provenance, and scopes of regeneration by dependence edges.

Synthesis (10) is the logical composition of Synthesis (1) through Synthesis (9). Synthesis (1) through Synthesis (4) supply the generation dependence from nontemporal



physical inputs through temporal comparison, temporal order, time representation, dynamical description, law time, and differential-law forms. Synthesis (5) and Synthesis (6) separate their validity domains, invalidation positions, and unresolved ranges through the six failure series, scale boundaries, invalidation signatures, and five-valued audit reports. Synthesis (7) supplies the persistence of primitive-time roots in the 72 audited sources and concrete connections to representative theories; Synthesis (8) supplies the objects retained from existing theories, the separation of invalidation of physical-time qualification from conditional invalidation of use of downstream computational results, and the conditions for safe reuse. Synthesis (9) supplies these as a single mathematical and logical structure connecting types, generation, use qualification, audit states, evidence provenance, ranges of effects, and scopes of regeneration.

In a classical continuous-approximation regime, Corollary 8.89 establishes that a classical continuous-approximation time representation is generated from a complete evaluation specification. Theorem 8.88 establishes that it holds at every scale in the maximal effective interval for approximation. Through this generation and validity-domain guarantee, LCTR changes the stage of establishment of classical continuous time from a primitive premise of law description to a generated approximation representation that holds within the maximal effective interval for approximation.

Hence, in a physical case in which every stage holds, the physical grounds and range of safe use of a time variable are obtained; in a case of invalidation, the invalidation positions and unresolved range become auditable. A time variable in a physical law is not established as a primitive premise. When a time variable presupposed as a black box without a generation path is replaced by a generated time representation, the basis on which problems presupposing ungenerated time appear as independent mysteries is lost, and each problem is transferred to generation conditions, validity domains, failure classifications, and the audit evidence to be acquired next.  $\square$

**Corollary 8.110** (Conditional physical use qualification for continuous approximation and differential forms). For a scale-indexed complete evaluation specification  $(\mathbf{Ev}, \lambda) \in \mathbf{EvSpec}_\Lambda$  that uses a differential equation employing continuous approximation as a physical description,

$$\begin{aligned} (\mathbf{Ev}, \lambda) &\in \mathcal{V}^{\text{full}} \\ \iff [\mathbf{Ev} \in \mathcal{V}^{\text{strict}} \wedge (\mathbf{Ev}, \lambda) \in \mathcal{V}^{\text{approx}}] \\ \iff K_{\text{full}}^{\text{fail}}[\mathbf{Ev}; \lambda] &= \top \end{aligned}$$

holds. The strict-establishment side includes the five series of comparison, representation, dynamical description, law description, and differentiation; the approximation-establishment side includes the nine continuous-approximation conditions. The physical use qualification of a differential equation employing continuous approximation therefore requires establishment of both continuous approximation and differential form.

The set of scales at which the qualification holds is given by the full evaluation-validity domain, and the maximal effective interval of its approximation component is the largest initial subset of the approximation-validity scale domain. Under the hypotheses of Theorem 6.163, if the complement of that scale domain has a least element, that element is the first approximation-failure scale. When condition invalidation occurs, the first-invalidation report and structural burden give each minimal invalidation position and the affected downstream objects, and the invalidation-generation dependence order gives the regeneration order after repair. When blockage by a prior condition, an unformed

state, or indeterminacy occurs, the state and its reason are retained in the five-valued audit report.

*Proof.* The first equivalence follows from Definition 6.4 and Proposition 6.162. The second equivalence follows from Definition 6.27 and Definition 6.4. The separation of the five exact series from the continuous-approximation series follows from Definition 6.10 and Proposition 6.154. That the differential stage includes the law-description stage and the five universal differential conditions follows from Definition 5.132.

The maximal effective interval for approximation and first exceedance are obtained from Theorem 6.97; the minimal invalidation positions and approximation scale boundary, from Theorem 6.164; structural burden, from Proposition 6.171; and the five-valued audit states and their reasons, from Theorem 6.183 and Corollary 6.187.  $\square$

**Proposition 8.111** (Connection of the final synthesis to engineering audit and classical world representations). The theoretical-establishment audit layer of Theorem 8.109 has as components the six failure series, invalidation positions, invalidation signatures, scale-indexed validity domains, and five-valued audit reports. The engineering-audit layer of Definition 6.194 has as components engineering-measurement type tags, engineering-measurement references, typed engineering-measurement values, and evidence-provenance certificates.

Through the verified engineering formal-condition evidence of Proposition 6.204, the invalidation positions of theoretical conditions corresponding to engineering-audit results for physical-carrier realization, record field of view, record invasiveness, observer-side information coupling, joint relations, and common law domains are determined. The classical multi-object audit example of Definition 8.104 and the classical world representation of Proposition 8.102 make concrete the evidence correspondence between these two layers.

For each engineering implementation, if a role-typed realization by physical carriers, an engineering evaluation specification, typed engineering measurements, evidence-provenance certificates, and verified evidence for engineering formal conditions are given, then the use of time-dependent laws in that implementation is evaluated by the same six-series, five-valued audit.

*Proof.* Definition 6.194 and Definition 6.195 supply engineering-measurement references, typed engineering-measurement values, engineering-implementation auxiliary-audit flags, and their evidence references as components of the engineering-audit layer. Definition 6.10 supplies the invalidation positions of theoretical conditions belonging to comparison, representation, dynamical description, continuous approximation, law description, and differentiation as components of the theoretical-establishment audit layer.

Proposition 6.199 treats verified engineering auxiliary-audit-effect evidence as the three components of an engineering auxiliary-audit flag, an engineering auxiliary-audit effect, and a target invalidation-position token, and maps the set of target components to the invalidation-position-token set. Proposition 6.204 localizes only the target components of verified engineering formal-condition evidence to the invalidation positions of theoretical conditions. Engineering measurements, auxiliary-audit flags, and auxiliary-audit effects therefore belong to the engineering-audit layer, while the invalidation positions of the corresponding theoretical conditions belong to the theoretical-establishment audit layer.

Table 22 and Proposition 8.107 show that the bulk-recording and following-recording types can have different invalidation signatures. Proposition 6.206 separates the structural burdens of the two recording types relative to engineering evidence.

By Proposition 8.102, a classical world representation enters time-reference establishment conditions in advance as background conditions outside the expressions. By Proposition 8.103, the mathematical consistency, predictive performance, and empirical fit of classical physics do not supply their physical generation. The same proposition restricts the domain in which multi-object relations hold under the same time variable to the scale-indexed validity domain satisfying the conditions on physical-carrier realization,  $\mathfrak{D}$ -derived record field of view, record stability, observer-side information coupling,  $\mathfrak{D}$ -communication, multi-object relations, and scale-indexed defect quantities.

Thus, the measurements, flags, effects, and structural burdens of the engineering-audit layer are connected to the invalidation positions, invalidation signatures, scale-indexed validity domains, and five-valued audit reports of the theoretical-establishment audit layer through the target components of verified evidence. Every engineering implementation supplied with a role-typed realization by physical carriers, an engineering evaluation specification, typed engineering measurements, evidence-provenance certificates, and verified evidence for engineering formal conditions belongs to the domain of this connection. Its use of time-dependent laws is therefore evaluated by the same audit-report generation map.  $\square$

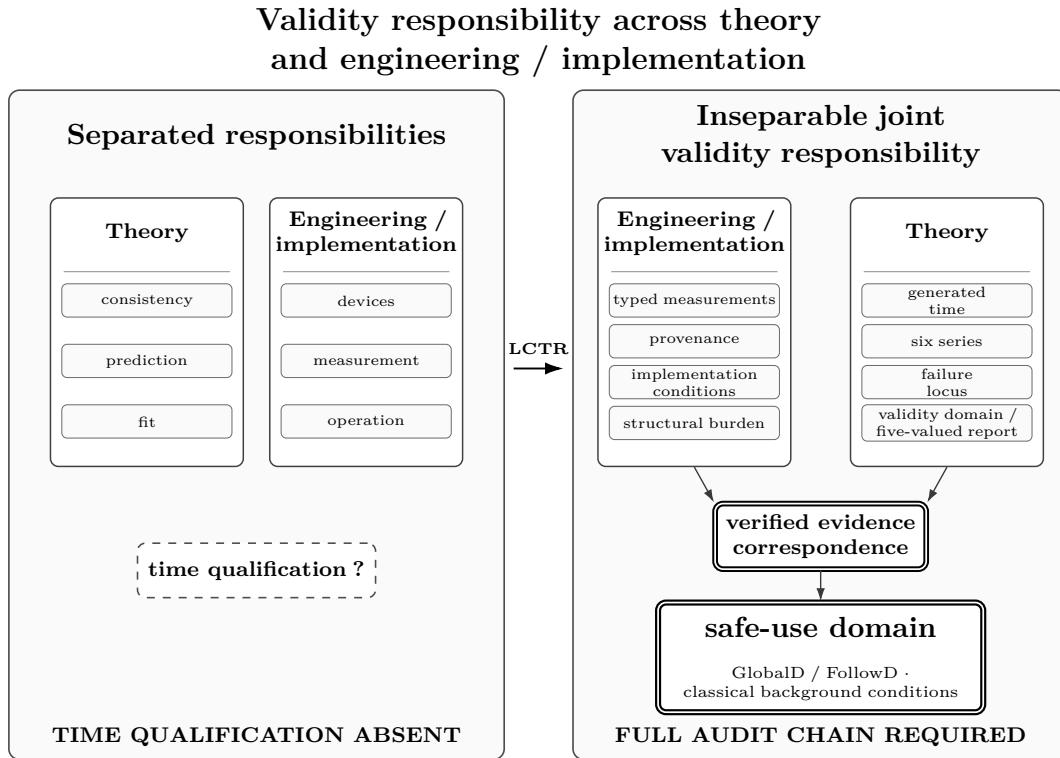


Figure 31: Connection of the final synthesis to engineering audit and classical world representations. The engineering measurements, auxiliary-audit flags, auxiliary-audit effects, and structural burdens of the engineering-audit layer are distinguished from the six failure series, invalidation positions, invalidation signatures, scale-indexed validity domains, and five-valued audit reports of the theoretical-establishment audit layer, and the two layers are connected through target components of verified evidence.

**Consequence after the final synthesis.** By Theorem 8.109, Corollary 8.110, and Proposition 8.111, the establishment of a time variable appearing in a physical law requires

the generation conditions of comparison, representation, dynamical description, and law description needed for the use of that temporal content. Use of differential law forms requires the differential-stage conditions, and claims using continuous approximation require a scale-indexed validity domain. A differential equation employing continuous approximation requires the dual qualification of continuous approximation and differential form. An invalidated or unresolved position is localized to the six failure series, invalidation signature, approximation boundary, structural burden, and five-valued audit report; condition invalidation, blockage by a prior condition, an unformed state, indeterminacy, and an engineering auxiliary-audit cause are separated by type.

By Theorem 7.203, a time variable in a physical law is not established as a primitive premise. By Theorem 7.190, every one of the 72 audited sources contains primitive-time roots and results dependent on primitive-time roots. Safe reuse of results from existing theories therefore requires generation of time content and its grounding structure from nontemporal physical inputs and audit of their validity domains, invalidation positions, and unresolved ranges.

# Formal-Verification Assets

The supplementary archive `lctr-verification.zip` contains formal-verification assets for Sections 2–6. Its top-level directory after extraction is `lctr-verification/`. The assets include a complete inventory of the mathematical definitions, formulas, claims, and proof obligations in this scope for the same version as the published PDF, the Lean and Isabelle formalization sources, and a correspondence table linking locations in the Japanese and English PDFs to both formalizations. The table records types, quantification, premises, domains, all conclusions, and dependencies component by component, and links each formula occurrence to the mathematical object it expresses.

The mathematical proof obligations in this scope are formalized in both Lean and Isabelle, and their proofs are checked by the Lean kernel and the Isabelle/Pure inference kernel. Lean proofs are additionally rechecked by the separate implementations `nanoda` and `con-leche`. For each formalized object in Lean and Isabelle, the assets include verification results, the axioms used, and theorem dependencies.

For each proof obligation in the text, the assets include a comparison tracing the proof bodies, definitions used, actual theorem dependencies, and local assumptions in both formalizations. The comparison links the premises and conclusions in the text to the dependencies and dependent-type or carrier conditions in both formalizations.

The partial output map of a law-candidate relation has machine-readable representations in MMT, OpenMath, and Content MathML, together with the results of checking these representations. The assets also include SMT proofs and checks of their proof traces, symbolic computations using computer algebra systems, unit and dimension checks, and results from checking counterexamples and boundary cases, together with the conditions under which each check applies. Each result is linked to its object in the text and records the domain and conditions checked.

The release package contains pinned version information for the verifiers and their library dependencies, the inputs and outputs of every included check, reports for each object, and instructions for running each standard tool directly. To establish file identity, the accompanying file manifest associates the name of each published PDF, formalization source, check input, verification result, and rerun script with its SHA-256 hash. The `reproduce.sh` script runs the included checks and regenerates their reports in the pinned verification environment. The release metadata identifies the verification-assets version corresponding to this PDF, its scope, and the checks performed.

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